

GRADE 6 COMMON CORE MATH

Answer Key

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Write a ratio that describes the relationship between two quantities

MAIN	Grade 6 main question	6.RP.A.1
BACKWARD 1	Reading counts from a table or graph	3.MD.B.3
BACKWARD 2	Parts and totals	2.OA.A.1
BACKWARD 3	Fraction of a whole	3.NF.A.1
FORWARD 1	Ratios and unit rates with fractional quantities	7.RP.A.1
FORWARD 2	The unit rate as the slope of a graph	8.EE.B.5

SET 1 • 6.RP.A.1

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.1

A fruit bowl has 5 pears and 8 plums.

Write the ratio of pears to plums.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q1/5

9

QUESTION ID

S1-M-Q1

TYPE

Short answer

CORRECT ANSWER

5 : 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q1/5

A choir has 14 sopranos and 9 altos.

Write the ratio of altos to sopranos.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q2/5

10

QUESTION ID

S1-M-Q2

TYPE

Short answer

CORRECT ANSWER

9 : 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q2/5

The table shows the pets owned by students in a class.

Write the ratio of dogs to fish.

Pet	Dogs	Cats	Fish
Students	11	6	4

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q3/5

11

QUESTION ID

S1-M-Q3

TYPE

Short answer

CORRECT ANSWER

11 : 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q3/5

A parking lot has 13 cars and 7 trucks. There are no other vehicles.

Write the ratio of trucks to all of the vehicles.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q4/5

12

QUESTION ID

S1-M-Q4

TYPE

Short answer

CORRECT ANSWER

7 : 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q4/5

A bag has 20 marbles. 12 are green and the rest are yellow.

Write the ratio of yellow marbles to green marbles.

Ratio: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q5/5

13

QUESTION ID

S1-M-Q5

TYPE

Short answer

CORRECT ANSWER

8 : 12

GRADING NOTE Equivalent ratio 2 : 3 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 1 • MAIN • Q5/5

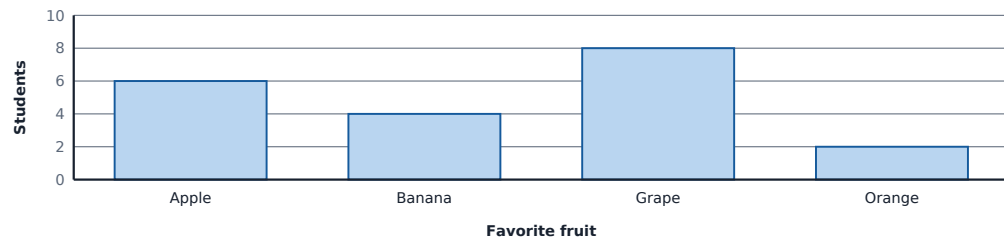
SET 1 • 6.RP.A.1

BACKWARD 1

Reading counts from a table or graph • Grade 3 • Measurement & Data •
3.MD.B.3

The graph shows the favorite fruits of a class.

How many students chose grape?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q1/5

15

QUESTION ID

S1-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

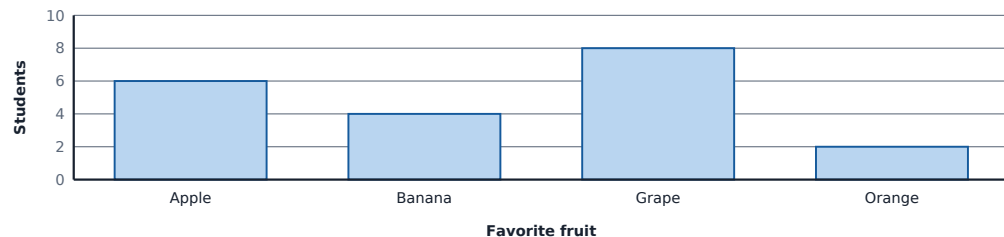
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q1/5

The graph shows the favorite fruits of a class.

How many students chose banana?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q2/5

16

QUESTION ID

S1-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q2/5

The table shows the pets owned by students.

How many students own cats?

Pet	Dogs	Cats	Fish
Students	11	6	4

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q3/5

17

QUESTION ID

S1-B1-Q3

TYPE

Short answer

CORRECT ANSWER

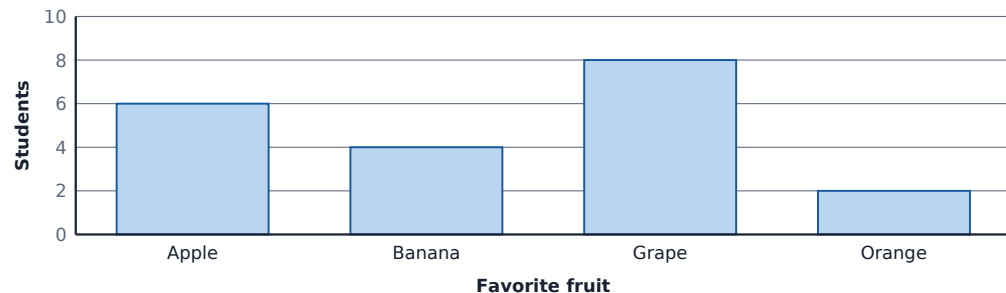
6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q3/5

Which fruit was chosen by 2 students?



A. Grape B. Orange C. Banana D. Apple

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q4/5

18

QUESTION ID

S1-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Orange

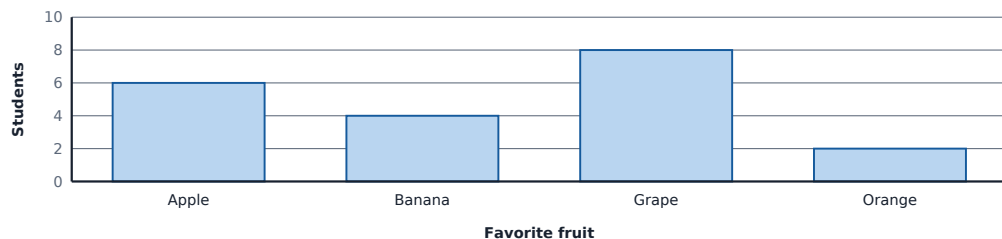
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q4/5

True or false?

6 students chose apple.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q5/5

19

QUESTION ID

S1-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 1 • BACKWARD 1 • Q5/5

SET 1 • 6.RP.A.1

BACKWARD 2

Parts and totals • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

A team has 9 boys and 6 girls.

How many players are on the team?

Players: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q1/5

21

QUESTION ID

S1-B2-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q1/5

A box holds 24 crayons. 15 are broken.

How many crayons are NOT broken?

Crayons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q2/5

22

QUESTION ID

S1-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q2/5

True or false?

A shelf has 7 red books and 12 blue books. There are 19 books in all.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q3/5

23

QUESTION ID

S1-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q3/5

A garden has 30 plants. 18 are flowers and the rest are herbs.

How many herbs are there?

A. 12

B. 18

C. 22

D. 48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q4/5

24

QUESTION ID

S1-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q4/5

A bus has 17 adults and 26 children.

How many people are on the bus?

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q5/5

25

QUESTION ID

S1-B2-Q5

TYPE

Short answer

CORRECT ANSWER

43

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 1 • BACKWARD 2 • Q5/5

SET 1 • 6.RP.A.1

BACKWARD 3

Fraction of a whole • Grade 3 • Number & Operations–Fractions • 3.NF.A.1

The bar is divided into equal parts.

What fraction of the bar is shaded?



Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q1/5

27

QUESTION ID

S1-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3/7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q1/5

A pizza is cut into 6 equal slices. Ava eats 5 slices.

What fraction of the pizza does Ava eat?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q2/5

28

QUESTION ID

S1-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q2/5

True or false?

A bar is cut into 4 equal parts and 1 part is shaded. The shaded part is $\frac{1}{4}$ of the bar.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q3/5

29

QUESTION ID

S1-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q3/5

What fraction of the bar is shaded?



A. $\frac{2}{5}$

B. $\frac{3}{5}$

C. $\frac{2}{3}$

D. $\frac{5}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q4/5

30

QUESTION ID

S1-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q4/5

A field has 8 equal sections. Corn grows in 3 of the sections.

What fraction of the field has corn?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q5/5

31

QUESTION ID

S1-B3-Q5

TYPE

Short answer

CORRECT ANSWER

3/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • 6.RP.A.1

FORWARD 1

Ratios and unit rates with fractional quantities • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A dressing uses $\frac{1}{2}$ cup of oil for every $\frac{3}{4}$ cup of vinegar.

How many cups of oil are used per cup of vinegar?

Cups of oil: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q1/5

33

QUESTION ID

S1-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{2}{3}$ cup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q1/5

Mia paints $\frac{2}{3}$ of a fence in $\frac{1}{2}$ hour.

How much of the fence does she paint per hour?

Fence per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q2/5

34

QUESTION ID

S1-F1-Q2

TYPE

Short answer

CORRECT ANSWER

**$\frac{4}{3} = 1 \frac{1}{3}$ fences
per hour**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q2/5

A garden uses $\frac{3}{4}$ bag of soil for every $\frac{3}{8}$ bag of mulch.

How many bags of soil are used per bag of mulch?

A. $\frac{9}{32}$

B. $\frac{1}{2}$

C. $1\frac{1}{8}$

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q3/5

35

QUESTION ID

S1-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q3/5

A cyclist rides $2\frac{1}{2}$ miles in $\frac{1}{6}$ hour.

What is the speed in miles per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q4/5

36

QUESTION ID

S1-F1-Q4

TYPE

Short answer

CORRECT ANSWER

15 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q4/5

A pump moves $\frac{5}{8}$ gallon of water every $\frac{1}{4}$ minute.

How many gallons does it move per minute?

Gallons per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q5/5

37

QUESTION ID

S1-F1-Q5

TYPE

Short answer

CORRECT ANSWER

**$5/2 = 2 \frac{1}{2}$ gallons
per minute**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 1 • FORWARD 1 • Q5/5

SET 1 • 6.RP.A.1

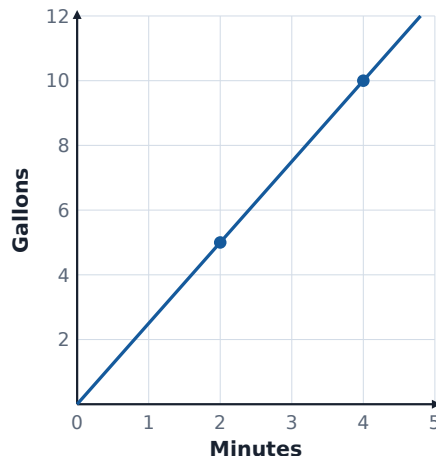
FORWARD 2

The unit rate as the slope of a graph • Grade 8 • Expressions & Equations •
8.EE.B.5

The graph shows a proportional relationship between minutes and gallons.

What is the slope of the line?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q1/5

39

QUESTION ID

S1-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$5/2 = 2.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q1/5

The graph of a proportional relationship passes through (0, 0) and (6, 9).

What is the slope?

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q2/5

40

QUESTION ID

S1-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$3/2 = 1.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

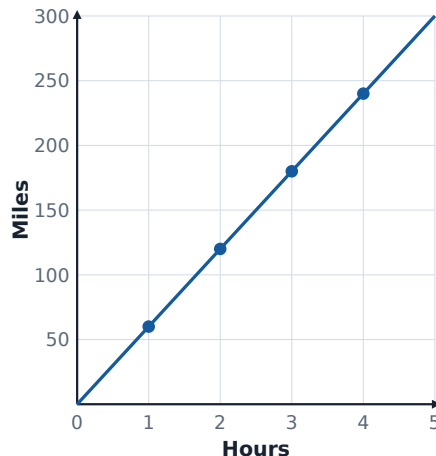
ID S1-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q2/5

The graph shows the distance a car travels.

What is the slope of the line in miles per hour?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q3/5

41

QUESTION ID

S1-F2-Q3

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q3/5

A proportional relationship has a unit rate of 4 dollars per pound.

Which point is on its graph?

- A.** (3, 12) **B.** (4, 1)
C. (3, 7) **D.** (12, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q4/5

42

QUESTION ID

S1-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. (3, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q4/5

True or false?

For $y = 2.5x$, the unit rate is 2.5, and the slope of its graph is also 2.5.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q5/5

43

QUESTION ID

S1-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 1 • FORWARD 2 • Q5/5

Interpret ratio language ("for every," "to")

MAIN	Grade 6 main question	6.RP.A.1
BACKWARD 1	Equal groups	3.OA.A.1
BACKWARD 2	Multiplicative comparison statements	4.OA.A.1
FORWARD 1	Constant of proportionality from a description	7.RP.A.2.b
FORWARD 2	Compare proportional relationships represented in different ways	8.EE.B.5

SET 2 • 6.RP.A.1

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.1

A recipe uses 5 cups of flour for every 2 cups of sugar.

Which statement describes the same ratio?

- A.** For every 7 cups of sugar, there are 5 cups of flour.
- B.** The ratio of sugar to flour is 2 : 5.
- C.** For every 2 cups of flour, there are 5 cups of sugar.
- D.** The ratio of flour to sugar is 2 : 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q1/5

46

QUESTION ID

S2-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. The ratio of sugar to flour is 2 : 5.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q1/5

At an aquarium, the ratio of sharks to rays is 4 : 9.

Which statement is true?

- A. For every 4 rays, there are 9 sharks.
- B. For every 9 sharks, there are 4 rays.
- C. For every 13 sharks, there are 4 rays.
- D. For every 9 rays, there are 4 sharks.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q2/5

47

QUESTION ID

S2-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

**D. For every 9 rays,
there are 4 sharks.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q2/5

The ratio of red beads to blue beads is 3 : 7.

Which statement is true?

- A.** For every 10 blue beads, there are 3 red beads.
- B.** For every 7 red beads, there are 3 blue beads.
- C.** For every 3 red beads, there are 7 blue beads.
- D.** For every 3 blue beads, there are 7 red beads.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q3/5

48

QUESTION ID

S2-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. For every 3 red beads, there are 7 blue beads.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q3/5

At a camp, there are 2 counselors for every 9 campers.

What is the ratio of campers to counselors?

- A. 2 : 9
- B. 11 : 9
- C. 9 : 2
- D. 2 : 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q4/5

49

QUESTION ID

S2-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 9 : 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q4/5

In a class vote, there were 4 votes for pizza for every 3 votes for tacos.

Which statement is true?

- A.** The ratio of pizza votes to taco votes is 3 : 4. **B.** The ratio of pizza votes to all votes is 4 : 3.
- C.** There were 4 taco votes for every 3 pizza votes. **D.** The ratio of taco votes to pizza votes is 3 : 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q5/5

50

QUESTION ID

S2-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. The ratio of taco votes to pizza votes is 3 : 4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.1 • Set 2 • MAIN • Q5/5

SET 2 • 6.RP.A.1

BACKWARD 1

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

4×6 can mean 4 groups with 6 in each group.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q1/5

52

QUESTION ID

S2-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q1/5

There are 3 bags with 5 apples in each bag.

Which expression shows the total?

A. 3×5

B. $5 - 3$

C. $3 + 5$

D. $5 \div 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q2/5

53

QUESTION ID

S2-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 3×5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q2/5

There are 7 tables with 4 chairs at each table.

How many chairs are there?

Chairs: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q3/5

54

QUESTION ID

S2-B1-Q3

TYPE

Short answer

CORRECT ANSWER

28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q3/5

True or false?

"2 wheels for every bike" means each bike has 2 wheels.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q4/5

55

QUESTION ID

S2-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q4/5

What does 6×2 mean?

- A.** 6 groups of 2
- B.** 62
- C.** 6 more than 2
- D.** 6 minus 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q5/5

56

QUESTION ID

S2-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 6 groups of 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • 6.RP.A.1

BACKWARD 2

Multiplicative comparison statements • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1

True or false?

24 is 3 times as many as 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q1/5

58

QUESTION ID

S2-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q1/5

Which equation shows "42 is 6 times as many as 7"?

A. $42 = 6 + 7$

B. $42 = 6 \times 7$

C. $42 - 7 = 6$

D. $7 = 6 \times 42$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q2/5

59

QUESTION ID

S2-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. $42 = 6 \times 7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q2/5

True or false?

18 is 6 times as many as 3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q3/5

60

QUESTION ID

S2-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q3/5

Fill in the blank.

45 is ____ times as many as 9.

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q4/5

61

QUESTION ID

S2-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q4/5

True or false?

20 is 4 times as many as 16.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q5/5

62

QUESTION ID

S2-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.1 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • 6.RP.A.1

FORWARD 1

Constant of proportionality from a description • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

A candle burns 2 inches every 3 hours at a constant rate.

What is the constant of proportionality in inches per hour?

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q1/5

64

QUESTION ID

S2-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2/3 inch per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q1/5

For every 5 laps, Jo swims 250 meters.

What is the constant of proportionality in meters per lap?

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q2/5

65

QUESTION ID

S2-F1-Q2

TYPE

Short answer

CORRECT ANSWER

50 meters per lap

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q2/5

A machine fills 18 bottles for every 4 minutes.

What is the constant of proportionality in bottles per minute?

A. $\frac{2}{9}$

B. 4.5

C. 14

D. 72

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q3/5

66

QUESTION ID

S2-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q3/5

For every 8 cups of water, a recipe uses 3 cups of rice.
What is the constant of proportionality in cups of rice
per cup of water?

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q4/5

67

QUESTION ID

S2-F1-Q4

TYPE

Short answer

CORRECT ANSWER

3/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q4/5

True or false?

For every 6 pounds of apples, the cost is \$9. The constant of proportionality is 1.5 dollars per pound.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q5/5

68

QUESTION ID

S2-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 2 • FORWARD 1 • Q5/5

SET 2 • 6.RP.A.1

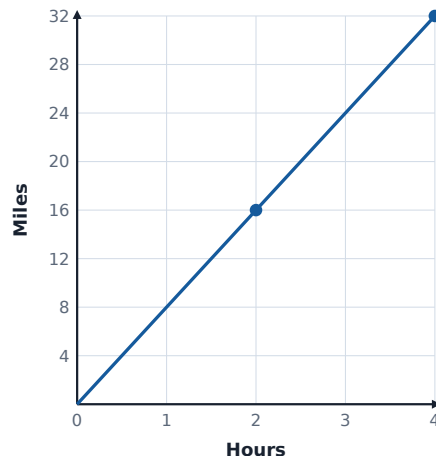
FORWARD 2

Compare proportional relationships represented in different ways • Grade 8 •
Expressions & Equations • 8.EE.B.5

Cyclist A's distance is shown in the graph. Cyclist B rides according to $d = 7.5t$, where d is miles and t is hours.

Which cyclist is faster?

- A. Cyclist A
- B. Cyclist B
- C. They ride at the same speed.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q1/5

70

QUESTION ID

S2-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Cyclist A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q1/5

Tank A fills according to $y = 12x$, where x is minutes and y is gallons. The graph of Tank B is a line through $(0, 0)$ and $(3, 45)$.

Which tank fills faster, and by how many gallons per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q2/5

71

QUESTION ID

S2-F2-Q2

TYPE

Short answer

CORRECT ANSWER

**Tank B is faster, by 3
gallons per minute**

GRADING NOTE Tank B rate = $45/3 = 15$ gal/min; Tank A = 12 gal/min

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q2/5

Store A: 2 pounds of apples cost \$3.50.

Store B: the cost is $c = 1.6p$ for p pounds.

Which store charges less per pound?

- A. Store A
- B. Store B
- C. They charge the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q3/5

72

QUESTION ID

S2-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Store B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q3/5

Printer A prints 3 pages for every 2 seconds. Printer B prints according to $p = 1.4s$.

Which printer is faster?

- A. Printer A
- B. Printer B
- C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q4/5

73

QUESTION ID

S2-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Printer A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q4/5

Car A travels 150 miles in 3 hours at a constant speed.

Car B travels according to $d = 48t$.

How much faster is Car A, in miles per hour?

Miles per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q5/5

74

QUESTION ID

S2-F2-Q5

TYPE

Short answer

CORRECT ANSWER

2 miles per hour

GRADING NOTE Car A = $150/3 = 50$ mph; Car B = 48 mph

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 2 • FORWARD 2 • Q5/5

Find and describe a unit rate

MAIN

Grade 6 main question

6.RP.A.2

BACKWARD 1

Whole-number division

4.NBT.B.6

BACKWARD 2

Fraction as division

5.NF.B.3

BACKWARD 3

Meaning of "per" (the amount for each one)

3.OA.A.2

FORWARD 1

Unit rates with fractions

7.RP.A.1

FORWARD 2

Rate of change of a linear function

8.F.B.4

SET 3 • 6.RP.A.2

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.2

A car travels 150 miles in 3 hours at a constant speed.

What is the unit rate in miles per hour?

Unit rate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q1/5

77

QUESTION ID

S3-M-Q1

TYPE

Short answer

CORRECT ANSWER

50 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q1/5

A 4-pound bag of rice costs \$12.

What is the price per pound?

Price per pound: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q2/5

78

QUESTION ID

S3-M-Q2

TYPE

Short answer

CORRECT ANSWER

\$3 per pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q2/5

A punch recipe uses 5 cups of juice for every 8 cups of water.

How many cups of juice are used for each cup of water?

Cups of juice: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q3/5

79

QUESTION ID

S3-M-Q3

TYPE

Short answer

CORRECT ANSWER

5/8 cup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q3/5

There are 2 teachers for every 30 students on a field trip.

How many students are there per teacher?

A. 15

B. 28

C. 30

D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q4/5

80

QUESTION ID

S3-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q4/5

A printer prints 90 pages in 6 minutes.

How many pages does it print per minute?

Pages per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q5/5

81

QUESTION ID

S3-M-Q5

TYPE

Short answer

CORRECT ANSWER

15 pages per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.2 • Set 3 • MAIN • Q5/5

SET 3 • 6.RP.A.2

BACKWARD 1

Whole-number division • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Find the quotient.

$$150 \div 3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q1/5

83

QUESTION ID

S3-B1-Q1

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q1/5

Which is equal to $96 \div 4$?

- A.** 22
- B.** 23
- C.** 24
- D.** 26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q2/5

84

QUESTION ID

S3-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q2/5

True or false?

$$84 \div 6 = 14$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q3/5

85

QUESTION ID

S3-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q3/5

Find the quotient.

$$212 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q4/5

86

QUESTION ID

S3-B1-Q4

TYPE

Short answer

CORRECT ANSWER

53

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q4/5

Find the quotient.

$$135 \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q5/5

87

QUESTION ID

S3-B1-Q5

TYPE

Short answer

CORRECT ANSWER

27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • 6.RP.A.2

BACKWARD 2

Fraction as division • Grade 5 • Number & Operations–Fractions • 5.NF.B.3

Write $5 \div 8$ as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q1/5

89

QUESTION ID

S3-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q1/5

True or false?

$\frac{2}{5}$ means the same as $2 \div 5$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q2/5

90

QUESTION ID

S3-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q2/5

Four friends share 3 sandwiches equally.
How much sandwich does each friend get?

Each friend gets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q3/5

91

QUESTION ID

S3-B2-Q3

TYPE

Short answer

CORRECT ANSWER

3/4 of a sandwich

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q3/5

Which division expression is equal to $\frac{7}{8}$?

- A. $8 \div 7$
- B. $8 - 7$
- C. $7 \div 8$
- D. 7×8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q4/5

92

QUESTION ID

S3-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $7 \div 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q4/5

A 5-yard ribbon is cut into 6 equal pieces.

How long is each piece?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q5/5

93

QUESTION ID

S3-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5/6 yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.3 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • 6.RP.A.2

BACKWARD 3

Meaning of "per" (the amount for each one) • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

12 apples are packed equally into 4 bags.

How many apples are in each bag?

Apples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q1/5

95

QUESTION ID

S3-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q1/5

True or false?

"5 miles per hour" means 5 miles for each 1 hour.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q2/5

96

QUESTION ID

S3-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q2/5

Which phrase means the same as "\$2 per pound"?

- A.** \$2 for each pound
- B.** \$2 for all of the pounds
- C.** 2 pounds for each dollar
- D.** \$2 more than a pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q3/5

97

QUESTION ID

S3-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. \$2 for each pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q3/5

A farmer puts 24 eggs into cartons, 6 eggs per carton.

How many cartons does the farmer fill?

Cartons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q4/5

98

QUESTION ID

S3-B3-Q4

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q4/5

Jen reads 40 pages in 5 days. She reads the same number of pages each day.

How many pages does she read per day?

Pages: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q5/5

99

QUESTION ID

S3-B3-Q5

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 3 • BACKWARD 3 • Q5/5

SET 3 • 6.RP.A.2

FORWARD 1

Unit rates with fractions • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.1

Kim walks $\frac{3}{4}$ mile in $\frac{1}{4}$ hour.

What is her speed in miles per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q1/5

101

QUESTION ID

S3-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q1/5

A painter paints $\frac{2}{3}$ of a wall in $\frac{1}{3}$ hour.

At this rate, how many walls can the painter paint per hour?

Walls per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q2/5

102

QUESTION ID

S3-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2 walls per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q2/5

A snail crawls $\frac{5}{6}$ foot in $\frac{1}{3}$ minute.

How many feet does it crawl per minute?

Feet per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q3/5

103

QUESTION ID

S3-F1-Q3

TYPE

Short answer

CORRECT ANSWER

$5/2 = 2 \frac{1}{2}$ feet per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q3/5

A machine fills $\frac{3}{4}$ of a tank in $\frac{1}{2}$ hour.

How many tanks does it fill per hour?

A. $\frac{1}{4}$

B. $\frac{3}{8}$

C. $1\frac{1}{4}$

D. $1\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q4/5

104

QUESTION ID

S3-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q4/5

Leo mows $\frac{7}{8}$ of an acre in $\frac{1}{2}$ hour.

How many acres does he mow per hour?

Acres per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q5/5

105

QUESTION ID

S3-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$7/4 = 1 \frac{3}{4}$ acres per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 3 • FORWARD 1 • Q5/5

SET 3 • 6.RP.A.2

FORWARD 2

Rate of change of a linear function • Grade 8 • Functions • 8.F.B.4

The table shows a linear function.

What is the rate of change?

x	0	1	2	3
y	5	8	11	14

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q1/5

107

QUESTION ID

S3-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q1/5

A plumber charges a \$40 fee plus \$65 per hour.

Write a function for the cost C after h hours.

$C =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q2/5

108

QUESTION ID

S3-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$C = 65h + 40$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q2/5

A line passes through (2, 7) and (6, 19).

What is the rate of change?

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q3/5

109

QUESTION ID

S3-F2-Q3

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q3/5

A tank drains at a constant rate.

What is the rate of change in gallons per minute?

Minutes	0	5	10
Gallons	200	170	140

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q4/5

110

QUESTION ID

S3-F2-Q4

TYPE

Short answer

CORRECT ANSWER

**–6 gallons per
minute**

GRADING NOTE The tank loses 6 gallons each minute; "6 gallons per minute, decreasing" is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q4/5

A taxi ride costs \$3 plus \$2.50 per mile.

Which function gives the cost y for a ride of x miles?

A. $y = 3x + 2.5$

B. $y = 2.5x - 3$

C. $y = 2.5x + 3$

D. $y = 5.5x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q5/5

111

QUESTION ID

S3-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. $y = 2.5x + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 3 • FORWARD 2 • Q5/5

Find missing values in tables of equivalent ratios

MAIN Grade 6 main question

BACKWARD 1 Equivalent fractions

BACKWARD 2 Two related number patterns

BACKWARD 3 Multiples of a number

FORWARD 1 Constant of proportionality from a table

FORWARD 2 Rate of change and initial value from a table

6.RP.A.3.a

4.NF.A.1

5.OA.B.3

4.OA.B.4

7.RP.A.2.b

8.F.B.4

SET 4 • 6.RP.A.3.a

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

The table shows equivalent ratios.

Find the missing value.

Cups of juice	2	4	6	10
Cups of water	3	6	9	?

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q1/5

114

QUESTION ID

S4-M-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q1/5

Paint is mixed in the same ratio in every column.

Find the missing value.

Blue (cups)	3	6	?	15
Yellow (cups)	2	4	8	10

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q2/5

115

QUESTION ID

S4-M-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q2/5

A swimmer keeps the same pace.

Find the missing value.

Laps	3	6	?	15
Minutes	8	16	24	40

Missing value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q3/5

116

QUESTION ID

S4-M-Q3

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q3/5

The table shows equivalent ratios.

What is the missing value?

Tickets	4	8	12	?
Cost (\$)	10	20	30	45

A. 16

B. 18

C. 20

D. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q4/5

117

QUESTION ID

S4-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q4/5

A recipe uses 5 cups of oats for every 2 cups of raisins.

Find the missing values A and B.

Oats (cups)	5	10	A	25
Raisins (cups)	2	B	6	10

A = _____

B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q5/5

118

QUESTION ID

S4-M-Q5

TYPE

Short answer

CORRECT ANSWER

A = 15, B = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 4 • MAIN • Q5/5

SET 4 • 6.RP.A.3.a

BACKWARD 1

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

True or false?

$\frac{2}{3}$ is equivalent to $\frac{8}{12}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q1/5

120

QUESTION ID

S4-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q1/5

Find the missing number.

$$\frac{3}{4} = \frac{?}{20}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q2/5

121

QUESTION ID

S4-B1-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q2/5

Which fraction is equivalent to $\frac{5}{6}$?

A. $\frac{5}{12}$

B. $\frac{10}{12}$

C. $\frac{6}{7}$

D. $\frac{10}{11}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q3/5

122

QUESTION ID

S4-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 10/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q3/5

True or false?

$\frac{4}{5}$ is equivalent to $\frac{6}{7}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q4/5

123

QUESTION ID

S4-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q4/5

Find the missing number.

$$\frac{2}{5} = \frac{6}{?}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q5/5

124

QUESTION ID

S4-B1-Q5

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 4 • BACKWARD 1 • Q5/5

SET 4 • 6.RP.A.3.a

BACKWARD 2

Two related number patterns • Grade 5 • Operations & Algebraic Thinking •
5.OA.B.3

Pattern A starts at 0 and adds 2.

Pattern B starts at 0 and adds 5.

What is the 5th number in Pattern B?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q1/5

126

QUESTION ID

S4-B2-Q1

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q1/5

True or false?

Each number in Pattern B is 2 times the matching number in Pattern A.

Pattern A (add 4)	0	4	8	12
Pattern B (add 8)	0	8	16	24

A. True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q2/5

127

QUESTION ID

S4-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q2/5

Pattern A starts at 0 and adds 7.

Pattern B starts at 0 and adds 14.

What is the 4th number in Pattern A?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q3/5

128

QUESTION ID

S4-B2-Q3

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q3/5

Pattern A: 0, 2, 4, 6, 8

Pattern B: 0, 6, 12, 18, 24

How does each number in Pattern B compare to the matching number in Pattern A?

- A.** 2 times as large **B.** 3 times as large
C. 4 more **D.** 6 times as large

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q4/5

129

QUESTION ID

S4-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 3 times as large

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q4/5

Pattern A: 0, 1, 2, 3, ...

Pattern B: 0, 4, 8, 12, ...

What number in Pattern B matches 5 in Pattern A?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q5/5

130

QUESTION ID

S4-B2-Q5

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • 6.RP.A.3.a

BACKWARD 3

Multiples of a number • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first four multiples of 3.

Multiples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q1/5

132

QUESTION ID

S4-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3, 6, 9, 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q1/5

True or false?

40 is a multiple of 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q2/5

133

QUESTION ID

S4-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q2/5

Which number is a multiple of 5?

- A.** 32
- B.** 35
- C.** 48
- D.** 52

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q3/5

134

QUESTION ID

S4-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q3/5

What is the 6th multiple of 4?

Multiple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q4/5

135

QUESTION ID

S4-B3-Q4

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q4/5

True or false?

27 is a multiple of 6.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q5/5

136

QUESTION ID

S4-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 4 • BACKWARD 3 • Q5/5

SET 4 • 6.RP.A.3.a

FORWARD 1

Constant of proportionality from a table • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b

The table shows a proportional relationship.

What is the constant of proportionality ($\frac{y}{x}$)?

x	2	5	8
y	7	17.5	28

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q1/5

138

QUESTION ID

S4-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q1/5

The table shows the cost of cheese.

What is the constant of proportionality in dollars per pound?

Pounds	3	6	9
Cost (\$)	7.50	15.00	22.50

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q2/5

139

QUESTION ID

S4-F1-Q2

TYPE

Short answer

CORRECT ANSWER

\$2.50 per pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q2/5

The table shows a proportional relationship.

What is the constant of proportionality?

x	4	8	12
y	3	6	9

A. $\frac{3}{4}$

B. $\frac{4}{3}$

C. 3

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q3/5

140

QUESTION ID

S4-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q3/5

The table shows how far a toy car rolls.

What is the constant of proportionality in meters per second?

Seconds	2	3	5
Meters	1.2	1.8	3.0

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q4/5

141

QUESTION ID

S4-F1-Q4

TYPE

Short answer

CORRECT ANSWER

0.6 meter per second

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q4/5

True or false?

The constant of proportionality for the table is 6.

x	1	2	4
y	6	12	24

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q5/5

142

QUESTION ID

S4-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.b • Set 4 • FORWARD 1 • Q5/5

SET 4 • 6.RP.A.3.a

FORWARD 2

Rate of change and initial value from a table • Grade 8 • Functions • 8.F.B.4

The table shows a linear function.

What are the rate of change and the initial value?

x	0	1	2	3
y	4	7	10	13

Rate of change: _____

Initial value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q1/5

144

QUESTION ID

S4-F2-Q1

TYPE

Short answer

CORRECT ANSWER

**Rate of change: 3;
Initial value: 4**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q1/5

The table shows the height of a plant.

What is the rate of change in cm per week?

Weeks	0	2	4
Height (cm)	6	9	12

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q2/5

145

QUESTION ID

S4-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1.5 cm per week

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q2/5

The table shows a linear function.

Which equation matches the table?

x	0	1	2	3
y	5	7	9	11

A. $y = x + 5$

B. $y = 5x + 2$

C. $y = 2x + 5$

D. $y = 2x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q3/5

146

QUESTION ID

S4-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $y = 2x + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q3/5

The table shows a linear function.

Write an equation for the function.

x	1	2	3	4
y	9	13	17	21

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q4/5

147

QUESTION ID

S4-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$$y = 4x + 5$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q4/5

True or false?

The initial value of the function in the table is 20.

x	0	1	2
y	20	17	14

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q5/5

148

QUESTION ID

S4-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 4 • FORWARD 2 • Q5/5

Plot pairs of values from a ratio table on the coordinate plane

MAIN	Grade 6 main question	6.RP.A.3.a
BACKWARD 1	Ordered pairs from two related patterns	5.OA.B.3
BACKWARD 2	Axes and ordered pairs	5.G.A.1
BACKWARD 3	Plotting points in the first quadrant	5.G.A.2
BACKWARD 4	Reading an axis marked by 2s, 3s, or 5s	3.MD.B.3
FORWARD 1	Use a graph to decide whether a relationship is proportional	7.RP.A.2.a
FORWARD 2	Graph proportional relationships; interpret the slope	8.EE.B.5

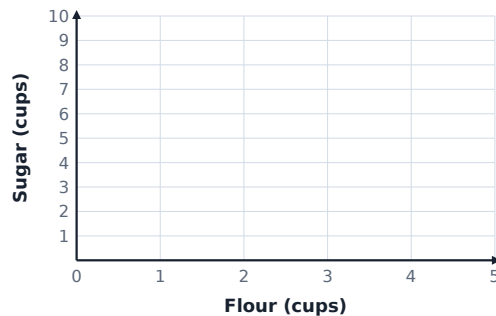
SET 5 • 6.RP.A.3.a

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

Plot the pairs of values from the table on the coordinate plane.

Flour (cups)	1	2	3	4
Sugar (cups)	2	4	6	8



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q1/5

151

QUESTION ID

S5-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Points plotted at (1, 2), (2, 4), (3, 6), (4, 8)

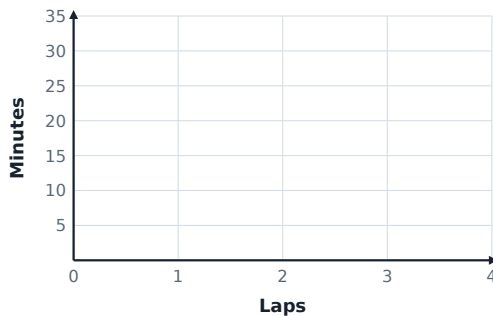
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q1/5

Plot the pairs of values from the table on the coordinate plane.

Laps	1	2	3
Minutes	10	20	30



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q2/5

152

QUESTION ID

S5-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Points plotted at (1, 10), (2, 20), (3, 30)

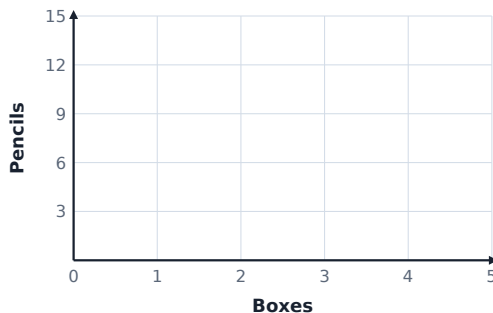
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q2/5

Plot the pairs of values from the table on the coordinate plane.

Boxes	1	2	3	4
Pencils	3	6	9	12



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q3/5

153

QUESTION ID

S5-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Points plotted at (1, 3), (2, 6), (3, 9), (4, 12)

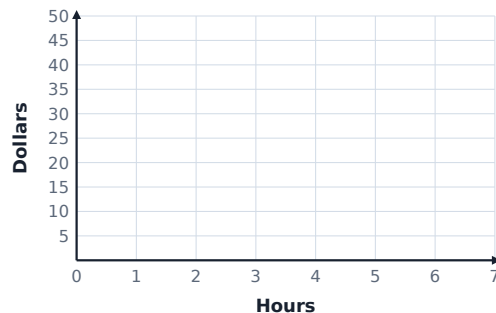
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q3/5

Plot the pairs of values from the table on the coordinate plane.

Hours	2	4	6
Dollars	15	30	45



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q4/5

154

QUESTION ID

S5-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Points plotted at (2, 15), (4, 30), (6, 45)

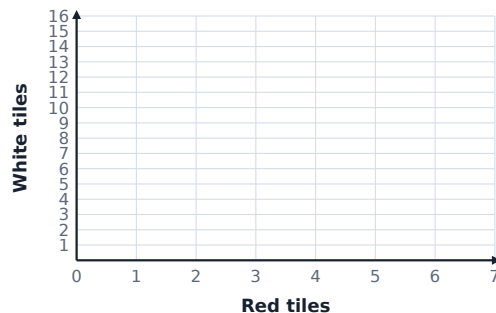
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q4/5

Plot the pairs of values from the table on the coordinate plane.

Red tiles	2	4	6
White tiles	5	10	15



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q5/5

155

QUESTION ID

S5-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Points plotted at (2, 5), (4, 10), (6, 15)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 5 • MAIN • Q5/5

SET 5 • 6.RP.A.3.a

BACKWARD 1

Ordered pairs from two related patterns • Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3

Pattern A: 0, 1, 2, 3

Pattern B: 0, 3, 6, 9

Write the ordered pair formed by the 3rd numbers of the patterns.

Ordered pair: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q1/5

157

QUESTION ID

S5-B1-Q1

TYPE

Short answer

CORRECT ANSWER

(2, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q1/5

Which ordered pairs come from the table?

x	1	2	3
y	5	10	15

- A.** (1, 5), (2, 10), (3, 15) **B.** (5, 1), (10, 2), (15, 3)
C. (1, 2), (5, 10), (15, 3) **D.** (1, 10), (2, 5), (3, 15)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q2/5

158

QUESTION ID

S5-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. (1, 5), (2, 10), (3, 15)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q2/5

True or false?

The table gives the ordered pair (4, 12).

x	2	4	6
y	6	12	18

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q3/5

159

QUESTION ID

S5-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q3/5

Write the ordered pair from the last column of the table.

x	1	2	3	4
y	7	14	21	28

Ordered pair: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q4/5

160

QUESTION ID

S5-B1-Q4

TYPE

Short answer

CORRECT ANSWER

(4, 28)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q4/5

True or false?

In an ordered pair, the first number comes from the top row of the table.

x	1	2
y	4	8

A. True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q5/5

161

QUESTION ID

S5-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • 6.RP.A.3.a

BACKWARD 2

Axes and ordered pairs • Grade 5 • Geometry • 5.G.A.1

True or false?

To plot (3, 7), start at (0, 0), move 3 units right, then 7 units up.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q1/5

163

QUESTION ID

S5-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q1/5

In the ordered pair (6, 2), which number tells how far to move up?

- A.** 2
- B.** 4
- C.** 6
- D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q2/5

164

QUESTION ID

S5-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q2/5

What are the coordinates of the origin?

Origin: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q3/5

165

QUESTION ID

S5-B2-Q3

TYPE

Short answer

CORRECT ANSWER

(0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q3/5

True or false?

(4, 9) and (9, 4) name the same point.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q4/5

166

QUESTION ID

S5-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q4/5

Which axis is vertical?

- A.** The x-axis
- B.** The y-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q5/5

167

QUESTION ID

S5-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. The y-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

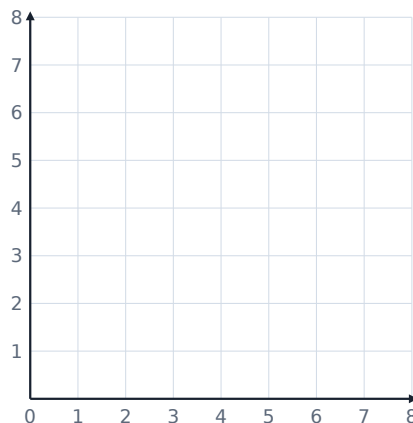
Grade 5 • Geometry • 5.G.A.1 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • 6.RP.A.3.a

BACKWARD 3

Plotting points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

Plot and label point A at (3, 5).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q1/5

169

QUESTION ID

S5-B3-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

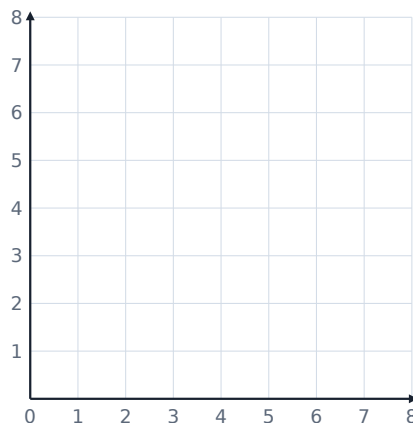
Point A plotted at (3, 5) and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q1/5

Plot and label point B at (6, 2).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q2/5

170

QUESTION ID

S5-B3-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

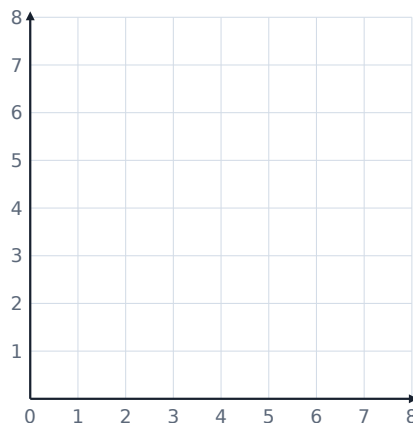
Point B plotted at (6, 2) and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q2/5

Plot and label point C at (0, 4).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q3/5

171

QUESTION ID

S5-B3-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Point C plotted at (0, 4) on the y-axis and labeled

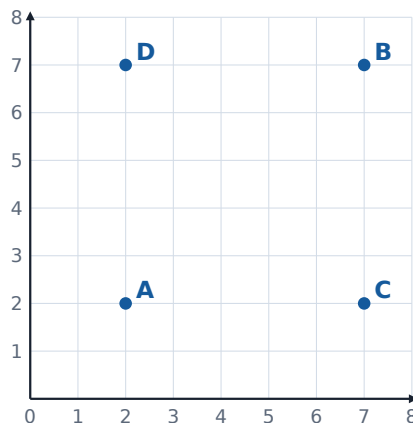
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q3/5

Which point is at (2, 7)?

- A. Point A
- B. Point B
- C. Point C
- D. Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q4/5

172

QUESTION ID

S5-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

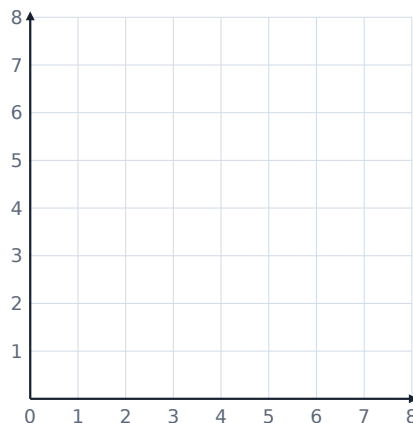
D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q4/5

Plot and label point D at (5, 5).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q5/5

173

QUESTION ID

S5-B3-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Point D plotted at (5, 5) and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • 6.RP.A.3.a

BACKWARD 4

Reading an axis marked by 2s, 3s, or 5s • Grade 3 • Measurement & Data •
3.MD.B.3

On the y-axis, the grid lines are numbered 0, 5, 10, 15.

Where would 20 go?

- A. At the 2nd grid line
- B. At the 20th grid line
- C. One grid line above 15
- D. Halfway between 10 and 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q1/5

175

QUESTION ID

S5-B4-Q1

TYPE

Multiple choice

CORRECT ANSWER

**C. One grid line
above 15**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q1/5

True or false?

On an axis numbered 0, 3, 6, 9, the value 12 is one grid line above 9.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q2/5

176

QUESTION ID

S5-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q2/5

On an axis numbered 0, 2, 4, 6, 8, where does the value 5 go?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q3/5

177

QUESTION ID

S5-B4-Q3

TYPE

Short answer

CORRECT ANSWER

**Halfway between 4
and 6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q3/5

An axis is numbered 0, 5, 10, 15, 20.

How many units does each grid line stand for?

- A.** 1
- B.** 5
- C.** 10
- D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q4/5

178

QUESTION ID

S5-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q4/5

True or false?

On an axis numbered by 10s, the value 35 is halfway between 30 and 40.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q5/5

179

QUESTION ID

S5-B4-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B4-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 5 • BACKWARD 4 • Q5/5

SET 5 • 6.RP.A.3.a

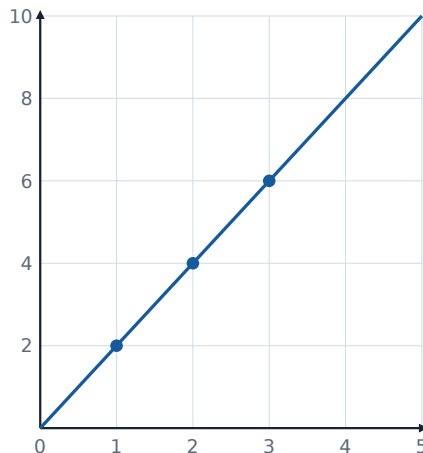
FORWARD 1

Use a graph to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

True or false?

The graph shows a proportional relationship.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q1/5

181

QUESTION ID

S5-F1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

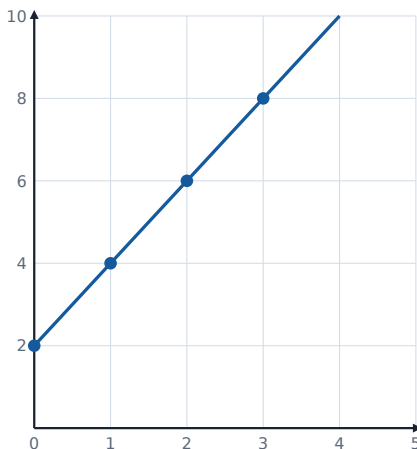
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q1/5

Which statement about the graph is true?

- A. It is proportional because y increases by 2.
- B. It is not proportional because the line does not pass through $(0, 0)$.
- C. It is not proportional because the points are not on a line.
- D. It is proportional because it is a straight line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q2/5

182

QUESTION ID

S5-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. It is not proportional because the line does not pass through $(0, 0)$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

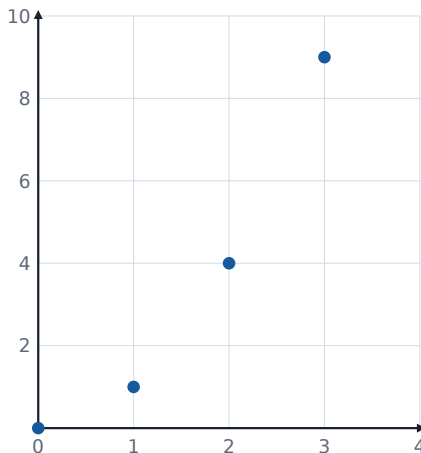
ID S5-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q2/5

True or false?

The graph shows a proportional relationship.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q3/5

183

QUESTION ID

S5-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q3/5

The points (2, 5), (4, 10), and (6, 15) are graphed.

Do they lie on a straight line through the origin? Write yes or no.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q4/5

184

QUESTION ID

S5-F1-Q4

TYPE

Short answer

CORRECT ANSWER

Yes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q4/5

Which set of points lies on a straight line through the origin?

- A. (0, 2), (1, 4), (2, 6)
- B. (1, 3), (2, 4), (3, 5)
- C. (2, 3), (4, 6), (6, 9)
- D. (1, 1), (2, 4), (3, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q5/5

185

QUESTION ID

S5-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. (2, 3), (4, 6), (6, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 5 • FORWARD 1 • Q5/5

SET 5 • 6.RP.A.3.a

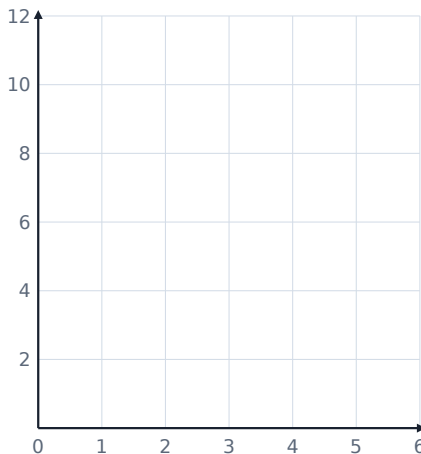
FORWARD 2

Graph proportional relationships; interpret the slope • Grade 8 • Expressions & Equations • 8.EE.B.5

Graph $y = 2x$ on the coordinate plane.

What is the slope of the line?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q1/5

187

QUESTION ID

S5-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Slope: 2

GRADING NOTE Graph: straight line through (0, 0), (1, 2), (3, 6), (6, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

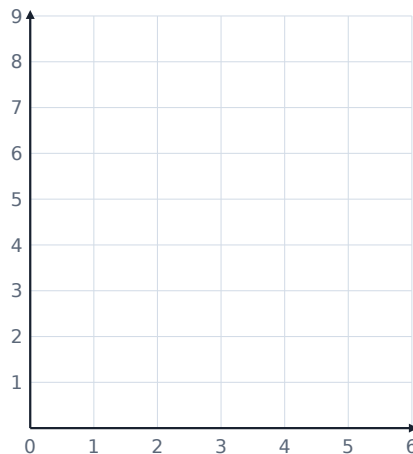
ID S5-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q1/5

Graph $y = 1.5x$ on the coordinate plane.

What is the slope of the line?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q2/5

188

QUESTION ID

S5-F2-Q2

TYPE

Short answer

CORRECT ANSWER

Slope: 1.5

GRADING NOTE Graph: straight line through (0, 0), (2, 3), (4, 6), (6, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

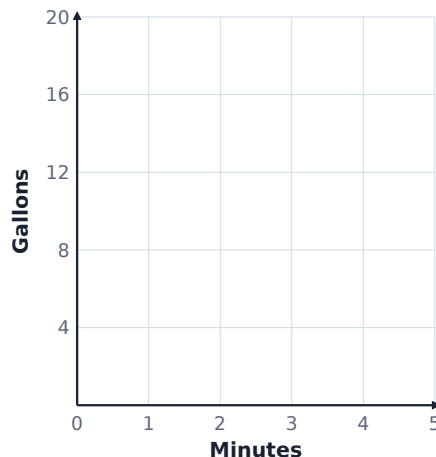
ID S5-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q2/5

Water flows at 4 gallons per minute. Graph the gallons y after x minutes.

What is the slope of the line?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q3/5

189

QUESTION ID

S5-F2-Q3

TYPE

Short answer

CORRECT ANSWER

Slope: 4

GRADING NOTE Graph: straight line through (0, 0), (1, 4), (5, 20)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q3/5

The graph of a proportional relationship passes through (5, 2).

What is the slope?

A. $\frac{2}{5}$

B. 2

C. $\frac{5}{2}$

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q4/5

190

QUESTION ID

S5-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q4/5

True or false?

The graph of $y = 3x$ passes through $(0, 0)$ and $(2, 6)$, and its slope is 3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q5/5

191

QUESTION ID

S5-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 5 • FORWARD 2 • Q5/5

Use tables of equivalent ratios to compare ratios

MAIN	Grade 6 main question	6.RP.A.3.a
BACKWARD 1	Comparing fractions	4.NF.A.2
BACKWARD 2	Common multiples	4.OA.B.4
BACKWARD 3	Equivalent fractions	4.NF.A.1
FORWARD 1	Compare unit rates involving fractions	7.RP.A.1
FORWARD 2	Compare proportional relationships represented in different ways	8.EE.B.5

SET 6 • 6.RP.A.3.a

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.a

Lemonade A uses 2 cups of mix for every 5 cups of water. Lemonade B uses 3 cups of mix for every 7 cups of water.

Use tables of equivalent ratios. Which lemonade has a stronger mix flavor?

A: Mix	2	4	6
A: Water	5	10	15

B: Mix	3	6	9
B: Water	7	14	21

- A.** Lemonade A **B.** Lemonade B **C.** They taste the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q1/5

194

QUESTION ID

S6-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Lemonade B

GRADING NOTE $A = 2/5 = 0.4$; $B = 3/7 \approx 0.43$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q1/5

Class A has 4 boys for every 6 girls. Class B has 6 boys for every 9 girls.

Use tables of equivalent ratios. Which statement is true?

- A. Class A has a greater ratio of boys to girls.
- B. Class B has a greater ratio of boys to girls.
- C. The ratios are equivalent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q2/5

195

QUESTION ID

S6-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. The ratios are equivalent.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q2/5

Runner A runs 3 laps in 10 minutes. Runner B runs 4 laps in 12 minutes.

Use tables of equivalent ratios. Who runs faster?

- A. Runner A
- B. Runner B
- C. They run at the same speed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q3/5

196

QUESTION ID

S6-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Runner B

GRADING NOTE A: 0.3 lap/min; B: 1/3 lap/min

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q3/5

Paint A uses 1 part blue for every 3 parts white. Paint B uses 2 parts blue for every 5 parts white.

Use tables of equivalent ratios. Which paint is a darker blue?

- A. Paint A
- B. Paint B
- C. They are the same color.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q4/5

197

QUESTION ID

S6-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Paint B

GRADING NOTE A: $\frac{1}{3}$; B: $\frac{2}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q4/5

Store A sells 5 pens for \$3. Store B sells 8 pens for \$5.

Use tables of equivalent ratios. Which store gives more pens for the same amount of money?

- A. Store A
- B. Store B
- C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q5/5

198

QUESTION ID

S6-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Store A

GRADING NOTE A: $\frac{5}{3}$ pens per \$; B: $\frac{8}{5}$ pens per \$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.a • Set 6 • MAIN • Q5/5

SET 6 • 6.RP.A.3.a

BACKWARD 1

Comparing fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.2

Which fraction is greater?

A. $\frac{3}{7}$

B. $\frac{3}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q1/5

200

QUESTION ID

S6-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. 3/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q1/5

True or false?

$$\frac{5}{8} > \frac{4}{8}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q2/5

201

QUESTION ID

S6-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q2/5

Which symbol makes the statement true?

$$\frac{2}{3} \text{ — } \frac{3}{4}$$

A. <

B. >

C. =

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q3/5

202

QUESTION ID

S6-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. <

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q3/5

True or false?

$$\frac{1}{3} = \frac{2}{6}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q4/5

203

QUESTION ID

S6-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q4/5

Which fraction is greatest?

A. $\frac{1}{2}$

B. $\frac{7}{12}$

C. $\frac{2}{3}$

D. $\frac{5}{6}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q5/5

204

QUESTION ID

S6-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. 5/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.2 • Set 6 • BACKWARD 1 • Q5/5

SET 6 • 6.RP.A.3.a

BACKWARD 2

Common multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the multiples of 5 and the multiples of 7 up to 40.

Which number is on both lists?

Common multiple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q1/5

206

QUESTION ID

S6-B2-Q1

TYPE

Short answer

CORRECT ANSWER

35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q1/5

True or false?

30 is a multiple of both 5 and 6.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q2/5

207

QUESTION ID

S6-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q2/5

Which number is a multiple of both 3 and 4?

- A.** 9
- B.** 12
- C.** 14
- D.** 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q3/5

208

QUESTION ID

S6-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q3/5

Name a number that is a multiple of both 6 and 9.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q4/5

209

QUESTION ID

S6-B2-Q4

TYPE

Short answer

CORRECT ANSWER

18

GRADING NOTE Any common multiple of 18 (18, 36, 54, ...) is correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q4/5

True or false?

20 is a multiple of both 4 and 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q5/5

210

QUESTION ID

S6-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • 6.RP.A.3.a

BACKWARD 3

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

Find the missing number.

$$\frac{2}{5} = \frac{?}{35}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q1/5

212

QUESTION ID

S6-B3-Q1

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q1/5

True or false?

$$\frac{3}{7} = \frac{6}{14}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q2/5

213

QUESTION ID

S6-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q2/5

Which fraction is equivalent to $\frac{4}{6}$?

A. $\frac{4}{9}$

B. $\frac{6}{9}$

C. $\frac{6}{8}$

D. $\frac{8}{10}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q3/5

214

QUESTION ID

S6-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q3/5

Find the missing number.

$$\frac{1}{3} = \frac{5}{?}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q4/5

215

QUESTION ID

S6-B3-Q4

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q4/5

True or false?

$$\frac{3}{10} = \frac{9}{20}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q5/5

216

QUESTION ID

S6-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • 6.RP.A.3.a

FORWARD 1

Compare unit rates involving fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

Ana walks $\frac{1}{2}$ mile in $\frac{1}{5}$ hour. Ben walks $\frac{3}{4}$ mile in $\frac{1}{3}$ hour.

Who walks faster?

- A. Ana
- B. Ben
- C. They walk at the same speed.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q1/5

218

QUESTION ID

S6-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Ana

GRADING NOTE Ana 2 1/2 mph; Ben 2 1/4 mph

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q1/5

Mix A uses $\frac{2}{3}$ cup of sugar per $\frac{1}{2}$ cup of water. Mix B uses $\frac{3}{4}$ cup of sugar per $\frac{2}{3}$ cup of water.

Which mix is sweeter?

- A. Mix A
- B. Mix B
- C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q2/5

219

QUESTION ID

S6-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Mix A

GRADING NOTE A: $\frac{4}{3}$; B: $\frac{9}{8}$ cup sugar per cup water

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q2/5

Pump A moves $\frac{3}{8}$ gallon in $\frac{1}{4}$ minute. Pump B moves $\frac{1}{2}$ gallon in $\frac{1}{2}$ minute.

Which pump is faster, and what is its rate in gallons per minute?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q3/5

220

QUESTION ID

S6-F1-Q3

TYPE

Short answer

CORRECT ANSWER

**Pump A, 1 1/2
gallons per minute**

GRADING NOTE Pump B = 1 gallon per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q3/5

Snail A crawls $\frac{1}{6}$ foot in $\frac{1}{2}$ minute. Snail B crawls $\frac{1}{4}$ foot in $\frac{2}{3}$ minute.

Which snail is faster?

- A. Snail A
- B. Snail B
- C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q4/5

221

QUESTION ID

S6-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Snail B

GRADING NOTE A: $\frac{1}{3}$ ft/min; B: $\frac{3}{8}$ ft/min

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q4/5

True or false?

$\frac{1}{2}$ mile in $\frac{1}{4}$ hour is a faster speed than $\frac{3}{4}$ mile in $\frac{1}{2}$ hour.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q5/5

222

QUESTION ID

S6-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 6 • FORWARD 1 • Q5/5

SET 6 • 6.RP.A.3.a

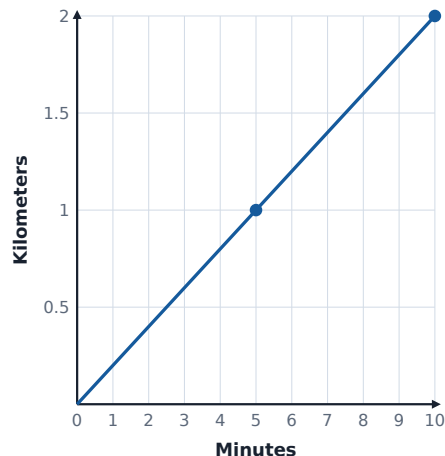
FORWARD 2

Compare proportional relationships represented in different ways • Grade 8 •
Expressions & Equations • 8.EE.B.5

Runner A's distance is shown in the graph. Runner B runs according to $d = 0.15t$, where d is kilometers and t is minutes.

Which runner is faster?

- A. Runner A
- B. Runner B
- C. They are the same.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q1/5

224

QUESTION ID

S6-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Runner A

GRADING NOTE Runner A = 0.2 km per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q1/5

Machine A: $y = 45x$ bottles after x minutes. Machine B: the table shows 3 minutes $\rightarrow$ 120 bottles, 6 minutes $\rightarrow$ 240 bottles.

Which machine is faster?

- A. Machine A
- B. Machine B
- C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q2/5

225

QUESTION ID

S6-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Machine A

GRADING NOTE Machine B = 40 bottles per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q2/5

Recipe A uses flour and sugar in the ratio $y = 1.5x$.
Recipe B's graph passes through (0, 0) and (4, 7).
Which recipe uses more flour y per cup of sugar x ?

Recipe: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q3/5

226

QUESTION ID

S6-F2-Q3

TYPE

Short answer

CORRECT ANSWER

Recipe B

GRADING NOTE B: $7/4 = 1.75$ per cup vs A: 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q3/5

Two proportional relationships: P is $y = 6x$. Q is a line through (0, 0) and (2, 13).

Which has the greater slope?

- A. P
- B. Q
- C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q4/5

227

QUESTION ID

S6-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Q

GRADING NOTE Q slope = 6.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q4/5

Plan A: 20 minutes of calls cost \$3.00. Plan B:
 $c = 0.12m$, where m is minutes.

Which plan costs less per minute?

Plan: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q5/5

228

QUESTION ID

S6-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Plan B

GRADING NOTE Plan A = \$0.15 per minute; Plan B = \$0.12 per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 6 • FORWARD 2 • Q5/5

Solve unit pricing problems

- MAIN** Grade 6 main question
- BACKWARD 1** Dividing money amounts
- BACKWARD 2** Multiplying money amounts
- BACKWARD 3** Finding the cost of one when each costs the same
- FORWARD 1** Equations for proportional relationships
- FORWARD 2** Build a linear function for a pricing situation

6.RP.A.3.b

5.NBT.B.7

5.NBT.B.7

3.OA.A.2

7.RP.A.2.c

8.F.B.4

SET 7 • 6.RP.A.3.b

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

4 notebooks cost \$10.

At this rate, how much do 7 notebooks cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q1/5

231

QUESTION ID

S7-M-Q1

TYPE

Short answer

CORRECT ANSWER

\$17.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q1/5

3 pounds of apples cost \$5.40.

At this rate, how much do 5 pounds cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q2/5

232

QUESTION ID

S7-M-Q2

TYPE

Short answer

CORRECT ANSWER

\$9.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q2/5

6 cans of soup cost \$7.50.

At this rate, how much do 10 cans cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q3/5

233

QUESTION ID

S7-M-Q3

TYPE

Short answer

CORRECT ANSWER

\$12.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q3/5

5 movie tickets cost \$42.50.

At this rate, how much do 3 tickets cost?

- A. \$8.50
- B. \$25.50
- C. \$28.33
- D. \$127.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q4/5

234

QUESTION ID

S7-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. \$25.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q4/5

8 yards of fabric cost \$36.

At this rate, how much do 3 yards cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q5/5

235

QUESTION ID

S7-M-Q5

TYPE

Short answer

CORRECT ANSWER

\$13.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 7 • MAIN • Q5/5

SET 7 • 6.RP.A.3.b

BACKWARD 1

Dividing money amounts • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Find the quotient.

$$\$10.00 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q1/5

237

QUESTION ID

S7-B1-Q1

TYPE

Short answer

CORRECT ANSWER

\$2.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find the quotient.

$$\$5.40 \div 3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q2/5

238

QUESTION ID

S7-B1-Q2

TYPE

Short answer

CORRECT ANSWER

\$1.80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q2/5

$$\$7.50 \div 6 = ?$$

- A.** \$0.80
- B.** \$1.20
- C.** \$1.25
- D.** \$12.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q3/5

239

QUESTION ID

S7-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. \$1.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find the quotient.

$$\$36 \div 8$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q4/5

240

QUESTION ID

S7-B1-Q4

TYPE

Short answer

CORRECT ANSWER

\$4.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q4/5

240

True or false?

$$\$42.50 \div 5 = \$8.50$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q5/5

241

QUESTION ID

S7-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • 6.RP.A.3.b

BACKWARD 2

Multiplying money amounts • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$\$2.50 \times 7$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q1/5

243

QUESTION ID

S7-B2-Q1

TYPE

Short answer

CORRECT ANSWER

\$17.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q1/5

243

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$\$1.80 \times 5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q2/5

244

QUESTION ID

S7-B2-Q2

TYPE

Short answer

CORRECT ANSWER

\$9.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q2/5

244

True or false?

$$\$1.25 \times 10 = \$12.50$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q3/5

245

QUESTION ID

S7-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q3/5

$$\$4.50 \times 3 = ?$$

- A.** \$1.50
- B.** \$7.50
- C.** \$12.50
- D.** \$13.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q4/5

246

QUESTION ID

S7-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. \$13.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$\$8.50 \times 3$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q5/5

247

QUESTION ID

S7-B2-Q5

TYPE

Short answer

CORRECT ANSWER

\$25.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • 6.RP.A.3.b

BACKWARD 3

Finding the cost of one when each costs the same • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2

4 toys cost \$12 in all. Each toy costs the same.

How much does 1 toy cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q1/5

249

QUESTION ID

S7-B3-Q1

TYPE

Short answer

CORRECT ANSWER

\$3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q1/5

5 pencils cost 30 cents in all. Each pencil costs the same.

How much does 1 pencil cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q2/5

250

QUESTION ID

S7-B3-Q2

TYPE

Short answer

CORRECT ANSWER

6 cents

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q2/5

True or false?

6 stickers cost 18 cents in all. Each costs the same, so 1 sticker costs 3 cents.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q3/5

251

QUESTION ID

S7-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q3/5

8 cards cost \$16. Each card costs the same.

How much is 1 card?

- A.** \$2
- B.** \$8
- C.** \$24
- D.** \$128

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q4/5

252

QUESTION ID

S7-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. \$2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q4/5

3 books cost \$27 in all. Each book costs the same.

How much does 1 book cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q5/5

253

QUESTION ID

S7-B3-Q5

TYPE

Short answer

CORRECT ANSWER

\$9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 7 • BACKWARD 3 • Q5/5

SET 7 • 6.RP.A.3.b

FORWARD 1

Equations for proportional relationships • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Apples cost \$2.50 per pound.

Write an equation for the cost c of p pounds of apples.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q1/5

255

QUESTION ID

S7-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$c = 2.5p$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q1/5

The table shows a proportional relationship.

Write an equation for the earnings e after h hours.

Hours (h)	2	3	5
Earnings (e)	36	54	90

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q2/5

256

QUESTION ID

S7-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$e = 18h$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q2/5

Pens cost \$0.75 each.

Which equation gives the cost c of n pens?

A. $c = \frac{n}{0.75}$

B. $c = n + 0.75$

C. $c = 0.75n$

D. $n = 0.75c$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q3/5

257

QUESTION ID

S7-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $c = 0.75n$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q3/5

5 pounds of rice cost \$8.

Write an equation for the cost c of p pounds of rice.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q4/5

258

QUESTION ID

S7-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$c = 1.6p$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q4/5

True or false?

Grapes cost \$3.20 per pound. The equation $c = 3.2p$ gives the cost c of p pounds.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q5/5

259

QUESTION ID

S7-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 7 • FORWARD 1 • Q5/5

SET 7 • 6.RP.A.3.b

FORWARD 2

Build a linear function for a pricing situation • Grade 8 • Functions • 8.F.B.4

A store charges \$2.50 per pound of cherries plus a \$4 box fee.

Write a function for the total cost C of p pounds.

$C =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q1/5

261

QUESTION ID

S7-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$C = 2.5p + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q1/5

Tickets cost \$12 each plus a \$5 service fee per order.

Which function gives the cost C of t tickets?

A. $C = 5t + 12$

B. $C = 12t - 5$

C. $C = 12t + 5$

D. $C = 17t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q2/5

262

QUESTION ID

S7-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. $C = 12t + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q2/5

A delivery costs \$6 plus \$1.75 per mile.

What are the rate of change and the initial value?

Rate of change: _____

Initial value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q3/5

263

QUESTION ID

S7-F2-Q3

TYPE

Short answer

CORRECT ANSWER

**Rate of change:
\$1.75 per mile;
Initial value: \$6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q3/5

The table shows the cost of a gym membership.

Write a function for the cost C after m months.

Months (m)	0	1	2	3
Cost (C)	25	55	85	115

$C =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q4/5

264

QUESTION ID

S7-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$C = 30m + 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q4/5

True or false?

A function for "\$3 per pound plus a \$2 bag fee" is
 $C = 3p + 2$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q5/5

265

QUESTION ID

S7-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 7 • FORWARD 2 • Q5/5

Solve constant speed problems

MAIN	Grade 6 main question
BACKWARD 1	Whole-number division
BACKWARD 2	Multiplying by a one-digit number
BACKWARD 3	Hours and minutes
FORWARD 1	Speed as a unit rate with fractions
FORWARD 2	Compare speeds from graphs and equations

6.RP.A.3.b

4.NBT.B.6

4.NBT.B.5

4.MD.A.1

7.RP.A.1

8.EE.B.5

SET 8 • 6.RP.A.3.b

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

A runner runs 12 kilometers in 60 minutes at a constant speed.

How far does the runner go in 25 minutes?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q1/5

268

QUESTION ID

S8-M-Q1

TYPE

Short answer

CORRECT ANSWER

5 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q1/5

A train travels 180 miles in 3 hours at a constant speed.

How far does it travel in 5 hours?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q2/5

269

QUESTION ID

S8-M-Q2

TYPE

Short answer

CORRECT ANSWER

300 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q2/5

A cyclist rides 45 miles in 3 hours at a constant speed.

How long does it take to ride 75 miles?

Time: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q3/5

270

QUESTION ID

S8-M-Q3

TYPE

Short answer

CORRECT ANSWER

5 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q3/5

A snail moves 18 cm in 6 minutes at a constant speed.

How far does it move in 20 minutes?

- A. 3 cm
- B. 54 cm
- C. 60 cm
- D. 120 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q4/5

271

QUESTION ID

S8-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 60 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q4/5

A car travels 220 miles in 4 hours at a constant speed.

How long does it take to travel 385 miles?

Time: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q5/5

272

QUESTION ID

S8-M-Q5

TYPE

Short answer

CORRECT ANSWER

7 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 8 • MAIN • Q5/5

SET 8 • 6.RP.A.3.b

BACKWARD 1

Whole-number division • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$180 \div 3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q1/5

274

QUESTION ID

S8-B1-Q1

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$220 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q2/5

275

QUESTION ID

S8-B1-Q2

TYPE

Short answer

CORRECT ANSWER

55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q2/5

275

True or false?

$$385 \div 55 = 7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q3/5

276

QUESTION ID

S8-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

$$45 \div 3 = ?$$

- A.** 12
- B.** 14
- C.** 15
- D.** 135

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q4/5

277

QUESTION ID

S8-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q4/5

277

Divide.

$$375 \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q5/5

278

QUESTION ID

S8-B1-Q5

TYPE

Short answer

CORRECT ANSWER

75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 8 • BACKWARD 1 • Q5/5

SET 8 • 6.RP.A.3.b

BACKWARD 2

Multiplying by a one-digit number • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$60 \times 5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q1/5

280

QUESTION ID

S8-B2-Q1

TYPE

Short answer

CORRECT ANSWER

300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$15 \times 7$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q2/5

281

QUESTION ID

S8-B2-Q2

TYPE

Short answer

CORRECT ANSWER

105

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q2/5

281

True or false?

$$55 \times 4 = 220$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q3/5

282

QUESTION ID

S8-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q3/5

$$48 \times 6 = ?$$

- A.** 54
- B.** 248
- C.** 288
- D.** 2,448

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q4/5

283

QUESTION ID

S8-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 288

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$125 \times 3$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q5/5

284

QUESTION ID

S8-B2-Q5

TYPE

Short answer

CORRECT ANSWER

375

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.5 • Set 8 • BACKWARD 2 • Q5/5

284

SET 8 • 6.RP.A.3.b

BACKWARD 3

Hours and minutes • Grade 4 • Measurement & Data • 4.MD.A.1

How many minutes are in 3 hours?

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q1/5

286

QUESTION ID

S8-B3-Q1

TYPE

Short answer

CORRECT ANSWER

180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q1/5

True or false?

There are 60 minutes in 1 hour.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q2/5

287

QUESTION ID

S8-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

Complete.

5 hours = ____ minutes

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q3/5

288

QUESTION ID

S8-B3-Q3

TYPE

Short answer

CORRECT ANSWER

300

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q3/5

Which is longer?

- A.** 1 hour
- B.** 90 minutes
- C.** They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q4/5

289

QUESTION ID

S8-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 90 minutes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q4/5

How many minutes are in $\frac{1}{2}$ hour?

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q5/5

290

QUESTION ID

S8-B3-Q5

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 8 • BACKWARD 3 • Q5/5

SET 8 • 6.RP.A.3.b

FORWARD 1

Speed as a unit rate with fractions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A hiker walks $\frac{3}{4}$ mile in $\frac{3}{8}$ hour.

What is the hiker's speed in miles per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q1/5

292

QUESTION ID

S8-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q1/5

A turtle crawls $\frac{1}{5}$ mile in $\frac{1}{2}$ hour.

What is its speed in miles per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q2/5

293

QUESTION ID

S8-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{2}{5}$ mile per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q2/5

A boat travels $2\frac{1}{2}$ miles in $\frac{1}{4}$ hour.

What is its speed in miles per hour?

A. $\frac{5}{8}$

B. $2\frac{3}{4}$

C. 5

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q3/5

294

QUESTION ID

S8-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q3/5

A skater glides $\frac{2}{3}$ kilometer in $\frac{1}{6}$ hour.

What is the skater's speed in kilometers per hour?

Speed: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q4/5

295

QUESTION ID

S8-F1-Q4

TYPE

Short answer

CORRECT ANSWER

**4 kilometers per
hour**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q4/5

True or false?

Going $\frac{1}{2}$ mile in $\frac{1}{10}$ hour is a speed of 5 miles per hour.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q5/5

296

QUESTION ID

S8-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 8 • FORWARD 1 • Q5/5

SET 8 • 6.RP.A.3.b

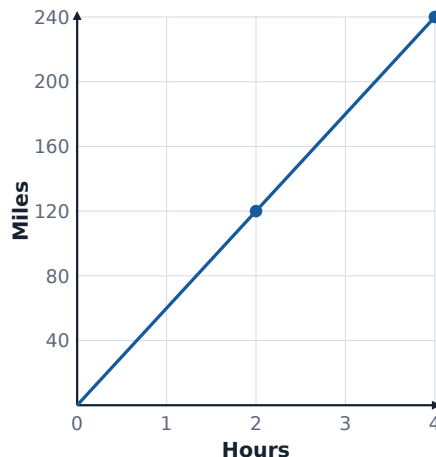
FORWARD 2

Compare speeds from graphs and equations • Grade 8 • Expressions & Equations • 8.EE.B.5

Car A's distance is shown in the graph. Car B travels according to $d = 55t$.

Which car is faster?

- A. Car A
- B. Car B
- C. They are the same.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q1/5

ID S8-F2-Q1

298

QUESTION ID

S8-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Car A

GRADING NOTE Car A = 60 mph

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q1/5

ID S8-F2-Q1

Train A travels according to $d = 80t$. Train B's graph is a line through $(0, 0)$ and $(3, 210)$.

Which train is faster, and by how many miles per hour?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q2/5

299

QUESTION ID

S8-F2-Q2

TYPE

Short answer

CORRECT ANSWER

**Train A is faster, by
10 miles per hour**

GRADING NOTE Train B = $210/3 = 70$ mph

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q2/5

Walker A: $d = 3.5t$. Walker B: the graph passes through (0, 0) and (2, 8).

Who is faster?

- A. Walker A
- B. Walker B
- C. They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q3/5

300

QUESTION ID

S8-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Walker B

GRADING NOTE Walker B = 4 per hour

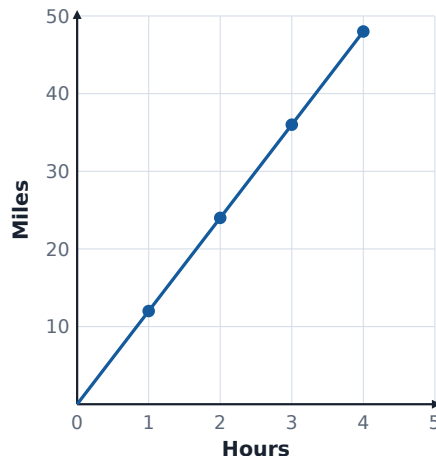
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q3/5

What is the speed shown by the graph in miles per hour?

Speed: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q4/5

301

QUESTION ID

S8-F2-Q4

TYPE

Short answer

CORRECT ANSWER

12 miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q4/5

True or false?

A line through (0, 0) and (5, 300) shows a greater speed than $d = 65t$.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q5/5

302

QUESTION ID

S8-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 8 • FORWARD 2 • Q5/5

Compare unit rates to find the better buy or the faster rate

MAIN	Grade 6 main question	6.RP.A.3.b
BACKWARD 1	Dividing money amounts	5.NBT.B.7
BACKWARD 2	Comparing decimals	5.NBT.A.3.b
BACKWARD 3	Finding the amount for one	3.OA.A.2
FORWARD 1	Compare deals using percents	7.RP.A.3
FORWARD 2	Compare two functions represented in different ways	8.F.A.2

SET 9 • 6.RP.A.3.b

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.b

Which is the better buy?

A: 6 bottles of water for \$4.50

B: 10 bottles of water for \$7.00

A. A

B. B

C. Both are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q1/5

305

QUESTION ID

S9-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. B

GRADING NOTE A: \$0.75 per bottle; B: \$0.70 per bottle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q1/5

Which is the better buy?

A: a 12-ounce box of cereal for \$3.60

B: an 18-ounce box of cereal for \$4.95

A. A

B. B

C. Both are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q2/5

306

QUESTION ID

S9-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. B

GRADING NOTE A: \$0.30 per oz; B: \$0.275 per oz

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q2/5

Which store has the lower price per pound?

A: 4 pounds of grapes for \$7.00

B: 3 pounds of grapes for \$5.40

A. A

B. B

C. Both are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q3/5

307

QUESTION ID

S9-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A

GRADING NOTE A: \$1.75 per lb; B: \$1.80 per lb

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q3/5

Which is the better buy?

A: 8 batteries for \$10.00

B: 12 batteries for \$14.40

A. A

B. B

C. Both are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q4/5

308

QUESTION ID

S9-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. B

GRADING NOTE A: \$1.25 each; B: \$1.20 each

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q4/5

Which printer is faster?

A: 90 pages in 6 minutes

B: 140 pages in 10 minutes

A. A

B. B

C. Both are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q5/5

309

QUESTION ID

S9-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. A

GRADING NOTE A: 15 pages/min; B: 14 pages/min

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.b • Set 9 • MAIN • Q5/5

SET 9 • 6.RP.A.3.b

BACKWARD 1

Dividing money amounts • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Find the quotient.

$$\$4.50 \div 6$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q1/5

311

QUESTION ID

S9-B1-Q1

TYPE

Short answer

CORRECT ANSWER

\$0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find the quotient.

$$\$3.60 \div 12$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q2/5

312

QUESTION ID

S9-B1-Q2

TYPE

Short answer

CORRECT ANSWER

\$0.30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q2/5

312

$$\$7.00 \div 10 = ?$$

- A.** \$0.07
- B.** \$0.70
- C.** \$7.00
- D.** \$70.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q3/5

313

QUESTION ID

S9-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. \$0.70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find the quotient.

$$\$14.40 \div 12$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q4/5

314

QUESTION ID

S9-B1-Q4

TYPE

Short answer

CORRECT ANSWER

\$1.20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q4/5

314

True or false?

$$\$5.40 \div 3 = \$1.80$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q5/5

315

QUESTION ID

S9-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 9 • BACKWARD 1 • Q5/5

SET 9 • 6.RP.A.3.b

BACKWARD 2

Comparing decimals • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.b

Which amount is less?

A. \$0.70

B. \$0.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q1/5

317

QUESTION ID

S9-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. \$0.70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q1/5

True or false?

$0.8 > 0.75$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q2/5

318

QUESTION ID

S9-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

Write $>$, $<$, or $=$ to compare.

0.275 ____ 0.3

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q3/5

319

QUESTION ID

S9-B2-Q3

TYPE

Short answer

CORRECT ANSWER

 $<$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q3/5

319

Which number is greatest?

- A.** 1.075
- B.** 1.705
- C.** 1.75
- D.** 1.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q4/5

320

QUESTION ID

S9-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q4/5

True or false?

$$1.25 < 1.2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q5/5

321

QUESTION ID

S9-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 9 • BACKWARD 2 • Q5/5

SET 9 • 6.RP.A.3.b

BACKWARD 3

Finding the amount for one • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

3 bags hold 21 oranges in all, the same number in each.

How many oranges are in 1 bag?

Oranges: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q1/5

323

QUESTION ID

S9-B3-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q1/5

A worker packs 32 boxes in 4 hours, the same number each hour.

How many boxes per hour?

Boxes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q2/5

324

QUESTION ID

S9-B3-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q2/5

True or false?

9 muffins cost \$18 in all, the same price each, so 1 muffin costs \$2.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q3/5

325

QUESTION ID

S9-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q3/5

40 pages are read in 8 days, the same number each day.

How many pages per day?

A. 5

B. 32

C. 48

D. 320

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q4/5

326

QUESTION ID

S9-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q4/5

6 packs hold 54 cards in all, the same number in each.

How many cards are in 1 pack?

Cards: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q5/5

327

QUESTION ID

S9-B3-Q5

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • 6.RP.A.3.b

FORWARD 1

Compare deals using percents • Grade 7 • Ratios & Proportional Relationships
• 7.RP.A.3

A \$60 jacket is on sale. Which deal saves more money?

- A. \$10 off
- B. 20% off
- C. They save the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q1/5

329

QUESTION ID

S9-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. 20% off

GRADING NOTE 20% of \$60 = \$12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q1/5

Which costs less after the discount?

A: a \$45 game at 30% off

B: a \$40 game at 20% off

A. A

B. B

C. They cost the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q2/5

330

QUESTION ID

S9-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. A

GRADING NOTE A: \$31.50; B: \$32.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q2/5

Store A sells a \$80 bike at 15% off. Store B sells the same bike for \$70.

Which store has the lower price, and by how much?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q3/5

331

QUESTION ID

S9-F1-Q3

TYPE

Short answer

CORRECT ANSWER

Store A, by \$2

GRADING NOTE Store A price = \$68

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q3/5

Which is the better deal on a \$120 phone?

- A.** 25% off
- B.** \$25 off
- C.** They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q4/5

332

QUESTION ID

S9-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 25% off

GRADING NOTE 25% of \$120 = \$30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q4/5

True or false?

A \$50 item at 10% off costs the same as a \$55 item at 20% off.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q5/5

333

QUESTION ID

S9-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 9 • FORWARD 1 • Q5/5

SET 9 • 6.RP.A.3.b

FORWARD 2

Compare two functions represented in different ways • Grade 8 • Functions •
8.F.A.2

Plan A costs $y = 12x$ dollars for x months. Plan B is shown in the table.

Which plan has the greater rate of change?

Months (x)	2	4	6
Cost (y)	26	52	78

- A.** Plan A **B.** Plan B **C.** They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q1/5

335

QUESTION ID

S9-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Plan B

GRADING NOTE Plan B = \$13 per month

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

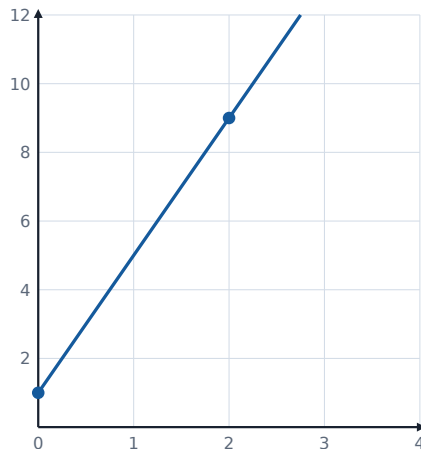
ID S9-F2-Q1

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q1/5

Function A is $y = 3x + 4$. Function B is shown in the graph.

Which function has the greater rate of change?

- A. Function A
- B. Function B
- C. They are the same.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q2/5

336

QUESTION ID

S9-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Function B**GRADING NOTE** Function B slope = 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q2/5

Function A is shown in the table. Function B is $y = -3x + 2$.

Which function has the greater y-intercept?

x	0	1	2
y	5	3	1

- A.** Function A **B.** Function B **C.** They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q3/5

337

GRADING NOTE Function A y-intercept = 5

QUESTION ID

S9-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Function A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q3/5

Phone plan A costs \$30 plus \$0.05 per text. Phone plan B is a line through (0, 20) and (100, 35).

Which plan has the greater cost per text?

Plan: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q4/5

338

QUESTION ID

S9-F2-Q4

TYPE

Short answer

CORRECT ANSWER

Plan B

GRADING NOTE Plan B = \$0.15 per text; Plan A = \$0.05 per text

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q4/5

True or false?

$y = 4x + 1$ has a greater rate of change than the function in the table.

x	0	1	2
y	0	5	10

A. True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q5/5

339

QUESTION ID

S9-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

Grade 8 • Functions • 8.F.A.2 • Set 9 • FORWARD 2 • Q5/5

Find a percent of a quantity as a rate per 100

MAIN	Grade 6 main question	6.RP.A.3.c
BACKWARD 1	Hundredths as decimals	4.NF.C.6
BACKWARD 2	Tenths as hundredths	4.NF.C.5
BACKWARD 3	Fraction of a whole number	5.NF.B.4.a
BACKWARD 4	Multiplying a decimal by a whole number	5.NBT.B.7
FORWARD 1	Multi-step percent problems	7.RP.A.3
FORWARD 2	Build a linear function that uses a percent rate	8.F.B.4

SET 10 • 6.RP.A.3.c

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

What is 35% of 60?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q1/5

342

QUESTION ID

S10-M-Q1

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q1/5

A jacket costs \$60. The sales tax is 8%.

How much is the tax?

Tax: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q2/5

343

QUESTION ID

S10-M-Q2

TYPE

Short answer

CORRECT ANSWER

\$4.80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q2/5

A class has 25 students. 40% of the students walk to school.

How many students walk to school?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q3/5

344

QUESTION ID

S10-M-Q3

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q3/5

A \$80 pair of shoes is 15% off.

How many dollars are taken off the price?

- A. \$1.20
- B. \$12
- C. \$15
- D. \$68

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q4/5

345

QUESTION ID

S10-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. \$12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q4/5

A school has 450 students. 12% of them play in the band.

How many students play in the band?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q5/5

346

QUESTION ID

S10-M-Q5

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 10 • MAIN • Q5/5

SET 10 • 6.RP.A.3.c

BACKWARD 1

Hundredths as decimals • Grade 4 • Number & Operations–Fractions • 4.NF.C.6

Write $\frac{45}{100}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q1/5

348

QUESTION ID

S10-B1-Q1

TYPE

Short answer

CORRECT ANSWER

0.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q1/5

True or false?

$$0.07 = \frac{7}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q2/5

349

QUESTION ID

S10-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q2/5

Which decimal is equal to $\frac{30}{100}$?

- A. 0.03
- B. 0.30
- C. 3.0
- D. 30.0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q3/5

350

QUESTION ID

S10-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 0.30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q3/5

Write 0.58 as a fraction with a denominator of 100.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q4/5

351

QUESTION ID

S10-B1-Q4

TYPE

Short answer

CORRECT ANSWER

58/100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q4/5

True or false?

$$\frac{8}{100} = 0.8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q5/5

352

QUESTION ID

S10-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • 6.RP.A.3.c

BACKWARD 2

Tenths as hundredths • Grade 4 • Number & Operations–Fractions • 4.NF.C.5

Find the missing number.

$$\frac{3}{10} = \frac{?}{100}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q1/5

354

QUESTION ID

S10-B2-Q1

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q1/5

True or false?

$$\frac{6}{10} + \frac{3}{100} = \frac{63}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q2/5

355

QUESTION ID

S10-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q2/5

Which fraction is equal to $\frac{7}{10}$?

A. $\frac{7}{100}$

B. $\frac{10}{70}$

C. $\frac{17}{100}$

D. $\frac{70}{100}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q3/5

356

QUESTION ID

S10-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 70/100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$\frac{2}{10} + \frac{45}{100}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q4/5

357

QUESTION ID

S10-B2-Q4

TYPE

Short answer

CORRECT ANSWER

65/100**GRADING NOTE** Equivalent 13/20 or 0.65 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q4/5

357

True or false?

$$\frac{9}{10} = \frac{90}{100}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q5/5

358

QUESTION ID

S10-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.5 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • 6.RP.A.3.c

BACKWARD 3

Fraction of a whole number • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{35}{100} \times 60$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q1/5

360

QUESTION ID

S10-B3-Q1

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find $\frac{1}{4}$ of 60.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q2/5

361

QUESTION ID

S10-B3-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q2/5

361

True or false?

$$\frac{2}{5} \times 25 = 10$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q3/5

362

QUESTION ID

S10-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q3/5

$$\frac{3}{4} \times 20 = ?$$

- A. 5
- B. 12
- C. 15
- D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q4/5

363

QUESTION ID

S10-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q4/5

Find $\frac{7}{10}$ of 50.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q5/5

364

QUESTION ID

S10-B3-Q5

TYPE

Short answer

CORRECT ANSWER

35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 10 • BACKWARD 3 • Q5/5

SET 10 • 6.RP.A.3.c

BACKWARD 4

Multiplying a decimal by a whole number • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.35 \times 60$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q1/5

366

QUESTION ID

S10-B4-Q1

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q1/5

366

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.08 \times 60$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q2/5

367

QUESTION ID

S10-B4-Q2

TYPE

Short answer

CORRECT ANSWER

4.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q2/5

367

True or false?

$$0.12 \times 450 = 54$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q3/5

368

QUESTION ID

S10-B4-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q3/5

$$0.4 \times 25 = ?$$

- A.** 0.1
- B.** 1
- C.** 10
- D.** 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q4/5

369

QUESTION ID

S10-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.15 \times 80$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q5/5

370

QUESTION ID

S10-B4-Q5

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B4-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 10 • BACKWARD 4 • Q5/5

370

SET 10 • 6.RP.A.3.c

FORWARD 1

Multi-step percent problems • Grade 7 • Ratios & Proportional Relationships •
7.RP.A.3

A meal costs \$40. You leave a 15% tip.

What is the total cost of the meal with the tip?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q1/5

372

QUESTION ID

S10-F1-Q1

TYPE

Short answer

CORRECT ANSWER

\$46

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q1/5

A pair of shoes costs \$80. They are on sale for 25% off.

What is the sale price?

Sale price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q2/5

373

QUESTION ID

S10-F1-Q2

TYPE

Short answer

CORRECT ANSWER

\$60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q2/5

A town's population grew from 2,000 to 2,300.

What is the percent increase?

Percent increase: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q3/5

374

QUESTION ID

S10-F1-Q3

TYPE

Short answer

CORRECT ANSWER

15%

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q3/5

A \$50 game is marked up 20%. Then a 5% sales tax is added to the new price.

What is the final cost?

- A.** \$60.00 **B.** \$62.00
C. \$62.50 **D.** \$63.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q4/5

375

QUESTION ID

S10-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. \$63.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q4/5

A salesperson earns a 6% commission. She sells \$3,500 in products.

How much commission does she earn?

Commission: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q5/5

376

QUESTION ID

S10-F1-Q5

TYPE

Short answer

CORRECT ANSWER

\$210

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 10 • FORWARD 1 • Q5/5

SET 10 • 6.RP.A.3.c

FORWARD 2

Build a linear function that uses a percent rate • Grade 8 • Functions • 8.F.B.4

A salesperson earns \$300 per week plus 5% of her sales s .

Write a function for her weekly earnings E .

$E =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q1/5

378

QUESTION ID

S10-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$E = 0.05s + 300$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q1/5

A server earns \$40 per shift plus 18% of the food sales f.

Which function gives the earnings E?

A. $E = 0.18f - 40$

B. $E = 0.18f + 40$

C. $E = 18f + 40$

D. $E = 40f + 0.18$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q2/5

379

QUESTION ID

S10-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. $E = 0.18f + 40$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q2/5

An agent earns \$500 per month plus 3% of her sales r .
What are the rate of change and the initial value of this function?

Rate of change: _____

Initial value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q3/5

380

QUESTION ID

S10-F2-Q3

TYPE

Short answer

CORRECT ANSWER

**Rate of change: 0.03
(3% of sales); Initial
value: \$500**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q3/5

A delivery driver earns \$60 per day plus 10% of the orders d delivered in dollars.

Write a function for the daily pay P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q4/5

381

QUESTION ID

S10-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$P = 0.1d + 60$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q4/5

True or false?

For $E = 0.04s + 250$, earnings increase by \$0.04 for each \$1 of sales s .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q5/5

382

QUESTION ID

S10-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 10 • FORWARD 2 • Q5/5

Find the whole, given a part and the percent

MAIN	Grade 6 main question	6.RP.A.3.c
BACKWARD 1	Benchmark percents as fractions	4.NF.A.1
BACKWARD 2	Finding an unknown factor	3.OA.A.4
BACKWARD 3	Dividing a whole number by a unit fraction	5.NF.B.7.b
FORWARD 1	Percent problems: find the original amount	7.RP.A.3
FORWARD 2	Linear equations with decimal coefficients	8.EE.C.7.b

SET 11 • 6.RP.A.3.c

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

12 is 25% of what number?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q1/5

385

QUESTION ID

S11-M-Q1

TYPE

Short answer

CORRECT ANSWER

48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q1/5

A student answered 18 questions correctly. This was 90% of the questions on the test.

How many questions were on the test?

Questions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q2/5

386

QUESTION ID

S11-M-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q2/5

Maria has saved \$45. This is 30% of the cost of a bike.
How much does the bike cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q3/5

387

QUESTION ID

S11-M-Q3

TYPE

Short answer

CORRECT ANSWER

\$150

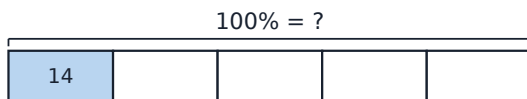
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q3/5

The tape diagram shows that 20% of a number is 14.

What is the number?



A. 28

B. 34

C. 70

D. 280

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q4/5

388

QUESTION ID

S11-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q4/5

In a survey, 36 students chose pizza. This was 40% of the students surveyed.

How many students were surveyed?

Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q5/5

389

QUESTION ID

S11-M-Q5

TYPE

Short answer

CORRECT ANSWER

90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.c • Set 11 • MAIN • Q5/5

SET 11 • 6.RP.A.3.c

BACKWARD 1

Benchmark percents as fractions • Grade 4 • Number & Operations–Fractions •
4.NF.A.1

Find the missing number.

$$\frac{25}{100} = \frac{1}{?}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q1/5

391

QUESTION ID

S11-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q1/5

Find the missing number.

$$\frac{50}{100} = \frac{?}{2}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q2/5

392

QUESTION ID

S11-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q2/5

True or false?

$$\frac{20}{100} = \frac{1}{5}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q3/5

393

QUESTION ID

S11-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q3/5

Which fraction is equivalent to $\frac{75}{100}$?

A. $\frac{1}{75}$

B. $\frac{7}{10}$

C. $\frac{3}{4}$

D. $\frac{4}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q4/5

394

QUESTION ID

S11-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q4/5

Find the missing number.

$$\frac{10}{100} = \frac{1}{?}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q5/5

395

QUESTION ID

S11-B1-Q5

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • 6.RP.A.3.c

BACKWARD 2

Finding an unknown factor • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.4

What number goes in the box?

$$\square \times 4 = 36$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q1/5

397

QUESTION ID

S11-B2-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q1/5

What number goes in the box?

$$5 \times \square = 45$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q2/5

398

QUESTION ID

S11-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q2/5

True or false?

If $6 \times \square = 42$, then $\square = 7$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q3/5

399

QUESTION ID

S11-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q3/5

$$\square \times 8 = 56$$

What number goes in the box?

- A. 6
- B. 7
- C. 8
- D. 48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q4/5

400

QUESTION ID

S11-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q4/5

What number goes in the box?

$$9 \times \square = 63$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q5/5

401

QUESTION ID

S11-B2-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • 6.RP.A.3.c

BACKWARD 3

Dividing a whole number by a unit fraction • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.b

Divide.

$$12 \div \frac{1}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q1/5

403

QUESTION ID

S11-B3-Q1

TYPE

Short answer

CORRECT ANSWER

48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q1/5

Divide.

$$5 \div \frac{1}{3}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q2/5

404

QUESTION ID

S11-B3-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q2/5

True or false?

$$6 \div \frac{1}{2} = 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q3/5

405

QUESTION ID

S11-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q3/5

How many fifths are in 3?

- A. $\frac{3}{5}$
- B. 5
- C. 8
- D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q4/5

406

QUESTION ID

S11-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q4/5

Divide.

$$8 \div \frac{1}{5}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q5/5

407

QUESTION ID

S11-B3-Q5

TYPE

Short answer

CORRECT ANSWER

40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 11 • BACKWARD 3 • Q5/5

SET 11 • 6.RP.A.3.c

FORWARD 1

Percent problems: find the original amount • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3

After a 20% discount, a jacket costs \$48.

What was the original price?

Original price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q1/5

409

QUESTION ID

S11-F1-Q1

TYPE

Short answer

CORRECT ANSWER

\$60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q1/5

A price increased by 10%. The new price is \$55.

What was the original price?

Original price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q2/5

410

QUESTION ID

S11-F1-Q2

TYPE

Short answer

CORRECT ANSWER

\$50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q2/5

The sales tax on an item was \$3.60. The tax rate is 6%.
What was the price before tax?

Price: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q3/5

411

QUESTION ID

S11-F1-Q3

TYPE

Short answer

CORRECT ANSWER

\$60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q3/5

A town's population grew by 25% to 1,500 people.

What was the population before the increase?

- A. 1,125
- B. 1,200
- C. 1,475
- D. 1,875

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q4/5

412

QUESTION ID

S11-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1,200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q4/5

A store marks up the price of a bike by 40%. The selling price is \$210.

How much did the store pay for the bike?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q5/5

413

QUESTION ID

S11-F1-Q5

TYPE

Short answer

CORRECT ANSWER

\$150

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.3 • Set 11 • FORWARD 1 • Q5/5

SET 11 • 6.RP.A.3.c

FORWARD 2

Linear equations with decimal coefficients • Grade 8 • Expressions & Equations
• 8.EE.C.7.b

Solve for x.

$$x - 0.2x = 48$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q1/5

415

QUESTION ID

S11-F2-Q1

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q1/5

Solve for x.

$$0.4x + 12 = 0.1x + 30$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q2/5

416

QUESTION ID

S11-F2-Q2

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q2/5

Solve for x.

$$0.25(x + 40) = 30$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q3/5

417

QUESTION ID

S11-F2-Q3

TYPE

Short answer

CORRECT ANSWER

80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q3/5

Which value of n solves the equation?

$$n + 0.15n = 92$$

- A. 78.2
- B. 80
- C. 92.15
- D. 105.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q4/5

418

QUESTION ID

S11-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q4/5

Solve for y.

$$2.5y - 7 = 1.5y + 3$$

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q5/5

419

QUESTION ID

S11-F2-Q5

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 11 • FORWARD 2 • Q5/5

Convert measurement units using ratio reasoning

MAIN	Grade 6 main question	6.RP.A.3.d
BACKWARD 1	Measurement equivalents	4.MD.A.1
BACKWARD 2	Multiplying and dividing by powers of 10	5.NBT.A.2
BACKWARD 3	Multiplying a decimal by a whole number	5.NBT.B.7
FORWARD 1	Rates with fractions and unit conversions	7.RP.A.1
FORWARD 2	Convert units for very large and very small quantities	8.EE.A.4

SET 12 • 6.RP.A.3.d

MAIN

Grade 6 main question • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.d

There are 3 feet in 1 yard.

How many feet are in 14 yards?

Feet: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q1/5

422

QUESTION ID

S12-M-Q1

TYPE

Short answer

CORRECT ANSWER

42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q1/5

There are 12 inches in 1 foot.

How many inches are in 4.5 feet?

Inches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q2/5

423

QUESTION ID

S12-M-Q2

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q2/5

There are 16 ounces in 1 pound.

How many pounds are in 56 ounces?

Pounds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q3/5

424

QUESTION ID

S12-M-Q3

TYPE

Short answer

CORRECT ANSWER

3.5

GRADING NOTE 3 1/2 pounds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q3/5

A recipe needs 3 quarts of milk. There are 4 cups in 1 quart.

How many cups of milk are needed?

- A. $\frac{3}{4}$ cup B. $1\frac{1}{3}$ cups
C. 7 cups D. 12 cups

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q4/5

425

QUESTION ID

S12-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 12 cups

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q4/5

1 inch = 2.54 centimeters.

A book is 10 inches tall. How many centimeters tall is it?

Centimeters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q5/5

426

QUESTION ID

S12-M-Q5

TYPE

Short answer

CORRECT ANSWER

25.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

Grade 6 • Ratios & Proportional Relationships • 6.RP.A.3.d • Set 12 • MAIN • Q5/5

SET 12 • 6.RP.A.3.d

BACKWARD 1

Measurement equivalents • Grade 4 • Measurement & Data • 4.MD.A.1

How many ounces are in 1 pound?

Ounces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q1/5

428

QUESTION ID

S12-B1-Q1

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q1/5

True or false?

There are 12 inches in 1 yard.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q2/5

429

QUESTION ID

S12-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q2/5

How many cups are in 1 quart?

- A.** 2
- B.** 4
- C.** 8
- D.** 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q3/5

430

QUESTION ID

S12-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q3/5

How many grams are in 1 kilogram?

Grams: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q4/5

431

QUESTION ID

S12-B1-Q4

TYPE

Short answer

CORRECT ANSWER

1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q4/5

True or false?

1 meter = 100 centimeters

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q5/5

432

QUESTION ID

S12-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • 6.RP.A.3.d

BACKWARD 2

Multiplying and dividing by powers of 10 • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$3.4 \times 1,000$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q1/5

434

QUESTION ID

S12-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3,400

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q1/5

434

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$2.54 \times 10$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q2/5

435

QUESTION ID

S12-B2-Q2

TYPE

Short answer

CORRECT ANSWER

25.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q2/5

435

True or false?

$$0.6 \times 1,000 = 60$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q3/5

436

QUESTION ID

S12-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q3/5

$$450 \div 100 = ?$$

- A.** 0.45
- B.** 4.5
- C.** 45
- D.** 45,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q4/5

437

QUESTION ID

S12-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$7.25 \times 100$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q5/5

438

QUESTION ID

S12-B2-Q5

TYPE

Short answer

CORRECT ANSWER

725

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 12 • BACKWARD 2 • Q5/5

438

SET 12 • 6.RP.A.3.d

BACKWARD 3

Multiplying a decimal by a whole number • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$4.5 \times 12$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q1/5

440

QUESTION ID

S12-B3-Q1

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q1/5

440

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$3.5 \times 16$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q2/5

441

QUESTION ID

S12-B3-Q2

TYPE

Short answer

CORRECT ANSWER

56

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q2/5

True or false?

$$1.5 \times 4 = 6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q3/5

442

QUESTION ID

S12-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q3/5

$$2.25 \times 4 = ?$$

- A.** 0.9
- B.** 8.1
- C.** 9
- D.** 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q4/5

443

QUESTION ID

S12-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$6 \times 0.75$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q5/5

444

QUESTION ID

S12-B3-Q5

TYPE

Short answer

CORRECT ANSWER

4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 12 • BACKWARD 3 • Q5/5

SET 12 • 6.RP.A.3.d

FORWARD 1

Rates with fractions and unit conversions • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1

A faucet drips $\frac{1}{4}$ cup of water every $\frac{1}{3}$ minute.

How many cups does it drip per hour?

Cups per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q1/5

446

QUESTION ID

S12-F1-Q1

TYPE

Short answer

CORRECT ANSWER

45 cups per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q1/5

A snail travels $\frac{3}{8}$ foot every $\frac{1}{2}$ minute.

How many feet does it travel per hour?

Feet per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q2/5

447

QUESTION ID

S12-F1-Q2

TYPE

Short answer

CORRECT ANSWER

45 feet per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q2/5

A plant grows $\frac{1}{2}$ inch in $\frac{2}{3}$ week.

How many inches does it grow per week?

A. $\frac{1}{6}$

B. $\frac{1}{3}$

C. $\frac{3}{4}$

D. $1\frac{1}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q3/5

448

QUESTION ID

S12-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q3/5

A hose fills $\frac{3}{4}$ gallon in $\frac{1}{6}$ minute.

How many gallons does it fill per minute?

Gallons per minute: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q4/5

449

QUESTION ID

S12-F1-Q4

TYPE

Short answer

CORRECT ANSWER

**4 1/2 gallons per
minute**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q4/5

A worker paints $\frac{2}{5}$ of a room in $\frac{1}{2}$ hour.

How many rooms can the worker paint in an 8-hour day?

Rooms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q5/5

450

QUESTION ID

S12-F1-Q5

TYPE

Short answer

CORRECT ANSWER

6 $\frac{2}{5}$ rooms

GRADING NOTE $(\frac{2}{5} \div \frac{1}{2}) \times 8 = 32/5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.1 • Set 12 • FORWARD 1 • Q5/5

SET 12 • 6.RP.A.3.d

FORWARD 2

Convert units for very large and very small quantities • Grade 8 • Expressions & Equations • 8.EE.A.4

$$1 \text{ km} = 10^6 \text{ mm.}$$

Write 3.2×10^6 millimeters in kilometers.

Kilometers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q1/5

452

QUESTION ID

S12-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3.2 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q1/5

Earth is about 1.5×10^8 km from the Sun. $1 \text{ km} = 10^3$ m.

How many meters is this? Write the answer in scientific notation.

Meters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q2/5

453

QUESTION ID

S12-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$1.5 \times 10^{11} \text{ m}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q2/5

A sheet of paper is about 1×10^{-4} meter thick.

Which unit is the most appropriate for this measurement?

- A. Miles
- B. Meters
- C. Millimeters
- D. Kilometers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q3/5

454

QUESTION ID

S12-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Millimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q3/5

A virus is 1.2×10^{-7} meter long. 1 nanometer = 10^{-9} meter.

How many nanometers long is the virus?

Nanometers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q4/5

455

QUESTION ID

S12-F2-Q4

TYPE

Short answer

CORRECT ANSWER

120 nm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q4/5

A glacier moves 3×10^{-1} meter per day. $1 \text{ m} = 10^2 \text{ cm}$.

How many centimeters does it move per day?

Centimeters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q5/5

456

QUESTION ID

S12-F2-Q5

TYPE

Short answer

CORRECT ANSWER

30 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 12 • FORWARD 2 • Q5/5

Compute quotients of fractions and mixed numbers

MAIN	Grade 6 main question	6.NS.A.1
BACKWARD 1	Multiplying fractions	5.NF.B.4.a
BACKWARD 2	Whole number $\div$ unit fraction	5.NF.B.7.b
BACKWARD 3	Unit fraction $\div$ whole number	5.NF.B.7.a
BACKWARD 4	Equivalent fractions	4.NF.A.1
BACKWARD 5	Mixed numbers as fractions	4.NF.B.3.b
FORWARD 1	Multiply and divide rational numbers	7.NS.A.2.c
FORWARD 2	Equations with fraction coefficients	8.EE.C.7.b

SET 13 • 6.NS.A.1

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$\frac{3}{5} \div \frac{2}{3}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q1/5

459

QUESTION ID

S13-M-Q1

TYPE

Short answer

CORRECT ANSWER

9/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q1/5

Divide.

$$\frac{5}{6} \div \frac{1}{3}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q2/5

460

QUESTION ID

S13-M-Q2

TYPE

Short answer

CORRECT ANSWER

$5/2 = 2 \frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q2/5

Divide.

$$1\frac{1}{2} \div \frac{3}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q3/5

461

QUESTION ID

S13-M-Q3

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q3/5

Which is equal to $\frac{4}{5} \div \frac{2}{3}$?

A. $\frac{2}{15}$

B. $\frac{8}{15}$

C. $\frac{5}{6}$

D. $1\frac{1}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q4/5

462

QUESTION ID

S13-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1 1/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q4/5

Divide.

$$3\frac{1}{3} \div \frac{5}{6}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q5/5

463

QUESTION ID

S13-M-Q5

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 13 • MAIN • Q5/5

SET 13 • 6.NS.A.1

BACKWARD 1

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{3}{5} \times \frac{2}{7}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q1/5

465

QUESTION ID

S13-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6/35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q1/5

True or false?

$$\frac{3}{4} \times \frac{2}{5} = \frac{6}{20}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q2/5

466

QUESTION ID

S13-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q2/5

$$\frac{1}{2} \times \frac{5}{6} = ?$$

A. $\frac{1}{3}$

B. $\frac{5}{12}$

C. $\frac{5}{8}$

D. $\frac{6}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q3/5

467

QUESTION ID

S13-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q3/5

Multiply.

$$\frac{4}{5} \times \frac{3}{2}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q4/5

468

QUESTION ID

S13-B1-Q4

TYPE

Short answer

CORRECT ANSWER

$6/5 = 1 \frac{1}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q4/5

Multiply.

$$\frac{5}{6} \times 3$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q5/5

469

QUESTION ID

S13-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$5/2 = 2 \frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 13 • BACKWARD 1 • Q5/5

SET 13 • 6.NS.A.1

BACKWARD 2

Whole number $\div$ unit fraction • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.b

Divide.

$$3 \div \frac{1}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q1/5

471

QUESTION ID

S13-B2-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q1/5

True or false?

$$4 \div \frac{1}{2} = 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q2/5

472

QUESTION ID

S13-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q2/5

How many thirds are in 2?

A. $\frac{1}{6}$

B. $\frac{2}{3}$

C. 5

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q3/5

473

QUESTION ID

S13-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q3/5

Divide.

$$6 \div \frac{1}{5}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q4/5

474

QUESTION ID

S13-B2-Q4

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q4/5

Divide.

$$2 \div \frac{1}{8}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q5/5

475

QUESTION ID

S13-B2-Q5

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.b • Set 13 • BACKWARD 2 • Q5/5

SET 13 • 6.NS.A.1

BACKWARD 3

Unit fraction $\div$ whole number • Grade 5 • Number & Operations–Fractions •
5.NF.B.7.a

Divide.

$$\frac{1}{4} \div 3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q1/5

477

QUESTION ID

S13-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q1/5

True or false?

$$\frac{1}{2} \div 3 = \frac{1}{6}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q2/5

478

QUESTION ID

S13-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q2/5

$$\frac{1}{5} \div 2 = ?$$

A. $\frac{1}{10}$

B. $\frac{2}{5}$

C. $\frac{5}{2}$

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q3/5

479

QUESTION ID

S13-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q3/5

Divide.

$$\frac{1}{6} \div 3$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q4/5

480

QUESTION ID

S13-B3-Q4

TYPE

Short answer

CORRECT ANSWER

1/18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q4/5

True or false?

$$\frac{1}{6} \div 2 = \frac{2}{6}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q5/5

481

QUESTION ID

S13-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.a • Set 13 • BACKWARD 3 • Q5/5

SET 13 • 6.NS.A.1

BACKWARD 4

Equivalent fractions • Grade 4 • Number & Operations–Fractions • 4.NF.A.1

True or false?

$\frac{8}{12}$ is equivalent to $\frac{2}{3}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q1/5

483

QUESTION ID

S13-B4-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q1/5

Find the missing number.

$$\frac{10}{15} = \frac{?}{3}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q2/5

484

QUESTION ID

S13-B4-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q2/5

Which fraction is equivalent to $\frac{9}{12}$?

A. $\frac{2}{3}$

B. $\frac{3}{4}$

C. $\frac{12}{9}$

D. $\frac{9}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q3/5

485

QUESTION ID

S13-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q3/5

Find the missing number.

$$\frac{6}{8} = \frac{?}{4}$$

Missing number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q4/5

486

QUESTION ID

S13-B4-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q4/5

True or false?

$\frac{4}{6}$ is equivalent to $\frac{6}{8}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q5/5

487

QUESTION ID

S13-B4-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B4-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.A.1 • Set 13 • BACKWARD 4 • Q5/5

SET 13 • 6.NS.A.1

BACKWARD 5

Mixed numbers as fractions • Grade 4 • Number & Operations–Fractions •
4.NF.B.3.b

Write $1\frac{1}{2}$ as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q1/5

489

QUESTION ID

S13-B5-Q1

TYPE

Short answer

CORRECT ANSWER

3/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q1/5

Write $3\frac{1}{3}$ as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q2/5

490

QUESTION ID

S13-B5-Q2

TYPE

Short answer

CORRECT ANSWER

10/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q2/5

True or false?

$$2\frac{1}{4} = \frac{9}{4}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q3/5

491

QUESTION ID

S13-B5-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q3/5

Which fraction is equal to $1\frac{2}{3}$?

A. $\frac{3}{3}$

B. $\frac{5}{3}$

C. $\frac{7}{3}$

D. $\frac{12}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q4/5

492

QUESTION ID

S13-B5-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q4/5

Write $\frac{11}{4}$ as a mixed number.

Mixed number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q5/5

493

QUESTION ID

S13-B5-Q5

TYPE

Short answer

CORRECT ANSWER

2 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B5-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.B.3.b • Set 13 • BACKWARD 5 • Q5/5

SET 13 • 6.NS.A.1

FORWARD 1

Multiply and divide rational numbers • Grade 7 • The Number System •
7.NS.A.2.c

Find the value.

$$\left(-\frac{2}{3}\right) \div \frac{4}{5}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q1/5

495

QUESTION ID

S13-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−5/6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q1/5

Find the value.

$$-\frac{3}{4} \times \left(-\frac{8}{9}\right)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q2/5

496

QUESTION ID

S13-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q2/5

$$(-1\frac{1}{2}) \div (-\frac{3}{8}) = ?$$

A. -4

B. $-\frac{9}{16}$

C. $\frac{9}{16}$

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q3/5

497

QUESTION ID

S13-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q3/5

Find the value.

$$\frac{5}{6} \div \left(-\frac{5}{12}\right)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q4/5

498

QUESTION ID

S13-F1-Q4

TYPE

Short answer

CORRECT ANSWER

-2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q4/5

True or false?

$$\left(-\frac{1}{2}\right) \times \frac{4}{5} = -\frac{2}{5}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q5/5

499

QUESTION ID

S13-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 13 • FORWARD 1 • Q5/5

SET 13 • 6.NS.A.1

FORWARD 2

Equations with fraction coefficients • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$\frac{3}{4}(x - 2) = \frac{1}{2}x + 5$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q1/5

501

QUESTION ID

S13-F2-Q1

TYPE

Short answer

CORRECT ANSWER

26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q1/5

Solve for x.

$$\frac{2}{3}x + 4 = \frac{1}{6}x + 7$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q2/5

502

QUESTION ID

S13-F2-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q2/5

Solve for x.

$$\frac{x}{2} + \frac{x}{3} = 10$$

- A. 5
- B. 6
- C. 12
- D. 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q3/5

503

QUESTION ID

S13-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q3/5

Solve for x.

$$\frac{1}{2}(4x + 6) = 3x - 5$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q4/5

504

QUESTION ID

S13-F2-Q4

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q4/5

Solve for y.

$$\frac{5}{8}y - 1 = \frac{3}{8}y + 2$$

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q5/5

505

QUESTION ID

S13-F2-Q5

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 13 • FORWARD 2 • Q5/5

Solve word problems involving division of fractions

MAIN	Grade 6 main question	6.NS.A.1
BACKWARD 1	Division as "how many groups?"	3.OA.A.2
BACKWARD 2	Unit fractions build a fraction	3.NF.A.1
BACKWARD 3	Area with fractional side lengths	5.NF.B.4.b
BACKWARD 4	Word problems: dividing with unit fractions	5.NF.B.7.c
FORWARD 1	Real-world problems with rational numbers	7.NS.A.3
FORWARD 2	Solve linear equations from contexts	8.EE.C.7.b

SET 14 • 6.NS.A.1

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.A.1

How many $\frac{2}{3}$ -cup scoops are in $\frac{5}{6}$ cup of flour?

Scoops: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q1/5

508

QUESTION ID

S14-M-Q1

TYPE

Short answer

CORRECT ANSWER

$5/4 = 1 \frac{1}{4}$ scoops

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q1/5

A path is $\frac{3}{4}$ mile long. It is divided into sections that are each $\frac{1}{8}$ mile long.

How many sections are there?

Sections: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q2/5

509

QUESTION ID

S14-M-Q2

TYPE

Short answer

CORRECT ANSWER

6 sections

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q2/5

A rectangular patio has an area of $\frac{3}{5}$ square yard. Its length is $\frac{9}{10}$ yard.

What is its width?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q3/5

510

QUESTION ID

S14-M-Q3

TYPE

Short answer

CORRECT ANSWER

$\frac{2}{3}$ yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q3/5

A pitcher holds $\frac{9}{10}$ liter of juice. Each glass holds $\frac{3}{10}$ liter.

How many glasses can be filled?

A. $\frac{27}{100}$

B. $\frac{1}{3}$

C. 3

D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q4/5

511

QUESTION ID

S14-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q4/5

Ana has $\frac{5}{6}$ pound of trail mix. She fills bags that each hold $\frac{5}{12}$ pound.

How many bags can she fill?

Bags: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q5/5

512

QUESTION ID

S14-M-Q5

TYPE

Short answer

CORRECT ANSWER

2 bags

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 14 • MAIN • Q5/5

SET 14 • 6.NS.A.1

BACKWARD 1

Division as "how many groups?" • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.2

How many groups of 4 are in 20?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q1/5

514

QUESTION ID

S14-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q1/5

True or false?

$12 \div 3$ can mean "How many groups of 3 are in 12?"

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q2/5

515

QUESTION ID

S14-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q2/5

Tom has 18 cookies. He puts 6 cookies in each bag.

Which equation finds the number of bags?

A. $18 \times 6 = \square$

B. $18 - 6 = \square$

C. $18 \div 6 = \square$

D. $6 \div 18 = \square$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q3/5

516

QUESTION ID

S14-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $18 \div 6 = \square$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q3/5

24 students are split into teams of 8.

How many teams are there?

Teams: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q4/5

517

QUESTION ID

S14-B1-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q4/5

How many 5s are in 35?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q5/5

518

QUESTION ID

S14-B1-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.2 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • 6.NS.A.1

BACKWARD 2

Unit fractions build a fraction • Grade 3 • Number & Operations–Fractions •
3.NF.A.1

How many $\frac{1}{4}$ s make $\frac{3}{4}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q1/5

520

QUESTION ID

S14-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q1/5

How many $\frac{1}{8}$ s make 1 whole?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q2/5

521

QUESTION ID

S14-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q2/5

True or false?

$\frac{5}{6}$ is 5 parts of size $\frac{1}{6}$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q3/5

522

QUESTION ID

S14-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q3/5

$\frac{3}{8}$ is made of how many parts of size $\frac{1}{8}$?

- A. $\frac{1}{3}$
- B. 3
- C. 5
- D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q4/5

523

QUESTION ID

S14-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q4/5

How many $\frac{1}{3}$ s make $\frac{2}{3}$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q5/5

524

QUESTION ID

S14-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.1 • Set 14 • BACKWARD 2 • Q5/5

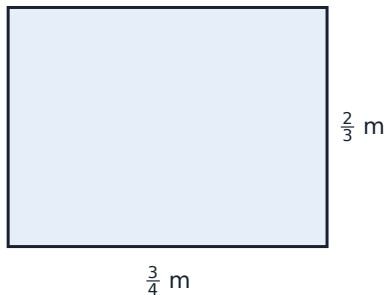
SET 14 • 6.NS.A.1

BACKWARD 3

Area with fractional side lengths • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.b

What is the area of the rectangle?

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q1/5

526

QUESTION ID

S14-B3-Q1

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{2}$ square meter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q1/5

True or false?

A rectangle that is $\frac{1}{2}$ foot by $\frac{1}{3}$ foot has an area of $\frac{1}{6}$ square foot.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q2/5

527

QUESTION ID

S14-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q2/5

A rectangle is $\frac{2}{5}$ unit by $\frac{1}{2}$ unit.

What is its area?

- A.** $\frac{1}{5}$ square unit **B.** $\frac{2}{7}$ square unit
C. $\frac{3}{7}$ square unit **D.** $\frac{9}{10}$ square unit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q3/5

528

QUESTION ID

S14-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 1/5 square unit

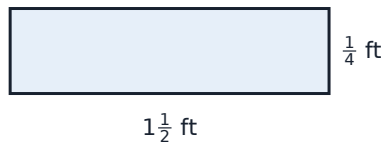
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q3/5

What is the area of the rectangle?

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q4/5

529

QUESTION ID

S14-B3-Q4

TYPE

Short answer

CORRECT ANSWER

$\frac{3}{8}$ square foot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q4/5

A garden is $\frac{5}{6}$ yard long and $\frac{3}{5}$ yard wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q5/5

530

QUESTION ID

S14-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{2}$ square yard

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.b • Set 14 • BACKWARD 3 • Q5/5

SET 14 • 6.NS.A.1

BACKWARD 4

Word problems: dividing with unit fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.7.c

How many $\frac{1}{4}$ -pound burgers can be made from 3 pounds of beef?

Burgers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q1/5

532

QUESTION ID

S14-B4-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q1/5

A 4-pound bag of rice is divided into $\frac{1}{2}$ -pound portions.

How many portions are there?

Portions: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q2/5

533

QUESTION ID

S14-B4-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q2/5

Which expression finds how many $\frac{1}{5}$ -mile laps are in 3 miles?

A. $\frac{1}{5} \div 3$

B. $3 - \frac{1}{5}$

C. $3 \div \frac{1}{5}$

D. $3 \times \frac{1}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q3/5

534

QUESTION ID

S14-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $3 \div 1/5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q3/5

4 friends share $\frac{1}{3}$ pound of cheese equally.

How much cheese does each friend get?

Each friend: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q4/5

535

QUESTION ID

S14-B4-Q4

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{12}$ pound

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q4/5

True or false?

There are 10 half-hours in 5 hours.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q5/5

536

QUESTION ID

S14-B4-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B4-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.7.c • Set 14 • BACKWARD 4 • Q5/5

SET 14 • 6.NS.A.1

FORWARD 1

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A diver descends $\frac{3}{4}$ foot each second.

How many seconds does it take to descend $5\frac{1}{4}$ feet?

Seconds: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q1/5

538

QUESTION ID

S14-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7 seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q1/5

The temperature was 4°F . It dropped $2\frac{1}{2}^{\circ}\text{F}$ each hour for 3 hours.

What was the final temperature?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q2/5

539

QUESTION ID

S14-F1-Q2

TYPE

Short answer

CORRECT ANSWER

-3.5°F

GRADING NOTE $-3\frac{1}{2}$ degrees

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q2/5

A stock's value changed by $-\$1\frac{1}{4}$ each day for 4 days.

What was the total change?

- A. $-\$5$
- B. $-\$4\frac{1}{4}$
- C. $-\$1$
- D. $\$5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q3/5

540

QUESTION ID

S14-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $-\$5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q3/5

A recipe needs $\frac{2}{3}$ cup of oil per batch. Jo has $3\frac{1}{3}$ cups of oil.

How many batches can Jo make?

Batches: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q4/5

541

QUESTION ID

S14-F1-Q4

TYPE

Short answer

CORRECT ANSWER

5 batches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q4/5

A hiker descends 1,250 feet in $\frac{5}{6}$ hour at a constant rate.

What is the change in elevation per hour?

Change per hour: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q5/5

542

QUESTION ID

S14-F1-Q5

TYPE

Short answer

CORRECT ANSWER

–1,500 feet per hour

GRADING NOTE A descent of 1,500 feet per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 14 • FORWARD 1 • Q5/5

SET 14 • 6.NS.A.1

FORWARD 2

Solve linear equations from contexts • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$\frac{1}{2}(x + 6) = \frac{1}{3}(x + 12)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q1/5

544

QUESTION ID

S14-F2-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q1/5

Lee's age plus $\frac{1}{3}$ of his age is 24.

Write and solve an equation to find Lee's age.

Age: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q2/5

545

QUESTION ID

S14-F2-Q2

TYPE

Short answer

CORRECT ANSWER

18 years old

GRADING NOTE Equation: $a + (1/3)a = 24$, or equivalent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q2/5

Tank A has 40 gallons and drains $\frac{3}{4}$ gallon per minute.

Tank B has 10 gallons and fills at $1\frac{1}{4}$ gallons per minute.

After how many minutes will the tanks hold the same amount?

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q3/5

546

QUESTION ID

S14-F2-Q3

TYPE

Short answer

CORRECT ANSWER

15 minutes

GRADING NOTE $40 - (3/4)t = 10 + (5/4)t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q3/5

Solve for x.

$$\frac{2}{5}x - 3 = \frac{1}{5}x + 1$$

A. -10

B. 4

C. 10

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q4/5

547

QUESTION ID

S14-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q4/5

Solve for n.

$$\frac{3}{4}n + \frac{1}{2} = \frac{1}{4}n + 4$$

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q5/5

548

QUESTION ID

S14-F2-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 14 • FORWARD 2 • Q5/5

Divide multi-digit numbers using the standard algorithm

MAIN	Grade 6 main question	6.NS.B.2
BACKWARD 1	Multiplying multi-digit numbers	5.NBT.B.5
BACKWARD 2	Dividing by a one-digit number	4.NBT.B.6
BACKWARD 3	Estimating quotients	5.NBT.B.6
BACKWARD 4	Multi-digit subtraction	4.NBT.B.4
FORWARD 1	Long division: fractions to decimals	7.NS.A.2.d
FORWARD 2	Repeating decimals and rational numbers	8.NS.A.1

SET 15 • 6.NS.B.2

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.2

Use the standard algorithm to divide.

$$4,508 \div 23$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q1/5

551

QUESTION ID

S15-M-Q1

TYPE

Short answer

CORRECT ANSWER

196

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q1/5

Use the standard algorithm to divide.

$$7,236 \div 36$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q2/5

552

QUESTION ID

S15-M-Q2

TYPE

Short answer

CORRECT ANSWER

201

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q2/5

Use the standard algorithm to divide.

$$9,975 \div 75$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q3/5

553

QUESTION ID

S15-M-Q3

TYPE

Short answer

CORRECT ANSWER

133

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q3/5

Use the standard algorithm to divide.

$$1,680 \div 48$$

- A. 30
- B. 35
- C. 36
- D. 350

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q4/5

554

QUESTION ID

S15-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q4/5

Use the standard algorithm to divide.

$$25,494 \div 42$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q5/5

555

QUESTION ID

S15-M-Q5

TYPE

Short answer

CORRECT ANSWER

607

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

Grade 6 • The Number System • 6.NS.B.2 • Set 15 • MAIN • Q5/5

SET 15 • 6.NS.B.2

BACKWARD 1

Multiplying multi-digit numbers • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$46 \times 23$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q1/5

557

QUESTION ID

S15-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1,058

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q1/5

557

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$196 \times 23$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q2/5

558

QUESTION ID

S15-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4,508

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q2/5

558

True or false?

$$35 \times 48 = 1,680$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q3/5

559

QUESTION ID

S15-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q3/5

$$133 \times 75 = ?$$

- A.** 1,596
- B.** 9,775
- C.** 9,875
- D.** 9,975

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q4/5

560

QUESTION ID

S15-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 9,975

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$607 \times 42$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q5/5

561

QUESTION ID

S15-B1-Q5

TYPE

Short answer

CORRECT ANSWER

25,494

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 15 • BACKWARD 1 • Q5/5

561

SET 15 • 6.NS.B.2

BACKWARD 2

Dividing by a one-digit number • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.6

Divide.

$$852 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q1/5

563

QUESTION ID

S15-B2-Q1

TYPE

Short answer

CORRECT ANSWER

213

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$1,256 \div 8$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q2/5

564

QUESTION ID

S15-B2-Q2

TYPE

Short answer

CORRECT ANSWER

157

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q2/5

True or false?

$$936 \div 3 = 312$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q3/5

565

QUESTION ID

S15-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q3/5

$$714 \div 7 = ?$$

- A.** 12
- B.** 102
- C.** 120
- D.** 1,002

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q4/5

566

QUESTION ID

S15-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 102

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$2,345 \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q5/5

567

QUESTION ID

S15-B2-Q5

TYPE

Short answer

CORRECT ANSWER

469

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 15 • BACKWARD 2 • Q5/5

567

SET 15 • 6.NS.B.2

BACKWARD 3

Estimating quotients • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6

Which is the best estimate for $4,508 \div 23$?

- A. 2
- B. 20
- C. 200
- D. 2,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q1/5

569

QUESTION ID

S15-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. 200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q1/5

Which is the best estimate for $7,236 \div 36$?

- A. 2
- B. 20
- C. 200
- D. 2,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q2/5

570

QUESTION ID

S15-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q2/5

True or false?

$9,975 \div 75$ is between 100 and 200.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q3/5

571

QUESTION ID

S15-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q3/5

Estimate $6,120 \div 29$ by rounding 29 to 30.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q4/5

572

QUESTION ID

S15-B3-Q4

TYPE

Short answer

CORRECT ANSWER

204

GRADING NOTE $6,120 \div 30 = 204$; "about 200" is also acceptable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q4/5

How many digits are in the quotient of $1,680 \div 48$?

- A. 1
- B. 2
- C. 3
- D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q5/5

573

QUESTION ID

S15-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • 6.NS.B.2

BACKWARD 4

Multi-digit subtraction • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$5,012 - 4,698$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q1/5

575

QUESTION ID

S15-B4-Q1

TYPE

Short answer

CORRECT ANSWER

314

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q1/5

575

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$3,000 - 1,847$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q2/5

576

QUESTION ID

S15-B4-Q2

TYPE

Short answer

CORRECT ANSWER

1,153

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q2/5

576

True or false?

$$7,236 - 7,200 = 36$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q3/5

577

QUESTION ID

S15-B4-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q3/5

$$9,975 - 9,000 = ?$$

- A.** 75
- B.** 975
- C.** 1,975
- D.** 9,075

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q4/5

578

QUESTION ID

S15-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 975

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$6,203 - 2,875$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q5/5

579

QUESTION ID

S15-B4-Q5

TYPE

Short answer

CORRECT ANSWER

3,328

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B4-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 15 • BACKWARD 4 • Q5/5

579

SET 15 • 6.NS.B.2

FORWARD 1

Long division: fractions to decimals • Grade 7 • The Number System •
7.NS.A.2.d

Use long division to write $\frac{5}{11}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q1/5

581

QUESTION ID

S15-F1-Q1

TYPE

Short answer

CORRECT ANSWER

**0.454545... (0.45
repeating)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q1/5

Use long division to write $\frac{3}{8}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q2/5

582

QUESTION ID

S15-F1-Q2

TYPE

Short answer

CORRECT ANSWER

0.375

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q2/5

Which fraction has a repeating decimal?

A. $\frac{1}{5}$

B. $\frac{2}{3}$

C. $\frac{3}{4}$

D. $\frac{7}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q3/5

583

QUESTION ID

S15-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q3/5

True or false?

The decimal form of $\frac{7}{20}$ terminates.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q4/5

584

QUESTION ID

S15-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q4/5

Use long division to write $\frac{5}{6}$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q5/5

585

QUESTION ID

S15-F1-Q5

TYPE

Short answer

CORRECT ANSWER

**0.8333... (0.83, with
the 3 repeating)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.d • Set 15 • FORWARD 1 • Q5/5

SET 15 • 6.NS.B.2

FORWARD 2

Repeating decimals and rational numbers • Grade 8 • The Number System •
8.NS.A.1

Write 0.272727... as a fraction in simplest form.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q1/5

587

QUESTION ID

S15-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3/11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q1/5

Write 0.444... as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q2/5

588

QUESTION ID

S15-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q2/5

Which number is irrational?

- A. 0.333...
- B. $\frac{5}{8}$
- C. $\sqrt{16}$
- D. $\sqrt{7}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q3/5

589

QUESTION ID

S15-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $\sqrt{7}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q3/5

True or false?

Every repeating decimal is a rational number.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q4/5

590

QUESTION ID

S15-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q4/5

Write 1.1666... as a fraction.

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q5/5

591

QUESTION ID

S15-F2-Q5

TYPE

Short answer

CORRECT ANSWER

7/6

GRADING NOTE 1 1/6 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

Grade 8 • The Number System • 8.NS.A.1 • Set 15 • FORWARD 2 • Q5/5

Add multi-digit decimals

MAIN	Grade 6 main question
BACKWARD 1	Place value of decimal digits
BACKWARD 2	Equivalent decimals
BACKWARD 3	Lining up decimal points to add
BACKWARD 4	Multi-digit addition
FORWARD 1	Add rational numbers
FORWARD 2	Linear equations with decimal coefficients

6.NS.B.3

5.NBT.A.3.a

5.NBT.A.3.b

5.NBT.B.7

4.NBT.B.4

7.NS.A.1.d

8.EE.C.7.b

SET 16 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Find the sum.

$$315.4 + 28.75 + 6.309$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q1/5

594

QUESTION ID

S16-M-Q1

TYPE

Short answer

CORRECT ANSWER

350.459

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q1/5

Find the sum.

$$47.08 + 9.6$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q2/5

595

QUESTION ID

S16-M-Q2

TYPE

Short answer

CORRECT ANSWER

56.68

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q2/5

Find the sum.

$$128.5 + 3.47 + 60.009$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q3/5

596

QUESTION ID

S16-M-Q3

TYPE

Short answer

CORRECT ANSWER

191.979

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q3/5

Find the sum.

$$6.75 + 12.9 + 0.48$$

- A.** 7.23
- B.** 19.23
- C.** 20.03
- D.** 20.13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q4/5

597

QUESTION ID

S16-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 20.13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q4/5

Find the sum.

$$0.875 + 14.2 + 3.06$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q5/5

598

QUESTION ID

S16-M-Q5

TYPE

Short answer

CORRECT ANSWER

18.135

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 16 • MAIN • Q5/5

SET 16 • 6.NS.B.3

BACKWARD 1

Place value of decimal digits • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

In 7.358, what place value does the digit 5 have?

- A.** Tenths
- B.** Hundredths
- C.** Ones
- D.** Thousandths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q1/5

600

QUESTION ID

S16-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Hundredths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q1/5

In 42.906, which digit is in the tenths place?

Digit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q2/5

601

QUESTION ID

S16-B1-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q2/5

True or false?

In 3.47, the digit 7 is in the hundredths place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q3/5

602

QUESTION ID

S16-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q3/5

What is the value of the 8 in 0.584?

- A. 0.008
- B. 0.08
- C. 0.8
- D. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q4/5

603

QUESTION ID

S16-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.08

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q4/5

Write $5 + 0.3 + 0.07$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q5/5

604

QUESTION ID

S16-B1-Q5

TYPE

Short answer

CORRECT ANSWER

5.37

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 16 • BACKWARD 1 • Q5/5

SET 16 • 6.NS.B.3

BACKWARD 2

Equivalent decimals • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.b

True or false?

2.4 and 2.40 have the same value.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q1/5

606

QUESTION ID

S16-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q1/5

True or false?

5.6 and 5.06 have the same value.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q2/5

607

QUESTION ID

S16-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q2/5

Which decimal is equal to 7.5?

- A.** 0.75
- B.** 7.005
- C.** 7.05
- D.** 7.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q3/5

608

QUESTION ID

S16-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 7.50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q3/5

Write 9.6 with a zero in the hundredths place.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q4/5

609

QUESTION ID

S16-B2-Q4

TYPE

Short answer

CORRECT ANSWER

9.60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q4/5

True or false?

$$0.300 = 0.3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q5/5

610

QUESTION ID

S16-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 16 • BACKWARD 2 • Q5/5

SET 16 • 6.NS.B.3

BACKWARD 3

Lining up decimal points to add • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Which setup lines up the decimal points for $36.2 + 4.58$?

A.

$$\begin{array}{r} 36.2 \\ +4.58 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 36.20 \\ + 4.58 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q1/5

612

QUESTION ID

S16-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Choice B (picture)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q1/5

Which setup lines up the decimal points for $5.07 + 12.3$?

A.

$$\begin{array}{r} 5.07 \\ +12.30 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 5.07 \\ +12.3 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q2/5

613

QUESTION ID

S16-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Choice A (picture)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q2/5

True or false?

The decimal points are lined up correctly.

$$\begin{array}{r} 8.4 \\ +2.75 \\ \hline \end{array}$$

- A.** True
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q3/5

614

QUESTION ID

S16-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q3/5

Which setup lines up the decimal points for $128.5 + 3.47$?

A.

$$\begin{array}{r} 128.50 \\ + 3.47 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 128.5 \\ + 3.47 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q4/5

615

QUESTION ID

S16-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Choice A (picture)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q4/5

True or false?

The decimal points are lined up correctly.

- A.** True
- B.** False

$$\begin{array}{r} 9.30 \\ + 0.45 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q5/5

616

QUESTION ID

S16-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 16 • BACKWARD 3 • Q5/5

SET 16 • 6.NS.B.3

BACKWARD 4

Multi-digit addition • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$3,154 + 2,875$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q1/5

618

QUESTION ID

S16-B4-Q1

TYPE

Short answer

CORRECT ANSWER

6,029

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q1/5

618

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$4,605 + 2,798$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q2/5

619

QUESTION ID

S16-B4-Q2

TYPE

Short answer

CORRECT ANSWER

7,403

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q2/5

619

True or false?

$$1,975 + 3,489 = 5,464$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q3/5

620

QUESTION ID

S16-B4-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q3/5

$$6,758 + 1,265 = ?$$

- A. 7,013
- B. 7,923
- C. 8,013
- D. 8,023

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q4/5

621

QUESTION ID

S16-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 8,023

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$12,480 + 9,635$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q5/5

622

QUESTION ID

S16-B4-Q5

TYPE

Short answer

CORRECT ANSWER

22,115

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B4-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 16 • BACKWARD 4 • Q5/5

622

SET 16 • 6.NS.B.3

FORWARD 1

Add rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$-18.6 + 4.35 =$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q1/5

624

QUESTION ID

S16-F1-Q1

TYPE

Short answer

CORRECT ANSWER

-14.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q1/5

Find the value.

$$-24.5 + 3.27 =$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q2/5

625

QUESTION ID

S16-F1-Q2

TYPE

Short answer

CORRECT ANSWER

-21.23

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q2/5

Find the value.

$$-3.4 + (-9.15) =$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q3/5

626

QUESTION ID

S16-F1-Q3

TYPE

Short answer

CORRECT ANSWER

-12.55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q3/5

Find the value.

$$-6.25 + (-2.8) = ?$$

- A. -9.05
- B. -3.45
- C. 3.45
- D. 9.05

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q4/5

627

QUESTION ID

S16-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. -9.05

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q4/5

Find the value.

$$-15.75 + 20.3 =$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q5/5

628

QUESTION ID

S16-F1-Q5

TYPE

Short answer

CORRECT ANSWER

4.55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 16 • FORWARD 1 • Q5/5

SET 16 • 6.NS.B.3

FORWARD 2

Linear equations with decimal coefficients • Grade 8 • Expressions & Equations
• 8.EE.C.7.b

Solve for x.

$$-2.4x + 7.8 = 1.2x - 1.2$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q1/5

630

QUESTION ID

S16-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q1/5

Solve for x.

$$-1.5x + 6.4 = 0.5x - 1.6$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q2/5

631

QUESTION ID

S16-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q2/5

Solve for x.

$$0.8x + 7.2 = -1.2x + 1.2$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q3/5

632

QUESTION ID

S16-F2-Q3

TYPE

Short answer

CORRECT ANSWER

-3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q3/5

Solve for x.

$$-4.5x - 3.25 = -2.5x + 0.75$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q4/5

633

QUESTION ID

S16-F2-Q4

TYPE

Short answer

CORRECT ANSWER

-2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q4/5

Solve for x.

$$2.4(x + 1.5) = 1.4x + 6.6$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q5/5

634

QUESTION ID

S16-F2-Q5

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 16 • FORWARD 2 • Q5/5

Subtract multi-digit decimals

MAIN

Grade 6 main question

6.NS.B.3

BACKWARD 1

Place value of decimal digits

5.NBT.A.3.a

BACKWARD 2

Writing zeros to make equal decimal places

5.NBT.A.3.b

BACKWARD 3

Lining up decimal points to subtract

5.NBT.B.7

BACKWARD 4

Subtracting across zeros

4.NBT.B.4

FORWARD 1

Subtract rational numbers by adding the additive inverse

7.NS.A.1.c

FORWARD 2

Solve equations by collecting like terms

8.EE.C.7.b

SET 17 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Find the difference.

$$52.3 - 18.47$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q1/5

637

QUESTION ID

S17-M-Q1

TYPE

Short answer

CORRECT ANSWER

33.83

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q1/5

Find the difference.

$$100 - 36.58$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q2/5

638

QUESTION ID

S17-M-Q2

TYPE

Short answer

CORRECT ANSWER

63.42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q2/5

Find the difference.

$$7.4 - 2.068$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q3/5

639

QUESTION ID

S17-M-Q3

TYPE

Short answer

CORRECT ANSWER

5.332

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q3/5

Find the difference.

$$15.02 - 6.9$$

- A.** 8.12
- B.** 8.93
- C.** 9.12
- D.** 14.33

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q4/5

640

QUESTION ID

S17-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 8.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q4/5

Find the difference.

$$243.6 - 87.95$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q5/5

641

QUESTION ID

S17-M-Q5

TYPE

Short answer

CORRECT ANSWER

155.65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 17 • MAIN • Q5/5

SET 17 • 6.NS.B.3

BACKWARD 1

Place value of decimal digits • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

In 2.068, what place value does the digit 6 have?

- A.** Tenths
- B.** Thousandths
- C.** Hundredths
- D.** Ones

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q1/5

643

QUESTION ID

S17-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Hundredths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q1/5

In 87.95, which digit is in the hundredths place?

Digit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q2/5

644

QUESTION ID

S17-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q2/5

True or false?

In 36.58, the digit 5 is in the tenths place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q3/5

645

QUESTION ID

S17-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q3/5

What is the value of the 4 in 7.04?

- A. 0.004
- B. 0.04
- C. 0.4
- D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q4/5

646

QUESTION ID

S17-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.04

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q4/5

Write $20 + 0.6 + 0.009$ as a decimal.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q5/5

647

QUESTION ID

S17-B1-Q5

TYPE

Short answer

CORRECT ANSWER

20.609

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 17 • BACKWARD 1 • Q5/5

SET 17 • 6.NS.B.3

BACKWARD 2

Writing zeros to make equal decimal places • Grade 5 • Number & Operations
in Base Ten • 5.NBT.A.3.b

True or false?

$$7.4 = 7.400$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q1/5

649

QUESTION ID

S17-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q1/5

Write 100 with two decimal places.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q2/5

650

QUESTION ID

S17-B2-Q2

TYPE

Short answer

CORRECT ANSWER

100.00

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q2/5

Which decimal is equal to 15.1?

- A.** 1.51
- B.** 15.001
- C.** 15.01
- D.** 15.10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q3/5

651

QUESTION ID

S17-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 15.10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q3/5

Write 52.3 with a digit in the hundredths place, without changing its value.

Decimal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q4/5

652

QUESTION ID

S17-B2-Q4

TYPE

Short answer

CORRECT ANSWER

52.30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q4/5

True or false?

$$6.9 = 6.09$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q5/5

653

QUESTION ID

S17-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.b • Set 17 • BACKWARD 2 • Q5/5

SET 17 • 6.NS.B.3

BACKWARD 3

Lining up decimal points to subtract • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Which setup lines up the decimal points for $15.6 - 3.42$?

A.

$$\begin{array}{r} 15.60 \\ - 3.42 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 15.6 \\ - 3.42 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q1/5

655

QUESTION ID

S17-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Choice A (picture)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q1/5

Which setup lines up the decimal points for
 $52.3 - 18.47$?

A.

$$\begin{array}{r} 52.3 \\ -18.47 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 52.30 \\ -18.47 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q2/5

656

QUESTION ID

S17-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Choice B (picture)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q2/5

True or false?

The decimal points are lined up correctly.

- A.** True
- B.** False

$$\begin{array}{r} 7.4 \\ -2.068 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q3/5

657

QUESTION ID

S17-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q3/5

True or false?

The decimal points are lined up correctly.

$$\begin{array}{r} 100.00 \\ - 36.58 \\ \hline \end{array}$$

- A.** True
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q4/5

658

QUESTION ID

S17-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q4/5

Which setup lines up the decimal points for
 $243.6 - 87.95$?

A.

$$\begin{array}{r} 243.60 \\ - 87.95 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 243.6 \\ - 87.95 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q5/5

659

QUESTION ID

S17-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Choice A (picture)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 17 • BACKWARD 3 • Q5/5

SET 17 • 6.NS.B.3

BACKWARD 4

Subtracting across zeros • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$5,000 - 2,847$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q1/5

661

QUESTION ID

S17-B4-Q1

TYPE

Short answer

CORRECT ANSWER

2,153

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$3,006 - 1,478$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q2/5

662

QUESTION ID

S17-B4-Q2

TYPE

Short answer

CORRECT ANSWER

1,528

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q2/5

True or false?

$$6,000 - 2,458 = 3,542$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q3/5

663

QUESTION ID

S17-B4-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q3/5

$$8,014 - 3,276 = ?$$

- A.** 4,738
- B.** 4,748
- C.** 4,838
- D.** 5,262

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q4/5

664

QUESTION ID

S17-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 4,738

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$10,000 - 3,658$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q5/5

665

QUESTION ID

S17-B4-Q5

TYPE

Short answer

CORRECT ANSWER

6,342

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B4-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 17 • BACKWARD 4 • Q5/5

665

SET 17 • 6.NS.B.3

FORWARD 1

Subtract rational numbers by adding the additive inverse • Grade 7 • The Number System • 7.NS.A.1.c

Find the value.

$$-4.2 - 3.75$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q1/5

667

QUESTION ID

S17-F1-Q1

TYPE

Short answer

CORRECT ANSWER

-7.95

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q1/5

Find the value.

$$2.5 - (-6.8)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q2/5

668

QUESTION ID

S17-F1-Q2

TYPE

Short answer

CORRECT ANSWER

9.3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q2/5

Which expression is equal to $8.1 - 12.4$?

- A. $12.4 - 8.1$
- B. $-8.1 + 12.4$
- C. $8.1 + (-12.4)$
- D. $8.1 + 12.4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q3/5

669

QUESTION ID

S17-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $8.1 + (-12.4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find the value.

$$-0.65 - (-1.2)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q4/5

670

QUESTION ID

S17-F1-Q4

TYPE

Short answer

CORRECT ANSWER

0.55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q4/5

670

True or false?

$$-7.5 - 2.5 = -7.5 + (-2.5)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q5/5

671

QUESTION ID

S17-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 17 • FORWARD 1 • Q5/5

SET 17 • 6.NS.B.3

FORWARD 2

Solve equations by collecting like terms • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$5.2x - 3.75 = 2.2x + 8.25$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q1/5

673

QUESTION ID

S17-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q1/5

Solve for x.

$$0.9x + 12.5 = 0.4x + 10$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q2/5

674

QUESTION ID

S17-F2-Q2

TYPE

Short answer

CORRECT ANSWER

-5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q2/5

Solve for x.

$$6.3 - 1.2x = 0.8x - 1.7$$

- A. -4
- B. 0.4
- C. 2.5
- D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q3/5

675

QUESTION ID

S17-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q3/5

Solve for x.

$$4.75 - x = 2x - 1.25$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q4/5

676

QUESTION ID

S17-F2-Q4

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q4/5

Solve for x.

$$3.4x - 9.6 = 1.4x - 2.6$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q5/5

677

QUESTION ID

S17-F2-Q5

TYPE

Short answer

CORRECT ANSWER

3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 17 • FORWARD 2 • Q5/5

Find and correct decimal alignment errors in addition and subtraction

MAIN	Grade 6 main question	6.NS.B.3
BACKWARD 1	Lining up decimal points	5.NBT.B.7
BACKWARD 2	Place value of decimal digits	5.NBT.A.3.a
BACKWARD 3	Estimating to check reasonableness	5.NBT.A.4
FORWARD 1	Find errors when adding and subtracting rational numbers	7.NS.A.1.d
FORWARD 2	Find errors in solving linear equations	8.EE.C.7.b

SET 18 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

Jon put this example on the board.

What mistake was made?

What should the answer be?

Mistake: _____

Correct answer: _____

$$\begin{array}{r} 315.2 \\ 24.8 \\ +7.936 \\ \hline 11.336 \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q1/5

680

QUESTION ID

S18-M-Q1

TYPE

Short answer

CORRECT ANSWER

Mistake: the numbers were lined up by their last digits instead of by their decimal points. Correct answer: 347.936

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q1/5

Ava put this example on the board.

What mistake was made?
What should the answer be?

$$\begin{array}{r} 56.4 \\ +3.25 \\ \hline 8.89 \end{array}$$

Mistake: _____

Correct answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q2/5

681

QUESTION ID

S18-M-Q2

TYPE

Short answer

CORRECT ANSWER

Mistake: the numbers were lined up by their last digits instead of by their decimal points. Correct answer: 59.65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q2/5

Leo put this example on the board.

What mistake was made?
What should the answer be?

$$\begin{array}{r} 48.6 \\ -2.15 \\ \hline 2.71 \end{array}$$

Mistake: _____

Correct answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q3/5

682

QUESTION ID

S18-M-Q3

TYPE

Short answer

CORRECT ANSWER

Mistake: the numbers were lined up by their last digits instead of by their decimal points. Correct answer: 46.45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q3/5

Mia put this example on the board.

What mistake was made?
What should the answer be?

$$\begin{array}{r} 7.08 \\ +12.6 \\ \hline 8.34 \end{array}$$

Mistake: _____

Correct answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q4/5

683

QUESTION ID

S18-M-Q4

TYPE

Short answer

CORRECT ANSWER

Mistake: the numbers were lined up by their last digits instead of by their decimal points. Correct answer: 19.68

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q4/5

Ben put this example on the board.

What mistake was made?
What should the answer be?

$$\begin{array}{r} 125.6 \\ -3.48 \\ \hline 9.08 \end{array}$$

Mistake: _____

Correct answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q5/5

684

QUESTION ID

S18-M-Q5

TYPE

Short answer

CORRECT ANSWER

Mistake: the numbers were lined up by their last digits instead of by their decimal points. Correct answer: 122.12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 18 • MAIN • Q5/5

SET 18 • 6.NS.B.3

BACKWARD 1

Lining up decimal points • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.7

Which setup lines up the decimal points for $56.4 + 3.25$?

A.

$$\begin{array}{r} 56.4 \\ +3.25 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 56.40 \\ + 3.25 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q1/5

686

QUESTION ID

S18-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Choice B (picture)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q1/5

Which setup lines up the decimal points for $48.6 - 2.15$?

A.

$$\begin{array}{r} 48.60 \\ - 2.15 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 48.6 \\ - 2.15 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q2/5

687

QUESTION ID

S18-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Choice A (picture)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q2/5

True or false?

The decimal points are lined up correctly.

$$\begin{array}{r} 7.08 \\ +12.6 \\ \hline \end{array}$$

- A.** True
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q3/5

688

QUESTION ID

S18-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q3/5

True or false?

The decimal points are lined up correctly.

- A.** True
- B.** False

$$\begin{array}{r} 125.60 \\ - 3.48 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q4/5

689

QUESTION ID

S18-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q4/5

Which setup lines up the decimal points for
 $24.8 + 7.936$?

A.

$$\begin{array}{r} 24.8 \\ +7.936 \\ \hline \end{array}$$

B.

$$\begin{array}{r} 24.800 \\ + 7.936 \\ \hline \end{array}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q5/5

690

QUESTION ID

S18-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Choice B (picture)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • 6.NS.B.3

BACKWARD 2

Place value of decimal digits • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.3.a

In 7.936, what place value does the digit 3 have?

- A.** Hundredths
- B.** Tenths
- C.** Thousandths
- D.** Ones

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q1/5

692

QUESTION ID

S18-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Hundredths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q1/5

In 315.2, which digit is in the ones place?

Digit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q2/5

693

QUESTION ID

S18-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q2/5

True or false?

In 3.25, the digit 2 is in the tenths place.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q3/5

694

QUESTION ID

S18-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q3/5

What is the value of the 6 in 12.56?

- A.** 0.06
- B.** 0.6
- C.** 6
- D.** 60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q4/5

695

QUESTION ID

S18-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0.06

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q4/5

In 48.6, which digit is in the tenths place?

Digit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q5/5

696

QUESTION ID

S18-B2-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.3.a • Set 18 • BACKWARD 2 • Q5/5

SET 18 • 6.NS.B.3

BACKWARD 3

Estimating to check reasonableness • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4

Round each number to the nearest whole number. Then add to estimate.

$$315.2 + 24.8 + 7.936$$

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q1/5

698

QUESTION ID

S18-B3-Q1

TYPE

Short answer

CORRECT ANSWER

348

GRADING NOTE $315 + 25 + 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q1/5

Round 48.6 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q2/5

699

QUESTION ID

S18-B3-Q2

TYPE

Short answer

CORRECT ANSWER

49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q2/5

True or false?

7.08 + 12.6 is about 20.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q3/5

700

QUESTION ID

S18-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q3/5

Which is the best estimate for $125.6 - 3.48$?

- A. 9
- B. 90
- C. 122
- D. 130

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q4/5

701

QUESTION ID

S18-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 122

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Round 56.4 and 3.25 to the nearest whole numbers.
Then add.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q5/5

702

QUESTION ID

S18-B3-Q5

TYPE

Short answer

CORRECT ANSWER

59**GRADING NOTE** $56 + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 18 • BACKWARD 3 • Q5/5

702

SET 18 • 6.NS.B.3

FORWARD 1

Find errors when adding and subtracting rational numbers • Grade 7 • The Number System • 7.NS.A.1.d

Ray wrote: $-3.5 + 1.25 = -4.75$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q1/5

704

QUESTION ID

S18-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Mistake: the numbers have different signs, so their absolute values should be subtracted, not added. Correct value: -2.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q1/5

Tia wrote: $-6.2 - 2.4 = -3.8$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q2/5

705

QUESTION ID

S18-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Mistake: subtracting 2.4 means adding -2.4 ; the result should be more negative, not closer to 0. Correct value: -8.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q2/5

Sam wrote: $4.5 - (-1.5) = 3$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q3/5

706

QUESTION ID

S18-F1-Q3

TYPE

Short answer

CORRECT ANSWER

**Mistake: subtracting
–1.5 means adding
1.5, not subtracting
1.5. Correct value: 6**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q3/5

Kai wrote: $-7.25 + (-2.5) = -4.75$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q4/5

707

QUESTION ID

S18-F1-Q4

TYPE

Short answer

CORRECT ANSWER

Mistake: both numbers are negative, so their absolute values should be added, not subtracted. Correct value: -9.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q4/5

Lia wrote: $-0.8 + 2.3 = -1.5$

What mistake was made? What is the correct value?

Mistake: _____

Correct answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q5/5

708

QUESTION ID

S18-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Mistake: the sign is wrong; 2.3 has the greater absolute value, so the sum is positive. Correct value: 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 18 • FORWARD 1 • Q5/5

SET 18 • 6.NS.B.3

FORWARD 2

Find errors in solving linear equations • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Kim solved $2.4x + 1.2 = 0.4x - 3.8$.

Step 1: $2x + 1.2 = -3.8$

Step 2: $2x = -2.6$

Step 3: $x = -1.3$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q1/5

710

QUESTION ID

S18-F2-Q1

TYPE

Short answer

CORRECT ANSWER

**Step 2 (subtracting
1.2 gives $2x = -5$).
Correct solution:
 $x = -2.5$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q1/5

Jo solved $3.5x - 2 = 1.5x + 6$.

Step 1: $2x - 2 = 6$

Step 2: $2x = 4$

Step 3: $x = 2$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q2/5

711

QUESTION ID

S18-F2-Q2

TYPE

Short answer

CORRECT ANSWER

**Step 2 (adding 2
gives $2x = 8$).**

**Correct solution:
 $x = 4$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q2/5

Ali solved $-1.2x + 5 = 0.8x - 3$.

Step 1: $-2x + 5 = -3$

Step 2: $-2x = -8$

Step 3: $x = -4$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q3/5

712

QUESTION ID

S18-F2-Q3

TYPE

Short answer

CORRECT ANSWER

**Step 3 (dividing -8
by -2 gives 4).**

**Correct solution:
 $x = 4$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q3/5

Ned solved $0.5(x + 6) = 2x - 3$.

Step 1: $0.5x + 6 = 2x - 3$

Step 2: $9 = 1.5x$

Step 3: $x = 6$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q4/5

713

QUESTION ID

S18-F2-Q4

TYPE

Short answer

CORRECT ANSWER

Step 1 (0.5 must multiply both terms: $0.5x + 3$). Correct solution: $x = 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q4/5

Eve solved $4.2x - 1.5 = 2.2x + 2.5$.

Step 1: $2x - 1.5 = 2.5$

Step 2: $2x = 1$

Step 3: $x = 0.5$

Which step has the mistake? What is the correct solution?

Step with the mistake: _____

Correct solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q5/5

714

QUESTION ID

S18-F2-Q5

TYPE

Short answer

CORRECT ANSWER

**Step 2 (adding 1.5
gives $2x = 4$).
Correct solution:
 $x = 2$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 18 • FORWARD 2 • Q5/5

Multiply multi-digit decimals

MAIN	Grade 6 main question
BACKWARD 1	Multiplying whole numbers
BACKWARD 2	Powers of 10 and the decimal point
BACKWARD 3	Multiplying tenths and hundredths
BACKWARD 4	Rounding decimals to estimate
FORWARD 1	Multiply rational numbers
FORWARD 2	Multiply numbers in scientific notation

6.NS.B.3

5.NBT.B.5

5.NBT.A.2

5.NBT.B.7

5.NBT.A.4

7.NS.A.2.c

8.EE.A.4

SET 19 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$3.6 \times 0.45$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q1/5

717

QUESTION ID

S19-M-Q1

TYPE

Short answer

CORRECT ANSWER

1.62

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q1/5

717

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$12.5 \times 3.8$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q2/5

718

QUESTION ID

S19-M-Q2

TYPE

Short answer

CORRECT ANSWER

47.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q2/5

718

Multiply.

$$4.25 \times 2.4$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q3/5

719

QUESTION ID

S19-M-Q3

TYPE

Short answer

CORRECT ANSWER

10.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q3/5

Multiply.

$$0.07 \times 0.6$$

- A.** 0.0042
- B.** 0.042
- C.** 0.42
- D.** 4.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q4/5

720

QUESTION ID

S19-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.042

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$18.4 \times 0.35$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q5/5

721

QUESTION ID

S19-M-Q5

TYPE

Short answer

CORRECT ANSWER

6.44

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 19 • MAIN • Q5/5

721

SET 19 • 6.NS.B.3

BACKWARD 1

Multiplying whole numbers • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$36 \times 45$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q1/5

723

QUESTION ID

S19-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1,620

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q1/5

723

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$125 \times 38$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q2/5

724

QUESTION ID

S19-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4,750

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q2/5

724

True or false?

$$425 \times 24 = 10,200$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q3/5

725

QUESTION ID

S19-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q3/5

$$184 \times 35 = ?$$

- A.** 1,472
- B.** 5,440
- C.** 6,340
- D.** 6,440

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q4/5

726

QUESTION ID

S19-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 6,440

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$208 \times 15$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q5/5

727

QUESTION ID

S19-B1-Q5

TYPE

Short answer

CORRECT ANSWER

3,120

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.5 • Set 19 • BACKWARD 1 • Q5/5

727

SET 19 • 6.NS.B.3

BACKWARD 2

Powers of 10 and the decimal point • Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.45 \times 10$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q1/5

729

QUESTION ID

S19-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q1/5

729

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$3.6 \times 100$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q2/5

730

QUESTION ID

S19-B2-Q2

TYPE

Short answer

CORRECT ANSWER

360

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q2/5

730

True or false?

$$24 \div 10 = 2.4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q3/5

731

QUESTION ID

S19-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q3/5

$$1,620 \div 1,000 = ?$$

- A.** 0.162
- B.** 1.62
- C.** 16.2
- D.** 162

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q4/5

732

QUESTION ID

S19-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1.62

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q4/5

Divide.

$$475 \div 10$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q5/5

733

QUESTION ID

S19-B2-Q5

TYPE

Short answer

CORRECT ANSWER

47.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • 6.NS.B.3

BACKWARD 3

Multiplying tenths and hundredths • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.3 \times 0.2$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q1/5

735

QUESTION ID

S19-B3-Q1

TYPE

Short answer

CORRECT ANSWER

0.06

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q1/5

735

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.5 \times 0.8$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q2/5

736

QUESTION ID

S19-B3-Q2

TYPE

Short answer

CORRECT ANSWER

0.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q2/5

736

True or false?

$$0.4 \times 0.4 = 1.6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q3/5

737

QUESTION ID

S19-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q3/5

$$0.2 \times 0.03 = ?$$

- A.** 0.006
- B.** 0.06
- C.** 0.6
- D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q4/5

738

QUESTION ID

S19-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0.006

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$1.2 \times 0.3$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q5/5

739

QUESTION ID

S19-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0.36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 19 • BACKWARD 3 • Q5/5

SET 19 • 6.NS.B.3

BACKWARD 4

Rounding decimals to estimate • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.4

Round 3.62 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q1/5

741

QUESTION ID

S19-B4-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q1/5

Round 12.48 to the nearest tenth.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q2/5

742

QUESTION ID

S19-B4-Q2

TYPE

Short answer

CORRECT ANSWER

12.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q2/5

True or false?

4.25 rounded to the nearest whole number is 4.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q3/5

743

QUESTION ID

S19-B4-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q3/5

Round 0.456 to the nearest hundredth.

- A. 0.4
- B. 0.45
- C. 0.46
- D. 0.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q4/5

744

QUESTION ID

S19-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 0.46

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q4/5

Round 8.75 to the nearest whole number.

Rounded: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q5/5

745

QUESTION ID

S19-B4-Q5

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B4-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.4 • Set 19 • BACKWARD 4 • Q5/5

SET 19 • 6.NS.B.3

FORWARD 1

Multiply rational numbers • Grade 7 • The Number System • 7.NS.A.2.c

Find the value.

$$(-3.6)(0.45)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q1/5

747

QUESTION ID

S19-F1-Q1

TYPE

Short answer

CORRECT ANSWER

-1.62

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q1/5

Find the value.

$$(-2.5)(-4.2)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q2/5

748

QUESTION ID

S19-F1-Q2

TYPE

Short answer

CORRECT ANSWER

10.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q2/5

$$-0.8 \times 1.5 = ?$$

- A.** -1.2
- B.** -0.12
- C.** 0.12
- D.** 1.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q3/5

749

QUESTION ID

S19-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. -1.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q3/5

Find the value.

$$(-1.2)(-0.5)(-3)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q4/5

750

QUESTION ID

S19-F1-Q4

TYPE

Short answer

CORRECT ANSWER

-1.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q4/5

True or false?

$$(-0.6)(0.7) = 0.42$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q5/5

751

QUESTION ID

S19-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.c • Set 19 • FORWARD 1 • Q5/5

SET 19 • 6.NS.B.3

FORWARD 2

Multiply numbers in scientific notation • Grade 8 • Expressions & Equations •
8.EE.A.4

Multiply. Write the answer in scientific notation.

$$(3.6 \times 10^4)(4.5 \times 10^{-2})$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q1/5

753

QUESTION ID

S19-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1.62×10^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q1/5

Multiply. Write the answer in scientific notation.

$$(2.5 \times 10^3)(4 \times 10^5)$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q2/5

754

QUESTION ID

S19-F2-Q2

TYPE

Short answer

CORRECT ANSWER

1×10^9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q2/5

$$(1.2 \times 10^6)(3 \times 10^{-4}) = ?$$

- A. 4.2×10^2
- B. 3.6×10^{10}
- C. 3.6×10^2
- D. 3.6×10^{-2}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q3/5

755

QUESTION ID

S19-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3.6×10^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q3/5

A light-year is about 9.5×10^{12} km.

About how many kilometers are in 4.2 light-years? Write the answer in scientific notation.

Kilometers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q4/5

756

QUESTION ID

S19-F2-Q4

TYPE

Short answer

CORRECT ANSWER

3.99×10^{13} km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q4/5

True or false?

$$(5 \times 10^3)(6 \times 10^2) = 3 \times 10^6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q5/5

757

QUESTION ID

S19-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 19 • FORWARD 2 • Q5/5

Divide multi-digit decimals

MAIN

Grade 6 main question

6.NS.B.3

BACKWARD 1

Dividing by a two-digit number

5.NBT.B.6

BACKWARD 2

Using powers of 10 to shift the decimal point

5.NBT.A.2

BACKWARD 3

Dividing tenths and hundredths

5.NBT.B.7

FORWARD 1

Divide rational numbers

7.NS.A.2.b

FORWARD 2

Divide numbers in scientific notation

8.EE.A.4

SET 20 • 6.NS.B.3

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.3

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$15.75 \div 0.35$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q1/5

760

QUESTION ID

S20-M-Q1

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q1/5

760

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$8.64 \div 2.4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q2/5

761

QUESTION ID

S20-M-Q2

TYPE

Short answer

CORRECT ANSWER

3.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q2/5

761

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$12.6 \div 0.45$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q3/5

762

QUESTION ID

S20-M-Q3

TYPE

Short answer

CORRECT ANSWER

28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q3/5

762

Divide.

$$0.72 \div 0.08$$

- A. 0.09
- B. 0.9
- C. 9
- D. 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q4/5

763

QUESTION ID

S20-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$283.5 \div 13.5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q5/5

764

QUESTION ID

S20-M-Q5

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

Grade 6 • The Number System • 6.NS.B.3 • Set 20 • MAIN • Q5/5

764

SET 20 • 6.NS.B.3

BACKWARD 1

Dividing by a two-digit number • Grade 5 • Number & Operations in Base Ten •
5.NBT.B.6

Divide.

$$1,575 \div 35$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q1/5

766

QUESTION ID

S20-B1-Q1

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$864 \div 24$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q2/5

767

QUESTION ID

S20-B1-Q2

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q2/5

767

True or false?

$$1,260 \div 45 = 28$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q3/5

768

QUESTION ID

S20-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q3/5

$$672 \div 32 = ?$$

- A.** 12
- B.** 21
- C.** 22
- D.** 210

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q4/5

769

QUESTION ID

S20-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$2,835 \div 15$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q5/5

770

QUESTION ID

S20-B1-Q5

TYPE

Short answer

CORRECT ANSWER

189

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.6 • Set 20 • BACKWARD 1 • Q5/5

770

SET 20 • 6.NS.B.3

BACKWARD 2

Using powers of 10 to shift the decimal point • Grade 5 • Number & Operations
in Base Ten • 5.NBT.A.2

Which has the same quotient as $15.75 \div 0.35$?

- A. $1,575 \div 35$
- B. $157.5 \div 35$
- C. $1,575 \div 3.5$
- D. $15.75 \div 35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q1/5

772

QUESTION ID

S20-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. 1,575 ÷ 35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q1/5

True or false?

$8.64 \div 2.4$ has the same quotient as $86.4 \div 24$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q2/5

773

QUESTION ID

S20-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$0.45 \times 100$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q3/5

774

QUESTION ID

S20-B2-Q3

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q3/5

774

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$12.6 \times 100$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q4/5

775

QUESTION ID

S20-B2-Q4

TYPE

Short answer

CORRECT ANSWER

1,260

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q4/5

775

$$0.08 \times \underline{\quad} = 8$$

- A. 0.1
- B. 10
- C. 100
- D. 1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q5/5

776

QUESTION ID

S20-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • 6.NS.B.3

BACKWARD 3

Dividing tenths and hundredths • Grade 5 • Number & Operations in Base Ten
• 5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$2.4 \div 0.6$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q1/5

778

QUESTION ID

S20-B3-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q1/5

778

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$3.5 \div 0.5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q2/5

779

QUESTION ID

S20-B3-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q2/5

779

True or false?

$$1.2 \div 0.3 = 0.4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q3/5

780

QUESTION ID

S20-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q3/5

$$0.9 \div 0.3 = ?$$

- A.** 0.03
- B.** 0.3
- C.** 3
- D.** 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q4/5

781

QUESTION ID

S20-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$4.8 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q5/5

782

QUESTION ID

S20-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 20 • BACKWARD 3 • Q5/5

SET 20 • 6.NS.B.3

FORWARD 1

Divide rational numbers • Grade 7 • The Number System • 7.NS.A.2.b

Find the value.

$$(-15.75) \div 0.35$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q1/5

784

QUESTION ID

S20-F1-Q1

TYPE

Short answer

CORRECT ANSWER

-45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q1/5

Find the value.

$$-8.64 \div (-2.4)$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q2/5

785

QUESTION ID

S20-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q2/5

True or false?

$$-(12 \div 4) = (-12) \div 4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q3/5

786

QUESTION ID

S20-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q3/5

$$7.2 \div (-0.9) = ?$$

- A.** -8
- B.** -0.8
- C.** 0.8
- D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q4/5

787

QUESTION ID

S20-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. -8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find the value.

$$-0.72 \div 0.08$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q5/5

788

QUESTION ID

S20-F1-Q5

TYPE

Short answer

CORRECT ANSWER

-9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.b • Set 20 • FORWARD 1 • Q5/5

788

SET 20 • 6.NS.B.3

FORWARD 2

Divide numbers in scientific notation • Grade 8 • Expressions & Equations •
8.EE.A.4

Divide. Write the answer in scientific notation.

$$(6.3 \times 10^8) \div (2.1 \times 10^3)$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q1/5

790

QUESTION ID

S20-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3×10^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q1/5

Divide. Write the answer in scientific notation.

$$(4.8 \times 10^{-2}) \div (1.6 \times 10^{-5})$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q2/5

791

QUESTION ID

S20-F2-Q2

TYPE

Short answer

CORRECT ANSWER

3×10^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q2/5

$$(9 \times 10^6) \div (3 \times 10^9) = ?$$

- A. 6×10^{-3}
- B. 3×10^{15}
- C. 3×10^3
- D. 3×10^{-3}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q3/5

792

QUESTION ID

S20-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 3×10^{-3}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q3/5

Divide. Write the answer in scientific notation.

$$(7.5 \times 10^4) \div (2.5 \times 10^{-1})$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q4/5

793

QUESTION ID

S20-F2-Q4

TYPE

Short answer

CORRECT ANSWER

3×10^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q4/5

True or false?

$$(8.4 \times 10^5) \div (4.2 \times 10^2) = 2 \times 10^3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q5/5

794

QUESTION ID

S20-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.4 • Set 20 • FORWARD 2 • Q5/5

Find the greatest common factor of two whole numbers

MAIN	Grade 6 main question	6.NS.B.4
BACKWARD 1	Factors	4.OA.B.4
BACKWARD 2	Prime and composite numbers	4.OA.B.4
BACKWARD 3	Division facts	3.OA.C.7
FORWARD 1	Factor linear expressions using a common factor	7.EE.A.1
FORWARD 2	Solve equations using the distributive property	8.EE.C.7.b

SET 21 • 6.NS.B.4

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 36.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q1/5

797

QUESTION ID

S21-M-Q1

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q1/5

What is the greatest common factor of 18 and 45?

- A.** 3
- B.** 9
- C.** 15
- D.** 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q2/5

798

QUESTION ID

S21-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q2/5

Find the greatest common factor of 28 and 42.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q3/5

799

QUESTION ID

S21-M-Q3

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q3/5

Find the greatest common factor of 30 and 75.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q4/5

800

QUESTION ID

S21-M-Q4

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q4/5

Find the greatest common factor of 16 and 40.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q5/5

801

QUESTION ID

S21-M-Q5

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 21 • MAIN • Q5/5

SET 21 • 6.NS.B.4

BACKWARD 1

Factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factors of 18.

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q1/5

803

QUESTION ID

S21-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1, 2, 3, 6, 9, 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q1/5

True or false?

7 is a factor of 42.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q2/5

804

QUESTION ID

S21-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q2/5

Which number is NOT a factor of 36?

- A.** 4
- B.** 8
- C.** 9
- D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q3/5

805

QUESTION ID

S21-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q3/5

True or false?

5 is a factor of 32.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q4/5

806

QUESTION ID

S21-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q4/5

List all the factors of 24.

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q5/5

807

QUESTION ID

S21-B1-Q5

TYPE

Short answer

CORRECT ANSWER

**1, 2, 3, 4, 6, 8, 12,
24**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • 6.NS.B.4

BACKWARD 2

Prime and composite numbers • Grade 4 • Operations & Algebraic Thinking •
4.OA.B.4

True or false?

13 is a prime number.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q1/5

809

QUESTION ID

S21-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q1/5

Which number is composite?

- A.** 2
- B.** 11
- C.** 17
- D.** 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q2/5

810

QUESTION ID

S21-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q2/5

True or false?

1 is a prime number.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q3/5

811

QUESTION ID

S21-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q3/5

Is 29 prime or composite?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q4/5

812

QUESTION ID

S21-B2-Q4

TYPE

Short answer

CORRECT ANSWER

Prime

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q4/5

Which number is prime?

- A.** 27
- B.** 31
- C.** 33
- D.** 39

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q5/5

813

QUESTION ID

S21-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 31

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • 6.NS.B.4

BACKWARD 3

Division facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Divide.

$$42 \div 7$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q1/5

815

QUESTION ID

S21-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q1/5

Divide.

$$36 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q2/5

816

QUESTION ID

S21-B3-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q2/5

True or false?

$$45 \div 9 = 5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q3/5

817

QUESTION ID

S21-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q3/5

$$56 \div 8 = ?$$

- A.** 6
- B.** 7
- C.** 8
- D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q4/5

818

QUESTION ID

S21-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$63 \div 9$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q5/5

819

QUESTION ID

S21-B3-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 21 • BACKWARD 3 • Q5/5

819

SET 21 • 6.NS.B.4

FORWARD 1

Factor linear expressions using a common factor • Grade 7 • Expressions & Equations • 7.EE.A.1

Factor using the greatest common factor.

$$18x + 24$$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q1/5

821

QUESTION ID

S21-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6(3x + 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q1/5

Factor using the greatest common factor.

$$20a - 35$$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q2/5

822

QUESTION ID

S21-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5(4a - 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q2/5

Which shows $16m + 40$ factored using the greatest common factor?

- A. $8(2m + 5)$
- B. $2(8m + 20)$
- C. $8(2m + 40)$
- D. $4(4m + 10)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q3/5

823

QUESTION ID

S21-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $8(2m + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q3/5

Factor using the greatest common factor.

$$27x + 45y$$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q4/5

824

QUESTION ID

S21-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$9(3x + 5y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q4/5

True or false?

$$14n - 21 = 7(2n - 3)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q5/5

825

QUESTION ID

S21-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 21 • FORWARD 1 • Q5/5

SET 21 • 6.NS.B.4

FORWARD 2

Solve equations using the distributive property • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$12x + 18 = 6(x + 5)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q1/5

827

QUESTION ID

S21-F2-Q1

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q1/5

Solve for x.

$$4(3x - 2) = 8(x + 1)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q2/5

828

QUESTION ID

S21-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q2/5

Solve for x.

$$15x - 10 = 5(2x + 3)$$

- A.** -1
- B.** 1
- C.** 5
- D.** 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q3/5

829

QUESTION ID

S21-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q3/5

Solve for x.

$$9(x + 2) = 6(x + 5)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q4/5

830

QUESTION ID

S21-F2-Q4

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q4/5

Solve for x.

$$10x + 25 = 5(x + 10)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q5/5

831

QUESTION ID

S21-F2-Q5

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 21 • FORWARD 2 • Q5/5

Find the least common multiple of two whole numbers

MAIN	Grade 6 main question	6.NS.B.4
BACKWARD 1	Multiples	4.OA.B.4
BACKWARD 2	Multiplication facts	3.OA.C.7
FORWARD 1	Add and subtract rational numbers with unlike denominators	7.NS.A.1.d
FORWARD 2	Equations with fractions: clearing denominators	8.EE.C.7.b

SET 22 • 6.NS.B.4

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.4

Find the least common multiple of 6 and 8.

LCM: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q1/5

834

QUESTION ID

S22-M-Q1

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q1/5

Find the least common multiple of 4 and 10.

LCM: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q2/5

835

QUESTION ID

S22-M-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q2/5

What is the least common multiple of 9 and 12?

- A.** 3
- B.** 36
- C.** 72
- D.** 108

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q3/5

836

QUESTION ID

S22-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q3/5

Find the least common multiple of 5 and 7.

LCM: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q4/5

837

QUESTION ID

S22-M-Q4

TYPE

Short answer

CORRECT ANSWER

35

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q4/5

Find the least common multiple of 6 and 15.

LCM: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q5/5

838

QUESTION ID

S22-M-Q5

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 22 • MAIN • Q5/5

SET 22 • 6.NS.B.4

BACKWARD 1

Multiples • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List the first five multiples of 6.

Multiples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q1/5

840

QUESTION ID

S22-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6, 12, 18, 24, 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q1/5

True or false?

36 is a multiple of 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q2/5

841

QUESTION ID

S22-B1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q2/5

Which number is a multiple of both 4 and 6?

- A.** 8
- B.** 12
- C.** 18
- D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q3/5

842

QUESTION ID

S22-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q3/5

True or false?

45 is a multiple of 9.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q4/5

843

QUESTION ID

S22-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q4/5

What is the 4th multiple of 7?

Multiple: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q5/5

844

QUESTION ID

S22-B1-Q5

TYPE

Short answer

CORRECT ANSWER

28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • 6.NS.B.4

BACKWARD 2

Multiplication facts • Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7

Multiply.

$$6 \times 8$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q1/5

846

QUESTION ID

S22-B2-Q1

TYPE

Short answer

CORRECT ANSWER

48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$9 \times 4$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q2/5

847

QUESTION ID

S22-B2-Q2

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q2/5

True or false?

$$7 \times 5 = 35$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q3/5

848

QUESTION ID

S22-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q3/5

$$3 \times 12 = ?$$

- A.** 15
- B.** 33
- C.** 36
- D.** 39

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q4/5

849

QUESTION ID

S22-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$5 \times 6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q5/5

850

QUESTION ID

S22-B2-Q5

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 22 • BACKWARD 2 • Q5/5

850

SET 22 • 6.NS.B.4

FORWARD 1

Add and subtract rational numbers with unlike denominators • Grade 7 • The Number System • 7.NS.A.1.d

Find the value.

$$-\frac{5}{6} + \frac{3}{8}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q1/5

852

QUESTION ID

S22-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−11/24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q1/5

Find the value.

$$\frac{3}{4} - (-\frac{1}{6})$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q2/5

853

QUESTION ID

S22-F1-Q2

TYPE

Short answer

CORRECT ANSWER

11/12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q2/5

$$-\frac{2}{3} - \frac{1}{4} = ?$$

A. $-\frac{11}{12}$

B. $-\frac{3}{7}$

C. $-\frac{1}{12}$

D. $\frac{5}{12}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q3/5

854

QUESTION ID

S22-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $-\frac{11}{12}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q3/5

Find the value.

$$-1\frac{1}{2} + \frac{5}{6}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q4/5

855

QUESTION ID

S22-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$-2/3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q4/5

True or false?

$$\frac{3}{10} + \left(-\frac{2}{15}\right) = \frac{1}{6}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q5/5

856

QUESTION ID

S22-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.d • Set 22 • FORWARD 1 • Q5/5

SET 22 • 6.NS.B.4

FORWARD 2

Equations with fractions: clearing denominators • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$\frac{x}{6} + \frac{x}{8} = 7$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q1/5

858

QUESTION ID

S22-F2-Q1

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q1/5

Solve for x.

$$\frac{x}{4} - 1 = \frac{x}{6} + 2$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q2/5

859

QUESTION ID

S22-F2-Q2

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q2/5

Solve for x.

$$\frac{2x}{3} + \frac{x}{2} = 14$$

- A. 7
- B. 12
- C. $\frac{84}{5}$
- D. 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q3/5

860

QUESTION ID

S22-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q3/5

Solve for x.

$$\frac{x}{5} + 3 = \frac{x}{2}$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q4/5

861

QUESTION ID

S22-F2-Q4

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q4/5

Solve for x.

$$\frac{(x+1)}{4} = \frac{(x-2)}{3}$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q5/5

862

QUESTION ID

S22-F2-Q5

TYPE

Short answer

CORRECT ANSWER

11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 22 • FORWARD 2 • Q5/5

Use the distributive property to write a sum as the GCF times a sum

MAIN	Grade 6 main question	6.NS.B.4
BACKWARD 1	Common factors	4.OA.B.4
BACKWARD 2	Distributive property with whole numbers	3.OA.B.5
BACKWARD 3	Parentheses and order of operations	5.OA.A.1
FORWARD 1	Factor and expand linear expressions	7.EE.A.1
FORWARD 2	One, none, or infinitely many solutions	8.EE.C.7.a

SET 23 • 6.NS.B.4

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.B.4

Use the greatest common factor to write $45 + 20$ as a product of the GCF and a sum of two whole numbers.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q1/5

865

QUESTION ID

S23-M-Q1

TYPE

Short answer

CORRECT ANSWER

5(9 + 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q1/5

Which expression is equal to $42 + 56$ and uses the greatest common factor?

- A. $2(21 + 28)$
- B. $14(3 + 4)$
- C. $14(3 + 8)$
- D. $7(6 + 8)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q2/5

866

QUESTION ID

S23-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. $14(3 + 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q2/5

Use the greatest common factor to write $27 + 45$ as a product.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q3/5

867

QUESTION ID

S23-M-Q3

TYPE

Short answer

CORRECT ANSWER

9(3 + 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q3/5

True or false?

$$24 + 60 = 12(2 + 5)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q4/5

868

QUESTION ID

S23-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q4/5

Complete the equation.

$$30 + 75 = 15(\underline{\quad} + \underline{\quad})$$

First blank: _____

Second blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q5/5

869

QUESTION ID

S23-M-Q5

TYPE

Short answer

CORRECT ANSWER

**First blank: 2;
Second blank: 5**

GRADING NOTE 5 and 2 in either order is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-M-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 23 • MAIN • Q5/5

SET 23 • 6.NS.B.4

BACKWARD 1

Common factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

True or false?

5 is a factor of both 45 and 20.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q1/5

871

QUESTION ID

S23-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q1/5

List the factors of 12 and the factors of 18.

Which factors do they share?

Shared factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q2/5

872

QUESTION ID

S23-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1, 2, 3, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q2/5

Which number is a factor of both 27 and 45?

- A.** 5
- B.** 6
- C.** 9
- D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q3/5

873

QUESTION ID

S23-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q3/5

List all the factors of 42.

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q4/5

874

QUESTION ID

S23-B1-Q4

TYPE

Short answer

CORRECT ANSWER

**1, 2, 3, 6, 7, 14, 21,
42**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q4/5

True or false?

6 is a factor of 56.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q5/5

875

QUESTION ID

S23-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 23 • BACKWARD 1 • Q5/5

SET 23 • 6.NS.B.4

BACKWARD 2

Distributive property with whole numbers • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$9 \times 6 = 9 \times 4 + 9 \times 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q1/5

877

QUESTION ID

S23-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q1/5

True or false?

$$4 \times (9 + 2) = 4 \times 9 + 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q2/5

878

QUESTION ID

S23-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q2/5

Which is equal to 6×12 ?

- A. $6 \times 10 + 2$
- B. $6 \times 10 \times 2$
- C. $6 \times 10 + 6 \times 2$
- D. $6 + 10 \times 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q3/5

879

QUESTION ID

S23-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $6 \times 10 + 6 \times 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q3/5

$$5 \times 13 = 5 \times 10 + 5 \times \underline{\quad}$$

Blank: _____

ID S23-B2-Q4

880

S23-B2-Q4

Short answer

3

ID S23-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q4/5

$$3 \times (4 + 5) = 3 \times 4 + 3 \times \underline{\quad}$$

Blank: _____

ID S23-B2-Q5

881

S23-B2-Q5

Short answer

5

ID S23-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 23 • BACKWARD 2 • Q5/5

SET 23 • 6.NS.B.4

BACKWARD 3

Parentheses and order of operations • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$5 \times (9 + 4)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q1/5

883

QUESTION ID

S23-B3-Q1

TYPE

Short answer

CORRECT ANSWER

65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q1/5

Evaluate.

$$14 \times (3 + 4)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q2/5

884

QUESTION ID

S23-B3-Q2

TYPE

Short answer

CORRECT ANSWER

98

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q2/5

True or false?

$$9 \times (3 + 5) = 32$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q3/5

885

QUESTION ID

S23-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q3/5

$$12 \times (2 + 5) = ?$$

- A.** 17
- B.** 29
- C.** 60
- D.** 84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q4/5

886

QUESTION ID

S23-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q4/5

Evaluate.

$$(20 - 5) \times 3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q5/5

887

QUESTION ID

S23-B3-Q5

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-B3-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 23 • BACKWARD 3 • Q5/5

SET 23 • 6.NS.B.4

FORWARD 1

Factor and expand linear expressions • Grade 7 • Expressions & Equations •
7.EE.A.1

Factor.

$$12x - 18$$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q1/5

889

QUESTION ID

S23-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$6(2x - 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Expand.

$$-3(2x - 5)$$

Expanded form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q2/5

890

QUESTION ID

S23-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$$-6x + 15$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q2/5

890

Which expression is equivalent to $8y + 20$?

- A. $4(2y + 20)$
- B. $8(y + 20)$
- C. $2(4y + 20)$
- D. $4(2y + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q3/5

891

QUESTION ID

S23-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $4(2y + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q3/5

Factor.

$$15a + 10b - 5$$

Factored form: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q4/5

892

QUESTION ID

S23-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$5(3a + 2b - 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q4/5

True or false?

$$\frac{1}{2}(6x - 4) = 3x - 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q5/5

893

QUESTION ID

S23-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 23 • FORWARD 1 • Q5/5

SET 23 • 6.NS.B.4

FORWARD 2

One, none, or infinitely many solutions • Grade 8 • Expressions & Equations •
8.EE.C.7.a

How many solutions does the equation have?

$$4(x + 3) = 4x + 12$$

- A. One solution
- B. No solution
- C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q1/5

895

QUESTION ID

S23-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q1/5

How many solutions does the equation have?

$$2(3x - 1) = 6x + 5$$

- A. One solution
- B. No solution
- C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q2/5

896

QUESTION ID

S23-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q2/5

How many solutions does the equation have?

$$3(x + 2) = 2x + 9$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q3/5

897

QUESTION ID

S23-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. One solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q3/5

What number goes in the box so that the equation has infinitely many solutions?

$$5(x - 2) = 5x - \square$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q4/5

898

QUESTION ID

S23-F2-Q4

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q4/5

Which equation has no solution?

- A. $2(x + 4) = 2x + 5$
- B. $2(x + 4) = x + 8$
- C. $2(x + 4) = 2x + 8$
- D. $2(x + 4) = 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q5/5

899

QUESTION ID

S23-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. $2(x + 4) = 2x + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S23-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 23 • FORWARD 2 • Q5/5

Use positive and negative numbers to represent quantities in context

MAIN	Grade 6 main question	6.NS.C.5
BACKWARD 1	Whole numbers as distances from 0	2.MD.B.6
BACKWARD 2	Increase and decrease situations	2.OA.A.1
FORWARD 1	Adding integers as moving on a number line	7.NS.A.1.b
FORWARD 2	The sign of the rate of change in context	8.F.B.4

SET 24 • 6.NS.C.5

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.5

A diver is 30 feet below sea level.

Write an integer to represent the diver's position.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q1/5

902

QUESTION ID

S24-M-Q1

TYPE

Short answer

CORRECT ANSWER

−30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q1/5

A company lost \$2,500 this month.

Write an integer to represent this change.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q2/5

903

QUESTION ID

S24-M-Q2

TYPE

Short answer

CORRECT ANSWER

−2,500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q2/5

A bank account shows a withdrawal of \$45.

Which number represents this change?

- A. -45
- B. -4.5
- C. 0
- D. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q3/5

904

QUESTION ID

S24-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. -45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q3/5

The temperature rose 12 degrees.

Write an integer to represent this change.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q4/5

905

QUESTION ID

S24-M-Q4

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE +12 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q4/5

A hawk flies 120 feet above sea level. A submarine is 120 feet below sea level.

Write an integer for each position.

Hawk: _____

Submarine: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q5/5

906

QUESTION ID

S24-M-Q5

TYPE

Short answer

CORRECT ANSWER

**Hawk: 120;
Submarine: -120**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-M-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 24 • MAIN • Q5/5

SET 24 • 6.NS.C.5

BACKWARD 1

Whole numbers as distances from 0 • Grade 2 • Measurement & Data •
2.MD.B.6

What number does point A represent?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q1/5

908

QUESTION ID

S24-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q1/5

How many units from 0 is the number 6?

Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q2/5

909

QUESTION ID

S24-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q2/5

True or false?

On a number line, 9 is farther from 0 than 4 is.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q3/5

910

QUESTION ID

S24-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q3/5

Which point is 3 units from 0?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q4/5

911

QUESTION ID

S24-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Point B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q4/5

What number does point B represent?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q5/5

912

QUESTION ID

S24-B1-Q5

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 24 • BACKWARD 1 • Q5/5

SET 24 • 6.NS.C.5

BACKWARD 2

Increase and decrease situations • Grade 2 • Operations & Algebraic Thinking •
2.OA.A.1

Kim had \$20. She spent \$8.

How much money does she have now?

Money: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q1/5

914

QUESTION ID

S24-B2-Q1

TYPE

Short answer

CORRECT ANSWER

\$12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q1/5

A plant was 12 cm tall. It grew 5 cm.

How tall is the plant now?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q2/5

915

QUESTION ID

S24-B2-Q2

TYPE

Short answer

CORRECT ANSWER

17 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q2/5

Which situation describes a decrease?

- A. A plant grows 4 inches.
- B. Tom earns \$10.
- C. A team gains 5 yards.
- D. A balloon loses 6 feet of height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q3/5

916

QUESTION ID

S24-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A balloon loses 6 feet of height.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q3/5

True or false?

"The temperature fell 7 degrees" describes a decrease.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q4/5

917

QUESTION ID

S24-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q4/5

Which word describes an increase in money?

- A.** Deposit
- B.** Loss
- C.** Drop
- D.** Withdrawal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q5/5

918

QUESTION ID

S24-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Deposit

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 24 • BACKWARD 2 • Q5/5

SET 24 • 6.NS.C.5

FORWARD 1

Adding integers as moving on a number line • Grade 7 • The Number System •
7.NS.A.1.b

The temperature is -4°F . It rises 9°F .

What is the new temperature?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q1/5

920

QUESTION ID

S24-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q1/5

A submarine is at -120 feet. It rises 45 feet.

What is its new position?

Position: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q2/5

921

QUESTION ID

S24-F1-Q2

TYPE

Short answer

CORRECT ANSWER

-75 feet

GRADING NOTE 75 feet below sea level

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q2/5

On a number line, where is $6 + (-4)$ compared to 6?

- A. 4 units to the right of 6
- B. 4 units to the left of 6
- C. At 0
- D. 6 units to the left of 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q3/5

922

QUESTION ID

S24-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 4 units to the left of 6

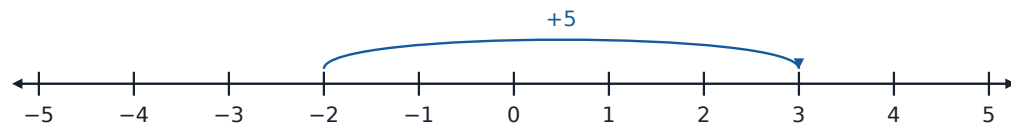
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q3/5

The number line shows $-2 + 5$.

What is the sum?



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q4/5

923

QUESTION ID

S24-F1-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q4/5

True or false?

A \$30 debt plus a \$30 deposit makes a balance of \$0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q5/5

924

QUESTION ID

S24-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 24 • FORWARD 1 • Q5/5

SET 24 • 6.NS.C.5

FORWARD 2

The sign of the rate of change in context • Grade 8 • Functions • 8.F.B.4

A tank holds 60 gallons. It drains 3 gallons per minute.

Write a function for the amount A after t minutes.

$A =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q1/5

926

QUESTION ID

S24-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$A = 60 - 3t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q1/5

An airplane at 12,000 feet descends 800 feet per minute.

What is the rate of change of its altitude?

- A.** $-12,000$ feet per minute **B.** -800 feet per minute
C. 800 feet per minute **D.** $12,000$ feet per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q2/5

927

QUESTION ID

S24-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. -800 feet per minute

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q2/5

The table shows a linear function.

What is the rate of change in degrees per hour?

Hours	0	2	4
Temperature (°F)	5	1	-3

Rate of change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q3/5

928

QUESTION ID

S24-F2-Q3

TYPE

Short answer

CORRECT ANSWER

-2 degrees per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q3/5

A diver starts at -10 feet and descends 4 feet per second.

Write a function for the diver's position d after t seconds.

$d =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q4/5

929

QUESTION ID

S24-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$d = -10 - 4t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q4/5

True or false?

A line through $(0, 8)$ and $(4, 0)$ has a rate of change of -2 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q5/5

930

QUESTION ID

S24-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S24-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 24 • FORWARD 2 • Q5/5

Explain the meaning of 0 in a real-world situation

MAIN	Grade 6 main question	6.NS.C.5
BACKWARD 1	Zero as the starting point on a number line	2.MD.B.6
BACKWARD 2	A change of zero means no change	1.OA.D.8
FORWARD 1	Opposite quantities combine to make 0	7.NS.A.1.a
FORWARD 2	Interpret the initial value (the value when $x = 0$)	8.F.B.4

SET 25 • 6.NS.C.5

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.5

On a Celsius thermometer, what does 0°C represent?

- A. No temperature at all
- B. The temperature at which water freezes
- C. The hottest possible temperature
- D. The temperature at which water boils

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q1/5

933

QUESTION ID

S25-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

**B. The temperature
at which water
freezes**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q1/5

When numbers describe elevation, what does 0 represent?

- A.** Sea level
- B.** No elevation at all
- C.** The top of a mountain
- D.** The ocean floor

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q2/5

934

QUESTION ID

S25-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Sea level

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q2/5

A bank account has a balance of \$0.

What does this mean?

- A.** The account has no money and owes nothing.
- B.** The account owes money.
- C.** The account has money in it.
- D.** The account earns interest.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q3/5

935

QUESTION ID

S25-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

**A. The account has
no money and owes
nothing.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q3/5

A football team's change in yards on a play is 0.

What does 0 mean?

- A.** The team scored.
- B.** The team gained 10 yards.
- C.** The team neither gained nor lost yards.
- D.** The team lost all its yards.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q4/5

936

QUESTION ID

S25-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

**C. The team neither
gained nor lost
yards.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q4/5

A scale shows that your weight changed by 0 pounds since last month.

What does 0 mean in this situation?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q5/5

937

QUESTION ID

S25-M-Q5

TYPE

Short answer

CORRECT ANSWER

The weight did not change: there was no gain and no loss.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-M-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 25 • MAIN • Q5/5

SET 25 • 6.NS.C.5

BACKWARD 1

Zero as the starting point on a number line • Grade 2 • Measurement & Data •
2.MD.B.6

True or false?

On a number line, 0 is the starting point for measuring distance.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q1/5

939

QUESTION ID

S25-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q1/5

What number is at point A?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q2/5

940

QUESTION ID

S25-B1-Q2

TYPE

Short answer

CORRECT ANSWER

0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q2/5

How far is 0 from itself on a number line?

- A.** 0 units
- B.** 1 unit
- C.** 2 units
- D.** 10 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q3/5

941

QUESTION ID

S25-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 0 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q3/5

How far is 5 from 0 on a number line?

Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q4/5

942

QUESTION ID

S25-B1-Q4

TYPE

Short answer

CORRECT ANSWER

5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q4/5

True or false?

A ruler starts measuring at 1.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q5/5

943

QUESTION ID

S25-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 25 • BACKWARD 1 • Q5/5

SET 25 • 6.NS.C.5

BACKWARD 2

A change of zero means no change • Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8

What number goes in the box?

$$8 + \square = 8$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q1/5

945

QUESTION ID

S25-B2-Q1

TYPE

Short answer

CORRECT ANSWER

0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q1/5

What number goes in the box?

$$15 - \square = 15$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q2/5

946

QUESTION ID

S25-B2-Q2

TYPE

Short answer

CORRECT ANSWER

0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q2/5

True or false?

A plant was 12 cm tall last week and is 12 cm tall now. It grew 0 cm.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q3/5

947

QUESTION ID

S25-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q3/5

A score went from 45 to 45.

What was the change?

- A.** 0
- B.** 1
- C.** 45
- D.** 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q4/5

948

QUESTION ID

S25-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q4/5

What number goes in the box?

$$\square + 23 = 23$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q5/5

949

QUESTION ID

S25-B2-Q5

TYPE

Short answer

CORRECT ANSWER

0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-B2-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8 • Set 25 • BACKWARD 2 • Q5/5

SET 25 • 6.NS.C.5

FORWARD 1

Opposite quantities combine to make 0 • Grade 7 • The Number System •
7.NS.A.1.a

Maya earns \$25 and then spends \$25.

What is the total change in her money?

Total change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q1/5

951

QUESTION ID

S25-F1-Q1

TYPE

Short answer

CORRECT ANSWER

\$0

GRADING NOTE No change

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q1/5

True or false?

$$-8 + 8 = 0$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q2/5

952

QUESTION ID

S25-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q2/5

Which sum is equal to 0?

- A. $-3.5 + 3.5$
- B. $-3.5 + 0$
- C. $-3.5 + (-3.5)$
- D. $3.5 + 3.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q3/5

953

QUESTION ID

S25-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $-3.5 + 3.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q3/5

A balloon rises 40 feet and then falls 40 feet.

What is its total change in height?

Total change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q4/5

954

QUESTION ID

S25-F1-Q4

TYPE

Short answer

CORRECT ANSWER

0 feet

GRADING NOTE No change

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q4/5

What number added to 14.2 gives 0?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q5/5

955

QUESTION ID

S25-F1-Q5

TYPE

Short answer

CORRECT ANSWER

−14.2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.a • Set 25 • FORWARD 1 • Q5/5

SET 25 • 6.NS.C.5

FORWARD 2

Interpret the initial value (the value when $x = 0$) • Grade 8 • Functions •
8.F.B.4

The water in a tank is $W = 60 - 3t$, where t is minutes.

What does 60 represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q1/5

957

QUESTION ID

S25-F2-Q1

TYPE

Short answer

CORRECT ANSWER

**The starting amount:
the tank holds 60
gallons at $t = 0$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q1/5

A hiker's elevation is $E = -40 + 15t$, where t is hours.

What does -40 represent?

- A. The hiker starts 40 m below sea level.
- B. The hiker ends 40 m below sea level.
- C. The hiker climbs 40 m each hour.
- D. The hike lasts 40 hours.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q2/5

958

QUESTION ID

S25-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

**A. The hiker starts
40 m below sea
level.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q2/5

A phone's battery charge is $B = 100 - 8h$ after h hours.

What is B when $h = 0$, and what does it mean?

Value: _____

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q3/5

959

QUESTION ID

S25-F2-Q3

TYPE

Short answer

CORRECT ANSWER

**Value: 100; Meaning:
the battery starts
fully charged (100%)
at 0 hours**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q3/5

A savings account has $y = 25x + 150$ dollars after x weeks.

What does 150 represent?

- A.** The starting amount
- B.** The amount saved each week
- C.** The number of weeks
- D.** The amount after 25 weeks

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q4/5

960

QUESTION ID

S25-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. The starting amount

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q4/5

True or false?

For the function $y = 4x$, the output is 0 when the input is 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q5/5

961

QUESTION ID

S25-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S25-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 25 • FORWARD 2 • Q5/5

Find opposites; the opposite of the opposite of a number

MAIN

Grade 6 main question

6.NS.C.6.a

BACKWARD 1

Distance from 0 on a number line

2.MD.B.6

BACKWARD 2

Positive and negative numbers in context

6.NS.C.5

FORWARD 1

Subtract by adding the opposite

7.NS.A.1.c

FORWARD 2

Solve equations by adding opposites to both sides

8.EE.C.7.b

SET 26 • 6.NS.C.6.a

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of -7 ?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q1/5

964

QUESTION ID

S26-M-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Simplify.

 $-(-12)$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q2/5

965

QUESTION ID

S26-M-Q2

TYPE

Short answer

CORRECT ANSWER

12

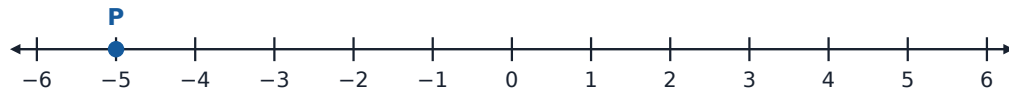
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q2/5

965

Which number is the opposite of the number at point P?



A. -5

B. 0

C. $\frac{1}{5}$

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q3/5

966

QUESTION ID

S26-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q3/5

True or false?

The opposite of 0 is 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q4/5

967

QUESTION ID

S26-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q4/5

What is the opposite of the opposite of 9?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q5/5

968

QUESTION ID

S26-M-Q5

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-M-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 26 • MAIN • Q5/5

SET 26 • 6.NS.C.6.a

BACKWARD 1

Distance from 0 on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

How far is point A from 0?



Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q1/5

970

QUESTION ID

S26-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q1/5

How many units from 0 is the number 8?

Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q2/5

971

QUESTION ID

S26-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q2/5

Which number is 2 units from 0 on this number line?



A. 0

B. 2

C. 4

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q3/5

972

QUESTION ID

S26-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q3/5

What number does point K represent?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q4/5

973

QUESTION ID

S26-B1-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q4/5

True or false?

The distance from 0 to 7 is 7 units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q5/5

974

QUESTION ID

S26-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 26 • BACKWARD 1 • Q5/5

SET 26 • 6.NS.C.6.a

BACKWARD 2

Positive and negative numbers in context • Grade 6 • The Number System •
6.NS.C.5

True or false?

–5 can represent 5 degrees below zero.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q1/5

976

QUESTION ID

S26-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q1/5

Write an integer for a gain of 12 yards.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q2/5

977

QUESTION ID

S26-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12

GRADING NOTE +12 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q2/5

Which integer represents 20 feet below sea level?

- A.** -20
- B.** -2
- C.** 0
- D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q3/5

978

QUESTION ID

S26-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. -20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q3/5

Write an integer for a loss of \$7.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q4/5

979

QUESTION ID

S26-B2-Q4

TYPE

Short answer

CORRECT ANSWER

-7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q4/5

True or false?

A deposit of \$15 is written as -15 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q5/5

980

QUESTION ID

S26-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-B2-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 26 • BACKWARD 2 • Q5/5

SET 26 • 6.NS.C.6.a

FORWARD 1

Subtract by adding the opposite • Grade 7 • The Number System • 7.NS.A.1.c

Rewrite $5 - 8$ as an addition expression.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q1/5

982

QUESTION ID

S26-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$5 + (-8)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q1/5

Find the value.

$$-3 - (-7)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q2/5

983

QUESTION ID

S26-F1-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q2/5

True or false?

$$6 - (-2) = 6 + 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q3/5

984

QUESTION ID

S26-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q3/5

Which expression is equal to $-4 - 9$?

- A. $4 + 9$
- B. $4 - (-9)$
- C. $-4 + 9$
- D. $-4 + (-9)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q4/5

985

QUESTION ID

S26-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $-4 + (-9)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q4/5

Find the value.

$$2.5 - 6$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q5/5

986

QUESTION ID

S26-F1-Q5

TYPE

Short answer

CORRECT ANSWER

-3.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 26 • FORWARD 1 • Q5/5

SET 26 • 6.NS.C.6.a

FORWARD 2

Solve equations by adding opposites to both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$5x - 7 = 2x + 8$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q1/5

988

QUESTION ID

S26-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q1/5

Solve for x.

$$-3x + 4 = 3x - 8$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q2/5

989

QUESTION ID

S26-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q2/5

Solve for x.

$$6 - 2x = 4x - 12$$

- A.** -3
- B.** 1
- C.** 3
- D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q3/5

990

QUESTION ID

S26-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q3/5

Solve for x.

$$7x + 3 = -2x + 21$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q4/5

991

QUESTION ID

S26-F2-Q4

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q4/5

Solve for x.

$$4(x - 1) = 2x + 10$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q5/5

992

QUESTION ID

S26-F2-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S26-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 26 • FORWARD 2 • Q5/5

Use signs of coordinates to identify quadrants

MAIN	Grade 6 main question	6.NS.C.6.b
BACKWARD 1	Axes, origin, and ordered pairs	5.G.A.1
BACKWARD 2	Negative numbers are on the opposite side of 0	6.NS.C.6.a
BACKWARD 3	Reading points in the first quadrant	5.G.A.2
FORWARD 1	Signs of products of rational numbers	7.NS.A.2.a
FORWARD 2	Transformations and the quadrant of the image	8.G.A.3

SET 27 • 6.NS.C.6.b

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.b

In which quadrant is the point $(-4, 7)$?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q1/5

995

QUESTION ID

S27-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Quadrant II

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q1/5

In which quadrant is the point $(3, -8)$?

- A. Quadrant I
- B. Quadrant II
- C. Quadrant III
- D. Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q2/5

996

QUESTION ID

S27-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q2/5

In which quadrant is the point $(-2.5, -6)$?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q3/5

997

QUESTION ID

S27-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Quadrant III

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q3/5

A point has a negative x-coordinate and a positive y-coordinate.

In which quadrant is it?

Quadrant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q4/5

998

QUESTION ID

S27-M-Q4

TYPE

Short answer

CORRECT ANSWER

Quadrant II

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q4/5

Which point is in Quadrant III?

- A.** $(-5, -1)$
- B.** $(5, 1)$
- C.** $(5, -1)$
- D.** $(-5, 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q5/5

999

QUESTION ID

S27-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. $(-5, -1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-M-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 27 • MAIN • Q5/5

SET 27 • 6.NS.C.6.b

BACKWARD 1

Axes, origin, and ordered pairs • Grade 5 • Geometry • 5.G.A.1

True or false?

The origin is the point (0, 0).

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q1/5

1001

QUESTION ID

S27-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q1/5

In the ordered pair (5, 2), which number tells how far to move along the x-axis?

- A.** 2
- B.** 3
- C.** 5
- D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q2/5

1002

QUESTION ID

S27-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q2/5

What is the name of the point where the x-axis and the y-axis meet?

Name: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q3/5

1003

QUESTION ID

S27-B1-Q3

TYPE

Short answer

CORRECT ANSWER

The origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q3/5

True or false?

In the ordered pair (3, 8), the y-coordinate is 3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q4/5

1004

QUESTION ID

S27-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q4/5

Which axis is horizontal?

- A.** The x-axis
- B.** The y-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q5/5

1005

QUESTION ID

S27-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. The x-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B1-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 27 • BACKWARD 1 • Q5/5

SET 27 • 6.NS.C.6.b

BACKWARD 2

Negative numbers are on the opposite side of 0 • Grade 6 • The Number System • 6.NS.C.6.a

True or false?

On a horizontal number line, -4 is to the left of 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q1/5

1007

QUESTION ID

S27-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q1/5

On a vertical number line, where is -3 ?

- A.** Above 0
- B.** Above 3
- C.** Below 0
- D.** At 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q2/5

1008

QUESTION ID

S27-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Below 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q2/5

True or false?

On a vertical number line, 5 is below 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q3/5

1009

QUESTION ID

S27-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q3/5

On a horizontal number line, is 2.5 to the left or the right of 0?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q4/5

1010

QUESTION ID

S27-B2-Q4

TYPE

Short answer

CORRECT ANSWER

To the right of 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q4/5

Which number is to the left of 0 on a horizontal number line?

- A. -6
- B. 0
- C. $\frac{1}{2}$
- D. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q5/5

1011

QUESTION ID

S27-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. -6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 27 • BACKWARD 2 • Q5/5

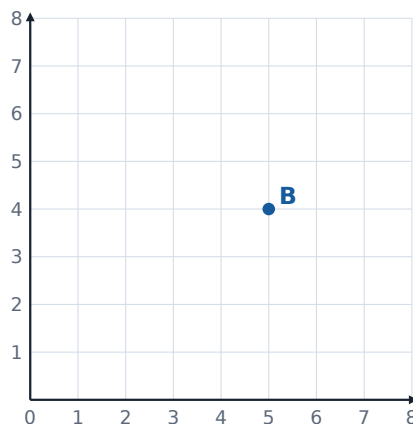
SET 27 • 6.NS.C.6.b

BACKWARD 3

Reading points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

What are the coordinates of point B?

B = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q1/5

1013

QUESTION ID

S27-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(5, 4)

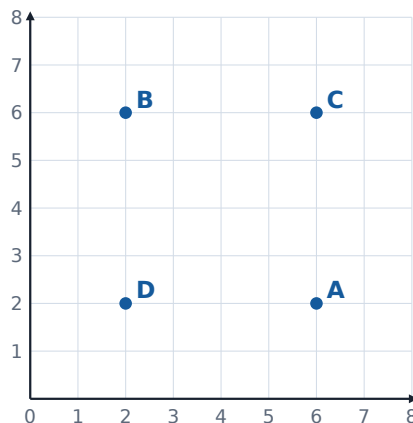
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q1/5

Which point is located at (2, 6)?

- A. Point A
- B. Point B
- C. Point C
- D. Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q2/5

1014

QUESTION ID

S27-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Point B

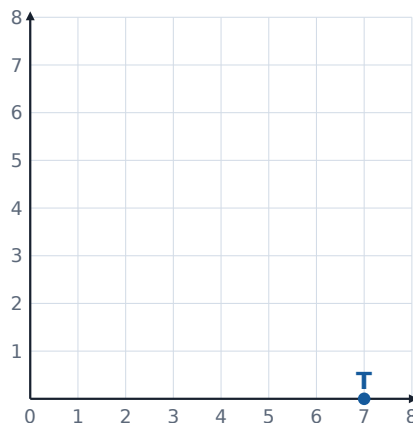
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q2/5

What are the coordinates of point T?

T = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q3/5

1015

QUESTION ID

S27-B3-Q3

TYPE

Short answer

CORRECT ANSWER

(7, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

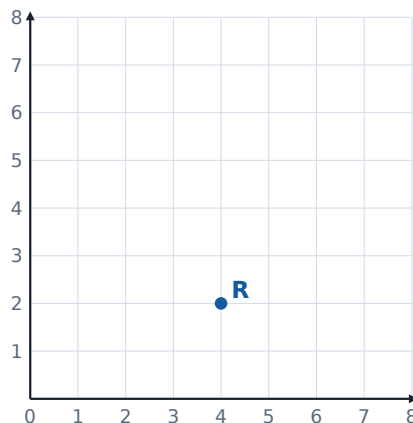
ID S27-B3-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q3/5

True or false?

Point R is located at (2, 4).

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q4/5

1016

QUESTION ID

S27-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

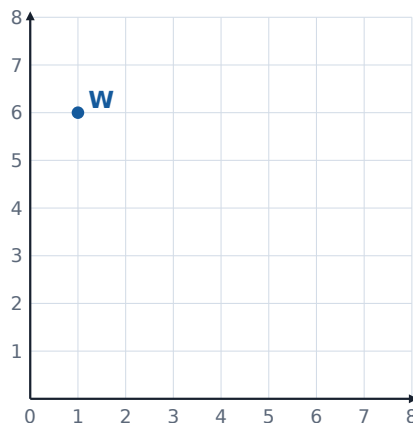
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q4/5

What are the coordinates of point W?

W = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q5/5

1017

QUESTION ID

S27-B3-Q5

TYPE

Short answer

CORRECT ANSWER

(1, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-B3-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 27 • BACKWARD 3 • Q5/5

SET 27 • 6.NS.C.6.b

FORWARD 1

Signs of products of rational numbers • Grade 7 • The Number System •
7.NS.A.2.a

The point (x, y) is in Quadrant II.
Is the product xy positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q1/5

1019

QUESTION ID

S27-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Negative

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q1/5

The point (x, y) is in Quadrant III.
Is the product xy positive or negative?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q2/5

1020

QUESTION ID

S27-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Positive

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q2/5

For the point (x, y) , the product xy is positive and x is negative.

In which quadrant is the point?

- A. Quadrant I
- B. Quadrant II
- C. Quadrant III
- D. Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q3/5

1021

QUESTION ID

S27-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Quadrant III

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q3/5

True or false?

If (x, y) is in Quadrant IV, then the product xy is negative.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q4/5

1022

QUESTION ID

S27-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q4/5

Find the product of the coordinates of the point $(-3, 5)$.

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q5/5

1023

QUESTION ID

S27-F1-Q5

TYPE

Short answer

CORRECT ANSWER

-15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 27 • FORWARD 1 • Q5/5

SET 27 • 6.NS.C.6.b

FORWARD 2

Transformations and the quadrant of the image • Grade 8 • Geometry •
8.G.A.3

The point (2, 5) is rotated 180° about the origin.

What are the coordinates of the image, and in which quadrant is it?

Image: _____

Quadrant: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q1/5

1025

QUESTION ID

S27-F2-Q1

TYPE

Short answer

CORRECT ANSWER

**Image: $(-2, -5)$;
Quadrant: III**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q1/5

A point in Quadrant II is reflected across the y-axis.

In which quadrant is the image?

- A. Quadrant I
- B. Quadrant II
- C. Quadrant III
- D. Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q2/5

1026

QUESTION ID

S27-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Quadrant I

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q2/5

The point $(-3, -4)$ is rotated 90° counterclockwise about the origin.

What are the coordinates of the image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q3/5

1027

QUESTION ID

S27-F2-Q3

TYPE

Short answer

CORRECT ANSWER

(4, -3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q3/5

True or false?

Translating (1, 2) 5 units to the left moves it into Quadrant II.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q4/5

1028

QUESTION ID

S27-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q4/5

The point $(4, -1)$ is reflected across the x-axis.

In which quadrant is the image?

- A. Quadrant I
- B. Quadrant II
- C. Quadrant III
- D. Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q5/5

1029

QUESTION ID

S27-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Quadrant I

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S27-F2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 27 • FORWARD 2 • Q5/5

Reflect points across one or both axes

MAIN	Grade 6 main question	6.NS.C.6.b
BACKWARD 1	Opposite numbers	6.NS.C.6.a
BACKWARD 2	Lines of symmetry	4.G.A.3
BACKWARD 3	Reading coordinates in the first quadrant	5.G.A.2
FORWARD 1	Distance between a point and its reflection	7.NS.A.1.c
FORWARD 2	Reflections and rotations with coordinates	8.G.A.3

SET 28 • 6.NS.C.6.b

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.b

The point $(3, -5)$ is reflected across the x-axis.

What are the coordinates of its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q1/5

1032

QUESTION ID

S28-M-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q1/5

The point $(-2, 6)$ is reflected across the y -axis.

What are the coordinates of its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q2/5

1033

QUESTION ID

S28-M-Q2

TYPE

Short answer

CORRECT ANSWER

(2, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

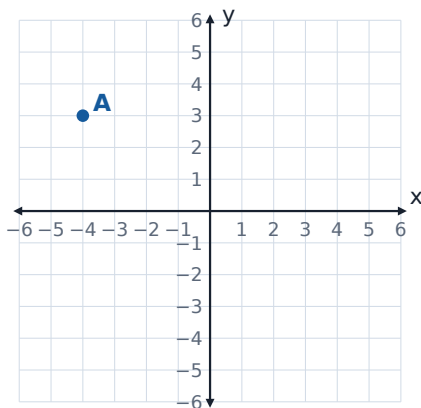
ID S28-M-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q2/5

Point A is reflected across the y-axis.

What are the coordinates of its image?

Image: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q3/5

1034

QUESTION ID

S28-M-Q3

TYPE

Short answer

CORRECT ANSWER

(4, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q3/5

Which point is the reflection of $(-7, -2)$ across the x-axis?

- A. $(7, 2)$
- B. $(-2, -7)$
- C. $(-7, 2)$
- D. $(7, -2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q4/5

1035

QUESTION ID

S28-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $(-7, 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q4/5

Point B(5, 1) is reflected across the y-axis and then across the x-axis.

What are the coordinates of the final image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q5/5

1036

QUESTION ID

S28-M-Q5

TYPE

Short answer

CORRECT ANSWER

(-5, -1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-M-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 28 • MAIN • Q5/5

SET 28 • 6.NS.C.6.b

BACKWARD 1

Opposite numbers • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of 6?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q1/5

1038

QUESTION ID

S28-B1-Q1

TYPE

Short answer

CORRECT ANSWER

−6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q1/5

True or false?

The opposite of -9 is 9.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q2/5

1039

QUESTION ID

S28-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q2/5

Which pair of numbers are opposites?

- A. 0 and 4
- B. -4 and -4
- C. 4 and $\frac{1}{4}$
- D. -4 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q3/5

1040

QUESTION ID

S28-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. -4 and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q3/5

What is the opposite of -2.5 ?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q4/5

1041

QUESTION ID

S28-B1-Q4

TYPE

Short answer

CORRECT ANSWER

2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q4/5

True or false?

$$-(-3) = -3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q5/5

1042

QUESTION ID

S28-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 28 • BACKWARD 1 • Q5/5

SET 28 • 6.NS.C.6.b

BACKWARD 2

Lines of symmetry • Grade 4 • Geometry • 4.G.A.3

True or false?

A line of symmetry divides a figure into two matching halves.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q1/5

1044

QUESTION ID

S28-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q1/5

How many lines of symmetry does a square have?

- A.** 0
- B.** 1
- C.** 2
- D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q2/5

1045

QUESTION ID

S28-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 4

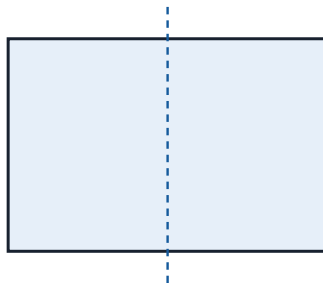
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q2/5

True or false?

The dashed line is a line of symmetry of the rectangle.



- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q3/5

1046

QUESTION ID

S28-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q3/5

Which letter has a vertical line of symmetry?

- A.** F
- B.** A
- C.** J
- D.** R

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q4/5

1047

QUESTION ID

S28-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q4/5

How many lines of symmetry does a rectangle that is not a square have?

Lines: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q5/5

1048

QUESTION ID

S28-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B2-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 28 • BACKWARD 2 • Q5/5

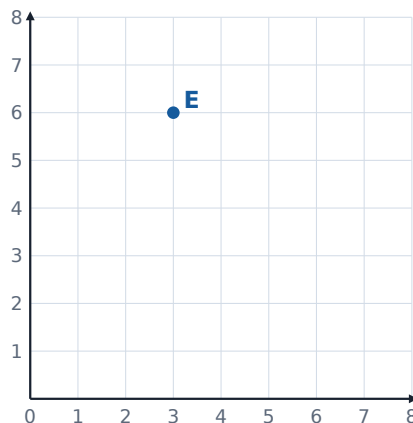
SET 28 • 6.NS.C.6.b

BACKWARD 3

Reading coordinates in the first quadrant • Grade 5 • Geometry • 5.G.A.2

What are the coordinates of point E?

E = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q1/5

1050

QUESTION ID

S28-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 6)

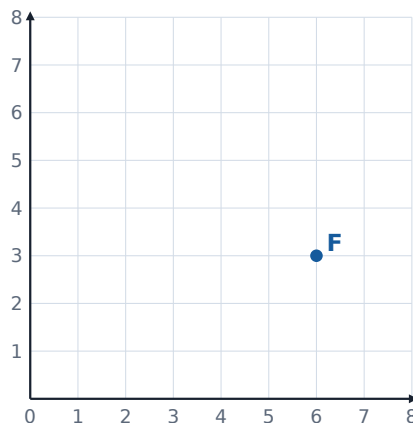
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q1/5

What are the coordinates of point F?

F = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q2/5

1051

QUESTION ID

S28-B3-Q2

TYPE

Short answer

CORRECT ANSWER

(6, 3)

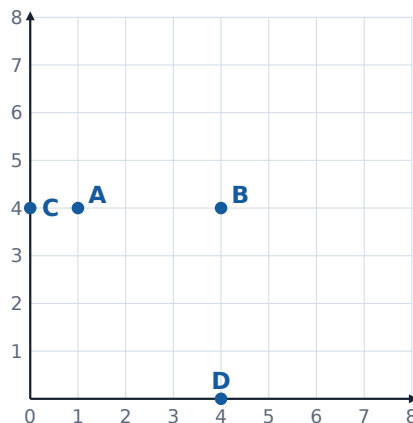
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q2/5

Which point is located at (4, 0)?

- A. Point A
- B. Point B
- C. Point C
- D. Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q3/5

1052

QUESTION ID

S28-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

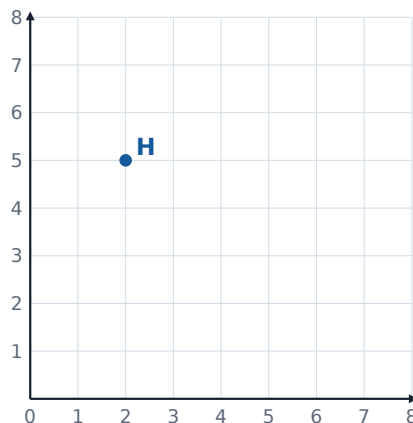
ID S28-B3-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q3/5

True or false?

Point H is located at (5, 2).

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q4/5

1053

QUESTION ID

S28-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

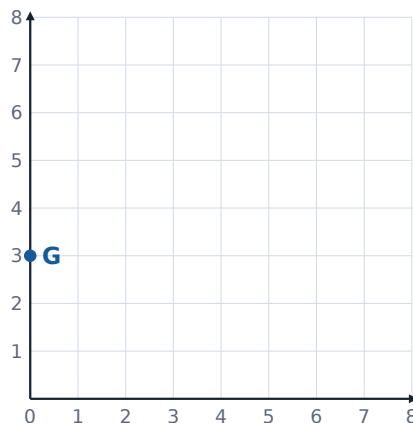
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q4/5

What are the coordinates of point G?

G = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q5/5

1054

QUESTION ID

S28-B3-Q5

TYPE

Short answer

CORRECT ANSWER

(0, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-B3-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 28 • BACKWARD 3 • Q5/5

SET 28 • 6.NS.C.6.b

FORWARD 1

Distance between a point and its reflection • Grade 7 • The Number System •
7.NS.A.1.c

The point $(3, -4.5)$ is reflected across the x-axis.

How far apart are the point and its image?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q1/5

1056

QUESTION ID

S28-F1-Q1

TYPE

Short answer

CORRECT ANSWER

9 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q1/5

The point $(-6.5, 2)$ is reflected across the y -axis.

How far apart are the point and its image?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q2/5

1057

QUESTION ID

S28-F1-Q2

TYPE

Short answer

CORRECT ANSWER

13 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q2/5

The point $(1, -2\frac{1}{2})$ is reflected across the x-axis.

How far apart are the point and its image?

- A. -5
- B. 0
- C. $2\frac{1}{2}$
- D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q3/5

1058

QUESTION ID

S28-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q3/5

A point in Quadrant I and its reflection across the y-axis are 11 units apart. The point's y-coordinate is 7.

What is the point's x-coordinate?

X = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q4/5

1059

QUESTION ID

S28-F1-Q4

TYPE

Short answer

CORRECT ANSWER

5.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q4/5

True or false?

The point $(-4, 3)$ and its reflection across the x-axis are 6 units apart.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q5/5

1060

QUESTION ID

S28-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 28 • FORWARD 1 • Q5/5

SET 28 • 6.NS.C.6.b

FORWARD 2

Reflections and rotations with coordinates • Grade 8 • Geometry • 8.G.A.3

A triangle has a vertex at $(2, -3)$. The triangle is reflected across the y -axis.

What are the coordinates of the image of that vertex?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q1/5

1062

QUESTION ID

S28-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$(-2, -3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q1/5

Which transformation is described by $(x, y) \rightarrow (x, -y)$?

- A. A rotation of 180°
- B. A reflection across the x-axis
- C. A translation
- D. A reflection across the y-axis

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q2/5

1063

QUESTION ID

S28-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

**B. A reflection
across the x-axis**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q2/5

The point $(4, 1)$ is rotated 180° about the origin.

What are the coordinates of its image?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q3/5

1064

QUESTION ID

S28-F2-Q3

TYPE

Short answer

CORRECT ANSWER

$(-4, -1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

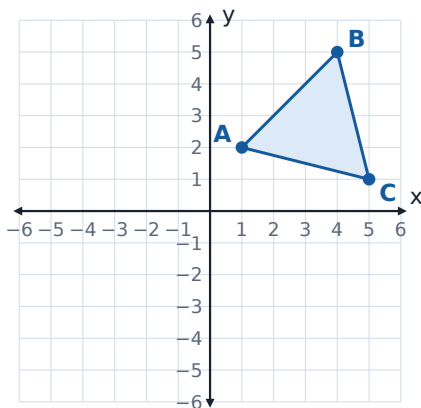
ID S28-F2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q3/5

Triangle ABC is reflected
across the x-axis.

What are the coordinates of
B'?

B' = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q4/5

ID S28-F2-Q4

1065

QUESTION ID

S28-F2-Q4

TYPE

Short answer

CORRECT ANSWER

(4, -5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q4/5

ID S28-F2-Q4

True or false?

Reflecting $(-3, 5)$ across the y -axis gives $(3, 5)$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q5/5

1066

QUESTION ID

S28-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S28-F2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 28 • FORWARD 2 • Q5/5

Name the rational number at a point on a number line

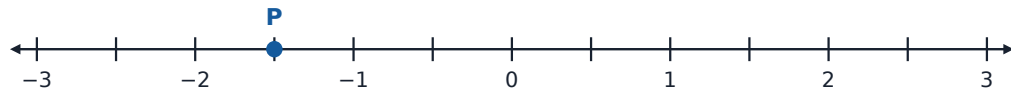
MAIN	Grade 6 main question	6.NS.C.6.c
BACKWARD 1	Fractions on a number line	3.NF.A.2
BACKWARD 2	Decimals on a number line	4.NF.C.6
BACKWARD 3	Opposites on a number line	6.NS.C.6.a
FORWARD 1	Addition shown on a number line	7.NS.A.1.b
FORWARD 2	Approximate irrational numbers on a number line	8.NS.A.2

SET 29 • 6.NS.C.6.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.c

What number does point P represent?



P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q1/5

1069

QUESTION ID

S29-M-Q1

TYPE

Short answer

CORRECT ANSWER

-1.5

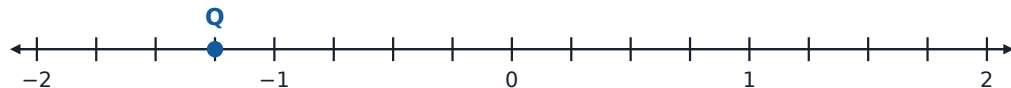
GRADING NOTE -1 1/2 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q1/5

Write the number at point Q as a mixed number.



Q = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q2/5

1070

QUESTION ID

S29-M-Q2

TYPE

Short answer

CORRECT ANSWER

$-1 \frac{1}{4}$

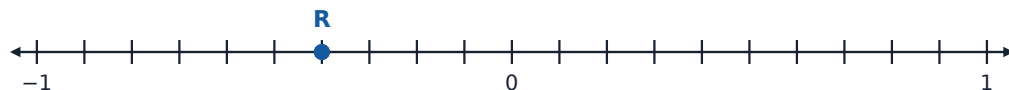
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q2/5

True or false?

Point R is located at -0.4 .



- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q3/5

1071

QUESTION ID

S29-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

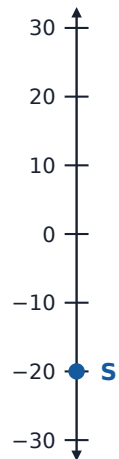
ID S29-M-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q3/5

The number line shows elevations in meters.

What elevation does point S show?

S = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q4/5

1072

QUESTION ID

S29-M-Q4

TYPE

Short answer

CORRECT ANSWER

–20 meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q4/5

Write the number at point T as a fraction.



T = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q5/5

1073

QUESTION ID

S29-M-Q5

TYPE

Short answer

CORRECT ANSWER

$-\frac{2}{3}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-M-Q5

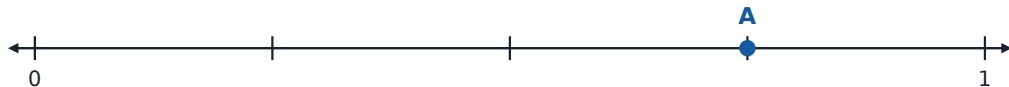
Grade 6 • The Number System • 6.NS.C.6.c • Set 29 • MAIN • Q5/5

SET 29 • 6.NS.C.6.c

BACKWARD 1

Fractions on a number line • Grade 3 • Number & Operations–Fractions •
3.NF.A.2

What fraction does point A represent?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q1/5

1075

QUESTION ID

S29-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q1/5

What fraction does point B represent?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q2/5

1076

QUESTION ID

S29-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/3

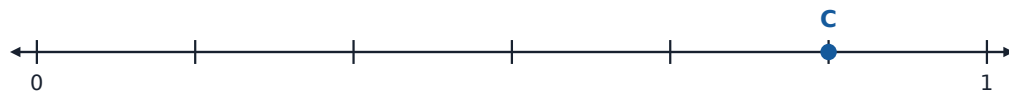
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q2/5

True or false?

Point C is located at $\frac{5}{6}$.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q3/5

1077

QUESTION ID

S29-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q3/5

The space from 0 to 1 is split into equal parts.

How many equal parts are there?



A. 7

B. 8

C. 9

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q4/5

1078

QUESTION ID

S29-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

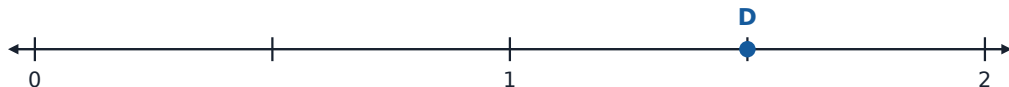
B. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q4/5

Write the number at point D as a fraction.



D = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q5/5

1079

QUESTION ID

S29-B1-Q5

TYPE

Short answer

CORRECT ANSWER

3/2

GRADING NOTE 1 1/2 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B1-Q5

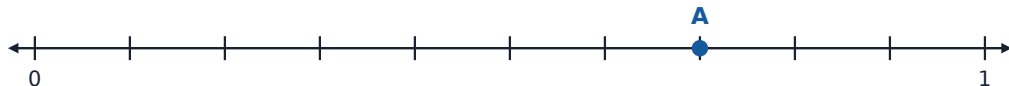
Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 29 • BACKWARD 1 • Q5/5

SET 29 • 6.NS.C.6.c

BACKWARD 2

Decimals on a number line • Grade 4 • Number & Operations–Fractions •
4.NF.C.6

What decimal does point A represent?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q1/5

1081

QUESTION ID

S29-B2-Q1

TYPE

Short answer

CORRECT ANSWER

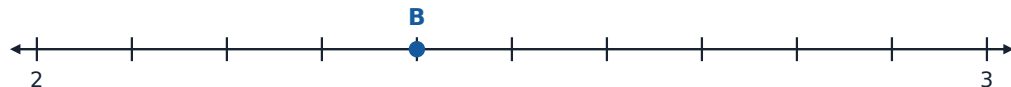
0.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q1/5

What decimal does point B represent?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q2/5

1082

QUESTION ID

S29-B2-Q2

TYPE

Short answer

CORRECT ANSWER

2.4

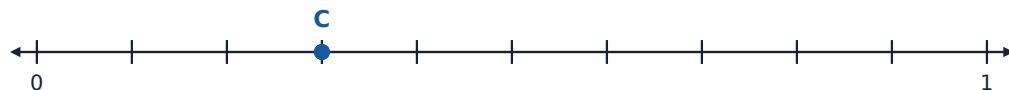
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q2/5

True or false?

Point C is located at 0.3.



- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q3/5

1083

QUESTION ID

S29-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q3/5

What decimal does point D represent?



A. 0.206

B. 0.26

C. 0.62

D. 2.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q4/5

1084

QUESTION ID

S29-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q4/5

What decimal does point E represent?



E = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q5/5

1085

QUESTION ID

S29-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5.9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 29 • BACKWARD 2 • Q5/5

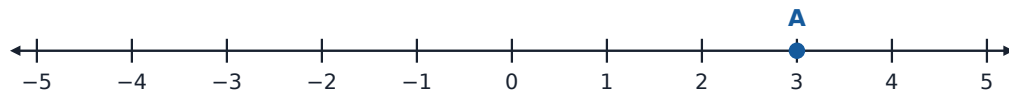
SET 29 • 6.NS.C.6.c

BACKWARD 3

Opposites on a number line • Grade 6 • The Number System • 6.NS.C.6.a

Point B (not shown) is the opposite of point A.

What number is at point B?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q1/5

1087

QUESTION ID

S29-B3-Q1

TYPE

Short answer

CORRECT ANSWER

-3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q1/5

True or false?

–2.5 and 2.5 are the same distance from 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q2/5

1088

QUESTION ID

S29-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q2/5

What is the opposite of $1\frac{3}{4}$?

A. $-1\frac{3}{4}$

B. $-\frac{4}{7}$

C. $\frac{4}{7}$

D. $1\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q3/5

1089

QUESTION ID

S29-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $-1\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q3/5

What is the opposite of -0.8 ?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q4/5

1090

QUESTION ID

S29-B3-Q4

TYPE

Short answer

CORRECT ANSWER

0.8

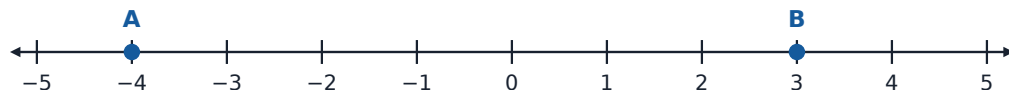
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q4/5

True or false?

Points A and B represent opposite numbers.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q5/5

1091

QUESTION ID

S29-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 29 • BACKWARD 3 • Q5/5

SET 29 • 6.NS.C.6.c

FORWARD 1

Addition shown on a number line • Grade 7 • The Number System • 7.NS.A.1.b

Start at -2.5 on a number line and move 4 units to the right.

Where do you end?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q1/5

1093

QUESTION ID

S29-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1.5

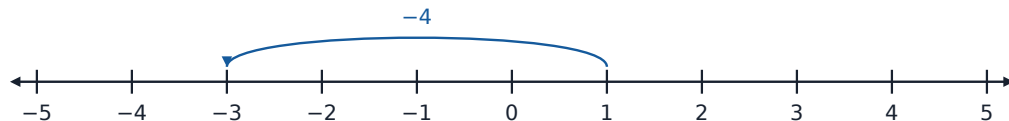
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q1/5

The number line shows $1 + (-4)$.

What is the sum?



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q2/5

1094

QUESTION ID

S29-F1-Q2

TYPE

Short answer

CORRECT ANSWER

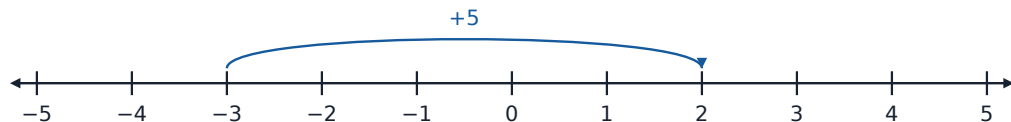
-3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q2/5

Which expression does the number line show?



A. $5 + 3$

B. $-3 - 5$

C. $-3 + 5$

D. $2 + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q3/5

1095

QUESTION ID

S29-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $-3 + 5$

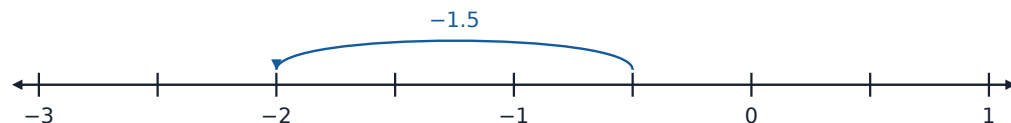
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q3/5

The number line shows $-0.5 + (-1.5)$.

What is the sum?



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q4/5

1096

QUESTION ID

S29-F1-Q4

TYPE

Short answer

CORRECT ANSWER

-2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q4/5

True or false?

On a number line, $-3 + 7$ is 7 units to the right of -3 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q5/5

1097

QUESTION ID

S29-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 29 • FORWARD 1 • Q5/5

SET 29 • 6.NS.C.6.c

FORWARD 2

Approximate irrational numbers on a number line • Grade 8 • The Number System • 8.NS.A.2

Between which two consecutive whole numbers is $\sqrt{20}$?

Between: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q1/5

1099

QUESTION ID

S29-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 and 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q1/5

Which point best shows $\sqrt{10}$?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q2/5

1100

QUESTION ID

S29-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Point C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q2/5

Estimate $\sqrt{50}$ to the nearest tenth.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q3/5

1101

QUESTION ID

S29-F2-Q3

TYPE

Short answer

CORRECT ANSWER

7.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q3/5

Which list is ordered from least to greatest?

- A. 2.5, $\sqrt{5}$, $\sqrt{8}$
- B. $\sqrt{5}$, 2.5, $\sqrt{8}$
- C. $\sqrt{8}$, $\sqrt{5}$, 2.5
- D. $\sqrt{5}$, $\sqrt{8}$, 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q4/5

1102

QUESTION ID

S29-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $\sqrt{5}$, 2.5, $\sqrt{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q4/5

True or false?

π is between 3.1 and 3.2 on the number line.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q5/5

1103

QUESTION ID

S29-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S29-F2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 29 • FORWARD 2 • Q5/5

Plot rational numbers on a horizontal or vertical number line

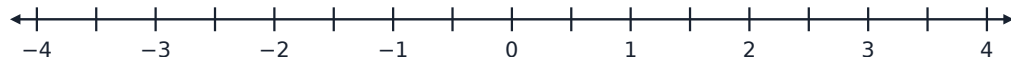
MAIN	Grade 6 main question	6.NS.C.6.c
BACKWARD 1	Placing fractions on a number line	3.NF.A.2
BACKWARD 2	Placing decimals on a number line	4.NF.C.6
BACKWARD 3	Opposites on a number line	6.NS.C.6.a
FORWARD 1	Represent addition on a number line	7.NS.A.1.b
FORWARD 2	Locate irrational numbers approximately on a number line	8.NS.A.2

SET 30 • 6.NS.C.6.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.c

Plot and label point A at -2.5 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q1/5

1106

QUESTION ID

S30-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

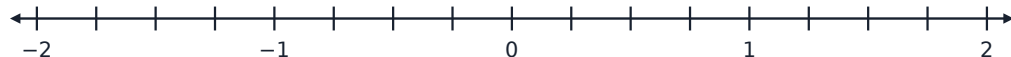
**Point A plotted at
 -2.5 (halfway
between -3 and -2)
and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q1/5

Plot and label point B at $-\frac{3}{4}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q2/5

1107

QUESTION ID

S30-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Point B plotted at $-\frac{3}{4}$ (third tick to the left of 0) and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

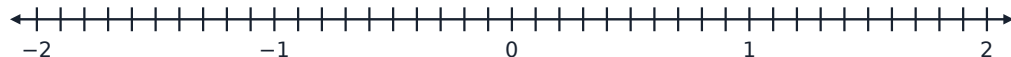
ID S30-M-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

Plot and label -1.2 and its opposite on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q3/5

1108

QUESTION ID

S30-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

**Points plotted at
 -1.2 and 1.2 and
labeled**

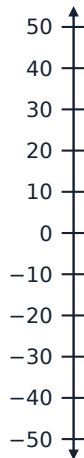
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q3/5

The number line shows elevations in feet.

Plot and label point C at -35 feet.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q4/5

1109

QUESTION ID

S30-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

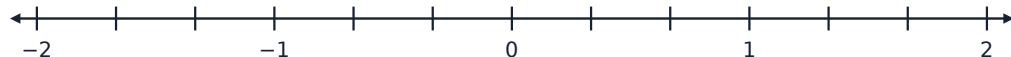
Point C plotted at -35 (halfway between -40 and -30 , below 0) and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q4/5

Plot and label point D at $1\frac{2}{3}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q5/5

1110

QUESTION ID

S30-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Point D plotted at $1\frac{2}{3}$ (second tick to the right of 1) and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-M-Q5

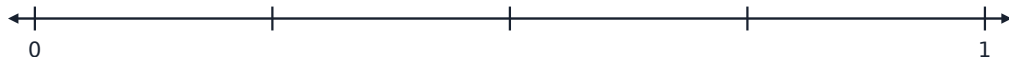
Grade 6 • The Number System • 6.NS.C.6.c • Set 30 • MAIN • Q5/5

SET 30 • 6.NS.C.6.c

BACKWARD 1

Placing fractions on a number line • Grade 3 • Number & Operations–Fractions
• 3.NF.A.2

Plot and label $\frac{3}{4}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q1/5

1112

QUESTION ID

S30-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**$\frac{3}{4}$ plotted at the
third tick after 0 and
labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q1

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q1/5

Plot and label $\frac{2}{3}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q2/5

1113

QUESTION ID

S30-B1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

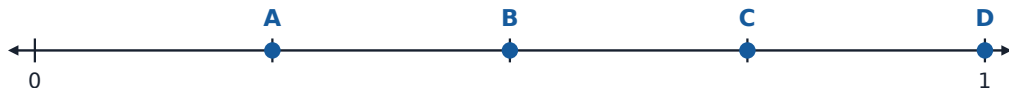
**$\frac{2}{3}$ plotted at the
second tick after 0
and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q2

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q2/5

Which point is at $\frac{1}{2}$?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q3/5

1114

QUESTION ID

S30-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

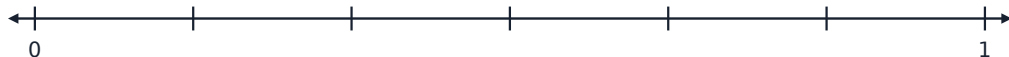
B. Point B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q3

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q3/5

Plot and label $\frac{5}{6}$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q4/5

1115

QUESTION ID

S30-B1-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

**5/6 plotted at the
fifth tick after 0 and
labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q4

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q4/5

True or false?

To plot $\frac{3}{8}$, count 3 of the 8 equal parts from 0.



- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q5/5

1116

QUESTION ID

S30-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B1-Q5

Grade 3 • Number & Operations-Fractions • 3.NF.A.2 • Set 30 • BACKWARD 1 • Q5/5

SET 30 • 6.NS.C.6.c

BACKWARD 2

Placing decimals on a number line • Grade 4 • Number & Operations–Fractions
• 4.NF.C.6

ANSWER KEY

Reduced copy of the question page, with its answer

Plot and label 0.7 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q1/5

1118

QUESTION ID

S30-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**0.7 plotted at the
seventh tick after 0
and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q1

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q1/5

1118

Plot and label 2.4 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q2/5

1119

QUESTION ID

S30-B2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**2.4 plotted at the
fourth tick after 2
and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q2

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q2/5

Which point is at 0.6?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q3/5

1120

QUESTION ID

S30-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Point C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q3

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Plot and label 0.35 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q4/5

1121

QUESTION ID

S30-B2-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

**0.35 plotted at the
fifth tick after 0.3
and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q4

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q4/5

1121

True or false?

On a number line marked in tenths, 0.9 is one tick mark to the left of 1.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q5/5

1122

QUESTION ID

S30-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B2-Q5

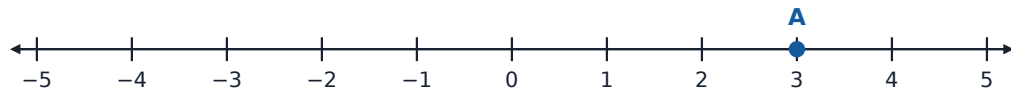
Grade 4 • Number & Operations-Fractions • 4.NF.C.6 • Set 30 • BACKWARD 2 • Q5/5

SET 30 • 6.NS.C.6.c

BACKWARD 3

Opposites on a number line • Grade 6 • The Number System • 6.NS.C.6.a

Point A is at 3. Plot and label its opposite.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q1/5

1124

QUESTION ID

S30-B3-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Point plotted at -3
and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

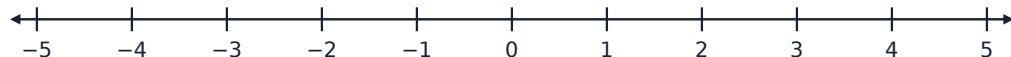
ID S30-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Plot and label 2 and -2 on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q2/5

1125

QUESTION ID

S30-B3-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Points plotted at 2
and -2 and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q2/5

1125

True or false?

A number and its opposite are the same distance from 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q3/5

1126

QUESTION ID

S30-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q3/5

Point B is at -1.5 . Where is its opposite?

- A.** At -1.5
- B.** At 1.5
- C.** At 0
- D.** At 0.15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q4/5

1127

QUESTION ID

S30-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. At 1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

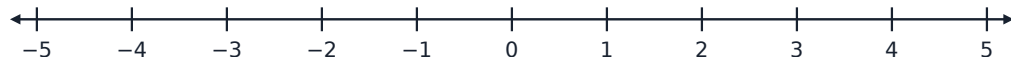
ID S30-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Plot and label the opposite of -4 .



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q5/5

1128

QUESTION ID

S30-B3-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

**Point plotted at 4
and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 30 • BACKWARD 3 • Q5/5

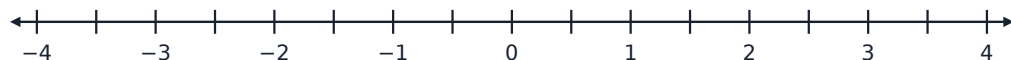
SET 30 • 6.NS.C.6.c

FORWARD 1

Represent addition on a number line • Grade 7 • The Number System •
7.NS.A.1.b

Draw an arrow on the number line to show $-2 + 3.5$.

What is the sum?



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q1/5

1130

QUESTION ID

S30-F1-Q1

TYPE

Short answer

CORRECT ANSWER

1.5

GRADING NOTE Arrow from -2 to 1.5

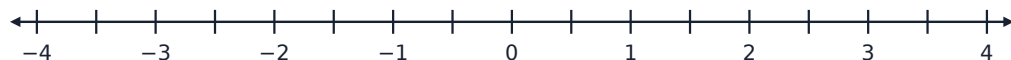
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q1/5

Draw arrows on the number line to show $1.5 + (-3)$.

What is the sum?



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q2/5

1131

QUESTION ID

S30-F1-Q2

TYPE

Short answer

CORRECT ANSWER

-1.5

GRADING NOTE Arrows: 0 to 1.5, then 1.5 to -1.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q2

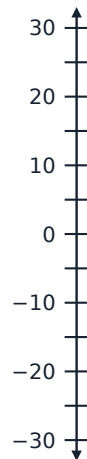
Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q2/5

Draw arrows on the vertical
number line to show

$$-10 + 25.$$

What is the sum?

Sum: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q3/5

1132

QUESTION ID

S30-F1-Q3

TYPE

Short answer

CORRECT ANSWER

15

GRADING NOTE Arrows: 0 down to -10 , then up 25 to 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q3/5

An arrow on a number line starts at $-\frac{1}{2}$ and moves $\frac{3}{4}$ unit to the right.

Where does it end?

A. $-1\frac{1}{4}$

B. $-\frac{1}{4}$

C. $\frac{1}{4}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q4/5

1133

QUESTION ID

S30-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q4/5

Draw arrows on the number line to show $-1\frac{1}{4} + (-\frac{3}{4})$.

What is the sum?



Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q5/5

1134

QUESTION ID

S30-F1-Q5

TYPE

Short answer

CORRECT ANSWER

-2

GRADING NOTE Arrows: 0 to $-1\frac{1}{4}$, then $-1\frac{1}{4}$ to -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.b • Set 30 • FORWARD 1 • Q5/5

SET 30 • 6.NS.C.6.c

FORWARD 2

Locate irrational numbers approximately on a number line • Grade 8 • The Number System • 8.NS.A.2

Plot $\sqrt{10}$ at its approximate location on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q1/5

1136

QUESTION ID

S30-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Point just past 3, at
about 3.16 (between
3 and 3.5, closer to
3)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Plot $\sqrt{2}$ at its approximate location on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q2/5

1137

QUESTION ID

S30-F2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Point at about 1.41
(between 1.4 and
1.5)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q2/5

1137

ANSWER KEY

Reduced copy of the question page, with its answer

Plot π at its approximate location on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q3/5

1138

QUESTION ID

S30-F2-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

**Point at about 3.14
(between 3.1 and
3.2)**

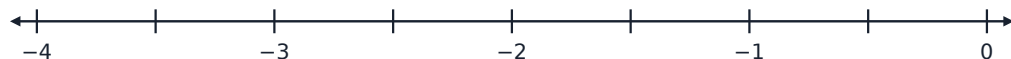
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q3/5

1138

Plot $-\sqrt{5}$ at its approximate location on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q4/5

1139

QUESTION ID

S30-F2-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

**Point at about -2.24
(between -2.5 and
 -2 , closer to -2)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Plot $\sqrt{30}$ at its approximate location on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q5/5

1140

QUESTION ID

S30-F2-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

**Point at about 5.48
(just to the left of
5.5)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S30-F2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 30 • FORWARD 2 • Q5/5

Name the coordinates of points in all four quadrants

MAIN	Grade 6 main question	6.NS.C.6.c
BACKWARD 1	Ordered pairs and the axes	5.G.A.1
BACKWARD 2	Reading points in the first quadrant	5.G.A.2
BACKWARD 3	Quadrants and signs of coordinates	6.NS.C.6.b
FORWARD 1	Interpret a point on the graph of a proportional relationship	7.RP.A.2.d
FORWARD 2	A function as a set of ordered pairs	8.F.A.1

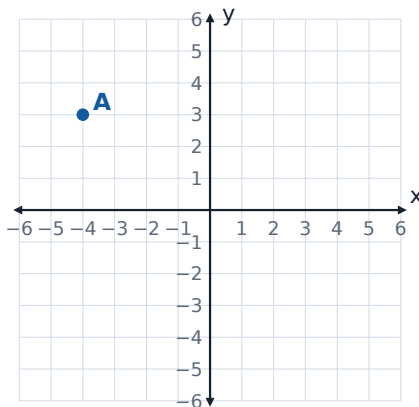
SET 31 • 6.NS.C.6.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.c

What are the coordinates of point A?

A = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q1/5

1143

QUESTION ID

S31-M-Q1

TYPE

Short answer

CORRECT ANSWER

(-4, 3)

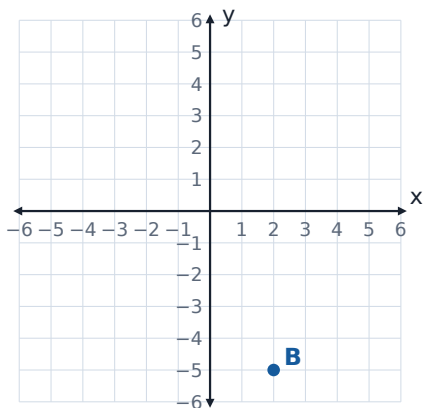
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q1/5

What are the coordinates of point B?

B = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q2/5

1144

QUESTION ID

S31-M-Q2

TYPE

Short answer

CORRECT ANSWER

(2, -5)

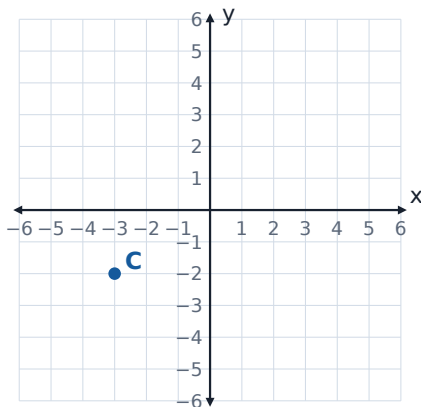
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q2/5

What are the coordinates of point C?

C = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q3/5

1145

QUESTION ID

S31-M-Q3

TYPE

Short answer

CORRECT ANSWER

(-3, -2)

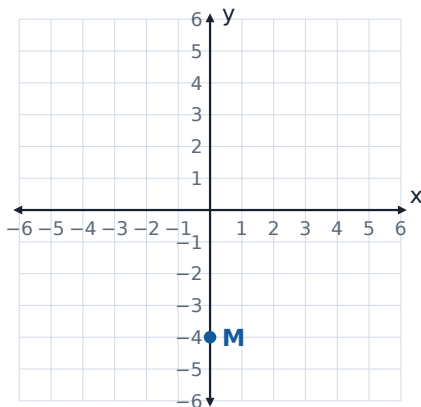
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q3/5

Which ordered pair names point M?

- A. $(0, 4)$
- B. $(4, 0)$
- C. $(0, -4)$
- D. $(-4, 0)$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q4/5

1146

QUESTION ID

S31-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $(0, -4)$

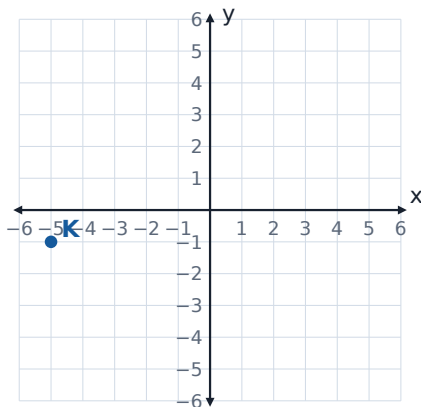
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q4/5

What are the coordinates of point K?

K = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q5/5

1147

QUESTION ID

S31-M-Q5

TYPE

Short answer

CORRECT ANSWER

(-5, -1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-M-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 31 • MAIN • Q5/5

SET 31 • 6.NS.C.6.c

BACKWARD 1

Ordered pairs and the axes • Grade 5 • Geometry • 5.G.A.1

True or false?

The x-coordinate tells how far to move from the origin along the x-axis.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q1/5

1149

QUESTION ID

S31-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q1/5

In the ordered pair (7, 3), what is the y-coordinate?

- A.** 3
- B.** 4
- C.** 7
- D.** 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q2/5

1150

QUESTION ID

S31-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q2/5

The x-axis and the y-axis cross at a point.

What are its name and coordinates?

Name: _____

Coordinates: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q3/5

1151

QUESTION ID

S31-B1-Q3

TYPE

Short answer

CORRECT ANSWER

**Name: the origin;
Coordinates: (0, 0)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q3/5

True or false?

(2, 5) and (5, 2) name the same point.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q4/5

1152

QUESTION ID

S31-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q4/5

In an ordered pair, which number is written first?

- A.** The smaller number
- B.** The x-coordinate
- C.** The y-coordinate
- D.** The larger number

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q5/5

1153

QUESTION ID

S31-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. The x-coordinate

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B1-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 31 • BACKWARD 1 • Q5/5

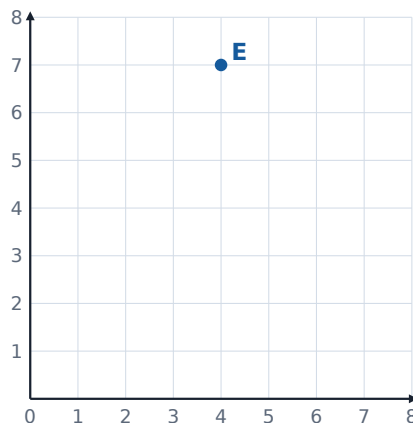
SET 31 • 6.NS.C.6.c

BACKWARD 2

Reading points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

What are the coordinates of point E?

E = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q1/5

1155

QUESTION ID

S31-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(4, 7)

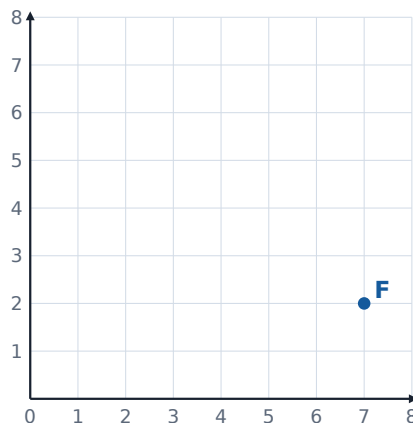
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q1/5

What are the coordinates of point F?

F = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q2/5

1156

QUESTION ID

S31-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(7, 2)

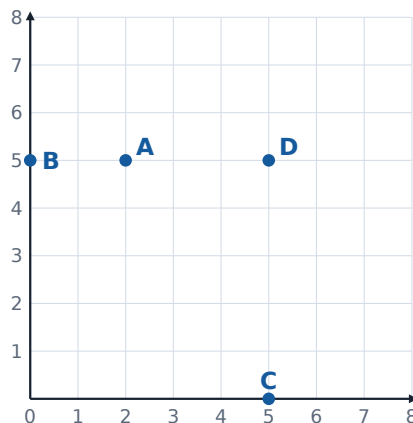
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q2/5

Which point is located at (0, 5)?

- A. Point A
- B. Point B
- C. Point C
- D. Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q3/5

1157

QUESTION ID

S31-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Point B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

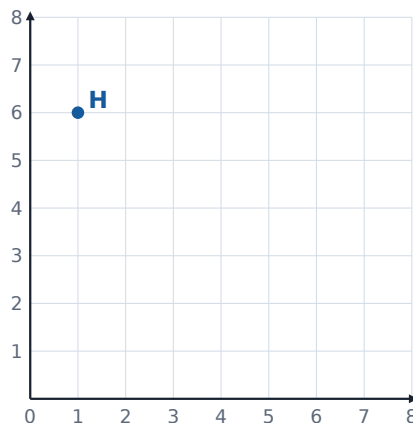
ID S31-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q3/5

True or false?

Point H is located at (6, 1).

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q4/5

1158

QUESTION ID

S31-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

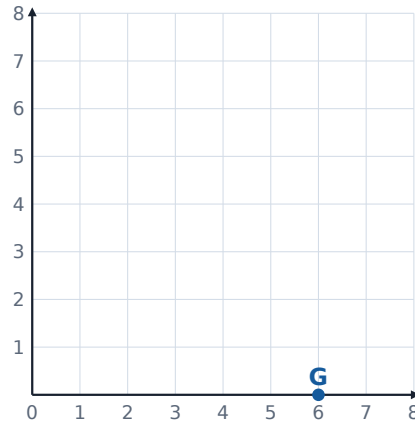
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q4/5

What are the coordinates of point G?

G = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q5/5

1159

QUESTION ID

S31-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(6, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 31 • BACKWARD 2 • Q5/5

SET 31 • 6.NS.C.6.c

BACKWARD 3

Quadrants and signs of coordinates • Grade 6 • The Number System •
6.NS.C.6.b

In which quadrant are both coordinates negative?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q1/5

1161

QUESTION ID

S31-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Quadrant III

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q1/5

In which quadrant is $(5, -2)$?

- A.** Quadrant I
- B.** Quadrant II
- C.** Quadrant III
- D.** Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q2/5

1162

QUESTION ID

S31-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Quadrant IV

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q2/5

True or false?

$(-3, 4)$ is in Quadrant II.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q3/5

1163

QUESTION ID

S31-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q3/5

A point is in Quadrant IV.

Write "positive" or "negative" for each coordinate.

x-coordinate: _____

y-coordinate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q4/5

1164

QUESTION ID

S31-B3-Q4

TYPE

Short answer

CORRECT ANSWER

**x-coordinate:
positive;
y-coordinate:
negative**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q4/5

True or false?

The point $(0, -6)$ is in Quadrant III.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q5/5

1165

QUESTION ID

S31-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 31 • BACKWARD 3 • Q5/5

SET 31 • 6.NS.C.6.c

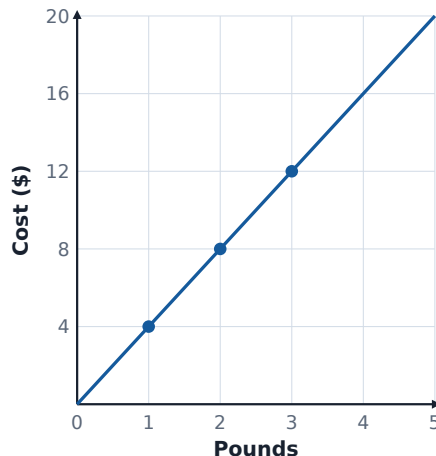
FORWARD 1

Interpret a point on the graph of a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d

The graph shows the cost of apples.

What does the point (2, 8) represent?

Meaning: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q1/5

1167

QUESTION ID

S31-F1-Q1

TYPE

Short answer

CORRECT ANSWER

**2 pounds of apples
cost \$8.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

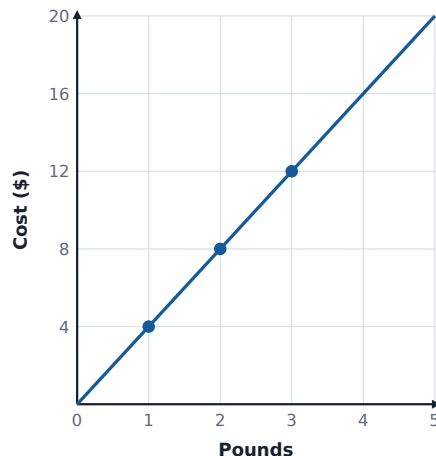
ID S31-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q1/5

The graph shows the cost of apples.

What does the point (1, 4) represent?

- A. 4 pounds cost \$1.
- B. 1 pound costs \$4.
- C. \$1 buys 4 pounds.
- D. The cost starts at \$4.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q2/5

1168

QUESTION ID

S31-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 1 pound costs \$4.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q2/5

True or false?

On the graph of a proportional relationship between pounds and cost, the point (0, 0) means 0 pounds cost \$0.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q3/5

1169

QUESTION ID

S31-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q3/5

The graph of a proportional relationship passes through (1, r) and (5, 30).

What is r ?

$r =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q4/5

1170

QUESTION ID

S31-F1-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q4/5

A graph shows the distance y in miles after x hours. It passes through $(3, 150)$.

What does this point mean?

- A.** In 3 hours, the distance is 150 miles.
- B.** The speed is 150 miles per hour.
- C.** In 150 hours, the distance is 3 miles.
- D.** The speed is 3 miles per hour.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q5/5

1171

QUESTION ID

S31-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. In 3 hours, the distance is 150 miles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.d • Set 31 • FORWARD 1 • Q5/5

SET 31 • 6.NS.C.6.c

FORWARD 2

A function as a set of ordered pairs • Grade 8 • Functions • 8.F.A.1

True or false?

The set of ordered pairs (1, 2), (2, 4), (3, 6), (4, 8) represents a function.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q1/5

1173

QUESTION ID

S31-F2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q1/5

Which set of ordered pairs does NOT represent a function?

- A. $(-1, 3), (0, 5), (1, 7)$
- B. $(1, 4), (2, 4), (3, 4)$
- C. $(0, 0), (1, 1), (2, 2)$
- D. $(2, 5), (2, 7), (3, 9)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q2/5

1174

QUESTION ID

S31-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $(2, 5), (2, 7), (3, 9)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q2/5

Does the set $(4, 1), (5, 2), (4, 3)$ represent a function?
Write yes or no.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q3/5

1175

QUESTION ID

S31-F2-Q3

TYPE

Short answer

CORRECT ANSWER

No

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

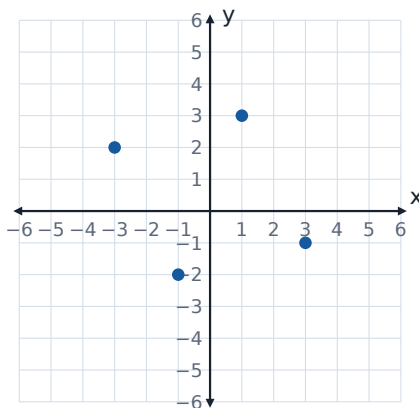
ID S31-F2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q3/5

True or false?

The graph represents a function.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q4/5

1176

QUESTION ID

S31-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q4/5

A function assigns to each input ____.

- A.** at least two outputs
- B.** no output
- C.** the same output as the input
- D.** exactly one output

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q5/5

1177

QUESTION ID

S31-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

**D. exactly one
output**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S31-F2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 31 • FORWARD 2 • Q5/5

Plot points in all four quadrants

MAIN

Grade 6 main question

6.NS.C.6.c

BACKWARD 1

Which direction to move first

5.G.A.1

BACKWARD 2

Plotting points in the first quadrant

5.G.A.2

BACKWARD 3

Direction from the sign of a coordinate

6.NS.C.6.b

FORWARD 1

Graph pairs to decide whether a relationship is proportional

7.RP.A.2.a

FORWARD 2

Graph proportional relationships

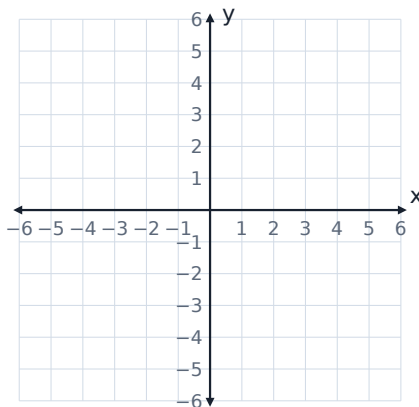
8.EE.B.5

SET 32 • 6.NS.C.6.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.6.c

Plot and label point $P(-3, 4)$
and point $Q(2, -5)$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q1/5

1180

QUESTION ID

S32-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

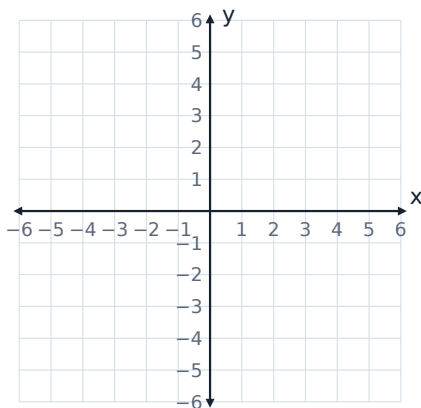
**P plotted at $(-3, 4)$
and Q plotted at $(2, -5)$, both labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q1/5

Plot and label point R($-5, -2$) and point S($4, 1$).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q2/5

1181

QUESTION ID

S32-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

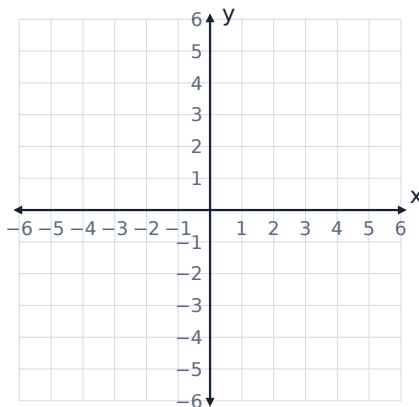
**R plotted at $(-5, -2)$
and S plotted at $(4, 1)$, both labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q2/5

Plot and label point $T(0, -3)$
and point $U(-4, 0)$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q3/5

1182

QUESTION ID

S32-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

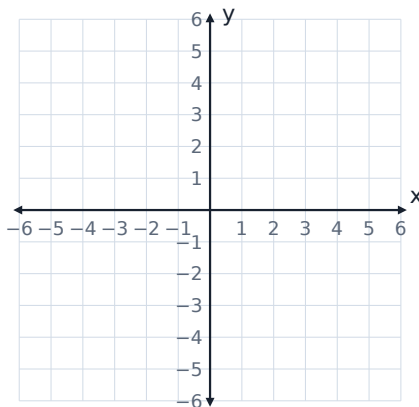
**T plotted at $(0, -3)$
on the y-axis and U
plotted at $(-4, 0)$ on
the x-axis, both
labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q3/5

Plot and label point $V(-2.5, 3)$ and point $W(1.5, -2)$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q4/5

1183

QUESTION ID

S32-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

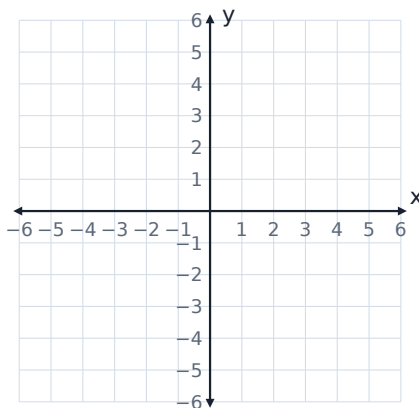
V plotted at $(-2.5, 3)$ and W plotted at $(1.5, -2)$, both labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q4/5

Plot and label point $X(-6, -6)$
and point $Y(5, -4)$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q5/5

1184

QUESTION ID

S32-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

**X plotted at $(-6, -6)$
and Y plotted at $(5, -4)$, both labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-M-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 32 • MAIN • Q5/5

SET 32 • 6.NS.C.6.c

BACKWARD 1

Which direction to move first • Grade 5 • Geometry • 5.G.A.1

True or false?

To plot (6, 1), first move 6 units along the x-axis.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q1/5

1186

QUESTION ID

S32-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q1/5

To plot (2, 7) from the origin, what do you do first?

- A. Move 7 units up
- B. Move 2 units right
- C. Move 2 units up
- D. Move 7 units right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q2/5

1187

QUESTION ID

S32-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Move 2 units right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q2

Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q2/5

True or false?

To plot (0, 5), you do not move left or right.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q3/5

1188

QUESTION ID

S32-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q3

Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q3/5

To plot (4, 3), how many units up do you move?

Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q4/5

1189

QUESTION ID

S32-B1-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q4/5

Which ordered pair is 3 units right and 6 units up from the origin?

- A. (3, 3)
- B. (3, 6)
- C. (6, 3)
- D. (6, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q5/5

1190

QUESTION ID

S32-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B1-Q5

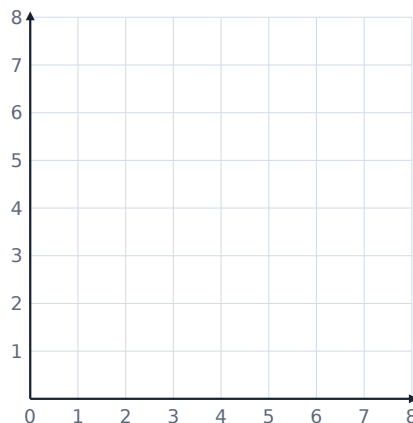
Grade 5 • Geometry • 5.G.A.1 • Set 32 • BACKWARD 1 • Q5/5

SET 32 • 6.NS.C.6.c

BACKWARD 2

Plotting points in the first quadrant • Grade 5 • Geometry • 5.G.A.2

Plot and label point A at (2, 6).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q1/5

1192

QUESTION ID

S32-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

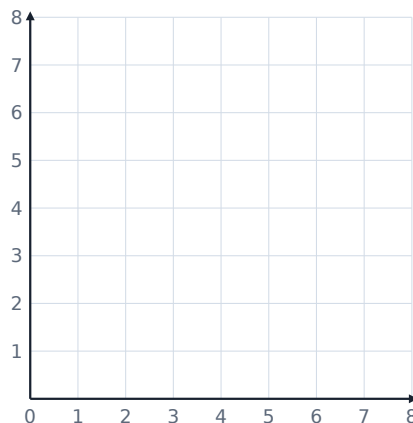
Point A plotted at (2, 6) and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q1/5

Plot and label point B at (5, 1).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q2/5

1193

QUESTION ID

S32-B2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

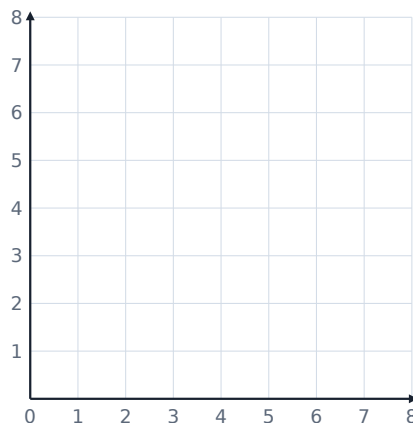
Point B plotted at (5, 1) and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q2/5

Plot and label point C at (0, 7).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q3/5

1194

QUESTION ID

S32-B2-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

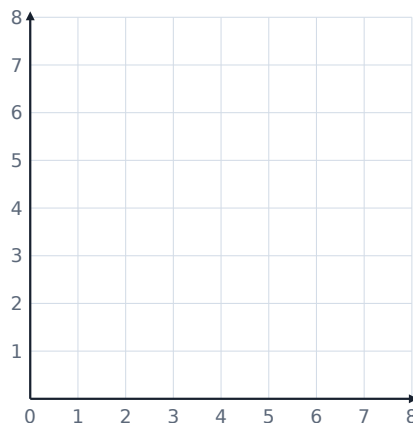
Point C plotted at (0, 7) on the y-axis and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q3/5

Plot and label point D at (7, 7).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q4/5

1195

QUESTION ID

S32-B2-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

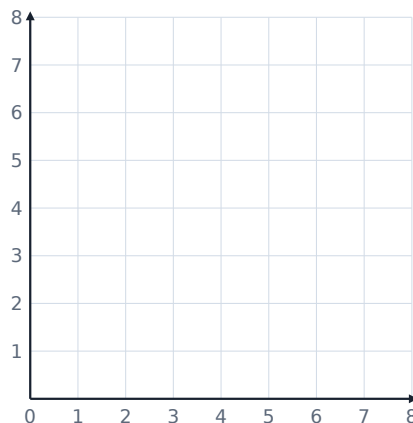
Point D plotted at (7, 7) and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q4/5

Plot and label point E at (5, 0).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q5/5

1196

QUESTION ID

S32-B2-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Point E plotted at (5, 0) on the x-axis and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 32 • BACKWARD 2 • Q5/5

SET 32 • 6.NS.C.6.c

BACKWARD 3

Direction from the sign of a coordinate • Grade 6 • The Number System •
6.NS.C.6.b

True or false?

To plot a point with a negative x-coordinate, you move left from the origin.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q1/5

1198

QUESTION ID

S32-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q1/5

To plot $(-2, -5)$, which directions do you move?

- A. Right 2, down 5
- B. Right 2, up 5
- C. Left 2, down 5
- D. Left 5, down 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q2/5

1199

QUESTION ID

S32-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Left 2, down 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q2/5

True or false?

To plot $(3, -4)$, you move up 4 units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q3/5

1200

QUESTION ID

S32-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

To plot $(-6, 1)$, do you move left or right?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q4/5

1201

QUESTION ID

S32-B3-Q4

TYPE

Short answer

CORRECT ANSWER

Left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q4/5

1201

Which point is left of the y-axis and above the x-axis?

- A. $(-3, 2)$
- B. $(3, -2)$
- C. $(3, 2)$
- D. $(-3, -2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q5/5

1202

QUESTION ID

S32-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. $(-3, 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.b • Set 32 • BACKWARD 3 • Q5/5

SET 32 • 6.NS.C.6.c

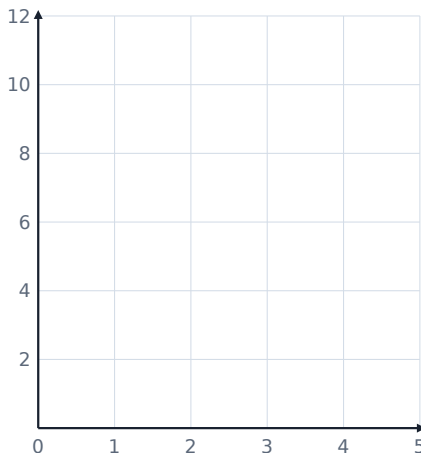
FORWARD 1

Graph pairs to decide whether a relationship is proportional • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a

Graph the pairs (1, 3), (2, 6), and (3, 9).

Do they lie on a straight line through the origin?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q1/5

1204

QUESTION ID

S32-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Points (1, 3), (2, 6), (3, 9) graphed; they lie on a line through (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

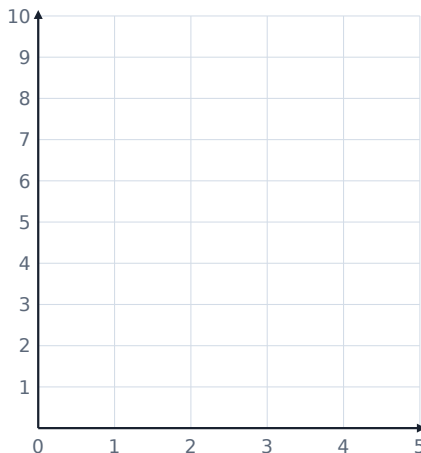
ID S32-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q1/5

Graph the pairs (1, 3), (2, 5), and (3, 7).

Is the relationship proportional?

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q2/5

1205

QUESTION ID

S32-F1-Q2

TYPE

Short answer

CORRECT ANSWER

No

GRADING NOTE Points (1, 3), (2, 5), (3, 7) graphed; the line does not pass through (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

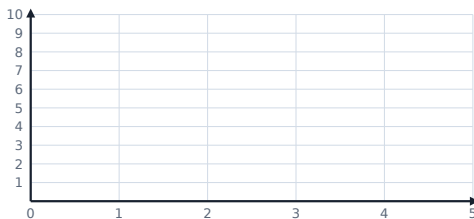
ID S32-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q2/5

Graph the pairs from the table.

Is the relationship proportional?

x	1	2	4
y	2.5	5	10



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q3/5

1206

QUESTION ID

S32-F1-Q3

TYPE

Short answer

CORRECT ANSWER

Yes

GRADING NOTE Points (1, 2.5), (2, 5), (4, 10) graphed; they lie on a line through (0, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q3/5

True or false?

The points (2, 1), (4, 2), and (6, 3) lie on a line through the origin, so the relationship is proportional.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q4/5

1207

QUESTION ID

S32-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q4/5

Which set of pairs would graph as a proportional relationship?

- A. (1, 1), (2, 4), (3, 9)
- B. (1, 3), (2, 4), (3, 5)
- C. (0, 2), (1, 4), (2, 6)
- D. (2, 3), (4, 6), (6, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q5/5

1208

QUESTION ID

S32-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. (2, 3), (4, 6), (6, 9)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.a • Set 32 • FORWARD 1 • Q5/5

SET 32 • 6.NS.C.6.c

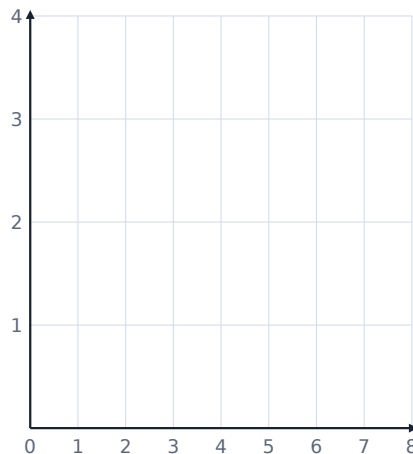
FORWARD 2

Graph proportional relationships • Grade 8 • Expressions & Equations •
8.EE.B.5

Graph $y = 0.5x$ on the coordinate plane.

What is the slope of the line?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q1/5

1210

QUESTION ID

S32-F2-Q1

TYPE

Short answer

CORRECT ANSWER

0.5

GRADING NOTE Graph: line through (0, 0), (2, 1), (4, 2), (8, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

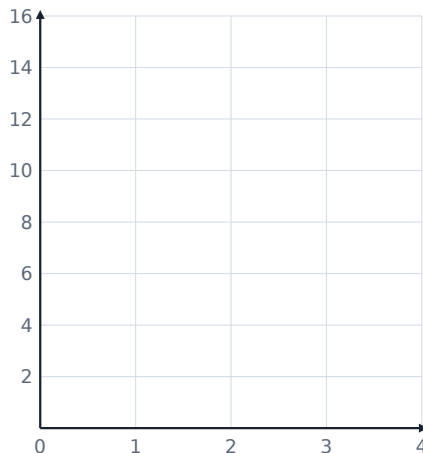
ID S32-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q1/5

Graph $y = 4x$ on the coordinate plane.

What is the slope of the line?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q2/5

1211

QUESTION ID

S32-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

GRADING NOTE Graph: line through (0, 0), (1, 4), (4, 16)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

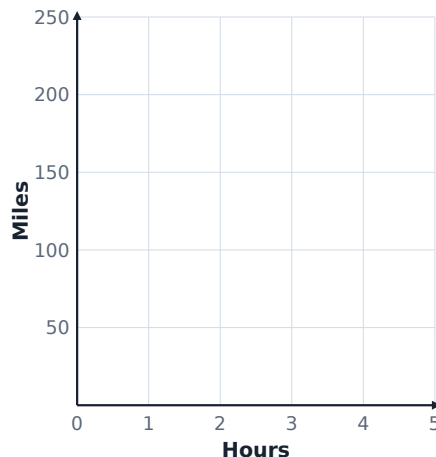
ID S32-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q2/5

A car travels 50 miles per hour. Graph the distance y after x hours.

What is the slope?

Slope: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q3/5

1212

QUESTION ID

S32-F2-Q3

TYPE

Short answer

CORRECT ANSWER

50

GRADING NOTE Graph: line through (0, 0), (1, 50), (5, 250)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q3/5

Which point is on the graph of $y = 3x$?

- A. (3, 1)
- B. (12, 4)
- C. (4, 12)
- D. (4, 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q4/5

1213

QUESTION ID

S32-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. (4, 12)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q4/5

True or false?

The graph of $y = \frac{2}{3}x$ passes through (3, 2).

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q5/5

1214

QUESTION ID

S32-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S32-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.5 • Set 32 • FORWARD 2 • Q5/5

Interpret inequalities as relative position on a number line

MAIN	Grade 6 main question
BACKWARD 1	Comparison symbols
BACKWARD 2	Position on a number line
BACKWARD 3	Negative numbers on a number line
FORWARD 1	Solve and graph inequalities
FORWARD 2	Compare and order irrational numbers

6.NS.C.7.a

2.NBT.A.4

2.MD.B.6

6.NS.C.6.c

7.EE.B.4.b

8.NS.A.2

SET 33 • 6.NS.C.7.a

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.a

$$-4 > -10$$

Which statement explains this inequality?

- A. -4 is to the left of -10 on a number line.
- B. 4 is less than 10 .
- C. -4 is farther from 0 than -10 .
- D. -4 is to the right of -10 on a number line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q1/5

1217

QUESTION ID

S33-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. -4 is to the right of -10 on a number line.

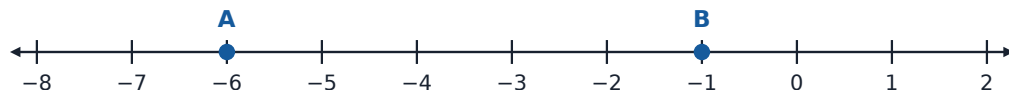
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q1/5

True or false?

The number line shows that $-6 > -1$.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q2/5

1218

QUESTION ID

S33-M-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q2/5

Write an inequality that compares -4 and -9 .

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q3/5

1219

QUESTION ID

S33-M-Q3

TYPE

Short answer

CORRECT ANSWER

$-4 > -9$

GRADING NOTE $-9 < -4$ is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q3/5

Which statement is true?

- A.** $-5 < -2$
- B.** $-2 < -5$
- C.** $-5 > 0$
- D.** $-5 > -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q4/5

1220

QUESTION ID

S33-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

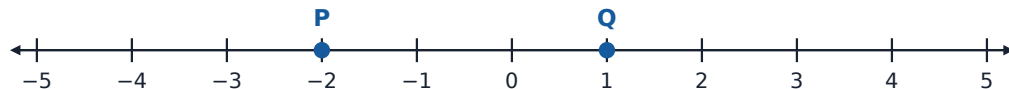
A. $-5 < -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q4/5

Write an inequality using the numbers at points P and Q.



Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q5/5

1221

QUESTION ID

S33-M-Q5

TYPE

Short answer

CORRECT ANSWER

$-2 < 1$

GRADING NOTE $1 > -2$ is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-M-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 33 • MAIN • Q5/5

SET 33 • 6.NS.C.7.a

BACKWARD 1

Comparison symbols • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Write $>$, $<$, or $=$ to compare.

452 ____ 425

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q1/5

1223

QUESTION ID

S33-B1-Q1

TYPE

Short answer

CORRECT ANSWER

$>$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q1/5

True or false?

$$67 < 76$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q2/5

1224

QUESTION ID

S33-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q2/5

Which statement is true?

- A.** $309 < 290$
- B.** $309 = 290$
- C.** $309 > 290$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q3/5

1225

QUESTION ID

S33-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $309 > 290$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q3/5

Write $>$, $<$, or $=$ to compare.

138 ____ 183

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q4/5

1226

QUESTION ID

S33-B1-Q4

TYPE

Short answer

CORRECT ANSWER

$<$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q4/5

True or false?

500 > 505

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q5/5

1227

QUESTION ID

S33-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 33 • BACKWARD 1 • Q5/5

SET 33 • 6.NS.C.7.a

BACKWARD 2

Position on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

True or false?

On a number line, 8 is to the right of 3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q1/5

1229

QUESTION ID

S33-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q1/5

Which number is farthest to the left on a number line?

- A.** 2
- B.** 5
- C.** 7
- D.** 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q2/5

1230

QUESTION ID

S33-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q2/5

Which point is farther to the left, A or B?



Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q3/5

1231

QUESTION ID

S33-B2-Q3

TYPE

Short answer

CORRECT ANSWER

B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q3/5

True or false?

Numbers get greater as you move to the left on a number line.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q4/5

1232

QUESTION ID

S33-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q4/5

Which number is to the right of 14 on a number line?

- A.** 9
- B.** 11
- C.** 14
- D.** 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q5/5

1233

QUESTION ID

S33-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B2-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 33 • BACKWARD 2 • Q5/5

SET 33 • 6.NS.C.7.a

BACKWARD 3

Negative numbers on a number line • Grade 6 • The Number System •
6.NS.C.6.c

What number is at point A?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q1/5

1235

QUESTION ID

S33-B3-Q1

TYPE

Short answer

CORRECT ANSWER

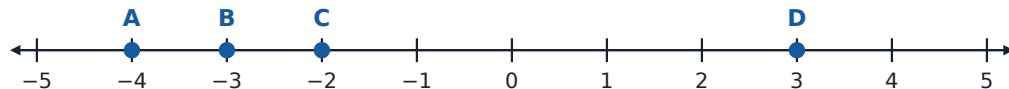
-6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q1/5

Which point is at -3 ?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q2/5

1236

QUESTION ID

S33-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Point B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q2/5

True or false?

–4 is to the left of –1 on a number line.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q3/5

1237

QUESTION ID

S33-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q3/5

What number is at point B?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q4/5

1238

QUESTION ID

S33-B3-Q4

TYPE

Short answer

CORRECT ANSWER

-8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q4/5

Which point is at -5 ?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q5/5

1239

QUESTION ID

S33-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Point B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-B3-Q5

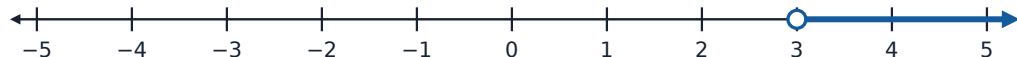
Grade 6 • The Number System • 6.NS.C.6.c • Set 33 • BACKWARD 3 • Q5/5

SET 33 • 6.NS.C.7.a

FORWARD 1

Solve and graph inequalities • Grade 7 • Expressions & Equations • 7.EE.B.4.b

Which inequality does the graph show?



A. $x > 3$

B. $x \leq 3$

C. $x < 3$

D. $x \geq 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q1/5

1241

QUESTION ID

S33-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. $x > 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q1/5

Solve the inequality.

$$2x + 5 > 11$$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q2/5

1242

QUESTION ID

S33-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$$x > 3$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q2/5

Solve the inequality.

$$-3x + 4 \leq 16$$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q3/5

1243

QUESTION ID

S33-F1-Q3

TYPE

Short answer

CORRECT ANSWER

$$x \geq -4$$

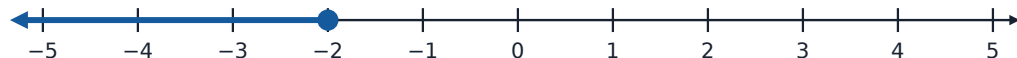
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q3/5

True or false?

The graph shows the solutions of $4x - 1 \leq -9$.



- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q4/5

1244

QUESTION ID

S33-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q4/5

Solve the inequality.

$$5x - 2 < 18$$

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q5/5

1245

QUESTION ID

S33-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$$x < 4$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 33 • FORWARD 1 • Q5/5

SET 33 • 6.NS.C.7.a

FORWARD 2

Compare and order irrational numbers • Grade 8 • The Number System •
8.NS.A.2

Which is greater, $\sqrt{50}$ or 7.2?

Greater: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q1/5

1247

QUESTION ID

S33-F2-Q1

TYPE

Short answer

CORRECT ANSWER

7.2

GRADING NOTE $\sqrt{50} \approx 7.07$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q1/5

Which number is between 3 and 4?

- A. $\sqrt{12}$
- B. $\sqrt{8}$
- C. $\sqrt{3}$
- D. $\sqrt{17}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q2/5

1248

QUESTION ID

S33-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $\sqrt{12}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q2/5

True or false?

$$\sqrt{30} < 5.5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q3/5

1249

QUESTION ID

S33-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q3/5

Order from least to greatest.

π , $\sqrt{8}$, 3.1

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q4/5

1250

QUESTION ID

S33-F2-Q4

TYPE

Short answer

CORRECT ANSWER

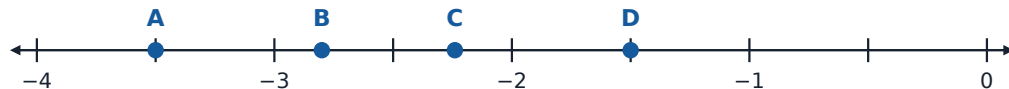
$\sqrt{8}$, 3.1, π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q4/5

Which point best shows $-\sqrt{5}$?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q5/5

1251

QUESTION ID

S33-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Point C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S33-F2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 33 • FORWARD 2 • Q5/5

Write, interpret, and explain order statements in real-world contexts

MAIN	Grade 6 main question	6.NS.C.7.b
BACKWARD 1	Negative numbers in context	6.NS.C.5
BACKWARD 2	Inequality symbols	1.NBT.B.3
BACKWARD 3	Order as position on a number line	6.NS.C.7.a
FORWARD 1	Write and solve inequalities for real-world situations	7.EE.B.4.b
FORWARD 2	Compare quantities written with powers of 10	8.EE.A.3

SET 34 • 6.NS.C.7.b

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.b

A temperature of -2°F is warmer than a temperature of -11°F .

Write an inequality to show this.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q1/5

1254

QUESTION ID

S34-M-Q1

TYPE

Short answer

CORRECT ANSWER

$-2 > -11$

GRADING NOTE $-11 < -2$ is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q1/5

Account A has a balance of $-\$25$. Account B has a balance of $-\$40$.

Which statement is true?

- A.** $-25 > -40$, so Account A has the greater balance.
- B.** $-40 > -25$, so Account B has the greater balance.
- C.** $-25 < -40$, so Account A owes more.
- D.** The balances are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q2/5

1255

QUESTION ID

S34-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $-25 > -40$, so Account A has the greater balance.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q2/5

A fish swims at -12 feet. A turtle swims at -5 feet.

Write an inequality to compare the positions. Which animal is deeper?

Inequality: _____

Deeper animal: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q3/5

1256

QUESTION ID

S34-M-Q3

TYPE

Short answer

CORRECT ANSWER

**Inequality: $-12 < -5$
(or $-5 > -12$);
Deeper animal: the
fish**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q3/5

True or false?

$-15 < -10$ means that -15°F is colder than -10°F .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q4/5

1257

QUESTION ID

S34-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q4/5

The lowest point in a valley is -86 meters. A beach is at 0 meters.

Which statement is true?

- A.** $-86 < 0$, so the valley's lowest point is lower. **B.** $-86 = 0$
- C.** $0 < -86$, so the beach is lower. **D.** $-86 > 0$, so the valley's lowest point is higher.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q5/5

1258

QUESTION ID

S34-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. $-86 < 0$, so the valley's lowest point is lower.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-M-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 34 • MAIN • Q5/5

SET 34 • 6.NS.C.7.b

BACKWARD 1

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

Write an integer for 12 degrees below zero.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q1/5

1260

QUESTION ID

S34-B1-Q1

TYPE

Short answer

CORRECT ANSWER

-12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q1/5

True or false?

A debt of \$30 can be written as -30 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q2/5

1261

QUESTION ID

S34-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q2/5

Which situation could be represented by -6 ?

- A.** A loss of 6 yards
- B.** A deposit of \$6
- C.** A gain of 6 yards
- D.** 6 feet above sea level

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q3/5

1262

QUESTION ID

S34-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A loss of 6 yards

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q3/5

Write an integer for 50 feet below sea level.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q4/5

1263

QUESTION ID

S34-B1-Q4

TYPE

Short answer

CORRECT ANSWER

−50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q4/5

True or false?

A temperature of 4 degrees above zero is written as -4 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q5/5

1264

QUESTION ID

S34-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 34 • BACKWARD 1 • Q5/5

SET 34 • 6.NS.C.7.b

BACKWARD 2

Inequality symbols • Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3

Which symbol makes this true?

42 ____ 24

A. <

B. >

C. =

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q1/5

1266

QUESTION ID

S34-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. >

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q1/5

True or false?

15 > 51

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q2/5

1267

QUESTION ID

S34-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q2/5

Write $>$, $<$, or $=$ to compare.

73 ____ 37

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q3/5

1268

QUESTION ID

S34-B2-Q3

TYPE

Short answer

CORRECT ANSWER

$>$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q3/5

Which symbol means "is less than"?

A. $<$

B. $>$

C. $=$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q4/5

1269

QUESTION ID

S34-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $<$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q4

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q4/5

True or false?

89 < 98

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q5/5

1270

QUESTION ID

S34-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B2-Q5

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 34 • BACKWARD 2 • Q5/5

SET 34 • 6.NS.C.7.b

BACKWARD 3

Order as position on a number line • Grade 6 • The Number System •
6.NS.C.7.a

True or false?

$-2 > -6$ because -2 is to the right of -6 on a number line.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q1/5

1272

QUESTION ID

S34-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q1/5

Which number is greatest?

- A.** -9
- B.** -7
- C.** -4
- D.** -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q2/5

1273

QUESTION ID

S34-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

–5 is to the left of 3 on a number line.

Write an inequality to show this.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q3/5

1274

QUESTION ID

S34-B3-Q3

TYPE

Short answer

CORRECT ANSWER

–5 < 3**GRADING NOTE** $3 > -5$ is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q3/5

1274

True or false?

–8 is to the right of –3 on a number line.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q4/5

1275

QUESTION ID

S34-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q4/5

Which number is to the left of -4 on a number line?

- A.** -6
- B.** -2
- C.** 0
- D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q5/5

1276

QUESTION ID

S34-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. -6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-B3-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 34 • BACKWARD 3 • Q5/5

SET 34 • 6.NS.C.7.b

FORWARD 1

Write and solve inequalities for real-world situations • Grade 7 • Expressions & Equations • 7.EE.B.4.b

A submarine at -40 m descends 12 m per minute. It must stay above -160 m.

Write and solve an inequality for the number of minutes t it can descend.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q1/5

1278

QUESTION ID

S34-F1-Q1

TYPE

Short answer

CORRECT ANSWER

**$-40 - 12t > -160$,
so $t < 10$ (fewer than
10 minutes)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q1/5

You have \$25 and spend \$4 per day. You want to keep at least \$5.

Write and solve an inequality for the number of days d .

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q2/5

1279

QUESTION ID

S34-F1-Q2

TYPE

Short answer

CORRECT ANSWER

**$25 - 4d \geq 5$, so $d \leq 5$
(at most 5 days)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q2/5

A freezer is at 2°C . It cools 3°C per hour.

For which numbers of hours h is its temperature below -10°C ?

- A.** $h > 4$ **B.** $h > -4$
C. $h < -4$ **D.** $h < 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q3/5

1280

QUESTION ID

S34-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $h > 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q3/5

A diver at -8 ft descends 5 ft per minute.

Write and solve an inequality to find when the diver is deeper than -48 ft.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q4/5

1281

QUESTION ID

S34-F1-Q4

TYPE

Short answer

CORRECT ANSWER

**$-8 - 5t < -48$, so
 $t > 8$ (after more
than 8 minutes)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q4/5

True or false?

The solution of $-2x + 1 > 9$ is $x < -4$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q5/5

1282

QUESTION ID

S34-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 34 • FORWARD 1 • Q5/5

SET 34 • 6.NS.C.7.b

FORWARD 2

Compare quantities written with powers of 10 • Grade 8 • Expressions & Equations • 8.EE.A.3

A mountain is about 9×10^3 m tall. An ocean trench is about -1.1×10^4 m deep.

Which is farther from sea level?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q1/5

1284

QUESTION ID

S34-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The ocean trench

GRADING NOTE $|-1.1 \times 10^4| = 11,000 \text{ m} > 9,000 \text{ m}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q1/5

Which number is greater?

A. 3×10^{-2}

B. 8×10^{-3}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q2/5

1285

QUESTION ID

S34-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 3×10^{-2}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q2/5

A city has about 2×10^6 people. A town has about 5×10^4 people.

How many times as many people live in the city?

Times as many: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q3/5

1286

QUESTION ID

S34-F2-Q3

TYPE

Short answer

CORRECT ANSWER

40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q3/5

Order from least to greatest.

$$4 \times 10^3, 9 \times 10^2, 1 \times 10^4$$

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q4/5

1287

QUESTION ID

S34-F2-Q4

TYPE

Short answer

CORRECT ANSWER

**$9 \times 10^2, 4 \times 10^3,$
 1×10^4**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q4/5

True or false?

$$-5 \times 10^3 < -2 \times 10^3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q5/5

1288

QUESTION ID

S34-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S34-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 34 • FORWARD 2 • Q5/5

Find absolute value as distance from 0

MAIN	Grade 6 main question	6.NS.C.7.c
BACKWARD 1	Distance on a number line	2.MD.B.6
BACKWARD 2	Opposite numbers	6.NS.C.6.a
FORWARD 1	Distance between rational numbers as $ p - q $	7.NS.A.1.c
FORWARD 2	Square roots and cube roots as solutions	8.EE.A.2

SET 35 • 6.NS.C.7.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$|-18|$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q1/5

1291

QUESTION ID

S35-M-Q1

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q1/5

Which number has the greatest absolute value?

- A.** -12
- B.** -5
- C.** 8
- D.** 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q2/5

1292

QUESTION ID

S35-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. -12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q2/5

How far is -7.5 from 0 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q3/5

1293

QUESTION ID

S35-M-Q3

TYPE

Short answer

CORRECT ANSWER

7.5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q3/5

True or false?

$$|-6| = |6|$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q4/5

1294

QUESTION ID

S35-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q4/5

Which two numbers have an absolute value of 11?

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q5/5

1295

QUESTION ID

S35-M-Q5

TYPE

Short answer

CORRECT ANSWER

11 and -11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-M-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 35 • MAIN • Q5/5

SET 35 • 6.NS.C.7.c

BACKWARD 1

Distance on a number line • Grade 2 • Measurement & Data • 2.MD.B.6

How many units apart are points A and B?



Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q1/5

1297

QUESTION ID

S35-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q1/5

True or false?

The distance from 0 to 5 is the same as the distance from 5 to 10.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q2/5

1298

QUESTION ID

S35-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q2/5

How many units from 0 is the number 11?

- A.** 0
- B.** 1
- C.** 10
- D.** 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q3/5

1299

QUESTION ID

S35-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q3/5

What is the distance from 4 to 10 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q4/5

1300

QUESTION ID

S35-B1-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q4/5

How many units is point C from 0?



Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q5/5

1301

QUESTION ID

S35-B1-Q5

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B1-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 35 • BACKWARD 1 • Q5/5

SET 35 • 6.NS.C.7.c

BACKWARD 2

Opposite numbers • Grade 6 • The Number System • 6.NS.C.6.a

What is the opposite of 15?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q1/5

1303

QUESTION ID

S35-B2-Q1

TYPE

Short answer

CORRECT ANSWER

–15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q1/5

True or false?

Opposite numbers are the same distance from 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q2/5

1304

QUESTION ID

S35-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q2/5

Which number is the opposite of $-\frac{2}{3}$?

A. $-\frac{3}{2}$

B. $-\frac{2}{3}$

C. $\frac{2}{3}$

D. $\frac{3}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q3/5

1305

QUESTION ID

S35-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q3/5

What is the opposite of 0?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q4/5

1306

QUESTION ID

S35-B2-Q4

TYPE

Short answer

CORRECT ANSWER

0

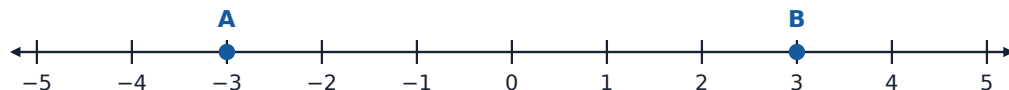
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q4/5

True or false?

Points A and B represent opposite numbers.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q5/5

1307

QUESTION ID

S35-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 35 • BACKWARD 2 • Q5/5

SET 35 • 6.NS.C.7.c

FORWARD 1

Distance between rational numbers as $|p - q|$ • Grade 7 • The Number System
• 7.NS.A.1.c

Find the distance between -4 and 7 on a number line.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q1/5

1309

QUESTION ID

S35-F1-Q1

TYPE

Short answer

CORRECT ANSWER

11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q1/5

Find the value.

$$|-3.5 - 2|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q2/5

1310

QUESTION ID

S35-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q2/5

Which expression gives the distance between -9 and -2 ?

- A. $-9 - 2$
- B. $|-9 - (-2)|$
- C. $|-9 + (-2)|$
- D. $-9 + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q3/5

1311

QUESTION ID

S35-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $|-9 - (-2)|$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q3/5

True or false?

The distance between -6 and 6 is 0 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q4/5

1312

QUESTION ID

S35-F1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q4/5

Find the distance between $-2\frac{1}{2}$ and $1\frac{1}{4}$ on a number line.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q5/5

1313

QUESTION ID

S35-F1-Q5

TYPE

Short answer

CORRECT ANSWER

3 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 35 • FORWARD 1 • Q5/5

SET 35 • 6.NS.C.7.c

FORWARD 2

Square roots and cube roots as solutions • Grade 8 • Expressions & Equations •
8.EE.A.2

Solve.

$$x^2 = 49$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q1/5

1315

QUESTION ID

S35-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$x = 7$ or $x = -7$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q1/5

Solve.

$$x^3 = 64$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q2/5

1316

QUESTION ID

S35-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q2/5

What are the solutions of $x^2 = 81$?

- A. $x = -9$ only
- B. $x = 9$ only
- C. $x = 40.5$
- D. $x = 9$ or $x = -9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q3/5

1317

QUESTION ID

S35-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x = 9$ or $x = -9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q3/5

Solve. Write the answer using a square root symbol.

$$x^2 = 20$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q4/5

1318

QUESTION ID

S35-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$x = \sqrt{20}$ or $x = -\sqrt{20}$

GRADING NOTE $\pm\sqrt{20}$; also accept $\pm 2\sqrt{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q4/5

True or false?

Both 6 and -6 are solutions of $x^2 = 36$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q5/5

1319

QUESTION ID

S35-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S35-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.2 • Set 35 • FORWARD 2 • Q5/5

Interpret absolute value as magnitude in a real-world situation

MAIN	Grade 6 main question	6.NS.C.7.c
BACKWARD 1	Negative numbers in context	6.NS.C.5
BACKWARD 2	Distance from 0	2.MD.B.6
BACKWARD 3	Opposites are the same distance from 0	6.NS.C.6.a
FORWARD 1	Real-world problems with rational numbers	7.NS.A.3
FORWARD 2	Compare magnitudes with powers of 10	8.EE.A.3

SET 36 • 6.NS.C.7.c

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.c

An account balance is $-\$45$.

What is the absolute value of the balance, and what does it describe?

Absolute value: _____

It describes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q1/5

1322

QUESTION ID

S36-M-Q1

TYPE

Short answer

CORRECT ANSWER

**Absolute value: 45;
It describes: a debt
of \$45 (the amount
owed)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q1/5

A diver is at an elevation of -42 feet.

How far is the diver from sea level?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q2/5

1323

QUESTION ID

S36-M-Q2

TYPE

Short answer

CORRECT ANSWER

42 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q2/5

A stock's value changed by $-\$8.50$ today.

What is the size of the change?

- A. $-\$8.50$
- B. $\$0$
- C. $\$8.50$
- D. $\$17.00$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q3/5

1324

QUESTION ID

S36-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $\$8.50$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q3/5

The temperature is -13°F .

How many degrees below zero is it?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q4/5

1325

QUESTION ID

S36-M-Q4

TYPE

Short answer

CORRECT ANSWER

13 degrees

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q4/5

A submarine is at -275 m. A plane is at 275 m.

Which is farther from sea level? Explain using absolute value.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q5/5

1326

QUESTION ID

S36-M-Q5

TYPE

Short answer

CORRECT ANSWER

**Neither: both are
275 m from sea
level, because
 $|-275| = |275|$
 $= 275$.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-M-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 36 • MAIN • Q5/5

SET 36 • 6.NS.C.7.c

BACKWARD 1

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

True or false?

–30 can represent owing \$30.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q1/5

1328

QUESTION ID

S36-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q1/5

Write an integer for 42 feet below sea level.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q2/5

1329

QUESTION ID

S36-B1-Q2

TYPE

Short answer

CORRECT ANSWER

-42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q2/5

Which situation is represented by +25?

- A.** A loss of 25 yards
- B.** 25 feet below sea level
- C.** A debt of \$25
- D.** A deposit of \$25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q3/5

1330

QUESTION ID

S36-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. A deposit of \$25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q3/5

Write an integer for a debt of \$18.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q4/5

1331

QUESTION ID

S36-B1-Q4

TYPE

Short answer

CORRECT ANSWER

−18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q4/5

True or false?

A temperature of 0°F means there is no temperature.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q5/5

1332

QUESTION ID

S36-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B1-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 36 • BACKWARD 1 • Q5/5

SET 36 • 6.NS.C.7.c

BACKWARD 2

Distance from 0 • Grade 2 • Measurement & Data • 2.MD.B.6

How many units from 0 is point A?



Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q1/5

1334

QUESTION ID

S36-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q1/5

True or false?

The distance from 0 to 13 is 13 units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q2/5

1335

QUESTION ID

S36-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q2/5

Which number is 5 units from 0 on this number line?



A. 0

B. 5

C. 10

D. 50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q3/5

1336

QUESTION ID

S36-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q3/5

How far is 20 from 0 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q4/5

1337

QUESTION ID

S36-B2-Q4

TYPE

Short answer

CORRECT ANSWER

20 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q4/5

True or false?

The number that is 0 units from 0 is 1.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q5/5

1338

QUESTION ID

S36-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B2-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 36 • BACKWARD 2 • Q5/5

SET 36 • 6.NS.C.7.c

BACKWARD 3

Opposites are the same distance from 0 • Grade 6 • The Number System •
6.NS.C.6.a

True or false?

–42 and 42 are the same distance from 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q1/5

1340

QUESTION ID

S36-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q1/5

What is the opposite of -275 ?

Opposite: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q2/5

1341

QUESTION ID

S36-B3-Q2

TYPE

Short answer

CORRECT ANSWER

275

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q2/5

Which number is the same distance from 0 as -8.5 ?

- A. -5.8
- B. 0
- C. 8.5
- D. 85

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q3/5

1342

QUESTION ID

S36-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 8.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q3/5

True or false?

The opposite of 13 is -13 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q4/5

1343

QUESTION ID

S36-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q4/5

Name the number that is the same distance from 0 as -45 , on the other side of 0.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q5/5

1344

QUESTION ID

S36-B3-Q5

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.a • Set 36 • BACKWARD 3 • Q5/5

SET 36 • 6.NS.C.7.c

FORWARD 1

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

A company lost \$2.5 million in one year and earned a profit of \$1.8 million the next year.

What is the total for the two years?

Total: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q1/5

1346

QUESTION ID

S36-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−\$0.7 million

GRADING NOTE A loss of \$700,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q1/5

The temperature was -6°F . It dropped 11°F .

What is the new temperature?

Temperature: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q2/5

1347

QUESTION ID

S36-F1-Q2

TYPE

Short answer

CORRECT ANSWER

-17°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q2/5

A diver at -35.5 feet rises 12.25 feet.

What is the diver's new position?

- A. -47.75 feet
- B. -23.25 feet
- C. -22.75 feet
- D. 23.25 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q3/5

1348

QUESTION ID

S36-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. -23.25 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q3/5

A hiker goes from an elevation of 1,200 feet to an elevation of -80 feet.

What is the change in elevation?

Change: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q4/5

1349

QUESTION ID

S36-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$-1,280$ feet

GRADING NOTE A decrease of 1,280 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q4/5

True or false?

If the temperature drops from -4°C to -9°C , it gets 5°C colder.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q5/5

1350

QUESTION ID

S36-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 36 • FORWARD 1 • Q5/5

SET 36 • 6.NS.C.7.c

FORWARD 2

Compare magnitudes with powers of 10 • Grade 8 • Expressions & Equations •
8.EE.A.3

One company's debt is 2×10^6 dollars. Another company's debt is 4×10^4 dollars.

How many times as large is the first debt?

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q1/5

1352

QUESTION ID

S36-F2-Q1

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q1/5

The deepest ocean point is about -1×10^4 meters. A lake bottom is about -5×10^1 meters.

About how many times as deep is the ocean point?

- A. 5 B. 20
C. 200 D. 2,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q2/5

1353

QUESTION ID

S36-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q2/5

Estimate 7,980,000 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q3/5

1354

QUESTION ID

S36-F2-Q3

TYPE

Short answer

CORRECT ANSWER

8×10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q3/5

True or false?

6×10^9 is 300 times as large as 2×10^7 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q4/5

1355

QUESTION ID

S36-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q4/5

Estimate 0.0000031 as a single digit times a power of 10.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q5/5

1356

QUESTION ID

S36-F2-Q5

TYPE

Short answer

CORRECT ANSWER

3×10^{-6}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S36-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.3 • Set 36 • FORWARD 2 • Q5/5

Distinguish comparisons of absolute value from statements about order

MAIN	Grade 6 main question	6.NS.C.7.d
BACKWARD 1	Absolute value	6.NS.C.7.c
BACKWARD 2	Ordering negative numbers	6.NS.C.7.a
BACKWARD 3	Negative numbers in context	6.NS.C.5
FORWARD 1	Real-world problems with rational numbers	7.NS.A.3
FORWARD 2	Compare the size and order of irrational numbers	8.NS.A.2

SET 37 • 6.NS.C.7.d

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.7.d

True or false?

An account balance less than $-\$50$ means a debt greater than $\$50$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q1/5

1359

QUESTION ID

S37-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q1

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q1/5

Which account balance shows the greatest debt?

- A.** $-\$55$
- B.** $-\$38$
- C.** $-\$20$
- D.** $\$10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q2/5

1360

QUESTION ID

S37-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $-\$55$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q2

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q2/5

True or false?

$-12 < -5$, so $|-12| < |-5|$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q3/5

1361

QUESTION ID

S37-M-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q3

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q3/5

Two temperatures are -15°F and -9°F .

Which temperature is colder? Which has the greater absolute value?

Colder: _____

Greater absolute value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q4

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q4/5

1362

QUESTION ID

S37-M-Q4

TYPE

Short answer

CORRECT ANSWER

**Colder: -15°F ;
Greater absolute
value: -15**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q4

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q4/5

A submarine is at -300 feet. A diver is at -45 feet.

Which statement is true?

- A. The diver is deeper because $-45 > -300$.
- B. The submarine is deeper because $-300 > -45$.
- C. The submarine is deeper because $|-300| > |-45|$.
- D. They are at the same depth.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q5/5

1363

QUESTION ID

S37-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. The submarine is deeper because $|-300| > |-45|$.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-M-Q5

Grade 6 • The Number System • 6.NS.C.7.d • Set 37 • MAIN • Q5/5

SET 37 • 6.NS.C.7.d

BACKWARD 1

Absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$|-25|$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q1/5

1365

QUESTION ID

S37-B1-Q1

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q1/5

Find the value.

|14|

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q2/5

1366

QUESTION ID

S37-B1-Q2

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q2/5

True or false?

$$|-3| = -3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q3/5

1367

QUESTION ID

S37-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q3/5

Which number has an absolute value of 7?

- A. -7
- B. $-\frac{1}{7}$
- C. 0
- D. $\frac{1}{7}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q4/5

1368

QUESTION ID

S37-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. -7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q4/5

Find the value.

$$|-2.5|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q5/5

1369

QUESTION ID

S37-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 37 • BACKWARD 1 • Q5/5

SET 37 • 6.NS.C.7.d

BACKWARD 2

Ordering negative numbers • Grade 6 • The Number System • 6.NS.C.7.a

Which number is least?

- A.** -20
- B.** -12
- C.** -5
- D.** 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q1/5

1371

QUESTION ID

S37-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. -20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q1/5

True or false?

$$-9 > -4$$

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q2/5

1372

QUESTION ID

S37-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

Order from least to greatest.

$-3, -8, 1$

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q3/5

1373

QUESTION ID

S37-B2-Q3

TYPE

Short answer

CORRECT ANSWER

 $-8, -3, 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q3/5

1373

True or false?

$$-1 > -100$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q4/5

1374

QUESTION ID

S37-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q4/5

Which number is greater?

A. -16

B. -6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q5/5

1375

QUESTION ID

S37-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. -6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 37 • BACKWARD 2 • Q5/5

SET 37 • 6.NS.C.7.d

BACKWARD 3

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

Write an integer for a debt of \$40.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q1/5

1377

QUESTION ID

S37-B3-Q1

TYPE

Short answer

CORRECT ANSWER

-40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q1/5

What could -8 represent?

- A.** A deposit of \$8
- B.** A gain of 8 yards
- C.** 8 feet below sea level
- D.** 8 feet above sea level

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q2/5

1378

QUESTION ID

S37-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 8 feet below sea level

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q2/5

True or false?

An account balance of $-\$10$ means you owe \$10.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q3/5

1379

QUESTION ID

S37-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q3/5

Write an integer for 15 degrees below zero.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q4/5

1380

QUESTION ID

S37-B3-Q4

TYPE

Short answer

CORRECT ANSWER

-15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q4/5

True or false?

A 20-foot descent can be written as +20.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q5/5

1381

QUESTION ID

S37-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-B3-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 37 • BACKWARD 3 • Q5/5

SET 37 • 6.NS.C.7.d

FORWARD 1

Real-world problems with rational numbers • Grade 7 • The Number System •
7.NS.A.3

Two accounts have balances of $-\$42.50$ and $-\$18.75$.

How much greater is the larger debt?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q1/5

1383

QUESTION ID

S37-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$\$23.75$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q1/5

A diver at -24.5 m rises 9.75 m and then descends 15 m.

What is the diver's final position?

Position: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q2/5

1384

QUESTION ID

S37-F1-Q2

TYPE

Short answer

CORRECT ANSWER

-29.75 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q2/5

The temperature at 6 a.m. was -8.5°F . At noon it was 3.5°F .

How much did the temperature rise?

- A.** -12°F **B.** -5°F
C. 5°F **D.** 12°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q3/5

1385

QUESTION ID

S37-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 12°F

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q3/5

Maria's balance is $-\$36$. She pays back $\frac{1}{4}$ of what she owes.

What is her new balance?

Balance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q4/5

1386

QUESTION ID

S37-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$-\$27$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q4/5

True or false?

A change from -12 m to -30 m is a change of -18 m.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q5/5

1387

QUESTION ID

S37-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 37 • FORWARD 1 • Q5/5

SET 37 • 6.NS.C.7.d

FORWARD 2

Compare the size and order of irrational numbers • Grade 8 • The Number System • 8.NS.A.2

Which number has the greater absolute value?

- A. 4.3
- B. $-\sqrt{20}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q1/5

1389

QUESTION ID

S37-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. $-\sqrt{20}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q1

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q1/5

True or false?

$$-\sqrt{10} < -3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q2/5

1390

QUESTION ID

S37-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q2

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

Order from least to greatest.

$-\sqrt{5}$, -2.5 , -2

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q3/5

1391

QUESTION ID

S37-F2-Q3

TYPE

Short answer

CORRECT ANSWER

 -2.5 , $-\sqrt{5}$, -2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q3

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q3/5

1391

Which number is farther from 0 on a number line?

A. 3.1

B. $-\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q4/5

1392

QUESTION ID

S37-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $-\pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q4

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q4/5

Between which two consecutive integers is $-\sqrt{30}$?

Between: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q5/5

1393

QUESTION ID

S37-F2-Q5

TYPE

Short answer

CORRECT ANSWER

−6 and −5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S37-F2-Q5

Grade 8 • The Number System • 8.NS.A.2 • Set 37 • FORWARD 2 • Q5/5

Find distances between points with the same first or second coordinate

MAIN	Grade 6 main question	6.NS.C.8
BACKWARD 1	Absolute value	6.NS.C.7.c
BACKWARD 2	Naming points on the coordinate plane	6.NS.C.6.c
BACKWARD 3	Distance between whole numbers	2.MD.B.6
FORWARD 1	Distance between rational numbers	7.NS.A.1.c
FORWARD 2	Distance between points using the Pythagorean Theorem	8.G.B.8

SET 38 • 6.NS.C.8

MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.8

Find the distance between $(-3, 4)$ and $(5, 4)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q1/5

1396

QUESTION ID

S38-M-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q1/5

Find the distance between $(2, -6)$ and $(2, 5)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q2/5

1397

QUESTION ID

S38-M-Q2

TYPE

Short answer

CORRECT ANSWER

11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q2/5

What is the distance between $(-7, -2)$ and $(-1, -2)$?

- A.** -6
- B.** 3
- C.** 6
- D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q3/5

1398

QUESTION ID

S38-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6

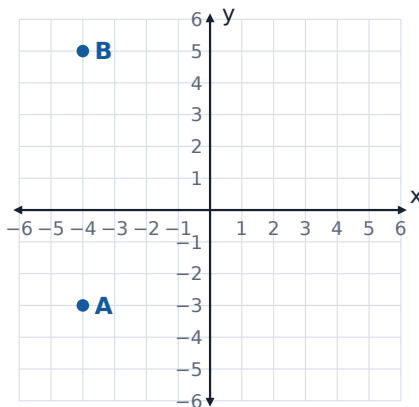
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ID S38-M-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q3/5

What is the distance between points A and B?

Distance: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q4/5

1399

QUESTION ID

S38-M-Q4

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q4/5

Find the distance between $(-2.5, 1)$ and $(3.5, 1)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q5/5

1400

QUESTION ID

S38-M-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-M-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 38 • MAIN • Q5/5

SET 38 • 6.NS.C.8

BACKWARD 1

Absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-7|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q1/5

1402

QUESTION ID

S38-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q1/5

True or false?

$$|-3| + |5| = 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q2/5

1403

QUESTION ID

S38-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q2/5

$$|-4| + |6| = ?$$

- A.** -10
- B.** -2
- C.** 2
- D.** 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q3/5

1404

QUESTION ID

S38-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q3/5

Find the value.

$$|-9| + |2|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q4/5

1405

QUESTION ID

S38-B1-Q4

TYPE

Short answer

CORRECT ANSWER

11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q4/5

True or false?

The distance from -5 to 0 is $|-5|$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q5/5

1406

QUESTION ID

S38-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 38 • BACKWARD 1 • Q5/5

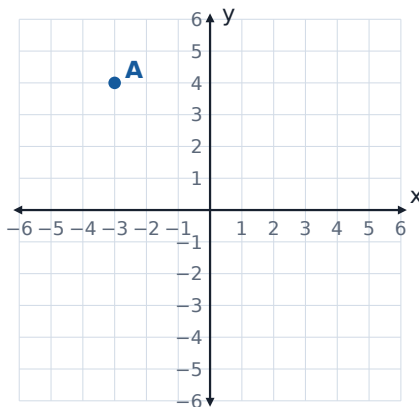
SET 38 • 6.NS.C.8

BACKWARD 2

Naming points on the coordinate plane • Grade 6 • The Number System •
6.NS.C.6.c

What are the coordinates of point A?

A = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q1/5

1408

QUESTION ID

S38-B2-Q1

TYPE

Short answer

CORRECT ANSWER

(-3, 4)

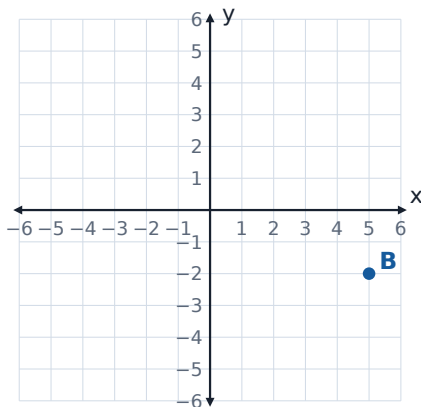
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q1/5

What are the coordinates of point B?

B = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q2/5

1409

QUESTION ID

S38-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(5, -2)

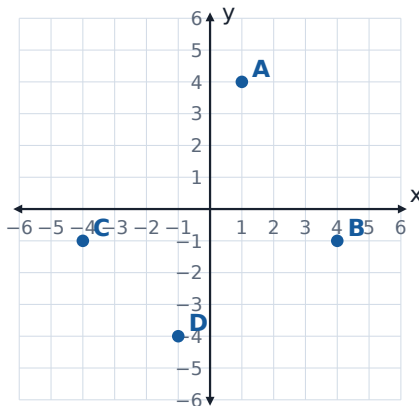
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q2/5

Which point is at $(-1, -4)$?

- A. Point A
- B. Point B
- C. Point C
- D. Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q3/5

1410

QUESTION ID

S38-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

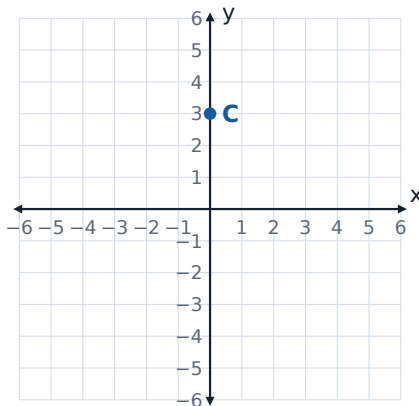
ID S38-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q3/5

True or false?

Point C is located at (3, 0).

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q4/5

1411

QUESTION ID

S38-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

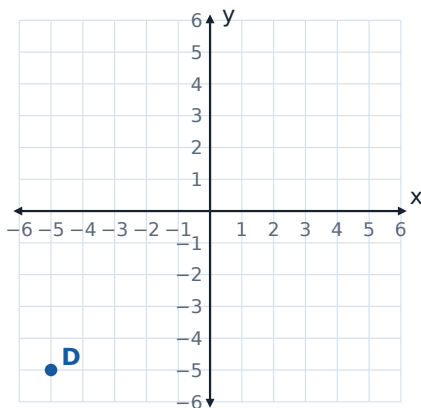
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q4/5

What are the coordinates of point D?

D = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q5/5

1412

QUESTION ID

S38-B2-Q5

TYPE

Short answer

CORRECT ANSWER

(-5, -5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 38 • BACKWARD 2 • Q5/5

SET 38 • 6.NS.C.8

BACKWARD 3

Distance between whole numbers • Grade 2 • Measurement & Data • 2.MD.B.6

What is the distance from 3 to 11 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q1/5

1414

QUESTION ID

S38-B3-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q1/5

How many units apart are points A and B?



Units: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q2/5

1415

QUESTION ID

S38-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q2/5

True or false?

The distance from 12 to 4 is 8 units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q3/5

1416

QUESTION ID

S38-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q3/5

What is the distance from 5 to 14?

- A.** 8
- B.** 9
- C.** 10
- D.** 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q4/5

1417

QUESTION ID

S38-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q4/5

What is the distance from 0 to 13 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q5/5

1418

QUESTION ID

S38-B3-Q5

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-B3-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 38 • BACKWARD 3 • Q5/5

SET 38 • 6.NS.C.8

FORWARD 1

Distance between rational numbers • Grade 7 • The Number System •
7.NS.A.1.c

Find the distance between -2.5 and 4.5 on a number line.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q1/5

1420

QUESTION ID

S38-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q1/5

Find the value.

$$|-8.4 - 3.6|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q2/5

1421

QUESTION ID

S38-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q2/5

What is the distance between $-3\frac{1}{2}$ and $2\frac{1}{4}$ on a number line?

A. $-5\frac{3}{4}$

B. $1\frac{1}{4}$

C. $5\frac{3}{4}$

D. $6\frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q3/5

1422

QUESTION ID

S38-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q3/5

True or false?

The distance between -7 and -2 is $|-7 - (-2)|$, which is 5.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q4/5

1423

QUESTION ID

S38-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q4/5

Find the distance between -4 and 2.5 on a number line.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q5/5

1424

QUESTION ID

S38-F1-Q5

TYPE

Short answer

CORRECT ANSWER

6.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 38 • FORWARD 1 • Q5/5

SET 38 • 6.NS.C.8

FORWARD 2

Distance between points using the Pythagorean Theorem • Grade 8 •
Geometry • 8.G.B.8

Find the distance between (1, 2) and (4, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q1/5

1426

QUESTION ID

S38-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5

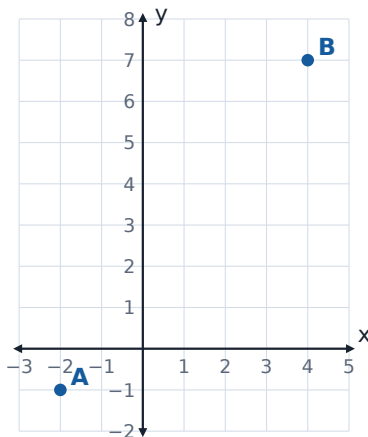
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q1/5

Find the distance between points A and B.

Distance: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q2/5

1427

QUESTION ID

S38-F2-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q2/5

What is the distance between (0, 0) and (5, 12)?

- A. $\sqrt{119}$
- B. 17
- C. 13
- D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q3/5

1428

QUESTION ID

S38-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q3/5

Find the distance between $(-3, 5)$ and $(1, 2)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q4/5

1429

QUESTION ID

S38-F2-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q4/5

True or false?

The distance between (2, 1) and (5, 5) is 5.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q5/5

1430

QUESTION ID

S38-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S38-F2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 38 • FORWARD 2 • Q5/5

Solve real-world problems by graphing points in all four quadrants

MAIN	Grade 6 main question	6.NS.C.8
BACKWARD 1	Absolute value as distance	6.NS.C.7.c
BACKWARD 2	Plotting points in all four quadrants	6.NS.C.6.c
BACKWARD 3	Perimeter of rectangles	3.MD.D.8
BACKWARD 4	Area of rectangles	3.MD.C.7.b
FORWARD 1	Use a map scale on a coordinate grid	7.G.A.1
FORWARD 2	Straight-line distances on a map	8.G.B.8

SET 39 • 6.NS.C.8

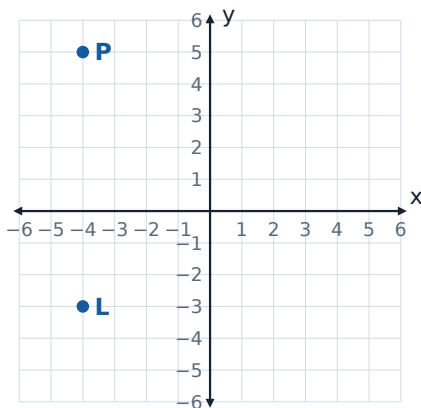
MAIN

Grade 6 main question • Grade 6 • The Number System • 6.NS.C.8

On the map, each unit is 1 block.

How many blocks is it from the library L to the park P?

Blocks: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q1/5

1433

QUESTION ID

S39-M-Q1

TYPE

Short answer

CORRECT ANSWER

8 blocks

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

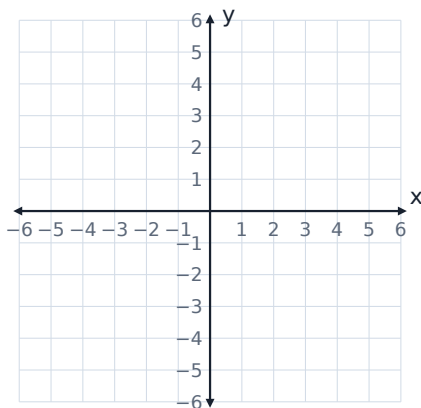
ID S39-M-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q1/5

A park is a rectangle with corners at $(-2, 3)$, $(4, 3)$, $(4, -1)$, and $(-2, -1)$. Each unit is 1 meter.

Graph the park. What is its perimeter?

Perimeter: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q2/5

1434

QUESTION ID

S39-M-Q2

TYPE

Short answer

CORRECT ANSWER

20 meters

GRADING NOTE Rectangle graphed with corners $(-2, 3)$, $(4, 3)$, $(4, -1)$, $(-2, -1)$; sides 6 m and 4 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

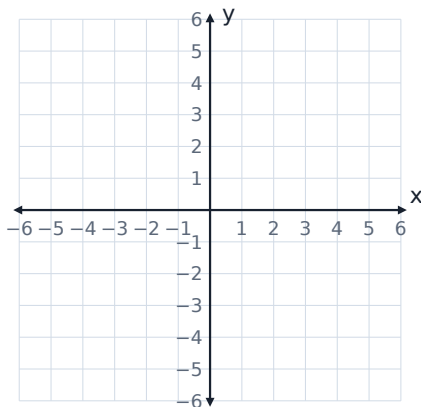
ID S39-M-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q2/5

The school is at $(-3, 2)$ and the pool is at $(-3, -4)$. Each unit is 1 block.

Graph both places. How many blocks apart are they?

Blocks: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q3/5

1435

QUESTION ID

S39-M-Q3

TYPE

Short answer

CORRECT ANSWER

6 blocks

GRADING NOTE Points graphed at $(-3, 2)$ and $(-3, -4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

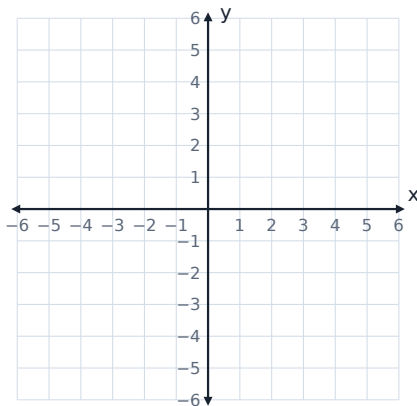
ID S39-M-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q3/5

A garden is a rectangle with corners at $(-5, 1)$, $(1, 1)$, $(1, -3)$, and $(-5, -3)$. Each unit is 1 meter.

What is the area of the garden?

- A.** 10 square meters **B.** 12 square meters
C. 20 square meters **D.** 24 square meters



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q4/5

1436

QUESTION ID

S39-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 24 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

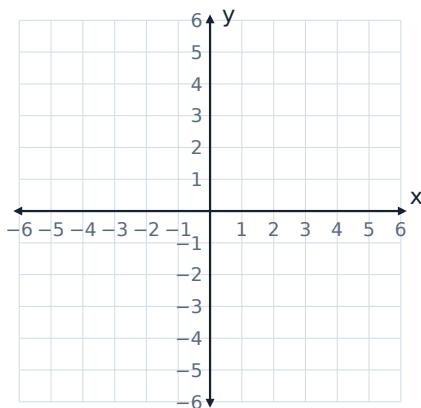
ID S39-M-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q4/5

A hiker starts at $(-5, 0)$, walks to $(3, 0)$, and then walks to $(3, 6)$. Each unit is 1 km.

Graph the path. How far does the hiker walk?

Distance: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q5/5

1437

QUESTION ID

S39-M-Q5

TYPE

Short answer

CORRECT ANSWER

14 km

GRADING NOTE Path graphed from $(-5, 0)$ to $(3, 0)$ to $(3, 6)$; $8 \text{ km} + 6 \text{ km}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-M-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 39 • MAIN • Q5/5

SET 39 • 6.NS.C.8

BACKWARD 1

Absolute value as distance • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-4| + |5|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q1/5

1439

QUESTION ID

S39-B1-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q1/5

True or false?

The distance from -3 to 0 is 3 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q2/5

1440

QUESTION ID

S39-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q2/5

How far is -6 from 0 ?

- A.** -6
- B.** 0
- C.** 6
- D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q3/5

1441

QUESTION ID

S39-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q3/5

Find the value.

$$|-2| + |4|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q4/5

1442

QUESTION ID

S39-B1-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q4/5

True or false?

$$|-5| = -5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q5/5

1443

QUESTION ID

S39-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B1-Q5

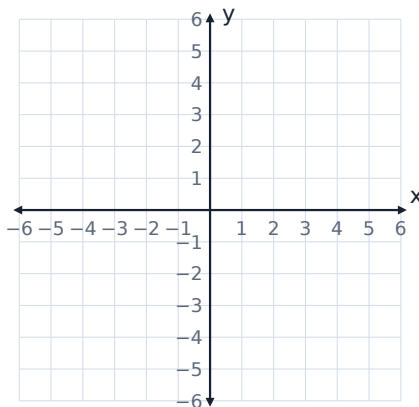
Grade 6 • The Number System • 6.NS.C.7.c • Set 39 • BACKWARD 1 • Q5/5

SET 39 • 6.NS.C.8

BACKWARD 2

Plotting points in all four quadrants • Grade 6 • The Number System •
6.NS.C.6.c

Plot and label point A($-2, 3$).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q1/5

1445

QUESTION ID

S39-B2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

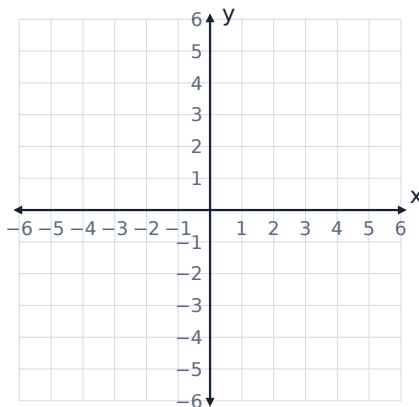
**Point A plotted at
($-2, 3$) and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q1/5

Plot and label point B(4, -1).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q2/5

1446

QUESTION ID

S39-B2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Point B plotted at (4, -1) and labeled

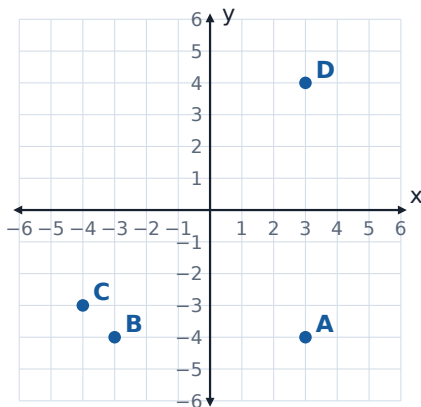
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q2/5

Which point is at $(-3, -4)$?

- A. Point A
- B. Point B
- C. Point C
- D. Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q3/5

1447

QUESTION ID

S39-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

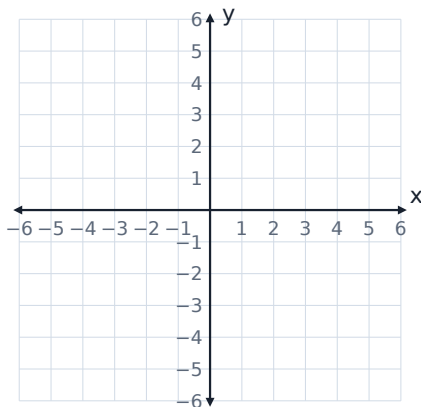
B. Point B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q3/5

Plot and label point C(0, -5).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q4/5

1448

QUESTION ID

S39-B2-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Point C plotted at (0, -5) on the y-axis and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

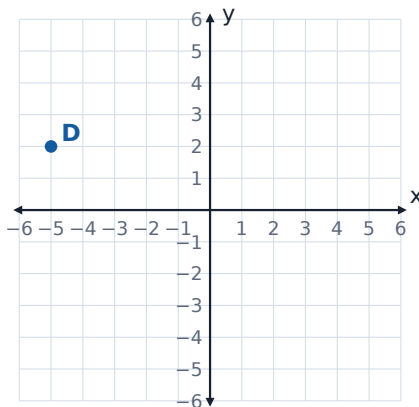
ID S39-B2-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q4/5

True or false?

Point D is located at $(-5, 2)$.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q5/5

1449

QUESTION ID

S39-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B2-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 39 • BACKWARD 2 • Q5/5

SET 39 • 6.NS.C.8

BACKWARD 3

Perimeter of rectangles • Grade 3 • Measurement & Data • 3.MD.D.8

What is the perimeter of the rectangle?

Perimeter: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q1/5

1451

QUESTION ID

S39-B3-Q1

TYPE

Short answer

CORRECT ANSWER

20 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q1/5

A rectangle is 9 cm long and 3 cm wide.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q2/5

1452

QUESTION ID

S39-B3-Q2

TYPE

Short answer

CORRECT ANSWER

24 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q2/5

True or false?

A square with 5-meter sides has a perimeter of 25 meters.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q3/5

1453

QUESTION ID

S39-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q3/5

A rectangle is 8 inches long and 2 inches wide.

What is its perimeter?

- A. 10 inches
- B. 16 inches
- C. 18 inches
- D. 20 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q4/5

1454

QUESTION ID

S39-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 20 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q4/5

A rectangle has a perimeter of 18 feet and a length of 5 feet.

What is its width?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q5/5

1455

QUESTION ID

S39-B3-Q5

TYPE

Short answer

CORRECT ANSWER

4 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B3-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 39 • BACKWARD 3 • Q5/5

SET 39 • 6.NS.C.8

BACKWARD 4

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

A rectangle is 6 m long and 4 m wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q1/5

1457

QUESTION ID

S39-B4-Q1

TYPE

Short answer

CORRECT ANSWER

24 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q1/5

True or false?

A rectangle that is 7 units by 3 units has an area of 21 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q2/5

1458

QUESTION ID

S39-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q2/5

A rectangle is 5 ft by 8 ft.

What is its area?

- A. 13 square feet
- B. 26 square feet
- C. 40 square feet
- D. 80 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q3/5

1459

QUESTION ID

S39-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 40 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q3/5

What is the area of the rectangle?

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q4/5

1460

QUESTION ID

S39-B4-Q4

TYPE

Short answer

CORRECT ANSWER

**18 square
centimeters**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q4/5

A square has 6-inch sides.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q5/5

1461

QUESTION ID

S39-B4-Q5

TYPE

Short answer

CORRECT ANSWER

36 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-B4-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 39 • BACKWARD 4 • Q5/5

SET 39 • 6.NS.C.8

FORWARD 1

Use a map scale on a coordinate grid • Grade 7 • Geometry • 7.G.A.1

On a map grid, each unit is $\frac{1}{2}$ mile. The school is at $(-3, 2)$ and the park is at $(5, 2)$.

What is the actual distance between them?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q1/5

1463

QUESTION ID

S39-F1-Q1

TYPE

Short answer

CORRECT ANSWER

4 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q1

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q1/5

On a trail map, each unit is 250 meters. The trail goes straight from $(0, -4)$ to $(0, 6)$.

How long is the actual trail?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q2/5

1464

QUESTION ID

S39-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2,500 meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q2

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q2/5

On a map, each unit is 1.5 km. A park has corners at $(-2, 1)$, $(4, 1)$, $(4, -3)$, and $(-2, -3)$.

What is the actual perimeter of the park?

- A.** 13.3 km **B.** 20 km
C. 30 km **D.** 36 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q3/5

1465

QUESTION ID

S39-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 30 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q3

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q3/5

On a plan, each unit is 20 feet. A garden has corners at $(-1, -1)$, $(3, -1)$, $(3, 2)$, and $(-1, 2)$.

What are the actual length and width?

Length and width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q4/5

1466

QUESTION ID

S39-F1-Q4

TYPE

Short answer

CORRECT ANSWER

**Length: 80 feet;
Width: 60 feet**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q4

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q4/5

True or false?

On a map where each unit is 5 miles, the points $(-3, 0)$ and $(3, 0)$ are 30 miles apart.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q5/5

1467

QUESTION ID

S39-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F1-Q5

Grade 7 • Geometry • 7.G.A.1 • Set 39 • FORWARD 1 • Q5/5

SET 39 • 6.NS.C.8

FORWARD 2

Straight-line distances on a map • Grade 8 • Geometry • 8.G.B.8

On a map, the library is at $(-2, 1)$ and the pool is at $(4, 9)$. Each unit is 1 block.

What is the straight-line distance between them?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q1/5

1469

QUESTION ID

S39-F2-Q1

TYPE

Short answer

CORRECT ANSWER

10 blocks

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q1/5

A drone flies from $(0, 0)$ to $(9, 12)$. Each unit is 1 meter.

How far does it fly in a straight line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q2/5

1470

QUESTION ID

S39-F2-Q2

TYPE

Short answer

CORRECT ANSWER

15 meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q2/5

Home is at $(-3, -1)$ and school is at $(1, 2)$. Each unit is 1 km.

What is the straight-line distance?

- A.** 5 km **B.** 25 km
C. 7 km **D.** $\sqrt{7}$ km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q3/5

1471

QUESTION ID

S39-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 5 km

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q3/5

Find the straight-line distance from $(-4, 6)$ to $(2, -2)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q4/5

1472

QUESTION ID

S39-F2-Q4

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q4/5

True or false?

The straight-line distance from (1, 1) to (7, 9) is 10.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q5/5

1473

QUESTION ID

S39-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S39-F2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 39 • FORWARD 2 • Q5/5

Evaluate numerical expressions with whole-number exponents

MAIN	Grade 6 main question	6.EE.A.1
BACKWARD 1	Powers of 10 with exponents	5.NBT.A.2
BACKWARD 2	Order of operations with parentheses	5.OA.A.1
BACKWARD 3	Repeated multiplication	3.OA.C.7
FORWARD 1	Products of signed numbers	7.NS.A.2.a
FORWARD 2	Evaluate expressions with integer exponents	8.EE.A.1

SET 40 • 6.EE.A.1

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$2^4 + 3^2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q1/5

1476

QUESTION ID

S40-M-Q1

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q1/5

Evaluate.

$$3 \times 2^3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q2/5

1477

QUESTION ID

S40-M-Q2

TYPE

Short answer

CORRECT ANSWER

24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q2/5

Evaluate.

$$(5 - 2)^3 \div 9$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q3/5

1478

QUESTION ID

S40-M-Q3

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q3/5

Evaluate.

$$5^2 - 4 \times 2$$

- A. -3
- B. 17
- C. 21
- D. 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q4/5

1479

QUESTION ID

S40-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q4/5

Evaluate.

$$\left(\frac{2}{3}\right)^2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q5/5

1480

QUESTION ID

S40-M-Q5

TYPE

Short answer

CORRECT ANSWER

4/9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 40 • MAIN • Q5/5

SET 40 • 6.EE.A.1

BACKWARD 1

Powers of 10 with exponents • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Write 10^3 as a whole number.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q1/5

1482

QUESTION ID

S40-B1-Q1

TYPE

Short answer

CORRECT ANSWER

1,000

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q1/5

True or false?

$$10^4 = 10,000$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q2/5

1483

QUESTION ID

S40-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q2/5

Which is equal to 100,000?

- A. 10^5
- B. 5^{10}
- C. 10^4
- D. 10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q3/5

1484

QUESTION ID

S40-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 10^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q3/5

Write 1,000,000 as a power of 10.

Power of 10: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q4/5

1485

QUESTION ID

S40-B1-Q4

TYPE

Short answer

CORRECT ANSWER

10^6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q4/5

True or false?

$$10^2 = 20$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q5/5

1486

QUESTION ID

S40-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 40 • BACKWARD 1 • Q5/5

SET 40 • 6.EE.A.1

BACKWARD 2

Order of operations with parentheses • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$(8 + 4) \div 3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q1/5

1488

QUESTION ID

S40-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q1/5

Evaluate.

$$20 - (3 \times 4)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q2/5

1489

QUESTION ID

S40-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q2/5

True or false?

$$2 \times (3 + 5) = 11$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q3/5

1490

QUESTION ID

S40-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q3/5

Evaluate.

$$18 \div (2 + 4)$$

- A.** 3
- B.** 6
- C.** 9
- D.** 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q4/5

1491

QUESTION ID

S40-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q4/5

Evaluate.

$$(9 - 3) \times 5$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q5/5

1492

QUESTION ID

S40-B2-Q5

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 40 • BACKWARD 2 • Q5/5

SET 40 • 6.EE.A.1

BACKWARD 3

Repeated multiplication • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$2 \times 2 \times 2$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q1/5

1494

QUESTION ID

S40-B3-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q1/5

1494

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$5 \times 5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q2/5

1495

QUESTION ID

S40-B3-Q2

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q2/5

1495

True or false?

$$3 \times 3 \times 3 = 9$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q3/5

1496

QUESTION ID

S40-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q3/5

$$4 \times 4 \times 4 = ?$$

- A.** 12
- B.** 16
- C.** 44
- D.** 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q4/5

1497

QUESTION ID

S40-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$6 \times 6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q5/5

1498

QUESTION ID

S40-B3-Q5

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 40 • BACKWARD 3 • Q5/5

1498

SET 40 • 6.EE.A.1

FORWARD 1

Products of signed numbers • Grade 7 • The Number System • 7.NS.A.2.a

Evaluate.

$$(-3)^2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q1/5

1500

QUESTION ID

S40-F1-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Evaluate.

$$(-2)^3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q2/5

1501

QUESTION ID

S40-F1-Q2

TYPE

Short answer

CORRECT ANSWER

-8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q2/5

1501

True or false?

$$-3^2 = (-3)^2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q3/5

1502

QUESTION ID

S40-F1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q3/5

$$(-1)^5 = ?$$

- A.** -5
- B.** -1
- C.** 1
- D.** 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q4/5

1503

QUESTION ID

S40-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q4/5

Evaluate.

$$\left(-\frac{1}{2}\right)^2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q5/5

1504

QUESTION ID

S40-F1-Q5

TYPE

Short answer

CORRECT ANSWER

1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 40 • FORWARD 1 • Q5/5

SET 40 • 6.EE.A.1

FORWARD 2

Evaluate expressions with integer exponents • Grade 8 • Expressions & Equations • 8.EE.A.1

Evaluate.

$$2^{-3}$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q1/5

1506

QUESTION ID

S40-F2-Q1

TYPE

Short answer

CORRECT ANSWER

1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q1/5

Evaluate.

$$5^{-2} \times 5^4$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q2/5

1507

QUESTION ID

S40-F2-Q2

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q2/5

$$10^{-2} = ?$$

- A.** -100
- B.** -20
- C.** $\frac{1}{100}$
- D.** 0.1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q3/5

1508

QUESTION ID

S40-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 1/100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q3/5

Evaluate.

$$\frac{4^5}{4^3}$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q4/5

1509

QUESTION ID

S40-F2-Q4

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q4/5

True or false?

$$7^0 = 1$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q5/5

1510

QUESTION ID

S40-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S40-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 40 • FORWARD 2 • Q5/5

Write numerical expressions with whole-number exponents

MAIN	Grade 6 main question	6.EE.A.1
BACKWARD 1	Powers of 10 with exponents	5.NBT.A.2
BACKWARD 2	Multiplying with repeated factors	3.OA.C.7
FORWARD 1	Powers of negative numbers	7.NS.A.2.a
FORWARD 2	Write expressions as a single power	8.EE.A.1

SET 41 • 6.EE.A.1

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.1

Write using an exponent.

$$7 \times 7 \times 7 \times 7$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q1/5

1513

QUESTION ID

S41-M-Q1

TYPE

Short answer

CORRECT ANSWER

7⁴

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q1/5

Which expression is equal to $3 \times 3 \times 3 \times 3 \times 3$?

- A. 5^3
- B. 15
- C. 3^5
- D. 3×5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q2/5

1514

QUESTION ID

S41-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 3^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q2/5

Write using an exponent.

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q3/5

1515

QUESTION ID

S41-M-Q3

TYPE

Short answer

CORRECT ANSWER

$(1/2)^3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q3/5

Write using an exponent.

$$0.4 \times 0.4$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q4/5

1516

QUESTION ID

S41-M-Q4

TYPE

Short answer

CORRECT ANSWER

0.4^2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q4/5

Write using exponents.

$$2 \times 2 \times 2 \times 5 \times 5$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q5/5

1517

QUESTION ID

S41-M-Q5

TYPE

Short answer

CORRECT ANSWER

$$2^3 \times 5^2$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 41 • MAIN • Q5/5

SET 41 • 6.EE.A.1

BACKWARD 1

Powers of 10 with exponents • Grade 5 • Number & Operations in Base Ten •
5.NBT.A.2

Write $10 \times 10 \times 10$ as a power of 10.

Power of 10: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q1/5

1519

QUESTION ID

S41-B1-Q1

TYPE

Short answer

CORRECT ANSWER

10^3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q1/5

True or false?

$$10 \times 10 \times 10 \times 10 = 10^4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q2/5

1520

QUESTION ID

S41-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q2/5

Which is equal to 10^5 ?

- A. $5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5 \times 5$
- B. $10 \times 10 \times 10 \times 10 \times 10$
- C. 100,000,000
- D. 10×5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q3/5

1521

QUESTION ID

S41-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B.

$10 \times 10 \times 10 \times 10 \times 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q3/5

How many factors of 10 are in 10^6 ?

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q4/5

1522

QUESTION ID

S41-B1-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q4/5

True or false?

$$10^2 = 10 \times 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q5/5

1523

QUESTION ID

S41-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B1-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.A.2 • Set 41 • BACKWARD 1 • Q5/5

SET 41 • 6.EE.A.1

BACKWARD 2

Multiplying with repeated factors • Grade 3 • Operations & Algebraic Thinking •
3.OA.C.7

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$2 \times 2 \times 2 \times 2$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q1/5

1525

QUESTION ID

S41-B2-Q1

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q1/5

1525

Multiply.

$$3 \times 3$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q2/5

1526

QUESTION ID

S41-B2-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q2/5

True or false?

$$5 \times 5 \times 5 = 15$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q3/5

1527

QUESTION ID

S41-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q3/5

How many times is 7 used as a factor in $7 \times 7 \times 7$?

- A. 1
- B. 3
- C. 7
- D. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q4/5

1528

QUESTION ID

S41-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$4 \times 4$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q5/5

1529

QUESTION ID

S41-B2-Q5

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.C.7 • Set 41 • BACKWARD 2 • Q5/5

1529

SET 41 • 6.EE.A.1

FORWARD 1

Powers of negative numbers • Grade 7 • The Number System • 7.NS.A.2.a

Write using an exponent.

$$(-3)(-3)(-3)(-3)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q1/5

1531

QUESTION ID

S41-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$(-3)^4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q1

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q1/5

Write $(-2)(-2)(-2)$ using an exponent. Is its value positive or negative?

Expression: _____

Sign: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q2/5

1532

QUESTION ID

S41-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(-2)^3$; negative

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q2

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q2/5

Which expression is equal to $(-5)^2$?

- A. -5×2
- B. $(-5)(-5)$
- C. -5×5
- D. $-(5 \times 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q3/5

1533

QUESTION ID

S41-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $(-5)(-5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q3

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q3/5

True or false?

$(-1)^6$ is positive.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q4/5

1534

QUESTION ID

S41-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q4

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q4/5

Write using an exponent.

$$\left(-\frac{2}{3}\right)\left(-\frac{2}{3}\right)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q5/5

1535

QUESTION ID

S41-F1-Q5

TYPE

Short answer

CORRECT ANSWER

$$\left(-\frac{2}{3}\right)^2$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F1-Q5

Grade 7 • The Number System • 7.NS.A.2.a • Set 41 • FORWARD 1 • Q5/5

SET 41 • 6.EE.A.1

FORWARD 2

Write expressions as a single power • Grade 8 • Expressions & Equations •
8.EE.A.1

Write as a single power.

$$4^3 \times 4^5$$

Single power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q1/5

1537

QUESTION ID

S41-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4⁸

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q1/5

Write as a single power.

$$(2^3)^4$$

Single power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q2/5

1538

QUESTION ID

S41-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2¹²

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q2/5

Which is equal to $\frac{6^7}{6^2}$?

- A. 6^{14}
- B. 6^5
- C. 6^9
- D. 1^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q3/5

1539

QUESTION ID

S41-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6^5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q3/5

Write as a single power.

$$3^4 \times 3^{-6}$$

Single power: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q4/5

1540

QUESTION ID

S41-F2-Q4

TYPE

Short answer

CORRECT ANSWER

3^{-2}

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q4/5

True or false?

$$5^2 \times 5^3 = 5^6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q5/5

1541

QUESTION ID

S41-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S41-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.A.1 • Set 41 • FORWARD 2 • Q5/5

Write expressions in which letters stand for numbers

MAIN	Grade 6 main question	6.EE.A.2.a
BACKWARD 1	Words for operations	5.OA.A.2
BACKWARD 2	A symbol for an unknown number	3.OA.A.4
BACKWARD 3	Order matters in subtraction and division	3.OA.B.5
FORWARD 1	Write and solve equations of the form $px + q = r$	7.EE.B.4.a
FORWARD 2	Construct a function from a description	8.F.B.4

SET 42 • 6.EE.A.2.a

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.2.a

Write an expression for "subtract m from 12."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q1/5

1544

QUESTION ID

S42-M-Q1

TYPE

Short answer

CORRECT ANSWER

12 – m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q1/5

Write an expression for "5 more than twice a number n ."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q2/5

1545

QUESTION ID

S42-M-Q2

TYPE

Short answer

CORRECT ANSWER

$2n + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q2/5

Which expression means "the quotient of x and 4"?

A. $4x$

B. $\frac{4}{x}$

C. $x - 4$

D. $\frac{x}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q3/5

1546

QUESTION ID

S42-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $x/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q3/5

Write an expression for "3 less than the product of 7 and a number m ."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q4/5

1547

QUESTION ID

S42-M-Q4

TYPE

Short answer

CORRECT ANSWER

$7m - 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q4/5

True or false?

"8 less than k" can be written as $8 - k$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q5/5

1548

QUESTION ID

S42-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.a • Set 42 • MAIN • Q5/5

SET 42 • 6.EE.A.2.a

BACKWARD 1

Words for operations • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

Which expression shows "add 6 and 9, then multiply by 4"?

- A. $6 + 9 \times 4$
- B. $4 \times (6 + 9)$
- C. $(4 + 6) \times 9$
- D. $4 \times 6 + 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q1/5

1550

QUESTION ID

S42-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. $4 \times (6 + 9)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q1/5

Write an expression for "subtract 4 from 10, then divide by 2."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q2/5

1551

QUESTION ID

S42-B1-Q2

TYPE

Short answer

CORRECT ANSWER

$(10 - 4) \div 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q2/5

True or false?

"The product of 6 and 9" means 6×9 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q3/5

1552

QUESTION ID

S42-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q3/5

Which word describes the answer to a subtraction problem?

- A.** Quotient
- B.** Difference
- C.** Product
- D.** Sum

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q4/5

1553

QUESTION ID

S42-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Difference

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q4/5

Write an expression for "the sum of 12 and 5, multiplied by 3."

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q5/5

1554

QUESTION ID

S42-B1-Q5

TYPE

Short answer

CORRECT ANSWER

$(12 + 5) \times 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 42 • BACKWARD 1 • Q5/5

SET 42 • 6.EE.A.2.a

BACKWARD 2

A symbol for an unknown number • Grade 3 • Operations & Algebraic Thinking
• 3.OA.A.4

What number goes in the box?

$$\square + 7 = 12$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q1/5

1556

QUESTION ID

S42-B2-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q1/5

True or false?

In $7 \times \square = 42$, the box stands for 6.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q2/5

1557

QUESTION ID

S42-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q2/5

$$\square - 5 = 9$$

What number goes in the box?

- A. 4
- B. 9
- C. 14
- D. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q3/5

1558

QUESTION ID

S42-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q3/5

What number goes in the box?

$$20 \div \square = 4$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q4/5

1559

QUESTION ID

S42-B2-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q4/5

What number goes in the box?

$$\square \times 3 = 27$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q5/5

1560

QUESTION ID

S42-B2-Q5

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.4 • Set 42 • BACKWARD 2 • Q5/5

SET 42 • 6.EE.A.2.a

BACKWARD 3

Order matters in subtraction and division • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$7 - 3 = 3 - 7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q1/5

1562

QUESTION ID

S42-B3-Q1

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q1/5

True or false?

$$6 \times 4 = 4 \times 6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q2/5

1563

QUESTION ID

S42-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q2/5

True or false?

$$12 \div 3 = 3 \div 12$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q3/5

1564

QUESTION ID

S42-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q3/5

Which statement is true?

- A. $10 \div 2 = 2 \div 10$
- B. $5 + 9 = 9 + 5$
- C. $8 - 2 = 2 - 8$
- D. $9 - 5 = 5 - 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q4/5

1565

QUESTION ID

S42-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $5 + 9 = 9 + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q4/5

True or false?

$$3 + 11 = 11 + 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q5/5

1566

QUESTION ID

S42-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 42 • BACKWARD 3 • Q5/5

SET 42 • 6.EE.A.2.a

FORWARD 1

Write and solve equations of the form $px + q = r$ • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A rectangle has a perimeter of 46 inches. Its width is 8 inches.

Write and solve an equation to find its length.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q1/5

1568

QUESTION ID

S42-F1-Q1

TYPE

Short answer

CORRECT ANSWER

**$2l + 2(8) = 46; l = 15$
inches**

GRADING NOTE Any equivalent equation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q1/5

Jo has \$40. She buys 3 shirts that cost the same amount and has \$4 left.

How much does each shirt cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q2/5

1569

QUESTION ID

S42-F1-Q2

TYPE

Short answer

CORRECT ANSWER

\$12

GRADING NOTE $3s + 4 = 40$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q2/5

Which equation represents "5 more than 3 times a number is 26"?

- A. $3(n + 5) = 26$
- B. $5n + 3 = 26$
- C. $3n + 5 = 26$
- D. $3n - 5 = 26$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q3/5

1570

QUESTION ID

S42-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $3n + 5 = 26$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q3/5

Twice a number minus 7 is 15.

Write and solve an equation to find the number.

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q4/5

1571

QUESTION ID

S42-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$2n - 7 = 15; n = 11$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q4/5

A phone plan costs \$30 per month plus \$0.10 per text.
One month's bill is \$42.

How many texts were sent?

Texts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q5/5

1572

QUESTION ID

S42-F1-Q5

TYPE

Short answer

CORRECT ANSWER

120 texts

GRADING NOTE $30 + 0.10t = 42$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 42 • FORWARD 1 • Q5/5

SET 42 • 6.EE.A.2.a

FORWARD 2

Construct a function from a description • Grade 8 • Functions • 8.F.B.4

A pool has 500 gallons of water. It is filled at 25 gallons per minute.

Write a function for the gallons G after t minutes.

$G =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q1/5

1574

QUESTION ID

S42-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$G = 500 + 25t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q1/5

Tickets cost \$4 each plus a \$6 fee for the order.

Which function gives the total cost y for x tickets?

- A. $y = 10x$
- B. $y = 4x + 6$
- C. $y = 6x + 4$
- D. $y = 4x - 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q2/5

1575

QUESTION ID

S42-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. $y = 4x + 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q2/5

The table shows a linear function.

Write an equation for the function.

x	0	1	2
y	7	10	13

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q3/5

1576

QUESTION ID

S42-F2-Q3

TYPE

Short answer

CORRECT ANSWER

$$y = 3x + 7$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q3/5

A line passes through $(0, -2)$ and $(3, 4)$.

Write an equation for the line.

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q4/5

1577

QUESTION ID

S42-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$y = 2x - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q4/5

True or false?

The function $y = 15x + 20$ has an initial value of 15.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q5/5

1578

QUESTION ID

S42-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S42-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 42 • FORWARD 2 • Q5/5

Identify parts of an expression: term, sum, product, factor, quotient, coefficient

MAIN	Grade 6 main question	6.EE.A.2.b
BACKWARD 1	Sum, difference, product, and quotient	5.OA.A.2
BACKWARD 2	Factors	4.OA.B.4
BACKWARD 3	Interpreting expressions without evaluating	5.OA.A.2
FORWARD 1	Combine like terms with rational coefficients	7.EE.A.1
FORWARD 2	Interpret the parts of a linear function	8.F.B.4

SET 43 • 6.EE.A.2.b

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.2.b

In the expression $4(9 + 3)$, how can the 4 be described?

- A.** A sum
- B.** A factor
- C.** A term of $9 + 3$
- D.** A quotient

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q1/5

1581

QUESTION ID

S43-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. A factor

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q1/5

How many terms are in the expression $3x + 5y - 2$?

Terms: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q2/5

1582

QUESTION ID

S43-M-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q2/5

What is the coefficient of y in $4x + 9y$?

Coefficient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q3/5

1583

QUESTION ID

S43-M-Q3

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q3/5

True or false?

In $5(m + n)$, the quantity $(m + n)$ is a single factor and also a sum of two terms.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q4/5

1584

QUESTION ID

S43-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q4/5

In the expression $7m + 3$, which part is the constant term?

- A. 3
- B. $7m$
- C. m
- D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q5/5

1585

QUESTION ID

S43-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.b • Set 43 • MAIN • Q5/5

SET 43 • 6.EE.A.2.b

BACKWARD 1

Sum, difference, product, and quotient • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

True or false?

8 + 7 is a sum.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q1/5

1587

QUESTION ID

S43-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q1/5

What is 6×9 called?

- A.** A quotient
- B.** A product
- C.** A sum
- D.** A difference

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q2/5

1588

QUESTION ID

S43-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. A product

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q2/5

ANSWER KEY

Reduced copy of the question page, with its answer

Is $24 \div 6$ a sum, difference, product, or quotient?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q3/5

1589

QUESTION ID

S43-B1-Q3

TYPE

Short answer

CORRECT ANSWER

Quotient

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q3/5

1589

True or false?

15 – 4 is called a product.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q4/5

1590

QUESTION ID

S43-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q4/5

Which expression is a quotient?

- A. 20×5
- B. $20 + 5$
- C. $20 - 5$
- D. $20 \div 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q5/5

1591

QUESTION ID

S43-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. $20 \div 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B1-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 1 • Q5/5

SET 43 • 6.EE.A.2.b

BACKWARD 2

Factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

True or false?

In $4 \times 7 = 28$, the numbers 4 and 7 are factors of 28.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q1/5

1593

QUESTION ID

S43-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q1/5

Name the two factors in 5×9 .

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q2/5

1594

QUESTION ID

S43-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5 and 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q2/5

Which number is NOT a factor of 20?

- A.** 4
- B.** 5
- C.** 6
- D.** 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q3/5

1595

QUESTION ID

S43-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q3/5

True or false?

3 is a factor of 12.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q4/5

1596

QUESTION ID

S43-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q4/5

Write one factor pair of 18.

Factor pair: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q5/5

1597

QUESTION ID

S43-B2-Q5

TYPE

Short answer

CORRECT ANSWER

1 and 18

GRADING NOTE Also correct: 2 and 9, or 3 and 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 43 • BACKWARD 2 • Q5/5

SET 43 • 6.EE.A.2.b

BACKWARD 3

Interpreting expressions without evaluating • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2

True or false?

$4 \times (2,365 + 718)$ is four times as large as $2,365 + 718$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q1/5

1599

QUESTION ID

S43-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q1/5

How does $5 \times (26 + 4)$ compare to $26 + 4$?

- A. It is the same.
- B. It is 5 times as large.
- C. It is 5 more.
- D. It is 5 less.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q2/5

1600

QUESTION ID

S43-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. It is 5 times as large.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q2/5

True or false?

$(9 + 4) \times 2$ is 2 more than $9 + 4$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q3/5

1601

QUESTION ID

S43-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q3/5

Which expression is 4 times as large as $125 - 25$?

- A. $(125 - 25) \div 4$
- B. $4 \times (125 - 25)$
- C. $125 - 25 \times 4$
- D. $4 + (125 - 25)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q4/5

1602

QUESTION ID

S43-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $4 \times (125 - 25)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q4/5

Without calculating, how many times as large is
 $10 \times (63 + 7)$ as $63 + 7$?

Times as large: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q5/5

1603

QUESTION ID

S43-B3-Q5

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-B3-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.2 • Set 43 • BACKWARD 3 • Q5/5

SET 43 • 6.EE.A.2.b

FORWARD 1

Combine like terms with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

ANSWER KEY

Reduced copy of the question page, with its answer

Simplify.

$$4x + 3 - 2x + 5$$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q1/5

1605

QUESTION ID

S43-F1-Q1

TYPE

Short answer

CORRECT ANSWER

 $2x + 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q1/5

1605

ANSWER KEY

Reduced copy of the question page, with its answer

Simplify.

$$0.5y + 1.25y$$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q2/5

1606

QUESTION ID

S43-F1-Q2

TYPE

Short answer

CORRECT ANSWER

1.75y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q2/5

1606

Which pair shows like terms?

- A.** $3a$ and $-7a$
- B.** x^2 and x
- C.** $3a$ and $3b$
- D.** $4x$ and 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q3/5

1607

QUESTION ID

S43-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $3a$ and $-7a$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q3/5

Simplify.

$$-3(x - 2) + 5x$$

Simplified: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q4/5

1608

QUESTION ID

S43-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$2x + 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q4/5

True or false?

$$\frac{1}{2}x + \frac{1}{4}x = \frac{3}{4}x$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q5/5

1609

QUESTION ID

S43-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 43 • FORWARD 1 • Q5/5

SET 43 • 6.EE.A.2.b

FORWARD 2

Interpret the parts of a linear function • Grade 8 • Functions • 8.F.B.4

The cost to bowl is $C = 4.5n + 12$, where n is the number of games.

What does 12 represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q1/5

1611

QUESTION ID

S43-F2-Q1

TYPE

Short answer

CORRECT ANSWER

The fixed \$12 cost that is paid no matter how many games are bowled (for example, a shoe rental or entry fee)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q1/5

In $y = -3x + 7$, what is the rate of change?

- A. x
- B. 3
- C. -3
- D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q2/5

1612

QUESTION ID

S43-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q2/5

The gas left in a car's tank is $G = 14 - 0.04m$, where m is miles driven.

What is the initial value?

Initial value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q3/5

1613

QUESTION ID

S43-F2-Q3

TYPE

Short answer

CORRECT ANSWER

14 gallons

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q3/5

A car's distance from home is $d = 60t + 25$, where t is hours.

What does 60 represent?

- A.** The speed in miles per hour **B.** The starting distance
C. The total distance **D.** The total time

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q4/5

1614

QUESTION ID

S43-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. The speed in miles per hour

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q4/5

True or false?

In $y = 8x$, the initial value is 0.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q5/5

1615

QUESTION ID

S43-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S43-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 43 • FORWARD 2 • Q5/5

Evaluate expressions and formulas for given values

MAIN	Grade 6 main question	6.EE.A.2.c
BACKWARD 1	Whole-number exponents	6.EE.A.1
BACKWARD 2	Order of operations with grouping symbols	5.OA.A.1
BACKWARD 3	A letter stands for a number	3.OA.D.8
BACKWARD 4	Multiplying fractions	5.NF.B.4.a
FORWARD 1	Evaluate expressions with rational numbers	7.NS.A.3
FORWARD 2	Volume formulas for cylinders, cones, and spheres	8.G.C.9

SET 44 • 6.EE.A.2.c

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $3x^2 + 2$ when $x = 4$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q1/5

1618

QUESTION ID

S44-M-Q1

TYPE

Short answer

CORRECT ANSWER

50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q1/5

The volume of a cube is $V = s^3$.

Find V when $s = \frac{2}{3}$ foot.

$V =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q2/5

1619

QUESTION ID

S44-M-Q2

TYPE

Short answer

CORRECT ANSWER

8/27 cubic foot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q2/5

The perimeter of a rectangle is $P = 2l + 2w$.

Find P when $l = 7.5$ cm and $w = 4$ cm.

$P =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q3/5

1620

QUESTION ID

S44-M-Q3

TYPE

Short answer

CORRECT ANSWER

23 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q3/5

Evaluate $2(a + b)$ when $a = 3.5$ and $b = 4$.

- A.** 7.5
- B.** 9.5
- C.** 11.5
- D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q4/5

1621

QUESTION ID

S44-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q4/5

Evaluate $\frac{x}{4} + 5y$ when $x = 12$ and $y = 2$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q5/5

1622

QUESTION ID

S44-M-Q5

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 44 • MAIN • Q5/5

SET 44 • 6.EE.A.2.c

BACKWARD 1

Whole-number exponents • Grade 6 • Expressions & Equations • 6.EE.A.1

Evaluate.

$$4^2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q1/5

1624

QUESTION ID

S44-B1-Q1

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q1/5

Evaluate.

$$\left(\frac{1}{2}\right)^3$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q2/5

1625

QUESTION ID

S44-B1-Q2

TYPE

Short answer

CORRECT ANSWER

1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q2/5

True or false?

$$3^2 = 6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q3/5

1626

QUESTION ID

S44-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q3/5

$$2^5 = ?$$

- A.** 10
- B.** 16
- C.** 25
- D.** 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q4/5

1627

QUESTION ID

S44-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q4/5

Evaluate.

$$5^2 + 1$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q5/5

1628

QUESTION ID

S44-B1-Q5

TYPE

Short answer

CORRECT ANSWER

26

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.1 • Set 44 • BACKWARD 1 • Q5/5

SET 44 • 6.EE.A.2.c

BACKWARD 2

Order of operations with grouping symbols • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$3 \times (16 + 2)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q1/5

1630

QUESTION ID

S44-B2-Q1

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Evaluate.

$$2 \times (6 + 4)$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q2/5

1631

QUESTION ID

S44-B2-Q2

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q2/5

1631

True or false?

$$(3 + 4) \times 2 = 11$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q3/5

1632

QUESTION ID

S44-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q3/5

Evaluate.

$$(12 \div 4) + 10$$

- A.** 1
- B.** 6
- C.** 13
- D.** 22

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q4/5

1633

QUESTION ID

S44-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Evaluate.

$$[2 \times (5 + 3)] - 4$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q5/5

1634

QUESTION ID

S44-B2-Q5

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1 • Set 44 • BACKWARD 2 • Q5/5

1634

SET 44 • 6.EE.A.2.c

BACKWARD 3

A letter stands for a number • Grade 3 • Operations & Algebraic Thinking •
3.OA.D.8

If $n = 6$, what is $n + 9$?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q1/5

1636

QUESTION ID

S44-B3-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q1/5

True or false?

If $a = 4$, then $a \times 5 = 20$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q2/5

1637

QUESTION ID

S44-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q2/5

If $k = 7$, what is $3 \times k$?

- A.** 10
- B.** 21
- C.** 37
- D.** 73

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q3/5

1638

QUESTION ID

S44-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q3/5

If $m = 20$, what is $m - 8$?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q4/5

1639

QUESTION ID

S44-B3-Q4

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q4/5

If $t = 3$, what is $4 \times t + 2$?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q5/5

1640

QUESTION ID

S44-B3-Q5

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 44 • BACKWARD 3 • Q5/5

SET 44 • 6.EE.A.2.c

BACKWARD 4

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{2}{3} \times \frac{2}{3} \times \frac{2}{3}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q1/5

1642

QUESTION ID

S44-B4-Q1

TYPE

Short answer

CORRECT ANSWER

8/27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q1/5

Multiply.

$$6 \times \frac{1}{3}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q2/5

1643

QUESTION ID

S44-B4-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q2/5

True or false?

$$\frac{2}{3} \times \frac{2}{3} = \frac{4}{6}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q3/5

1644

QUESTION ID

S44-B4-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q3/5

$$\frac{3}{4} \times 8 = ?$$

- A. $\frac{3}{32}$
- B. 2
- C. 6
- D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q4/5

1645

QUESTION ID

S44-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q4/5

Multiply.

$$\frac{1}{2} \times 12$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q5/5

1646

QUESTION ID

S44-B4-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-B4-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 44 • BACKWARD 4 • Q5/5

SET 44 • 6.EE.A.2.c

FORWARD 1

Evaluate expressions with rational numbers • Grade 7 • The Number System •
7.NS.A.3

Use $C = \frac{5}{9}(F - 32)$ to find C when $F = -4$.

$C =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q1/5

1648

QUESTION ID

S44-F1-Q1

TYPE

Short answer

CORRECT ANSWER

−20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q1

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q1/5

Evaluate $-2x + 5$ when $x = -3.5$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q2/5

1649

QUESTION ID

S44-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q2

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q2/5

Evaluate $a^2 - b$ when $a = -3$ and $b = 4$.

- A.** -13
- B.** -5
- C.** 5
- D.** 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q3/5

1650

QUESTION ID

S44-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q3

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q3/5

Evaluate $\frac{(x+y)}{2}$ when $x = -6.5$ and $y = 2.5$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q4/5

1651

QUESTION ID

S44-F1-Q4

TYPE

Short answer

CORRECT ANSWER

-2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q4

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q4/5

True or false?

When $x = -2$, the value of $x^3 + 1$ is -7 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q5/5

1652

QUESTION ID

S44-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F1-Q5

Grade 7 • The Number System • 7.NS.A.3 • Set 44 • FORWARD 1 • Q5/5

SET 44 • 6.EE.A.2.c

FORWARD 2

Volume formulas for cylinders, cones, and spheres • Grade 8 • Geometry •
8.G.C.9

A cylinder has a radius of 3 cm and a height of 10 cm.

Find its volume. Use 3.14 for π .

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q1/5

1654

QUESTION ID

S44-F2-Q1

TYPE

Short answer

CORRECT ANSWER

282.6 cubic cm

GRADING NOTE 90π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

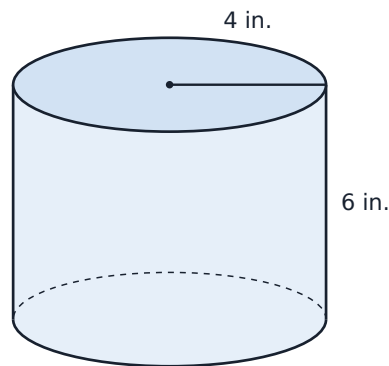
ID S44-F2-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q1/5

Find the volume of the cylinder.

Use 3.14 for π .

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q2/5

1655

QUESTION ID

S44-F2-Q2

TYPE

Short answer

CORRECT ANSWER

301.44 cubic inches

GRADING NOTE 96π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q2/5

A cone has a radius of 3 units and a height of 4 units.

What is its volume in terms of π ?

- A. 36π
- B. 48π
- C. 4π
- D. 12π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q3/5

1656

QUESTION ID

S44-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 12π

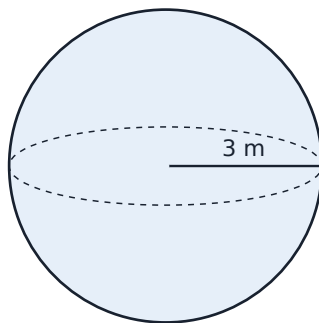
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q3/5

Find the volume of the sphere
in terms of π .

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q4/5

1657

QUESTION ID

S44-F2-Q4

TYPE

Short answer

CORRECT ANSWER

36π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q4/5

True or false?

A cylinder with a radius of 2 and a height of 5 has a volume of 20π .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q5/5

1658

QUESTION ID

S44-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S44-F2-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 44 • FORWARD 2 • Q5/5

Use the distributive property to expand expressions

MAIN	Grade 6 main question	6.EE.A.3
BACKWARD 1	Distributive property with numbers	3.OA.B.5
BACKWARD 2	Equal groups	3.OA.A.1
BACKWARD 3	Multiplying a decimal by a whole number	5.NBT.B.7
FORWARD 1	Expand linear expressions with rational coefficients	7.EE.A.1
FORWARD 2	Solve equations by expanding and collecting like terms	8.EE.C.7.b

SET 45 • 6.EE.A.3

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.3

Expand.

$$3(4 + x)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q1/5

1661

QUESTION ID

S45-M-Q1

TYPE

Short answer

CORRECT ANSWER

12 + 3x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Expand.

$$5(3a - 2)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q2/5

1662

QUESTION ID

S45-M-Q2

TYPE

Short answer

CORRECT ANSWER

15a – 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q2/5

1662

Which expression is equivalent to $7(2m + 5)$?

- A. $9m + 35$
- B. $2m + 35$
- C. $14m + 5$
- D. $14m + 35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q3/5

1663

QUESTION ID

S45-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $14m + 35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q3/5

Expand.

$$2(x + 3y + 4)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q4/5

1664

QUESTION ID

S45-M-Q4

TYPE

Short answer

CORRECT ANSWER

$$2x + 6y + 8$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Expand.

$$4(0.5y + 3)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q5/5

1665

QUESTION ID

S45-M-Q5

TYPE

Short answer

CORRECT ANSWER

 $2y + 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 45 • MAIN • Q5/5

1665

SET 45 • 6.EE.A.3

BACKWARD 1

Distributive property with numbers • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$6 \times (4 + 3) = 6 \times 4 + 6 \times 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 45 • BACKWARD 1 • Q1/5

1667

QUESTION ID

S45-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 45 • BACKWARD 1 • Q1/5

$$5 \times (2 + 7) = 5 \times 2 + 5 \times \underline{\quad}$$

Blank: _____

ID S45-B1-Q2

1668

S45-B1-Q2

Short answer

7

ID S45-B1-Q2

1668

Which is equal to 3×14 ?

- A. $30 + 4 + 3$
- B. $3 \times 10 + 4$
- C. $3 \times 10 + 3 \times 4$
- D. $3 + 10 \times 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 45 • BACKWARD 1 • Q3/5

1669

QUESTION ID

S45-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $3 \times 10 + 3 \times 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 45 • BACKWARD 1 • Q3/5

True or false?

$$7 \times 12 = 7 \times 10 + 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 45 • BACKWARD 1 • Q4/5

1670

QUESTION ID

S45-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 45 • BACKWARD 1 • Q4/5

Blank: _____

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 45 • BACKWARD 1 • Q5/5

1671

8 × 1

GRADING NOTE "8" is also correct

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 45 • BACKWARD 1 • Q5/5

SET 45 • 6.EE.A.3

BACKWARD 2

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$3 \times (4 + 2)$ means 3 groups of $(4 + 2)$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q1/5

1673

QUESTION ID

S45-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q1/5

Which means 4 groups of $(5 + 1)$?

- A. $4 + (5 + 1)$
- B. $4 \times (5 + 1)$
- C. $4 \times 5 + 1$
- D. $(4 + 5) \times 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q2/5

1674

QUESTION ID

S45-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. $4 \times (5 + 1)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q2/5

How many groups of $(2 + 6)$ are in $5 \times (2 + 6)$?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q3/5

1675

QUESTION ID

S45-B2-Q3

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q3/5

True or false?

$$2 \times (3 + 4) = (3 + 4) + (3 + 4)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q4/5

1676

QUESTION ID

S45-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q4/5

Write 6×3 as a sum of equal groups.

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q5/5

1677

QUESTION ID

S45-B2-Q5

TYPE

Short answer

CORRECT ANSWER

6 + 6 + 6

GRADING NOTE $3 + 3 + 3 + 3 + 3 + 3$ is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 45 • BACKWARD 2 • Q5/5

SET 45 • 6.EE.A.3

BACKWARD 3

Multiplying a decimal by a whole number • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$4 \times 0.5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q1/5

1679

QUESTION ID

S45-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q1/5

1679

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$3 \times 1.2$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q2/5

1680

QUESTION ID

S45-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q2/5

1680

True or false?

$$6 \times 0.25 = 1.5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q3/5

1681

QUESTION ID

S45-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q3/5

$$5 \times 0.4 = ?$$

- A.** 0.2
- B.** 2
- C.** 5.4
- D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q4/5

1682

QUESTION ID

S45-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Multiply.

$$8 \times 0.75$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q5/5

1683

QUESTION ID

S45-B3-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 45 • BACKWARD 3 • Q5/5

1683

SET 45 • 6.EE.A.3

FORWARD 1

Expand linear expressions with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Expand.

$$\frac{1}{2}(8x - 6)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q1/5

1685

QUESTION ID

S45-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$4x - 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q1/5

Expand.

$$-4(2y - 3)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q2/5

1686

QUESTION ID

S45-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$-8y + 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q2/5

Which is equivalent to $0.5(6a + 10)$?

- A. $3a + 5$
- B. $6a + 5$
- C. $3.5a + 5$
- D. $3a + 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q3/5

1687

QUESTION ID

S45-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $3a + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q3/5

Expand.

$$-\frac{2}{3}(9m + 3)$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q4/5

1688

QUESTION ID

S45-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$-6m - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q4/5

True or false?

$$-2(x - 7) = -2x + 14$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q5/5

1689

QUESTION ID

S45-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 45 • FORWARD 1 • Q5/5

SET 45 • 6.EE.A.3

FORWARD 2

Solve equations by expanding and collecting like terms • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$3(x + 2) = 5x - 4$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q1/5

1691

QUESTION ID

S45-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q1/5

Solve for x.

$$2(4x - 1) - 3x = 18$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q2/5

1692

QUESTION ID

S45-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q2/5

Solve for x.

$$-2(x - 5) = 3(x + 5)$$

- A. -5
- B. -1
- C. 1
- D. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q3/5

1693

QUESTION ID

S45-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q3/5

Solve for x.

$$6(x - 1) = 4(x + 2)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q4/5

1694

QUESTION ID

S45-F2-Q4

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q4/5

Solve for x.

$$5(2x + 1) = 3(3x + 4)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q5/5

1695

QUESTION ID

S45-F2-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S45-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 45 • FORWARD 2 • Q5/5

Use the distributive property to factor expressions

MAIN	Grade 6 main question	6.EE.A.3
BACKWARD 1	Greatest common factor	6.NS.B.4
BACKWARD 2	Distributive property with numbers	3.OA.B.5
BACKWARD 3	Factors	4.OA.B.4
FORWARD 1	Factor linear expressions with rational coefficients	7.EE.A.1
FORWARD 2	Use factoring to decide how many solutions an equation has	8.EE.C.7.a

SET 46 • 6.EE.A.3

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.3

Factor using the greatest common factor.

$$24x + 16$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q1/5

1698

QUESTION ID

S46-M-Q1

TYPE

Short answer

CORRECT ANSWER

8(3x + 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q1/5

Factor using the greatest common factor.

$$18y + 30$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q2/5

1699

QUESTION ID

S46-M-Q2

TYPE

Short answer

CORRECT ANSWER

$6(3y + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q2/5

Which shows $15a + 25b$ factored using the greatest common factor?

- A. $15(a + 10b)$
- B. $5(3a + 5b)$
- C. $3(5a + 25b)$
- D. $5(3a + 25b)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q3/5

1700

QUESTION ID

S46-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $5(3a + 5b)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q3/5

Factor using the greatest common factor.

$$36m + 27n$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q4/5

1701

QUESTION ID

S46-M-Q4

TYPE

Short answer

CORRECT ANSWER

9(4m + 3n)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q4/5

Factor using the greatest common factor.

$$14x + 21$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q5/5

1702

QUESTION ID

S46-M-Q5

TYPE

Short answer

CORRECT ANSWER

$7(2x + 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 46 • MAIN • Q5/5

SET 46 • 6.EE.A.3

BACKWARD 1

Greatest common factor • Grade 6 • The Number System • 6.NS.B.4

Find the greatest common factor of 24 and 16.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q1/5

1704

QUESTION ID

S46-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q1

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q1/5

Find the greatest common factor of 18 and 30.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q2/5

1705

QUESTION ID

S46-B1-Q2

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q2

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q2/5

True or false?

The greatest common factor of 15 and 25 is 5.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q3/5

1706

QUESTION ID

S46-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q3

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q3/5

What is the greatest common factor of 36 and 27?

- A.** 3
- B.** 6
- C.** 9
- D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q4/5

1707

QUESTION ID

S46-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q4

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q4/5

Find the greatest common factor of 14 and 21.

GCF: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q5/5

1708

QUESTION ID

S46-B1-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B1-Q5

Grade 6 • The Number System • 6.NS.B.4 • Set 46 • BACKWARD 1 • Q5/5

SET 46 • 6.EE.A.3

BACKWARD 2

Distributive property with numbers • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$8 \times 3 + 8 \times 2 = 8 \times (3 + 2)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q1/5

1710

QUESTION ID

S46-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q1/5

Fill in the blank.

$$6 \times 4 + 6 \times 5 = 6 \times (4 + \underline{\quad})$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q2/5

1711

QUESTION ID

S46-B2-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q2/5

Which is equal to $7 \times 2 + 7 \times 9$?

- A. $14 + 9$
- B. $7 \times 2 \times 9$
- C. $7 \times (2 + 9)$
- D. $7 + 2 + 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q3/5

1712

QUESTION ID

S46-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $7 \times (2 + 9)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q3/5

True or false?

$$5 \times 6 + 3 \times 6 = 6 \times (5 + 3)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q4/5

1713

QUESTION ID

S46-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q4/5

Fill in the blank.

$$9 \times 7 + 9 \times 3 = \underline{\hspace{1cm}} \times (7 + 3)$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q5/5

1714

QUESTION ID

S46-B2-Q5

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 46 • BACKWARD 2 • Q5/5

SET 46 • 6.EE.A.3

BACKWARD 3

Factors • Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4

List all the factors of 16.

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q1/5

1716

QUESTION ID

S46-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1, 2, 4, 8, 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q1/5

True or false?

6 is a factor of 30.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q2/5

1717

QUESTION ID

S46-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q2/5

Which number is a factor of 27?

- A.** 4
- B.** 6
- C.** 9
- D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q3/5

1718

QUESTION ID

S46-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q3/5

List all the factors of 21.

Factors: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q4/5

1719

QUESTION ID

S46-B3-Q4

TYPE

Short answer

CORRECT ANSWER

1, 3, 7, 21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q4/5

True or false?

8 is a factor of 36.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q5/5

1720

QUESTION ID

S46-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.B.4 • Set 46 • BACKWARD 3 • Q5/5

SET 46 • 6.EE.A.3

FORWARD 1

Factor linear expressions with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Factor out 2.5.

$$2.5x + 10$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q1/5

1722

QUESTION ID

S46-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$2.5(x + 4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q1/5

Factor out -4 .

$$-4x + 12$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q2/5

1723

QUESTION ID

S46-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$-4(x - 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q2/5

Which is equivalent to $\frac{1}{2}x + 3$?

A. $\frac{1}{2}(x + \frac{3}{2})$

B. $\frac{1}{2}(x + 3)$

C. $\frac{1}{2}(x + 6)$

D. $2(x + 6)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q3/5

1724

QUESTION ID

S46-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $(\frac{1}{2})(x + 6)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

Factor out 0.6.

$$0.6y - 1.8$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q4/5

1725

QUESTION ID

S46-F1-Q4

TYPE

Short answer

CORRECT ANSWER

 $0.6(y - 3)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q4/5

1725

True or false?

$$-5a - 15 = -5(a + 3)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q5/5

1726

QUESTION ID

S46-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 46 • FORWARD 1 • Q5/5

SET 46 • 6.EE.A.3

FORWARD 2

Use factoring to decide how many solutions an equation has • Grade 8 •
Expressions & Equations • 8.EE.C.7.a

Factor the left side to decide. How many solutions does the equation have?

$$6x + 9 = 3(2x + 3)$$

- A. One solution
- B. No solution
- C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q1/5

1728

QUESTION ID

S46-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q1/5

How many solutions does the equation have?

$$12x + 8 = 4(3x + 1)$$

- A. One solution
- B. No solution
- C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q2/5

1729

QUESTION ID

S46-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q2/5

How many solutions does the equation have?

$$10x - 15 = 5(2x - 3)$$

- A. One solution
- B. No solution
- C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q3/5

1730

QUESTION ID

S46-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q3/5

How many solutions does the equation have?

$$8x + 4 = 4(x + 1)$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q4/5

1731

QUESTION ID

S46-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. One solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q4/5

What number goes in the box so that the equation has infinitely many solutions?

$$14x + 21 = 7(2x + \square)$$

$\square =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q5/5

1732

QUESTION ID

S46-F2-Q5

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S46-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 46 • FORWARD 2 • Q5/5

Combine like terms to write equivalent expressions

MAIN	Grade 6 main question	6.EE.A.3
BACKWARD 1	Repeated addition as multiplication	3.OA.A.1
BACKWARD 2	Changing the order and grouping of addends	3.OA.B.5
FORWARD 1	Combine like terms with rational coefficients	7.EE.A.1
FORWARD 2	Solve equations by collecting like terms	8.EE.C.7.b

SET 47 • 6.EE.A.3

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.3

Write an equivalent expression with one term.

$$y + y + y + y$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q1/5

1735

QUESTION ID

S47-M-Q1

TYPE

Short answer

CORRECT ANSWER

4y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q1/5

Simplify.

$$5x + 3x$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q2/5

1736

QUESTION ID

S47-M-Q2

TYPE

Short answer

CORRECT ANSWER

8x

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q2/5

Which expression is equivalent to $7a + 2 + 3a$?

- A. $7a + 5a$
- B. $10a + 5$
- C. $10a + 2$
- D. $12a$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q3/5

1737

QUESTION ID

S47-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $10a + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q3/5

Simplify.

$$4m + 6 + 2m + 1$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q4/5

1738

QUESTION ID

S47-M-Q4

TYPE

Short answer

CORRECT ANSWER

6m + 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Simplify.

$$9n - 4n + 3$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q5/5

1739

QUESTION ID

S47-M-Q5

TYPE

Short answer

CORRECT ANSWER

 $5n + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.3 • Set 47 • MAIN • Q5/5

1739

SET 47 • 6.EE.A.3

BACKWARD 1

Repeated addition as multiplication • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$$4 + 4 + 4 = 3 \times 4$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q1/5

1741

QUESTION ID

S47-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q1/5

Write $7 + 7 + 7 + 7$ as a multiplication expression.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q2/5

1742

QUESTION ID

S47-B1-Q2

TYPE

Short answer

CORRECT ANSWER

4 × 7

GRADING NOTE 7×4 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q2/5

What does 4×9 mean?

- A.** 4 groups of 9
- B.** 9 minus 4
- C.** 4 more than 9
- D.** The number 49

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q3/5

1743

QUESTION ID

S47-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 4 groups of 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q3/5

True or false?

$$2 + 2 + 2 + 2 + 2 = 2 \times 2$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q4/5

1744

QUESTION ID

S47-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q4/5

Write $9 + 9$ as a multiplication expression.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q5/5

1745

QUESTION ID

S47-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2 × 9

GRADING NOTE 9×2 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 47 • BACKWARD 1 • Q5/5

SET 47 • 6.EE.A.3

BACKWARD 2

Changing the order and grouping of addends • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$4 + 6 + 2 = 4 + 2 + 6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 47 • BACKWARD 2 • Q1/5

1747

QUESTION ID

S47-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 47 • BACKWARD 2 • Q1/5

$$7 + 5 + 3 = 7 + 3 + \underline{\hspace{1cm}}$$

Blank: _____

ID S47-B2-Q2

1748

5

ID S47-B2-Q2

1748

Which is equal to $(8 + 2) + 5$?

- A. $8 + 2 \times 5$
- B. $8 \times (2 + 5)$
- C. $(8 - 2) + 5$
- D. $8 + (2 + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 47 • BACKWARD 2 • Q3/5

1749

QUESTION ID

S47-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $8 + (2 + 5)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 47 • BACKWARD 2 • Q3/5

True or false?

$$9 + 1 + 9 = 9 + 9 + 1$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 47 • BACKWARD 2 • Q4/5

1750

QUESTION ID

S47-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-B2-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 47 • BACKWARD 2 • Q4/5

$$6 + 3 + 4 = 6 + 4 + \underline{\quad}$$

Blank: _____

ID S47-B2-Q5

1751

S47-B2-Q5

Short answer

3

ID S47-B2-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 47 • BACKWARD 2 • Q5/5

SET 47 • 6.EE.A.3

FORWARD 1

Combine like terms with rational coefficients • Grade 7 • Expressions & Equations • 7.EE.A.1

Simplify.

$$2.25y + 0.75y$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q1/5

1753

QUESTION ID

S47-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q1/5

Simplify.

$$\frac{1}{2}x + \frac{1}{4}x$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q2/5

1754

QUESTION ID

S47-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$(\frac{3}{4})x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q2/5

Simplify.

$$6x - 2 + 3x + 7$$

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q3/5

1755

QUESTION ID

S47-F1-Q3

TYPE

Short answer

CORRECT ANSWER

$9x + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q3/5

Simplify.

$$-3a + 7a - 2$$

- A. $4a - 2$
- B. $4a + 2$
- C. $2a$
- D. $-10a - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q4/5

1756

QUESTION ID

S47-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. $4a - 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q4/5

True or false?

$$2.5m - m = 1.5m$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q5/5

1757

QUESTION ID

S47-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.1 • Set 47 • FORWARD 1 • Q5/5

SET 47 • 6.EE.A.3

FORWARD 2

Solve equations by collecting like terms • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$3x + 5x - 4 = 20$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q1/5

1759

QUESTION ID

S47-F2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q1/5

Solve for y.

$$7y - 2y + 3 = 2y + 18$$

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q2/5

1760

QUESTION ID

S47-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q2/5

Solve for a.

$$4a + 6 - a = 2a + 10$$

A. -4

B. $\frac{16}{5}$

C. 4

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q3/5

1761

QUESTION ID

S47-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q3/5

Solve for n.

$$0.5n + 1.5n - 3 = n + 5$$

n = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q4/5

1762

QUESTION ID

S47-F2-Q4

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q4/5

Solve for k.

$$6k - 2k + 1 = 3k - 4$$

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q5/5

1763

QUESTION ID

S47-F2-Q5

TYPE

Short answer

CORRECT ANSWER

-5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S47-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 47 • FORWARD 2 • Q5/5

Identify when two expressions are equivalent

MAIN	Grade 6 main question	6.EE.A.4
BACKWARD 1	Commutative and associative properties	3.OA.B.5
BACKWARD 2	Evaluate an expression for a given value	6.EE.A.2.c
BACKWARD 3	Equal groups	3.OA.A.1
FORWARD 1	Rewrite expressions to show how quantities are related	7.EE.A.2
FORWARD 2	One, none, or infinitely many solutions	8.EE.C.7.a

SET 48 • 6.EE.A.4

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.A.4

True or false?

$m + m + m + m$ and $4m$ are equivalent expressions.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q1/5

1766

QUESTION ID

S48-M-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q1

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q1/5

Which expression is equivalent to $2(x + 4)$?

- A. $x + 8$
- B. $2x + 4$
- C. $2x + 6$
- D. $2x + 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q2/5

1767

QUESTION ID

S48-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $2x + 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q2

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q2/5

True or false?

$3(x + 2)$ and $3x + 2$ are equivalent expressions.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q3/5

1768

QUESTION ID

S48-M-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q3

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q3/5

Which expression is equivalent to $6a + 4a$?

- A. $6a + 4$
- B. $10a$
- C. $24a$
- D. $10a^2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q4/5

1769

QUESTION ID

S48-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $10a$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q4

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q4/5

Which pair of expressions is equivalent?

- A. $5(a + 3)$ and $5a + 15$
- B. $5(a + 3)$ and $a + 15$
- C. $5(a + 3)$ and $5a + 3$
- D. $5(a + 3)$ and $8a$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q5/5

1770

QUESTION ID

S48-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

**A. $5(a + 3)$ and
 $5a + 15$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-M-Q5

Grade 6 • Expressions & Equations • 6.EE.A.4 • Set 48 • MAIN • Q5/5

SET 48 • 6.EE.A.4

BACKWARD 1

Commutative and associative properties • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5

True or false?

$$8 + 5 = 5 + 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q1/5

1772

QUESTION ID

S48-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q1/5

True or false?

$$(2 \times 3) \times 4 = 2 \times (3 \times 4)$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q2/5

1773

QUESTION ID

S48-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q2/5

Which equation shows the commutative property of multiplication?

- A. $6 \times 7 = 42$
- B. $6 \times 7 = 7 \times 6$
- C. $6 \times (7 + 1) = 6 \times 7 + 6$
- D. $6 + 0 = 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q3/5

1774

QUESTION ID

S48-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $6 \times 7 = 7 \times 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q3/5

True or false?

$$9 - 4 = 4 - 9$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q4/5

1775

QUESTION ID

S48-B1-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q4/5

Fill in the blank.

$$3 \times 8 = 8 \times \underline{\quad}$$

Blank: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q5/5

1776

QUESTION ID

S48-B1-Q5

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.5 • Set 48 • BACKWARD 1 • Q5/5

SET 48 • 6.EE.A.4

BACKWARD 2

Evaluate an expression for a given value • Grade 6 • Expressions & Equations •
6.EE.A.2.c

Evaluate $4x + 2x$ when $x = 3$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q1/5

1778

QUESTION ID

S48-B2-Q1

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q1/5

Evaluate $3(x + 2)$ when $x = 5$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q2/5

1779

QUESTION ID

S48-B2-Q2

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q2/5

True or false?

When $x = 2$, the value of $3x + 2$ is 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q3/5

1780

QUESTION ID

S48-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q3/5

Evaluate $2(x + 4)$ when $x = 1$.

- A.** 6
- B.** 9
- C.** 10
- D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q4/5

1781

QUESTION ID

S48-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q4/5

Evaluate $5a + 15$ when $a = 2$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q5/5

1782

QUESTION ID

S48-B2-Q5

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B2-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 48 • BACKWARD 2 • Q5/5

SET 48 • 6.EE.A.4

BACKWARD 3

Equal groups • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1

True or false?

$$6 + 6 + 6 + 6 = 4 \times 6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q1/5

1784

QUESTION ID

S48-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q1/5

Which expression means 3 groups of 8?

- A. $3 + 8$
- B. 38
- C. $8 - 3$
- D. 3×8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q2/5

1785

QUESTION ID

S48-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 3×8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q2/5

Write $5 + 5 + 5$ as a multiplication expression.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q3/5

1786

QUESTION ID

S48-B3-Q3

TYPE

Short answer

CORRECT ANSWER

5 × 3

GRADING NOTE 3×5 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q3/5

True or false?

$$7 \times 2 = 7 + 7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q4/5

1787

QUESTION ID

S48-B3-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q4/5

How many groups of 4 are in $4 + 4 + 4 + 4 + 4$?

Groups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q5/5

1788

QUESTION ID

S48-B3-Q5

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.1 • Set 48 • BACKWARD 3 • Q5/5

SET 48 • 6.EE.A.4

FORWARD 1

Rewrite expressions to show how quantities are related • Grade 7 •
Expressions & Equations • 7.EE.A.2

A price c plus an 8% tax can be written as $c + 0.08c$.

Which expression is equivalent?

- A. $0.08c$
- B. $1.08c$
- C. $1.8c$
- D. $8.08c$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q1/5

1790

QUESTION ID

S48-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. $1.08c$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q1/5

True or false?

$$a - 0.2a = 0.8a$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q2/5

1791

QUESTION ID

S48-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q2/5

Write an expression with one term that is equivalent to $n - 0.25n$.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q3/5

1792

QUESTION ID

S48-F1-Q3

TYPE

Short answer

CORRECT ANSWER

0.75n

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q3/5

Which expression shows the price p after a 30% discount?

- A. $0.3p$
- B. $0.7p$
- C. $1.3p$
- D. $p - 30$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q4/5

1793

QUESTION ID

S48-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $0.7p$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q4/5

The perimeter of a square is $s + s + s + s$.

Write an equivalent expression with one term.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q5/5

1794

QUESTION ID

S48-F1-Q5

TYPE

Short answer

CORRECT ANSWER

4s

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.A.2 • Set 48 • FORWARD 1 • Q5/5

SET 48 • 6.EE.A.4

FORWARD 2

One, none, or infinitely many solutions • Grade 8 • Expressions & Equations •
8.EE.C.7.a

True or false?

The equation $2(x + 3) = 2x + 6$ has infinitely many solutions.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q1/5

1796

QUESTION ID

S48-F2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q1/5

How many solutions does the equation have?

$$5x - 3 = 5x + 1$$

- A. One solution
- B. No solution
- C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q2/5

1797

QUESTION ID

S48-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q2/5

How many solutions does the equation have?

$$3(2x + 4) = 6x + 12$$

- A. One solution
- B. No solution
- C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q3/5

1798

QUESTION ID

S48-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q3/5

Write a number in the box so the equation has no solution.

$$4x + 7 = 4x + \square$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q4/5

1799

QUESTION ID

S48-F2-Q4

TYPE

Short answer

CORRECT ANSWER

**Any number except
7**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q4/5

Which equation has exactly one solution?

- A. $3x + 1 = 2x + 1$
- B. $3x + 1 = 3x + 2$
- C. $3x + 1 = 3x + 1$
- D. $3(x + 1) = 3x + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q5/5

1800

QUESTION ID

S48-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. $3x + 1 = 2x + 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S48-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 48 • FORWARD 2 • Q5/5

Use substitution to decide whether a number makes an equation or inequality true

MAIN	Grade 6 main question	6.EE.B.5
BACKWARD 1	Evaluate an expression	6.EE.A.2.c
BACKWARD 2	Comparing numbers	2.NBT.A.4
BACKWARD 3	Meaning of the equal sign	1.OA.D.7
FORWARD 1	Solve equations of the form $px + q = r$ and $p(x + q) = r$	7.EE.B.4.a
FORWARD 2	Solutions of a system of two linear equations	8.EE.C.8.a

SET 49 • 6.EE.B.5

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.5

Which value from the set $\{2, 4, 6, 8\}$ makes $3x + 1 = 19$ true?

- A.** 2
- B.** 4
- C.** 6
- D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q1/5

1803

QUESTION ID

S49-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q1/5

True or false?

$x = 5$ is a solution of $2x - 3 > 8$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q2/5

1804

QUESTION ID

S49-M-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q2/5

Which numbers from the set $\{1, 3, 5, 7\}$ make $x + 4 < 10$ true?

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q3/5

1805

QUESTION ID

S49-M-Q3

TYPE

Short answer

CORRECT ANSWER

1, 3, 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q3/5

Is $y = 2.5$ a solution of $4y = 10$?

A. Yes

B. No

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q4/5

1806

QUESTION ID

S49-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Yes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q4/5

Which value from the set $\{12, 15, 18\}$ makes $n \div 3 = 6$ true?

$n =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q5/5

1807

QUESTION ID

S49-M-Q5

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.5 • Set 49 • MAIN • Q5/5

SET 49 • 6.EE.B.5

BACKWARD 1

Evaluate an expression • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Evaluate $3x + 1$ when $x = 6$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q1/5

1809

QUESTION ID

S49-B1-Q1

TYPE

Short answer

CORRECT ANSWER

19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q1

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q1/5

Evaluate $2x - 3$ when $x = 5$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q2/5

1810

QUESTION ID

S49-B1-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q2

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q2/5

True or false?

When $y = 4$, the value of $5y$ is 20.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q3/5

1811

QUESTION ID

S49-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q3

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q3/5

Evaluate $x + 4$ when $x = 7$.

- A.** 3
- B.** 11
- C.** 28
- D.** 47

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q4/5

1812

QUESTION ID

S49-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q4

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q4/5

Evaluate $n \div 3$ when $n = 15$.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q5/5

1813

QUESTION ID

S49-B1-Q5

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B1-Q5

Grade 6 • Expressions & Equations • 6.EE.A.2.c • Set 49 • BACKWARD 1 • Q5/5

SET 49 • 6.EE.B.5

BACKWARD 2

Comparing numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Write $>$, $<$, or $=$ to compare.

19 ____ 21

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q1/5

1815

QUESTION ID

S49-B2-Q1

TYPE

Short answer

CORRECT ANSWER

$<$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q1/5

True or false?

$$7 > 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q2/5

1816

QUESTION ID

S49-B2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q2/5

Which statement is true?

- A.** $9 > 6$
- B.** $10 < 5$
- C.** $11 < 10$
- D.** $6 > 9$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q3/5

1817

QUESTION ID

S49-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $9 > 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q3/5

True or false?

$$24 = 24$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q4/5

1818

QUESTION ID

S49-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Write $>$, $<$, or $=$ to compare.

105 ____ 150

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q5/5

1819

QUESTION ID

S49-B2-Q5

TYPE

Short answer

CORRECT ANSWER

 $<$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 49 • BACKWARD 2 • Q5/5

1819

SET 49 • 6.EE.B.5

BACKWARD 3

Meaning of the equal sign • Grade 1 • Operations & Algebraic Thinking •
1.OA.D.7

True or false?

$$4 + 5 = 9 + 0$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q1/5

1821

QUESTION ID

S49-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q1/5

True or false?

$$12 = 15 - 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q2/5

1822

QUESTION ID

S49-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q2/5

True or false?

$$3 + 6 = 5 + 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q3/5

1823

QUESTION ID

S49-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q3/5

Which equation is true?

A. $4 + 3 = 5 + 3$

B. $4 + 3 = 5 + 2$

C. $7 = 4 - 3$

D. $4 + 3 = 8$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q4/5

1824

QUESTION ID

S49-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. $4 + 3 = 5 + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q4/5

True or false?

$$10 - 2 = 7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q5/5

1825

QUESTION ID

S49-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-B3-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.D.7 • Set 49 • BACKWARD 3 • Q5/5

SET 49 • 6.EE.B.5

FORWARD 1

Solve equations of the form $px + q = r$ and $p(x + q) = r$ • Grade 7 •
Expressions & Equations • 7.EE.B.4.a

Solve for x.

$$3x + 7 = 22$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q1/5

1827

QUESTION ID

S49-F1-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q1/5

Solve for x.

$$4(x - 2) = 20$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q2/5

1828

QUESTION ID

S49-F1-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q2/5

Solve for x.

$$0.5x + 6 = 10$$

- A.** 2
- B.** 8
- C.** 20
- D.** 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q3/5

1829

QUESTION ID

S49-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q3/5

Solve for x.

$$\frac{2}{3}x - 4 = 10$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q4/5

1830

QUESTION ID

S49-F1-Q4

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q4/5

A movie ticket costs \$12. Popcorn costs p dollars. Two tickets and one popcorn cost \$31.

What is p ?

$p =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q5/5

1831

QUESTION ID

S49-F1-Q5

TYPE

Short answer

CORRECT ANSWER

\$7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 49 • FORWARD 1 • Q5/5

SET 49 • 6.EE.B.5

FORWARD 2

Solutions of a system of two linear equations • Grade 8 • Expressions & Equations • 8.EE.C.8.a

True or false?

$(2, 3)$ is a solution of both $y = x + 1$ and $y = 2x - 1$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q1/5

1833

QUESTION ID

S49-F2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q1/5

Which point is a solution of both $y = 2x$ and $y = x + 3$?

- A. (0, 3)
- B. (2, 5)
- C. (1, 2)
- D. (3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q2/5

1834

QUESTION ID

S49-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. (3, 6)

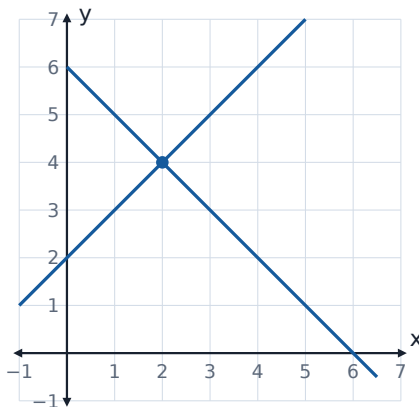
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q2/5

What is the solution of the system shown by the graph?

Solution: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q3/5

1835

QUESTION ID

S49-F2-Q3

TYPE

Short answer

CORRECT ANSWER

(2, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q3/5

True or false?

(1, 4) is a solution of both $y = 3x + 1$ and $y = -x + 5$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q4/5

1836

QUESTION ID

S49-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q4/5

Is $(0, 2)$ a solution of both $y = 2x + 2$ and $y = x - 2$?

A. Yes

B. No

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q5/5

1837

QUESTION ID

S49-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. No

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S49-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 49 • FORWARD 2 • Q5/5

Use variables to write expressions for real-world problems

MAIN	Grade 6 main question	6.EE.B.6
BACKWARD 1	Word problems with an unknown number	3.OA.A.3
BACKWARD 2	Multiplicative comparison word problems	4.OA.A.2
BACKWARD 3	Number patterns that follow a rule	4.OA.C.5
FORWARD 1	Write and solve equations for word problems	7.EE.B.4.a
FORWARD 2	Model a linear relationship with a function	8.F.B.4

SET 50 • 6.EE.B.6

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.6

Tickets cost \$8 each.

Write an expression for the cost of t tickets.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q1/5

1840

QUESTION ID

S50-M-Q1

TYPE

Short answer

CORRECT ANSWER

8t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q1/5

Kim is k years old. Sam is 3 years older than Kim.

Write an expression for Sam's age.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q2/5

1841

QUESTION ID

S50-M-Q2

TYPE

Short answer

CORRECT ANSWER

$k + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q2/5

A pizza is cut into s slices. 4 friends share the slices equally.

Which expression shows how many slices each friend gets?

A. $4s$

B. $s - 4$

C. $\frac{s}{4}$

D. $s + 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q3/5

1842

QUESTION ID

S50-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $s/4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q3/5

The length of a rectangle is 5 inches more than its width w .

Write an expression for the length.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q4/5

1843

QUESTION ID

S50-M-Q4

TYPE

Short answer

CORRECT ANSWER

$w + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q4/5

A gym charges \$20 to join plus \$15 per month.

Write an expression for the total cost after m months.

Expression: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q5/5

1844

QUESTION ID

S50-M-Q5

TYPE

Short answer

CORRECT ANSWER

20 + 15m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.6 • Set 50 • MAIN • Q5/5

SET 50 • 6.EE.B.6

BACKWARD 1

Word problems with an unknown number • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

There are 6 bags with the same number of apples.
There are 42 apples in all.
How many apples are in each bag?

Apples: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q1/5

1846

QUESTION ID

S50-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q1/5

There are 5 boxes with $\square$ pencils in each box. There are 35 pencils in all.

Which equation matches?

A. $35 \times 5 = \square$

B. $\square - 5 = 35$

C. $5 + \square = 35$

D. $5 \times \square = 35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q2/5

1847

QUESTION ID

S50-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $5 \times \square = 35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q2/5

True or false?

If $4 \times \square = 28$, then $\square = 7$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q3/5

1848

QUESTION ID

S50-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q3/5

There are 8 rows with $\square$ chairs in each row. There are 56 chairs in all.

What number is $\square$?

$\square =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q4/5

1849

QUESTION ID

S50-B1-Q4

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q4/5

Jan split 45 stickers equally among 9 friends.

How many stickers did each friend get?

Stickers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q5/5

1850

QUESTION ID

S50-B1-Q5

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 50 • BACKWARD 1 • Q5/5

SET 50 • 6.EE.B.6

BACKWARD 2

Multiplicative comparison word problems • Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2

A red ribbon is 7 inches long. A blue ribbon is 3 times as long.

How long is the blue ribbon?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q1/5

1852

QUESTION ID

S50-B2-Q1

TYPE

Short answer

CORRECT ANSWER

21 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q1/5

Ben has 6 books. Ali has 4 times as many books as Ben.

Which equation finds b , the number of books Ali has?

A. $b = 4 \times 6$

B. $b = 4 + 6$

C. $b = 6 \div 4$

D. $b = 6 - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q2/5

1853

QUESTION ID

S50-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $b = 4 \times 6$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q2/5

True or false?

A cat weighs 5 pounds. A dog weighs 8 times as much. The dog weighs 40 pounds.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q3/5

1854

QUESTION ID

S50-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q3/5

A bush is 4 feet tall. A tree is 5 times as tall.

How tall is the tree?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q4/5

1855

QUESTION ID

S50-B2-Q4

TYPE

Short answer

CORRECT ANSWER

20 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q4/5

Sara ran 24 miles. That is 3 times as far as Tom ran.

How far did Tom run?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q5/5

1856

QUESTION ID

S50-B2-Q5

TYPE

Short answer

CORRECT ANSWER

8 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B2-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.A.2 • Set 50 • BACKWARD 2 • Q5/5

SET 50 • 6.EE.B.6

BACKWARD 3

Number patterns that follow a rule • Grade 4 • Operations & Algebraic Thinking
• 4.OA.C.5

The rule is "add 4." The pattern starts at 3.

Write the first five numbers.

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q1/5

1858

QUESTION ID

S50-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3, 7, 11, 15, 19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q1/5

The rule is "multiply by 2." The pattern starts at 1.

What is the 5th number?

- A. 8
- B. 10
- C. 16
- D. 32

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q2/5

1859

QUESTION ID

S50-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q2/5

True or false?

The rule "add 5," starting at 5, gives 5, 10, 15, 20.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q3/5

1860

QUESTION ID

S50-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q3/5

What is the rule for this pattern?

7, 11, 15, 19, ...

Rule: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q4/5

1861

QUESTION ID

S50-B3-Q4

TYPE

Short answer

CORRECT ANSWER

Add 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q4/5

The rule is "add 6." The pattern starts at 2.

What is the 4th number?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q5/5

1862

QUESTION ID

S50-B3-Q5

TYPE

Short answer

CORRECT ANSWER

20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-B3-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 50 • BACKWARD 3 • Q5/5

SET 50 • 6.EE.B.6

FORWARD 1

Write and solve equations for word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Sam has \$60. He spends \$8 each week.

Write and solve an equation to find when he will have \$20 left.

Weeks: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q1/5

1864

QUESTION ID

S50-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$60 - 8w = 20; w = 5$
weeks

GRADING NOTE Any equivalent equation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q1/5

A rectangle has a perimeter of 50 cm and a length of 15 cm.

Which equation can be used to find its width w ?

A. $2(15 + w) = 50$

B. $2(15w) = 50$

C. $15 + 2w = 50$

D. $15 + w = 50$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q2/5

1865

QUESTION ID

S50-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $2(15 + w) = 50$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q2/5

A taxi charges \$3 plus \$2 per mile. The fare was \$17.

How many miles was the ride?

Miles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q3/5

1866

QUESTION ID

S50-F1-Q3

TYPE

Short answer

CORRECT ANSWER

7 miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q3/5

You buy 4 notebooks and a \$6 backpack. You spend \$18 in all.

How much does each notebook cost?

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q4/5

1867

QUESTION ID

S50-F1-Q4

TYPE

Short answer

CORRECT ANSWER

\$3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q4/5

True or false?

"Five less than twice a number is 17" can be written as $2n - 5 = 17$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q5/5

1868

QUESTION ID

S50-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 50 • FORWARD 1 • Q5/5

SET 50 • 6.EE.B.6

FORWARD 2

Model a linear relationship with a function • Grade 8 • Functions • 8.F.B.4

A plant is 4 cm tall. It grows 1.5 cm each week.

Write a function for its height h after w weeks.

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q1/5

1870

QUESTION ID

S50-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$h = 4 + 1.5w$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q1/5

Which situation matches $y = 2x + 10$, where y is dollars in savings after x weeks?

- A. Save \$12 each week.
- B. Start with \$10 and spend \$2 each week.
- C. Start with \$10 and save \$2 each week.
- D. Start with \$2 and save \$10 each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q2/5

1871

QUESTION ID

S50-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Start with \$10 and save \$2 each week.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q2/5

The table shows savings over time.

Write a function for the savings y after x weeks.

Weeks (x)	0	1	2
Savings (y)	30	45	60

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q3/5

1872

QUESTION ID

S50-F2-Q3

TYPE

Short answer

CORRECT ANSWER

$$y = 15x + 30$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q3/5

A rental truck costs \$50 plus \$0.25 per mile.

Write a function for the cost C of driving m miles.

$C =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q4/5

1873

QUESTION ID

S50-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$C = 50 + 0.25m$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q4/5

True or false?

A line through (0, 5) and (2, 9) has an initial value of 5 and a rate of change of 2.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q5/5

1874

QUESTION ID

S50-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S50-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 50 • FORWARD 2 • Q5/5

Solve one-step equations of the form

$$x + p = q$$

MAIN	Grade 6 main question	6.EE.B.7
BACKWARD 1	Subtraction as an unknown-addend problem	1.OA.B.4
BACKWARD 2	Adding and subtracting decimals	5.NBT.B.7
BACKWARD 3	Adding and subtracting fractions and mixed numbers	5.NF.A.1
FORWARD 1	Solve two-step equations	7.EE.B.4.a
FORWARD 2	Solve equations with variables on both sides	8.EE.C.7.b

SET 51 • 6.EE.B.7

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$x + 7.5 = 12$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q1/5

1877

QUESTION ID

S51-M-Q1

TYPE

Short answer

CORRECT ANSWER

4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q1/5

Solve for n.

$$n - \frac{3}{4} = 5\frac{1}{2}$$

A. $4\frac{3}{4}$

B. $5\frac{1}{4}$

C. $6\frac{1}{4}$

D. $6\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q2/5

1878

QUESTION ID

S51-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 6 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q2/5

Solve for y.

$$y + 2\frac{1}{3} = 6$$

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q3/5

1879

QUESTION ID

S51-M-Q3

TYPE

Short answer

CORRECT ANSWER

3 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q3/5

Solve for m.

$$18.25 + m = 30$$

m = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q4/5

1880

QUESTION ID

S51-M-Q4

TYPE

Short answer

CORRECT ANSWER

11.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q4/5

Solve for k.

$$k - 4.8 = 9.2$$

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q5/5

1881

QUESTION ID

S51-M-Q5

TYPE

Short answer

CORRECT ANSWER

14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 51 • MAIN • Q5/5

SET 51 • 6.EE.B.7

BACKWARD 1

Subtraction as an unknown-addend problem • Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4

True or false?

$12 - 7 = \square$ can be solved by thinking

$7 + \square = 12$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q1/5

1883

QUESTION ID

S51-B1-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q1

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q1/5

What number goes in the box?

$$7 + \square = 15$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q2/5

1884

QUESTION ID

S51-B1-Q2

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q2

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q2/5

To find $16 - 9$, think $9 + \square = 16$.

What number goes in the box?

- A. 6
- B. 7
- C. 8
- D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q3/5

1885

QUESTION ID

S51-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q3

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q3/5

What number goes in the box?

$$\square + 8 = 13$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q4/5

1886

QUESTION ID

S51-B1-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q4

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q4/5

True or false?

To find $11 - 5$, you can think $5 + 6 = 11$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q5/5

1887

QUESTION ID

S51-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B1-Q5

Grade 1 • Operations & Algebraic Thinking • 1.OA.B.4 • Set 51 • BACKWARD 1 • Q5/5

SET 51 • 6.EE.B.7

BACKWARD 2

Adding and subtracting decimals • Grade 5 • Number & Operations in Base Ten
• 5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$12 - 7.5$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q1/5

1889

QUESTION ID

S51-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q1/5

1889

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$30 - 18.25$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q2/5

1890

QUESTION ID

S51-B2-Q2

TYPE

Short answer

CORRECT ANSWER

11.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q2/5

1890

True or false?

$$9.2 + 4.8 = 14$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q3/5

1891

QUESTION ID

S51-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q3/5

$$6.25 + 2.8 = ?$$

- A.** 6.53
- B.** 8.33
- C.** 9.05
- D.** 9.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q4/5

1892

QUESTION ID

S51-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 9.05

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$3.7 + 5.45$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q5/5

1893

QUESTION ID

S51-B2-Q5

TYPE

Short answer

CORRECT ANSWER

9.15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 51 • BACKWARD 2 • Q5/5

1893

SET 51 • 6.EE.B.7

BACKWARD 3

Adding and subtracting fractions and mixed numbers • Grade 5 • Number & Operations–Fractions • 5.NF.A.1

Subtract.

$$6 - 2\frac{1}{3}$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q1/5

1895

QUESTION ID

S51-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3 2/3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q1/5

Add.

$$5\frac{1}{2} + \frac{3}{4}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q2/5

1896

QUESTION ID

S51-B3-Q2

TYPE

Short answer

CORRECT ANSWER

6 1/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q2/5

True or false?

$$\frac{2}{3} + \frac{1}{6} = \frac{5}{6}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q3/5

1897

QUESTION ID

S51-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q3/5

$$\frac{7}{8} - \frac{1}{4} = ?$$

A. $\frac{1}{2}$

B. $\frac{5}{8}$

C. $\frac{3}{4}$

D. $\frac{6}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q4/5

1898

QUESTION ID

S51-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q4/5

Add.

$$1\frac{2}{5} + \frac{1}{2}$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q5/5

1899

QUESTION ID

S51-B3-Q5

TYPE

Short answer

CORRECT ANSWER

1 9/10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.A.1 • Set 51 • BACKWARD 3 • Q5/5

SET 51 • 6.EE.B.7

FORWARD 1

Solve two-step equations • Grade 7 • Expressions & Equations • 7.EE.B.4.a

Solve for x.

$$3x + 8 = 29$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q1/5

1901

QUESTION ID

S51-F1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Solve for x.

$$-2x + 5 = 17$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q2/5

1902

QUESTION ID

S51-F1-Q2

TYPE

Short answer

CORRECT ANSWER

-6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q2/5

1902

Solve for x.

$$\frac{x}{4} - 3 = 2$$

A. -4

B. $\frac{5}{4}$

C. 5

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q3/5

1903

QUESTION ID

S51-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q3/5

Solve for x.

$$0.6x + 1.5 = 4.5$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q4/5

1904

QUESTION ID

S51-F1-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q4/5

Solve for x.

$$5(x + 2) = 45$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q5/5

1905

QUESTION ID

S51-F1-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 51 • FORWARD 1 • Q5/5

SET 51 • 6.EE.B.7

FORWARD 2

Solve equations with variables on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Solve for x.

$$7x - 4 = 3x + 12$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q1/5

1907

QUESTION ID

S51-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q1/5

Solve for x.

$$-5x + 9 = x - 3$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q2/5

1908

QUESTION ID

S51-F2-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q2/5

Solve for x.

$$2.5x + 4 = 1.5x + 9$$

- A.** -5
- B.** 5
- C.** 6.5
- D.** 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q3/5

1909

QUESTION ID

S51-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q3/5

Solve for x.

$$\frac{1}{2}x + 6 = 2x - 3$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q4/5

1910

QUESTION ID

S51-F2-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q4/5

Solve for x.

$$3(2x - 5) = 4x + 1$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q5/5

1911

QUESTION ID

S51-F2-Q5

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S51-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 51 • FORWARD 2 • Q5/5

Solve one-step equations of the form

$$px = q$$

MAIN	Grade 6 main question	6.EE.B.7
BACKWARD 1	Division as an unknown-factor problem	3.OA.B.6
BACKWARD 2	Dividing by a fraction	6.NS.A.1
BACKWARD 3	Dividing decimals	5.NBT.B.7
FORWARD 1	Solve equations of the form $p(x + q) = r$	7.EE.B.4.a
FORWARD 2	Solve multi-step linear equations	8.EE.C.7.b

SET 52 • 6.EE.B.7

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.7

Solve for x.

$$4x = 30$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q1/5

1914

QUESTION ID

S52-M-Q1

TYPE

Short answer

CORRECT ANSWER

7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q1/5

Solve for y.

$$\frac{2}{3}y = 12$$

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q2/5

1915

QUESTION ID

S52-M-Q2

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q2/5

Solve for m.

$$0.6m = 4.2$$

$$m = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q3/5

1916

QUESTION ID

S52-M-Q3

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q3/5

Solve for n.

$$5n = 3.5$$

- A. -1.5
- B. 0.7
- C. 1.43
- D. 17.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q4/5

1917

QUESTION ID

S52-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 0.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q4/5

Solve for k.

$$\frac{3}{4}k = \frac{9}{10}$$

k = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q5/5

1918

QUESTION ID

S52-M-Q5

TYPE

Short answer

CORRECT ANSWER

6/5

GRADING NOTE 1 1/5 or 1.2 is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 52 • MAIN • Q5/5

SET 52 • 6.EE.B.7

BACKWARD 1

Division as an unknown-factor problem • Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6

Find $48 \div 6$ by thinking $6 \times \square = 48$.

$\square =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q1/5

1920

QUESTION ID

S52-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q1/5

True or false?

$63 \div 7 = 9$ because $7 \times 9 = 63$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q2/5

1921

QUESTION ID

S52-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q2/5

Which fact helps you find $45 \div 5$?

- A. $9 - 5 = 4$
- B. $45 \times 5 = 225$
- C. $5 \times 9 = 45$
- D. $5 + 40 = 45$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q3/5

1922

QUESTION ID

S52-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $5 \times 9 = 45$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q3/5

What number goes in the box?

$$\square \times 6 = 42$$

$$\square = \underline{\hspace{2cm}}$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q4/5

1923

QUESTION ID

S52-B1-Q4

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q4/5

True or false?

$24 \div 4 = 5$ because $4 \times 5 = 24$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q5/5

1924

QUESTION ID

S52-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.B.6 • Set 52 • BACKWARD 1 • Q5/5

SET 52 • 6.EE.B.7

BACKWARD 2

Dividing by a fraction • Grade 6 • The Number System • 6.NS.A.1

Divide.

$$12 \div \frac{2}{3}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q1/5

1926

QUESTION ID

S52-B2-Q1

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q1

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q1/5

Divide.

$$\frac{9}{10} \div \frac{3}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q2/5

1927

QUESTION ID

S52-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q2

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q2/5

True or false?

$$5 \div \frac{5}{8} = 8$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q3/5

1928

QUESTION ID

S52-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q3

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q3/5

$$\frac{9}{10} \div \frac{3}{5} = ?$$

A. $\frac{27}{50}$

B. $\frac{2}{3}$

C. $1\frac{1}{3}$

D. $1\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q4/5

1929

QUESTION ID

S52-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 1 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q4

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q4/5

Divide.

$$6 \div \frac{3}{4}$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q5/5

1930

QUESTION ID

S52-B2-Q5

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B2-Q5

Grade 6 • The Number System • 6.NS.A.1 • Set 52 • BACKWARD 2 • Q5/5

SET 52 • 6.EE.B.7

BACKWARD 3

Dividing decimals • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$4.2 \div 0.6$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q1/5

1932

QUESTION ID

S52-B3-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$30 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q2/5

1933

QUESTION ID

S52-B3-Q2

TYPE

Short answer

CORRECT ANSWER

7.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q2/5

1933

True or false?

$$3.5 \div 5 = 0.7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q3/5

1934

QUESTION ID

S52-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

$$2.4 \div 0.3 = ?$$

- A.** 0.8
- B.** 7
- C.** 8
- D.** 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q4/5

1935

QUESTION ID

S52-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q4/5

1935

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$7.2 \div 9$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q5/5

1936

QUESTION ID

S52-B3-Q5

TYPE

Short answer

CORRECT ANSWER

0.8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 52 • BACKWARD 3 • Q5/5

1936

SET 52 • 6.EE.B.7

FORWARD 1

Solve equations of the form $p(x + q) = r$ • Grade 7 • Expressions & Equations •
7.EE.B.4.a

Solve for x.

$$3(x - 4) = 21$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q1/5

1938

QUESTION ID

S52-F1-Q1

TYPE

Short answer

CORRECT ANSWER

11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q1/5

Solve for x.

$$\frac{1}{2}(x + 6) = 9$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q2/5

1939

QUESTION ID

S52-F1-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q2/5

Solve for x.

$$3(x + 1.5) = 12$$

- A.** 2
- B.** 2.5
- C.** 4
- D.** 5.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q3/5

1940

QUESTION ID

S52-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q3/5

Solve for x.

$$-2(x - 3) = 14$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q4/5

1941

QUESTION ID

S52-F1-Q4

TYPE

Short answer

CORRECT ANSWER

-4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q4/5

Solve for x.

$$0.5(x + 8) = 7$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q5/5

1942

QUESTION ID

S52-F1-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 52 • FORWARD 1 • Q5/5

SET 52 • 6.EE.B.7

FORWARD 2

Solve multi-step linear equations • Grade 8 • Expressions & Equations •
8.EE.C.7.b

Solve for x.

$$3(2x - 1) = 4x + 7$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q1/5

1944

QUESTION ID

S52-F2-Q1

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q1/5

Solve for x.

$$\frac{2}{3}(x + 3) = x - 1$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q2/5

1945

QUESTION ID

S52-F2-Q2

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q2/5

Solve for x.

$$5x - 2(x + 1) = 10$$

A. 2

B. $\frac{8}{3}$

C. 4

D. 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q3/5

1946

QUESTION ID

S52-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q3/5

Solve for x.

$$0.4(10x - 5) = 2x + 4$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q4/5

1947

QUESTION ID

S52-F2-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q4/5

Solve for x.

$$6x = 2(x + 8)$$

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q5/5

1948

QUESTION ID

S52-F2-Q5

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S52-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 52 • FORWARD 2 • Q5/5

Write and solve one-step equations for real-world problems

MAIN	Grade 6 main question	6.EE.B.7
BACKWARD 1	Word problems with an unknown factor	3.OA.A.3
BACKWARD 2	Word problems with an unknown addend	2.OA.A.1
BACKWARD 3	A letter stands for an unknown quantity	3.OA.D.8
FORWARD 1	Write and solve two-step equations for word problems	7.EE.B.4.a
FORWARD 2	Solve real-world problems with variables on both sides	8.EE.C.7.b

SET 53 • 6.EE.B.7

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.7

A pack of 6 pens costs \$4.50.

Write and solve an equation to find the cost p of one pen.

Equation: _____

$p =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q1/5

1951

QUESTION ID

S53-M-Q1

TYPE

Short answer

CORRECT ANSWER

$6p = 4.50; p = \$0.75$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q1/5

After spending \$12.75, Sam has \$30.25 left.

Write and solve an equation to find how much money m Sam started with.

Equation: _____

$m =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q2/5

1952

QUESTION ID

S53-M-Q2

TYPE

Short answer

CORRECT ANSWER

$$m - 12.75 = 30.25;$$

$$m = \$43$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q2/5

A board is cut into 5 equal pieces. Each piece is 1.8 meters long.

Write and solve an equation to find the length L of the board.

Equation: _____

$L =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q3/5

1953

QUESTION ID

S53-M-Q3

TYPE

Short answer

CORRECT ANSWER

**$L \div 5 = 1.8$; $L = 9$
meters**

GRADING NOTE $L = 5 \times 1.8$ is also acceptable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q3/5

Maya is 1.35 m tall. She is 0.4 m taller than her brother.
Which equation can be used to find her brother's height b ?

A. $0.4b = 1.35$

B. $1.35 + 0.4 = b$

C. $b - 0.4 = 1.35$

D. $b + 0.4 = 1.35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q4/5

1954

QUESTION ID

S53-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $b + 0.4 = 1.35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q4/5

A recipe uses $\frac{3}{4}$ cup of sugar per batch. Kim used 6 cups of sugar.

Write and solve an equation to find the number of batches b .

Equation: _____

$b =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q5/5

1955

QUESTION ID

S53-M-Q5

TYPE

Short answer

CORRECT ANSWER

**$(\frac{3}{4})b = 6; b = 8$
batches**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.7 • Set 53 • MAIN • Q5/5

SET 53 • 6.EE.B.7

BACKWARD 1

Word problems with an unknown factor • Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3

4 friends share some grapes equally. Each gets 9 grapes.

How many grapes were there?

Grapes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q1/5

1957

QUESTION ID

S53-B1-Q1

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q1/5

There are 7 teams with $\square$ players each. There are 56 players in all.

Which equation matches?

A. $7 \times \square = 56$

B. $56 \times 7 = \square$

C. $\square - 7 = 56$

D. $7 + \square = 56$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q2/5

1958

QUESTION ID

S53-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. $7 \times \square = 56$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q2/5

True or false?

If $3 \times \square = 24$, then $\square = 8$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q3/5

1959

QUESTION ID

S53-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q3/5

A teacher puts 40 books into 5 equal stacks.

How many books are in each stack?

Books: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q4/5

1960

QUESTION ID

S53-B1-Q4

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q4/5

Each box holds 6 eggs. There are 54 eggs.

How many boxes are there?

Boxes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q5/5

1961

QUESTION ID

S53-B1-Q5

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B1-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.A.3 • Set 53 • BACKWARD 1 • Q5/5

SET 53 • 6.EE.B.7

BACKWARD 2

Word problems with an unknown addend • Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1

Lin had some stickers. She got 15 more. Now she has 42.

How many did she have at first?

Stickers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q1/5

1963

QUESTION ID

S53-B2-Q1

TYPE

Short answer

CORRECT ANSWER

27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q1

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q1/5

A plant grew 8 cm and is now 35 cm tall.

Which equation finds the starting height h ?

- A. $8h = 35$
- B. $h - 8 = 35$
- C. $h + 35 = 8$
- D. $h + 8 = 35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q2/5

1964

QUESTION ID

S53-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. $h + 8 = 35$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q2

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q2/5

True or false?

Tom spent \$17 and has \$26 left, so he started with \$43.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q3/5

1965

QUESTION ID

S53-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q3

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q3/5

A bus had 30 people. Some got off. Now there are 18.

How many got off?

People: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q4/5

1966

QUESTION ID

S53-B2-Q4

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q4

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q4/5

Ada read 24 pages on Monday and some more on Tuesday. She read 61 pages in all.

How many pages did she read on Tuesday?

Pages: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q5/5

1967

QUESTION ID

S53-B2-Q5

TYPE

Short answer

CORRECT ANSWER

37

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B2-Q5

Grade 2 • Operations & Algebraic Thinking • 2.OA.A.1 • Set 53 • BACKWARD 2 • Q5/5

SET 53 • 6.EE.B.7

BACKWARD 3

A letter stands for an unknown quantity • Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8

True or false?

In $5 \times n = 40$, the letter n stands for 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q1/5

1969

QUESTION ID

S53-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q1

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q1/5

In the equation $p + 9 = 20$, what number is p ?

$p =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q2/5

1970

QUESTION ID

S53-B3-Q2

TYPE

Short answer

CORRECT ANSWER

11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q2

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q2/5

Which equation matches "a number plus 6 is 15"?

A. $n + 15 = 6$

B. $n - 6 = 15$

C. $6n = 15$

D. $n + 6 = 15$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q3/5

1971

QUESTION ID

S53-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. $n + 6 = 15$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q3

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q3/5

In the equation $3 \times k = 27$, what number is k ?

$k =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q4/5

1972

QUESTION ID

S53-B3-Q4

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q4

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q4/5

True or false?

In $m - 4 = 10$, the letter m stands for 6.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q5/5

1973

QUESTION ID

S53-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-B3-Q5

Grade 3 • Operations & Algebraic Thinking • 3.OA.D.8 • Set 53 • BACKWARD 3 • Q5/5

SET 53 • 6.EE.B.7

FORWARD 1

Write and solve two-step equations for word problems • Grade 7 • Expressions & Equations • 7.EE.B.4.a

A gym charges a \$25 fee plus \$15 per month. Jen paid \$130.

Write and solve an equation to find the number of months m .

$m =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q1/5

1975

QUESTION ID

S53-F1-Q1

TYPE

Short answer

CORRECT ANSWER

**$25 + 15m = 130;$
 $m = 7$ months**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q1/5

A plumber charges \$45 plus \$60 per hour. A bill is \$225.

How many hours did the plumber work?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q2/5

1976

QUESTION ID

S53-F1-Q2

TYPE

Short answer

CORRECT ANSWER

3 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q2/5

A class buys 12 tickets and pays a \$15 group fee. The total is \$159.

Which equation finds the cost t of one ticket?

- A.** $12t - 15 = 159$ **B.** $12t + 15 = 159$
C. $12(t + 15) = 159$ **D.** $15t + 12 = 159$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q3/5

1977

QUESTION ID

S53-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. $12t + 15 = 159$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q3/5

Mo saves \$18 per week and already has \$54. He wants \$180.

How many weeks w will it take?

$w =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q4/5

1978

QUESTION ID

S53-F1-Q4

TYPE

Short answer

CORRECT ANSWER

7 weeks

GRADING NOTE $54 + 18w = 180$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q4/5

True or false?

A \$6 pizza plus 3 toppings costs \$10.50. Each topping costs \$1.50.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q5/5

1979

QUESTION ID

S53-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.a • Set 53 • FORWARD 1 • Q5/5

SET 53 • 6.EE.B.7

FORWARD 2

Solve real-world problems with variables on both sides • Grade 8 • Expressions & Equations • 8.EE.C.7.b

Store A charges \$20 plus \$3 per shirt. Store B charges \$5 per shirt.

Write and solve an equation to find when the costs are equal.

Shirts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q1/5

1981

QUESTION ID

S53-F2-Q1

TYPE

Short answer

CORRECT ANSWER

**$20 + 3s = 5s$; $s = 10$
shirts**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q1/5

Ana has \$120 and spends \$8 per week. Ben has \$40 and saves \$12 per week.

After how many weeks will they have the same amount?

Weeks: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q2/5

1982

QUESTION ID

S53-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4 weeks

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q2/5

Taxi A charges \$4 plus \$2.50 per mile. Taxi B charges \$1 plus \$3 per mile.

For how many miles do the rides cost the same?

A. 3

B. 5

C. 6

D. 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q3/5

1983

QUESTION ID

S53-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q3/5

A candle 30 cm tall burns 1.5 cm per hour. Another candle 20 cm tall burns 0.5 cm per hour.

After how many hours are they the same height?

Hours: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q4/5

1984

QUESTION ID

S53-F2-Q4

TYPE

Short answer

CORRECT ANSWER

10 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q4/5

Plan A: \$15 per month plus \$0.10 per text. Plan B: \$0.25 per text.

For how many texts do the plans cost the same?

Texts: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q5/5

1985

QUESTION ID

S53-F2-Q5

TYPE

Short answer

CORRECT ANSWER

100 texts

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S53-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.b • Set 53 • FORWARD 2 • Q5/5

Write an inequality $x > c$ or $x < c$ for a real-world condition

MAIN	Grade 6 main question	6.EE.B.8
BACKWARD 1	Inequality symbols	1.NBT.B.3
BACKWARD 2	Meaning of "at least," "at most," "more than," and "fewer than"	6.NS.C.7.b
BACKWARD 3	Negative numbers in context	6.NS.C.5
FORWARD 1	Write and solve inequalities for real-world problems	7.EE.B.4.b
FORWARD 2	Use two linear equations to decide when one quantity is greater...	8.EE.C.8.c

SET 54 • 6.EE.B.8

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.8

Riders must be at least 48 inches tall.

Write an inequality for the height h of a rider.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q1/5

1988

QUESTION ID

S54-M-Q1

TYPE

Short answer

CORRECT ANSWER

$h \geq 48$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q1/5

A bag can hold at most 20 pounds.

Which inequality describes the weight w the bag can hold?

A. $w \geq 20$

B. $w < 20$

C. $w \leq 20$

D. $w > 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q2/5

1989

QUESTION ID

S54-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. $w \leq 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q2/5

The temperature must stay below -3°C .

Write an inequality for the temperature t .

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q3/5

1990

QUESTION ID

S54-M-Q3

TYPE

Short answer

CORRECT ANSWER

$t < -3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q3/5

More than 25 people must sign up for a trip to happen.

Write an inequality for the number of people p .

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q4/5

1991

QUESTION ID

S54-M-Q4

TYPE

Short answer

CORRECT ANSWER

$p > 25$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q4/5

Children 12 years old and younger get a discount.

Write an inequality for the age a of a child who gets the discount.

Inequality: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q5/5

1992

QUESTION ID

S54-M-Q5

TYPE

Short answer

CORRECT ANSWER

$a \leq 12$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 54 • MAIN • Q5/5

SET 54 • 6.EE.B.8

BACKWARD 1

Inequality symbols • Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3

Which symbol makes this true?

58 ____ 85

A. <

B. >

C. =

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q1/5

1994

QUESTION ID

S54-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. <

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q1/5

True or false?

$30 > 29$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q2/5

1995

QUESTION ID

S54-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q2/5

Write $>$, $<$, or $=$ to compare.

64 ____ 46

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q3/5

1996

QUESTION ID

S54-B1-Q3

TYPE

Short answer

CORRECT ANSWER

$>$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q3/5

Which symbol means "is greater than"?

A. <

B. >

C. =

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q4

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q4/5

1997

QUESTION ID

S54-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. >

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q4

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q4/5

True or false?

$$17 < 12$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q5

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q5/5

1998

QUESTION ID

S54-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B1-Q5

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 54 • BACKWARD 1 • Q5/5

SET 54 • 6.EE.B.8

BACKWARD 2

Meaning of "at least," "at most," "more than," and "fewer than" • Grade 6 • The Number System • 6.NS.C.7.b

True or false?

"At least 10" includes the number 10.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q1/5

2000

QUESTION ID

S54-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q1/5

What does "at most 6" mean?

- A.** 6 or more
- B.** Less than 6
- C.** More than 6
- D.** 6 or less

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q2/5

2001

QUESTION ID

S54-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 6 or less

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q2/5

True or false?

"Fewer than 8" includes the number 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q3/5

2002

QUESTION ID

S54-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q3/5

Which number is NOT "at least 12"?

- A.** 11
- B.** 12
- C.** 13
- D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q4/5

2003

QUESTION ID

S54-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q4/5

True or false?

9 is "more than 7."

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q5/5

2004

QUESTION ID

S54-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.b • Set 54 • BACKWARD 2 • Q5/5

SET 54 • 6.EE.B.8

BACKWARD 3

Negative numbers in context • Grade 6 • The Number System • 6.NS.C.5

Write an integer for 3 degrees below zero.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q1/5

2006

QUESTION ID

S54-B3-Q1

TYPE

Short answer

CORRECT ANSWER

-3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q1

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q1/5

True or false?

-10°C is colder than -3°C .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q2/5

2007

QUESTION ID

S54-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q2

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q2/5

Which number represents 15 feet below sea level?

- A.** -15
- B.** -1.5
- C.** 0
- D.** 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q3/5

2008

QUESTION ID

S54-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. -15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q3

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q3/5

Write an integer for a temperature of 7 degrees below zero.

Integer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q4/5

2009

QUESTION ID

S54-B3-Q4

TYPE

Short answer

CORRECT ANSWER

-7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q4

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q4/5

True or false?

"Below -3°C " includes -5°C .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q5/5

2010

QUESTION ID

S54-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-B3-Q5

Grade 6 • The Number System • 6.NS.C.5 • Set 54 • BACKWARD 3 • Q5/5

SET 54 • 6.EE.B.8

FORWARD 1

Write and solve inequalities for real-world problems • Grade 7 • Expressions & Equations • 7.EE.B.4.b

You have \$50. Each ride costs \$6.

Write and solve an inequality to find how many rides r you can afford.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q1/5

2012

QUESTION ID

S54-F1-Q1

TYPE

Short answer

CORRECT ANSWER

**$6r \leq 50$; $r \leq 8 \frac{1}{3}$, so
at most 8 rides**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q1/5

A salesperson earns \$80 plus \$12 for each sale.

Write and solve an inequality for the number of sales s needed to earn more than \$200.

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q2/5

2013

QUESTION ID

S54-F1-Q2

TYPE

Short answer

CORRECT ANSWER

**$80 + 12s > 200;$
 $s > 10$ (at least 11 sales)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q2/5

A phone has 75% battery and uses 5% per hour. It must stay above 20%.

Which inequality gives the hours h ?

A. $75 - 5h > 20$

B. $75 + 5h > 20$

C. $75 - 5h < 20$

D. $5h - 75 > 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q3/5

2014

QUESTION ID

S54-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $75 - 5h > 20$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q3/5

A van can carry at most 1,200 pounds. The driver weighs 180 pounds. Each box weighs 60 pounds. Write and solve an inequality for the number of boxes b .

Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q4/5

2015

QUESTION ID

S54-F1-Q4

TYPE

Short answer

CORRECT ANSWER

**$180 + 60b \leq 1,200;$
 $b \leq 17$ (at most 17
boxes)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q4/5

True or false?

The solution of $3x + 2 \geq 17$ is $x \geq 5$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q5/5

2016

QUESTION ID

S54-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 54 • FORWARD 1 • Q5/5

FORWARD 2

Use two linear equations to decide when one quantity is greater • Grade 8 •
Expressions & Equations • 8.EE.C.8.c

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Plan A costs \$20 plus \$5 per month. Plan B costs \$35 plus \$2 per month.

After how many months does Plan A cost more than Plan B?

Months: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q1/5 • NEAREST RELATED

2018

QUESTION ID

S54-F2-Q1

TYPE

Short answer

CORRECT ANSWER

**After 5 months
($m > 5$)**

GRADING NOTE $20 + 5m > 35 + 2m$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q1/5 • NEAREST RELATED

Tank A has 100 gallons and drains 6 gallons per minute.
Tank B has 40 gallons and fills 4 gallons per minute.
After how many minutes does Tank B hold more water?

Minutes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q2/5 • NEAREST RELATED

2019

QUESTION ID

S54-F2-Q2

TYPE

Short answer

CORRECT ANSWER

**After 6 minutes
($t > 6$)**

GRADING NOTE $40 + 4t > 100 - 6t$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q2/5 • NEAREST RELATED

Sara has \$50 and saves \$8 per week. Tom has \$90 and saves \$3 per week.

When will Sara have more money than Tom?

- A.** After 8 weeks **B.** After 3 weeks
C. After 5 weeks **D.** Never

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q3/5 • NEAREST RELATED

2020

QUESTION ID

S54-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. After 8 weeks

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q3/5 • NEAREST RELATED

Gym X charges \$60 plus \$4 per class. Gym Y charges \$12 per class.

For how many classes is Gym X cheaper?

Classes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q4/5 • NEAREST RELATED

2021

QUESTION ID

S54-F2-Q4

TYPE

Short answer

CORRECT ANSWER

**8 or more classes
($c > 7.5$)**

GRADING NOTE $60 + 4c < 12c$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

$y = 3x + 2$ and $y = x + 10$ are equal when $x = 4$.
For $x > 4$, $3x + 2$ is greater.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q5/5 • NEAREST RELATED

2022

QUESTION ID

S54-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S54-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.c • Set 54 • FORWARD 2 • Q5/5 • NEAREST RELATED

Represent solutions of $x > c$ or $x < c$ on a number line

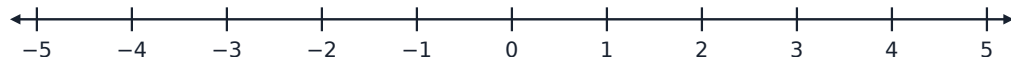
MAIN	Grade 6 main question	6.EE.B.8
BACKWARD 1	Locating numbers on a number line	6.NS.C.6.c
BACKWARD 2	Greater numbers are to the right	6.NS.C.7.a
BACKWARD 3	Meaning of inequality symbols	1.NBT.B.3
FORWARD 1	Solve and graph two-step inequalities	7.EE.B.4.b
FORWARD 2	Linear equations with one, no, or infinitely many solutions — near...	8.EE.C.7.a

SET 55 • 6.EE.B.8

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.B.8

Graph the solutions of $x > 3$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q1/5

2025

QUESTION ID

S55-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

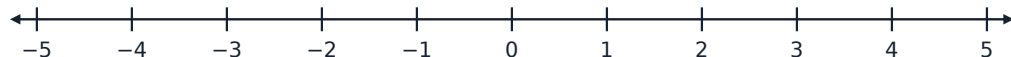
**Open circle at 3 with
the line shaded to
the right**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q1

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q1/5

Graph the solutions of $x \leq -1$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q2/5

2026

QUESTION ID

S55-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

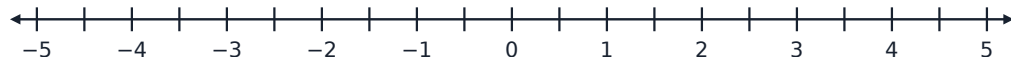
**Closed circle at -1
with the line shaded
to the left**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q2

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q2/5

Graph the solutions of $x < 2.5$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q3/5

2027

QUESTION ID

S55-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

**Open circle at 2.5
with the line shaded
to the left**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q3

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q3/5

Graph the solutions of $x \geq -4$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q4/5

2028

QUESTION ID

S55-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

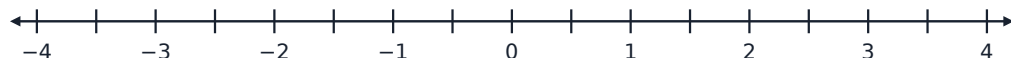
**Closed circle at -4
with the line shaded
to the right**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q4

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q4/5

Graph the solutions of $x > -1.5$ on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q5/5

2029

QUESTION ID

S55-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

**Open circle at -1.5
with the line shaded
to the right**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-M-Q5

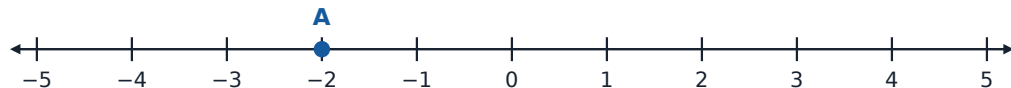
Grade 6 • Expressions & Equations • 6.EE.B.8 • Set 55 • MAIN • Q5/5

SET 55 • 6.EE.B.8

BACKWARD 1

Locating numbers on a number line • Grade 6 • The Number System •
6.NS.C.6.c

What number is at point A?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q1/5

2031

QUESTION ID

S55-B1-Q1

TYPE

Short answer

CORRECT ANSWER

-2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q1/5

Which point is at 3.5?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q2/5

2032

QUESTION ID

S55-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Point C

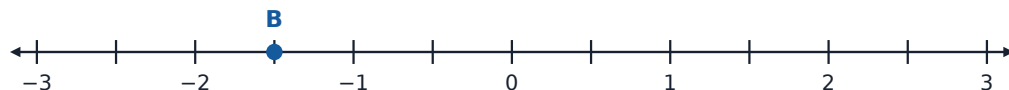
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q2/5

True or false?

Point B is at -1.5 .



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q3/5

2033

QUESTION ID

S55-B1-Q3

TYPE

True or false

CORRECT ANSWER

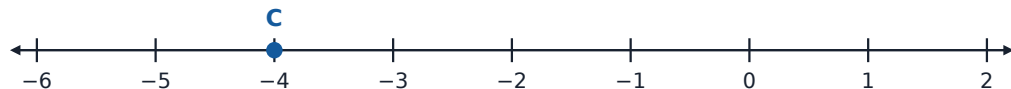
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q3/5

What number is at point C?



C = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q4/5

2034

QUESTION ID

S55-B1-Q4

TYPE

Short answer

CORRECT ANSWER

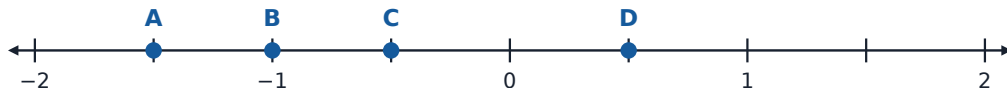
-4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q4/5

Which point is at $-\frac{1}{2}$?



A. Point A

B. Point B

C. Point C

D. Point D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q5/5

2035

QUESTION ID

S55-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. Point C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 55 • BACKWARD 1 • Q5/5

SET 55 • 6.EE.B.8

BACKWARD 2

Greater numbers are to the right • Grade 6 • The Number System • 6.NS.C.7.a

True or false?

Every number to the right of 3 on a number line is greater than 3.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q1/5

2037

QUESTION ID

S55-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q1/5

Which numbers are less than -1 ?

- A.** Numbers to the right of -1
- B.** Numbers to the left of -1
- C.** Only -2
- D.** Only 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q2/5

2038

QUESTION ID

S55-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Numbers to the left of -1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q2/5

True or false?

-3 is greater than -1 .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q3/5

2039

QUESTION ID

S55-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q3/5

Is 2.5 to the left or the right of 2 on a number line?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q4/5

2040

QUESTION ID

S55-B2-Q4

TYPE

Short answer

CORRECT ANSWER

To the right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q4/5

Which number is greater than -4 ?

- A. -6
- B. -5
- C. -4.5
- D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q5/5

2041

QUESTION ID

S55-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. -3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.a • Set 55 • BACKWARD 2 • Q5/5

SET 55 • 6.EE.B.8

BACKWARD 3

Meaning of inequality symbols • Grade 1 • Number & Operations in Base Ten •
1.NBT.B.3

What does the symbol $>$ mean?

- A.** Is equal to
- B.** Is greater than
- C.** Is less than

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q1/5

2043

QUESTION ID

S55-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Is greater than

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q1

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q1/5

True or false?

The symbol $<$ means "is less than."

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q2/5

2044

QUESTION ID

S55-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q2

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q2/5

Which statement is true?

- A.** $8 < 5$
- B.** $8 = 5$
- C.** $8 > 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q3/5

2045

QUESTION ID

S55-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $8 > 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q3

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q3/5

Write the symbol for "is less than."

Symbol: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q4

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q4/5

2046

QUESTION ID

S55-B3-Q4

TYPE

Short answer

CORRECT ANSWER

<

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q4

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q4/5

True or false?

12 > 21

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q5

Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q5/5

2047

QUESTION ID

S55-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-B3-Q5

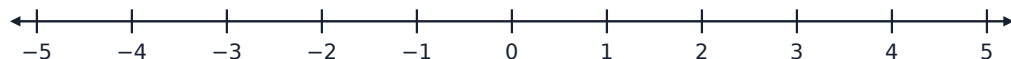
Grade 1 • Number & Operations in Base Ten • 1.NBT.B.3 • Set 55 • BACKWARD 3 • Q5/5

SET 55 • 6.EE.B.8

FORWARD 1

Solve and graph two-step inequalities • Grade 7 • Expressions & Equations •
7.EE.B.4.b

Solve $2x + 1 > 7$. Then graph the solutions.



Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q1/5

2049

QUESTION ID

S55-F1-Q1

TYPE

Short answer

CORRECT ANSWER

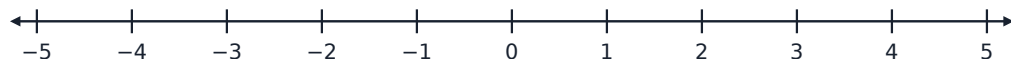
$x > 3$; open circle at 3, shaded to the right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q1

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q1/5

Solve $4x - 3 \leq 5$. Then graph the solutions.



Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q2/5

2050

QUESTION ID

S55-F1-Q2

TYPE

Short answer

CORRECT ANSWER

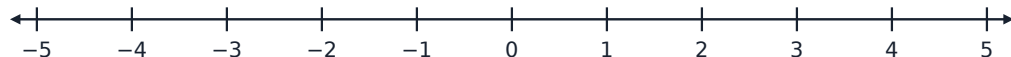
**$x \leq 2$; closed circle
at 2, shaded to the
left**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q2

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q2/5

Solve $-2x + 4 < 10$. Then graph the solutions.



Solution: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q3/5

2051

QUESTION ID

S55-F1-Q3

TYPE

Short answer

CORRECT ANSWER

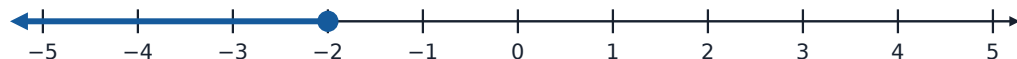
**$x > -3$; open circle
at -3 , shaded to the
right**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q3

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q3/5

Which inequality does the graph show?



A. $x < -2$

B. $x > -2$

C. $x \geq -2$

D. $x \leq -2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q4/5

2052

QUESTION ID

S55-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $x \leq -2$

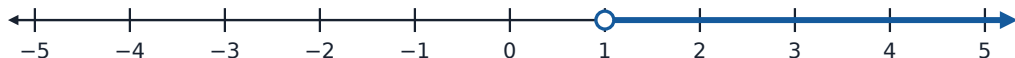
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q4

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q4/5

True or false?

The graph shows the solutions of $3x + 2 > 5$.



- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q5/5

2053

QUESTION ID

S55-F1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F1-Q5

Grade 7 • Expressions & Equations • 7.EE.B.4.b • Set 55 • FORWARD 1 • Q5/5

FORWARD 2

Linear equations with one, no, or infinitely many solutions • Grade 8 •
Expressions & Equations • 8.EE.C.7.a

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

True or false?

$x + 3 = x + 3$ has infinitely many solutions.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q1/5 • NEAREST RELATED

2055

QUESTION ID

S55-F2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q1/5 • NEAREST RELATED

How many solutions does the equation have?

$$2x + 5 = 2x - 1$$

- A. One solution
- B. No solution
- C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q2/5 • NEAREST RELATED

2056

QUESTION ID

S55-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. No solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q2/5 • NEAREST RELATED

Which equation has infinitely many solutions?

- A. $4x - 1 = x$
- B. $4x = 0$
- C. $4(x - 1) = 4x - 4$
- D. $4(x - 1) = 4x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q3/5 • NEAREST RELATED

2057

QUESTION ID

S55-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. $4(x - 1) = 4x - 4$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q3/5 • NEAREST RELATED

How many solutions does the equation have?

$$3x - 7 = 3x - 7$$

- A. One solution
- B. No solution
- C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q4/5 • NEAREST RELATED

2058

QUESTION ID

S55-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q4/5 • NEAREST RELATED

How many solutions does the equation have?

$$6x - 2 = 4x + 8$$

- A.** One solution
- B.** No solution
- C.** Infinitely many solutions

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q5/5 • NEAREST RELATED

2059

QUESTION ID

S55-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. One solution

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S55-F2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.7.a • Set 55 • FORWARD 2 • Q5/5 • NEAREST RELATED

Identify the independent and dependent variables

MAIN Grade 6 main question

6.EE.C.9

BACKWARD 1 Input-output rules

4.OA.C.5

BACKWARD 2 Two related number patterns

5.OA.B.3

FORWARD 1 Write an equation for a proportional relationship

7.RP.A.2.c

FORWARD 2 Functions: each input has exactly one output

8.F.A.1

SET 56 • 6.EE.C.9

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.C.9

The equation $c = 2.5p$ gives the cost c of p pounds of grapes.

Which is the independent variable?

- A.** p (pounds)
- B.** c (cost)
- C.** 2.5
- D.** There is none.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q1/5

2062

QUESTION ID

S56-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. p (pounds)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q1/5

Money earned m depends on the number of hours worked h .

Which is the dependent variable?

Dependent variable: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q2/5

2063

QUESTION ID

S56-M-Q2

TYPE

Short answer

CORRECT ANSWER

m (money earned)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q2/5

A plant's height depends on the number of days it grows.

Which is the independent variable?

- A.** The plant's height **B.** The number of days
C. Neither **D.** Both

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q3/5

2064

QUESTION ID

S56-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. The number of days

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q3/5

In $y = 4x + 3$, which variable is the dependent variable?

Dependent variable: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q4/5

2065

QUESTION ID

S56-M-Q4

TYPE

Short answer

CORRECT ANSWER

y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q4/5

True or false?

Ice cream sales s depend on the temperature t .
So t is the independent variable.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q5/5

2066

QUESTION ID

S56-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-M-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 56 • MAIN • Q5/5

SET 56 • 6.EE.C.9

BACKWARD 1

Input-output rules • Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5

The rule is "multiply by 3."

What is the output when the input is 7?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q1/5

2068

QUESTION ID

S56-B1-Q1

TYPE

Short answer

CORRECT ANSWER

21

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q1/5

True or false?

The rule is "add 6." When the input is 4, the output is 10.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q2/5

2069

QUESTION ID

S56-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q2/5

What is the rule?

In	2	4	6
Out	6	8	10

- A.** Multiply by 2 **B.** Multiply by 3
C. Add 2 **D.** Add 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q3/5

2070

QUESTION ID

S56-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Add 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q3/5

The rule is "subtract 2."

What is the output when the input is 15?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q4/5

2071

QUESTION ID

S56-B1-Q4

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q4/5

True or false?

In an input-output table, the output depends on the input.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q5/5

2072

QUESTION ID

S56-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 56 • BACKWARD 1 • Q5/5

SET 56 • 6.EE.C.9

BACKWARD 2

Two related number patterns • Grade 5 • Operations & Algebraic Thinking •
5.OA.B.3

True or false?

Pattern A: 0, 2, 4, 6

Pattern B: 0, 6, 12, 18

Each number in Pattern B is 3 times the matching number in Pattern A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q1/5

2074

QUESTION ID

S56-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q1/5

The rule is "add 5," starting at 0.

What number comes next? 0, 5, 10, ____

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q2/5

2075

QUESTION ID

S56-B2-Q2

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q2/5

Pattern A starts at 0 and adds 3. Pattern B starts at 0 and adds 9.

When Pattern A is 12, what is the matching number in Pattern B?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q3/5

2076

QUESTION ID

S56-B2-Q3

TYPE

Short answer

CORRECT ANSWER

36

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q3/5

Pattern A starts at 0 and adds 1. Pattern B starts at 0 and adds 4.

Which ordered pairs come from the patterns?

- A.** (0, 4), (1, 5), (2, 6) **B.** (0, 0), (1, 4), (2, 8)
C. (1, 4), (2, 5), (3, 6) **D.** (0, 0), (4, 1), (8, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q4/5

2077

QUESTION ID

S56-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. (0, 0), (1, 4), (2, 8)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q4/5

True or false?

Pattern A: 0, 1, 2, 3

Pattern B: 0, 5, 10, 15

Each number in Pattern B is 5 more than the matching number in Pattern A.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q5/5

2078

QUESTION ID

S56-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-B2-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 56 • BACKWARD 2 • Q5/5

SET 56 • 6.EE.C.9

FORWARD 1

Write an equation for a proportional relationship • Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c

Tomatoes cost \$3.20 per pound.

Write an equation for the cost c of p pounds.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q1/5

2080

QUESTION ID

S56-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$c = 3.2p$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q1/5

The table shows a proportional relationship.
Write an equation for the cost c of n ounces.

Ounces (n)	4	10
Cost (c)	\$1.20	\$3.00

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q2/5

2081

QUESTION ID

S56-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$c = 0.3n$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q2/5

Which equation represents a proportional relationship?

- A. $y = 4.5x$
- B. $y = 4.5x + 1$
- C. $y = x + 4.5$
- D. $y = 4.5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q3/5

2082

QUESTION ID

S56-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $y = 4.5x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q3/5

A car uses 3 gallons of gas to travel 84 miles.

Write an equation for the miles m traveled on g gallons.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q4/5

2083

QUESTION ID

S56-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$m = 28g$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q4/5

True or false?

$y = \frac{2}{3}x$ represents a proportional relationship.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q5/5

2084

QUESTION ID

S56-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 56 • FORWARD 1 • Q5/5

SET 56 • 6.EE.C.9

FORWARD 2

Functions: each input has exactly one output • Grade 8 • Functions • 8.F.A.1

True or false?

A function assigns exactly one output to each input.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q1/5

2086

QUESTION ID

S56-F2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q1

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q1/5

Which table does NOT represent a function?

- A. x: 0, 1, 2 y: 0, 2, 4
- B. x: 1, 1, 2 y: 3, 5, 7
- C. x: 2, 3, 4 y: 5, 6, 7
- D. x: 1, 2, 3 y: 4, 4, 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q2/5

2087

QUESTION ID

S56-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. x: 1, 1, 2 y: 3, 5, 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q2

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q2/5

In the function $C = 12t$, which variable is the input?

Input: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q3/5

2088

QUESTION ID

S56-F2-Q3

TYPE

Short answer

CORRECT ANSWER

t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q3

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q3/5

True or false?

The set of ordered pairs (1, 2), (2, 3), (1, 4) represents a function.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q4/5

2089

QUESTION ID

S56-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q4

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q4/5

A function gives the height of a ball after t seconds.
What is the input?

- A. The speed
- B. The ball
- C. The height
- D. The time t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q5/5

2090

QUESTION ID

S56-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. The time t

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S56-F2-Q5

Grade 8 • Functions • 8.F.A.1 • Set 56 • FORWARD 2 • Q5/5

Write an equation for the relationship shown in a table or graph

MAIN	Grade 6 main question	6.EE.C.9
BACKWARD 1	Finding a rule from a table	4.OA.C.5
BACKWARD 2	Reading points on a graph	5.G.A.2
BACKWARD 3	Ordered pairs from two patterns	5.OA.B.3
FORWARD 1	Equations for proportional relationships from tables and graphs	7.RP.A.2.c
FORWARD 2	Write a linear function from a table or graph	8.F.B.4

SET 57 • 6.EE.C.9

MAIN

Grade 6 main question • Grade 6 • Expressions & Equations • 6.EE.C.9

Write an equation that shows how y depends on x .

x	1	2	3	4
y	6	12	18	24

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q1/5

2093

QUESTION ID

S57-M-Q1

TYPE

Short answer

CORRECT ANSWER

$y = 6x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

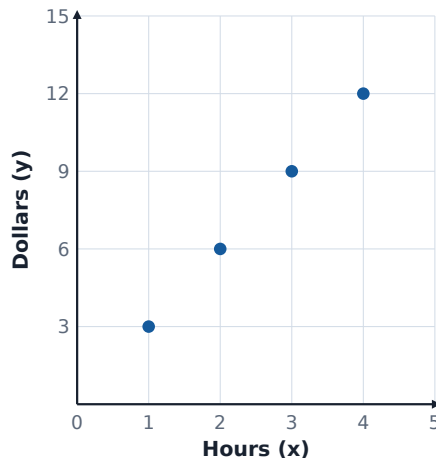
ID S57-M-Q1

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q1/5

The graph shows y dollars earned for x hours worked.

Write an equation for the relationship.

$y =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q2/5

2094

QUESTION ID

S57-M-Q2

TYPE

Short answer

CORRECT ANSWER

$y = 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q2

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q2/5

The table shows a plant's height h after w weeks.

Write an equation for h in terms of w .

Weeks (w)	0	1	2	3
Height (h)	5	7	9	11

$h =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q3/5

2095

QUESTION ID

S57-M-Q3

TYPE

Short answer

CORRECT ANSWER

$h = 2w + 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q3

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q3/5

Which equation matches the table?

x	1	2	3
y	4	5	6

A. $y = 4x$

B. $y = x + 4$

C. $y = x + 3$

D. $y = 3x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q4/5

2096

QUESTION ID

S57-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $y = x + 3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q4

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q4/5

The table shows the cost c of n bags of popcorn.

Write an equation for c in terms of n .

Bags (n)	2	4	6
Cost (c)	7	14	21

$C =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q5/5

2097

QUESTION ID

S57-M-Q5

TYPE

Short answer

CORRECT ANSWER

$c = 3.5n$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-M-Q5

Grade 6 • Expressions & Equations • 6.EE.C.9 • Set 57 • MAIN • Q5/5

SET 57 • 6.EE.C.9

BACKWARD 1

Finding a rule from a table • Grade 4 • Operations & Algebraic Thinking •
4.OA.C.5

What is the rule?

In	1	2	3	4
Out	5	10	15	20

Rule: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q1/5

2099

QUESTION ID

S57-B1-Q1

TYPE

Short answer

CORRECT ANSWER

Multiply by 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q1

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q1/5

What is the rule?

In	3	5	7
Out	7	9	11

Rule: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q2/5

2100

QUESTION ID

S57-B1-Q2

TYPE

Short answer

CORRECT ANSWER

Add 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q2

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q2/5

True or false?

The rule for the table is "multiply by 4."

In	1	2	3
Out	4	8	12

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q3/5

2101

QUESTION ID

S57-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q3

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q3/5

What is the rule?

In	3	6	9
Out	1	2	3

- A.** Multiply by 3 **B.** Subtract 2
C. Add 3 **D.** Divide by 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q4/5

2102

QUESTION ID

S57-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Divide by 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q4

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q4/5

The rule is "multiply by 6." What is the output when the input is 5?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q5/5

2103

QUESTION ID

S57-B1-Q5

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B1-Q5

Grade 4 • Operations & Algebraic Thinking • 4.OA.C.5 • Set 57 • BACKWARD 1 • Q5/5

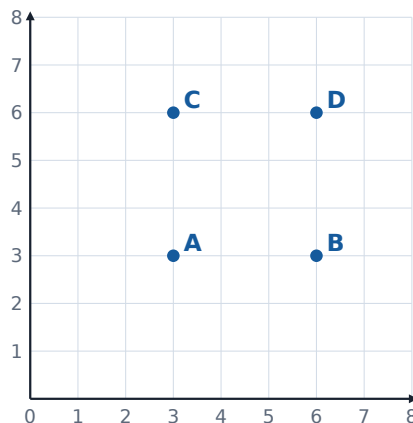
SET 57 • 6.EE.C.9

BACKWARD 2

Reading points on a graph • Grade 5 • Geometry • 5.G.A.2

Which point is at (3, 6)?

- A. Point A
- B. Point B
- C. Point C
- D. Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q1/5

2105

QUESTION ID

S57-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Point C

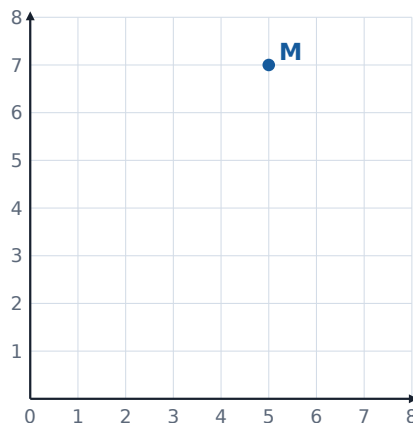
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q1

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q1/5

What are the coordinates of point M?

M = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q2/5

2106

QUESTION ID

S57-B2-Q2

TYPE

Short answer

CORRECT ANSWER

(5, 7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q2

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q2/5

True or false?

The point (2, 8) is 2 units right and 8 units up from the origin.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q3/5

2107

QUESTION ID

S57-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

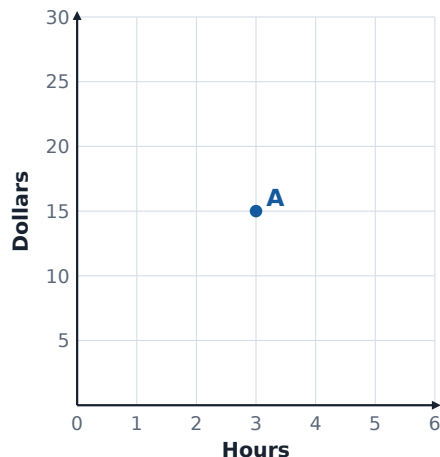
ID S57-B2-Q3

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q3/5

The graph shows dollars earned after working some hours.

How many dollars were earned after 3 hours?

Dollars: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q4/5

2108

QUESTION ID

S57-B2-Q4

TYPE

Short answer

CORRECT ANSWER

\$15

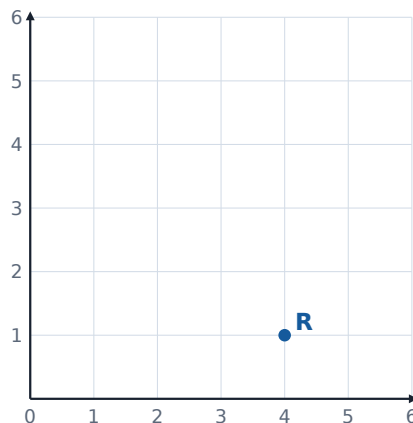
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q4

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q4/5

Which ordered pair names point R?

- A. (4, 4)
- B. (1, 4)
- C. (1, 1)
- D. (4, 1)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q5/5

2109

QUESTION ID

S57-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. (4, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B2-Q5

Grade 5 • Geometry • 5.G.A.2 • Set 57 • BACKWARD 2 • Q5/5

SET 57 • 6.EE.C.9

BACKWARD 3

Ordered pairs from two patterns • Grade 5 • Operations & Algebraic Thinking •
5.OA.B.3

Pattern A: 0, 1, 2, 3

Pattern B: 0, 5, 10, 15

Write the ordered pair formed by the 4th numbers.

Ordered pair: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q1/5

2111

QUESTION ID

S57-B3-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 15)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q1

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q1/5

True or false?

The pairs (1, 3), (2, 6), (3, 9) show that each y-value is 3 times the x-value.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q2/5

2112

QUESTION ID

S57-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q2

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q2/5

Pattern A adds 2. Pattern B adds 6. Both start at 0.
How does each B number compare to its A number?

- A. 2 more
- B. 3 times as large
- C. 4 more
- D. 6 times as large

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q3/5

2113

QUESTION ID

S57-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 3 times as large

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q3

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q3/5

Pattern A: 0, 2, 4

Pattern B: 0, 14, 28

What is the rule that changes each A number into its B number?

Rule: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q4/5

2114

QUESTION ID

S57-B3-Q4

TYPE

Short answer

CORRECT ANSWER

Multiply by 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q4

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q4/5

True or false?

The pairs (0, 0), (1, 4), (2, 8) come from Pattern A "add 4" and Pattern B "add 1."

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q5/5

2115

QUESTION ID

S57-B3-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-B3-Q5

Grade 5 • Operations & Algebraic Thinking • 5.OA.B.3 • Set 57 • BACKWARD 3 • Q5/5

SET 57 • 6.EE.C.9

FORWARD 1

Equations for proportional relationships from tables and graphs • Grade 7 •
Ratios & Proportional Relationships • 7.RP.A.2.c

Write an equation for the proportional relationship in the table.

x	2	5	8
y	7	17.5	28

$y =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q1/5

2117

QUESTION ID

S57-F1-Q1

TYPE

Short answer

CORRECT ANSWER

$y = 3.5x$

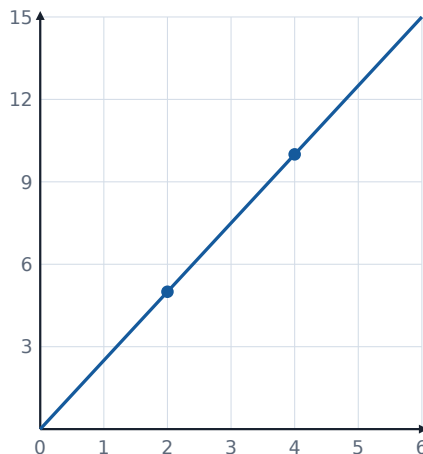
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q1

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q1/5

Write an equation for the proportional relationship in the graph.

$y =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q2/5

2118

QUESTION ID

S57-F1-Q2

TYPE

Short answer

CORRECT ANSWER

$y = 2.5x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q2

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q2/5

The table shows a proportional relationship.

Which equation matches?

x	4	8	12
y	3	6	9

A. $y = 0.75x$

B. $y = 1.33x$

C. $y = 3x$

D. $y = x - 1$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q3/5

2119

QUESTION ID

S57-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $y = 0.75x$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q3

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q3/5

Write an equation for the cost c of p pounds.

Pounds (p)	3	6	9
Cost (c)	7.50	15.00	22.50

$C =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q4/5

2120

QUESTION ID

S57-F1-Q4

TYPE

Short answer

CORRECT ANSWER

$c = 2.5p$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q4

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q4/5

True or false?

The table shows the equation $y = 6x$.

x	1	2	4
y	6	12	24

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q5/5

2121

QUESTION ID

S57-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F1-Q5

Grade 7 • Ratios & Proportional Relationships • 7.RP.A.2.c • Set 57 • FORWARD 1 • Q5/5

SET 57 • 6.EE.C.9

FORWARD 2

Write a linear function from a table or graph • Grade 8 • Functions • 8.F.B.4

Write a function for the table.

x	0	1	2	3
y	4	7	10	13

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q1/5

2123

QUESTION ID

S57-F2-Q1

TYPE

Short answer

CORRECT ANSWER

$$y = 3x + 4$$

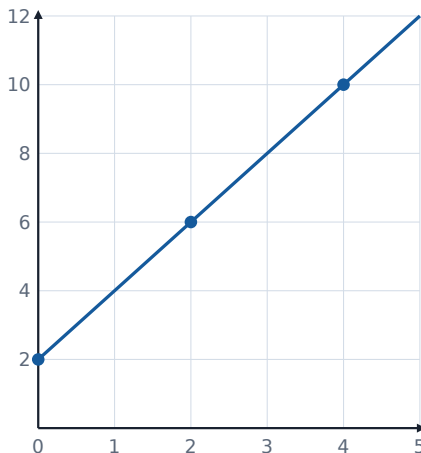
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q1

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q1/5

Write a function for the line in the graph.

$y =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q2/5

2124

QUESTION ID

S57-F2-Q2

TYPE

Short answer

CORRECT ANSWER

$y = 2x + 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q2

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q2/5

Which function matches the table?

x	0	1	2
y	10	8	6

A. $y = -2x + 10$

B. $y = 10x - 2$

C. $y = -2x$

D. $y = 2x + 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q3/5

2125

QUESTION ID

S57-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. $y = -2x + 10$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q3

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q3/5

Write a function for the table.

x	1	2	3	4
y	9	13	17	21

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q4/5

2126

QUESTION ID

S57-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$$y = 4x + 5$$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q4

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q4/5

True or false?

The line through (0, 1) and (3, 7) has the equation $y = 2x + 1$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q5/5

2127

QUESTION ID

S57-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S57-F2-Q5

Grade 8 • Functions • 8.F.B.4 • Set 57 • FORWARD 2 • Q5/5

Area of right triangles and other triangles

MAIN

Grade 6 main question

6.G.A.1

BACKWARD 1

Area of rectangles

3.MD.C.7.b

BACKWARD 2

Half of a product

5.NF.B.4.a

BACKWARD 3

Right angles and perpendicular segments

4.G.A.1

FORWARD 1

Area and volume of figures made of triangles

7.G.B.6

FORWARD 2

Use the Pythagorean Theorem in right triangles

8.G.B.7

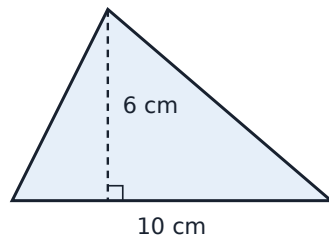
SET 58 • 6.G.A.1

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.1

Find the area of the triangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q1/5

2130

QUESTION ID

S58-M-Q1

TYPE

Short answer

CORRECT ANSWER

30 square cm

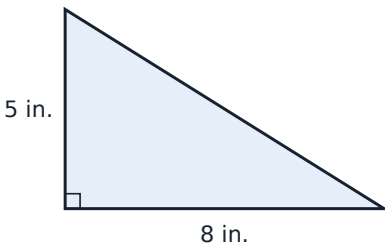
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q1/5

Find the area of the right triangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q2/5

2131

QUESTION ID

S58-M-Q2

TYPE

Short answer

CORRECT ANSWER

20 square inches

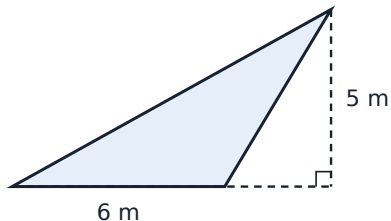
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q2/5

Find the area of the triangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q3/5

2132

QUESTION ID

S58-M-Q3

TYPE

Short answer

CORRECT ANSWER

15 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q3/5

A triangle has a base of 12 feet and a height of 7 feet.

What is its area?

- A. 19 square feet
- B. 38 square feet
- C. 42 square feet
- D. 84 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q4/5

2133

QUESTION ID

S58-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 42 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q4/5

2133

A triangular sign has a base of 3.5 feet and a height of 2 feet.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q5/5

2134

QUESTION ID

S58-M-Q5

TYPE

Short answer

CORRECT ANSWER

3.5 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-M-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 58 • MAIN • Q5/5

SET 58 • 6.G.A.1

BACKWARD 1

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

Find the area of the rectangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q1/5

2136

QUESTION ID

S58-B1-Q1

TYPE

Short answer

CORRECT ANSWER

28 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q1/5

A rectangle is 9 feet long and 6 feet wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q2/5

2137

QUESTION ID

S58-B1-Q2

TYPE

Short answer

CORRECT ANSWER

54 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q2/5

True or false?

A rectangle that is 5 cm long and 3 cm wide has an area of 15 square centimeters.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q3/5

2138

QUESTION ID

S58-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q3/5

A rectangle is 12 m long and 4 m wide.

What is its area?

- A.** 16 square meters
- B.** 24 square meters
- C.** 32 square meters
- D.** 48 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q4/5

2139

QUESTION ID

S58-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 48 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q4/5

Find the area of the rectangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q5/5

2140

QUESTION ID

S58-B1-Q5

TYPE

Short answer

CORRECT ANSWER

60 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 58 • BACKWARD 1 • Q5/5

SET 58 • 6.G.A.1

BACKWARD 2

Half of a product • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{1}{2} \times 10 \times 6$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q1/5

2142

QUESTION ID

S58-B2-Q1

TYPE

Short answer

CORRECT ANSWER

30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q1/5

Multiply.

$$\frac{1}{2} \times 14$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q2/5

2143

QUESTION ID

S58-B2-Q2

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q2/5

True or false?

$$\frac{1}{2} \times 8 \times 5 = 20$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q3/5

2144

QUESTION ID

S58-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q3/5

$$\frac{1}{2} \times 12 \times 7 = ?$$

- A. 19
- B. 21
- C. 42
- D. 84

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q4/5

2145

QUESTION ID

S58-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q4/5

Multiply.

$$\frac{1}{2} \times 9 \times 4$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q5/5

2146

QUESTION ID

S58-B2-Q5

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 58 • BACKWARD 2 • Q5/5

SET 58 • 6.G.A.1

BACKWARD 3

Right angles and perpendicular segments • Grade 4 • Geometry • 4.G.A.1

True or false?

Perpendicular lines meet to form right angles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q1/5

2148

QUESTION ID

S58-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q1/5

Which angle measure is a right angle?

- A.** 45°
- B.** 90°
- C.** 100°
- D.** 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q2/5

2149

QUESTION ID

S58-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

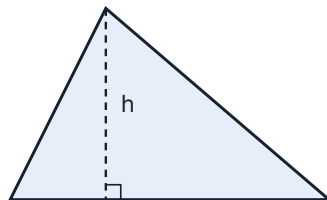
ID S58-B3-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q2/5

True or false?

The dashed segment is perpendicular to the base of the triangle.

- A. True
- B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q3/5

2150

QUESTION ID

S58-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q3/5

How many degrees are in a right angle?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q4/5

2151

QUESTION ID

S58-B3-Q4

TYPE

Short answer

CORRECT ANSWER

90 degrees

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q4/5

True or false?

A square has 4 right angles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q5/5

2152

QUESTION ID

S58-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-B3-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 58 • BACKWARD 3 • Q5/5

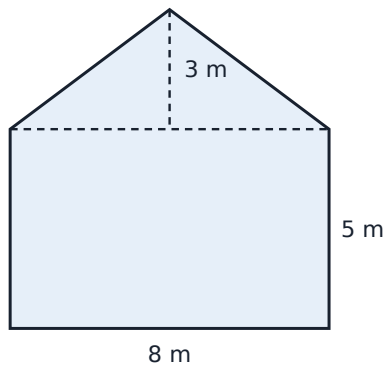
SET 58 • 6.G.A.1

FORWARD 1

Area and volume of figures made of triangles • Grade 7 • Geometry • 7.G.B.6

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q1/5

2154

QUESTION ID

S58-F1-Q1

TYPE

Short answer

CORRECT ANSWER

52 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q1/5

A kite is made of two triangles. Each triangle has a base of 10 inches and a height of 4 inches.

What is the area of the kite?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q2/5

2155

QUESTION ID

S58-F1-Q2

TYPE

Short answer

CORRECT ANSWER

40 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q2/5

A triangle with a base of 4 cm and a height of 3 cm is cut out of a rectangle that is 12 cm by 6 cm.

What is the area of the part that is left?

- A.** 60 square cm **B.** 66 square cm
C. 72 square cm **D.** 78 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q3/5

2156

QUESTION ID

S58-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 66 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q3/5

A triangular prism has triangle bases with a base of 6 cm and a height of 4 cm. The prism is 10 cm long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q4/5

2157

QUESTION ID

S58-F1-Q4

TYPE

Short answer

CORRECT ANSWER

120 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q4/5

True or false?

A figure made of a 6-by-4 rectangle and a triangle with a base of 3 and a height of 4 has an area of 30 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q5/5

2158

QUESTION ID

S58-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 58 • FORWARD 1 • Q5/5

SET 58 • 6.G.A.1

FORWARD 2

Use the Pythagorean Theorem in right triangles • Grade 8 • Geometry • 8.G.B.7

A right triangle has legs of 6 cm and 8 cm.

What is the length of the hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q1/5

2160

QUESTION ID

S58-F2-Q1

TYPE

Short answer

CORRECT ANSWER

10 cm

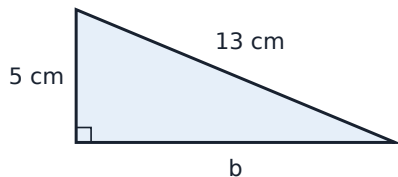
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q1/5

Find the length b .

$b =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q2/5

2161

QUESTION ID

S58-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q2/5

A 10-foot ladder leans against a wall. The top reaches 8 feet up the wall.

How far is the bottom of the ladder from the wall?

- A.** 2 feet **B.** 6 feet
C. 12.8 feet **D.** 18 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q3/5

2162

QUESTION ID

S58-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 6 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q3/5

An isosceles triangle has a base of 16 inches and two sides of 10 inches.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q4/5

2163

QUESTION ID

S58-F2-Q4

TYPE

Short answer

CORRECT ANSWER

48 square inches

GRADING NOTE Height = 6 in. by the Pythagorean Theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q4/5

A right triangle has legs of 9 m and 12 m.

What is the length of the hypotenuse?

Hypotenuse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q5/5

2164

QUESTION ID

S58-F2-Q5

TYPE

Short answer

CORRECT ANSWER

15 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S58-F2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 58 • FORWARD 2 • Q5/5

Area of parallelograms and trapezoids

MAIN	Grade 6 main question	6.G.A.1
BACKWARD 1	Area of rectangles	3.MD.C.7.b
BACKWARD 2	Area by splitting into rectangles	3.MD.C.7.d
BACKWARD 3	Classifying quadrilaterals	5.G.B.4
BACKWARD 4	Height is perpendicular to the base	4.G.A.1
FORWARD 1	Area of figures made of quadrilaterals in real-world problems	7.G.B.6
FORWARD 2	Use the Pythagorean Theorem to find a missing height	8.G.B.7

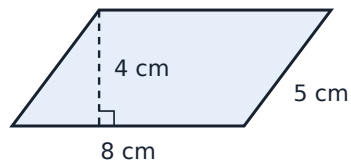
SET 59 • 6.G.A.1

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.1

Find the area of the parallelogram.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q1/5

2167

QUESTION ID

S59-M-Q1

TYPE

Short answer

CORRECT ANSWER

32 square cm

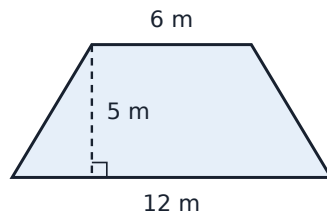
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q1/5

Find the area of the trapezoid.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q2/5

2168

QUESTION ID

S59-M-Q2

TYPE

Short answer

CORRECT ANSWER

45 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q2/5

A parallelogram has a base of 9 feet and a height of 5 feet.

What is its area?

- A.** 14 square feet **B.** 22.5 square feet
C. 28 square feet **D.** 45 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q3/5

2169

QUESTION ID

S59-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 45 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q3/5

A trapezoid has bases of 7 inches and 11 inches. Its height is 4 inches.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q4/5

2170

QUESTION ID

S59-M-Q4

TYPE

Short answer

CORRECT ANSWER

36 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q4/5

A parallelogram has a base of 12.5 meters and a height of 4 meters.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q5/5

2171

QUESTION ID

S59-M-Q5

TYPE

Short answer

CORRECT ANSWER

50 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-M-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 59 • MAIN • Q5/5

SET 59 • 6.G.A.1

BACKWARD 1

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

Find the area of the rectangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q1/5

2173

QUESTION ID

S59-B1-Q1

TYPE

Short answer

CORRECT ANSWER

24 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q1/5

A rectangle is 11 cm long and 5 cm wide.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q2/5

2174

QUESTION ID

S59-B1-Q2

TYPE

Short answer

CORRECT ANSWER

55 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q2/5

True or false?

A square with 6-meter sides has an area of 24 square meters.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q3/5

2175

QUESTION ID

S59-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q3/5

A rectangle is 15 inches long and 2 inches wide.

What is its area?

- A. 17 square inches
- B. 30 square inches
- C. 34 square inches
- D. 60 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q4/5

2176

QUESTION ID

S59-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 30 square inches

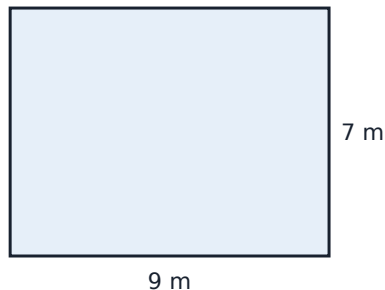
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q4/5

Find the area of the rectangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q5/5

2177

QUESTION ID

S59-B1-Q5

TYPE

Short answer

CORRECT ANSWER

63 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 59 • BACKWARD 1 • Q5/5

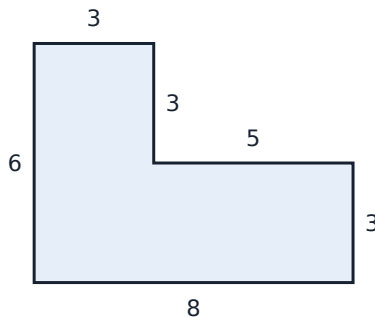
SET 59 • 6.G.A.1

BACKWARD 2

Area by splitting into rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.d

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q1/5

2179

QUESTION ID

S59-B2-Q1

TYPE

Short answer

CORRECT ANSWER

33 square units

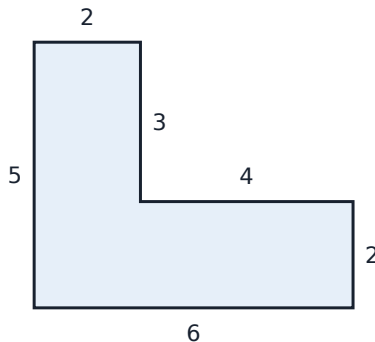
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q1/5

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q2/5

2180

QUESTION ID

S59-B2-Q2

TYPE

Short answer

CORRECT ANSWER

18 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q2/5

True or false?

The area of an L-shaped figure can be found by adding the areas of two rectangles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q3/5

2181

QUESTION ID

S59-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q3/5

A figure is made of a 4-by-5 rectangle and a 2-by-3 rectangle that do not overlap.

What is its area?

- A.** 14 square units **B.** 20 square units
C. 26 square units **D.** 120 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q4/5

2182

QUESTION ID

S59-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 26 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q4/5

A figure is made of a 10-by-2 rectangle and a 3-by-3 square that do not overlap.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q5/5

2183

QUESTION ID

S59-B2-Q5

TYPE

Short answer

CORRECT ANSWER

29 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B2-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 59 • BACKWARD 2 • Q5/5

SET 59 • 6.G.A.1

BACKWARD 3

Classifying quadrilaterals • Grade 5 • Geometry • 5.G.B.4

True or false?

Every rectangle is a parallelogram.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q1/5

2185

QUESTION ID

S59-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q1

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q1/5

True or false?

Every parallelogram is a rectangle.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q2/5

2186

QUESTION ID

S59-B3-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q2

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q2/5

Which quadrilateral always has four right angles?

- A.** Trapezoid
- B.** Rhombus
- C.** Parallelogram
- D.** Rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q3/5

2187

QUESTION ID

S59-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q3

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q3/5

How many pairs of parallel sides does a parallelogram have?

- A.** 0
- B.** 1
- C.** 2
- D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q4/5

2188

QUESTION ID

S59-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q4

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q4/5

True or false?

A square is a rhombus.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q5/5

2189

QUESTION ID

S59-B3-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B3-Q5

Grade 5 • Geometry • 5.G.B.4 • Set 59 • BACKWARD 3 • Q5/5

SET 59 • 6.G.A.1

BACKWARD 4

Height is perpendicular to the base • Grade 4 • Geometry • 4.G.A.1

True or false?

The height of a parallelogram is perpendicular to its base.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q1/5

2191

QUESTION ID

S59-B4-Q1

TYPE

True or false

CORRECT ANSWER

A. True

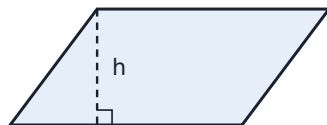
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q1/5

True or false?

The dashed segment shows the height of the parallelogram.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q2/5

2192

QUESTION ID

S59-B4-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q2/5

Which statement describes perpendicular lines?

- A. They meet at right angles.
- B. They never meet.
- C. They meet at a 45° angle.
- D. They are always the same length.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q3/5

2193

QUESTION ID

S59-B4-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. They meet at right angles.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q3/5

True or false?

The slanted side of a parallelogram is always its height.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q4/5

2194

QUESTION ID

S59-B4-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q4/5

What is the measure of the angle between a base and its height?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q5/5

2195

QUESTION ID

S59-B4-Q5

TYPE

Short answer

CORRECT ANSWER

90 degrees

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-B4-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 59 • BACKWARD 4 • Q5/5

SET 59 • 6.G.A.1

FORWARD 1

Area of figures made of quadrilaterals in real-world problems • Grade 7 •
Geometry • 7.G.B.6

A patio is made of a rectangle that is 10 ft by 6 ft and a trapezoid with bases of 6 ft and 4 ft and a height of 3 ft.
What is the area of the patio?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q1/5

2197

QUESTION ID

S59-F1-Q1

TYPE

Short answer

CORRECT ANSWER

75 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q1/5

A wall is 12 ft by 8 ft. It has a window shaped like a parallelogram with a base of 3 ft and a height of 2 ft.

What is the area of the wall that is not window?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q2/5

2198

QUESTION ID

S59-F1-Q2

TYPE

Short answer

CORRECT ANSWER

90 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q2/5

A sign is a trapezoid with bases of 1.5 m and 2.5 m and a height of 1.2 m.

What is its area?

- A.** 2.4 square meters **B.** 3.6 square meters
C. 4.8 square meters **D.** 5.2 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q3/5

2199

QUESTION ID

S59-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 2.4 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q3/5

A garden is made of two parallelograms that do not overlap. Each has a base of 7 m and a height of 4 m.
What is the total area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q4/5

2200

QUESTION ID

S59-F1-Q4

TYPE

Short answer

CORRECT ANSWER

56 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q4/5

True or false?

A trapezoid with bases of 6 and 10 and a height of 5 has the same area as a rectangle that is 8 by 5.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q5/5

2201

QUESTION ID

S59-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 59 • FORWARD 1 • Q5/5

SET 59 • 6.G.A.1

FORWARD 2

Use the Pythagorean Theorem to find a missing height • Grade 8 • Geometry •
8.G.B.7

A parallelogram has a slanted side of 5 cm. The slanted side, the height, and a 3 cm piece of the base form a right triangle.

What is the height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q1/5

2203

QUESTION ID

S59-F2-Q1

TYPE

Short answer

CORRECT ANSWER

4 cm

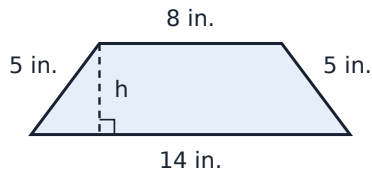
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q1/5

Find the height h of the trapezoid.

$h =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q2/5

2204

QUESTION ID

S59-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q2/5

A ramp forms a right triangle. The ramp is 13 ft long and its base is 12 ft long.

How tall is the ramp?

- A.** 1 ft **B.** 5 ft
C. 17.7 ft **D.** 25 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q3/5

2205

QUESTION ID

S59-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5 ft

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q3/5

A parallelogram has a slanted side of 10 m. The slanted side, the height, and a 6 m piece of the base form a right triangle.

What is the height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q4/5

2206

QUESTION ID

S59-F2-Q4

TYPE

Short answer

CORRECT ANSWER

8 m

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q4/5

True or false?

A trapezoid has a slanted side of 10 in. The slanted side, the height, and a 6 in. piece of the base form a right triangle. The height is 8 in.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q5/5

2207

QUESTION ID

S59-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S59-F2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 59 • FORWARD 2 • Q5/5

Area of polygons by composing and decomposing

MAIN	Grade 6 main question	6.G.A.1
BACKWARD 1	Area of rectilinear figures	3.MD.C.7.d
BACKWARD 2	Missing side lengths of rectangles	4.MD.A.3
BACKWARD 3	Area of triangles	6.G.A.1
FORWARD 1	Area of composite figures in real-world problems	7.G.B.6
FORWARD 2	Volume of composite solids	8.G.C.9

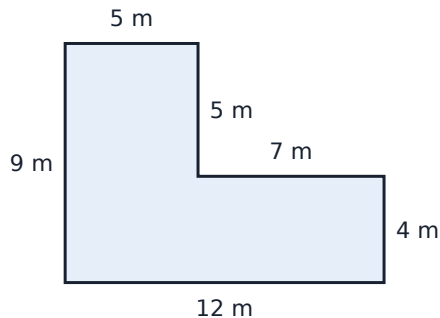
SET 60 • 6.G.A.1

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.1

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q1/5

2210

QUESTION ID

S60-M-Q1

TYPE

Short answer

CORRECT ANSWER

73 square meters

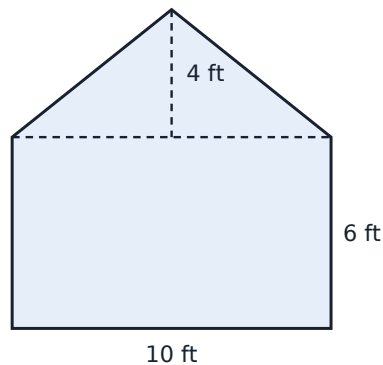
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q1/5

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q2/5

2211

QUESTION ID

S60-M-Q2

TYPE

Short answer

CORRECT ANSWER

80 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q2/5

A rectangular yard is 20 m by 15 m. A triangular garden in the yard has a base of 6 m and a height of 5 m.

What is the area of the yard that is NOT garden?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q3/5

2212

QUESTION ID

S60-M-Q3

TYPE

Short answer

CORRECT ANSWER

285 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q3/5

A polygon is split into a rectangle that is 8 by 3 and two triangles. Each triangle has a base of 2 and a height of 3.

What is the area of the polygon?

- A.** 24 square units **B.** 27 square units
C. 30 square units **D.** 36 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q4/5

2213

QUESTION ID

S60-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 30 square units

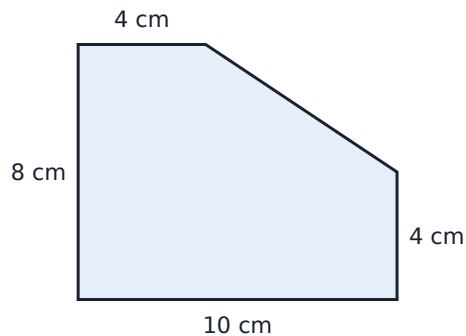
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q4/5

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q5/5

2214

QUESTION ID

S60-M-Q5

TYPE

Short answer

CORRECT ANSWER

68 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-M-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 60 • MAIN • Q5/5

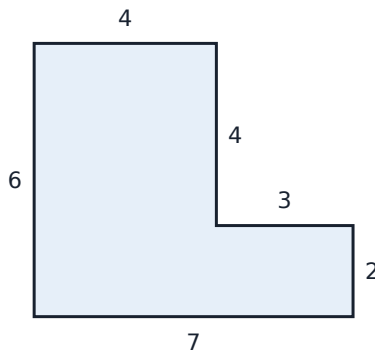
SET 60 • 6.G.A.1

BACKWARD 1

Area of rectilinear figures • Grade 3 • Measurement & Data • 3.MD.C.7.d

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q1/5

2216

QUESTION ID

S60-B1-Q1

TYPE

Short answer

CORRECT ANSWER

30 square units

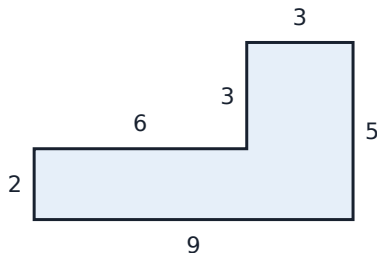
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q1/5

Find the area of the figure.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q2/5

2217

QUESTION ID

S60-B1-Q2

TYPE

Short answer

CORRECT ANSWER

27 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q2/5

True or false?

A figure made of a 5-by-2 rectangle and a 3-by-4 rectangle has an area of 22 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q3/5

2218

QUESTION ID

S60-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

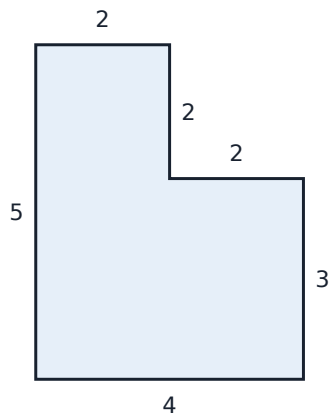
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q3/5

Which expression gives the area of the figure?

- A. 4×5
- B. $4 + 3 + 2 + 2$
- C. $4 \times 3 \times 2$
- D. $4 \times 3 + 2 \times 2$



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q4/5

2219

QUESTION ID

S60-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. $4 \times 3 + 2 \times 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q4/5

A figure is made of two rectangles: 6 by 3 and 2 by 4.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q5/5

2220

QUESTION ID

S60-B1-Q5

TYPE

Short answer

CORRECT ANSWER

26 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 60 • BACKWARD 1 • Q5/5

SET 60 • 6.G.A.1

BACKWARD 2

Missing side lengths of rectangles • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle has an area of 48 square feet and a width of 6 feet.

What is its length?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q1/5

2222

QUESTION ID

S60-B2-Q1

TYPE

Short answer

CORRECT ANSWER

8 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q1

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q1/5

A rectangle has a perimeter of 30 cm and a length of 9 cm.

What is its width?

Width: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q2/5

2223

QUESTION ID

S60-B2-Q2

TYPE

Short answer

CORRECT ANSWER

6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q2

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q2/5

True or false?

A rectangle with an area of 36 square cm and a length of 9 cm has a width of 4 cm.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q3/5

2224

QUESTION ID

S60-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q3

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q3/5

A rectangle has an area of 56 square inches and a width of 7 inches.

What is its length?

- A.** 8 inches
- B.** 9 inches
- C.** 49 inches
- D.** 63 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q4/5

2225

QUESTION ID

S60-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 8 inches

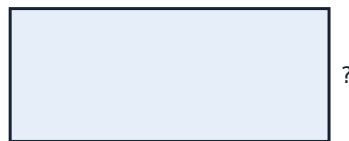
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q4

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q4/5

The rectangle has an area of 60 square meters.

What is the missing side length?



Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q5/5

2226

QUESTION ID

S60-B2-Q5

TYPE

Short answer

CORRECT ANSWER

5 meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B2-Q5

Grade 4 • Measurement & Data • 4.MD.A.3 • Set 60 • BACKWARD 2 • Q5/5

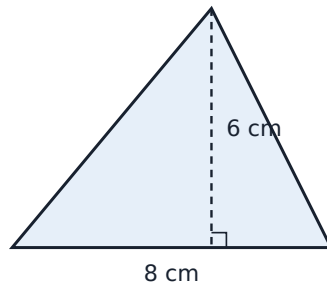
SET 60 • 6.G.A.1

BACKWARD 3

Area of triangles • Grade 6 • Geometry • 6.G.A.1

Find the area of the triangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q1/5

2228

QUESTION ID

S60-B3-Q1

TYPE

Short answer

CORRECT ANSWER

24 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q1/5

A triangle has a base of 10 m and a height of 3 m.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q2/5

2229

QUESTION ID

S60-B3-Q2

TYPE

Short answer

CORRECT ANSWER

15 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q2/5

True or false?

A right triangle with legs of 4 in. and 6 in. has an area of 24 square inches.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q3/5

2230

QUESTION ID

S60-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q3/5

A triangle has a base of 5 ft and a height of 8 ft.

What is its area?

- A. 13 square feet
- B. 20 square feet
- C. 26 square feet
- D. 40 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q4/5

2231

QUESTION ID

S60-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 20 square feet

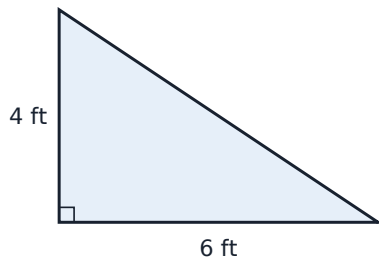
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q4/5

Find the area of the triangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q5/5

2232

QUESTION ID

S60-B3-Q5

TYPE

Short answer

CORRECT ANSWER

12 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-B3-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 60 • BACKWARD 3 • Q5/5

SET 60 • 6.G.A.1

FORWARD 1

Area of composite figures in real-world problems • Grade 7 • Geometry •
7.G.B.6

A deck is made of a rectangle that is 12 ft by 10 ft and a triangle with a base of 10 ft and a height of 6 ft.

What is the area of the deck?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q1/5

2234

QUESTION ID

S60-F1-Q1

TYPE

Short answer

CORRECT ANSWER

150 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q1/5

A trapezoid-shaped garden has bases of 8 m and 14 m and a height of 5 m. It has a square pond with 2 m sides.

What is the area of the garden without the pond?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q2/5

2235

QUESTION ID

S60-F1-Q2

TYPE

Short answer

CORRECT ANSWER

51 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q2/5

A regular hexagon is split into 6 equal triangles. Each triangle has a base of 4 cm and a height of 3.5 cm.

What is the area of the hexagon?

- A.** 21 square cm **B.** 24 square cm
C. 42 square cm **D.** 84 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q3/5

2236

QUESTION ID

S60-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 42 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q3/5

An L-shaped prism has a base area of 30 square feet and a height of 4 feet.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q4/5

2237

QUESTION ID

S60-F1-Q4

TYPE

Short answer

CORRECT ANSWER

120 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q4/5

True or false?

A figure made of a 5-by-5 square and a triangle with a base of 5 and a height of 2 has an area of 30 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q5/5

2238

QUESTION ID

S60-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 60 • FORWARD 1 • Q5/5

SET 60 • 6.G.A.1

FORWARD 2

Volume of composite solids • Grade 8 • Geometry • 8.G.C.9

A silo is a cylinder with a radius of 3 m and a height of 10 m, topped by a hemisphere with a radius of 3 m.

What is its volume in terms of π ?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q1/5

2240

QUESTION ID

S60-F2-Q1

TYPE

Short answer

CORRECT ANSWER

108π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q1/5

A cone has a radius of 3 in. and a height of 4 in. A hemisphere with a radius of 3 in. sits on top.

What is the total volume in terms of π ?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q2/5

2241

QUESTION ID

S60-F2-Q2

TYPE

Short answer

CORRECT ANSWER

30π cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q2/5

A cone with a radius of 2 and a height of 3 is cut out of a cylinder with a radius of 2 and a height of 6.

What volume is left?

- A.** 20π **B.** 28π
C. 24π **D.** 4π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q3/5

2242

QUESTION ID

S60-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 20π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q3/5

A cylinder and a cone each have a radius of 1 and a height of 2.

How much more volume does the cylinder have, in terms of π ?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q4/5

2243

QUESTION ID

S60-F2-Q4

TYPE

Short answer

CORRECT ANSWER

$4\pi/3$

GRADING NOTE $1 \frac{1}{3} \pi$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q4/5

True or false?

A sphere with a radius of 3 has the same volume as a cylinder with a radius of 3 and a height of 4.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q5/5

2244

QUESTION ID

S60-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S60-F2-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 60 • FORWARD 2 • Q5/5

Find volume by packing a prism with unit-fraction cubes

MAIN	Grade 6 main question	6.G.A.2
BACKWARD 1	Volume by counting unit cubes	5.MD.C.4
BACKWARD 2	The volume of one small cube	5.NF.B.4.a
BACKWARD 3	Volume as the number of unit cubes	5.MD.C.3.b
FORWARD 1	Volume of prisms in real-world problems	7.G.B.6
FORWARD 2	Volume of cylinders, cones, and spheres	8.G.C.9

SET 61 • 6.G.A.2

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.2

How many cubes with $\frac{1}{2}$ -inch edges fill a box that is 2 in. by $1\frac{1}{2}$ in. by 1 in.?

Cubes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q1/5

2247

QUESTION ID

S61-M-Q1

TYPE

Short answer

CORRECT ANSWER

24 cubes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q1

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q1/5

A box is 3 in. by 2 in. by $1\frac{1}{2}$ in.

How many cubes with $\frac{1}{2}$ -inch edges fill the box?

Cubes: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q2/5

2248

QUESTION ID

S61-M-Q2

TYPE

Short answer

CORRECT ANSWER

72 cubes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q2/5

True or false?

A prism is packed with 48 cubes that each have $\frac{1}{2}$ -unit edges. Its volume is 6 cubic units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q3/5

2249

QUESTION ID

S61-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q3/5

A prism is packed with 64 cubes that each have $\frac{1}{4}$ -inch edges.

What is the volume of the prism?

- A.** $\frac{1}{4}$ cubic inch **B.** 1 cubic inch
C. 4 cubic inches **D.** 16 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q4/5

2250

QUESTION ID

S61-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 1 cubic inch

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

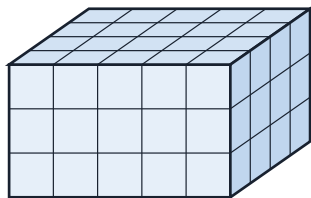
ID S61-M-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q4/5

The prism is packed with cubes that have $\frac{1}{2}$ -cm edges.

It is 5 cubes long, 4 cubes wide, and 3 cubes tall.

What is its volume?



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q5/5

2251

QUESTION ID

S61-M-Q5

TYPE

Short answer

CORRECT ANSWER

7.5 cubic cm

GRADING NOTE 60 cubes $\times$ $\frac{1}{8}$ cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-M-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 61 • MAIN • Q5/5

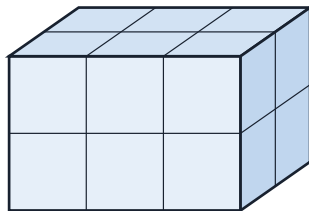
SET 61 • 6.G.A.2

BACKWARD 1

Volume by counting unit cubes • Grade 5 • Measurement & Data • 5.MD.C.4

Each cube is 1 cubic unit.

What is the volume of the prism?



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q1/5

2253

QUESTION ID

S61-B1-Q1

TYPE

Short answer

CORRECT ANSWER

12 cubic units

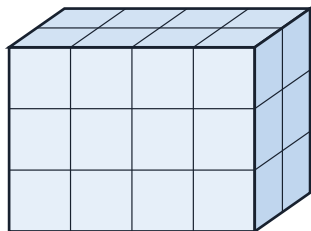
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q1/5

Each cube is 1 cubic unit.

What is the volume of the prism?



Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q2/5

2254

QUESTION ID

S61-B1-Q2

TYPE

Short answer

CORRECT ANSWER

24 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q2/5

True or false?

A prism made of 2 layers with 6 unit cubes in each layer has a volume of 12 cubic units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q3/5

2255

QUESTION ID

S61-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q3/5

A box holds 3 layers with 5 unit cubes in each layer.

What is its volume?

- A. 8 cubic units
- B. 15 cubic units
- C. 35 cubic units
- D. 53 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q4/5

2256

QUESTION ID

S61-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 15 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

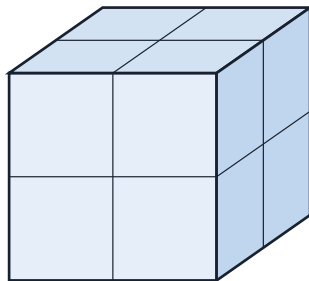
ID S61-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q4/5

Each cube is 1 cubic unit.

What is the volume of the prism?

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q5/5

2257

QUESTION ID

S61-B1-Q5

TYPE

Short answer

CORRECT ANSWER

8 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.4 • Set 61 • BACKWARD 1 • Q5/5

SET 61 • 6.G.A.2

BACKWARD 2

The volume of one small cube • Grade 5 • Number & Operations–Fractions •
5.NF.B.4.a

Multiply.

$$\frac{1}{2} \times \frac{1}{2} \times \frac{1}{2}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q1/5

2259

QUESTION ID

S61-B2-Q1

TYPE

Short answer

CORRECT ANSWER

1/8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q1/5

Multiply.

$$\frac{1}{4} \times \frac{1}{4} \times \frac{1}{4}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q2/5

2260

QUESTION ID

S61-B2-Q2

TYPE

Short answer

CORRECT ANSWER

1/64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q2/5

True or false?

$$\frac{1}{3} \times \frac{1}{3} \times \frac{1}{3} = \frac{1}{9}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q3/5

2261

QUESTION ID

S61-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q3/5

Multiply.

$$24 \times \frac{1}{8}$$

- A. $\frac{1}{3}$
- B. 3
- C. 32
- D. 192

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q4/5

2262

QUESTION ID

S61-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q4/5

Multiply.

$$64 \times \frac{1}{64}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q5/5

2263

QUESTION ID

S61-B2-Q5

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 61 • BACKWARD 2 • Q5/5

SET 61 • 6.G.A.2

BACKWARD 3

Volume as the number of unit cubes • Grade 5 • Measurement & Data •
5.MD.C.3.b

True or false?

A solid that can be packed with 20 unit cubes with no gaps has a volume of 20 cubic units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q1

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q1/5

2265

QUESTION ID

S61-B3-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q1

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q1/5

A box is filled with 36 unit cubes with no gaps or overlaps.

What is its volume?

- A.** 6 cubic units **B.** 12 cubic units
C. 36 cubic units **D.** 36 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q2

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q2/5

2266

QUESTION ID

S61-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. 36 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q2

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q2/5

A box is filled with 45 unit cubes with no gaps.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q3

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q3/5

2267

QUESTION ID

S61-B3-Q3

TYPE

Short answer

CORRECT ANSWER

45 cubic units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q3

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q3/5

True or false?

Volume is measured in square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q4

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q4/5

2268

QUESTION ID

S61-B3-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q4

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q4/5

Which unit could be used to measure volume?

- A.** Square centimeters
- B.** Cubic centimeters
- C.** Centimeters
- D.** Kilograms

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q5

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q5/5

2269

QUESTION ID

S61-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Cubic centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-B3-Q5

Grade 5 • Measurement & Data • 5.MD.C.3.b • Set 61 • BACKWARD 3 • Q5/5

SET 61 • 6.G.A.2

FORWARD 1

Volume of prisms in real-world problems • Grade 7 • Geometry • 7.G.B.6

A triangular prism has triangle bases with a base of 5 in. and a height of 4 in. The prism is 8 in. long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q1/5

2271

QUESTION ID

S61-F1-Q1

TYPE

Short answer

CORRECT ANSWER

80 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q1/5

A prism has a base area of $12\frac{1}{2}$ square cm and a height of 4 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q2/5

2272

QUESTION ID

S61-F1-Q2

TYPE

Short answer

CORRECT ANSWER

50 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q2/5

A cube has edges of 2.5 cm.

What is its volume?

- A.** 6.25 cubic cm
- B.** 7.5 cubic cm
- C.** 15 cubic cm
- D.** 15.625 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q3/5

2273

QUESTION ID

S61-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 15.625 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q3/5

An aquarium is 30 in. long, 12 in. wide, and 16 in. tall. It is filled with water to a height of 12 in.

What is the volume of the water?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q4/5

2274

QUESTION ID

S61-F1-Q4

TYPE

Short answer

CORRECT ANSWER

4,320 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q4/5

True or false?

A triangular prism with a base area of 15 square units and a length of 6 units has a volume of 90 cubic units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q5/5

2275

QUESTION ID

S61-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 61 • FORWARD 1 • Q5/5

SET 61 • 6.G.A.2

FORWARD 2

Volume of cylinders, cones, and spheres • Grade 8 • Geometry • 8.G.C.9

A cylinder has a radius of 5 and a height of 4.

What is its volume in terms of π ?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q1/5

2277

QUESTION ID

S61-F2-Q1

TYPE

Short answer

CORRECT ANSWER

100π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

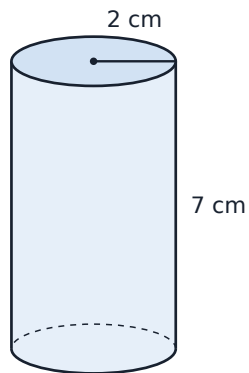
ID S61-F2-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q1/5

Find the volume of the cylinder.

Use 3.14 for π .

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q2/5

ID S61-F2-Q2

2278

QUESTION ID

S61-F2-Q2

TYPE

Short answer

CORRECT ANSWER

87.92 cubic cm

GRADING NOTE 28π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q2/5

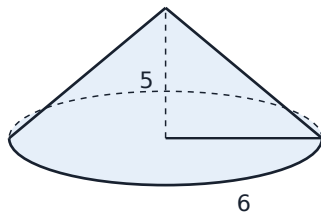
ID S61-F2-Q2

2278

A cone has a radius of 6 and a height of 5.

What is its volume?

- A. 150π B. 180π
C. 30π D. 60π



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q3/5

2279

QUESTION ID

S61-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 60π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q3/5

A sphere has a radius of 6.

What is its volume in terms of π ?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q4/5

2280

QUESTION ID

S61-F2-Q4

TYPE

Short answer

CORRECT ANSWER

288π

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q4/5

True or false?

A cylinder has a greater volume than a cone with the same radius and height.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q5/5

2281

QUESTION ID

S61-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S61-F2-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 61 • FORWARD 2 • Q5/5

Apply $V = l w h$ and $V = b h$ with fractional edge lengths

MAIN

Grade 6 main question

6.G.A.2

BACKWARD 1

Volume formula with whole numbers

5.MD.C.5.b

BACKWARD 2

Multiplying mixed numbers

5.NF.B.6

BACKWARD 3

Multiplying fractions

5.NF.B.4.a

FORWARD 1

Volume of triangular and rectangular prisms

7.G.B.6

FORWARD 2

Volume of cylinders using $V = Bh$

8.G.C.9

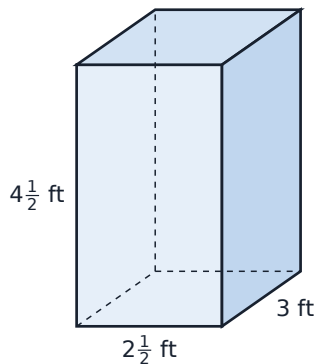
SET 62 • 6.G.A.2

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.2

Find the volume of the rectangular prism.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q1/5

ID S62-M-Q1

2284

QUESTION ID

S62-M-Q1

TYPE

Short answer

CORRECT ANSWER

33 $\frac{3}{4}$ cubic feet

GRADING NOTE 33.75

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q1/5

ID S62-M-Q1

2284

A box is $1\frac{1}{2}$ ft long, 2 ft wide, and $\frac{3}{4}$ ft tall.

What is its volume?

- A.** $2\frac{1}{4}$ cubic feet **B.** $2\frac{3}{4}$ cubic feet
C. 3 cubic feet **D.** $4\frac{1}{4}$ cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q2/5

2285

QUESTION ID

S62-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 2 1/4 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q2

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q2/5

Use $V = l \times w \times h$ to find the volume of a prism with
 $l = 5$ cm, $w = 2\frac{1}{2}$ cm, and $h = 1\frac{1}{5}$ cm.

$V =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q3/5

2286

QUESTION ID

S62-M-Q3

TYPE

Short answer

CORRECT ANSWER

15 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q3

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q3/5

A cube has edges of $1\frac{1}{2}$ feet.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q4/5

2287

QUESTION ID

S62-M-Q4

TYPE

Short answer

CORRECT ANSWER

3 $\frac{3}{8}$ cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q4

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q4/5

2287

A prism has a base area of $4\frac{1}{2}$ square inches and a height of $2\frac{2}{3}$ inches.

Use $V = b \times h$ to find its volume.

$V =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q5/5

2288

QUESTION ID

S62-M-Q5

TYPE

Short answer

CORRECT ANSWER

12 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-M-Q5

Grade 6 • Geometry • 6.G.A.2 • Set 62 • MAIN • Q5/5

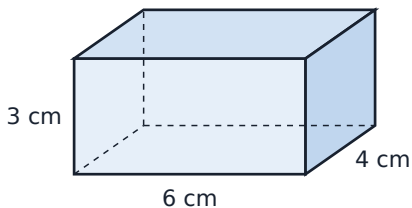
SET 62 • 6.G.A.2

BACKWARD 1

Volume formula with whole numbers • Grade 5 • Measurement & Data •
5.MD.C.5.b

Find the volume of the prism.

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q1/5

2290

QUESTION ID

S62-B1-Q1

TYPE

Short answer

CORRECT ANSWER

72 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q1

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q1/5

A prism has a length of 10 m, a width of 2 m, and a height of 5 m.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q2/5

2291

QUESTION ID

S62-B1-Q2

TYPE

Short answer

CORRECT ANSWER

100 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q2

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q2/5

True or false?

A cube with 3-inch edges has a volume of 27 cubic inches.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q3/5

2292

QUESTION ID

S62-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q3

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q3/5

A box is 8 in. by 5 in. by 2 in.

What is its volume?

- A. 15 cubic inches
- B. 40 cubic inches
- C. 80 cubic inches
- D. 160 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q4/5

2293

QUESTION ID

S62-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 80 cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q4

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q4/5

A prism has a base area of 20 square cm and a height of 6 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q5/5

2294

QUESTION ID

S62-B1-Q5

TYPE

Short answer

CORRECT ANSWER

120 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B1-Q5

Grade 5 • Measurement & Data • 5.MD.C.5.b • Set 62 • BACKWARD 1 • Q5/5

SET 62 • 6.G.A.2

BACKWARD 2

Multiplying mixed numbers • Grade 5 • Number & Operations–Fractions •
5.NF.B.6

Multiply.

$$2\frac{1}{2} \times 3$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q1/5

2296

QUESTION ID

S62-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q1/5

Multiply.

$$1\frac{1}{2} \times \frac{3}{4}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q2/5

2297

QUESTION ID

S62-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$9/8 = 1\frac{1}{8}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q2/5

True or false?

$$2\frac{1}{2} \times 2 = 5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q3/5

2298

QUESTION ID

S62-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q3/5

$$3 \times 4\frac{1}{2} = ?$$

A. $7\frac{1}{2}$

B. $12\frac{1}{6}$

C. $12\frac{1}{2}$

D. $13\frac{1}{2}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q4/5

2299

QUESTION ID

S62-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 13 1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q4/5

Multiply.

$$1\frac{1}{3} \times 1\frac{1}{2}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q5/5

2300

QUESTION ID

S62-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B2-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.6 • Set 62 • BACKWARD 2 • Q5/5

SET 62 • 6.G.A.2

BACKWARD 3

Multiplying fractions • Grade 5 • Number & Operations–Fractions • 5.NF.B.4.a

Multiply.

$$\frac{3}{4} \times \frac{2}{3}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q1/5

2302

QUESTION ID

S62-B3-Q1

TYPE

Short answer

CORRECT ANSWER

1/2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q1

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q1/5

Multiply.

$$\frac{5}{2} \times \frac{9}{2}$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q2/5

2303

QUESTION ID

S62-B3-Q2

TYPE

Short answer

CORRECT ANSWER

$45/4 = 11 \frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q2

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q2/5

True or false?

$$\frac{3}{2} \times \frac{3}{2} = \frac{9}{4}$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q3/5

2304

QUESTION ID

S62-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q3

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q3/5

$$\frac{1}{2} \times \frac{4}{5} = ?$$

A. $\frac{2}{5}$

B. $\frac{4}{7}$

C. $\frac{5}{7}$

D. $\frac{8}{5}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q4/5

2305

QUESTION ID

S62-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q4

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q4/5

Multiply.

$$\frac{6}{5} \times 5$$

Product: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q5/5

2306

QUESTION ID

S62-B3-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-B3-Q5

Grade 5 • Number & Operations-Fractions • 5.NF.B.4.a • Set 62 • BACKWARD 3 • Q5/5

SET 62 • 6.G.A.2

FORWARD 1

Volume of triangular and rectangular prisms • Grade 7 • Geometry • 7.G.B.6

A triangular prism has triangle bases with a base of 6 cm and a height of 4 cm. The prism is 9 cm long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q1/5

2308

QUESTION ID

S62-F1-Q1

TYPE

Short answer

CORRECT ANSWER

48 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q1/5

A tent is a triangular prism. Its triangle has a base of 2.4 m and a height of 1.5 m. The tent is 3 m long.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q2/5

2309

QUESTION ID

S62-F1-Q2

TYPE

Short answer

CORRECT ANSWER

5.4 cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q2/5

A prism has a base area of 7.5 square feet and a height of 2.4 feet.

What is its volume?

- A.** 9.9 cubic feet
- B.** 15 cubic feet
- C.** 18 cubic feet
- D.** 180 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q3/5

2310

QUESTION ID

S62-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 18 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q3/5

A box is 4.5 cm by 2 cm by 3.2 cm.

What is its volume?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q4/5

2311

QUESTION ID

S62-F1-Q4

TYPE

Short answer

CORRECT ANSWER

28.8 cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q4/5

True or false?

A triangular prism with a triangle area of 10 square inches and a length of 7 inches has a volume of 70 cubic inches.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q5/5

2312

QUESTION ID

S62-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 62 • FORWARD 1 • Q5/5

SET 62 • 6.G.A.2

FORWARD 2

Volume of cylinders using $V = Bh$ • Grade 8 • Geometry • 8.G.C.9

A cylinder has a base area of 12π square cm and a height of 5 cm.

What is its volume in terms of π ?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q1/5

2314

QUESTION ID

S62-F2-Q1

TYPE

Short answer

CORRECT ANSWER

60π cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q1/5

A can has a radius of 3 in. and a height of 8 in.

What is its volume in terms of π ?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q2/5

2315

QUESTION ID

S62-F2-Q2

TYPE

Short answer

CORRECT ANSWER

72π cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q2/5

A cylinder has a radius of 1.5 m and a height of 4 m.

What is its volume?

- A. 12π cubic meters
- B. 9π cubic meters
- C. 3π cubic meters
- D. 6π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q3/5

2316

QUESTION ID

S62-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 9π cubic meters

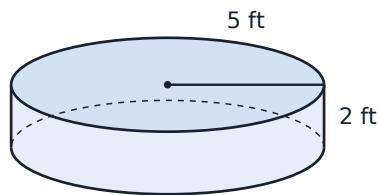
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q3/5

Find the volume of the cylinder in terms of π .

Volume: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q4/5

2317

QUESTION ID

S62-F2-Q4

TYPE

Short answer

CORRECT ANSWER

50π cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q4/5

True or false?

Doubling the height of a cylinder doubles its volume.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q5/5

2318

QUESTION ID

S62-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S62-F2-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 62 • FORWARD 2 • Q5/5

Draw polygons in the coordinate plane from the coordinates of their vertices

MAIN	Grade 6 main question	6.G.A.3
BACKWARD 1	Plotting points in all four quadrants	6.NS.C.6.c
BACKWARD 2	Attributes of quadrilaterals	3.G.A.1
BACKWARD 3	Naming shapes by sides and angles	2.G.A.1
FORWARD 1	Draw shapes with given conditions	7.G.A.2
FORWARD 2	Draw images of polygons under transformations	8.G.A.3

SET 63 • 6.G.A.3

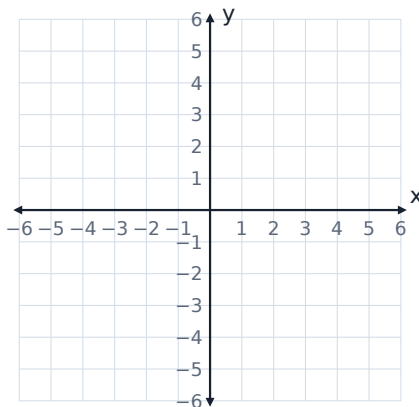
MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.3

Plot $A(-3, 2)$, $B(4, 2)$, $C(4, -3)$, and $D(-3, -3)$. Connect them in order.

What polygon did you draw?

Polygon: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q1/5

2321

QUESTION ID

S63-M-Q1

TYPE

Short answer

CORRECT ANSWER

A rectangle

GRADING NOTE Graph: $A(-3, 2)$, $B(4, 2)$, $C(4, -3)$, $D(-3, -3)$ connected in order

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

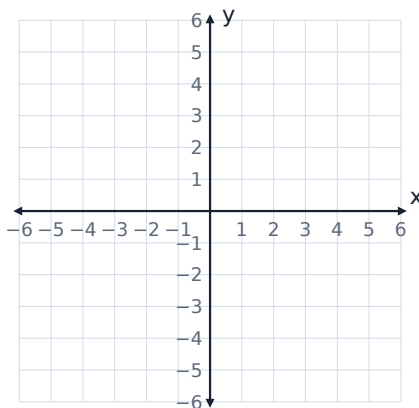
ID S63-M-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q1/5

Plot $P(0, 4)$, $Q(3, -2)$, and $R(-3, -2)$. Connect them in order.

What polygon did you draw?

Polygon: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q2/5

2322

QUESTION ID

S63-M-Q2

TYPE

Short answer

CORRECT ANSWER

A triangle (isosceles)

GRADING NOTE Graph: $P(0, 4)$, $Q(3, -2)$, $R(-3, -2)$ connected

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

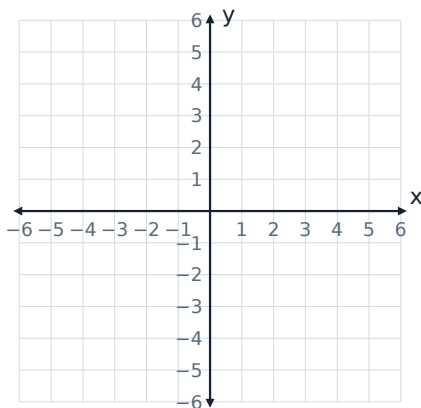
ID S63-M-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q2/5

Plot $E(-2, 1)$, $F(2, 1)$, $G(4, -3)$, and $H(-4, -3)$. Connect them in order.

What polygon did you draw?

Polygon: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q3/5

2323

QUESTION ID

S63-M-Q3

TYPE

Short answer

CORRECT ANSWER

A trapezoid

GRADING NOTE Graph: $E(-2, 1)$, $F(2, 1)$, $G(4, -3)$, $H(-4, -3)$ connected; EF is parallel to HG

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

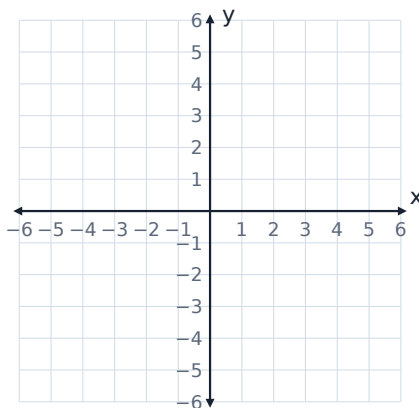
ID S63-M-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q3/5

Plot $J(-1, 3)$, $K(3, 3)$, $L(3, -1)$, and $M(-1, -1)$. Connect them in order.

What polygon did you draw?

Polygon: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q4/5

2324

QUESTION ID

S63-M-Q4

TYPE

Short answer

CORRECT ANSWER

A square

GRADING NOTE Graph: $J(-1, 3)$, $K(3, 3)$, $L(3, -1)$, $M(-1, -1)$ connected; each side is 4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

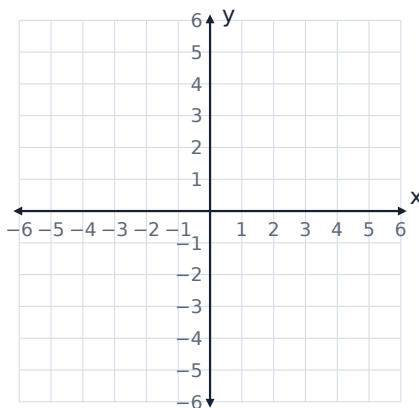
ID S63-M-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q4/5

Plot $W(-5, -2)$, $X(-2, 3)$, $Y(4, 3)$, and $Z(1, -2)$. Connect them in order.

What polygon did you draw?

Polygon: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q5/5

2325

QUESTION ID

S63-M-Q5

TYPE

Short answer

CORRECT ANSWER

A parallelogram

GRADING NOTE Graph: $W(-5, -2)$, $X(-2, 3)$, $Y(4, 3)$, $Z(1, -2)$ connected; WZ and XY are parallel and 6 units long

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-M-Q5

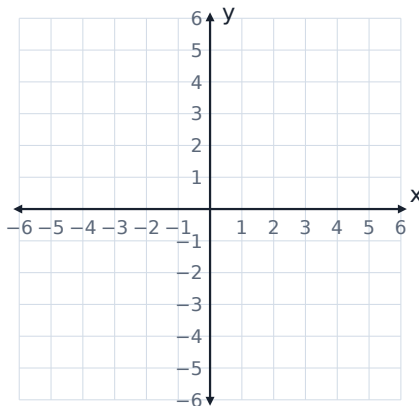
Grade 6 • Geometry • 6.G.A.3 • Set 63 • MAIN • Q5/5

SET 63 • 6.G.A.3

BACKWARD 1

Plotting points in all four quadrants • Grade 6 • The Number System •
6.NS.C.6.c

Plot and label point A($-3, 2$).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q1/5

2327

QUESTION ID

S63-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

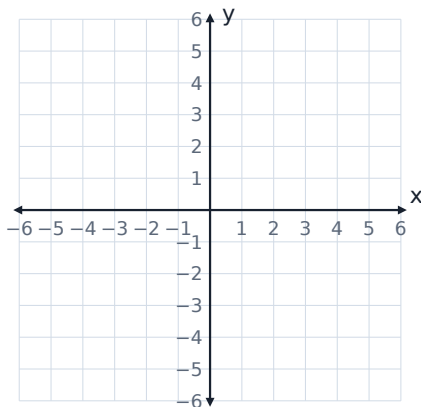
**Point A plotted at
($-3, 2$) and labeled**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q1/5

Plot and label point B(4, -3).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q2/5

2328

QUESTION ID

S63-B1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Point B plotted at (4, -3) and labeled

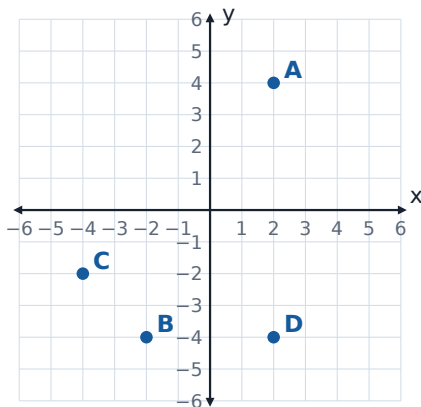
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q2/5

Which point is at $(-2, -4)$?

- A. Point A
- B. Point B
- C. Point C
- D. Point D



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q3/5

2329

QUESTION ID

S63-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

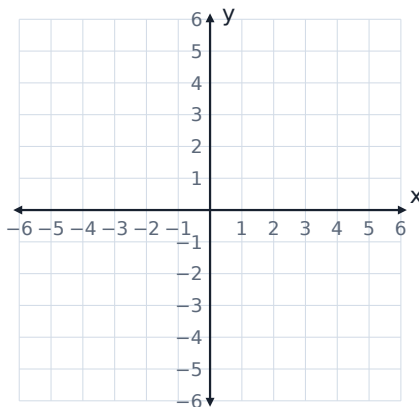
B. Point B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q3/5

Plot and label point C(0, 4).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q4/5

2330

QUESTION ID

S63-B1-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Point C plotted at (0, 4) on the y-axis and labeled

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

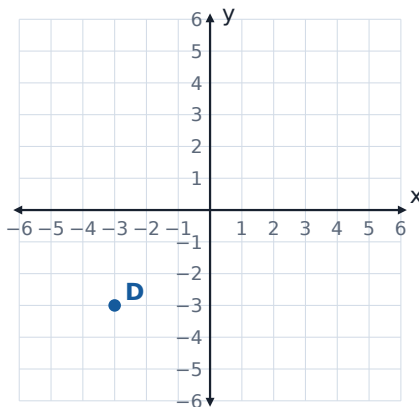
ID S63-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q4/5

True or false?

Point D is at $(-3, -3)$.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q5/5

2331

QUESTION ID

S63-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 63 • BACKWARD 1 • Q5/5

SET 63 • 6.G.A.3

BACKWARD 2

Attributes of quadrilaterals • Grade 3 • Geometry • 3.G.A.1

True or false?

A rectangle has 4 right angles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q1/5

2333

QUESTION ID

S63-B2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q1

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q1/5

True or false?

Opposite sides of a rectangle have the same length.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q2

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q2/5

2334

QUESTION ID

S63-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q2

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q2/5

Which shape always has 4 sides of equal length and 4 right angles?

- A.** Square
- B.** Rectangle
- C.** Trapezoid
- D.** Rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q3

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q3/5

2335

QUESTION ID

S63-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q3

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q3/5

True or false?

Every quadrilateral has 4 right angles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q4

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q4/5

2336

QUESTION ID

S63-B2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q4

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q4/5

How many sides does a quadrilateral have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q5

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q5/5

2337

QUESTION ID

S63-B2-Q5

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B2-Q5

Grade 3 • Geometry • 3.G.A.1 • Set 63 • BACKWARD 2 • Q5/5

SET 63 • 6.G.A.3

BACKWARD 3

Naming shapes by sides and angles • Grade 2 • Geometry • 2.G.A.1

A shape has 3 sides and 3 angles.

What is it called?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q1/5

2339

QUESTION ID

S63-B3-Q1

TYPE

Short answer

CORRECT ANSWER

A triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q1/5

How many sides does a pentagon have?

- A.** 4
- B.** 5
- C.** 6
- D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q2/5

2340

QUESTION ID

S63-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q2/5

True or false?

A hexagon has 6 sides.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q3/5

2341

QUESTION ID

S63-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q3/5

A shape has 4 sides and 4 angles.

What is it called?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q4/5

2342

QUESTION ID

S63-B3-Q4

TYPE

Short answer

CORRECT ANSWER

A quadrilateral

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q4/5

Which shape has the most sides?

- A.** Pentagon
- B.** Quadrilateral
- C.** Triangle
- D.** Hexagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q5/5

2343

QUESTION ID

S63-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Hexagon

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-B3-Q5

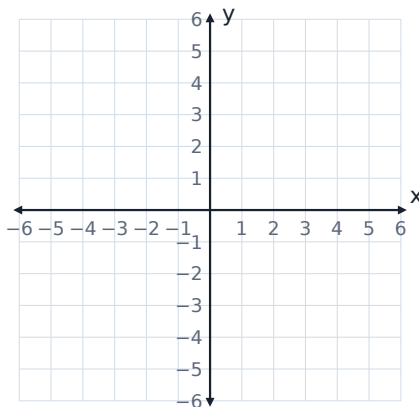
Grade 2 • Geometry • 2.G.A.1 • Set 63 • BACKWARD 3 • Q5/5

SET 63 • 6.G.A.3

FORWARD 1

Draw shapes with given conditions • Grade 7 • Geometry • 7.G.A.2

Draw a rectangle with an area of 12 square units. One vertex is at the origin.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q1/5

2345

QUESTION ID

S63-F1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

Any rectangle with one vertex at (0, 0) and area 12, such as (0, 0), (4, 0), (4, 3), (0, 3)

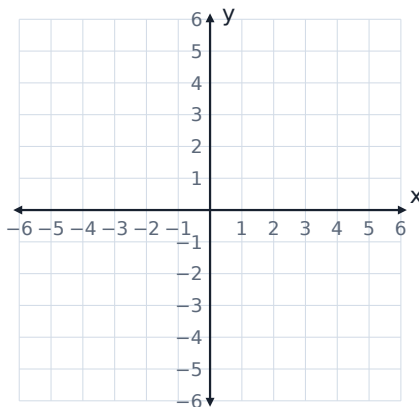
GRADING NOTE Side lengths 3×4 or 2×6 in any quadrant are correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q1/5

Draw a right triangle with legs of 4 units and 3 units. One vertex is at $(-2, -2)$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q2/5

2346

QUESTION ID

S63-F1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A right triangle with legs 4 and 3 meeting at a right angle, with one vertex at $(-2, -2)$, such as $(-2, -2)$, $(2, -2)$, $(-2, 1)$

GRADING NOTE Any orientation with legs of 4 and 3 units is correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q2/5

Can a triangle have side lengths of 2 cm, 3 cm, and 6 cm? Write yes or no.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q3/5

2347

QUESTION ID

S63-F1-Q3

TYPE

Short answer

CORRECT ANSWER

No

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q3/5

A triangle has angles of 90° and 45° .

What is the third angle?

- A.** 45°
- B.** 55°
- C.** 90°
- D.** 135°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q4/5

2348

QUESTION ID

S63-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

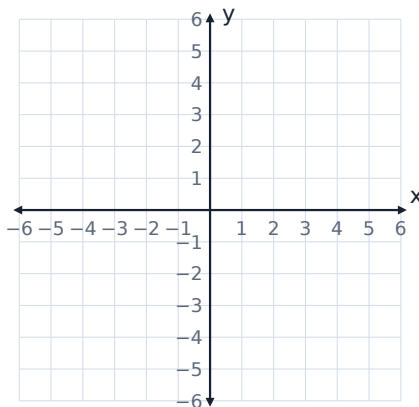
A. 45°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q4/5

Draw a square with a side length of 5 units. One vertex is at $(-3, 1)$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q5/5

2349

QUESTION ID

S63-F1-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

A square with 5-unit sides and one vertex at $(-3, 1)$, such as $(-3, 1)$, $(2, 1)$, $(2, 6)$, $(-3, 6)$

GRADING NOTE Any square with side 5 using $(-3, 1)$ as a vertex that fits on the grid is correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 63 • FORWARD 1 • Q5/5

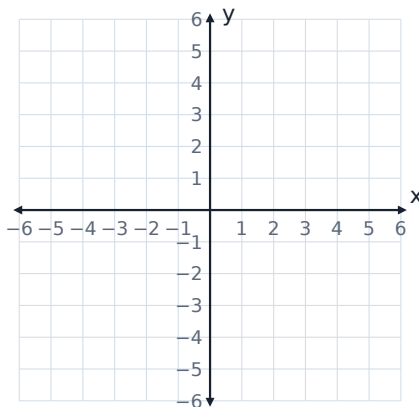
SET 63 • 6.G.A.3

FORWARD 2

Draw images of polygons under transformations • Grade 8 • Geometry •
8.G.A.3

Triangle ABC has vertices
 $A(1, 1)$, $B(4, 1)$, and $C(1, 3)$.

Draw its image after a
translation 5 units left and 2
units down.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q1/5

2351

QUESTION ID

S63-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Image triangle with
vertices $A'(-4, -1)$,
 $B'(-1, -1)$, $C'(-4, 1)$**

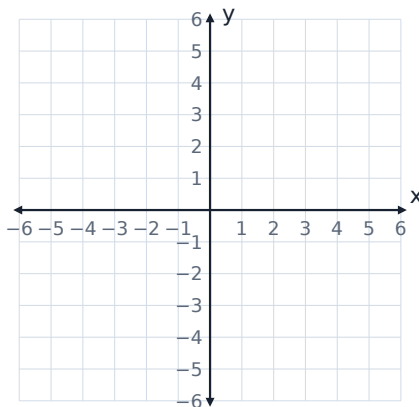
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q1/5

Rectangle PQRS has vertices $P(1, 2)$, $Q(4, 2)$, $R(4, 4)$, and $S(1, 4)$.

Draw its image after a reflection across the x-axis.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q2/5

2352

QUESTION ID

S63-F2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Image rectangle
with vertices $P'(1, -2)$, $Q'(4, -2)$, $R'(4, -4)$, $S'(1, -4)$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q2/5

Triangle ABC has vertices A(2, 1), B(4, 3), and C(2, 3). It is rotated 180° about the origin.

What are the coordinates of B'?

B' = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q3/5

2353

QUESTION ID

S63-F2-Q3

TYPE

Short answer

CORRECT ANSWER

(-4, -3)

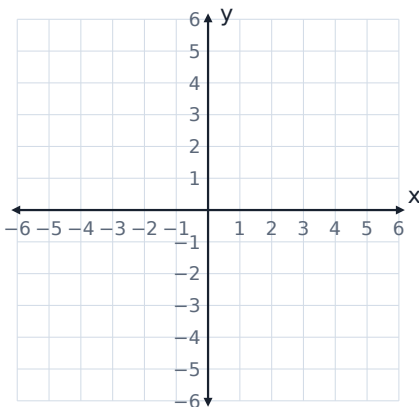
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q3/5

Square EFGH has vertices E(1, 1), F(2, 1), G(2, 2), and H(1, 2).

Draw its image after a dilation by a scale factor of 3 centered at the origin.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q4/5

2354

QUESTION ID

S63-F2-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Image square with vertices E'(3, 3), F'(6, 3), G'(6, 6), H'(3, 6)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q4/5

True or false?

A rectangle with vertices $(1, 1)$, $(3, 1)$, $(3, 2)$, $(1, 2)$ is reflected across the y -axis. One image vertex is $(-3, 2)$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q5/5

2355

QUESTION ID

S63-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S63-F2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 63 • FORWARD 2 • Q5/5

Use coordinates to find side lengths and perimeters of polygons

MAIN	Grade 6 main question	6.G.A.3
BACKWARD 1	Distance between points with the same x or y	6.NS.C.8
BACKWARD 2	Absolute value	6.NS.C.7.c
BACKWARD 3	Perimeter of polygons	3.MD.D.8
FORWARD 1	Lengths between points with rational coordinates	7.NS.A.1.c
FORWARD 2	Find slanted side lengths with the Pythagorean Theorem	8.G.B.8

SET 64 • 6.G.A.3

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.3

A rectangle has vertices at $(-3, 2)$, $(4, 2)$, $(4, -5)$, and $(-3, -5)$.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q1/5

2358

QUESTION ID

S64-M-Q1

TYPE

Short answer

CORRECT ANSWER

28 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q1/5

One side of a polygon goes from $(-2, -3)$ to $(-2, 5)$.

How long is this side?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q2/5

2359

QUESTION ID

S64-M-Q2

TYPE

Short answer

CORRECT ANSWER

8 units

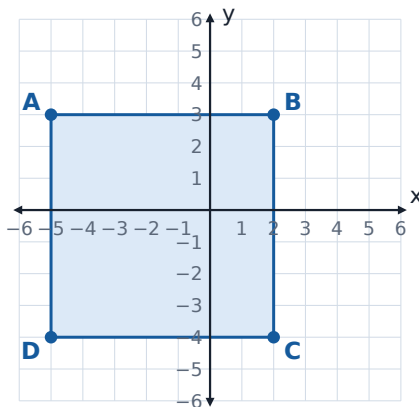
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q2/5

What is the length of side AB?

AB = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q3/5

2360

QUESTION ID

S64-M-Q3

TYPE

Short answer

CORRECT ANSWER

7 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q3/5

A triangle has vertices at $(-4, 1)$, $(2, 1)$, and $(2, -7)$.

How long is the horizontal side?

- A. 2 units
- B. 6 units
- C. 8 units
- D. 10 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q4/5

2361

QUESTION ID

S64-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 6 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q4/5

A rectangle has vertices at $(-5, -2)$, $(1, -2)$, $(1, 3)$, and $(-5, 3)$.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q5/5

2362

QUESTION ID

S64-M-Q5

TYPE

Short answer

CORRECT ANSWER

22 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-M-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 64 • MAIN • Q5/5

SET 64 • 6.G.A.3

BACKWARD 1

Distance between points with the same x or y • Grade 6 • The Number System
• 6.NS.C.8

Find the distance between $(-3, 2)$ and $(4, 2)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q1/5

2364

QUESTION ID

S64-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q1/5

Find the distance between $(1, -4)$ and $(1, 6)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q2/5

2365

QUESTION ID

S64-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q2/5

True or false?

The distance between $(-5, 0)$ and $(5, 0)$ is 10.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q3/5

2366

QUESTION ID

S64-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q3/5

What is the distance between $(2, -3)$ and $(2, -8)$?

- A. -11
- B. -5
- C. 5
- D. 11

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q4/5

2367

QUESTION ID

S64-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q4/5

Find the distance between $(-6, -1)$ and $(-2, -1)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q5/5

2368

QUESTION ID

S64-B1-Q5

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 64 • BACKWARD 1 • Q5/5

SET 64 • 6.G.A.3

BACKWARD 2

Absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|-5| + |2|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q1/5

2370

QUESTION ID

S64-B2-Q1

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q1/5

True or false?

$$|-7| = 7$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q2/5

2371

QUESTION ID

S64-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q2/5

$$|-3| + |4| = ?$$

- A.** -7
- B.** -1
- C.** 1
- D.** 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q3/5

2372

QUESTION ID

S64-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q3/5

Find the value.

$$|-6|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q4/5

2373

QUESTION ID

S64-B2-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q4/5

True or false?

$$|-2| + |-2| = 0$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q5/5

2374

QUESTION ID

S64-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 64 • BACKWARD 2 • Q5/5

SET 64 • 6.G.A.3

BACKWARD 3

Perimeter of polygons • Grade 3 • Measurement & Data • 3.MD.D.8

Find the perimeter of the rectangle.

Perimeter: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q1/5

2376

QUESTION ID

S64-B3-Q1

TYPE

Short answer

CORRECT ANSWER

20 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q1

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q1/5

A square has sides of 9 inches.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q2/5

2377

QUESTION ID

S64-B3-Q2

TYPE

Short answer

CORRECT ANSWER

36 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q2

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q2/5

True or false?

A rectangle that is 10 m by 4 m has a perimeter of 28 m.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q3/5

2378

QUESTION ID

S64-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q3

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q3/5

A triangle has sides of 5 cm, 6 cm, and 7 cm.

What is its perimeter?

- A. 11 cm
- B. 18 cm
- C. 30 cm
- D. 210 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q4/5

2379

QUESTION ID

S64-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 18 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q4

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q4/5

A rectangle is 12 ft long and 5 ft wide.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q5/5

2380

QUESTION ID

S64-B3-Q5

TYPE

Short answer

CORRECT ANSWER

34 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-B3-Q5

Grade 3 • Measurement & Data • 3.MD.D.8 • Set 64 • BACKWARD 3 • Q5/5

SET 64 • 6.G.A.3

FORWARD 1

Lengths between points with rational coordinates • Grade 7 • The Number System • 7.NS.A.1.c

A side of a polygon goes from $(-2.5, 3)$ to $(4, 3)$.

How long is the side?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q1/5

2382

QUESTION ID

S64-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6.5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q1

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q1/5

A rectangle has vertices at $(-1.5, 2)$, $(3.5, 2)$, $(3.5, -1)$, and $(-1.5, -1)$.

What is its perimeter?

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q2/5

2383

QUESTION ID

S64-F1-Q2

TYPE

Short answer

CORRECT ANSWER

16 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q2

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q2/5

How long is the segment from $(0, -3\frac{1}{2})$ to $(0, 1\frac{1}{4})$?

A. $-4\frac{3}{4}$

B. $2\frac{1}{4}$

C. $4\frac{3}{4}$

D. $5\frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q3/5

2384

QUESTION ID

S64-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 4 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q3

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q3/5

A square has vertices at $(-2.5, -2.5)$, $(2.5, -2.5)$, $(2.5, 2.5)$, and $(-2.5, 2.5)$.

What is the length of one side?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q4/5

2385

QUESTION ID

S64-F1-Q4

TYPE

Short answer

CORRECT ANSWER

5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q4

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q4/5

True or false?

The segment from $(-4.2, 1)$ to $(1.8, 1)$ is 6 units long.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q5/5

2386

QUESTION ID

S64-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F1-Q5

Grade 7 • The Number System • 7.NS.A.1.c • Set 64 • FORWARD 1 • Q5/5

SET 64 • 6.G.A.3

FORWARD 2

Find slanted side lengths with the Pythagorean Theorem • Grade 8 • Geometry
• 8.G.B.8

A triangle has vertices at $(0, 0)$, $(6, 0)$, and $(6, 8)$.

How long is the slanted side?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q1/5

2388

QUESTION ID

S64-F2-Q1

TYPE

Short answer

CORRECT ANSWER

10 units

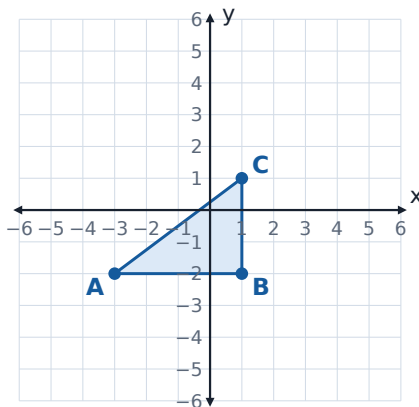
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q1/5

Find the length of side AC.

AC = _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q2/5

2389

QUESTION ID

S64-F2-Q2

TYPE

Short answer

CORRECT ANSWER

5 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q2/5

A parallelogram has a side from (1, 1) to (4, 5).

How long is that side?

- A. 3
- B. 5
- C. 7
- D. 25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q3/5

2390

QUESTION ID

S64-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q3/5

Find the perimeter of the triangle with vertices (0, 0), (5, 0), and (0, 12).

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q4/5

2391

QUESTION ID

S64-F2-Q4

TYPE

Short answer

CORRECT ANSWER

30 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q4/5

True or false?

The side from $(-2, 3)$ to $(4, -5)$ is 10 units long.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q5/5

2392

QUESTION ID

S64-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S64-F2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 64 • FORWARD 2 • Q5/5

Represent three-dimensional figures using nets

MAIN	Grade 6 main question	6.G.A.4
BACKWARD 1	Faces of three-dimensional shapes	2.G.A.1
BACKWARD 2	Shapes of faces	3.G.A.1
FORWARD 1	Cross sections of three-dimensional figures	7.G.A.3
FORWARD 2	Volume of a cylinder described by its net	8.G.C.9

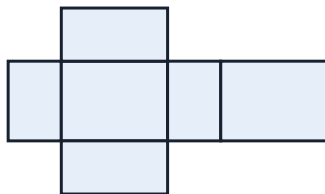
SET 65 • 6.G.A.4

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.4

What three-dimensional figure does the net form?

- A. Triangular prism
- B. Rectangular prism
- C. Square pyramid
- D. Cube



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q1/5

2395

QUESTION ID

S65-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Rectangular prism

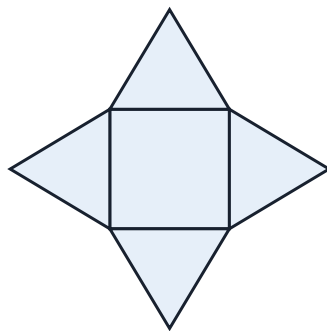
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q1/5

What three-dimensional figure does the net form?

- A. Cube
- B. Triangular pyramid
- C. Square pyramid
- D. Triangular prism



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q2/5

2396

QUESTION ID

S65-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Square pyramid

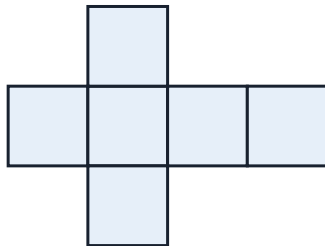
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q2/5

What three-dimensional figure does the net form?

- A. Rectangular prism
- B. Triangular prism
- C. Cube
- D. Square pyramid



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q3/5

2397

QUESTION ID

S65-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Cube

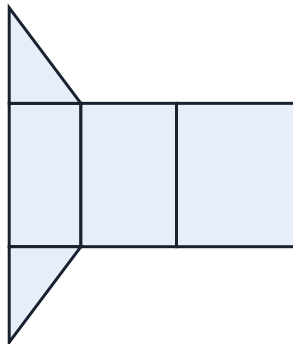
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q3/5

What three-dimensional figure does the net form?

- A. Triangular prism
- B. Triangular pyramid
- C. Square pyramid
- D. Rectangular prism



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q4/5

2398

QUESTION ID

S65-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Triangular prism

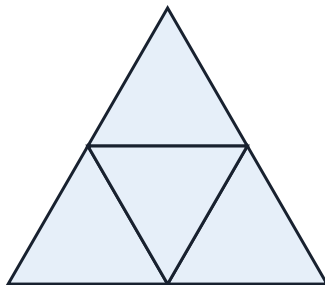
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q4/5

What three-dimensional figure does the net form?

- A. Cube
- B. Square pyramid
- C. Triangular pyramid
- D. Triangular prism



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q5/5

2399

QUESTION ID

S65-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

**C. Triangular
pyramid**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-M-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 65 • MAIN • Q5/5

SET 65 • 6.G.A.4

BACKWARD 1

Faces of three-dimensional shapes • Grade 2 • Geometry • 2.G.A.1

How many faces does a cube have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 65 • BACKWARD 1 • Q1/5

2401

QUESTION ID

S65-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q1

Grade 2 • Geometry • 2.G.A.1 • Set 65 • BACKWARD 1 • Q1/5

How many faces does a rectangular prism have?

- A.** 4
- B.** 6
- C.** 8
- D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 65 • BACKWARD 1 • Q2/5

2402

QUESTION ID

S65-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q2

Grade 2 • Geometry • 2.G.A.1 • Set 65 • BACKWARD 1 • Q2/5

True or false?

A square pyramid has 5 faces.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 65 • BACKWARD 1 • Q3/5

2403

QUESTION ID

S65-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q3

Grade 2 • Geometry • 2.G.A.1 • Set 65 • BACKWARD 1 • Q3/5

How many faces does a triangular prism have?

Faces: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 65 • BACKWARD 1 • Q4/5

2404

QUESTION ID

S65-B1-Q4

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q4

Grade 2 • Geometry • 2.G.A.1 • Set 65 • BACKWARD 1 • Q4/5

How many triangular faces does a square pyramid have?

- A.** 1
- B.** 3
- C.** 4
- D.** 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 65 • BACKWARD 1 • Q5/5

2405

QUESTION ID

S65-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B1-Q5

Grade 2 • Geometry • 2.G.A.1 • Set 65 • BACKWARD 1 • Q5/5

SET 65 • 6.G.A.4

BACKWARD 2

Shapes of faces • Grade 3 • Geometry • 3.G.A.1

What shape are the faces of a cube?

- A.** Triangles
- B.** Pentagons
- C.** Circles
- D.** Squares

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q1

Grade 3 • Geometry • 3.G.A.1 • Set 65 • BACKWARD 2 • Q1/5

2407

QUESTION ID

S65-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Squares

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q1

Grade 3 • Geometry • 3.G.A.1 • Set 65 • BACKWARD 2 • Q1/5

True or false?

Every square is also a rectangle.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q2

Grade 3 • Geometry • 3.G.A.1 • Set 65 • BACKWARD 2 • Q2/5

2408

QUESTION ID

S65-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q2

Grade 3 • Geometry • 3.G.A.1 • Set 65 • BACKWARD 2 • Q2/5

What shape is the base of a square pyramid?

- A.** Square
- B.** Circle
- C.** Triangle
- D.** Rectangle that is not a square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q3

Grade 3 • Geometry • 3.G.A.1 • Set 65 • BACKWARD 2 • Q3/5

2409

QUESTION ID

S65-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q3

Grade 3 • Geometry • 3.G.A.1 • Set 65 • BACKWARD 2 • Q3/5

True or false?

The two ends of a triangular prism are triangles.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q4

Grade 3 • Geometry • 3.G.A.1 • Set 65 • BACKWARD 2 • Q4/5

2410

QUESTION ID

S65-B2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q4

Grade 3 • Geometry • 3.G.A.1 • Set 65 • BACKWARD 2 • Q4/5

What shape are the side faces of a triangular prism?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q5

Grade 3 • Geometry • 3.G.A.1 • Set 65 • BACKWARD 2 • Q5/5

2411

QUESTION ID

S65-B2-Q5

TYPE

Short answer

CORRECT ANSWER

Rectangles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-B2-Q5

Grade 3 • Geometry • 3.G.A.1 • Set 65 • BACKWARD 2 • Q5/5

SET 65 • 6.G.A.4

FORWARD 1

Cross sections of three-dimensional figures • Grade 7 • Geometry • 7.G.A.3

A rectangular prism is sliced parallel to its base.

What shape is the cross section?

- A.** Rectangle
- B.** Circle
- C.** Pentagon
- D.** Triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 65 • FORWARD 1 • Q1/5

2413

QUESTION ID

S65-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Rectangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q1

Grade 7 • Geometry • 7.G.A.3 • Set 65 • FORWARD 1 • Q1/5

A square pyramid is sliced parallel to its base.

What shape is the cross section?

- A.** Square
- B.** Rectangle that is not a square
- C.** Circle
- D.** Triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 65 • FORWARD 1 • Q2/5

2414

QUESTION ID

S65-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q2

Grade 7 • Geometry • 7.G.A.3 • Set 65 • FORWARD 1 • Q2/5

A cube is sliced through the middle, parallel to one face.

What shape is the cross section?

Shape: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 65 • FORWARD 1 • Q3/5

2415

QUESTION ID

S65-F1-Q3

TYPE

Short answer

CORRECT ANSWER

A square

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q3

Grade 7 • Geometry • 7.G.A.3 • Set 65 • FORWARD 1 • Q3/5

A square pyramid is sliced vertically through its top vertex.

What shape is the cross section?

- A.** Pentagon
- B.** Square
- C.** Rectangle
- D.** Triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 65 • FORWARD 1 • Q4/5

2416

QUESTION ID

S65-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q4

Grade 7 • Geometry • 7.G.A.3 • Set 65 • FORWARD 1 • Q4/5

True or false?

A triangular prism sliced parallel to its triangular bases makes a triangle cross section.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 65 • FORWARD 1 • Q5/5

2417

QUESTION ID

S65-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F1-Q5

Grade 7 • Geometry • 7.G.A.3 • Set 65 • FORWARD 1 • Q5/5

SET 65 • 6.G.A.4

FORWARD 2

Volume of a cylinder described by its net • Grade 8 • Geometry • 8.G.C.9

A net of a cylinder has two circles with a radius of 3 cm and a rectangle that is 8 cm tall.

What is the volume of the cylinder in terms of π ?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 65 • FORWARD 2 • Q1/5

2419

QUESTION ID

S65-F2-Q1

TYPE

Short answer

CORRECT ANSWER

72π cubic cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q1

Grade 8 • Geometry • 8.G.C.9 • Set 65 • FORWARD 2 • Q1/5

A net of a cylinder has two circles with a radius of 5 in. and a rectangle that is 4 in. tall.

What is the volume in terms of π ?

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 65 • FORWARD 2 • Q2/5

2420

QUESTION ID

S65-F2-Q2

TYPE

Short answer

CORRECT ANSWER

100π cubic inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q2

Grade 8 • Geometry • 8.G.C.9 • Set 65 • FORWARD 2 • Q2/5

A net of a cylinder has circles with a diameter of 6 m.
The rectangle is 10 m tall.

What is the volume?

- A.** 30π cubic meters **B.** 90π cubic meters
C. 360π cubic meters **D.** 60π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 65 • FORWARD 2 • Q3/5

2421

QUESTION ID

S65-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 90π cubic meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q3

Grade 8 • Geometry • 8.G.C.9 • Set 65 • FORWARD 2 • Q3/5

A net of a cylinder has two circles with a radius of 2 ft and a rectangle that is 7 ft tall.

What is the volume? Use 3.14 for π .

Volume: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 65 • FORWARD 2 • Q4/5

2422

QUESTION ID

S65-F2-Q4

TYPE

Short answer

CORRECT ANSWER

87.92 cubic feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q4

Grade 8 • Geometry • 8.G.C.9 • Set 65 • FORWARD 2 • Q4/5

True or false?

In the net of a cylinder, the height of the rectangle is the height of the cylinder.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 65 • FORWARD 2 • Q5/5

2423

QUESTION ID

S65-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S65-F2-Q5

Grade 8 • Geometry • 8.G.C.9 • Set 65 • FORWARD 2 • Q5/5

Use nets to find surface area

MAIN	Grade 6 main question	6.G.A.4
BACKWARD 1	Area of rectangles	3.MD.C.7.b
BACKWARD 2	Area of triangles	6.G.A.1
BACKWARD 3	Adding the areas of parts	3.MD.C.7.d
FORWARD 1	Surface area of prisms in real-world problems	7.G.B.6
FORWARD 2	Pythagorean Theorem in three dimensions	8.G.B.7

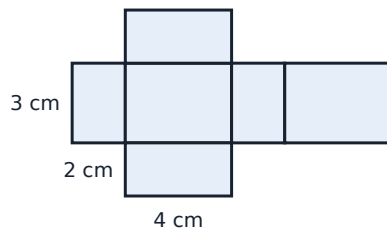
SET 66 • 6.G.A.4

MAIN

Grade 6 main question • Grade 6 • Geometry • 6.G.A.4

The net folds into a rectangular prism.

What is the surface area of the prism?



Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 66 • MAIN • Q1/5

2426

QUESTION ID

S66-M-Q1

TYPE

Short answer

CORRECT ANSWER

52 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

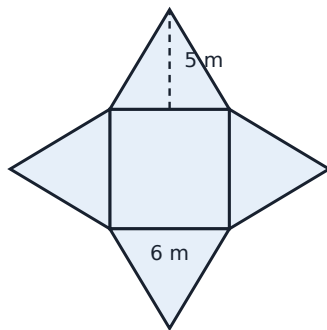
ID S66-M-Q1

Grade 6 • Geometry • 6.G.A.4 • Set 66 • MAIN • Q1/5

The net folds into a square pyramid.

What is the surface area of the pyramid?

Surface area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 66 • MAIN • Q2/5

2427

QUESTION ID

S66-M-Q2

TYPE

Short answer

CORRECT ANSWER

96 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

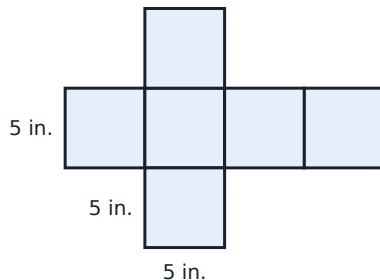
ID S66-M-Q2

Grade 6 • Geometry • 6.G.A.4 • Set 66 • MAIN • Q2/5

The net folds into a cube.

What is the surface area of the cube?

Surface area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 66 • MAIN • Q3/5

2428

QUESTION ID

S66-M-Q3

TYPE

Short answer

CORRECT ANSWER

150 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

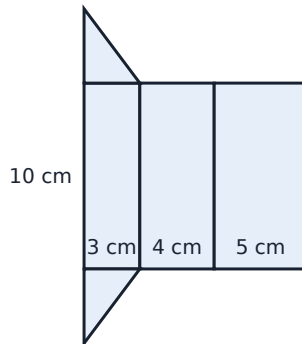
ID S66-M-Q3

Grade 6 • Geometry • 6.G.A.4 • Set 66 • MAIN • Q3/5

The net folds into a triangular prism.

What is the surface area of the prism?

Surface area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 66 • MAIN • Q4/5

2429

QUESTION ID

S66-M-Q4

TYPE

Short answer

CORRECT ANSWER

132 square cm

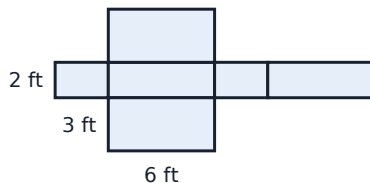
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q4

Grade 6 • Geometry • 6.G.A.4 • Set 66 • MAIN • Q4/5

The net folds into a rectangular prism.

What is the surface area of the prism?



Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 66 • MAIN • Q5/5

2430

QUESTION ID

S66-M-Q5

TYPE

Short answer

CORRECT ANSWER

72 square feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-M-Q5

Grade 6 • Geometry • 6.G.A.4 • Set 66 • MAIN • Q5/5

SET 66 • 6.G.A.4

BACKWARD 1

Area of rectangles • Grade 3 • Measurement & Data • 3.MD.C.7.b

Find the area of the rectangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 66 • BACKWARD 1 • Q1/5

2432

QUESTION ID

S66-B1-Q1

TYPE

Short answer

CORRECT ANSWER

8 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 66 • BACKWARD 1 • Q1/5

A rectangle is 5 m by 7 m.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 66 • BACKWARD 1 • Q2/5

2433

QUESTION ID

S66-B1-Q2

TYPE

Short answer

CORRECT ANSWER

35 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 66 • BACKWARD 1 • Q2/5

True or false?

A rectangle that is 3 units by 6 units has an area of 18 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 66 • BACKWARD 1 • Q3/5

2434

QUESTION ID

S66-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 66 • BACKWARD 1 • Q3/5

A rectangle is 6 m by 2 m.

What is its area?

- A. 8 square meters
- B. 12 square meters
- C. 16 square meters
- D. 24 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 66 • BACKWARD 1 • Q4/5

2435

QUESTION ID

S66-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 12 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 66 • BACKWARD 1 • Q4/5

A square face has 5-inch edges.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 66 • BACKWARD 1 • Q5/5

2436

QUESTION ID

S66-B1-Q5

TYPE

Short answer

CORRECT ANSWER

25 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B1-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.b • Set 66 • BACKWARD 1 • Q5/5

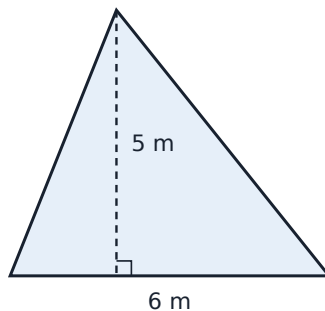
SET 66 • 6.G.A.4

BACKWARD 2

Area of triangles • Grade 6 • Geometry • 6.G.A.1

Find the area of the triangle.

Area: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 66 • BACKWARD 2 • Q1/5

2438

QUESTION ID

S66-B2-Q1

TYPE

Short answer

CORRECT ANSWER

15 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q1

Grade 6 • Geometry • 6.G.A.1 • Set 66 • BACKWARD 2 • Q1/5

A triangle has a base of 8 cm and a height of 3 cm.

What is its area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 66 • BACKWARD 2 • Q2/5

2439

QUESTION ID

S66-B2-Q2

TYPE

Short answer

CORRECT ANSWER

12 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q2

Grade 6 • Geometry • 6.G.A.1 • Set 66 • BACKWARD 2 • Q2/5

True or false?

A triangle with a base of 10 and a height of 4 has an area of 40 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 66 • BACKWARD 2 • Q3/5

2440

QUESTION ID

S66-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q3

Grade 6 • Geometry • 6.G.A.1 • Set 66 • BACKWARD 2 • Q3/5

A right triangle has legs of 3 cm and 4 cm.

What is its area?

- A. 6 square cm
- B. 7 square cm
- C. 10 square cm
- D. 12 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 66 • BACKWARD 2 • Q4/5

2441

QUESTION ID

S66-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 6 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q4

Grade 6 • Geometry • 6.G.A.1 • Set 66 • BACKWARD 2 • Q4/5

Four triangles each have a base of 6 m and a height of 5 m.

What is their total area?

Total area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 66 • BACKWARD 2 • Q5/5

2442

QUESTION ID

S66-B2-Q5

TYPE

Short answer

CORRECT ANSWER

60 square meters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B2-Q5

Grade 6 • Geometry • 6.G.A.1 • Set 66 • BACKWARD 2 • Q5/5

SET 66 • 6.G.A.4

BACKWARD 3

Adding the areas of parts • Grade 3 • Measurement & Data • 3.MD.C.7.d

A figure is made of a 4-by-3 rectangle and a 4-by-2 rectangle that do not overlap.

What is its total area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 66 • BACKWARD 3 • Q1/5

2444

QUESTION ID

S66-B3-Q1

TYPE

Short answer

CORRECT ANSWER

20 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q1

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 66 • BACKWARD 3 • Q1/5

True or false?

A figure made of two 5-by-5 squares has an area of 50 square units.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 66 • BACKWARD 3 • Q2/5

2445

QUESTION ID

S66-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q2

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 66 • BACKWARD 3 • Q2/5

A figure is made of rectangles with areas of 12, 8, and 6 square units.

What is its total area?

- A.** 14 square units **B.** 20 square units
C. 26 square units **D.** 576 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 66 • BACKWARD 3 • Q3/5

2446

QUESTION ID

S66-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 26 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q3

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 66 • BACKWARD 3 • Q3/5

Six squares each have an area of 9 square inches.

What is their total area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 66 • BACKWARD 3 • Q4/5

2447

QUESTION ID

S66-B3-Q4

TYPE

Short answer

CORRECT ANSWER

54 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q4

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 66 • BACKWARD 3 • Q4/5

A figure is made of rectangles that are 2 by 6, 2 by 6,
and 3 by 6.

What is its total area?

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 66 • BACKWARD 3 • Q5/5

2448

QUESTION ID

S66-B3-Q5

TYPE

Short answer

CORRECT ANSWER

42 square units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-B3-Q5

Grade 3 • Measurement & Data • 3.MD.C.7.d • Set 66 • BACKWARD 3 • Q5/5

SET 66 • 6.G.A.4

FORWARD 1

Surface area of prisms in real-world problems • Grade 7 • Geometry • 7.G.B.6

A cube has edges of 2.5 cm.

What is its surface area?

Surface area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 66 • FORWARD 1 • Q1/5

2450

QUESTION ID

S66-F1-Q1

TYPE

Short answer

CORRECT ANSWER

37.5 square cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q1

Grade 7 • Geometry • 7.G.B.6 • Set 66 • FORWARD 1 • Q1/5

A rectangular prism is 5 in. by 4 in. by 2.5 in.

What is its surface area?

- A.** 42.5 square inches
- B.** 50 square inches
- C.** 65 square inches
- D.** 85 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 66 • FORWARD 1 • Q2/5

2451

QUESTION ID

S66-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 85 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q2

Grade 7 • Geometry • 7.G.B.6 • Set 66 • FORWARD 1 • Q2/5

A gift box is 8 in. by 6 in. by 4 in.

How much wrapping paper covers the box exactly?

Paper: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 66 • FORWARD 1 • Q3/5

2452

QUESTION ID

S66-F1-Q3

TYPE

Short answer

CORRECT ANSWER

208 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q3

Grade 7 • Geometry • 7.G.B.6 • Set 66 • FORWARD 1 • Q3/5

A box with no lid is 10 in. long, 4 in. wide, and 3 in. tall.

How much cardboard is needed to make its 5 faces?

Cardboard: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 66 • FORWARD 1 • Q4/5

2453

QUESTION ID

S66-F1-Q4

TYPE

Short answer

CORRECT ANSWER

124 square inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q4

Grade 7 • Geometry • 7.G.B.6 • Set 66 • FORWARD 1 • Q4/5

True or false?

The surface area of a prism is the sum of the areas of all of its faces.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 66 • FORWARD 1 • Q5/5

2454

QUESTION ID

S66-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F1-Q5

Grade 7 • Geometry • 7.G.B.6 • Set 66 • FORWARD 1 • Q5/5

SET 66 • 6.G.A.4

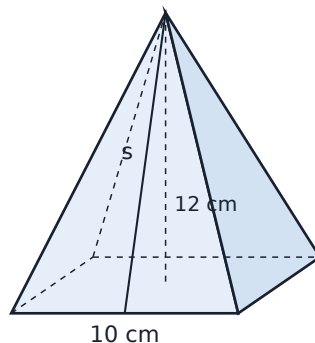
FORWARD 2

Pythagorean Theorem in three dimensions • Grade 8 • Geometry • 8.G.B.7

The square pyramid has a base edge of 10 cm and a height of 12 cm.

What is its slant height s ?

$S =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 66 • FORWARD 2 • Q1/5

2456

QUESTION ID

S66-F2-Q1

TYPE

Short answer

CORRECT ANSWER

13 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q1

Grade 8 • Geometry • 8.G.B.7 • Set 66 • FORWARD 2 • Q1/5

A box is 3 in. by 4 in. by 12 in.

How long is the longest diagonal from one corner to the opposite corner?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 66 • FORWARD 2 • Q2/5

2457

QUESTION ID

S66-F2-Q2

TYPE

Short answer

CORRECT ANSWER

13 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q2

Grade 8 • Geometry • 8.G.B.7 • Set 66 • FORWARD 2 • Q2/5

A square pyramid has a base edge of 16 cm and a slant height of 10 cm.

What is its height?

- A.** 6 cm **B.** 8 cm
C. 12.8 cm **D.** 18.9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 66 • FORWARD 2 • Q3/5

2458

QUESTION ID

S66-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. 6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q3

Grade 8 • Geometry • 8.G.B.7 • Set 66 • FORWARD 2 • Q3/5

A cone has a radius of 5 in. and a slant height of 13 in.

What is its height?

Height: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 66 • FORWARD 2 • Q4/5

2459

QUESTION ID

S66-F2-Q4

TYPE

Short answer

CORRECT ANSWER

12 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q4

Grade 8 • Geometry • 8.G.B.7 • Set 66 • FORWARD 2 • Q4/5

True or false?

The bottom of a box is 6 in. by 8 in. The diagonal across the bottom is 10 in.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 66 • FORWARD 2 • Q5/5

2460

QUESTION ID

S66-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S66-F2-Q5

Grade 8 • Geometry • 8.G.B.7 • Set 66 • FORWARD 2 • Q5/5

Recognize statistical questions

MAIN Grade 6 main question

BACKWARD 1 Reading a bar graph

BACKWARD 2 Reading a line plot

FORWARD 1 Random samples and representative samples

FORWARD 2 Statistical questions about two variables and scatter plots

6.SP.A.1

3.MD.B.3

4.MD.B.4

7.SP.A.1

8.SP.A.1

SET 67 • 6.SP.A.1

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.A.1

Which question is a statistical question?

- A. How many hours did I sleep last night?
- B. How many days are in a week?
- C. What time does our school start?
- D. How many hours of sleep do students in my class get each night?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q1

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 67 • MAIN • Q1/5

2463

QUESTION ID

S67-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. How many hours of sleep do students in my class get each night?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q1

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 67 • MAIN • Q1/5

True or false?

"How tall is the flagpole at our school?" is a statistical question.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q2

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 67 • MAIN • Q2/5

2464

QUESTION ID

S67-M-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q2

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 67 • MAIN • Q2/5

Which question is NOT a statistical question?

- A. How much do the dogs at the shelter weigh?
- B. What is the population of our city right now?
- C. How long do students spend on homework each day?
- D. How many pets do the students in grade 6 have?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q3

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 67 • MAIN • Q3/5

2465

QUESTION ID

S67-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

**B. What is the
population of our
city right now?**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q3

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 67 • MAIN • Q3/5

True or false?

"What are the ages of the teachers at our school?" is a statistical question.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q4

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 67 • MAIN • Q4/5

2466

QUESTION ID

S67-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q4

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 67 • MAIN • Q4/5

Which question is a statistical question?

- A.** How many minutes do the players on our team practice each week?
- B.** How many minutes are in an hour?
- C.** How many minutes did Coach Lee practice today?
- D.** How many players are on a soccer field at once?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q5

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 67 • MAIN • Q5/5

2467

QUESTION ID

S67-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. How many minutes do the players on our team practice each week?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-M-Q5

Grade 6 • Statistics & Probability • 6.SP.A.1 • Set 67 • MAIN • Q5/5

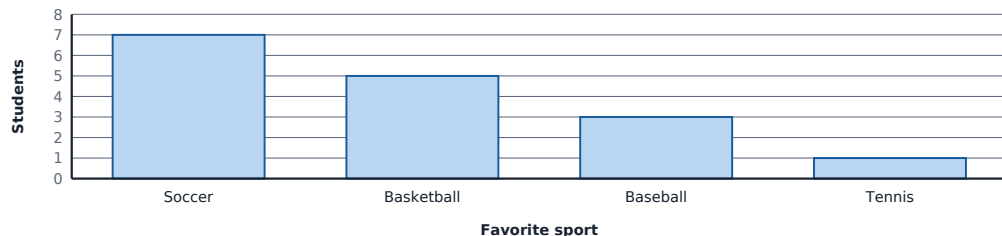
SET 67 • 6.SP.A.1

BACKWARD 1

Reading a bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

The graph shows the favorite sports of a class.

How many students chose basketball?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 67 • BACKWARD 1 • Q1/5

2469

QUESTION ID

S67-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5

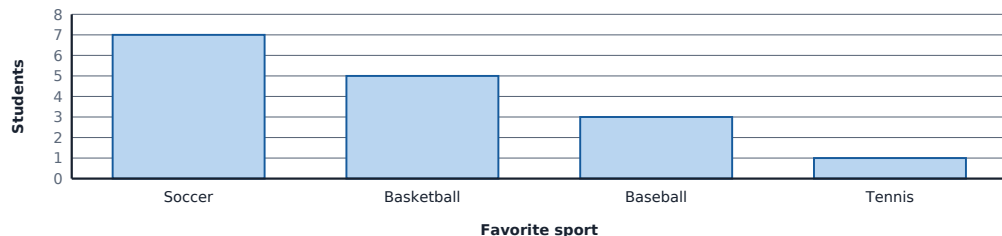
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 67 • BACKWARD 1 • Q1/5

The graph shows the favorite sports of a class.

How many more students chose soccer than baseball?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 67 • BACKWARD 1 • Q2/5

2470

QUESTION ID

S67-B1-Q2

TYPE

Short answer

CORRECT ANSWER

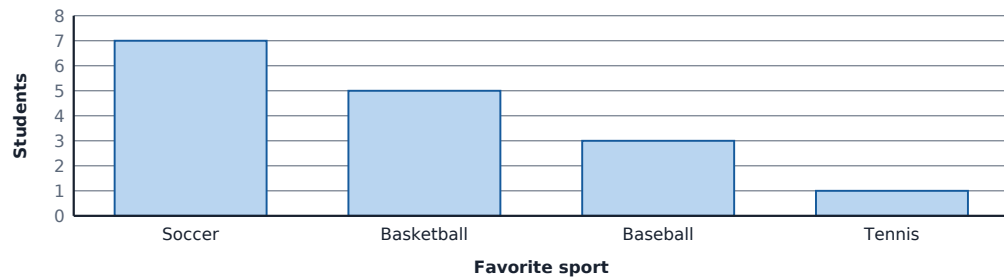
4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 67 • BACKWARD 1 • Q2/5

Which sport did 3 students choose?



- A.** Baseball **B.** Tennis **C.** Soccer **D.** Basketball

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 67 • BACKWARD 1 • Q3/5

2471

QUESTION ID

S67-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Baseball

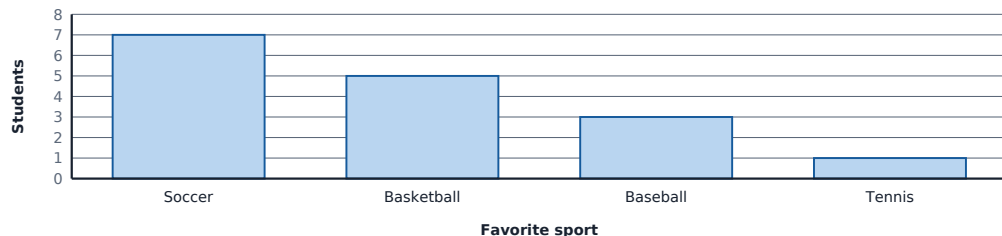
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 67 • BACKWARD 1 • Q3/5

The graph shows the favorite sports of a class.

How many students answered in all?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 67 • BACKWARD 1 • Q4/5

2472

QUESTION ID

S67-B1-Q4

TYPE

Short answer

CORRECT ANSWER

16

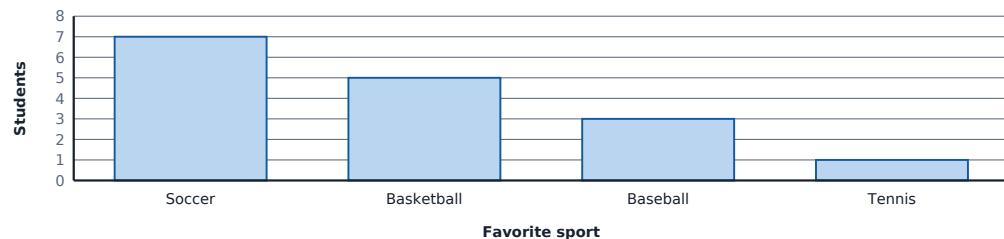
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 67 • BACKWARD 1 • Q4/5

True or false?

More students chose basketball than soccer.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 67 • BACKWARD 1 • Q5/5

2473

QUESTION ID

S67-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B1-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 67 • BACKWARD 1 • Q5/5

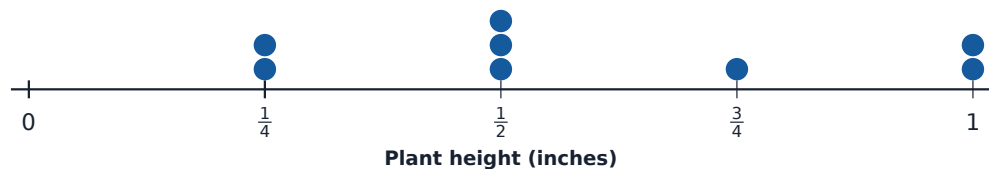
SET 67 • 6.SP.A.1

BACKWARD 2

Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows the heights of plants.

How many plants are $\frac{1}{2}$ inch tall?



Plants: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 67 • BACKWARD 2 • Q1/5

2475

QUESTION ID

S67-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

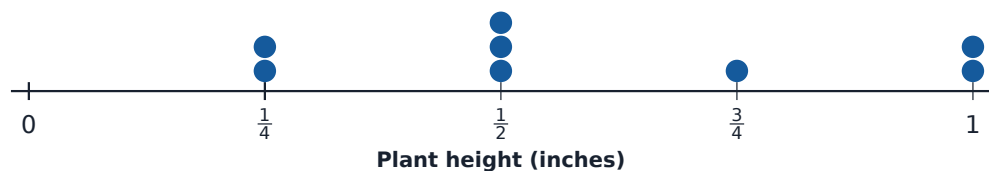
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 67 • BACKWARD 2 • Q1/5

The line plot shows the heights of plants.

How many plants were measured?



Plants: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 67 • BACKWARD 2 • Q2/5

2476

QUESTION ID

S67-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

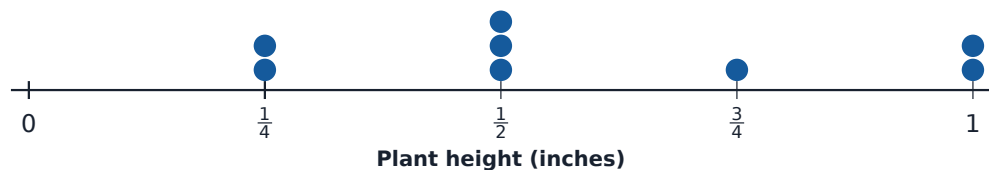
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 67 • BACKWARD 2 • Q2/5

True or false?

The most common plant height is $\frac{3}{4}$ inch.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 67 • BACKWARD 2 • Q3/5

2477

QUESTION ID

S67-B2-Q3

TYPE

True or false

CORRECT ANSWER

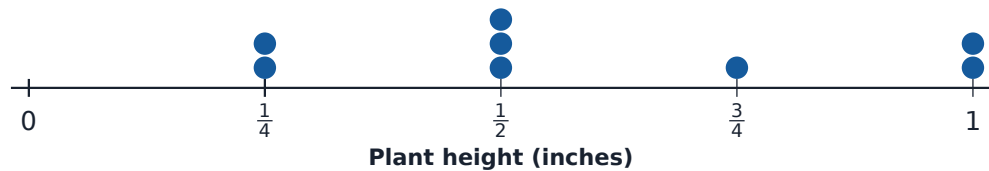
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 67 • BACKWARD 2 • Q3/5

Which height appears least often?



- A. $\frac{1}{4}$ inch B. $\frac{1}{2}$ inch C. $\frac{3}{4}$ inch D. 1 inch

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 67 • BACKWARD 2 • Q4/5

2478

QUESTION ID

S67-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

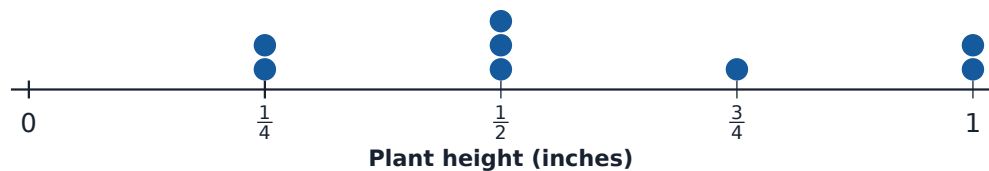
C. $\frac{3}{4}$ inch

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 67 • BACKWARD 2 • Q4/5

What is the difference between the tallest and the shortest plant heights?



Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 67 • BACKWARD 2 • Q5/5

2479

QUESTION ID

S67-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3/4 inch

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-B2-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 67 • BACKWARD 2 • Q5/5

SET 67 • 6.SP.A.1

FORWARD 1

Random samples and representative samples • Grade 7 • Statistics & Probability • 7.SP.A.1

A principal wants to know the favorite lunch of all 600 students.

Which sample is most likely to represent all students?

- A.** The students on the soccer team
- B.** 50 students chosen at random from the whole school
- C.** The 30 students in one class
- D.** The first 50 students in line on pizza day

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 67 • FORWARD 1 • Q1/5

2481

QUESTION ID

S67-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

**B. 50 students
chosen at random
from the whole
school**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 67 • FORWARD 1 • Q1/5

True or false?

A random sample is more likely to represent a population than a sample of your friends.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 67 • FORWARD 1 • Q2/5

2482

QUESTION ID

S67-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 67 • FORWARD 1 • Q2/5

A reporter asks only basketball players, "What is your favorite sport?"

Why is this sample biased?

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 67 • FORWARD 1 • Q3/5

2483

QUESTION ID

S67-F1-Q3

TYPE

Short answer

CORRECT ANSWER

Only basketball players were asked, so the sample favors basketball and does not represent all people.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 67 • FORWARD 1 • Q3/5

A random sample of 40 students out of 800 shows that 10 walk to school.

About how many of the 800 students walk to school?

- A.** 10 **B.** 40
C. 80 **D.** 200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 67 • FORWARD 1 • Q4/5

2484

QUESTION ID

S67-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 200

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 67 • FORWARD 1 • Q4/5

True or false?

Conclusions about a population from a sample are valid only if the sample represents the population.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 67 • FORWARD 1 • Q5/5

2485

QUESTION ID

S67-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 67 • FORWARD 1 • Q5/5

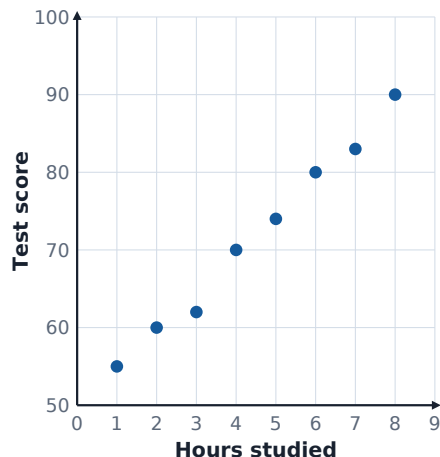
SET 67 • 6.SP.A.1

FORWARD 2

Statistical questions about two variables and scatter plots • Grade 8 • Statistics & Probability • 8.SP.A.1

What type of association does the scatter plot show?

- A. No association
- B. Nonlinear
- C. Positive linear
- D. Negative linear



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • FORWARD 2 • Q1/5

2487

QUESTION ID

S67-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Positive linear

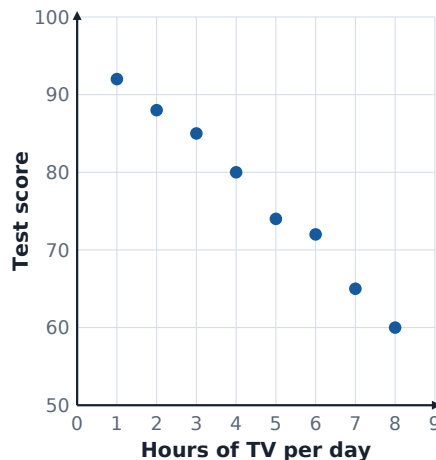
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • FORWARD 2 • Q1/5

What type of association does the scatter plot show?

- A. Negative linear
- B. Positive linear
- C. No association
- D. Nonlinear



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • FORWARD 2 • Q2/5

2488

QUESTION ID

S67-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Negative linear

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • FORWARD 2 • Q2/5

True or false?

A scatter plot shows data about two variables for each person or object.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • FORWARD 2 • Q3/5

2489

QUESTION ID

S67-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • FORWARD 2 • Q3/5

Which question would be best answered with a scatter plot?

- A. How many students ride the bus?
- B. What is the most common eye color?
- C. Is there a relationship between height and arm span?
- D. What is the median age in the class?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • FORWARD 2 • Q4/5

2490

QUESTION ID

S67-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Is there a relationship between height and arm span?

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

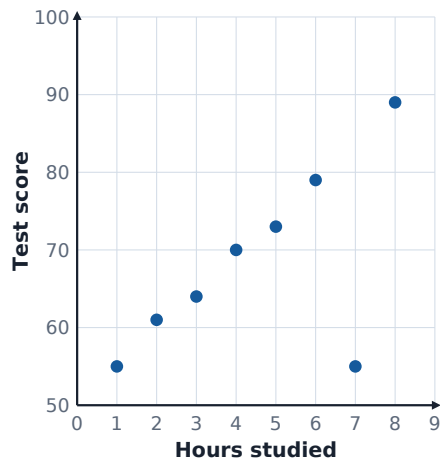
ID S67-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • FORWARD 2 • Q4/5

Which point is an outlier?

Write its coordinates.

Outlier: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • FORWARD 2 • Q5/5

2491

QUESTION ID

S67-F2-Q5

TYPE

Short answer

CORRECT ANSWER

(7, 55)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S67-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 67 • FORWARD 2 • Q5/5

Describe the overall shape of a distribution

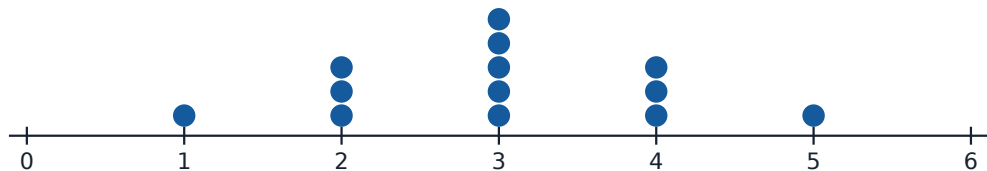
MAIN	Grade 6 main question	6.SP.A.2
BACKWARD 1	Reading a line plot	4.MD.B.4
BACKWARD 2	Most and least in a bar graph	3.MD.B.3
FORWARD 1	Compare two distributions visually	7.SP.B.3
FORWARD 2	Linear and nonlinear patterns in scatter plots	8.SP.A.1

SET 68 • 6.SP.A.2

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.A.2

Which word best describes the shape of the data?



- A.** Uniform **B.** Symmetric
C. Skewed left **D.** Skewed right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 68 • MAIN • Q1/5

2494

QUESTION ID

S68-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

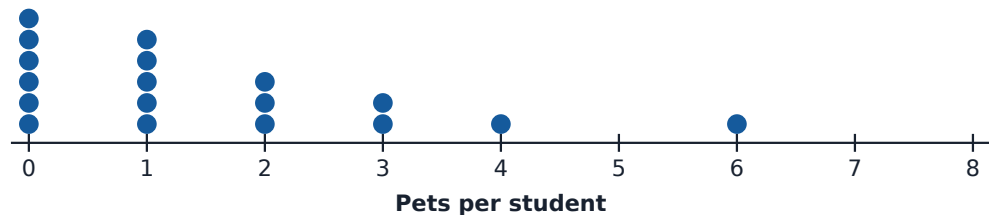
B. Symmetric

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 68 • MAIN • Q1/5

Which word best describes the shape of the data?



- A. Skewed left B. Symmetric
C. Skewed right D. Uniform

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 68 • MAIN • Q2/5

2495

QUESTION ID

S68-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

C. Skewed right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 68 • MAIN • Q2/5

Which word best describes the shape of the data?



- A.** Symmetric **B.** Skewed left
C. Skewed right **D.** Uniform

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 68 • MAIN • Q3/5

2496

QUESTION ID

S68-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

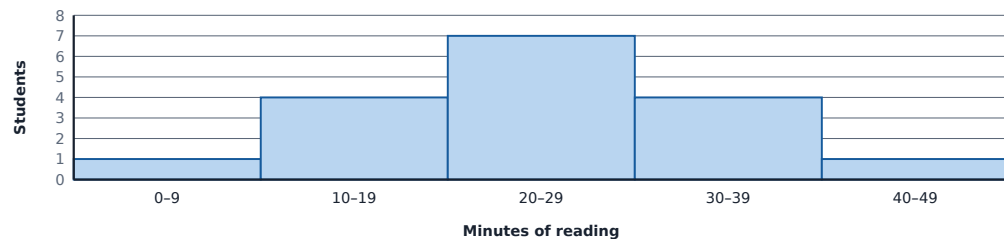
B. Skewed left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 68 • MAIN • Q3/5

Which word best describes the shape of the data in the histogram?



- A.** Skewed right **B.** Skewed left
C. Uniform **D.** Symmetric

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 68 • MAIN • Q4/5

2497

QUESTION ID

S68-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. Symmetric

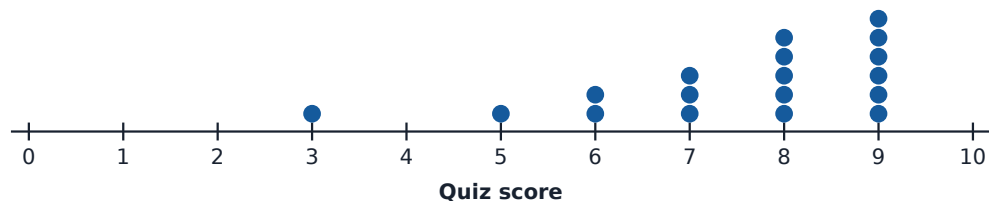
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 68 • MAIN • Q4/5

True or false?

The data are skewed right.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 68 • MAIN • Q5/5

2498

QUESTION ID

S68-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-M-Q5

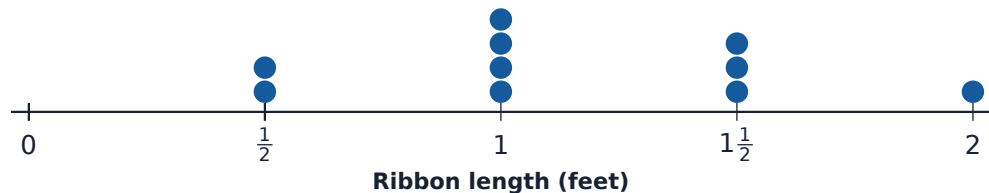
Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 68 • MAIN • Q5/5

SET 68 • 6.SP.A.2

BACKWARD 1

Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

How many ribbons are 1 foot long?



Ribbons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 68 • BACKWARD 1 • Q1/5

2500

QUESTION ID

S68-B1-Q1

TYPE

Short answer

CORRECT ANSWER

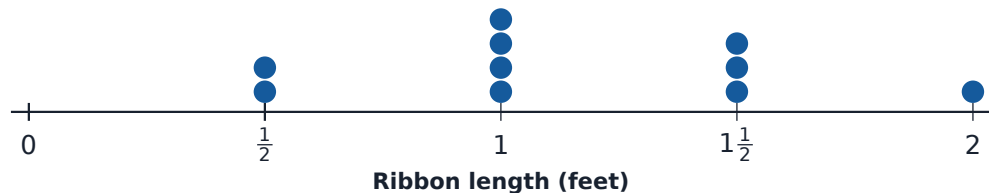
4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 68 • BACKWARD 1 • Q1/5

How many ribbons are there in all?



Ribbons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 68 • BACKWARD 1 • Q2/5

2501

QUESTION ID

S68-B1-Q2

TYPE

Short answer

CORRECT ANSWER

10

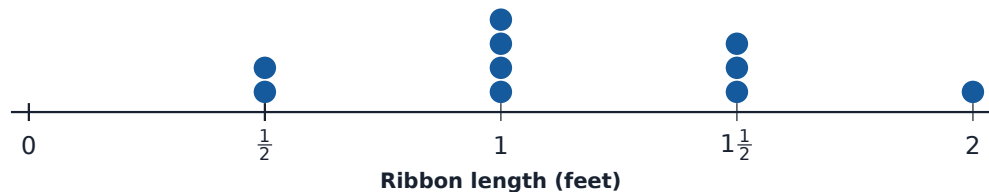
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 68 • BACKWARD 1 • Q2/5

True or false?

Two ribbons are $\frac{1}{2}$ foot long.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 68 • BACKWARD 1 • Q3/5

2502

QUESTION ID

S68-B1-Q3

TYPE

True or false

CORRECT ANSWER

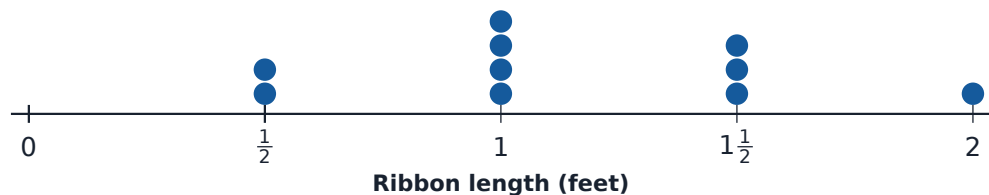
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 68 • BACKWARD 1 • Q3/5

Which ribbon length is most common?



- A. $\frac{1}{2}$ foot B. 1 foot C. $1\frac{1}{2}$ feet D. 2 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 68 • BACKWARD 1 • Q4/5

2503

QUESTION ID

S68-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

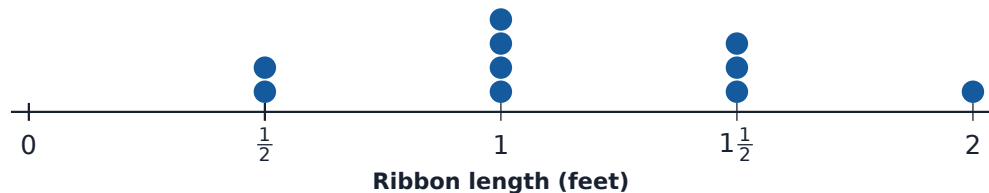
B. 1 foot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 68 • BACKWARD 1 • Q4/5

How many ribbons are longer than 1 foot?



Ribbons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 68 • BACKWARD 1 • Q5/5

2504

QUESTION ID

S68-B1-Q5

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B1-Q5

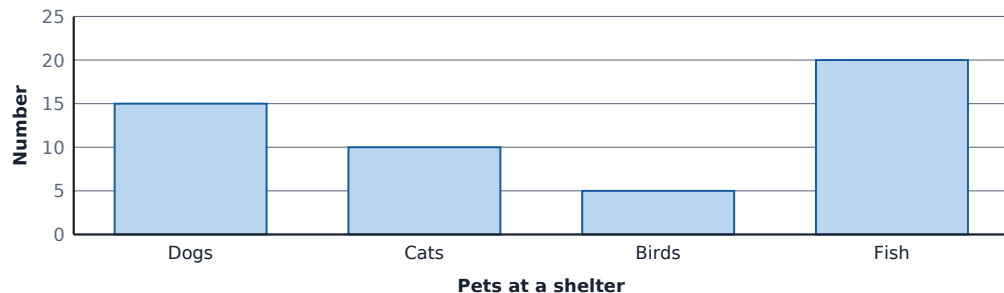
Grade 4 • Measurement & Data • 4.MD.B.4 • Set 68 • BACKWARD 1 • Q5/5

SET 68 • 6.SP.A.2

BACKWARD 2

Most and least in a bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

Which pet does the shelter have the most of?



- A.** Birds **B.** Fish **C.** Dogs **D.** Cats

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 68 • BACKWARD 2 • Q1/5

2506

QUESTION ID

S68-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

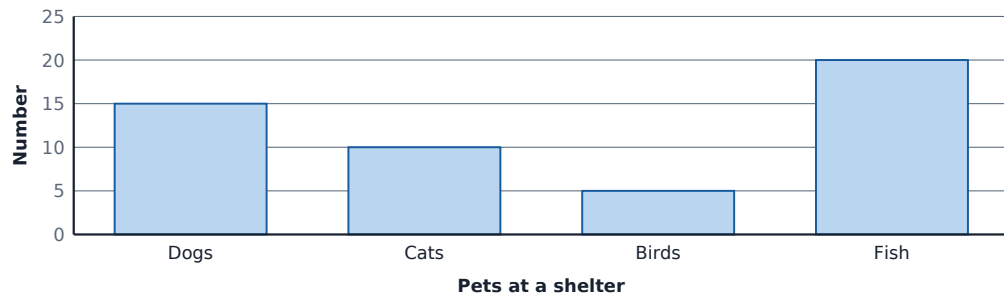
B. Fish

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 68 • BACKWARD 2 • Q1/5

Which pet does the shelter have the fewest of?



- A.** Dogs **B.** Birds **C.** Cats **D.** Fish

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 68 • BACKWARD 2 • Q2/5

2507

QUESTION ID

S68-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Birds

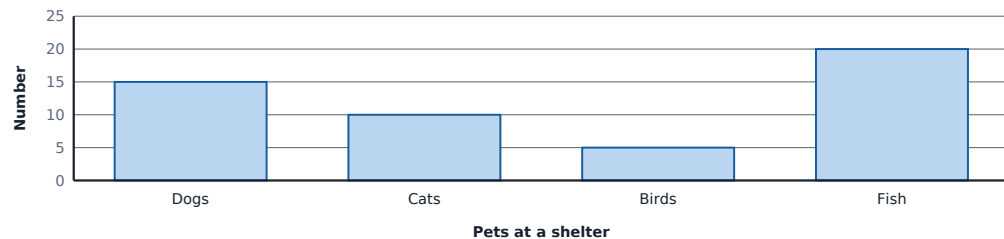
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 68 • BACKWARD 2 • Q2/5

True or false?

The shelter has more dogs than cats.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 68 • BACKWARD 2 • Q3/5

2508

QUESTION ID

S68-B2-Q3

TYPE

True or false

CORRECT ANSWER

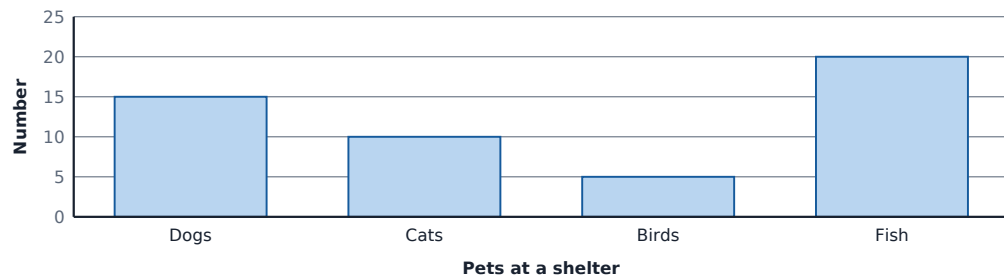
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 68 • BACKWARD 2 • Q3/5

How many more fish than birds are at the shelter?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 68 • BACKWARD 2 • Q4/5

2509

QUESTION ID

S68-B2-Q4

TYPE

Short answer

CORRECT ANSWER

15

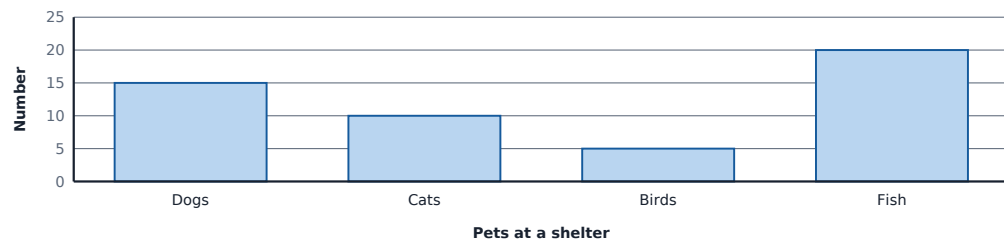
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 68 • BACKWARD 2 • Q4/5

True or false?

The tallest bar shows the smallest number.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 68 • BACKWARD 2 • Q5/5

2510

QUESTION ID

S68-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-B2-Q5

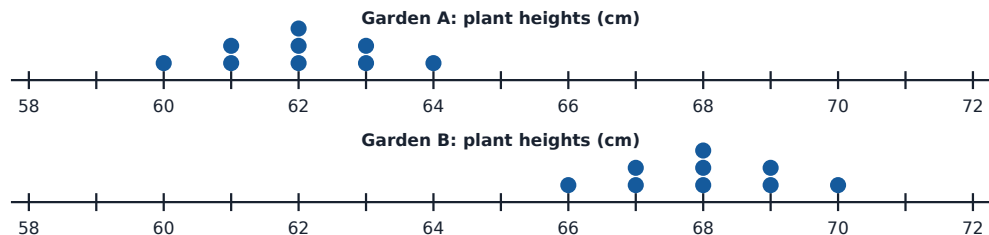
Grade 3 • Measurement & Data • 3.MD.B.3 • Set 68 • BACKWARD 2 • Q5/5

SET 68 • 6.SP.A.2

FORWARD 1

Compare two distributions visually • Grade 7 • Statistics & Probability •
7.SP.B.3

About how many centimeters greater is the center of Garden B's plant heights?



Centimeters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 68 • FORWARD 1 • Q1/5

2512

QUESTION ID

S68-F1-Q1

TYPE

Short answer

CORRECT ANSWER

About 6 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 68 • FORWARD 1 • Q1/5

True or false?

The two distributions have about the same shape and spread.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 68 • FORWARD 1 • Q2/5

2513

QUESTION ID

S68-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 68 • FORWARD 1 • Q2/5

Which statement is true?



- A.** Garden B has more variability. **B.** Garden A's plants are typically taller.
- C.** The distributions do not overlap. **D.** The centers are equal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 68 • FORWARD 1 • Q3/5

2514

QUESTION ID

S68-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. The distributions do not overlap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 68 • FORWARD 1 • Q3/5

Garden A's mean height is 62 cm and Garden B's is 68 cm. Each has a MAD of about 1 cm.

The difference in means is about how many times the MAD?

Times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 68 • FORWARD 1 • Q4/5

2515

QUESTION ID

S68-F1-Q4

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 68 • FORWARD 1 • Q4/5

Two classes have mean scores of 78 and 84. Each has a MAD of 6.

The difference in means is how many times the MAD?

- A.** 0.5 **B.** 1
C. 2 **D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 68 • FORWARD 1 • Q5/5

2516

QUESTION ID

S68-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 68 • FORWARD 1 • Q5/5

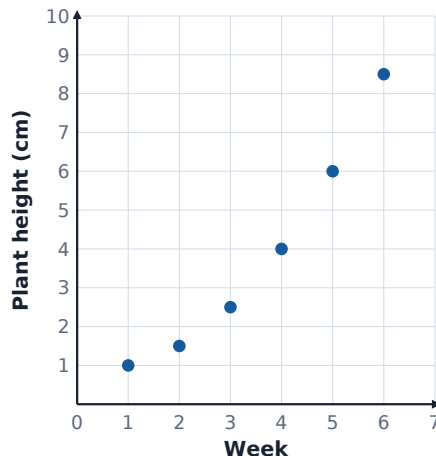
SET 68 • 6.SP.A.2

FORWARD 2

Linear and nonlinear patterns in scatter plots • Grade 8 • Statistics & Probability • 8.SP.A.1

Which describes the pattern in the scatter plot?

- A. Nonlinear
- B. No association
- C. Negative linear
- D. Constant



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 68 • FORWARD 2 • Q1/5

2518

QUESTION ID

S68-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Nonlinear

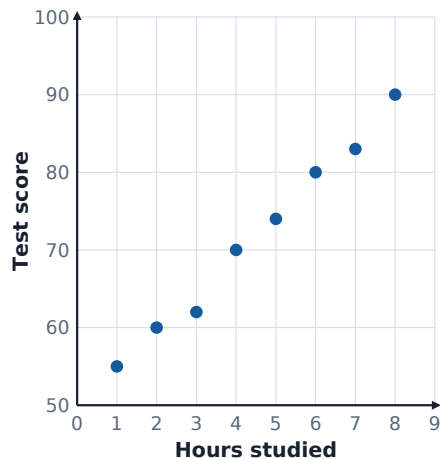
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 68 • FORWARD 2 • Q1/5

Is the pattern in the scatter plot linear or nonlinear?

- A. Linear
- B. Nonlinear



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 68 • FORWARD 2 • Q2/5

2519

QUESTION ID

S68-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. Linear

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

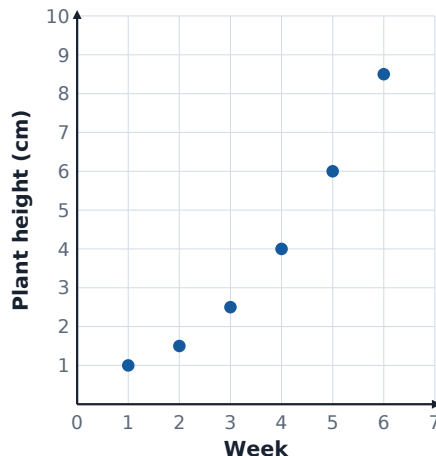
ID S68-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 68 • FORWARD 2 • Q2/5

True or false?

The pattern in the scatter plot is linear.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 68 • FORWARD 2 • Q3/5

2520

QUESTION ID

S68-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

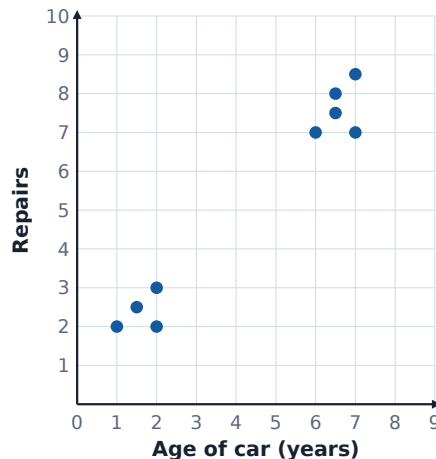
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 68 • FORWARD 2 • Q3/5

How many clusters does the scatter plot show?

Clusters: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 68 • FORWARD 2 • Q4/5

2521

QUESTION ID

S68-F2-Q4

TYPE

Short answer

CORRECT ANSWER

2

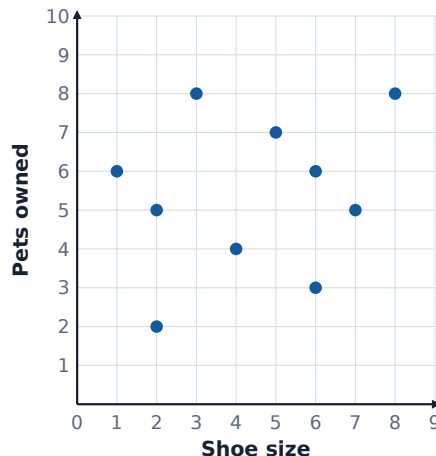
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 68 • FORWARD 2 • Q4/5

Which describes the scatter plot?

- A. Positive linear
- B. Nonlinear
- C. Negative linear
- D. No association



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 68 • FORWARD 2 • Q5/5

2522

QUESTION ID

S68-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. No association

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S68-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 68 • FORWARD 2 • Q5/5

Describe the center of a distribution

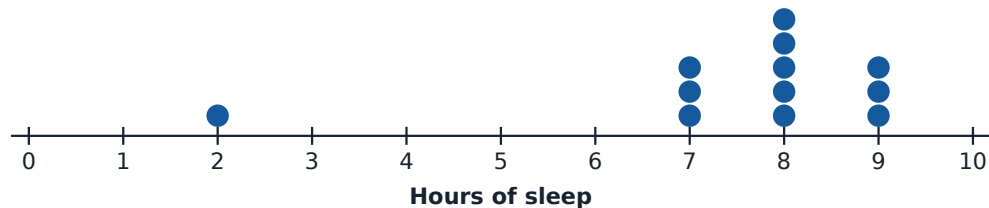
MAIN	Grade 6 main question	6.SP.A.2
BACKWARD 1	Reading a line plot	4.MD.B.4
BACKWARD 2	Ordering numbers	2.NBT.A.4
BACKWARD 3	Halfway between two numbers	2.MD.B.6
FORWARD 1	Compare the centers of two populations	7.SP.B.4
FORWARD 2	Use a line of best fit to make predictions — nearest related (no di...	8.SP.A.2

SET 69 • 6.SP.A.2

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.A.2

Around what value do most of the data cluster?



Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 69 • MAIN • Q1/5

2525

QUESTION ID

S69-M-Q1

TYPE

Short answer

CORRECT ANSWER

About 8 hours

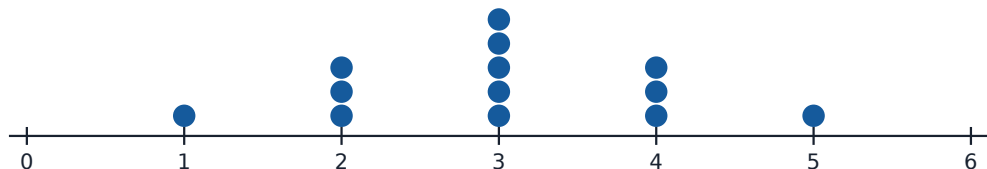
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 69 • MAIN • Q1/5

The data are symmetric.

What value is at the center of the distribution?



Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 69 • MAIN • Q2/5

2526

QUESTION ID

S69-M-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 69 • MAIN • Q2/5

In which interval is the center of the data?



- A.** 0-9 **B.** 10-19 **C.** 20-29 **D.** 30-39

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 69 • MAIN • Q3/5

2527

QUESTION ID

S69-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

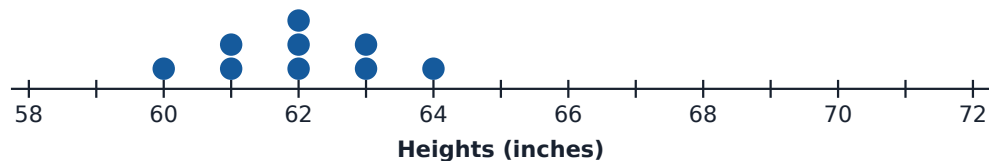
B. 10-19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 69 • MAIN • Q3/5

What value is at the center of the heights?



Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 69 • MAIN • Q4/5

2528

QUESTION ID

S69-M-Q4

TYPE

Short answer

CORRECT ANSWER

62 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 69 • MAIN • Q4/5

True or false?

The center of the distribution is about 12.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 69 • MAIN • Q5/5

2529

QUESTION ID

S69-M-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-M-Q5

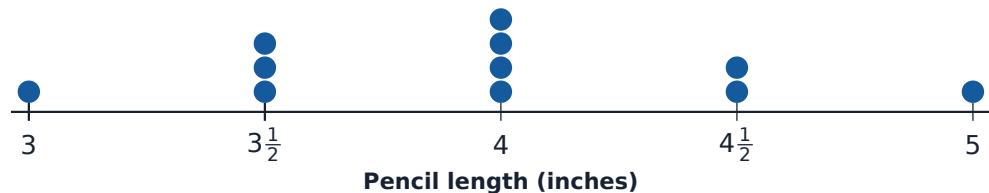
Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 69 • MAIN • Q5/5

SET 69 • 6.SP.A.2

BACKWARD 1

Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

How many pencils are 4 inches long?



Pencils: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 69 • BACKWARD 1 • Q1/5

2531

QUESTION ID

S69-B1-Q1

TYPE

Short answer

CORRECT ANSWER

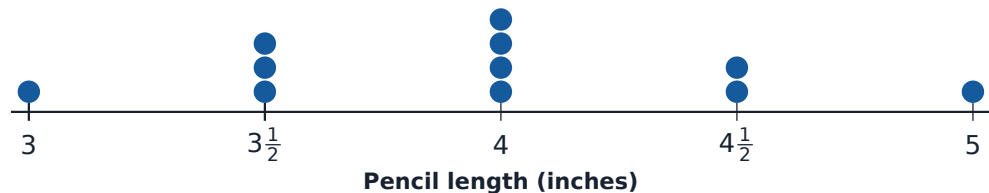
4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 69 • BACKWARD 1 • Q1/5

How many pencils were measured?



Pencils: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 69 • BACKWARD 1 • Q2/5

2532

QUESTION ID

S69-B1-Q2

TYPE

Short answer

CORRECT ANSWER

11

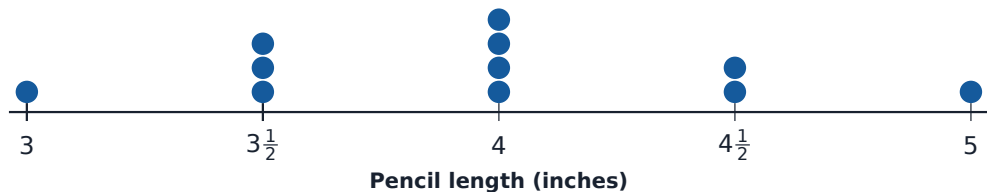
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 69 • BACKWARD 1 • Q2/5

True or false?

Three pencils are $3\frac{1}{2}$ inches long.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 69 • BACKWARD 1 • Q3/5

2533

QUESTION ID

S69-B1-Q3

TYPE

True or false

CORRECT ANSWER

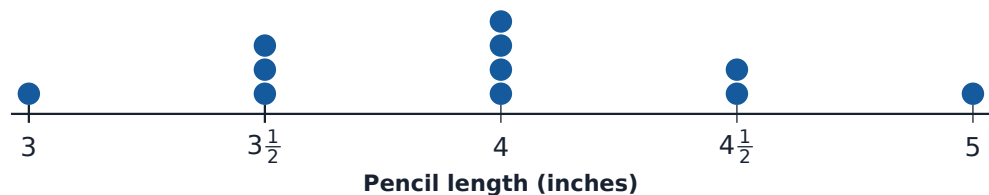
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 69 • BACKWARD 1 • Q3/5

Which pencil length is most common?



- A. 3 inches B. $3\frac{1}{2}$ inches C. 4 inches D. 5 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 69 • BACKWARD 1 • Q4/5

2534

QUESTION ID

S69-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

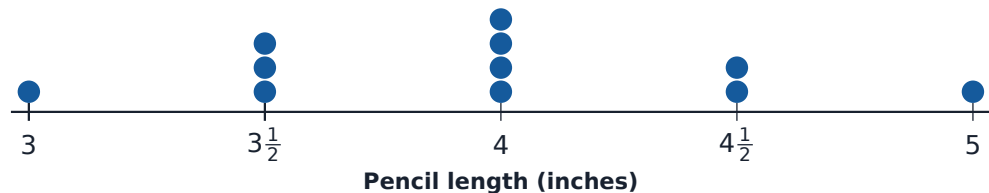
C. 4 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 69 • BACKWARD 1 • Q4/5

How many pencils are shorter than 4 inches?



Pencils: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 69 • BACKWARD 1 • Q5/5

2535

QUESTION ID

S69-B1-Q5

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B1-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 69 • BACKWARD 1 • Q5/5

SET 69 • 6.SP.A.2

BACKWARD 2

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

ANSWER KEY

Reduced copy of the question page, with its answer

Order from least to greatest.

42, 24, 36, 18

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 69 • BACKWARD 2 • Q1/5

2537

QUESTION ID

S69-B2-Q1

TYPE

Short answer

CORRECT ANSWER

18, 24, 36, 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 69 • BACKWARD 2 • Q1/5

2537

Which number is greatest?

- A.** 53
- B.** 299
- C.** 305
- D.** 350

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 69 • BACKWARD 2 • Q2/5

2538

QUESTION ID

S69-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 350

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 69 • BACKWARD 2 • Q2/5

True or false?

These numbers are in order from least to greatest: 12, 21, 19, 30

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 69 • BACKWARD 2 • Q3/5

2539

QUESTION ID

S69-B2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 69 • BACKWARD 2 • Q3/5

Order from least to greatest.

105, 150, 115

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 69 • BACKWARD 2 • Q4/5

2540

QUESTION ID

S69-B2-Q4

TYPE

Short answer

CORRECT ANSWER

105, 115, 150

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 69 • BACKWARD 2 • Q4/5

Which number is least?

- A.** 89
- B.** 91
- C.** 98
- D.** 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 69 • BACKWARD 2 • Q5/5

2541

QUESTION ID

S69-B2-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 89

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 69 • BACKWARD 2 • Q5/5

SET 69 • 6.SP.A.2

BACKWARD 3

Halfway between two numbers • Grade 2 • Measurement & Data • 2.MD.B.6

What number is halfway between 10 and 20?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 69 • BACKWARD 3 • Q1/5

2543

QUESTION ID

S69-B3-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 69 • BACKWARD 3 • Q1/5

What number is halfway between points A and B?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 69 • BACKWARD 3 • Q2/5

2544

QUESTION ID

S69-B3-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 69 • BACKWARD 3 • Q2/5

True or false?

The number halfway between 60 and 64 is 62.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 69 • BACKWARD 3 • Q3/5

2545

QUESTION ID

S69-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 69 • BACKWARD 3 • Q3/5

What number is halfway between 4 and 12?

- A.** 6
- B.** 8
- C.** 10
- D.** 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 69 • BACKWARD 3 • Q4/5

2546

QUESTION ID

S69-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 69 • BACKWARD 3 • Q4/5

What number is halfway between 0 and 18?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 69 • BACKWARD 3 • Q5/5

2547

QUESTION ID

S69-B3-Q5

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-B3-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 69 • BACKWARD 3 • Q5/5

SET 69 • 6.SP.A.2

FORWARD 1

Compare the centers of two populations • Grade 7 • Statistics & Probability •
7.SP.B.4

Class A has a median score of 82. Class B has a median score of 75. The variability is similar.

Which statement is true?

- A.** The classes scored the same.
- B.** Class A typically scored higher.
- C.** Class B typically scored higher.
- D.** Class A's scores vary more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • FORWARD 1 • Q1/5

2549

QUESTION ID

S69-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Class A typically scored higher.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • FORWARD 1 • Q1/5

Random samples: Town A has a mean age of 34 years.
Town B has a mean age of 42 years.
What is the difference in mean ages?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • FORWARD 1 • Q2/5

2550

QUESTION ID

S69-F1-Q2

TYPE

Short answer

CORRECT ANSWER

8 years

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • FORWARD 1 • Q2/5

In random samples, 7th graders sleep a median of 8 hours, and 6th graders sleep a median of 9 hours.

Which group typically sleeps more?

Group: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • FORWARD 1 • Q3/5

2551

QUESTION ID

S69-F1-Q3

TYPE

Short answer

CORRECT ANSWER

6th graders

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • FORWARD 1 • Q3/5

Sample A has a mean of 12 and Sample B has a mean of 15. The samples have similar variability.

Which conclusion is best?

- A.** The populations are the same.
- B.** Values in population A tend to be greater.
- C.** Values in population B tend to be greater.
- D.** No conclusion is possible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • FORWARD 1 • Q4/5

2552

QUESTION ID

S69-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Values in population B tend to be greater.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • FORWARD 1 • Q4/5

True or false?

To compare the centers of two populations, you can compare the medians of random samples.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • FORWARD 1 • Q5/5

2553

QUESTION ID

S69-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 69 • FORWARD 1 • Q5/5

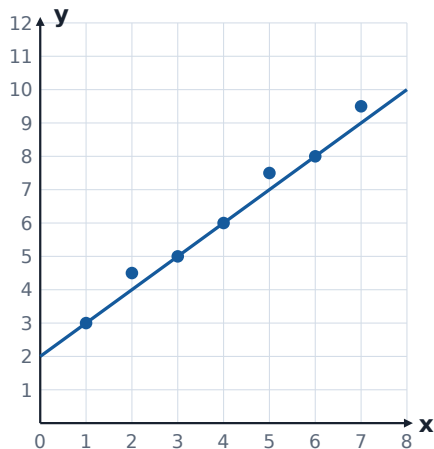
FORWARD 2

Use a line of best fit to make predictions • Grade 8 • Statistics & Probability •
8.SP.A.2

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Use the line of best fit to predict y when $x = 6$.

$y =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 69 • FORWARD 2 • Q1/5 • NEAREST RELATED

2555

QUESTION ID

S69-F2-Q1

TYPE

Short answer

CORRECT ANSWER

8

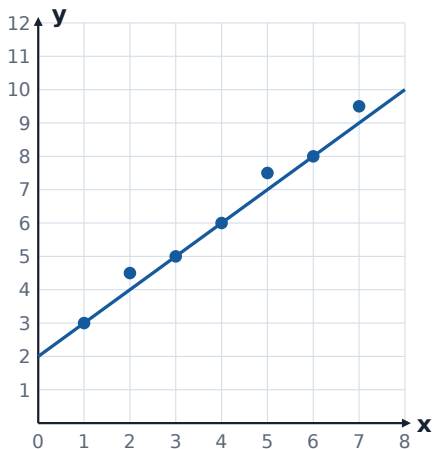
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 69 • FORWARD 2 • Q1/5 • NEAREST RELATED

Use the line of best fit to predict y when $x = 2$.

$y =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 69 • FORWARD 2 • Q2/5 • NEAREST RELATED

2556

QUESTION ID

S69-F2-Q2

TYPE

Short answer

CORRECT ANSWER

4

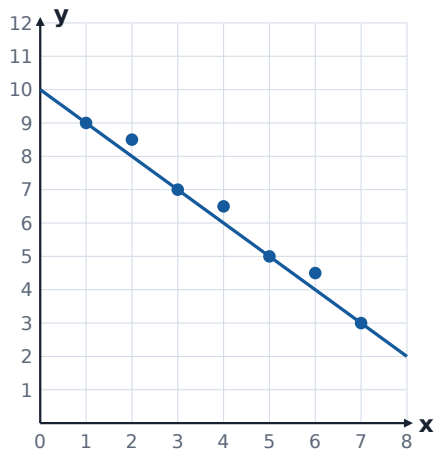
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 69 • FORWARD 2 • Q2/5 • NEAREST RELATED

Use the line of best fit to predict y when $x = 4$.

$y =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 69 • FORWARD 2 • Q3/5 • NEAREST RELATED

2557

QUESTION ID

S69-F2-Q3

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 69 • FORWARD 2 • Q3/5 • NEAREST RELATED

Which statement about a line of best fit is true?

- A. It must pass through every point.
- B. It shows the overall trend of the data.
- C. It must pass through the origin.
- D. It is always horizontal.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 69 • FORWARD 2 • Q4/5 • NEAREST RELATED

2558

QUESTION ID

S69-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. It shows the overall trend of the data.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 69 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A line of best fit usually has about the same number of points above it as below it.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 69 • FORWARD 2 • Q5/5 • NEAREST RELATED

2559

QUESTION ID

S69-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S69-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 69 • FORWARD 2 • Q5/5 • NEAREST RELATED

Describe the spread of a distribution

MAIN	Grade 6 main question	6.SP.A.2
BACKWARD 1	Finding a difference	2.NBT.B.5
BACKWARD 2	Reading a line plot	4.MD.B.4
BACKWARD 3	Greatest and least numbers	2.NBT.A.4
FORWARD 1	Compare variability of two distributions	7.SP.B.3
FORWARD 2	Judge how closely data fit a line	8.SP.A.2

SET 70 • 6.SP.A.2

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.A.2

What is the range of the data?



Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 70 • MAIN • Q1/5

2562

QUESTION ID

S70-M-Q1

TYPE

Short answer

CORRECT ANSWER

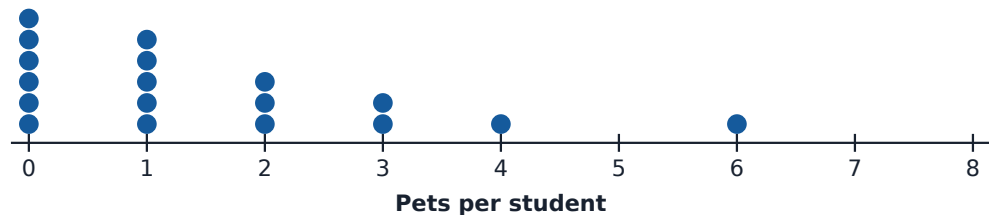
7 minutes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 70 • MAIN • Q1/5

What is the range of the data?



Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 70 • MAIN • Q2/5

2563

QUESTION ID

S70-M-Q2

TYPE

Short answer

CORRECT ANSWER

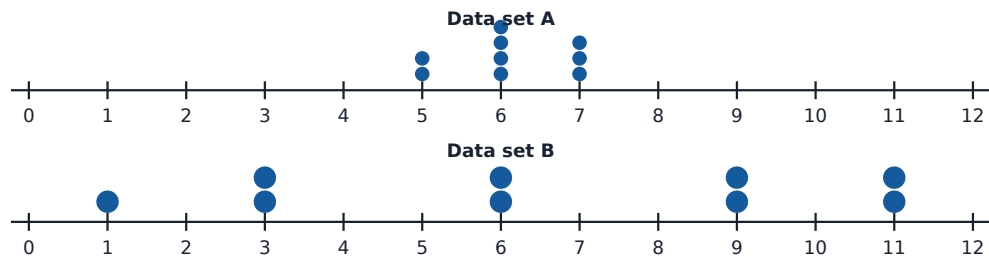
6 pets

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 70 • MAIN • Q2/5

Which data set has the greater spread?



- A.** Data set A **B.** Data set B **C.** They have the same spread.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 70 • MAIN • Q3/5

2564

QUESTION ID

S70-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

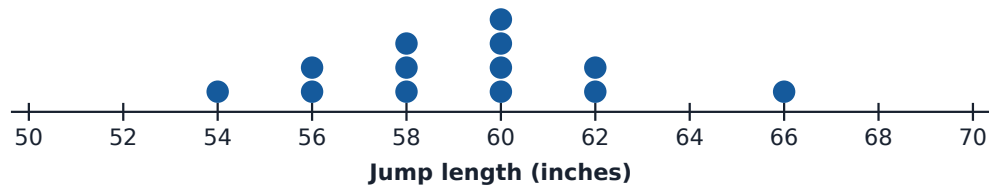
B. Data set B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 70 • MAIN • Q3/5

What is the range of the jump lengths?



Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 70 • MAIN • Q4/5

2565

QUESTION ID

S70-M-Q4

TYPE

Short answer

CORRECT ANSWER

12 inches

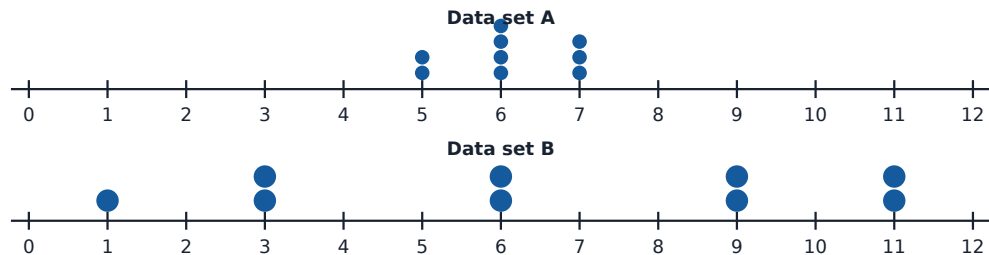
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 70 • MAIN • Q4/5

True or false?

Data set A has a range of 2.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 70 • MAIN • Q5/5

2566

QUESTION ID

S70-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-M-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 70 • MAIN • Q5/5

SET 70 • 6.SP.A.2

BACKWARD 1

Finding a difference • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

The greatest value is 47. The least value is 18.

What is the difference?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 70 • BACKWARD 1 • Q1/5

2568

QUESTION ID

S70-B1-Q1

TYPE

Short answer

CORRECT ANSWER

29

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 70 • BACKWARD 1 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$83 - 29$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 70 • BACKWARD 1 • Q2/5

2569

QUESTION ID

S70-B1-Q2

TYPE

Short answer

CORRECT ANSWER

54

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 70 • BACKWARD 1 • Q2/5

2569

True or false?

$$62 - 35 = 27$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 70 • BACKWARD 1 • Q3/5

2570

QUESTION ID

S70-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 70 • BACKWARD 1 • Q3/5

$$91 - 46 = ?$$

- A.** 45
- B.** 47
- C.** 55
- D.** 137

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 70 • BACKWARD 1 • Q4/5

2571

QUESTION ID

S70-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 70 • BACKWARD 1 • Q4/5

What is the difference between 66 and 54?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 70 • BACKWARD 1 • Q5/5

2572

QUESTION ID

S70-B1-Q5

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B1-Q5

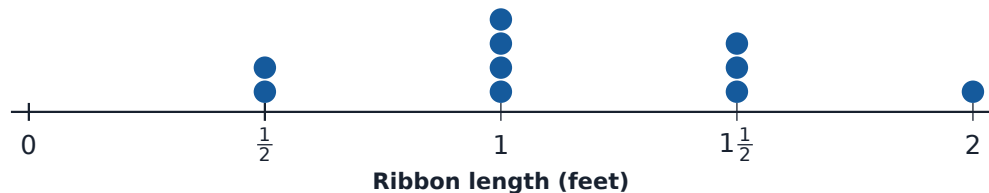
Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 70 • BACKWARD 1 • Q5/5

SET 70 • 6.SP.A.2

BACKWARD 2

Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

What is the longest ribbon length shown?



Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 70 • BACKWARD 2 • Q1/5

2574

QUESTION ID

S70-B2-Q1

TYPE

Short answer

CORRECT ANSWER

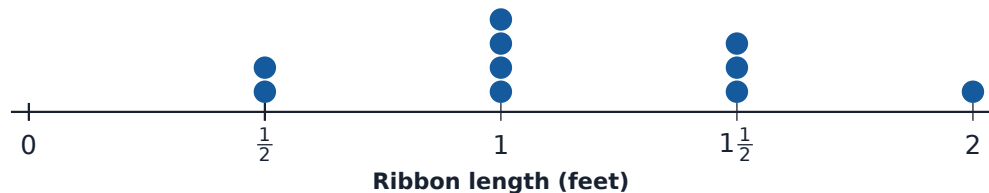
2 feet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 70 • BACKWARD 2 • Q1/5

What is the shortest ribbon length shown?



Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 70 • BACKWARD 2 • Q2/5

2575

QUESTION ID

S70-B2-Q2

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{2}$ foot

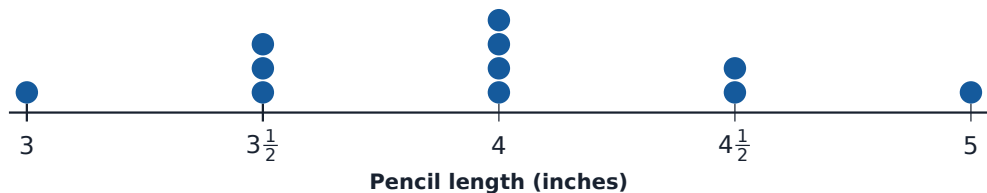
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 70 • BACKWARD 2 • Q2/5

True or false?

The longest pencil is 5 inches.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 70 • BACKWARD 2 • Q3/5

2576

QUESTION ID

S70-B2-Q3

TYPE

True or false

CORRECT ANSWER

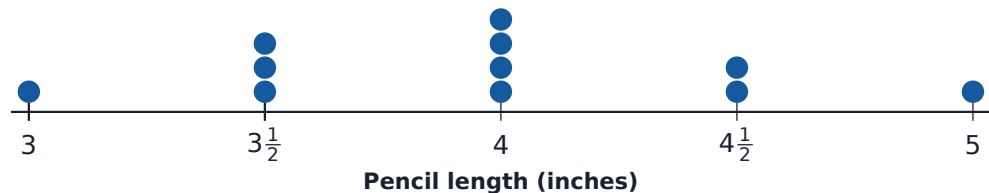
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 70 • BACKWARD 2 • Q3/5

What is the shortest pencil length?



- A. 3 inches B. $3\frac{1}{2}$ inches C. 4 inches D. 5 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 70 • BACKWARD 2 • Q4/5

2577

QUESTION ID

S70-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

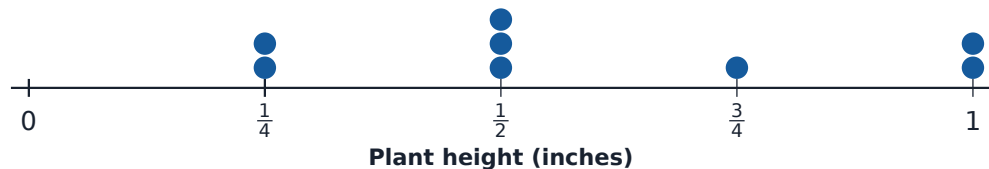
A. 3 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 70 • BACKWARD 2 • Q4/5

How many plants are 1 inch tall?



Plants: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 70 • BACKWARD 2 • Q5/5

2578

QUESTION ID

S70-B2-Q5

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B2-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 70 • BACKWARD 2 • Q5/5

SET 70 • 6.SP.A.2

BACKWARD 3

Greatest and least numbers • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.4

What is the greatest number?

18, 81, 58, 85

Greatest: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 70 • BACKWARD 3 • Q1/5

2580

QUESTION ID

S70-B3-Q1

TYPE

Short answer

CORRECT ANSWER

85

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 70 • BACKWARD 3 • Q1/5

What is the least number?

204, 240, 402, 42

Least: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 70 • BACKWARD 3 • Q2/5

2581

QUESTION ID

S70-B3-Q2

TYPE

Short answer

CORRECT ANSWER

42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 70 • BACKWARD 3 • Q2/5

True or false?

The least number in the list 33, 30, 13, 31 is 13.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 70 • BACKWARD 3 • Q3/5

2582

QUESTION ID

S70-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 70 • BACKWARD 3 • Q3/5

Which number is greatest?

- A.** 207
- B.** 270
- C.** 702
- D.** 720

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 70 • BACKWARD 3 • Q4/5

2583

QUESTION ID

S70-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

D. 720

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 70 • BACKWARD 3 • Q4/5

What is the least number?

66, 56, 65, 60

Least: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 70 • BACKWARD 3 • Q5/5

2584

QUESTION ID

S70-B3-Q5

TYPE

Short answer

CORRECT ANSWER

56

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-B3-Q5

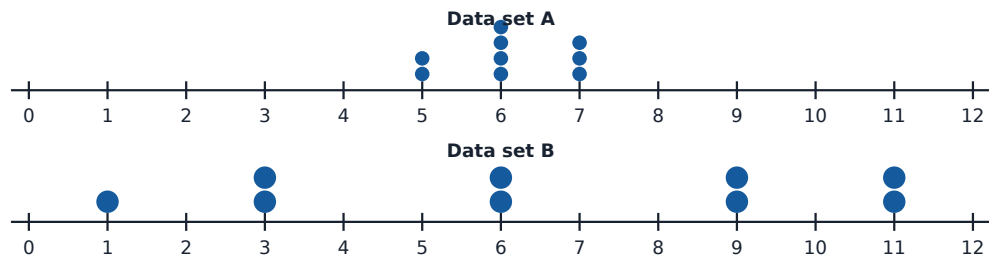
Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 70 • BACKWARD 3 • Q5/5

SET 70 • 6.SP.A.2

FORWARD 1

Compare variability of two distributions • Grade 7 • Statistics & Probability •
7.SP.B.3

Which data set has more variability?



- A.** Data set A **B.** Data set B **C.** They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 70 • FORWARD 1 • Q1/5

2586

QUESTION ID

S70-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Data set B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 70 • FORWARD 1 • Q1/5

Class A's mean is 60 and Class B's mean is 66. Each class has a MAD of 3.

The difference in means is how many times the MAD?

Times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 70 • FORWARD 1 • Q2/5

2587

QUESTION ID

S70-F1-Q2

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 70 • FORWARD 1 • Q2/5

True or false?

Two data sets with means of 50 and 56 and MADs of 3 have means that differ by 2 MADs.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 70 • FORWARD 1 • Q3/5

2588

QUESTION ID

S70-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 70 • FORWARD 1 • Q3/5

Data set A has a mean of 20 and a MAD of 2. Data set B has a mean of 21 and a MAD of 6.

Which statement is true?

- A.** The means differ by 3 MADs.
- B.** Data set A's values are more spread out.
- C.** Data set B's values are more spread out.
- D.** The data sets do not overlap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 70 • FORWARD 1 • Q4/5

2589

QUESTION ID

S70-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Data set B's values are more spread out.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 70 • FORWARD 1 • Q4/5

Two data sets have means of 8 and 14. Each has a MAD of 1.5.

The difference in means is how many times the MAD?

Times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 70 • FORWARD 1 • Q5/5

2590

QUESTION ID

S70-F1-Q5

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 70 • FORWARD 1 • Q5/5

SET 70 • 6.SP.A.2

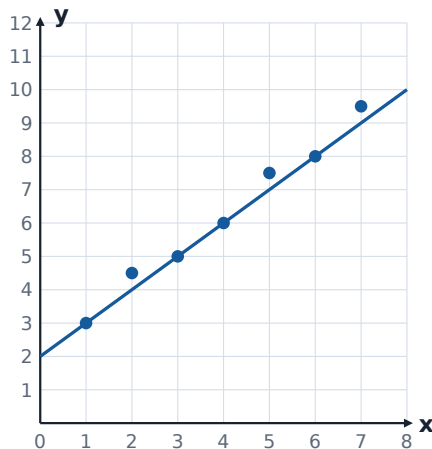
FORWARD 2

Judge how closely data fit a line • Grade 8 • Statistics & Probability • 8.SP.A.2

True or false?

The line is a good fit for the data.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 70 • FORWARD 2 • Q1/5

2592

QUESTION ID

S70-F2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

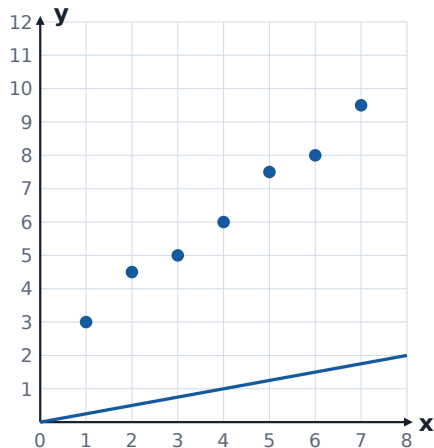
ID S70-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 70 • FORWARD 2 • Q1/5

True or false?

The line is a good fit for the data.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 70 • FORWARD 2 • Q2/5

2593

QUESTION ID

S70-F2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 70 • FORWARD 2 • Q2/5

Which is a sign of a good line of best fit?

- A. The line goes through only one point.
- B. All points are below the line.
- C. Points are close to the line on both sides.
- D. All points are above the line.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 70 • FORWARD 2 • Q3/5

2594

QUESTION ID

S70-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. Points are close to the line on both sides.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

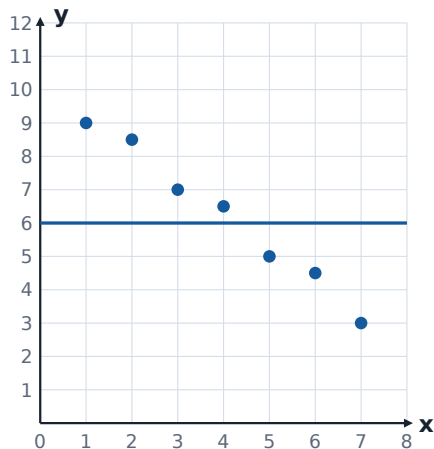
ID S70-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 70 • FORWARD 2 • Q3/5

True or false?

The line is a good fit for the data.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 70 • FORWARD 2 • Q4/5

2595

QUESTION ID

S70-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

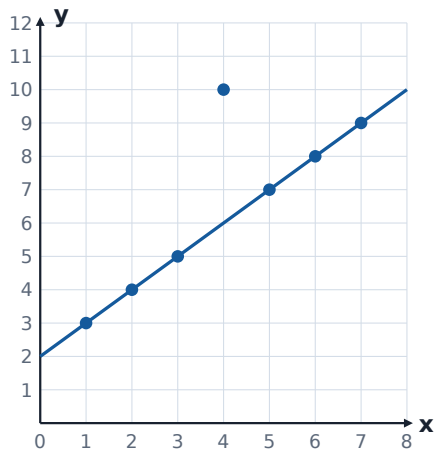
ID S70-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 70 • FORWARD 2 • Q4/5

Which point is farthest from the line?

Write its coordinates.

Point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 70 • FORWARD 2 • Q5/5

2596

QUESTION ID

S70-F2-Q5

TYPE

Short answer

CORRECT ANSWER

(4, 10)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 70 • FORWARD 2 • Q5/5

A measure of center vs. a measure of variation

MAIN	Grade 6 main question	6.SP.A.3
BACKWARD 1	Reading a line plot	4.MD.B.4
BACKWARD 2	Sharing a total equally	5.MD.B.2
BACKWARD 3	Finding a difference	2.NBT.B.5
FORWARD 1	Compare populations using center and variability	7.SP.B.4
FORWARD 2	Interpret the slope and intercept of a linear model — nearest rela...	8.SP.A.3

SET 71 • 6.SP.A.3

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.A.3

Which measure describes the center of a data set?

- A.** Interquartile range
- B.** Mean absolute deviation
- C.** Median
- D.** Range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 71 • MAIN • Q1/5

2599

QUESTION ID

S71-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Median

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q1

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 71 • MAIN • Q1/5

Which measure describes how much the values in a data set vary?

- A. Mean
- B. The middle value
- C. Median
- D. Interquartile range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 71 • MAIN • Q2/5

2600

QUESTION ID

S71-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Interquartile range

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q2

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 71 • MAIN • Q2/5

True or false?

A measure of center summarizes all of the values in a data set with a single number.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 71 • MAIN • Q3/5

2601

QUESTION ID

S71-M-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q3

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 71 • MAIN • Q3/5

True or false?

A measure of variation describes how the values in a data set vary with a single number.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 71 • MAIN • Q4/5

2602

QUESTION ID

S71-M-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q4

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 71 • MAIN • Q4/5

Which measure is a measure of variation?

- A.** Median
- B.** Mean
- C.** Mean absolute deviation
- D.** The middle value

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 71 • MAIN • Q5/5

2603

QUESTION ID

S71-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

**C. Mean absolute
deviation**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-M-Q5

Grade 6 • Statistics & Probability • 6.SP.A.3 • Set 71 • MAIN • Q5/5

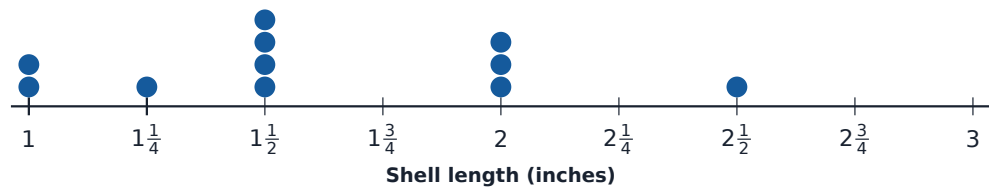
SET 71 • 6.SP.A.3

BACKWARD 1

Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows shell lengths.

How many shells are $1\frac{1}{2}$ inches long?



Shells: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 71 • BACKWARD 1 • Q1/5

2605

QUESTION ID

S71-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

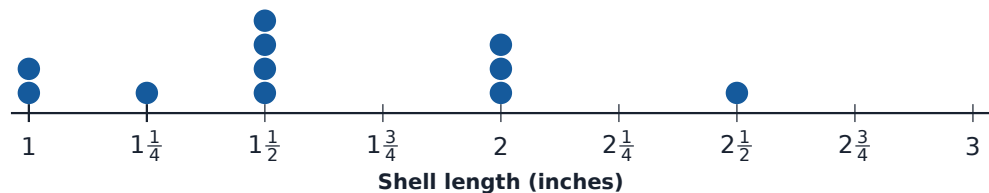
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 71 • BACKWARD 1 • Q1/5

The line plot shows shell lengths.

How many shells are there in all?



Shells: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 71 • BACKWARD 1 • Q2/5

2606

QUESTION ID

S71-B1-Q2

TYPE

Short answer

CORRECT ANSWER

11

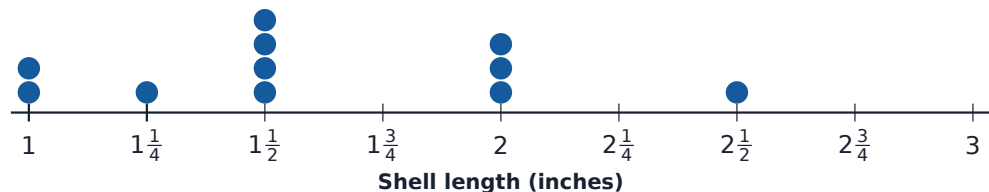
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 71 • BACKWARD 1 • Q2/5

True or false?

Three shells are 2 inches long.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 71 • BACKWARD 1 • Q3/5

2607

QUESTION ID

S71-B1-Q3

TYPE

True or false

CORRECT ANSWER

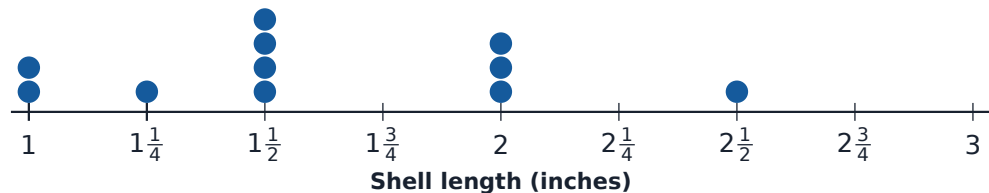
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 71 • BACKWARD 1 • Q3/5

Which shell length is most common?



- A. 1 inch B. $1\frac{1}{2}$ inches C. 2 inches D. $2\frac{1}{2}$ inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 71 • BACKWARD 1 • Q4/5

2608

QUESTION ID

S71-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

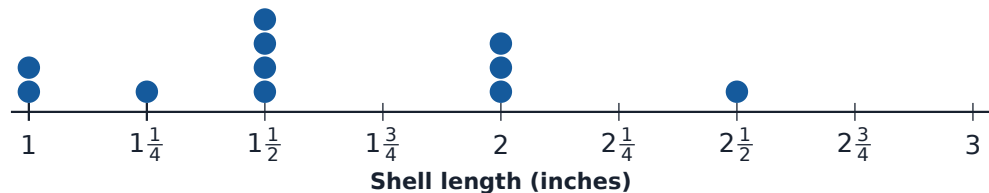
B. $1\frac{1}{2}$ inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 71 • BACKWARD 1 • Q4/5

What is the total length of the 2-inch shells?



Total length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 71 • BACKWARD 1 • Q5/5

2609

QUESTION ID

S71-B1-Q5

TYPE

Short answer

CORRECT ANSWER

6 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B1-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 71 • BACKWARD 1 • Q5/5

SET 71 • 6.SP.A.3

BACKWARD 2

Sharing a total equally • Grade 5 • Measurement & Data • 5.MD.B.2

Three friends have 4, 6, and 8 stickers. They share all of the stickers equally.

How many stickers does each friend get?

Stickers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q1

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 71 • BACKWARD 2 • Q1/5

2611

QUESTION ID

S71-B2-Q1

TYPE

Short answer

CORRECT ANSWER

6 stickers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q1

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 71 • BACKWARD 2 • Q1/5

Four jars hold 2, 3, 5, and 6 liters of water. The water is shared equally among the jars.

How much water is in each jar?

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q2

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 71 • BACKWARD 2 • Q2/5

2612

QUESTION ID

S71-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4 liters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q2

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 71 • BACKWARD 2 • Q2/5

True or false?

Cups holding 1, 2, and 3 liters are shared equally.
Each cup then holds 2 liters.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q3

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 71 • BACKWARD 2 • Q3/5

2613

QUESTION ID

S71-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q3

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 71 • BACKWARD 2 • Q3/5

Five friends collected 3, 5, 6, 7, and 9 cans. They share the cans equally.

How many cans does each friend get?

A. 5

B. 6

C. 7

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q4

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 71 • BACKWARD 2 • Q4/5

2614

QUESTION ID

S71-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q4

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 71 • BACKWARD 2 • Q4/5

Two bottles hold $\frac{1}{4}$ liter and $\frac{3}{4}$ liter. The juice is shared equally between the bottles.

How much juice is in each bottle?

Liters: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q5

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 71 • BACKWARD 2 • Q5/5

2615

QUESTION ID

S71-B2-Q5

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{2}$ liter

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B2-Q5

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 71 • BACKWARD 2 • Q5/5

SET 71 • 6.SP.A.3

BACKWARD 3

Finding a difference • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$76 - 38$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 71 • BACKWARD 3 • Q1/5

2617

QUESTION ID

S71-B3-Q1

TYPE

Short answer

CORRECT ANSWER

38

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 71 • BACKWARD 3 • Q1/5

2617

The highest score is 95 and the lowest is 58.

What is the difference?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 71 • BACKWARD 3 • Q2/5

2618

QUESTION ID

S71-B3-Q2

TYPE

Short answer

CORRECT ANSWER

37

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 71 • BACKWARD 3 • Q2/5

True or false?

$$100 - 47 = 53$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 71 • BACKWARD 3 • Q3/5

2619

QUESTION ID

S71-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 71 • BACKWARD 3 • Q3/5

$$84 - 29 = ?$$

- A.** 45
- B.** 55
- C.** 65
- D.** 113

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 71 • BACKWARD 3 • Q4/5

2620

QUESTION ID

S71-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 55

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 71 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$62 - 17$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 71 • BACKWARD 3 • Q5/5

2621

QUESTION ID

S71-B3-Q5

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-B3-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 71 • BACKWARD 3 • Q5/5

2621

SET 71 • 6.SP.A.3

FORWARD 1

Compare populations using center and variability • Grade 7 • Statistics & Probability • 7.SP.B.4

Class A has a median score of 82 and an IQR of 10.
Class B has a median score of 75 and an IQR of 10.

Which statement is true?

- A.** Class A typically scored higher.
- B.** Class A's scores vary more.
- C.** Class B typically scored higher.
- D.** The classes scored the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 71 • FORWARD 1 • Q1/5

2623

QUESTION ID

S71-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Class A typically scored higher.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 71 • FORWARD 1 • Q1/5

True or false?

Two samples have the same median. The sample with the greater IQR has more variability.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 71 • FORWARD 1 • Q2/5

2624

QUESTION ID

S71-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 71 • FORWARD 1 • Q2/5

Which measure is best for comparing the variability of two samples?

- A.** IQR
- B.** Sample size
- C.** Median
- D.** Mean

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 71 • FORWARD 1 • Q3/5

2625

QUESTION ID

S71-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. IQR

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 71 • FORWARD 1 • Q3/5

Sample A: mean 40, MAD 2. Sample B: mean 40, MAD 7.

Which sample has values that vary more?

Sample: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 71 • FORWARD 1 • Q4/5

2626

QUESTION ID

S71-F1-Q4

TYPE

Short answer

CORRECT ANSWER

Sample B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 71 • FORWARD 1 • Q4/5

Store A's daily sales have a median of \$520. Store B's have a median of \$610. Their IQRs are similar.

Which store typically has greater daily sales?

Store: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 71 • FORWARD 1 • Q5/5

2627

QUESTION ID

S71-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Store B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 71 • FORWARD 1 • Q5/5

FORWARD 2

Interpret the slope and intercept of a linear model • Grade 8 • Statistics & Probability • 8.SP.A.3

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The model $y = 2.5x + 40$ predicts a test score y after x hours of study.

What does the slope 2.5 mean?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 71 • FORWARD 2 • Q1/5 • NEAREST RELATED

2629

QUESTION ID

S71-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Each additional hour of study adds 2.5 points to the predicted score.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 71 • FORWARD 2 • Q1/5 • NEAREST RELATED

The model $y = 3x + 12$ gives a plant's height y (cm) after x weeks.

What does 12 represent?

- A.** The number of weeks
- B.** The height at week 0
- C.** The height after 12 weeks
- D.** The growth per week

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 71 • FORWARD 2 • Q2/5 • NEAREST RELATED

2630

QUESTION ID

S71-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. The height at week 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 71 • FORWARD 2 • Q2/5 • NEAREST RELATED

True or false?

In the model $y = -0.5x + 30$, y decreases by 0.5 for each increase of 1 in x .

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 71 • FORWARD 2 • Q3/5 • NEAREST RELATED

2631

QUESTION ID

S71-F2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 71 • FORWARD 2 • Q3/5 • NEAREST RELATED

The model $y = 1.2x + 5$ gives the length y (cm) of a fish at x months old.

What does 5 represent?

Meaning: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 71 • FORWARD 2 • Q4/5 • NEAREST RELATED

2632

QUESTION ID

S71-F2-Q4

TYPE

Short answer

CORRECT ANSWER

The fish's length at 0 months: it starts at 5 cm.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 71 • FORWARD 2 • Q4/5 • NEAREST RELATED

In the model $y = 45x + 200$, what does 45 represent if y is dollars saved after x weeks?

- A. Total savings
- B. Starting savings
- C. Number of weeks
- D. Dollars saved per week

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 71 • FORWARD 2 • Q5/5 • NEAREST RELATED

2633

QUESTION ID

S71-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. Dollars saved per week

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S71-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 71 • FORWARD 2 • Q5/5 • NEAREST RELATED

Display numerical data in a dot plot

MAIN	Grade 6 main question	6.SP.B.4
BACKWARD 1	Line plots of whole-number measurements	2.MD.D.9
BACKWARD 2	Line plots with fractional measurements	4.MD.B.4
BACKWARD 3	Ordering numbers	2.NBT.A.4
FORWARD 1	Compare two dot plots	7.SP.B.3
FORWARD 2	Construct a scatter plot	8.SP.A.1

SET 72 • 6.SP.B.4

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.4

Data: 2, 4, 4, 5, 7, 7, 7, 9

Make a dot plot of the data on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 72 • MAIN • Q1/5

2636

QUESTION ID

S72-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Dot plot: 1 dot at 2,
2 dots at 4, 1 dot at
5, 3 dots at 7, 1 dot
at 9**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 72 • MAIN • Q1/5

Number of pets: 0, 1, 1, 2, 2, 2, 3, 5

Make a dot plot of the data on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 72 • MAIN • Q2/5

2637

QUESTION ID

S72-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Dot plot: 1 dot at 0,
2 dots at 1, 3 dots at
2, 1 dot at 3, 1 dot
at 5**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 72 • MAIN • Q2/5

Quiz scores: 6, 7, 7, 8, 8, 8, 9, 10, 10

Make a dot plot of the data on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 72 • MAIN • Q3/5

2638

QUESTION ID

S72-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

**Dot plot: 1 dot at 6,
2 dots at 7, 3 dots at
8, 1 dot at 9, 2 dots
at 10**

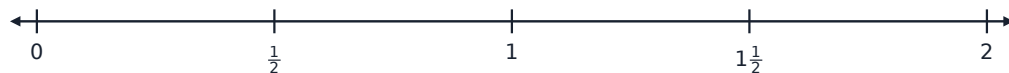
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 72 • MAIN • Q3/5

Ribbon lengths (feet): $\frac{1}{2}$, 1, 1, $1\frac{1}{2}$, 2, 1

Make a dot plot of the data on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 72 • MAIN • Q4/5

2639

QUESTION ID

S72-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Dot plot: 1 dot at $\frac{1}{2}$, 3 dots at 1, 1 dot at $1\frac{1}{2}$, 1 dot at 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 72 • MAIN • Q4/5

Hours of sleep: 7, 8, 9, 8, 10, 7, 8

Make a dot plot of the data on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 72 • MAIN • Q5/5

2640

QUESTION ID

S72-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

**Dot plot: 2 dots at 7,
3 dots at 8, 1 dot at
9, 1 dot at 10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 72 • MAIN • Q5/5

SET 72 • 6.SP.B.4

BACKWARD 1

Line plots of whole-number measurements • Grade 2 • Measurement & Data •
2.MD.D.9

Lengths (inches): 3, 4, 4, 5, 4

In a line plot, how many marks are drawn at 4?

Marks: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q1

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 72 • BACKWARD 1 • Q1/5

2642

QUESTION ID

S72-B1-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q1

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 72 • BACKWARD 1 • Q1/5

Lengths (cm): 2, 3, 3, 5

Make a line plot of the lengths.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q2

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 72 • BACKWARD 1 • Q2/5

2643

QUESTION ID

S72-B1-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Line plot: 1 mark at 2, 2 marks at 3, 1 mark at 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q2

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 72 • BACKWARD 1 • Q2/5

True or false?

In a line plot of 6, 6, 7, 9, there are two marks above 6.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q3

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 72 • BACKWARD 1 • Q3/5

2644

QUESTION ID

S72-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q3

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 72 • BACKWARD 1 • Q3/5

Which number line is best for a line plot of 12, 14, 15, 15?

- A. A number line from 0 to 100 by tens
- B. A number line from 20 to 30
- C. A number line from 10 to 16
- D. A number line from 0 to 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q4

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 72 • BACKWARD 1 • Q4/5

2645

QUESTION ID

S72-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

**C. A number line
from 10 to 16**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q4

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 72 • BACKWARD 1 • Q4/5

Lengths (feet): 8, 9, 9, 9, 11

Which value gets the most marks?

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q5

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 72 • BACKWARD 1 • Q5/5

2646

QUESTION ID

S72-B1-Q5

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B1-Q5

Grade 2 • Measurement & Data • 2.MD.D.9 • Set 72 • BACKWARD 1 • Q5/5

SET 72 • 6.SP.B.4

BACKWARD 2

Line plots with fractional measurements • Grade 4 • Measurement & Data •
4.MD.B.4

Lengths (inches): $\frac{1}{2}$, 1, 1, $1\frac{1}{2}$, 1, $\frac{1}{2}$

In a line plot, how many X's are drawn at 1?

X's: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 72 • BACKWARD 2 • Q1/5

2648

QUESTION ID

S72-B2-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 72 • BACKWARD 2 • Q1/5

True or false?

In a line plot of 2, $2\frac{1}{2}$, $2\frac{1}{2}$, 3, there are two X's above $2\frac{1}{2}$.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 72 • BACKWARD 2 • Q2/5

2649

QUESTION ID

S72-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 72 • BACKWARD 2 • Q2/5

Which number line would you use for these lengths?

$\frac{1}{4}$, $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{2}$

- A. A number line from 0 to 10 marked in ones
- B. A number line from 0 to 100 marked in tens
- C. A number line from 1 to 2 marked in halves
- D. A number line from 0 to 1 marked in fourths

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 72 • BACKWARD 2 • Q3/5

2650

QUESTION ID

S72-B2-Q3

TYPE

Multiple choice

CORRECT ANSWER

**D. A number line
from 0 to 1 marked
in fourths**

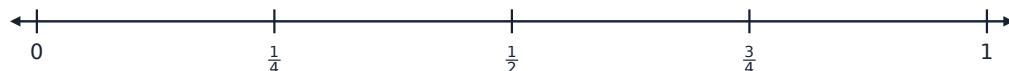
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 72 • BACKWARD 2 • Q3/5

Lengths (inches): $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$

Make a line plot of the lengths.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 72 • BACKWARD 2 • Q4/5

2651

QUESTION ID

S72-B2-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

**Line plot: 1 X at $\frac{1}{4}$,
2 X's at $\frac{1}{2}$, 1 X at
 $\frac{3}{4}$**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 72 • BACKWARD 2 • Q4/5

True or false?

A line plot shows each data value as a mark above a number line.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 72 • BACKWARD 2 • Q5/5

2652

QUESTION ID

S72-B2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B2-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 72 • BACKWARD 2 • Q5/5

SET 72 • 6.SP.B.4

BACKWARD 3

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

ANSWER KEY

Reduced copy of the question page, with its answer

Order from least to greatest.

7, 2, 9, 4, 4

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 72 • BACKWARD 3 • Q1/5

2654

QUESTION ID

S72-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2, 4, 4, 7, 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 72 • BACKWARD 3 • Q1/5

2654

Which list is in order from least to greatest?

- A.** 6, 7, 8, 10
- B.** 10, 8, 7, 6
- C.** 7, 6, 8, 10
- D.** 6, 8, 7, 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 72 • BACKWARD 3 • Q2/5

2655

QUESTION ID

S72-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 6, 7, 8, 10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 72 • BACKWARD 3 • Q2/5

True or false?

These numbers are in order from least to greatest: 1, 2, 5, 3

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 72 • BACKWARD 3 • Q3/5

2656

QUESTION ID

S72-B3-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 72 • BACKWARD 3 • Q3/5

What are the least and greatest values?

8, 10, 7, 9, 7

Least: _____

Greatest: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 72 • BACKWARD 3 • Q4/5

2657

QUESTION ID

S72-B3-Q4

TYPE

Short answer

CORRECT ANSWER

**Least: 7; Greatest:
10**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 72 • BACKWARD 3 • Q4/5

Which number is least?

- A.** 0
- B.** 1
- C.** 2
- D.** 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 72 • BACKWARD 3 • Q5/5

2658

QUESTION ID

S72-B3-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 0

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-B3-Q5

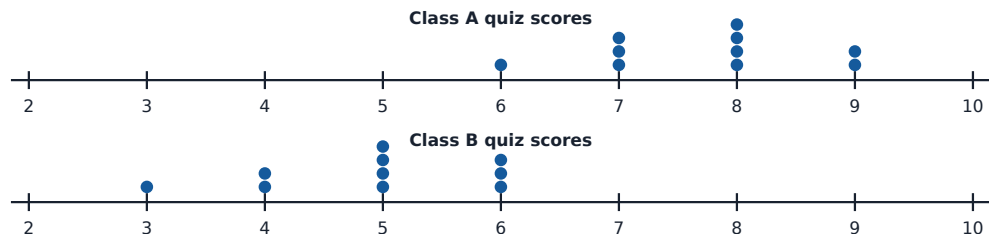
Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 72 • BACKWARD 3 • Q5/5

SET 72 • 6.SP.B.4

FORWARD 1

Compare two dot plots • Grade 7 • Statistics & Probability • 7.SP.B.3

About how many points greater is the center of Class A's scores than Class B's?



Points: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 72 • FORWARD 1 • Q1/5

2660

QUESTION ID

S72-F1-Q1

TYPE

Short answer

CORRECT ANSWER

About 3 points

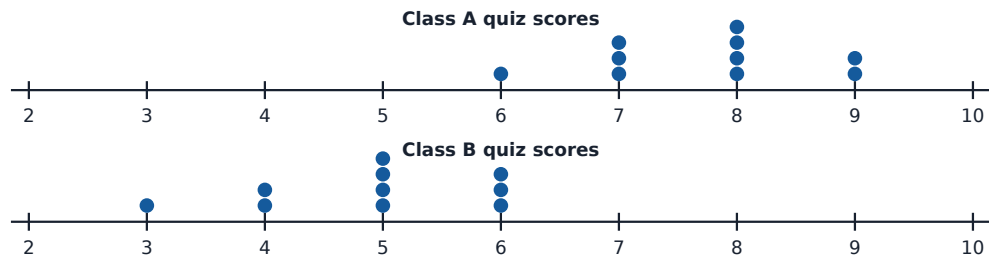
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 72 • FORWARD 1 • Q1/5

True or false?

The two dot plots overlap.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 72 • FORWARD 1 • Q2/5

2661

QUESTION ID

S72-F1-Q2

TYPE

True or false

CORRECT ANSWER

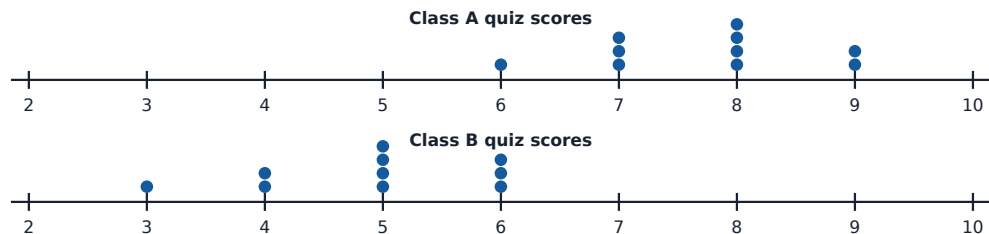
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 72 • FORWARD 1 • Q2/5

Which statement is true?



- A.** Class B has more variability. **B.** The medians are equal.
- C.** Class B's scores are typically higher. **D.** Class A's scores are typically higher.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 72 • FORWARD 1 • Q3/5

2662

QUESTION ID

S72-F1-Q3

TYPE

Multiple choice

CORRECT ANSWER

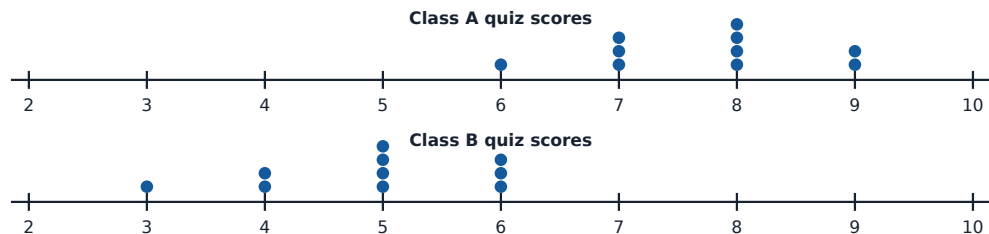
D. Class A's scores are typically higher.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 72 • FORWARD 1 • Q3/5

What is the range of each class's scores?



Class A: _____

Class B: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 72 • FORWARD 1 • Q4/5

2663

QUESTION ID

S72-F1-Q4

TYPE

Short answer

CORRECT ANSWER

Class A: 3; Class B: 3

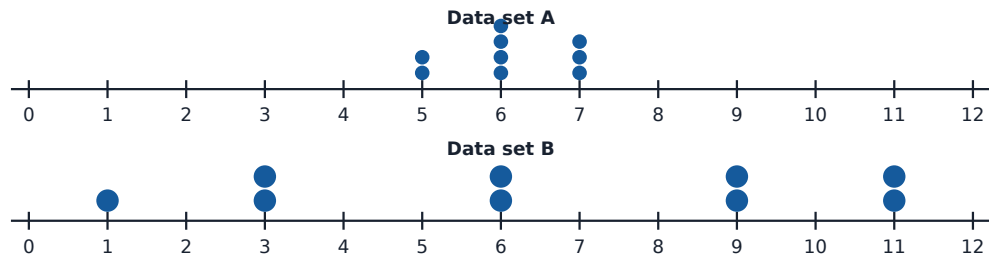
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 72 • FORWARD 1 • Q4/5

True or false?

Data set A and data set B have the same median.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 72 • FORWARD 1 • Q5/5

2664

QUESTION ID

S72-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 72 • FORWARD 1 • Q5/5

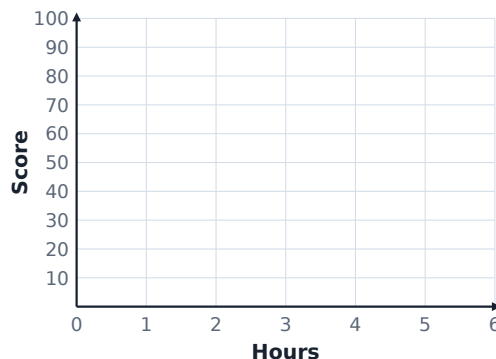
SET 72 • 6.SP.B.4

FORWARD 2

Construct a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

Make a scatter plot of the data.

Hours	1	2	3	4	5
Score	60	65	75	80	90



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 72 • FORWARD 2 • Q1/5

2666

QUESTION ID

S72-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Scatter plot with
points (1, 60), (2,
65), (3, 75), (4, 80),
(5, 90)**

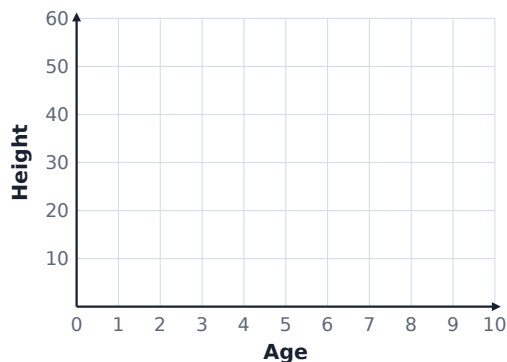
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 72 • FORWARD 2 • Q1/5

Make a scatter plot of the data.

Age	2	4	6	8
Height (in.)	34	40	46	50



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 72 • FORWARD 2 • Q2/5

2667

QUESTION ID

S72-F2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Scatter plot with
points (2, 34), (4,
40), (6, 46), (8, 50)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 72 • FORWARD 2 • Q2/5

A scatter plot shows age on the x-axis and height on the y-axis.

A 10-year-old is 54 inches tall. Which point shows this person?

- A.** (54, 10) **B.** (54, 54)
C. (10, 10) **D.** (10, 54)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 72 • FORWARD 2 • Q3/5

2668

QUESTION ID

S72-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. (10, 54)

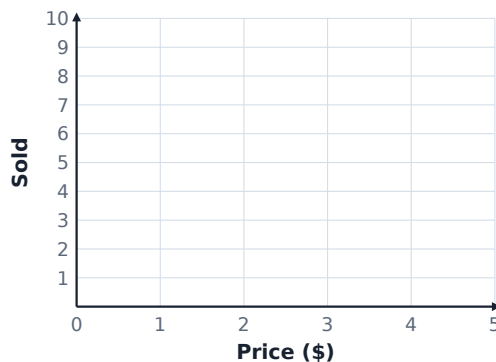
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 72 • FORWARD 2 • Q3/5

Make a scatter plot of the data.

Price (\$)	1	2	3	4
Sold	9	7	4	2



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 72 • FORWARD 2 • Q4/5

2669

QUESTION ID

S72-F2-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

**Scatter plot with
points (1, 9), (2, 7),
(3, 4), (4, 2)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 72 • FORWARD 2 • Q4/5

True or false?

To make a scatter plot of height and arm span, each person is shown as one point.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 72 • FORWARD 2 • Q5/5

2670

QUESTION ID

S72-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S72-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 72 • FORWARD 2 • Q5/5

Display numerical data in a histogram

MAIN	Grade 6 main question	6.SP.B.4
BACKWARD 1	Drawing a scaled bar graph	3.MD.B.3
BACKWARD 2	Placing numbers in intervals	2.NBT.A.4
BACKWARD 3	Organizing data into categories	1.MD.C.4
FORWARD 1	Compare two populations shown in histograms	7.SP.B.4
FORWARD 2	Display numerical data for two variables in a scatter plot — neares...	8.SP.A.1

SET 73 • 6.SP.B.4

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.4

Books read: 3, 12, 15, 8, 22, 17, 25, 11, 31, 19

Make a histogram of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 73 • MAIN • Q1/5

2673

QUESTION ID

S73-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Histogram bars: 0-9:
2; 10-19: 5; 20-29:
2; 30-39: 1**

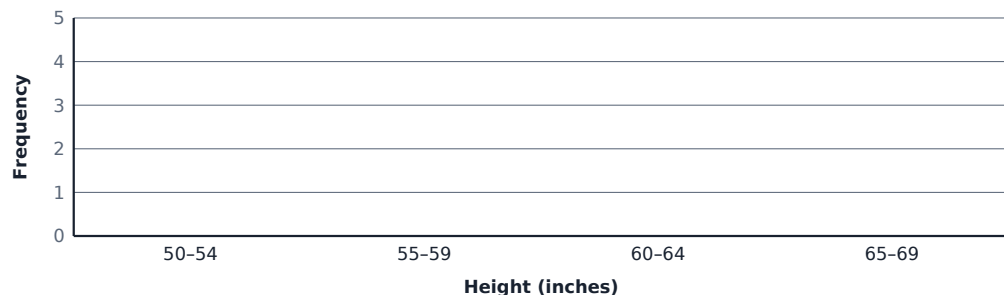
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 73 • MAIN • Q1/5

Heights (in.): 52, 58, 61, 55, 63, 57, 66, 60, 54, 59

Make a histogram of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 73 • MAIN • Q2/5

2674

QUESTION ID

S73-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

Histogram bars:

50-54: 2; 55-59: 4;

60-64: 3; 65-69: 1

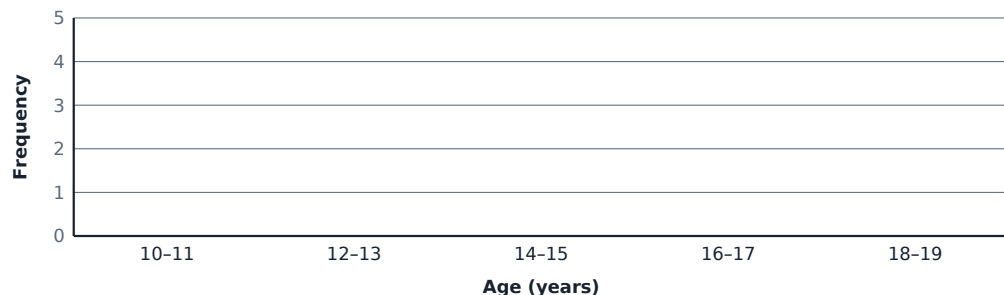
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 73 • MAIN • Q2/5

Ages: 11, 12, 12, 13, 14, 14, 14, 15, 16, 17, 19

Make a histogram of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 73 • MAIN • Q3/5

2675

QUESTION ID

S73-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

Histogram bars:

10-11: 1; 12-13: 3;

14-15: 4; 16-17: 2;

18-19: 1

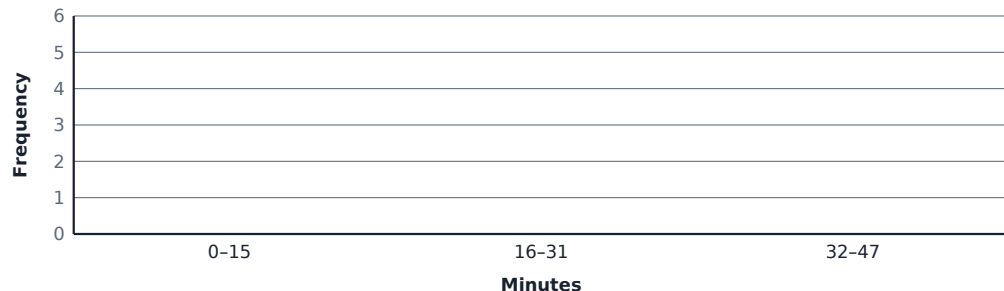
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 73 • MAIN • Q3/5

Minutes of exercise: 5, 18, 22, 25, 30, 34, 41, 45, 12, 28

Make a histogram of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 73 • MAIN • Q4/5

2676

QUESTION ID

S73-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

Histogram bars:
0-15: 2; 16-31: 5;
32-47: 3

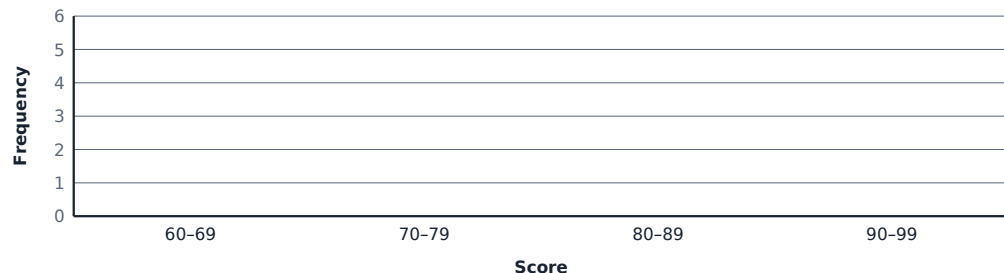
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 73 • MAIN • Q4/5

Test scores: 62, 75, 78, 81, 84, 88, 90, 93, 95, 71, 86, 79

Make a histogram of the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 73 • MAIN • Q5/5

2677

QUESTION ID

S73-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

Histogram bars:

60-69: 1; 70-79: 4;

80-89: 4; 90-99: 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 73 • MAIN • Q5/5

SET 73 • 6.SP.B.4

BACKWARD 1

Drawing a scaled bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

Draw a bar graph: Red 4, Blue 6, Green 2.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 73 • BACKWARD 1 • Q1/5

2679

QUESTION ID

S73-B1-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Bar graph bars: Red
4, Blue 6, Green 2**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 73 • BACKWARD 1 • Q1/5

On a bar graph, the bar for "Cats" ends at 5 on a scale that counts by 1s.

How many cats does the bar show?

Cats: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 73 • BACKWARD 1 • Q2/5

2680

QUESTION ID

S73-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 73 • BACKWARD 1 • Q2/5

True or false?

On a bar graph with a scale of 2, a bar that stops halfway between 6 and 8 shows 7.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 73 • BACKWARD 1 • Q3/5

2681

QUESTION ID

S73-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 73 • BACKWARD 1 • Q3/5

A bar graph has a scale that counts by 5s. A bar shows 20.

How many grid lines tall is the bar?

A. 4

B. 5

C. 15

D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 73 • BACKWARD 1 • Q4/5

2682

QUESTION ID

S73-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

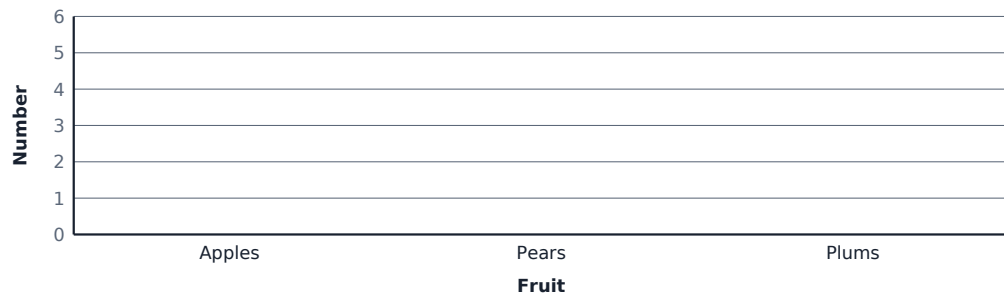
A. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 73 • BACKWARD 1 • Q4/5

Draw a bar graph: Apples 3, Pears 1, Plums 5.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 73 • BACKWARD 1 • Q5/5

2683

QUESTION ID

S73-B1-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

**Bar graph bars:
Apples 3, Pears 1,
Plums 5**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B1-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 73 • BACKWARD 1 • Q5/5

SET 73 • 6.SP.B.4

BACKWARD 2

Placing numbers in intervals • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.4

Which interval contains 17?

- A.** 0–9
- B.** 10–19
- C.** 20–29
- D.** 30–39

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 73 • BACKWARD 2 • Q1/5

2685

QUESTION ID

S73-B2-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. 10–19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 73 • BACKWARD 2 • Q1/5

True or false?

25 is in the interval 20–29.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 73 • BACKWARD 2 • Q2/5

2686

QUESTION ID

S73-B2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 73 • BACKWARD 2 • Q2/5

Data: 3, 12, 15, 18, 22

How many of the numbers are in the interval 10–19?

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 73 • BACKWARD 2 • Q3/5

2687

QUESTION ID

S73-B2-Q3

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 73 • BACKWARD 2 • Q3/5

Which number is in the interval 30–39?

- A.** 3
- B.** 29
- C.** 34
- D.** 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 73 • BACKWARD 2 • Q4/5

2688

QUESTION ID

S73-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 34

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 73 • BACKWARD 2 • Q4/5

True or false?

19 is in the interval 20–29.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 73 • BACKWARD 2 • Q5/5

2689

QUESTION ID

S73-B2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B2-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 73 • BACKWARD 2 • Q5/5

SET 73 • 6.SP.B.4

BACKWARD 3

Organizing data into categories • Grade 1 • Measurement & Data • 1.MD.C.4

Data: cat, dog, cat, fish, dog, cat

How many answers are "cat"?

Cat: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q1

Grade 1 • Measurement & Data • 1.MD.C.4 • Set 73 • BACKWARD 3 • Q1/5

2691

QUESTION ID

S73-B3-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q1

Grade 1 • Measurement & Data • 1.MD.C.4 • Set 73 • BACKWARD 3 • Q1/5

True or false?

Data: red, blue, red, red. There are 3 answers of "red."

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q2

Grade 1 • Measurement & Data • 1.MD.C.4 • Set 73 • BACKWARD 3 • Q2/5

2692

QUESTION ID

S73-B3-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q2

Grade 1 • Measurement & Data • 1.MD.C.4 • Set 73 • BACKWARD 3 • Q2/5

Data: A, B, A, C, A, B

Which category has the most?

- A.** A
- B.** B
- C.** C

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q3

Grade 1 • Measurement & Data • 1.MD.C.4 • Set 73 • BACKWARD 3 • Q3/5

2693

QUESTION ID

S73-B3-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q3

Grade 1 • Measurement & Data • 1.MD.C.4 • Set 73 • BACKWARD 3 • Q3/5

Data: yes, no, yes, yes, no

How many more "yes" answers than "no" answers are there?

More: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q4

Grade 1 • Measurement & Data • 1.MD.C.4 • Set 73 • BACKWARD 3 • Q4/5

2694

QUESTION ID

S73-B3-Q4

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q4

Grade 1 • Measurement & Data • 1.MD.C.4 • Set 73 • BACKWARD 3 • Q4/5

Data: 3, 7, 12, 14, 18

How many numbers are less than 10?

Numbers: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q5

Grade 1 • Measurement & Data • 1.MD.C.4 • Set 73 • BACKWARD 3 • Q5/5

2695

QUESTION ID

S73-B3-Q5

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-B3-Q5

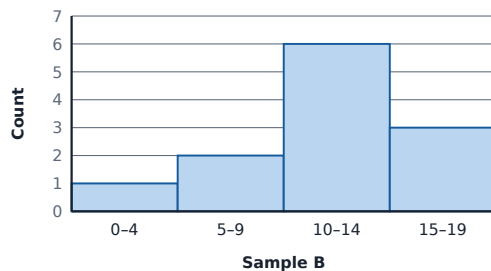
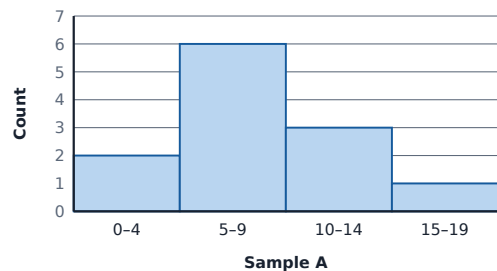
Grade 1 • Measurement & Data • 1.MD.C.4 • Set 73 • BACKWARD 3 • Q5/5

SET 73 • 6.SP.B.4

FORWARD 1

Compare two populations shown in histograms • Grade 7 • Statistics & Probability • 7.SP.B.4

Which sample tends to have greater values?



- A.** Sample A **B.** Sample B **C.** They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 73 • FORWARD 1 • Q1/5

2697

QUESTION ID

S73-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

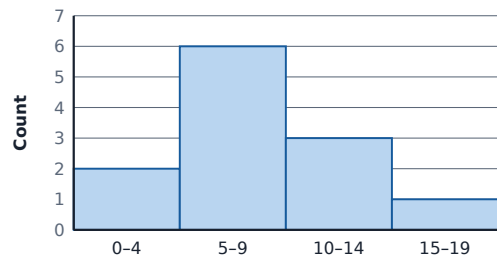
B. Sample B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

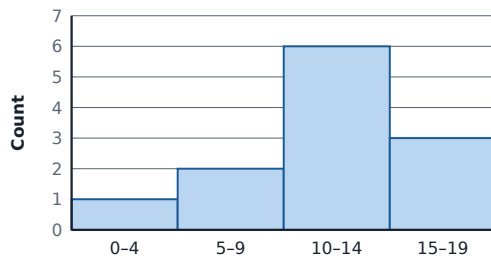
ID S73-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 73 • FORWARD 1 • Q1/5

In which interval is the median of Sample A?



Sample A



Sample B

- A. 0-4 B. 5-9 C. 10-14 D. 15-19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 73 • FORWARD 1 • Q2/5

2698

QUESTION ID

S73-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

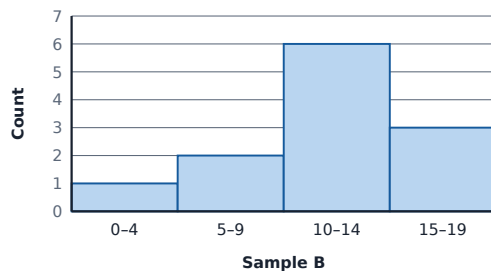
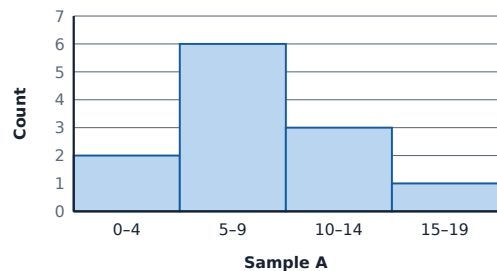
B. 5-9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 73 • FORWARD 1 • Q2/5

How many values are in each sample?



Values: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 73 • FORWARD 1 • Q3/5

2699

QUESTION ID

S73-F1-Q3

TYPE

Short answer

CORRECT ANSWER

12 values in each sample

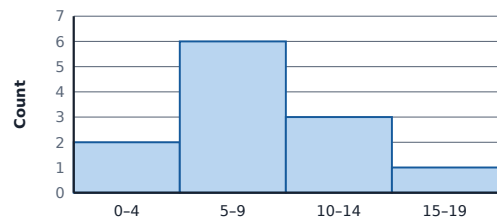
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q3

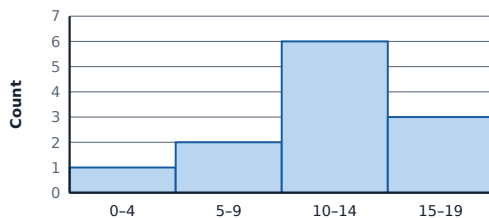
Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 73 • FORWARD 1 • Q3/5

True or false?

Sample B has more values from 15 to 19 than Sample A.



Sample A



Sample B

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 73 • FORWARD 1 • Q4/5

2700

QUESTION ID

S73-F1-Q4

TYPE

True or false

CORRECT ANSWER

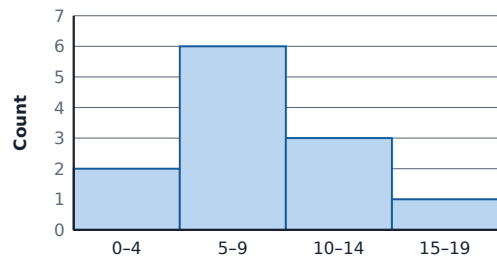
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

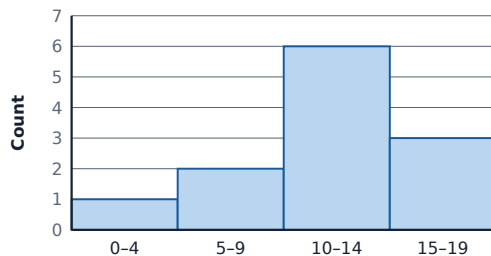
ID S73-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 73 • FORWARD 1 • Q4/5

In which interval is the median of Sample B?



Sample A



Sample B

- A. 0-4 B. 5-9 C. 10-14 D. 15-19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 73 • FORWARD 1 • Q5/5

2701

QUESTION ID

S73-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. 10-14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 73 • FORWARD 1 • Q5/5

FORWARD 2

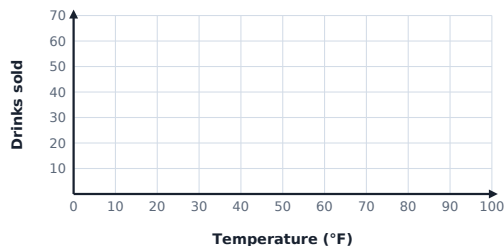
Display numerical data for two variables in a scatter plot • Grade 8 • Statistics & Probability • 8.SP.A.1

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Make a scatter plot of the data.

Describe the association between temperature and drinks sold.

Temp (°F)	60	70	75	85	90
Drinks sold	20	35	40	55	65



Association: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 73 • FORWARD 2 • Q1/5 • NEAREST RELATED

2703

QUESTION ID

S73-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Points plotted at (60, 20), (70, 35), (75, 40), (85, 55), (90, 65). Positive association: as the temperature increases, more drinks are sold.

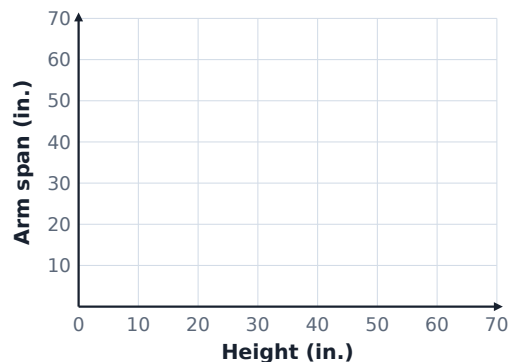
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 73 • FORWARD 2 • Q1/5 • NEAREST RELATED

Make a scatter plot of the data.

Height (in.)	50	54	58	62
Arm span (in.)	49	55	57	63



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 73 • FORWARD 2 • Q2/5 • NEAREST RELATED

2704

QUESTION ID

S73-F2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Points plotted at
(50, 49), (54, 55),
(58, 57), (62, 63)**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 73 • FORWARD 2 • Q2/5 • NEAREST RELATED

A data set gives the height and the shoe size of 20 students.

Which display shows how the two variables are related?

- A.** Dot plot
- B.** Scatter plot
- C.** Box plot
- D.** Histogram

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 73 • FORWARD 2 • Q3/5 • NEAREST RELATED

2705

QUESTION ID

S73-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Scatter plot

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 73 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

A histogram displays one numerical variable, while a scatter plot displays two numerical variables for each person or object.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 73 • FORWARD 2 • Q4/5 • NEAREST RELATED

2706

QUESTION ID

S73-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

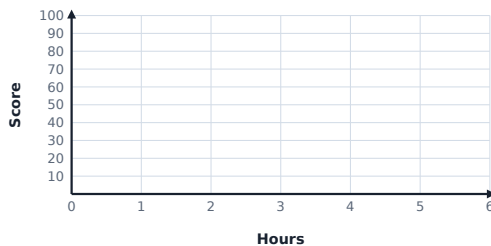
ID S73-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 73 • FORWARD 2 • Q4/5 • NEAREST RELATED

Make a scatter plot of the data.

Which point does not fit the pattern?

Hours	1	2	3	4	5
Score	62	68	40	80	86



Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 73 • FORWARD 2 • Q5/5 • NEAREST RELATED

2707

QUESTION ID

S73-F2-Q5

TYPE

Short answer

CORRECT ANSWER

Points plotted at (1, 62), (2, 68), (3, 40), (4, 80), (5, 86). The point (3, 40) does not fit the pattern.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S73-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 73 • FORWARD 2 • Q5/5 • NEAREST RELATED

Display numerical data in a box plot

MAIN	Grade 6 main question	6.SP.B.4
BACKWARD 1	Ordering numbers	2.NBT.A.4
BACKWARD 2	Halfway between two numbers	2.MD.B.6
BACKWARD 3	Reading a number line scale	4.MD.A.2
BACKWARD 4	Finding the median	6.SP.B.5.c
FORWARD 1	Compare populations with box plots	7.SP.B.4
FORWARD 2	Spread and outliers in scatter plots — nearest related (no direct G...	8.SP.A.1

SET 74 • 6.SP.B.4

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.4

Data: 3, 5, 7, 8, 10, 12, 15

Make a box plot of the data on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 74 • MAIN • Q1/5

2710

QUESTION ID

S74-M-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

**Box plot: minimum
3, Q1 5, median 8,
Q3 12, maximum 15**

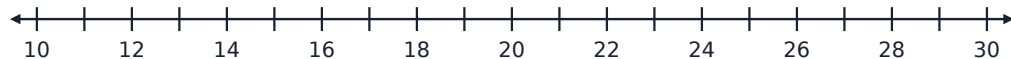
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 74 • MAIN • Q1/5

Data: 12, 14, 15, 18, 20, 21, 25, 28

Make a box plot of the data on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 74 • MAIN • Q2/5

2711

QUESTION ID

S74-M-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

**Box plot: minimum
12, Q1 14.5, median
19, Q3 23, maximum
28**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 74 • MAIN • Q2/5

Data: 40, 45, 45, 50, 55, 60, 70

Make a box plot of the data on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 74 • MAIN • Q3/5

2712

QUESTION ID

S74-M-Q3

TYPE

Drawing / plotting

CORRECT ANSWER

**Box plot: minimum
40, Q1 45, median
50, Q3 60, maximum
70**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 74 • MAIN • Q3/5

Data: 2, 4, 4, 5, 6, 8, 9, 11, 12

Make a box plot of the data on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 74 • MAIN • Q4/5

2713

QUESTION ID

S74-M-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

**Box plot: minimum
2, Q1 4, median 6,
Q3 10, maximum 12**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 74 • MAIN • Q4/5

Data: 21, 23, 24, 26, 28, 30, 31, 35

Make a box plot of the data on the number line.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 74 • MAIN • Q5/5

2714

QUESTION ID

S74-M-Q5

TYPE

Drawing / plotting

CORRECT ANSWER

**Box plot: minimum
21, Q1 23.5, median
27, Q3 30.5,
maximum 35**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.4 • Set 74 • MAIN • Q5/5

SET 74 • 6.SP.B.4

BACKWARD 1

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

ANSWER KEY

Reduced copy of the question page, with its answer

Order from least to greatest.

15, 8, 23, 4, 16

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 74 • BACKWARD 1 • Q1/5

2716

QUESTION ID

S74-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4, 8, 15, 16, 23

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 74 • BACKWARD 1 • Q1/5

2716

Which list is in order from least to greatest?

- A. 27, 21, 12, 7
- B. 7, 21, 12, 27
- C. 12, 7, 21, 27
- D. 7, 12, 21, 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 74 • BACKWARD 1 • Q2/5

2717

QUESTION ID

S74-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 7, 12, 21, 27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 74 • BACKWARD 1 • Q2/5

True or false?

These numbers are in order from least to greatest: 30, 33, 31, 35

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 74 • BACKWARD 1 • Q3/5

2718

QUESTION ID

S74-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 74 • BACKWARD 1 • Q3/5

Order from least to greatest.

62, 26, 60, 20, 66

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 74 • BACKWARD 1 • Q4/5

2719

QUESTION ID

S74-B1-Q4

TYPE

Short answer

CORRECT ANSWER

20, 26, 60, 62, 66

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 74 • BACKWARD 1 • Q4/5

Which number is least?

- A.** 108
- B.** 118
- C.** 180
- D.** 801

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 74 • BACKWARD 1 • Q5/5

2720

QUESTION ID

S74-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 108

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 74 • BACKWARD 1 • Q5/5

SET 74 • 6.SP.B.4

BACKWARD 2

Halfway between two numbers • Grade 2 • Measurement & Data • 2.MD.B.6

What number is halfway between 12 and 18?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 74 • BACKWARD 2 • Q1/5

2722

QUESTION ID

S74-B2-Q1

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q1

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 74 • BACKWARD 2 • Q1/5

What number is halfway between points A and B?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 74 • BACKWARD 2 • Q2/5

2723

QUESTION ID

S74-B2-Q2

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q2

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 74 • BACKWARD 2 • Q2/5

True or false?

The number halfway between 40 and 50 is 45.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 74 • BACKWARD 2 • Q3/5

2724

QUESTION ID

S74-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q3

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 74 • BACKWARD 2 • Q3/5

What number is halfway between 6 and 10?

- A.** 4
- B.** 7
- C.** 8
- D.** 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 74 • BACKWARD 2 • Q4/5

2725

QUESTION ID

S74-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q4

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 74 • BACKWARD 2 • Q4/5

What number is halfway between 0 and 30?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 74 • BACKWARD 2 • Q5/5

2726

QUESTION ID

S74-B2-Q5

TYPE

Short answer

CORRECT ANSWER

15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B2-Q5

Grade 2 • Measurement & Data • 2.MD.B.6 • Set 74 • BACKWARD 2 • Q5/5

SET 74 • 6.SP.B.4

BACKWARD 3

Reading a number line scale • Grade 4 • Measurement & Data • 4.MD.A.2

What number is at point A?



A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q1

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 74 • BACKWARD 3 • Q1/5

2728

QUESTION ID

S74-B3-Q1

TYPE

Short answer

CORRECT ANSWER

65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q1

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 74 • BACKWARD 3 • Q1/5

What number is at point B?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q2

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 74 • BACKWARD 3 • Q2/5

2729

QUESTION ID

S74-B3-Q2

TYPE

Short answer

CORRECT ANSWER

34

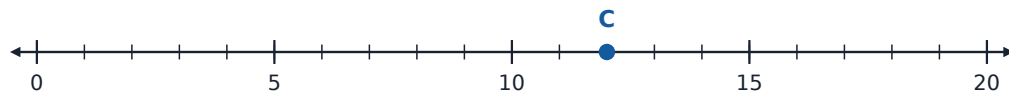
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q2

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 74 • BACKWARD 3 • Q2/5

True or false?

Point C is at 12.



- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q3

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 74 • BACKWARD 3 • Q3/5

2730

QUESTION ID

S74-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q3

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 74 • BACKWARD 3 • Q3/5

What number is at point D?



A. 150

B. 170

C. 175

D. 180

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q4

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 74 • BACKWARD 3 • Q4/5

2731

QUESTION ID

S74-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 175

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q4

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 74 • BACKWARD 3 • Q4/5

What number is at point E?



E = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q5

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 74 • BACKWARD 3 • Q5/5

2732

QUESTION ID

S74-B3-Q5

TYPE

Short answer

CORRECT ANSWER

450

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B3-Q5

Grade 4 • Measurement & Data • 4.MD.A.2 • Set 74 • BACKWARD 3 • Q5/5

SET 74 • 6.SP.B.4

BACKWARD 4

Finding the median • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median.

3, 5, 7, 8, 10, 12, 15

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B4-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 74 • BACKWARD 4 • Q1/5

2734

QUESTION ID

S74-B4-Q1

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B4-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 74 • BACKWARD 4 • Q1/5

Find the median.

12, 14, 15, 18, 20, 21, 25, 28

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B4-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 74 • BACKWARD 4 • Q2/5

2735

QUESTION ID

S74-B4-Q2

TYPE

Short answer

CORRECT ANSWER

19

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B4-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 74 • BACKWARD 4 • Q2/5

True or false?

The median of 40, 45, 45, 50, 55, 60, 70 is 50.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B4-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 74 • BACKWARD 4 • Q3/5

2736

QUESTION ID

S74-B4-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B4-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 74 • BACKWARD 4 • Q3/5

What is the median of 2, 4, 4, 5, 6, 8, 9, 11, 12?

- A.** 5
- B.** 6
- C.** 7
- D.** 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B4-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 74 • BACKWARD 4 • Q4/5

2737

QUESTION ID

S74-B4-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B4-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 74 • BACKWARD 4 • Q4/5

Find the median of the lower half.

3, 5, 7

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B4-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 74 • BACKWARD 4 • Q5/5

2738

QUESTION ID

S74-B4-Q5

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-B4-Q5

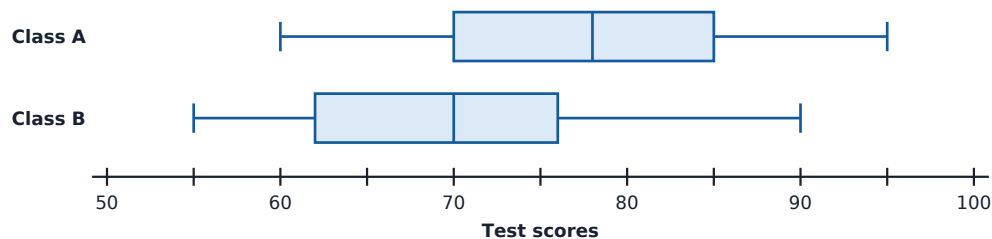
Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 74 • BACKWARD 4 • Q5/5

SET 74 • 6.SP.B.4

FORWARD 1

Compare populations with box plots • Grade 7 • Statistics & Probability •
7.SP.B.4

Which class has the greater median score?



- A.** Class A **B.** Class B **C.** They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 74 • FORWARD 1 • Q1/5

2740

QUESTION ID

S74-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

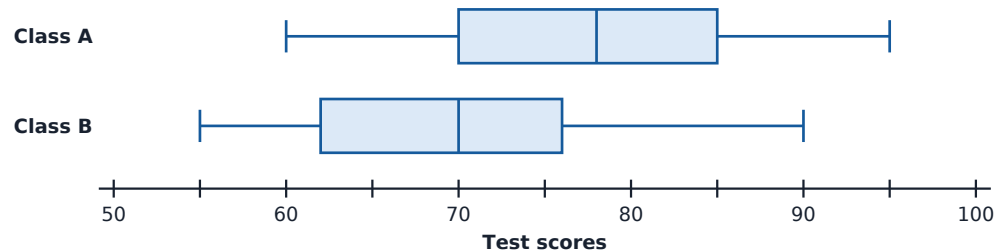
A. Class A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 74 • FORWARD 1 • Q1/5

By how many points do the medians differ?



- A. 3 B. 5 C. 8 D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 74 • FORWARD 1 • Q2/5

2741

QUESTION ID

S74-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

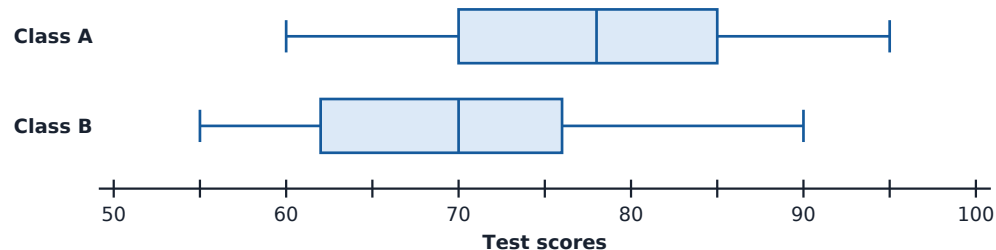
C. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 74 • FORWARD 1 • Q2/5

What is the interquartile range of Class A's scores?



IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 74 • FORWARD 1 • Q3/5

2742

QUESTION ID

S74-F1-Q3

TYPE

Short answer

CORRECT ANSWER

15

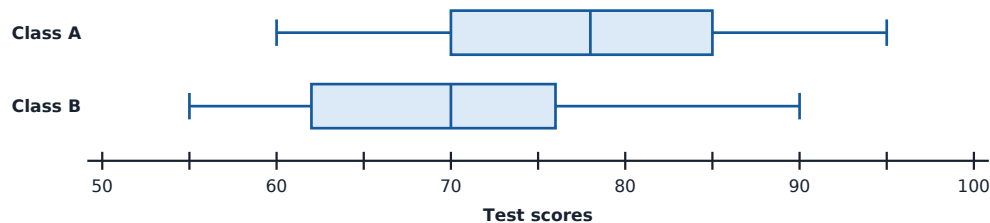
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 74 • FORWARD 1 • Q3/5

True or false?

At least half of Class A scored 78 or higher.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 74 • FORWARD 1 • Q4/5

2743

QUESTION ID

S74-F1-Q4

TYPE

True or false

CORRECT ANSWER

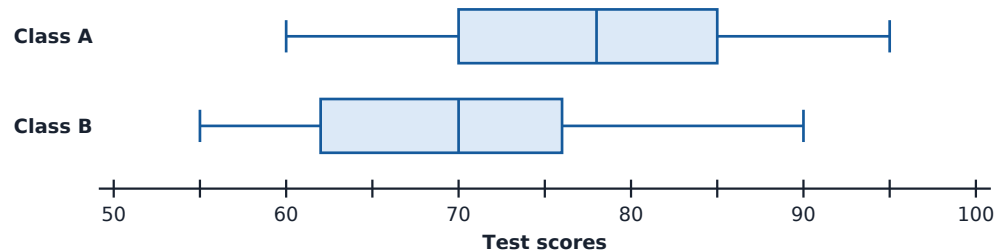
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 74 • FORWARD 1 • Q4/5

Which class had the highest single score?



Class: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 74 • FORWARD 1 • Q5/5

2744

QUESTION ID

S74-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Class A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 74 • FORWARD 1 • Q5/5

FORWARD 2

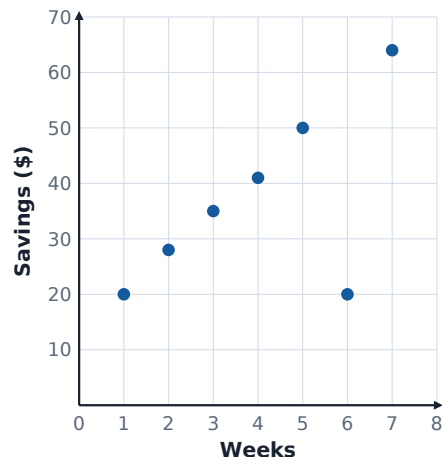
Spread and outliers in scatter plots • Grade 8 • Statistics & Probability •
8.SP.A.1

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Which point is an outlier?

Write its coordinates.

Outlier: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 74 • FORWARD 2 • Q1/5 • NEAREST RELATED

2746

QUESTION ID

S74-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(6, 20)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

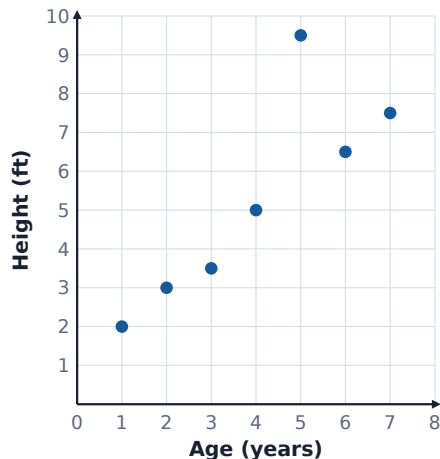
ID S74-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 74 • FORWARD 2 • Q1/5 • NEAREST RELATED

Which point is an outlier?

Write its coordinates.

Outlier: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 74 • FORWARD 2 • Q2/5 • NEAREST RELATED

2747

QUESTION ID

S74-F2-Q2

TYPE

Short answer

CORRECT ANSWER

(5, 9.5)

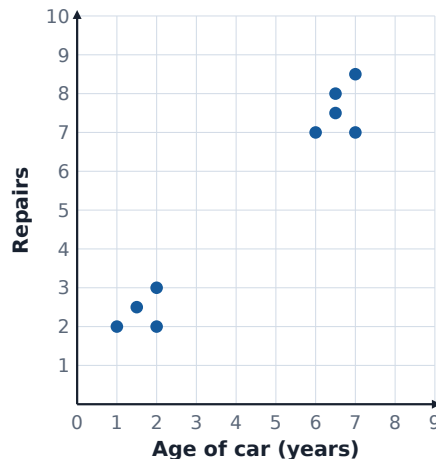
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 74 • FORWARD 2 • Q2/5 • NEAREST RELATED

Which describes the scatter plot?

- A. A single outlier
- B. A strong positive association
- C. No pattern
- D. Two clusters



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 74 • FORWARD 2 • Q3/5 • NEAREST RELATED

2748

QUESTION ID

S74-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. Two clusters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 74 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

An outlier in a scatter plot is a point that is far from the overall pattern.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 74 • FORWARD 2 • Q4/5 • NEAREST RELATED

2749

QUESTION ID

S74-F2-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

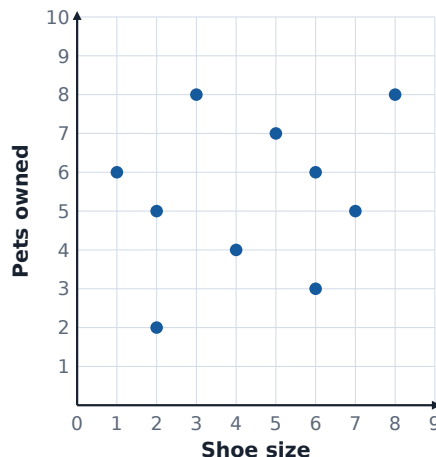
ID S74-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 74 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

The scatter plot shows no association.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 74 • FORWARD 2 • Q5/5 • NEAREST RELATED

2750

QUESTION ID

S74-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S74-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 74 • FORWARD 2 • Q5/5 • NEAREST RELATED

Report the number of observations

MAIN	Grade 6 main question	6.SP.B.5.a
BACKWARD 1	Reading a line plot	4.MD.B.4
BACKWARD 2	Totals from a bar graph	3.MD.B.3
BACKWARD 3	Adding several two-digit numbers	2.NBT.B.6
FORWARD 1	Use a random sample to make inferences	7.SP.A.2
FORWARD 2	Relative frequencies in two-way tables	8.SP.A.4

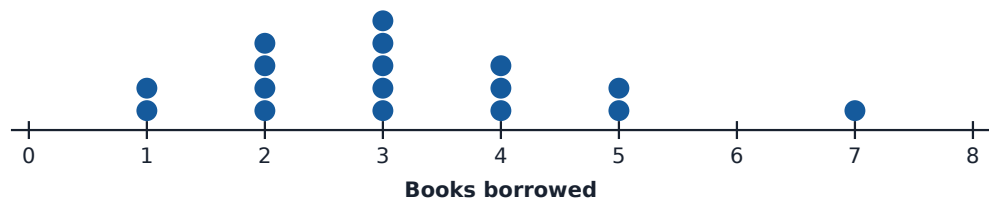
SET 75 • 6.SP.B.5.a

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.a

The dot plot shows how many books students borrowed.

How many observations are in the data set?



Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 75 • MAIN • Q1/5

2753

QUESTION ID

S75-M-Q1

TYPE

Short answer

CORRECT ANSWER

17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 75 • MAIN • Q1/5

How many students are represented in the histogram?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 75 • MAIN • Q2/5

2754

QUESTION ID

S75-M-Q2

TYPE

Short answer

CORRECT ANSWER

18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 75 • MAIN • Q2/5

The table shows how many siblings students have.

How many observations are in the data set?

Siblings	0	1	2	3
Students	4	9	6	2

Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 75 • MAIN • Q3/5

2755

QUESTION ID

S75-M-Q3

TYPE

Short answer

CORRECT ANSWER

21

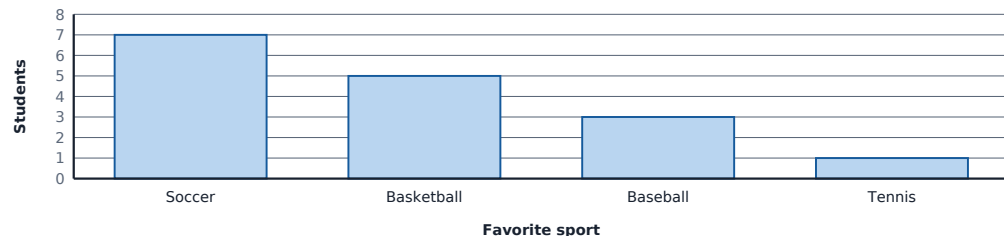
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 75 • MAIN • Q3/5

The bar graph shows survey answers.

How many observations are in the data set?



Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 75 • MAIN • Q4/5

2756

QUESTION ID

S75-M-Q4

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 75 • MAIN • Q4/5

Data: 12, 15, 9, 15, 20, 11, 13, 18, 15, 10

How many observations are in the data set?

Observations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 75 • MAIN • Q5/5

2757

QUESTION ID

S75-M-Q5

TYPE

Short answer

CORRECT ANSWER

10

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.a • Set 75 • MAIN • Q5/5

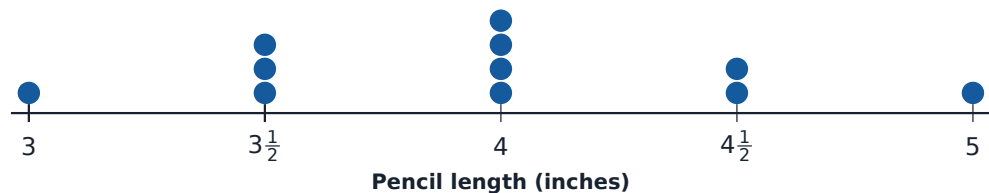
SET 75 • 6.SP.B.5.a

BACKWARD 1

Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows pencil lengths.

How many pencils are 4 inches long?



Pencils: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 75 • BACKWARD 1 • Q1/5

2759

QUESTION ID

S75-B1-Q1

TYPE

Short answer

CORRECT ANSWER

4

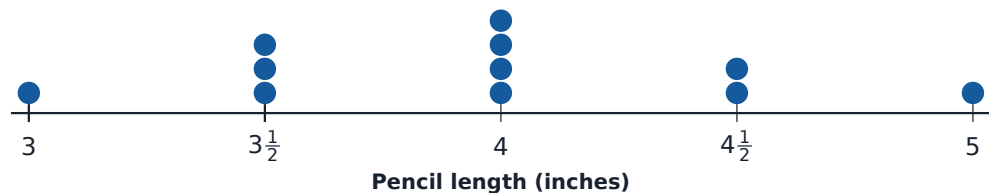
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 75 • BACKWARD 1 • Q1/5

The line plot shows pencil lengths.

How many pencils were measured?



Pencils: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 75 • BACKWARD 1 • Q2/5

2760

QUESTION ID

S75-B1-Q2

TYPE

Short answer

CORRECT ANSWER

11

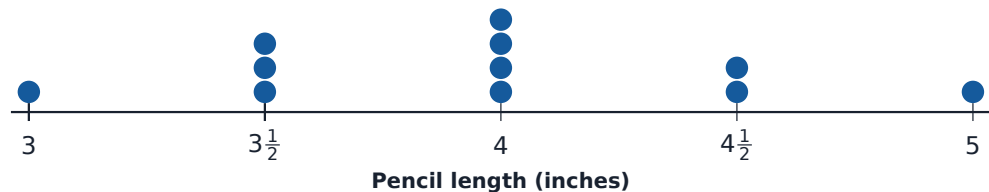
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 75 • BACKWARD 1 • Q2/5

True or false?

Two pencils are $4\frac{1}{2}$ inches long.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 75 • BACKWARD 1 • Q3/5

2761

QUESTION ID

S75-B1-Q3

TYPE

True or false

CORRECT ANSWER

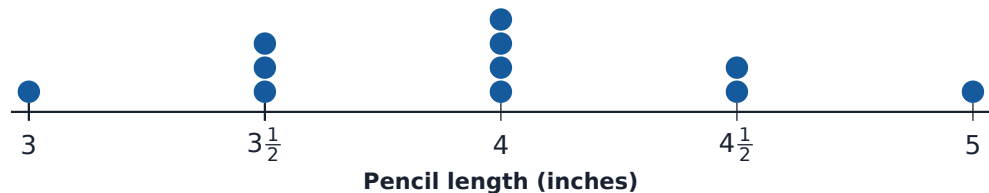
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 75 • BACKWARD 1 • Q3/5

How many pencils are longer than 4 inches?



A. 2

B. 3

C. 4

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 75 • BACKWARD 1 • Q4/5

2762

QUESTION ID

S75-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

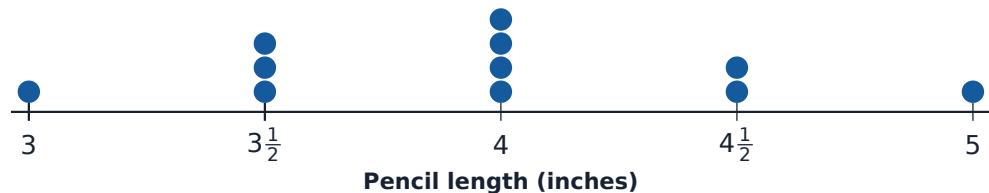
B. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 75 • BACKWARD 1 • Q4/5

How many pencils are 5 inches long?



Pencils: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 75 • BACKWARD 1 • Q5/5

2763

QUESTION ID

S75-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B1-Q5

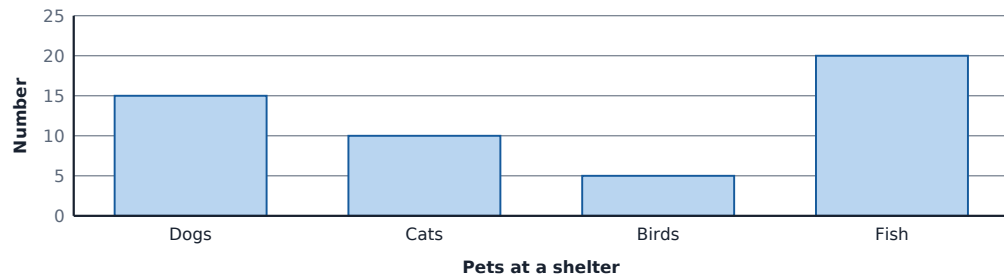
Grade 4 • Measurement & Data • 4.MD.B.4 • Set 75 • BACKWARD 1 • Q5/5

SET 75 • 6.SP.B.5.a

BACKWARD 2

Totals from a bar graph • Grade 3 • Measurement & Data • 3.MD.B.3

How many pets are at the shelter in all?



Pets: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 75 • BACKWARD 2 • Q1/5

2765

QUESTION ID

S75-B2-Q1

TYPE

Short answer

CORRECT ANSWER

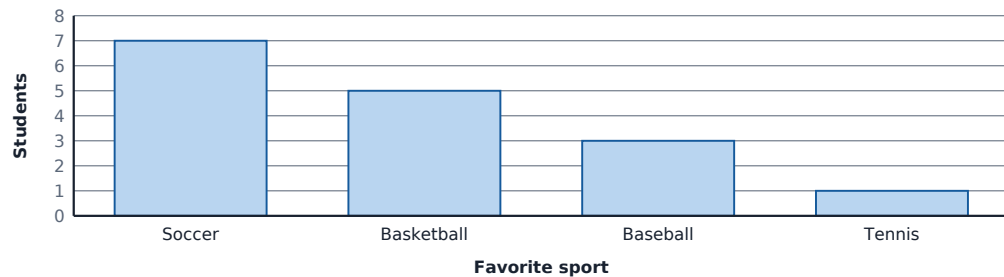
50

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 75 • BACKWARD 2 • Q1/5

How many students chose soccer or tennis?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 75 • BACKWARD 2 • Q2/5

2766

QUESTION ID

S75-B2-Q2

TYPE

Short answer

CORRECT ANSWER

8

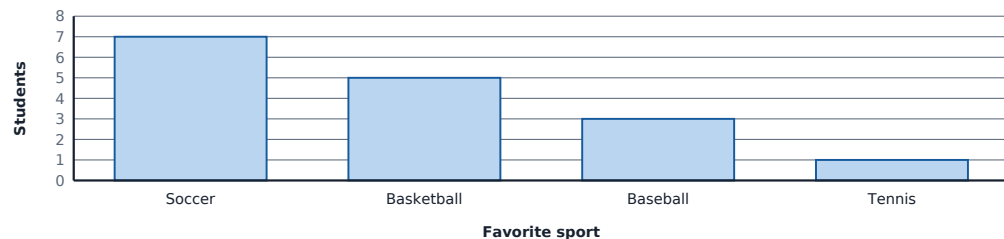
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 75 • BACKWARD 2 • Q2/5

True or false?

12 students chose soccer or basketball.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 75 • BACKWARD 2 • Q3/5

2767

QUESTION ID

S75-B2-Q3

TYPE

True or false

CORRECT ANSWER

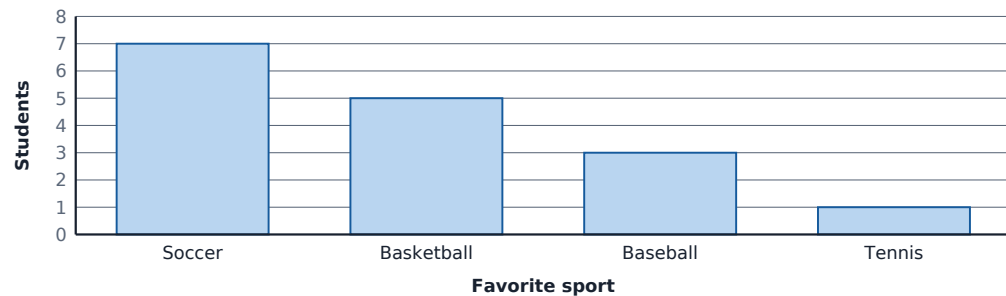
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 75 • BACKWARD 2 • Q3/5

How many students did NOT choose soccer?



- A. 5 B. 7 C. 9 D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 75 • BACKWARD 2 • Q4/5

2768

QUESTION ID

S75-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

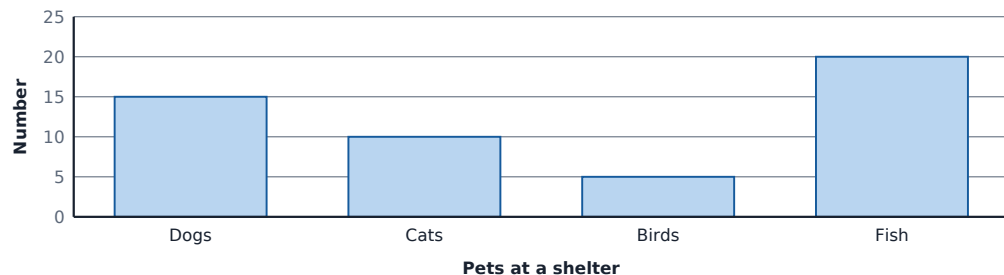
C. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 75 • BACKWARD 2 • Q4/5

How many dogs and cats are at the shelter together?



Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 75 • BACKWARD 2 • Q5/5

2769

QUESTION ID

S75-B2-Q5

TYPE

Short answer

CORRECT ANSWER

25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B2-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 75 • BACKWARD 2 • Q5/5

SET 75 • 6.SP.B.5.a

BACKWARD 3

Adding several two-digit numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6

Add.

$$14 + 23 + 31 + 12$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 75 • BACKWARD 3 • Q1/5

2771

QUESTION ID

S75-B3-Q1

TYPE

Short answer

CORRECT ANSWER

80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 75 • BACKWARD 3 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$25 + 18 + 30$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 75 • BACKWARD 3 • Q2/5

2772

QUESTION ID

S75-B3-Q2

TYPE

Short answer

CORRECT ANSWER

73

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 75 • BACKWARD 3 • Q2/5

2772

True or false?

$$16 + 16 + 20 + 8 = 60$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 75 • BACKWARD 3 • Q3/5

2773

QUESTION ID

S75-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 75 • BACKWARD 3 • Q3/5

$$41 + 27 + 13 + 9 = ?$$

- A.** 80
- B.** 90
- C.** 91
- D.** 100

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 75 • BACKWARD 3 • Q4/5

2774

QUESTION ID

S75-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 75 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$35 + 22 + 18 + 15$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 75 • BACKWARD 3 • Q5/5

2775

QUESTION ID

S75-B3-Q5

TYPE

Short answer

CORRECT ANSWER

90

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-B3-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.6 • Set 75 • BACKWARD 3 • Q5/5

2775

SET 75 • 6.SP.B.5.a

FORWARD 1

Use a random sample to make inferences • Grade 7 • Statistics & Probability •
7.SP.A.2

A researcher surveys 80 randomly chosen students out of 900.

What is the sample size?

- A.** 80 **B.** 820
C. 900 **D.** 980

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 75 • FORWARD 1 • Q1/5

2777

QUESTION ID

S75-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. 80

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 75 • FORWARD 1 • Q1/5

True or false?

A larger random sample generally gives a better estimate about a population.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 75 • FORWARD 1 • Q2/5

2778

QUESTION ID

S75-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 75 • FORWARD 1 • Q2/5

In a random sample of 25 students out of 500, 5 like jazz.

Estimate how many of the 500 students like jazz.

Estimate: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 75 • FORWARD 1 • Q3/5

2779

QUESTION ID

S75-F1-Q3

TYPE

Short answer

CORRECT ANSWER

100 students

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 75 • FORWARD 1 • Q3/5

About how many voters support the park?

- A.** 30 **B.** 600
- C.** 1,200 **D.** 1,500

ID S75-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 75 • FORWARD 1 • Q4/5

2780

S75-F1-Q4

Multiple choice

C. 1,200

ID S75-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 75 • FORWARD 1 • Q4/5

Two random samples of 20 students each found 12 and 14 students who like pizza best.

What is the mean of the two sample results?

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 75 • FORWARD 1 • Q5/5

2781

QUESTION ID

S75-F1-Q5

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.A.2 • Set 75 • FORWARD 1 • Q5/5

SET 75 • 6.SP.B.5.a

FORWARD 2

Relative frequencies in two-way tables • Grade 8 • Statistics & Probability •
8.SP.A.4

What fraction of 7th graders have a pet? What fraction of 8th graders have a pet?

	Has a pet	No pet	Total
Grade 7	18	12	30
Grade 8	20	10	30
Total	38	22	60

Grade 7: _____

Grade 8: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 75 • FORWARD 2 • Q1/5

2783

QUESTION ID

S75-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Grade 7:
 $18/30 = 3/5$; Grade
8: $20/30 = 2/3$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 75 • FORWARD 2 • Q1/5

Compare the relative frequencies. Which grade has a greater fraction of pet owners?

	Has a pet	No pet	Total
Grade 7	18	12	30
Grade 8	20	10	30
Total	38	22	60

- A.** Grade 7 **B.** Grade 8 **C.** They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 75 • FORWARD 2 • Q2/5

2784

QUESTION ID

S75-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Grade 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 75 • FORWARD 2 • Q2/5

What is the relative frequency of instrument players who also play sports?

	Plays sports	No sports	Total
Plays an instrument	14	6	20
No instrument	16	4	20
Total	30	10	40

- A. 0.35 B. 0.47 C. 0.7 D. 14

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 75 • FORWARD 2 • Q3/5

2785

QUESTION ID

S75-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

C. 0.7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 75 • FORWARD 2 • Q3/5

Among students with no instrument, what fraction play sports?

	Plays sports	No sports	Total
Plays an instrument	14	6	20
No instrument	16	4	20
Total	30	10	40

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 75 • FORWARD 2 • Q4/5

2786

QUESTION ID

S75-F2-Q4

TYPE

Short answer

CORRECT ANSWER

16/20 = 4/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 75 • FORWARD 2 • Q4/5

True or false?

Students with no instrument are more likely to play sports than students who play an instrument.

	Plays sports	No sports	Total
Plays an instrument	14	6	20
No instrument	16	4	20
Total	30	10	40

A. True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 75 • FORWARD 2 • Q5/5

2787

QUESTION ID

S75-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S75-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 75 • FORWARD 2 • Q5/5

Describe the attribute measured and its units

MAIN	Grade 6 main question	6.SP.B.5.b
BACKWARD 1	Choosing measurement units	4.MD.A.1
BACKWARD 2	What a bar graph shows	3.MD.B.3
BACKWARD 3	Choosing a measuring tool	2.MD.A.1
FORWARD 1	How data are collected: sampling methods	7.SP.A.1
FORWARD 2	Association between two categorical variables	8.SP.A.4

SET 76 • 6.SP.B.5.b

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.b

A data set lists the heights of 25 plants in centimeters.

What attribute is being measured, and in what unit?

Attribute: _____

Unit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.b • Set 76 • MAIN • Q1/5

2790

QUESTION ID

S76-M-Q1

TYPE

Short answer

CORRECT ANSWER

**Attribute: plant
height; Unit:
centimeters**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.b • Set 76 • MAIN • Q1/5

A data set records the time each student spends on homework.

Which unit is most likely?

- A.** Degrees
- B.** Pounds
- C.** Inches
- D.** Minutes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.b • Set 76 • MAIN • Q2/5

2791

QUESTION ID

S76-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Minutes

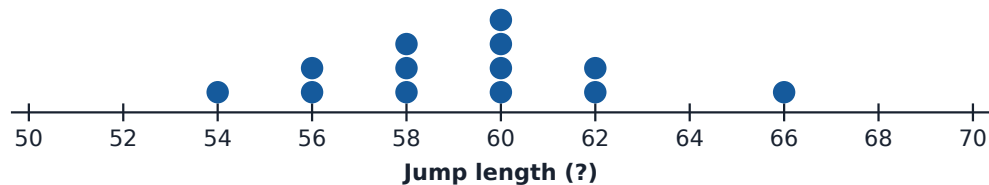
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.b • Set 76 • MAIN • Q2/5

The dot plot shows how far students jumped.

Which unit is most likely?



- A.** Inches **B.** Seconds **C.** Pounds **D.** Miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.b • Set 76 • MAIN • Q3/5

2792

QUESTION ID

S76-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

A. Inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.b • Set 76 • MAIN • Q3/5

A data set lists the mass of each of 30 apples.

What attribute is being measured, and what is a reasonable unit?

Attribute: _____

Unit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.b • Set 76 • MAIN • Q4/5

2793

QUESTION ID

S76-M-Q4

TYPE

Short answer

CORRECT ANSWER

**Attribute: the mass
(weight) of each
apple; a reasonable
unit: grams**

GRADING NOTE Ounces is also acceptable

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.b • Set 76 • MAIN • Q4/5

A data set records how long each student's trip to school takes.

Which describes the attribute and its unit?

- A.** Distance; minutes
- B.** Time; minutes
- C.** Speed; pounds
- D.** Time; miles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.b • Set 76 • MAIN • Q5/5

2794

QUESTION ID

S76-M-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Time; minutes

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.b • Set 76 • MAIN • Q5/5

SET 76 • 6.SP.B.5.b

BACKWARD 1

Choosing measurement units • Grade 4 • Measurement & Data • 4.MD.A.1

Which unit is best for measuring the length of a pencil?

- A.** Centimeters
- B.** Kilograms
- C.** Meters
- D.** Kilometers

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 76 • BACKWARD 1 • Q1/5

2796

QUESTION ID

S76-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Centimeters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q1

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 76 • BACKWARD 1 • Q1/5

True or false?

A kilogram is heavier than a gram.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 76 • BACKWARD 1 • Q2/5

2797

QUESTION ID

S76-B1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q2

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 76 • BACKWARD 1 • Q2/5

Which unit measures liquid volume?

- A.** Meters
- B.** Liters
- C.** Grams
- D.** Seconds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 76 • BACKWARD 1 • Q3/5

2798

QUESTION ID

S76-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Liters

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q3

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 76 • BACKWARD 1 • Q3/5

Name one unit used to measure time.

Unit: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 76 • BACKWARD 1 • Q4/5

2799

QUESTION ID

S76-B1-Q4

TYPE

Short answer

CORRECT ANSWER

Seconds

GRADING NOTE Any unit of time (seconds, minutes, hours, days, ...) is correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q4

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 76 • BACKWARD 1 • Q4/5

True or false?

An inch is longer than a foot.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q5

Grade 4 • Measurement & Data • 4.MD.A.1 • Set 76 • BACKWARD 1 • Q5/5

2800

QUESTION ID

S76-B1-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B1-Q5

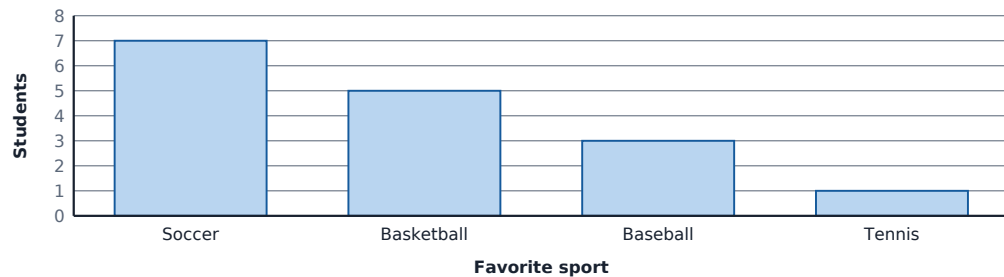
Grade 4 • Measurement & Data • 4.MD.A.1 • Set 76 • BACKWARD 1 • Q5/5

SET 76 • 6.SP.B.5.b

BACKWARD 2

What a bar graph shows • Grade 3 • Measurement & Data • 3.MD.B.3

How many categories does the graph show?



Categories: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 76 • BACKWARD 2 • Q1/5

2802

QUESTION ID

S76-B2-Q1

TYPE

Short answer

CORRECT ANSWER

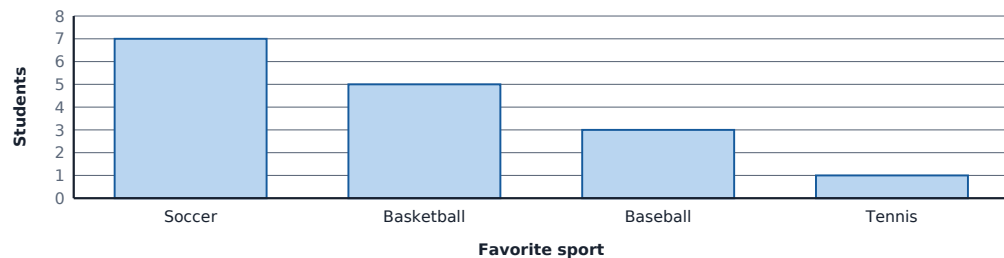
4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q1

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 76 • BACKWARD 2 • Q1/5

What does the height of each bar show?



- A.** The number of sports **B.** The score of a game
C. The length of a game **D.** The number of students

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 76 • BACKWARD 2 • Q2/5

2803

QUESTION ID

S76-B2-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. The number of students

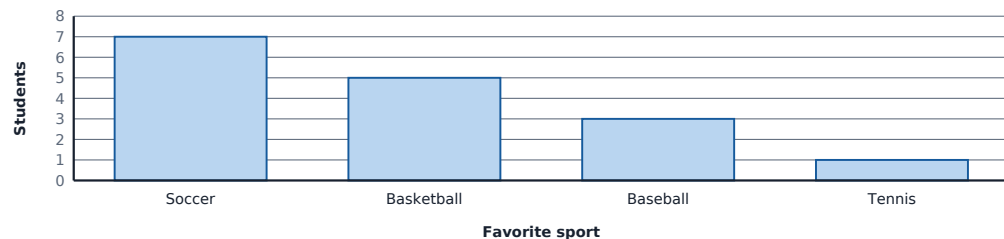
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q2

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 76 • BACKWARD 2 • Q2/5

True or false?

"Tennis" is one of the categories in the graph.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 76 • BACKWARD 2 • Q3/5

2804

QUESTION ID

S76-B2-Q3

TYPE

True or false

CORRECT ANSWER

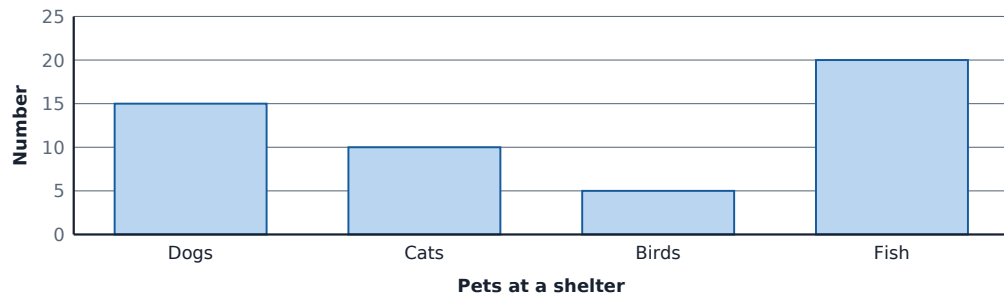
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q3

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 76 • BACKWARD 2 • Q3/5

Which is a category in the graph?



- A.** 25 **B.** Birds **C.** Horses **D.** Soccer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 76 • BACKWARD 2 • Q4/5

2805

QUESTION ID

S76-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

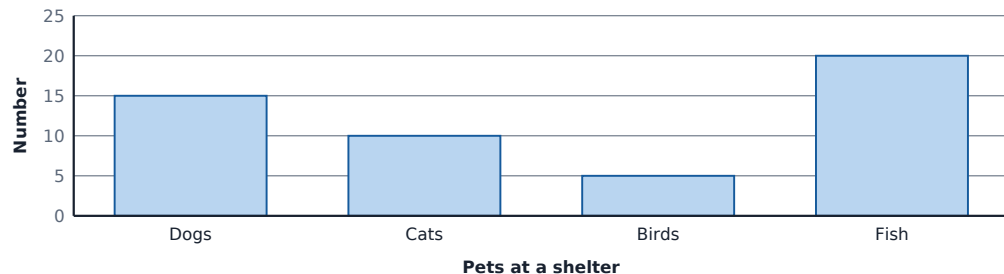
B. Birds

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q4

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 76 • BACKWARD 2 • Q4/5

What is being counted in the graph?



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 76 • BACKWARD 2 • Q5/5

2806

QUESTION ID

S76-B2-Q5

TYPE

Short answer

CORRECT ANSWER

**The pets (animals)
at the shelter, by
type**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B2-Q5

Grade 3 • Measurement & Data • 3.MD.B.3 • Set 76 • BACKWARD 2 • Q5/5

SET 76 • 6.SP.B.5.b

BACKWARD 3

Choosing a measuring tool • Grade 2 • Measurement & Data • 2.MD.A.1

Which tool measures the length of a table?

- A.** Tape measure
- B.** Thermometer
- C.** Scale
- D.** Clock

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q1

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 76 • BACKWARD 3 • Q1/5

2808

QUESTION ID

S76-B3-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Tape measure

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q1

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 76 • BACKWARD 3 • Q1/5

Which tool measures how heavy an apple is?

- A.** Ruler
- B.** Clock
- C.** Measuring cup
- D.** Scale

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q2

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 76 • BACKWARD 3 • Q2/5

2809

QUESTION ID

S76-B3-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Scale

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q2

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 76 • BACKWARD 3 • Q2/5

True or false?

A clock measures time.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q3

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 76 • BACKWARD 3 • Q3/5

2810

QUESTION ID

S76-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q3

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 76 • BACKWARD 3 • Q3/5

Which tool measures temperature?

- A.** Thermometer
- B.** Scale
- C.** Ruler
- D.** Yardstick

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q4

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 76 • BACKWARD 3 • Q4/5

2811

QUESTION ID

S76-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. Thermometer

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q4

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 76 • BACKWARD 3 • Q4/5

Name a tool that measures the length of a pencil.

Tool: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q5

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 76 • BACKWARD 3 • Q5/5

2812

QUESTION ID

S76-B3-Q5

TYPE

Short answer

CORRECT ANSWER

A ruler

GRADING NOTE A tape measure or yardstick is also correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-B3-Q5

Grade 2 • Measurement & Data • 2.MD.A.1 • Set 76 • BACKWARD 3 • Q5/5

SET 76 • 6.SP.B.5.b

FORWARD 1

How data are collected: sampling methods • Grade 7 • Statistics & Probability •
7.SP.A.1

Which method gives a random sample of the students in a school?

- A. Ask the students in the library
- B. Ask the first 20 students to arrive
- C. Ask your friends
- D. Draw names from a hat that holds every student's name

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 76 • FORWARD 1 • Q1/5

2814

QUESTION ID

S76-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

D. Draw names from a hat that holds every student's name

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 76 • FORWARD 1 • Q1/5

True or false?

Surveying people who walk out of a gym is a good way to learn how often all people in a town exercise.

- A. True
- B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 76 • FORWARD 1 • Q2/5

2815

QUESTION ID

S76-F1-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 76 • FORWARD 1 • Q2/5

2815

A town wants to know how residents feel about a new library.

Describe a way to choose a random sample.

Method: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 76 • FORWARD 1 • Q3/5

2816

QUESTION ID

S76-F1-Q3

TYPE

Short answer

CORRECT ANSWER

List every resident, then choose names at random (for example, draw names from a hat or use a random number generator).

GRADING NOTE Any method that gives every resident an equal chance is correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 76 • FORWARD 1 • Q3/5

Which sample is most likely to be biased?

- A. Asking only people at a dog park whether they like dogs
- B. Picking every 10th name from a school list
- C. Choosing 50 residents at random from a town list
- D. Drawing 30 names from a hat of all members

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 76 • FORWARD 1 • Q4/5

2817

QUESTION ID

S76-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

**A. Asking only
people at a dog park
whether they like
dogs**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 76 • FORWARD 1 • Q4/5

True or false?

In a random sample, every member of the population has an equal chance of being chosen.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 76 • FORWARD 1 • Q5/5

2818

QUESTION ID

S76-F1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.A.1 • Set 76 • FORWARD 1 • Q5/5

SET 76 • 6.SP.B.5.b

FORWARD 2

Association between two categorical variables • Grade 8 • Statistics & Probability • 8.SP.A.4

Which pair of variables is categorical and could be shown in a two-way table?

- A. Hours slept and test score
- B. Height and weight
- C. Age and arm span
- D. Grade level and way of getting to school

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 76 • FORWARD 2 • Q1/5

2820

QUESTION ID

S76-F2-Q1

TYPE

Multiple choice

CORRECT ANSWER

**D. Grade level and
way of getting to
school**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 76 • FORWARD 2 • Q1/5

What fraction of 6th graders walk to school?

	Walks to school	Rides to school	Total
Grade 6	12	18	30
Grade 8	15	5	20
Total	27	23	50

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 76 • FORWARD 2 • Q2/5

2821

QUESTION ID

S76-F2-Q2

TYPE

Short answer

CORRECT ANSWER

12/30 = 2/5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 76 • FORWARD 2 • Q2/5

What fraction of 8th graders walk to school?

	Walks to school	Rides to school	Total
Grade 6	12	18	30
Grade 8	15	5	20
Total	27	23	50

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 76 • FORWARD 2 • Q3/5

2822

QUESTION ID

S76-F2-Q3

TYPE

Short answer

CORRECT ANSWER

15/20 = 3/4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 76 • FORWARD 2 • Q3/5

What is the relative frequency of 8th graders who ride to school?

	Walks to school	Rides to school	Total
Grade 6	12	18	30
Grade 8	15	5	20
Total	27	23	50

A. $\frac{1}{10}$

B. $\frac{5}{23}$

C. $\frac{1}{4}$

D. $\frac{3}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 76 • FORWARD 2 • Q4/5

2823

QUESTION ID

S76-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. $\frac{1}{4}$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 76 • FORWARD 2 • Q4/5

True or false?

8th graders are more likely to walk to school than 6th graders.

	Walks to school	Rides to school	Total
Grade 6	12	18	30
Grade 8	15	5	20
Total	27	23	50

A. True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 76 • FORWARD 2 • Q5/5

2824

QUESTION ID

S76-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S76-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.4 • Set 76 • FORWARD 2 • Q5/5

Find the mean of a data set

MAIN	Grade 6 main question	6.SP.B.5.c
BACKWARD 1	Adding multi-digit numbers	4.NBT.B.4
BACKWARD 2	Dividing a total into equal groups	4.NBT.B.6
BACKWARD 3	Sharing a total equally	5.MD.B.2
FORWARD 1	Compare populations using means	7.SP.B.4
FORWARD 2	Use a linear model to make predictions — nearest related (no dire...	8.SP.A.3

SET 77 • 6.SP.B.5.c

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean.

4, 7, 9, 10, 15

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • MAIN • Q1/5

2827

QUESTION ID

S77-M-Q1

TYPE

Short answer

CORRECT ANSWER

9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • MAIN • Q1/5

Five test scores are 82, 90, 75, 88, and 95.

What is the mean score?

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • MAIN • Q2/5

2828

QUESTION ID

S77-M-Q2

TYPE

Short answer

CORRECT ANSWER

86

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • MAIN • Q2/5

Find the mean.

12, 15, 18, 19

- A.** 15
- B.** 16
- C.** 16.5
- D.** 64

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • MAIN • Q3/5

2829

QUESTION ID

S77-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • MAIN • Q3/5

The dot plot shows goals scored in 8 games.

What is the mean number of goals?



Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • MAIN • Q4/5

2830

QUESTION ID

S77-M-Q4

TYPE

Short answer

CORRECT ANSWER

4.75 goals

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • MAIN • Q4/5

Find the mean.

3.5, 4.2, 5.0, 6.1, 6.2

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • MAIN • Q5/5

2831

QUESTION ID

S77-M-Q5

TYPE

Short answer

CORRECT ANSWER

5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 77 • MAIN • Q5/5

SET 77 • 6.SP.B.5.c

BACKWARD 1

Adding multi-digit numbers • Grade 4 • Number & Operations in Base Ten •
4.NBT.B.4

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$82 + 90 + 75 + 88 + 95$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 77 • BACKWARD 1 • Q1/5

2833

QUESTION ID

S77-B1-Q1

TYPE

Short answer

CORRECT ANSWER

430

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 77 • BACKWARD 1 • Q1/5

2833

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$145 + 238 + 97$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 77 • BACKWARD 1 • Q2/5

2834

QUESTION ID

S77-B1-Q2

TYPE

Short answer

CORRECT ANSWER

480

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 77 • BACKWARD 1 • Q2/5

2834

True or false?

$$56 + 64 + 70 = 190$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 77 • BACKWARD 1 • Q3/5

2835

QUESTION ID

S77-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 77 • BACKWARD 1 • Q3/5

$$125 + 175 + 200 = ?$$

- A.** 400
- B.** 450
- C.** 500
- D.** 550

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 77 • BACKWARD 1 • Q4/5

2836

QUESTION ID

S77-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 500

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 77 • BACKWARD 1 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Add.

$$1,250 + 875$$

Sum: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 77 • BACKWARD 1 • Q5/5

2837

QUESTION ID

S77-B1-Q5

TYPE

Short answer

CORRECT ANSWER

2,125

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B1-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.4 • Set 77 • BACKWARD 1 • Q5/5

2837

SET 77 • 6.SP.B.5.c

BACKWARD 2

Dividing a total into equal groups • Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6

Divide.

$$430 \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 77 • BACKWARD 2 • Q1/5

2839

QUESTION ID

S77-B2-Q1

TYPE

Short answer

CORRECT ANSWER

86

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q1

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 77 • BACKWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$64 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 77 • BACKWARD 2 • Q2/5

2840

QUESTION ID

S77-B2-Q2

TYPE

Short answer

CORRECT ANSWER

16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q2

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 77 • BACKWARD 2 • Q2/5

2840

True or false?

$$108 \div 4 = 27$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 77 • BACKWARD 2 • Q3/5

2841

QUESTION ID

S77-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q3

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 77 • BACKWARD 2 • Q3/5

$$252 \div 6 = ?$$

- A.** 36
- B.** 41
- C.** 42
- D.** 48

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 77 • BACKWARD 2 • Q4/5

2842

QUESTION ID

S77-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 42

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q4

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 77 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$315 \div 7$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 77 • BACKWARD 2 • Q5/5

2843

QUESTION ID

S77-B2-Q5

TYPE

Short answer

CORRECT ANSWER

45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B2-Q5

Grade 4 • Number & Operations in Base Ten • 4.NBT.B.6 • Set 77 • BACKWARD 2 • Q5/5

2843

SET 77 • 6.SP.B.5.c

BACKWARD 3

Sharing a total equally • Grade 5 • Measurement & Data • 5.MD.B.2

Two friends have 5 and 9 shells. They share the shells equally.

How many shells does each friend get?

Shells: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q1

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 77 • BACKWARD 3 • Q1/5

2845

QUESTION ID

S77-B3-Q1

TYPE

Short answer

CORRECT ANSWER

7 shells

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q1

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 77 • BACKWARD 3 • Q1/5

Three boxes hold 2, 7, and 9 books. The books are shared equally among the boxes.

How many books are in each box?

Books: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q2

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 77 • BACKWARD 3 • Q2/5

2846

QUESTION ID

S77-B3-Q2

TYPE

Short answer

CORRECT ANSWER

6 books

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q2

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 77 • BACKWARD 3 • Q2/5

True or false?

Jars holding 4, 4, and 10 cups are shared equally.
Each jar then holds 6 cups.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q3

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 77 • BACKWARD 3 • Q3/5

2847

QUESTION ID

S77-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q3

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 77 • BACKWARD 3 • Q3/5

Four children have 1, 2, 6, and 7 cards. They share the cards equally.

How many cards does each child get?

A. 3

B. 4

C. 5

D. 16

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q4

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 77 • BACKWARD 3 • Q4/5

2848

QUESTION ID

S77-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q4

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 77 • BACKWARD 3 • Q4/5

Two cups hold $\frac{1}{3}$ cup and $\frac{2}{3}$ cup of milk. The milk is shared equally.

How much is in each cup?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q5

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 77 • BACKWARD 3 • Q5/5

2849

QUESTION ID

S77-B3-Q5

TYPE

Short answer

CORRECT ANSWER

$\frac{1}{2}$ cup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-B3-Q5

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 77 • BACKWARD 3 • Q5/5

SET 77 • 6.SP.B.5.c

FORWARD 1

Compare populations using means • Grade 7 • Statistics & Probability •
7.SP.B.4

A random sample of 6th graders has a mean height of 58 in. A random sample of 8th graders has a mean height of 64 in.

How much greater is the 8th graders' mean?

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 77 • FORWARD 1 • Q1/5

2851

QUESTION ID

S77-F1-Q1

TYPE

Short answer

CORRECT ANSWER

6 inches

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 77 • FORWARD 1 • Q1/5

Two random samples of plant heights have means of 24 cm and 30 cm. Each has a MAD of 3 cm.

How many MADs apart are the means?

- A.** 1 **B.** 2
C. 3 **D.** 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 77 • FORWARD 1 • Q2/5

2852

QUESTION ID

S77-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 77 • FORWARD 1 • Q2/5

True or false?

To compare two populations, you can compare the means of random samples from each population.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 77 • FORWARD 1 • Q3/5

2853

QUESTION ID

S77-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 77 • FORWARD 1 • Q3/5

Sample A: mean 15.2 minutes. Sample B: mean 18.7 minutes.

Which population probably has greater values?

Population: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 77 • FORWARD 1 • Q4/5

2854

QUESTION ID

S77-F1-Q4

TYPE

Short answer

CORRECT ANSWER

Population B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 77 • FORWARD 1 • Q4/5

Ten random samples of 20 students gave these mean hours of sleep: 8.1, 8.3, 7.9, 8.0, 8.2, 8.1, 8.4, 7.8, 8.0, 8.2.

What is a good estimate of the population mean?

- A.** About 20 hours
- B.** About 7 hours
- C.** About 8.1 hours
- D.** About 9 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 77 • FORWARD 1 • Q5/5

2855

QUESTION ID

S77-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

C. About 8.1 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 77 • FORWARD 1 • Q5/5

FORWARD 2

Use a linear model to make predictions • Grade 8 • Statistics & Probability •
8.SP.A.3

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

The equation $y = 2.5x + 40$ models test score y after x hours of study.

Predict the score for 8 hours of study.

Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 77 • FORWARD 2 • Q1/5 • NEAREST RELATED

2857

QUESTION ID

S77-F2-Q1

TYPE

Short answer

CORRECT ANSWER

60

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 77 • FORWARD 2 • Q1/5 • NEAREST RELATED

The model $y = 15x + 100$ gives the cost y of x tickets.
Predict the cost of 20 tickets.

Cost: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 77 • FORWARD 2 • Q2/5 • NEAREST RELATED

2858

QUESTION ID

S77-F2-Q2

TYPE

Short answer

CORRECT ANSWER

\$400

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 77 • FORWARD 2 • Q2/5 • NEAREST RELATED

The model $y = 3x + 12$ gives a plant's height y (cm) after x weeks.

What height does the model predict after 5 weeks?

- A.** 15 cm **B.** 27 cm
C. 36 cm **D.** 60 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 77 • FORWARD 2 • Q3/5 • NEAREST RELATED

2859

QUESTION ID

S77-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 27 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 77 • FORWARD 2 • Q3/5 • NEAREST RELATED

The model $y = -0.5x + 30$ gives the value y of a toy after x years.

Predict the value after 6 years.

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 77 • FORWARD 2 • Q4/5 • NEAREST RELATED

2860

QUESTION ID

S77-F2-Q4

TYPE

Short answer

CORRECT ANSWER

27

GRADING NOTE \$27

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 77 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

The model $y = 4x + 10$ predicts $y = 50$ when $x = 10$.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 77 • FORWARD 2 • Q5/5 • NEAREST RELATED

2861

QUESTION ID

S77-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S77-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.3 • Set 77 • FORWARD 2 • Q5/5 • NEAREST RELATED

Find the median of a data set

MAIN	Grade 6 main question	6.SP.B.5.c
BACKWARD 1	Ordering numbers	2.NBT.A.4
BACKWARD 2	The number halfway between two numbers	5.NBT.B.7
BACKWARD 3	Reading a line plot	4.MD.B.4
FORWARD 1	Compare populations using medians	7.SP.B.4
FORWARD 2	Draw and use a line of best fit — nearest related (no direct Grade...	8.SP.A.2

SET 78 • 6.SP.B.5.c

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the median.

12, 5, 9, 20, 7, 14

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 78 • MAIN • Q1/5

2864

QUESTION ID

S78-M-Q1

TYPE

Short answer

CORRECT ANSWER

10.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 78 • MAIN • Q1/5

True or false?

The median of 3, 8, 8, 10, 21 is 8.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 78 • MAIN • Q2/5

2865

QUESTION ID

S78-M-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 78 • MAIN • Q2/5

Heights (in inches): 58, 60, 61, 63, 63

What is the median height?

- A.** 60
- B.** 61
- C.** 61.5
- D.** 63

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 78 • MAIN • Q3/5

2866

QUESTION ID

S78-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

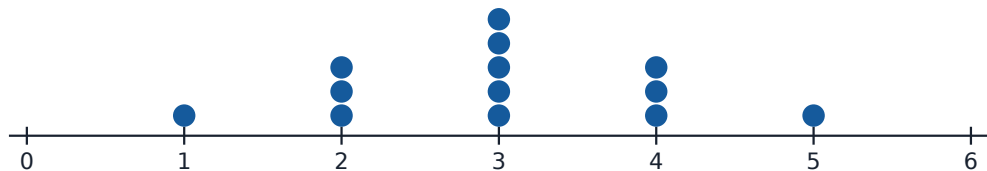
B. 61

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 78 • MAIN • Q3/5

What is the median of the data in the dot plot?



Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 78 • MAIN • Q4/5

2867

QUESTION ID

S78-M-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 78 • MAIN • Q4/5

Find the median.

4.5, 2.1, 3.8, 6.0

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 78 • MAIN • Q5/5

2868

QUESTION ID

S78-M-Q5

TYPE

Short answer

CORRECT ANSWER

4.15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 78 • MAIN • Q5/5

SET 78 • 6.SP.B.5.c

BACKWARD 1

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Order from least to greatest.

12, 5, 9, 20, 7, 14

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 78 • BACKWARD 1 • Q1/5

2870

QUESTION ID

S78-B1-Q1

TYPE

Short answer

CORRECT ANSWER

5, 7, 9, 12, 14, 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 78 • BACKWARD 1 • Q1/5

Which list is in order from least to greatest?

- A.** 58, 60, 61, 63
- B.** 58, 61, 60, 63
- C.** 60, 58, 61, 63
- D.** 63, 61, 60, 58

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 78 • BACKWARD 1 • Q2/5

2871

QUESTION ID

S78-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 58, 60, 61, 63

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 78 • BACKWARD 1 • Q2/5

True or false?

These numbers are in order from least to greatest: 75, 82, 88, 90, 95

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 78 • BACKWARD 1 • Q3/5

2872

QUESTION ID

S78-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 78 • BACKWARD 1 • Q3/5

Order from least to greatest.

47, 74, 44, 77

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 78 • BACKWARD 1 • Q4/5

2873

QUESTION ID

S78-B1-Q4

TYPE

Short answer

CORRECT ANSWER

44, 47, 74, 77

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 78 • BACKWARD 1 • Q4/5

When 3, 9, and 6 are put in order, which number is in the middle?

- A. 3
- B. 6
- C. 9
- D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 78 • BACKWARD 1 • Q5/5

2874

QUESTION ID

S78-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 78 • BACKWARD 1 • Q5/5

SET 78 • 6.SP.B.5.c

BACKWARD 2

The number halfway between two numbers • Grade 5 • Number & Operations
in Base Ten • 5.NBT.B.7

Find the value.

$$(9 + 12) \div 2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 78 • BACKWARD 2 • Q1/5

2876

QUESTION ID

S78-B2-Q1

TYPE

Short answer

CORRECT ANSWER

10.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 78 • BACKWARD 2 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find the value.

$$(3.8 + 4.5) \div 2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 78 • BACKWARD 2 • Q2/5

2877

QUESTION ID

S78-B2-Q2

TYPE

Short answer

CORRECT ANSWER

4.15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 78 • BACKWARD 2 • Q2/5

2877

True or false?

$$(6 + 9) \div 2 = 7.5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 78 • BACKWARD 2 • Q3/5

2878

QUESTION ID

S78-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 78 • BACKWARD 2 • Q3/5

$$(20 + 25) \div 2 = ?$$

- A.** 22
- B.** 22.5
- C.** 23.5
- D.** 45

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 78 • BACKWARD 2 • Q4/5

2879

QUESTION ID

S78-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 22.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 78 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find the value.

$$(3.5 + 4.5) \div 2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 78 • BACKWARD 2 • Q5/5

2880

QUESTION ID

S78-B2-Q5

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 78 • BACKWARD 2 • Q5/5

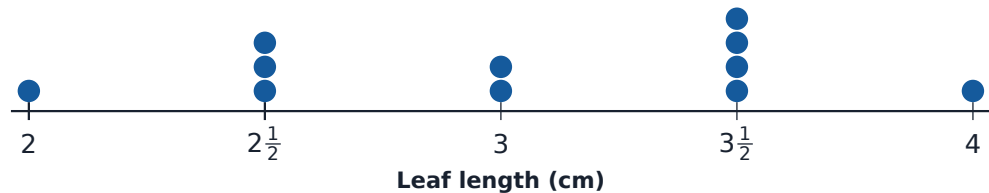
2880

SET 78 • 6.SP.B.5.c

BACKWARD 3

Reading a line plot • Grade 4 • Measurement & Data • 4.MD.B.4

How many leaves are $3\frac{1}{2}$ cm long?



Leaves: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 78 • BACKWARD 3 • Q1/5

2882

QUESTION ID

S78-B3-Q1

TYPE

Short answer

CORRECT ANSWER

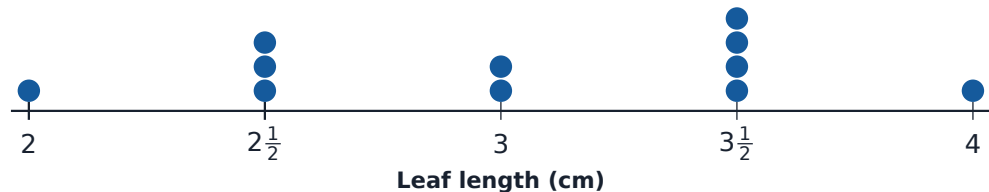
4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 78 • BACKWARD 3 • Q1/5

How many leaves were measured?



Leaves: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 78 • BACKWARD 3 • Q2/5

2883

QUESTION ID

S78-B3-Q2

TYPE

Short answer

CORRECT ANSWER

11

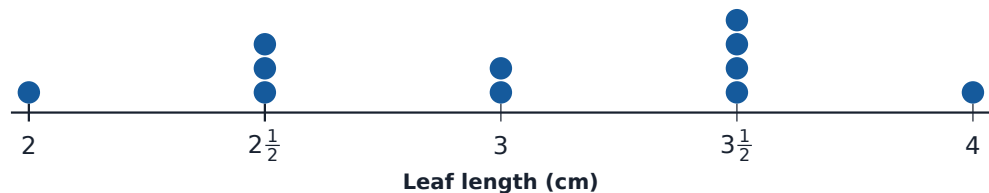
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 78 • BACKWARD 3 • Q2/5

True or false?

One leaf is 4 cm long.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 78 • BACKWARD 3 • Q3/5

2884

QUESTION ID

S78-B3-Q3

TYPE

True or false

CORRECT ANSWER

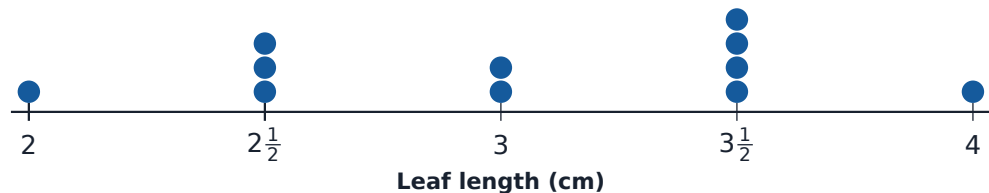
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 78 • BACKWARD 3 • Q3/5

Which leaf length is most common?



- A. $2\frac{1}{2}$ cm B. 3 cm C. $3\frac{1}{2}$ cm D. 4 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 78 • BACKWARD 3 • Q4/5

2885

QUESTION ID

S78-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

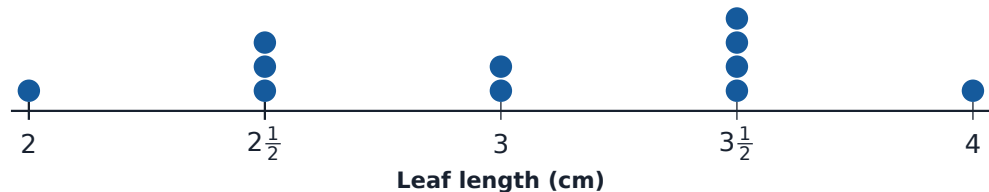
C. $3\frac{1}{2}$ cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 78 • BACKWARD 3 • Q4/5

How many leaves are shorter than 3 cm?



Leaves: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 78 • BACKWARD 3 • Q5/5

2886

QUESTION ID

S78-B3-Q5

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-B3-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 78 • BACKWARD 3 • Q5/5

SET 78 • 6.SP.B.5.c

FORWARD 1

Compare populations using medians • Grade 7 • Statistics & Probability •
7.SP.B.4

Sample A has a median of 12 and Sample B has a median of 15. The samples have similar variability.

Which conclusion is best?

- A.** Values in population A tend to be greater.
- B.** The populations are the same.
- C.** Values in population B tend to be greater.
- D.** No conclusion is possible.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 78 • FORWARD 1 • Q1/5

2888

QUESTION ID

S78-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

C. Values in population B tend to be greater.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 78 • FORWARD 1 • Q1/5

In random samples, the median commute is 25 minutes in Town X and 18 minutes in Town Y. The IQRs are similar.

In which town are commutes typically longer?

Town: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 78 • FORWARD 1 • Q2/5

2889

QUESTION ID

S78-F1-Q2

TYPE

Short answer

CORRECT ANSWER

Town X

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 78 • FORWARD 1 • Q2/5

Sample A: median 42, IQR 8. Sample B: median 50, IQR 8.

The difference in medians is how many times the IQR?

Times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 78 • FORWARD 1 • Q3/5

2890

QUESTION ID

S78-F1-Q3

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 78 • FORWARD 1 • Q3/5

True or false?

When data are skewed, comparing medians is often better than comparing means.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 78 • FORWARD 1 • Q4/5

2891

QUESTION ID

S78-F1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 78 • FORWARD 1 • Q4/5

Sample X has a median of 7.5 hours of sleep. Sample Y has a median of 8.5 hours. Both IQRs are 1 hour.

Which statement is best?

- A.** Population Y probably sleeps more.
- B.** Population X probably sleeps more.
- C.** Nothing can be said.
- D.** They sleep the same amount.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 78 • FORWARD 1 • Q5/5

2892

QUESTION ID

S78-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. Population Y probably sleeps more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F1-Q5

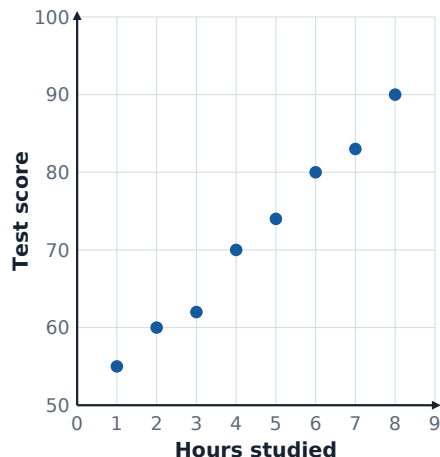
Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 78 • FORWARD 1 • Q5/5

FORWARD 2

Draw and use a line of best fit • Grade 8 • Statistics & Probability • 8.SP.A.2

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Draw a line of best fit for the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 78 • FORWARD 2 • Q1/5 • NEAREST RELATED

2894

QUESTION ID

S78-F2-Q1

TYPE

Drawing / plotting

CORRECT ANSWER

A straight rising line through the middle of the points, with about as many points above as below, for example from about (1, 55) to (8, 89) ($y \approx 5x + 50$)

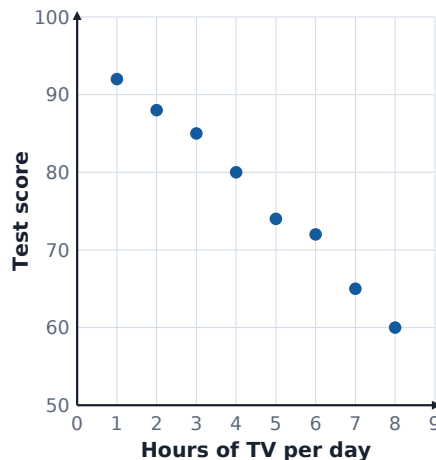
GRADING NOTE Accept any line with a positive slope of about 4.5 to 5.5 that follows the trend

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 78 • FORWARD 2 • Q1/5 • NEAREST RELATED

Draw a line of best fit for the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 78 • FORWARD 2 • Q2/5 • NEAREST RELATED

2895

QUESTION ID

S78-F2-Q2

TYPE

Drawing / plotting

CORRECT ANSWER

A straight falling line through the middle of the points, with about as many points above as below, for example from about (1, 93) to (8, 60) ($y \approx -4.7x + 97$)

GRADING NOTE Accept any line with a negative slope of about -4 to -5.5 that follows the trend

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 78 • FORWARD 2 • Q2/5 • NEAREST RELATED

A line of best fit should have ____.

- A. all points below it
- B. no points near it
- C. all points above it
- D. about the same number of points above and below it

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 78 • FORWARD 2 • Q3/5 • NEAREST RELATED

2896

QUESTION ID

S78-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

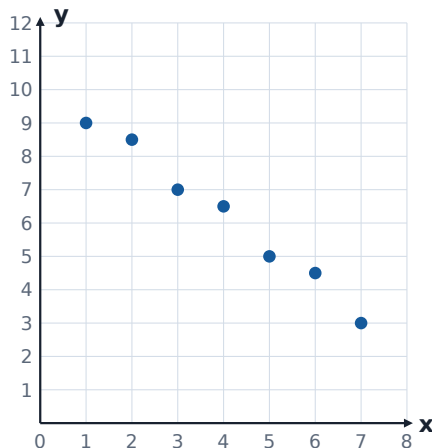
**D. about the same
number of points
above and below it**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 78 • FORWARD 2 • Q3/5 • NEAREST RELATED

Draw a line of best fit for the data.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 78 • FORWARD 2 • Q4/5 • NEAREST RELATED

2897

QUESTION ID

S78-F2-Q4

TYPE

Drawing / plotting

CORRECT ANSWER

A straight falling line through the middle of the points, for example $y = 10 - x$, from (1, 9) to (7, 3)

GRADING NOTE Accept any line with a slope of about -1 that follows the trend

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 78 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

A line of best fit must pass through the first and last points.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 78 • FORWARD 2 • Q5/5 • NEAREST RELATED

2898

QUESTION ID

S78-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S78-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 78 • FORWARD 2 • Q5/5 • NEAREST RELATED

Find the interquartile range of a data set

MAIN	Grade 6 main question	6.SP.B.5.c
BACKWARD 1	Ordering numbers	2.NBT.A.4
BACKWARD 2	The number halfway between two numbers	5.NBT.B.7
BACKWARD 3	Finding a difference	2.NBT.B.5
FORWARD 1	Difference in centers as a multiple of the IQR	7.SP.B.3
FORWARD 2	Outliers and clusters in scatter plots — nearest related (no direct...	8.SP.A.1

SET 79 • 6.SP.B.5.c

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the interquartile range.

2, 4, 5, 7, 9, 11, 12, 15

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 79 • MAIN • Q1/5

2901

QUESTION ID

S79-M-Q1

TYPE

Short answer

CORRECT ANSWER

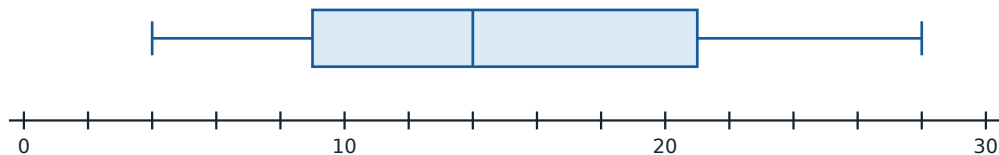
7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 79 • MAIN • Q1/5

What is the interquartile range of the data in the box plot?



IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 79 • MAIN • Q2/5

2902

QUESTION ID

S79-M-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 79 • MAIN • Q2/5

What is the interquartile range?

10, 12, 15, 18, 20, 25, 30

- A. 5
- B. 13
- C. 18
- D. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 79 • MAIN • Q3/5

2903

QUESTION ID

S79-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. 13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 79 • MAIN • Q3/5

Find the interquartile range.

3, 6, 6, 8, 9, 12, 14, 15, 18

IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 79 • MAIN • Q4/5

2904

QUESTION ID

S79-M-Q4

TYPE

Short answer

CORRECT ANSWER

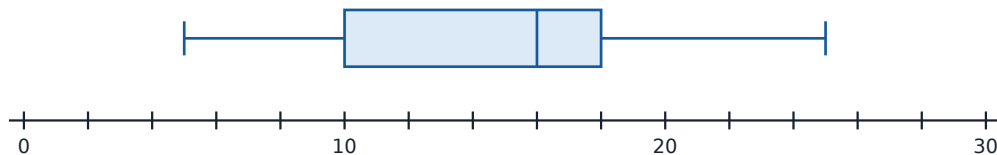
8.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 79 • MAIN • Q4/5

What is the interquartile range of the data in the box plot?



IQR: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 79 • MAIN • Q5/5

2905

QUESTION ID

S79-M-Q5

TYPE

Short answer

CORRECT ANSWER

8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 79 • MAIN • Q5/5

SET 79 • 6.SP.B.5.c

BACKWARD 1

Ordering numbers • Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4

Order from least to greatest.

15, 9, 2, 12, 5

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 79 • BACKWARD 1 • Q1/5

2907

QUESTION ID

S79-B1-Q1

TYPE

Short answer

CORRECT ANSWER

2, 5, 9, 12, 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 79 • BACKWARD 1 • Q1/5

Which list is in order from least to greatest?

- A.** 18, 15, 12, 10
- B.** 10, 15, 12, 18
- C.** 12, 10, 15, 18
- D.** 10, 12, 15, 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 79 • BACKWARD 1 • Q2/5

2908

QUESTION ID

S79-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. 10, 12, 15, 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 79 • BACKWARD 1 • Q2/5

True or false?

These numbers are in order from least to greatest: 3, 6, 9, 8

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 79 • BACKWARD 1 • Q3/5

2909

QUESTION ID

S79-B1-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 79 • BACKWARD 1 • Q3/5

Order from least to greatest.

25, 20, 30, 10

Order: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 79 • BACKWARD 1 • Q4/5

2910

QUESTION ID

S79-B1-Q4

TYPE

Short answer

CORRECT ANSWER

10, 20, 25, 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 79 • BACKWARD 1 • Q4/5

Which number is least?

- A.** 2
- B.** 5
- C.** 9
- D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 79 • BACKWARD 1 • Q5/5

2911

QUESTION ID

S79-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

A. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B1-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 79 • BACKWARD 1 • Q5/5

SET 79 • 6.SP.B.5.c

BACKWARD 2

The number halfway between two numbers • Grade 5 • Number & Operations
in Base Ten • 5.NBT.B.7

ANSWER KEY

Reduced copy of the question page, with its answer

Find the value.

$$(4 + 5) \div 2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 79 • BACKWARD 2 • Q1/5

2913

QUESTION ID

S79-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 79 • BACKWARD 2 • Q1/5

2913

ANSWER KEY

Reduced copy of the question page, with its answer

Find the value.

$$(11 + 12) \div 2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 79 • BACKWARD 2 • Q2/5

2914

QUESTION ID

S79-B2-Q2

TYPE

Short answer

CORRECT ANSWER

11.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 79 • BACKWARD 2 • Q2/5

2914

True or false?

$$(7 + 10) \div 2 = 8.5$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 79 • BACKWARD 2 • Q3/5

2915

QUESTION ID

S79-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 79 • BACKWARD 2 • Q3/5

$$(15 + 18) \div 2 = ?$$

- A.** 16
- B.** 16.5
- C.** 17.5
- D.** 33

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 79 • BACKWARD 2 • Q4/5

2916

QUESTION ID

S79-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 16.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 79 • BACKWARD 2 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Find the value.

$$(2.5 + 3.5) \div 2$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 79 • BACKWARD 2 • Q5/5

2917

QUESTION ID

S79-B2-Q5

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B2-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 79 • BACKWARD 2 • Q5/5

2917

SET 79 • 6.SP.B.5.c

BACKWARD 3

Finding a difference • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$25 - 12$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 79 • BACKWARD 3 • Q1/5

2919

QUESTION ID

S79-B3-Q1

TYPE

Short answer

CORRECT ANSWER

13

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 79 • BACKWARD 3 • Q1/5

2919

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$21 - 9$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 79 • BACKWARD 3 • Q2/5

2920

QUESTION ID

S79-B3-Q2

TYPE

Short answer

CORRECT ANSWER

12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 79 • BACKWARD 3 • Q2/5

2920

True or false?

$$30 - 14 = 16$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 79 • BACKWARD 3 • Q3/5

2921

QUESTION ID

S79-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 79 • BACKWARD 3 • Q3/5

$$18 - 10 = ?$$

- A.** 7
- B.** 8
- C.** 9
- D.** 28

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 79 • BACKWARD 3 • Q4/5

2922

QUESTION ID

S79-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 8

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 79 • BACKWARD 3 • Q4/5

ANSWER KEY

Reduced copy of the question page, with its answer

Subtract.

$$40 - 23$$

Difference: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 79 • BACKWARD 3 • Q5/5

2923

QUESTION ID

S79-B3-Q5

TYPE

Short answer

CORRECT ANSWER

17

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-B3-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5 • Set 79 • BACKWARD 3 • Q5/5

2923

SET 79 • 6.SP.B.5.c

FORWARD 1

Difference in centers as a multiple of the IQR • Grade 7 • Statistics & Probability • 7.SP.B.3

Two data sets have medians of 40 and 55. Each has an IQR of 5.

The difference in medians is how many times the IQR?

Times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 79 • FORWARD 1 • Q1/5

2925

QUESTION ID

S79-F1-Q1

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 79 • FORWARD 1 • Q1/5

Two data sets have medians of 20 and 26. Each has an IQR of 3.

How many IQRs apart are the medians?

A. 2

B. 3

C. 6

D. 9

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 79 • FORWARD 1 • Q2/5

2926

QUESTION ID

S79-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

A. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 79 • FORWARD 1 • Q2/5

True or false?

Two data sets with medians that differ by less than one IQR probably overlap a lot.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 79 • FORWARD 1 • Q3/5

2927

QUESTION ID

S79-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 79 • FORWARD 1 • Q3/5

Class A's median is 72 and Class B's median is 84. Each class has an IQR of 12.

The difference in medians is how many times the IQR?

Times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 79 • FORWARD 1 • Q4/5

2928

QUESTION ID

S79-F1-Q4

TYPE

Short answer

CORRECT ANSWER

1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 79 • FORWARD 1 • Q4/5

Which pair of data sets has the LEAST overlap?

- A. Medians 10 and 14, each IQR 8
- B. Medians 10 and 30, each IQR 4
- C. Medians 10 and 11, each IQR 3
- D. Medians 10 and 12, each IQR 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 79 • FORWARD 1 • Q5/5

2929

QUESTION ID

S79-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

B. Medians 10 and 30, each IQR 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 79 • FORWARD 1 • Q5/5

FORWARD 2

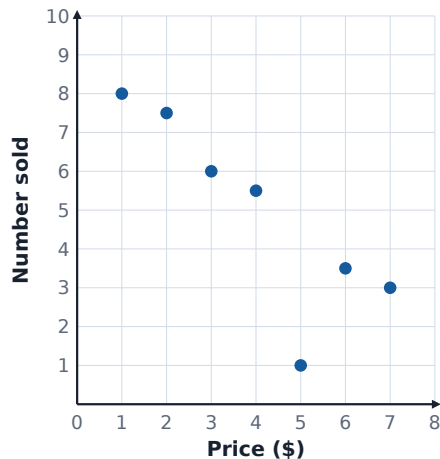
Outliers and clusters in scatter plots • Grade 8 • Statistics & Probability •
8.SP.A.1

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

Which point is an outlier?

Write its coordinates.

Outlier: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 79 • FORWARD 2 • Q1/5 • NEAREST RELATED

2931

QUESTION ID

S79-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(5, 1)

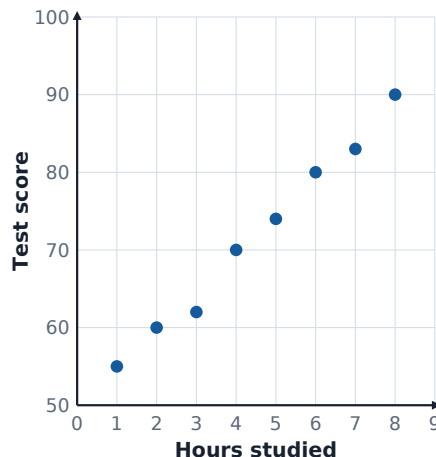
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 79 • FORWARD 2 • Q1/5 • NEAREST RELATED

Which point would be an outlier if it were added to the scatter plot?

- A. (5, 73)
- B. (7, 50)
- C. (7, 85)
- D. (3, 63)



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 79 • FORWARD 2 • Q2/5 • NEAREST RELATED

2932

QUESTION ID

S79-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. (7, 50)

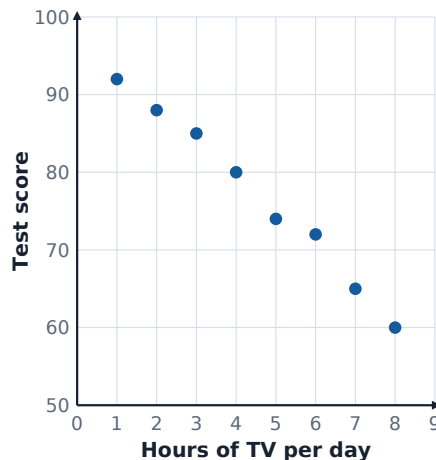
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 79 • FORWARD 2 • Q2/5 • NEAREST RELATED

Which describes the association in the scatter plot?

- A. No association
- B. Negative linear
- C. Nonlinear
- D. Positive linear



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 79 • FORWARD 2 • Q3/5 • NEAREST RELATED

2933

QUESTION ID

S79-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Negative linear

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

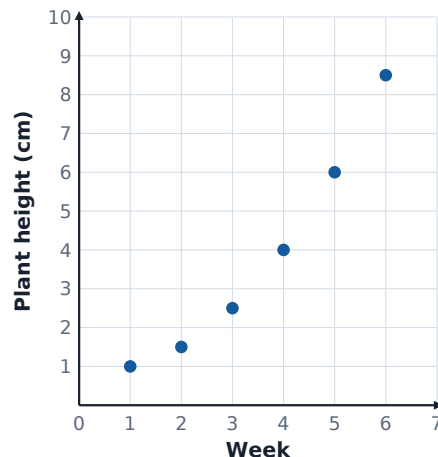
ID S79-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 79 • FORWARD 2 • Q3/5 • NEAREST RELATED

True or false?

The scatter plot has an outlier.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 79 • FORWARD 2 • Q4/5 • NEAREST RELATED

2934

QUESTION ID

S79-F2-Q4

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 79 • FORWARD 2 • Q4/5 • NEAREST RELATED

True or false?

Every scatter plot has at least one outlier.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 79 • FORWARD 2 • Q5/5 • NEAREST RELATED

2935

QUESTION ID

S79-F2-Q5

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S79-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 79 • FORWARD 2 • Q5/5 • NEAREST RELATED

Find the mean absolute deviation of a data set

MAIN	Grade 6 main question	6.SP.B.5.c
BACKWARD 1	Sharing a total equally (the mean)	5.MD.B.2
BACKWARD 2	Distance as absolute value	6.NS.C.7.c
BACKWARD 3	Dividing to find an average distance	5.NBT.B.7
FORWARD 1	Difference in means as a multiple of the MAD	7.SP.B.3
FORWARD 2	Judge model fit by distances from the line	8.SP.A.2

SET 80 • 6.SP.B.5.c

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean absolute deviation.

2, 4, 6, 8, 10

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 80 • MAIN • Q1/5

2938

QUESTION ID

S80-M-Q1

TYPE

Short answer

CORRECT ANSWER

2.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 80 • MAIN • Q1/5

What is the mean absolute deviation?

3, 3, 5, 7, 7

- A.** 0
- B.** 1.6
- C.** 2
- D.** 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 80 • MAIN • Q2/5

2939

QUESTION ID

S80-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 1.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 80 • MAIN • Q2/5

Find the mean absolute deviation.

5, 7, 8, 12

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 80 • MAIN • Q3/5

2940

QUESTION ID

S80-M-Q3

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 80 • MAIN • Q3/5

Find the mean absolute deviation.

6, 8, 10, 12, 14, 16

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 80 • MAIN • Q4/5

2941

QUESTION ID

S80-M-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 80 • MAIN • Q4/5

Find the mean absolute deviation.

20, 22, 25, 29

MAD: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 80 • MAIN • Q5/5

2942

QUESTION ID

S80-M-Q5

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 80 • MAIN • Q5/5

SET 80 • 6.SP.B.5.c

BACKWARD 1

Sharing a total equally (the mean) • Grade 5 • Measurement & Data • 5.MD.B.2

Five jars hold 2, 4, 6, 8, and 10 cups of rice. The rice is shared equally among the jars.

How much is in each jar?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q1

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 80 • BACKWARD 1 • Q1/5

2944

QUESTION ID

S80-B1-Q1

TYPE

Short answer

CORRECT ANSWER

6 cups

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q1

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 80 • BACKWARD 1 • Q1/5

Four friends have 3, 3, 7, and 7 marbles. They share the marbles equally.

How many marbles does each friend get?

Marbles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q2

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 80 • BACKWARD 1 • Q2/5

2945

QUESTION ID

S80-B1-Q2

TYPE

Short answer

CORRECT ANSWER

5 marbles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q2

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 80 • BACKWARD 1 • Q2/5

True or false?

Bags with 1, 5, and 6 pounds of flour are shared equally. Each bag then has 4 pounds.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q3

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 80 • BACKWARD 1 • Q3/5

2946

QUESTION ID

S80-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q3

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 80 • BACKWARD 1 • Q3/5

Six students have 2, 3, 3, 4, 5, and 7 pencils. They share equally.

How many pencils does each student get?

A. 3

B. 4

C. 5

D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q4

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 80 • BACKWARD 1 • Q4/5

2947

QUESTION ID

S80-B1-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q4

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 80 • BACKWARD 1 • Q4/5

Three cups hold $\frac{1}{2}$, $\frac{1}{2}$, and 2 cups of water. The water is shared equally.

How much is in each cup?

Cups: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q5

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 80 • BACKWARD 1 • Q5/5

2948

QUESTION ID

S80-B1-Q5

TYPE

Short answer

CORRECT ANSWER

1 cup

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B1-Q5

Grade 5 • Measurement & Data • 5.MD.B.2 • Set 80 • BACKWARD 1 • Q5/5

SET 80 • 6.SP.B.5.c

BACKWARD 2

Distance as absolute value • Grade 6 • The Number System • 6.NS.C.7.c

Find the value.

$$|2 - 6|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 80 • BACKWARD 2 • Q1/5

2950

QUESTION ID

S80-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q1

Grade 6 • The Number System • 6.NS.C.7.c • Set 80 • BACKWARD 2 • Q1/5

How far is 9 from 6 on a number line?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 80 • BACKWARD 2 • Q2/5

2951

QUESTION ID

S80-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q2

Grade 6 • The Number System • 6.NS.C.7.c • Set 80 • BACKWARD 2 • Q2/5

True or false?

$$|4 - 10| = 6$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 80 • BACKWARD 2 • Q3/5

2952

QUESTION ID

S80-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q3

Grade 6 • The Number System • 6.NS.C.7.c • Set 80 • BACKWARD 2 • Q3/5

What is the distance between 3 and 8 on a number line?

- A. -5
- B. 5
- C. 11
- D. 24

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 80 • BACKWARD 2 • Q4/5

2953

QUESTION ID

S80-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q4

Grade 6 • The Number System • 6.NS.C.7.c • Set 80 • BACKWARD 2 • Q4/5

Find the value.

$$|10 - 6|$$

Value: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 80 • BACKWARD 2 • Q5/5

2954

QUESTION ID

S80-B2-Q5

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B2-Q5

Grade 6 • The Number System • 6.NS.C.7.c • Set 80 • BACKWARD 2 • Q5/5

SET 80 • 6.SP.B.5.c

BACKWARD 3

Dividing to find an average distance • Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7

Divide.

$$12 \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 80 • BACKWARD 3 • Q1/5

2956

QUESTION ID

S80-B3-Q1

TYPE

Short answer

CORRECT ANSWER

2.4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q1

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 80 • BACKWARD 3 • Q1/5

ANSWER KEY

Reduced copy of the question page, with its answer

Divide.

$$8 \div 5$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 80 • BACKWARD 3 • Q2/5

2957

QUESTION ID

S80-B3-Q2

TYPE

Short answer

CORRECT ANSWER

1.6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q2

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 80 • BACKWARD 3 • Q2/5

2957

True or false?

$$18 \div 6 = 3$$

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 80 • BACKWARD 3 • Q3/5

2958

QUESTION ID

S80-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q3

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 80 • BACKWARD 3 • Q3/5

ANSWER KEY

Reduced copy of the question page, with its answer

$$10 \div 4 = ?$$

- A.** 0.4
- B.** 2.4
- C.** 2.5
- D.** 40

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 80 • BACKWARD 3 • Q4/5

2959

QUESTION ID

S80-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. 2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q4

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 80 • BACKWARD 3 • Q4/5

Divide.

$$9 \div 4$$

Quotient: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 80 • BACKWARD 3 • Q5/5

2960

QUESTION ID

S80-B3-Q5

TYPE

Short answer

CORRECT ANSWER

2.25

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-B3-Q5

Grade 5 • Number & Operations in Base Ten • 5.NBT.B.7 • Set 80 • BACKWARD 3 • Q5/5

SET 80 • 6.SP.B.5.c

FORWARD 1

Difference in means as a multiple of the MAD • Grade 7 • Statistics & Probability • 7.SP.B.3

Class A's mean is 72 and Class B's mean is 80. Each class has a MAD of 4.

The difference in means is how many times the MAD?

Times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 80 • FORWARD 1 • Q1/5

2962

QUESTION ID

S80-F1-Q1

TYPE

Short answer

CORRECT ANSWER

2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 80 • FORWARD 1 • Q1/5

Two data sets have means of 15 and 21. Each has a MAD of 2.

How many MADs apart are the means?

- A.** 2 **B.** 3
C. 6 **D.** 12

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 80 • FORWARD 1 • Q2/5

2963

QUESTION ID

S80-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 80 • FORWARD 1 • Q2/5

True or false?

Two data sets have means that differ by 1 MAD.
The data sets probably overlap a lot.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 80 • FORWARD 1 • Q3/5

2964

QUESTION ID

S80-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 80 • FORWARD 1 • Q3/5

Two teams have mean heights of 150 cm and 162 cm.
Each has a MAD of 4 cm.

The difference in means is how many times the MAD?

Times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 80 • FORWARD 1 • Q4/5

2965

QUESTION ID

S80-F1-Q4

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 80 • FORWARD 1 • Q4/5

Which pair of data sets has the LEAST overlap?

- A. Means 10 and 30, each MAD 2
- B. Means 10 and 12, each MAD 2
- C. Means 10 and 11, each MAD 3
- D. Means 10 and 14, each MAD 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 80 • FORWARD 1 • Q5/5

2966

QUESTION ID

S80-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

**A. Means 10 and 30,
each MAD 2**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 80 • FORWARD 1 • Q5/5

SET 80 • 6.SP.B.5.c

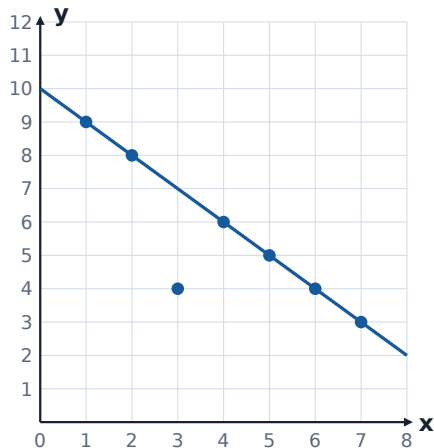
FORWARD 2

Judge model fit by distances from the line • Grade 8 • Statistics & Probability •
8.SP.A.2

Which point is farthest from the line?

Write its coordinates.

Point: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 80 • FORWARD 2 • Q1/5

2968

QUESTION ID

S80-F2-Q1

TYPE

Short answer

CORRECT ANSWER

(3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

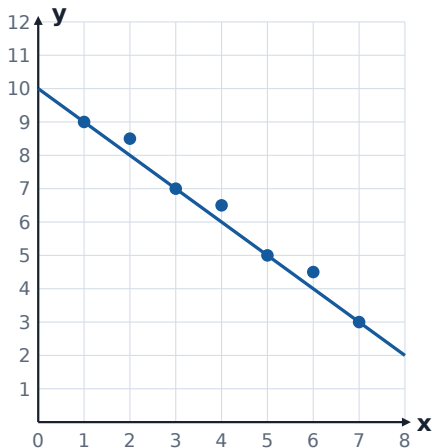
ID S80-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 80 • FORWARD 2 • Q1/5

True or false?

The points are close to the line, so it is a good fit.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 80 • FORWARD 2 • Q2/5

2969

QUESTION ID

S80-F2-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

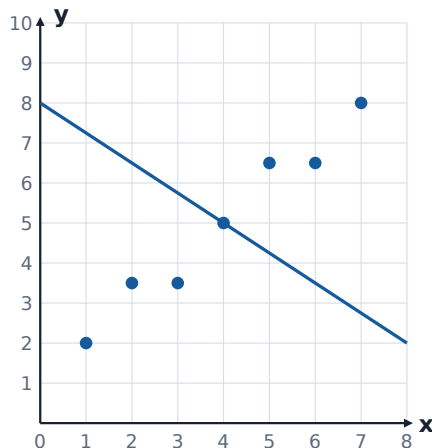
ID S80-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 80 • FORWARD 2 • Q2/5

True or false?

The line is a good fit for the data.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 80 • FORWARD 2 • Q3/5

2970

QUESTION ID

S80-F2-Q3

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 80 • FORWARD 2 • Q3/5

Which line fits data better?

- A. A line that touches only one point
- B. A line with all points far above it
- C. A line with points close to it on both sides
- D. A line with all points below it

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 80 • FORWARD 2 • Q4/5

2971

QUESTION ID

S80-F2-Q4

TYPE

Multiple choice

CORRECT ANSWER

**C. A line with points
close to it on both
sides**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

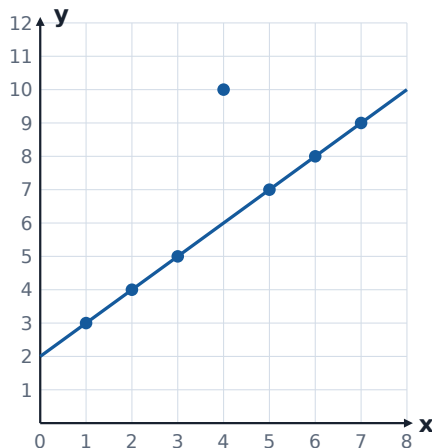
ID S80-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 80 • FORWARD 2 • Q4/5

For the point (4, 10), the line gives $y = 6$.

How far is the point from the line, measured vertically?

Distance: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 80 • FORWARD 2 • Q5/5

2972

QUESTION ID

S80-F2-Q5

TYPE

Short answer

CORRECT ANSWER

4 units

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S80-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 80 • FORWARD 2 • Q5/5

Describe the overall pattern and striking deviations in the context of the data

MAIN	Grade 6 main question	6.SP.B.5.c
BACKWARD 1	Where data cluster and where there are gaps	6.SP.A.2
BACKWARD 2	Reading a dot plot in context	4.MD.B.4
BACKWARD 3	Greatest and least values	2.NBT.A.4
FORWARD 1	Compare the patterns of two distributions	7.SP.B.3
FORWARD 2	Describe patterns and deviations in scatter plots in context	8.SP.A.1

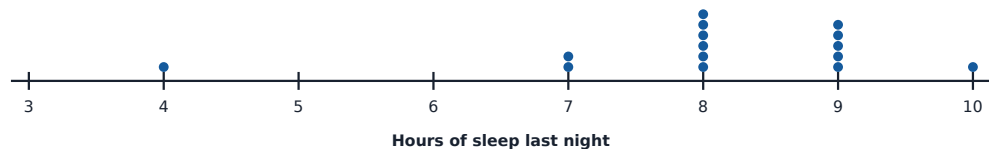
SET 81 • 6.SP.B.5.c

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.c

The dot plot shows how many hours students slept last night.

Describe the overall pattern. Which value stands apart, and what might explain it?



Overall pattern: _____

Value that stands apart: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 81 • MAIN • Q1/5

ID S81-M-Q1

2975

QUESTION ID

S81-M-Q1

TYPE

Short answer

CORRECT ANSWER

Most students slept 7 to 9 hours, clustering around 8 hours. One student slept only 4 hours, which stands apart; a reasonable explanation is that the student stayed up late, was sick, or had an early trip.

GRADING NOTE Must name 4 hours as the value that stands apart; any reasonable context explanation is correct

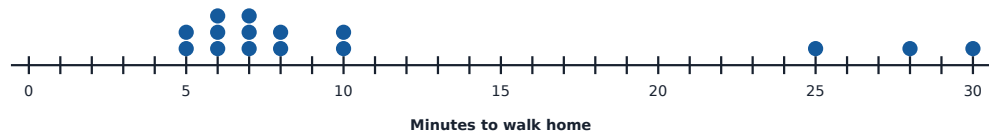
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 81 • MAIN • Q1/5

ID S81-M-Q1

The dot plot shows how long students take to walk home.

Which description best fits the data in context?



- A. The times are spread evenly from 5 to 30 minutes.
- B. All students take about the same amount of time.
- C. Most students take 25 to 30 minutes.
- D. Most students take 5 to 10 minutes; a few who live farther away take 25 to 30 minutes.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 81 • MAIN • Q2/5

2976

QUESTION ID

S81-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Most students take 5 to 10 minutes; a few who live farther away take 25 to 30 minutes.

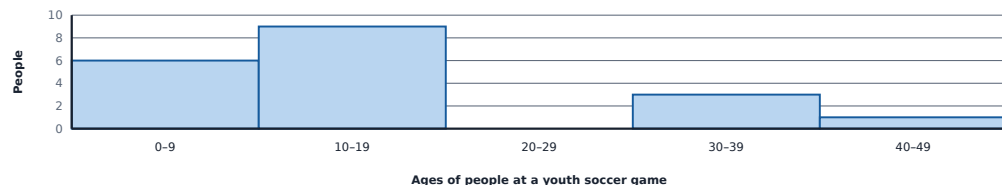
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 81 • MAIN • Q2/5

The histogram shows the ages of people at a youth soccer game.

Describe the overall pattern. Which ages stand apart, and who might they be?



Overall pattern: _____

Ages that stand apart: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 81 • MAIN • Q3/5

ID S81-M-Q3

2977

QUESTION ID

S81-M-Q3

TYPE

Short answer

CORRECT ANSWER

Most people are 0 to 19 years old (the young players). There are no people aged 20 to 29. A few people are 30 to 49; they are probably parents, coaches, or referees.

GRADING NOTE Must describe the cluster of young ages and name the 30–49 ages as standing apart, with a reasonable explanation

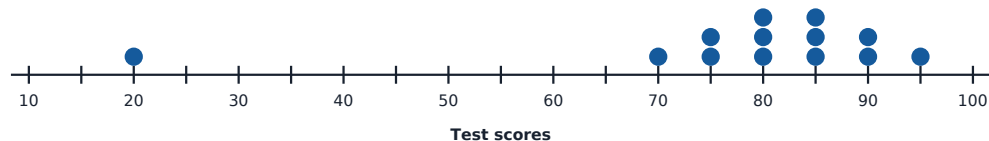
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 81 • MAIN • Q3/5

The dot plot shows the scores on a class test.

Which statement describes a striking deviation in context?



- A.** One student scored 20, far below the others, who scored from 70 to 95.
- B.** Most students scored below 50.
- C.** The scores are spread evenly from 20 to 95.
- D.** There are no unusual scores.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 81 • MAIN • Q4/5

2978

QUESTION ID

S81-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

A. One student scored 20, far below the others, who scored from 70 to 95.

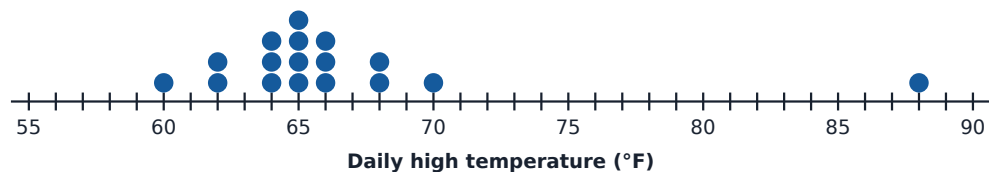
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 81 • MAIN • Q4/5

True or false?

The daily high temperatures cluster from 60°F to 70°F, with one unusually hot day at 88°F.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 81 • MAIN • Q5/5

2979

QUESTION ID

S81-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-M-Q5

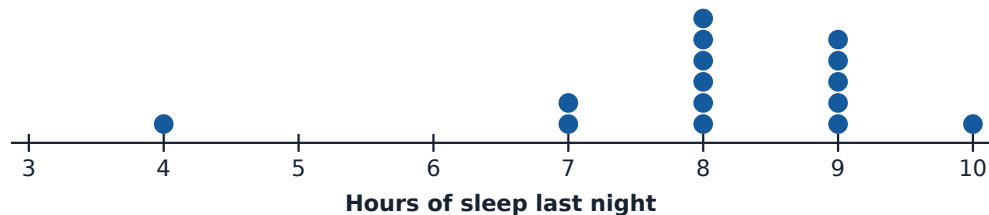
Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 81 • MAIN • Q5/5

SET 81 • 6.SP.B.5.c

BACKWARD 1

Where data cluster and where there are gaps • Grade 6 • Statistics & Probability • 6.SP.A.2

Between which two values do most of the data cluster?



Between: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 81 • BACKWARD 1 • Q1/5

2981

QUESTION ID

S81-B1-Q1

TYPE

Short answer

CORRECT ANSWER

7 and 9 hours

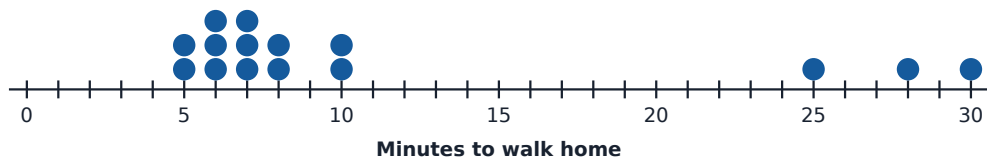
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 81 • BACKWARD 1 • Q1/5

True or false?

There is a gap in the data between 10 and 25 minutes.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 81 • BACKWARD 1 • Q2/5

2982

QUESTION ID

S81-B1-Q2

TYPE

True or false

CORRECT ANSWER

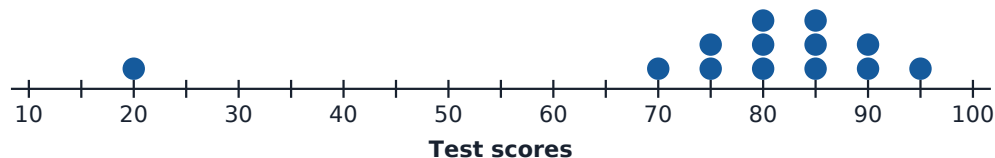
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 81 • BACKWARD 1 • Q2/5

Where is there a gap in the data?



- A.** Between 70 and 80 **B.** Between 20 and 70
C. Between 85 and 95 **D.** There is no gap.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 81 • BACKWARD 1 • Q3/5

2983

QUESTION ID

S81-B1-Q3

TYPE

Multiple choice

CORRECT ANSWER

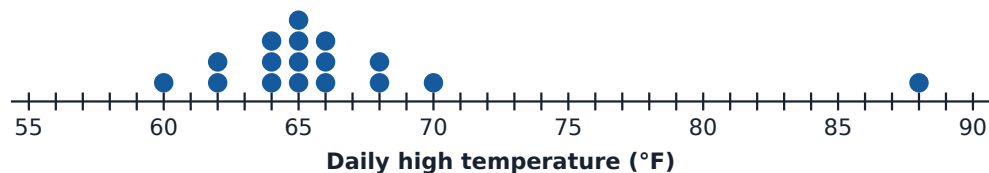
B. Between 20 and 70

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 81 • BACKWARD 1 • Q3/5

Between which two temperatures do most of the data cluster?



Between: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 81 • BACKWARD 1 • Q4/5

2984

QUESTION ID

S81-B1-Q4

TYPE

Short answer

CORRECT ANSWER

60°F and 70°F

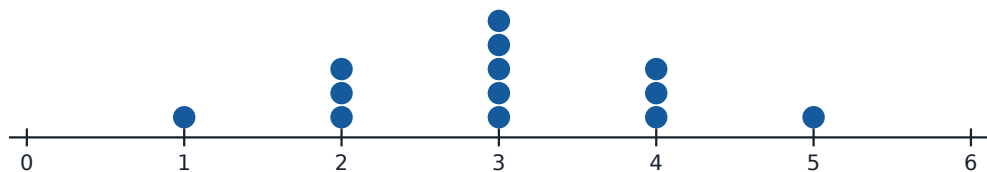
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 81 • BACKWARD 1 • Q4/5

True or false?

The data have no gaps.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 81 • BACKWARD 1 • Q5/5

2985

QUESTION ID

S81-B1-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B1-Q5

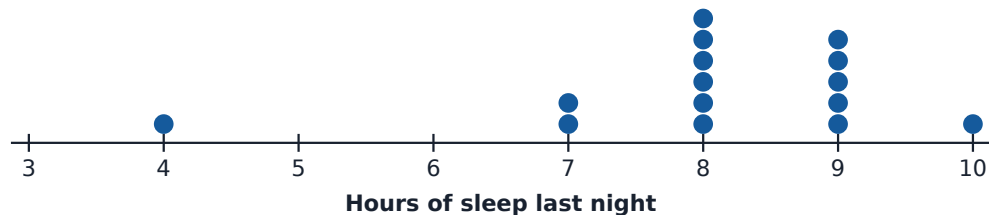
Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 81 • BACKWARD 1 • Q5/5

SET 81 • 6.SP.B.5.c

BACKWARD 2

Reading a dot plot in context • Grade 4 • Measurement & Data • 4.MD.B.4

How many students slept 8 hours last night?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 81 • BACKWARD 2 • Q1/5

2987

QUESTION ID

S81-B2-Q1

TYPE

Short answer

CORRECT ANSWER

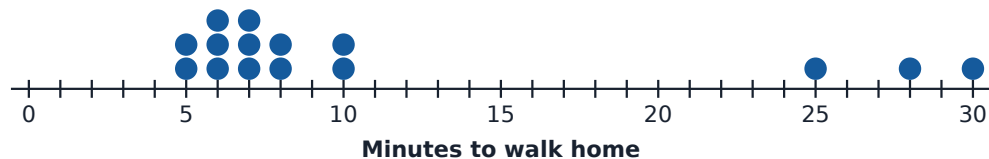
6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 81 • BACKWARD 2 • Q1/5

How many students take 7 minutes to walk home?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 81 • BACKWARD 2 • Q2/5

2988

QUESTION ID

S81-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

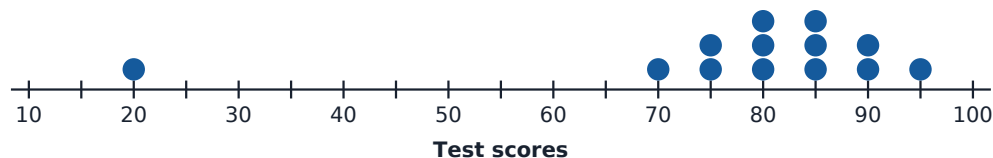
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 81 • BACKWARD 2 • Q2/5

True or false?

Three students scored 85 on the test.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 81 • BACKWARD 2 • Q3/5

2989

QUESTION ID

S81-B2-Q3

TYPE

True or false

CORRECT ANSWER

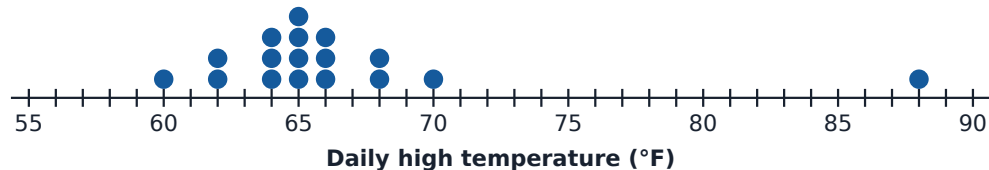
A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 81 • BACKWARD 2 • Q3/5

How many days had a high temperature of 65°F?



A. 3

B. 4

C. 5

D. 65

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 81 • BACKWARD 2 • Q4/5

2990

QUESTION ID

S81-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

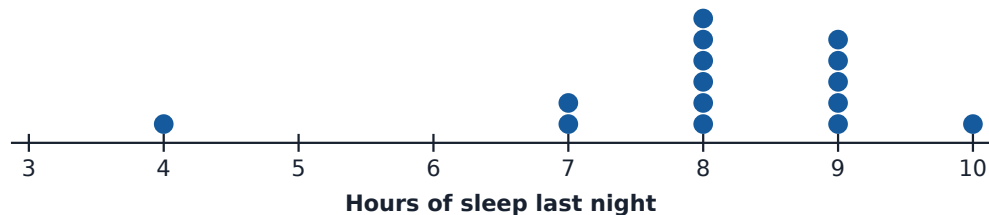
B. 4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 81 • BACKWARD 2 • Q4/5

What does each dot in the dot plot represent?



Each dot: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 81 • BACKWARD 2 • Q5/5

2991

QUESTION ID

S81-B2-Q5

TYPE

Short answer

CORRECT ANSWER

**One student's hours
of sleep last night**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B2-Q5

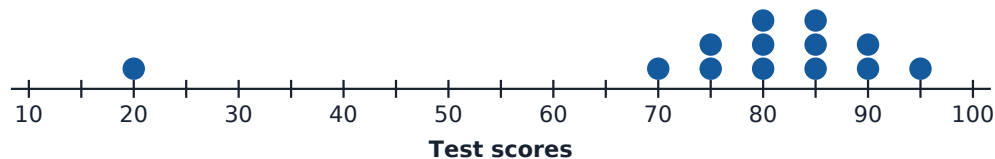
Grade 4 • Measurement & Data • 4.MD.B.4 • Set 81 • BACKWARD 2 • Q5/5

SET 81 • 6.SP.B.5.c

BACKWARD 3

Greatest and least values • Grade 2 • Number & Operations in Base Ten •
2.NBT.A.4

What is the least value in the data?



Least: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 81 • BACKWARD 3 • Q1/5

2993

QUESTION ID

S81-B3-Q1

TYPE

Short answer

CORRECT ANSWER

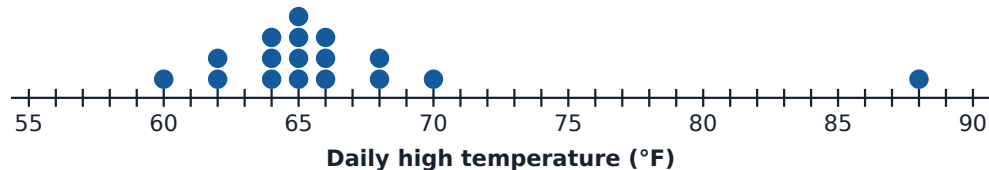
20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q1

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 81 • BACKWARD 3 • Q1/5

What is the greatest value in the data?



Greatest: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 81 • BACKWARD 3 • Q2/5

2994

QUESTION ID

S81-B3-Q2

TYPE

Short answer

CORRECT ANSWER

88°F

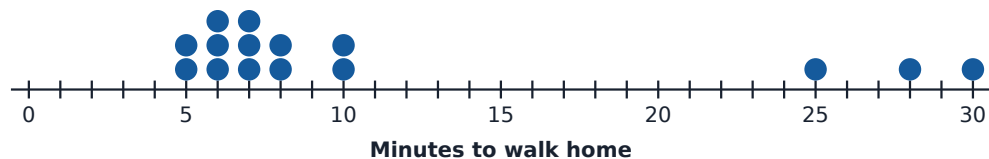
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q2

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 81 • BACKWARD 3 • Q2/5

True or false?

The greatest walking time is 30 minutes.

**A.** True**B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 81 • BACKWARD 3 • Q3/5

2995

QUESTION ID

S81-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q3

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 81 • BACKWARD 3 • Q3/5

Which number is least?

- A.** 20
- B.** 70
- C.** 75
- D.** 95

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 81 • BACKWARD 3 • Q4/5

2996

QUESTION ID

S81-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

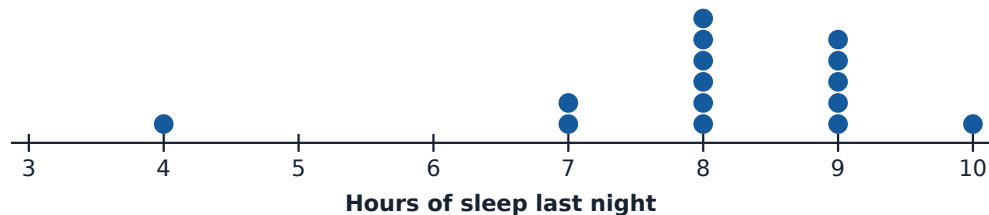
A. 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q4

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 81 • BACKWARD 3 • Q4/5

What is the least number of hours of sleep shown?



Least: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q5

Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 81 • BACKWARD 3 • Q5/5

2997

QUESTION ID

S81-B3-Q5

TYPE

Short answer

CORRECT ANSWER

4 hours

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-B3-Q5

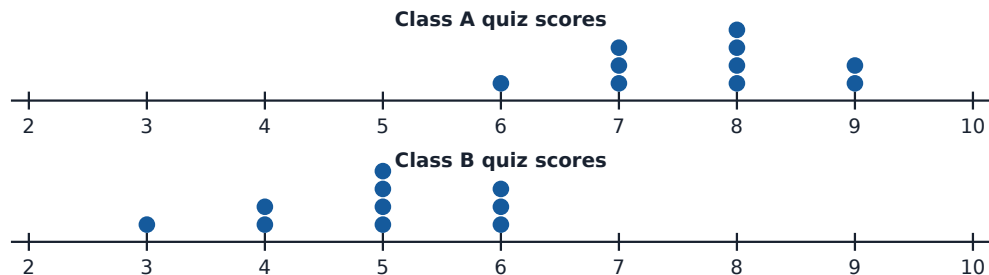
Grade 2 • Number & Operations in Base Ten • 2.NBT.A.4 • Set 81 • BACKWARD 3 • Q5/5

SET 81 • 6.SP.B.5.c

FORWARD 1

Compare the patterns of two distributions • Grade 7 • Statistics & Probability •
7.SP.B.3

Describe how the two distributions differ in context.



Comparison: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 81 • FORWARD 1 • Q1/5

2999

QUESTION ID

S81-F1-Q1

TYPE

Short answer

CORRECT ANSWER

Class A's scores cluster from 7 to 9 (center about 8). Class B's scores cluster from 4 to 6 (center about 5). Class A typically scored about 3 points higher; the spreads are similar, and the plots overlap only at 6.

GRADING NOTE Must compare the centers in context; mentioning spread or overlap is a plus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 81 • FORWARD 1 • Q1/5

Which statement describes both gardens?



- A.** The gardens have the same center. **B.** Garden A's plants are taller.
C. Garden B is much more spread out. **D.** Both cluster tightly, and Garden B's plants are about 6 cm taller.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 81 • FORWARD 1 • Q2/5

3000

QUESTION ID

S81-F1-Q2

TYPE

Multiple choice

CORRECT ANSWER

D. Both cluster tightly, and Garden B's plants are about 6 cm taller.

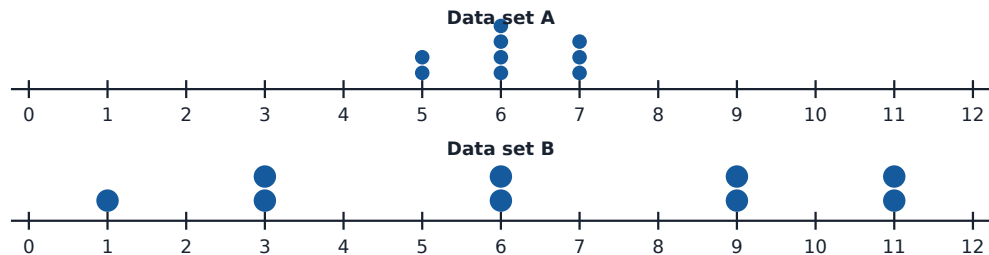
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 81 • FORWARD 1 • Q2/5

True or false?

Data set B is more spread out than data set A.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 81 • FORWARD 1 • Q3/5

3001

QUESTION ID

S81-F1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 81 • FORWARD 1 • Q3/5

Class A's quiz scores cluster from 7 to 9, with one score of 2. Class B's scores cluster from 4 to 6 with no unusual values.

Which class has a striking deviation?

Class: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 81 • FORWARD 1 • Q4/5

3002

QUESTION ID

S81-F1-Q4

TYPE

Short answer

CORRECT ANSWER

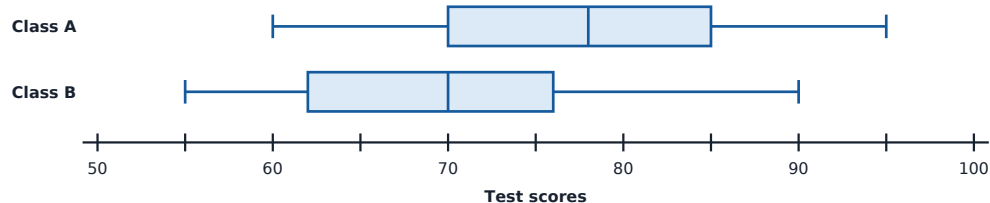
Class A

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 81 • FORWARD 1 • Q4/5

Which statement about the middle halves of the two box plots is true?



- A.** They overlap from 70 to 76. **B.** They do not overlap.
C. They overlap from 55 to 95. **D.** They are the same.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 81 • FORWARD 1 • Q5/5

3003

QUESTION ID

S81-F1-Q5

TYPE

Multiple choice

CORRECT ANSWER

**A. They overlap from
70 to 76.**

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.3 • Set 81 • FORWARD 1 • Q5/5

SET 81 • 6.SP.B.5.c

FORWARD 2

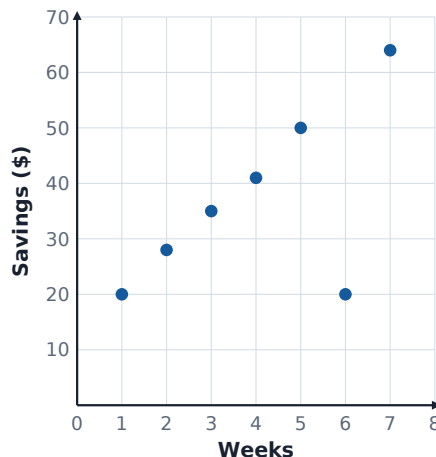
Describe patterns and deviations in scatter plots in context • Grade 8 •
Statistics & Probability • 8.SP.A.1

The scatter plot shows savings over several weeks.

Describe the overall pattern and the point that does not fit.

Pattern: _____

Point that does not fit: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 81 • FORWARD 2 • Q1/5

ID S81-F2-Q1

3005

QUESTION ID

S81-F2-Q1

TYPE

Short answer

CORRECT ANSWER

Savings grow steadily week by week (a positive, linear pattern of about \$7 to \$9 per week). The point (6, 20) does not fit; savings dropped that week, perhaps because money was spent.

GRADING NOTE Must describe the upward trend and name (6, 20)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

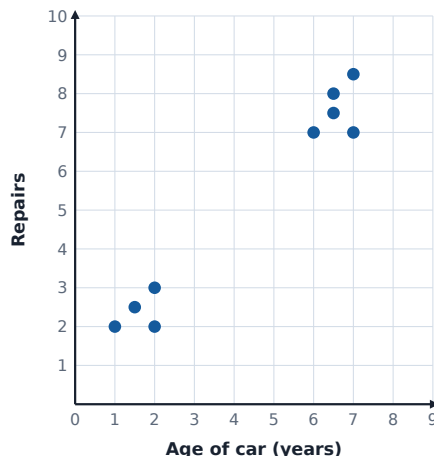
Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 81 • FORWARD 2 • Q1/5

ID S81-F2-Q1

The scatter plot shows the age of cars and the number of repairs.

Which description fits the data in context?

- A. Older cars have fewer repairs.
- B. Newer cars have few repairs and older cars have many repairs, forming two groups.
- C. All cars have the same number of repairs.
- D. There is no pattern.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 81 • FORWARD 2 • Q2/5

3006

QUESTION ID

S81-F2-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Newer cars have few repairs and older cars have many repairs, forming two groups.

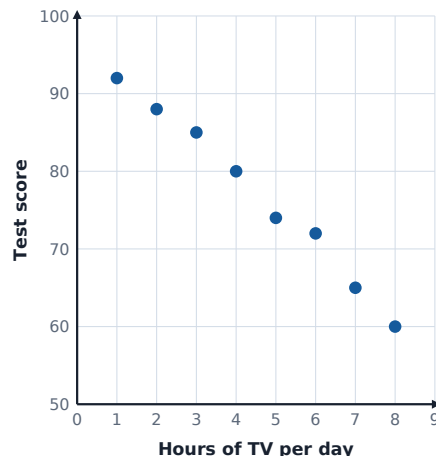
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 81 • FORWARD 2 • Q2/5

Which statement describes the association in context?

- A. As hours of TV increase, test scores tend to increase.
- B. Hours of TV and test scores are not related.
- C. Every student scored the same.
- D. As hours of TV increase, test scores tend to decrease.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 81 • FORWARD 2 • Q3/5

3007

QUESTION ID

S81-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

D. As hours of TV increase, test scores tend to decrease.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q3

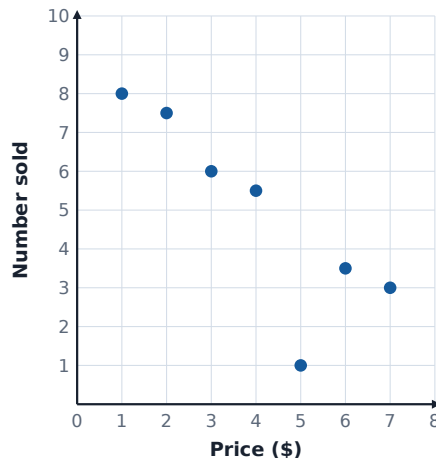
Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 81 • FORWARD 2 • Q3/5

The scatter plot shows the price of an item and the number sold.

Which point does not fit the pattern? What might explain it?

Point: _____

Possible reason: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 81 • FORWARD 2 • Q4/5

3008

QUESTION ID

S81-F2-Q4

TYPE

Short answer

CORRECT ANSWER

(5, 1). Far fewer items sold than the pattern predicts at \$5; a reasonable explanation is that the item was out of stock or it was a slow day.

GRADING NOTE Any reasonable explanation is correct

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

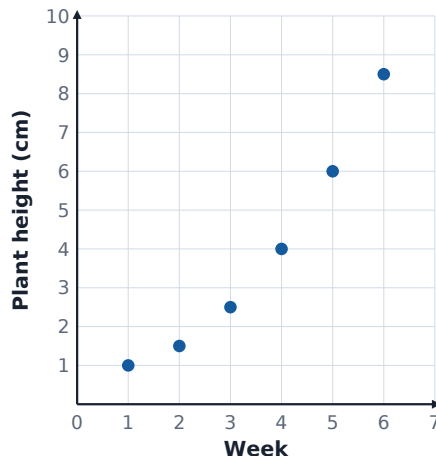
ID S81-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 81 • FORWARD 2 • Q4/5

True or false?

The plant grows faster each week, so the pattern in the scatter plot is nonlinear.

- A.** True
B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 81 • FORWARD 2 • Q5/5

3009

QUESTION ID

S81-F2-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S81-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.1 • Set 81 • FORWARD 2 • Q5/5

Choose measures of center and variability based on the shape of the data

MAIN	Grade 6 main question	6.SP.B.5.d
BACKWARD 1	Shape of a distribution	6.SP.A.2
BACKWARD 2	Computing the mean and median	6.SP.B.5.c
BACKWARD 3	Reading a dot plot	4.MD.B.4
FORWARD 1	Compare populations using appropriate measures	7.SP.B.4
FORWARD 2	Judge how well a line fits the data — nearest related (no direct Gra...	8.SP.A.2

SET 82 • 6.SP.B.5.d

MAIN

Grade 6 main question • Grade 6 • Statistics & Probability • 6.SP.B.5.d

Data: 20, 22, 23, 25, 90

Which measure of center better describes a typical value?

- A.** Mean
- B.** Median

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 82 • MAIN • Q1/5

3012

QUESTION ID

S82-M-Q1

TYPE

Multiple choice

CORRECT ANSWER

B. Median

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 82 • MAIN • Q1/5

A data set is skewed right and has an outlier.

Which pair of measures best describes its center and spread?

- A.** Mode and range
- B.** Median and IQR
- C.** Mean and range
- D.** Mean and MAD

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 82 • MAIN • Q2/5

3013

QUESTION ID

S82-M-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Median and IQR

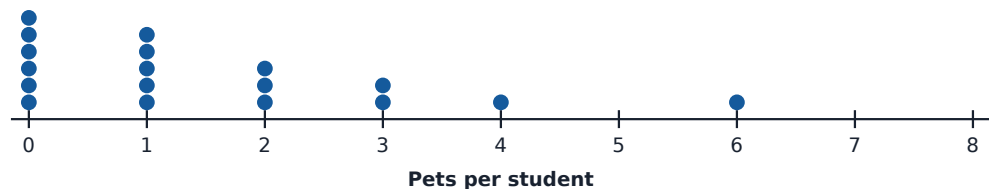
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 82 • MAIN • Q2/5

The dot plot is skewed right.

Which measure of center better describes the typical number of pets?



A. Mean

B. Median

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 82 • MAIN • Q3/5

3014

QUESTION ID

S82-M-Q3

TYPE

Multiple choice

CORRECT ANSWER

B. Median

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 82 • MAIN • Q3/5

Salaries: \$30,000; \$32,000; \$35,000; \$38,000;
\$250,000

Which measure of center better describes a typical salary?

- A.** Mean
- B.** Median

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 82 • MAIN • Q4/5

3015

QUESTION ID

S82-M-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. Median

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 82 • MAIN • Q4/5

True or false?

Heights of 20 sixth graders are symmetric with no outliers, so the mean is a good measure of center.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 82 • MAIN • Q5/5

3016

QUESTION ID

S82-M-Q5

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-M-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.d • Set 82 • MAIN • Q5/5

SET 82 • 6.SP.B.5.d

BACKWARD 1

Shape of a distribution • Grade 6 • Statistics & Probability • 6.SP.A.2

Which word describes the shape of the data?



- A.** Skewed right **B.** Symmetric
C. Skewed left **D.** Uniform

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 82 • BACKWARD 1 • Q1/5

3018

QUESTION ID

S82-B1-Q1

TYPE

Multiple choice

CORRECT ANSWER

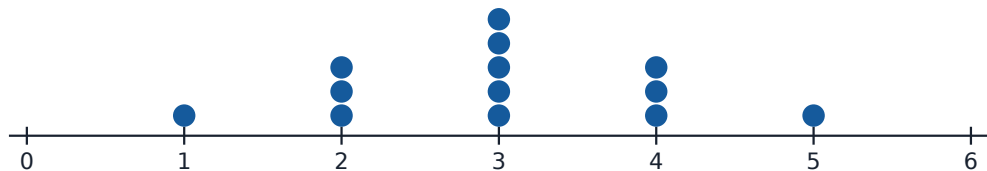
C. Skewed left

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q1

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 82 • BACKWARD 1 • Q1/5

Which word describes the shape of the data?



- A.** Skewed left **B.** Symmetric
C. Uniform **D.** Skewed right

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 82 • BACKWARD 1 • Q2/5

3019

QUESTION ID

S82-B1-Q2

TYPE

Multiple choice

CORRECT ANSWER

B. Symmetric

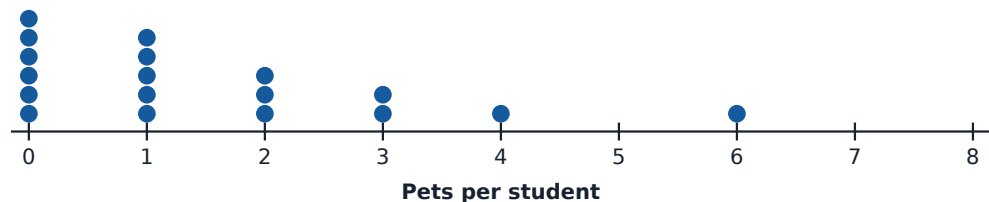
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q2

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 82 • BACKWARD 1 • Q2/5

True or false?

The data are skewed right.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 82 • BACKWARD 1 • Q3/5

3020

QUESTION ID

S82-B1-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q3

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 82 • BACKWARD 1 • Q3/5

True or false?

A data set with most values on the left and a long tail to the right is skewed right.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 82 • BACKWARD 1 • Q4/5

3021

QUESTION ID

S82-B1-Q4

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q4

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 82 • BACKWARD 1 • Q4/5

Which value in the data set is an outlier?

5, 6, 6, 7, 8, 30

- A.** 5
- B.** 6
- C.** 8
- D.** 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 82 • BACKWARD 1 • Q5/5

3022

QUESTION ID

S82-B1-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. 30

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B1-Q5

Grade 6 • Statistics & Probability • 6.SP.A.2 • Set 82 • BACKWARD 1 • Q5/5

SET 82 • 6.SP.B.5.d

BACKWARD 2

Computing the mean and median • Grade 6 • Statistics & Probability •
6.SP.B.5.c

Find the mean.

2, 3, 3, 4, 8

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 82 • BACKWARD 2 • Q1/5

3024

QUESTION ID

S82-B2-Q1

TYPE

Short answer

CORRECT ANSWER

4

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q1

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 82 • BACKWARD 2 • Q1/5

Find the median.

2, 3, 3, 4, 8

Median: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 82 • BACKWARD 2 • Q2/5

3025

QUESTION ID

S82-B2-Q2

TYPE

Short answer

CORRECT ANSWER

3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q2

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 82 • BACKWARD 2 • Q2/5

True or false?

The mean of 10, 20, 30 is 20.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 82 • BACKWARD 2 • Q3/5

3026

QUESTION ID

S82-B2-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q3

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 82 • BACKWARD 2 • Q3/5

What is the median of 1, 4, 6, 9?

- A.** 4
- B.** 5
- C.** 6
- D.** 20

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 82 • BACKWARD 2 • Q4/5

3027

QUESTION ID

S82-B2-Q4

TYPE

Multiple choice

CORRECT ANSWER

B. 5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q4

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 82 • BACKWARD 2 • Q4/5

Find the mean.

5, 5, 6, 8, 11

Mean: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 82 • BACKWARD 2 • Q5/5

3028

QUESTION ID

S82-B2-Q5

TYPE

Short answer

CORRECT ANSWER

7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B2-Q5

Grade 6 • Statistics & Probability • 6.SP.B.5.c • Set 82 • BACKWARD 2 • Q5/5

SET 82 • 6.SP.B.5.d

BACKWARD 3

Reading a dot plot • Grade 4 • Measurement & Data • 4.MD.B.4

How many students scored 9 on the quiz?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 82 • BACKWARD 3 • Q1/5

3030

QUESTION ID

S82-B3-Q1

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q1

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 82 • BACKWARD 3 • Q1/5

What is the lowest quiz score shown?



Score: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 82 • BACKWARD 3 • Q2/5

3031

QUESTION ID

S82-B3-Q2

TYPE

Short answer

CORRECT ANSWER

3

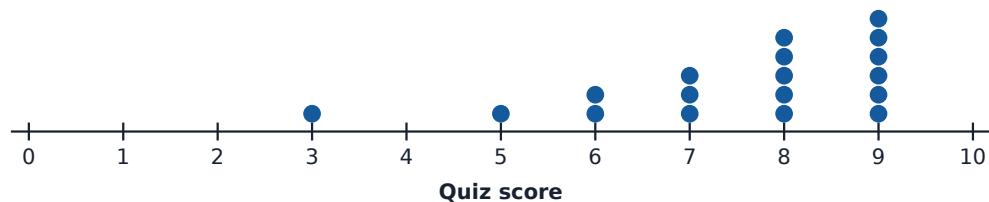
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q2

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 82 • BACKWARD 3 • Q2/5

True or false?

Five students scored 8.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 82 • BACKWARD 3 • Q3/5

3032

QUESTION ID

S82-B3-Q3

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q3

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 82 • BACKWARD 3 • Q3/5

How many students took the quiz?



A. 6

B. 9

C. 15

D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 82 • BACKWARD 3 • Q4/5

3033

QUESTION ID

S82-B3-Q4

TYPE

Multiple choice

CORRECT ANSWER

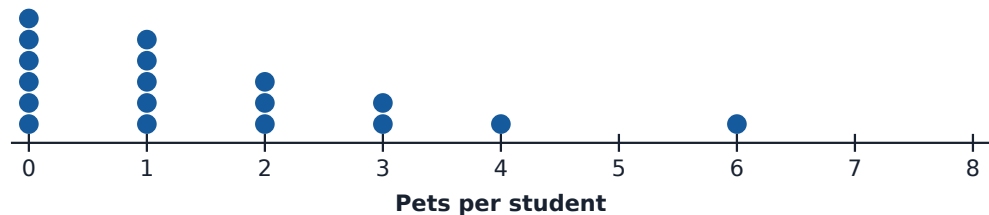
D. 18

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q4

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 82 • BACKWARD 3 • Q4/5

How many students had 0 pets?



Students: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 82 • BACKWARD 3 • Q5/5

3034

QUESTION ID

S82-B3-Q5

TYPE

Short answer

CORRECT ANSWER

6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-B3-Q5

Grade 4 • Measurement & Data • 4.MD.B.4 • Set 82 • BACKWARD 3 • Q5/5

SET 82 • 6.SP.B.5.d

FORWARD 1

Compare populations using appropriate measures • Grade 7 • Statistics & Probability • 7.SP.B.4

Two random samples of house prices are both skewed right with outliers.

Which measures are best for comparing them?

- A.** Medians and IQRs
- B.** Minimums
- C.** Means and MADs
- D.** Maximums

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 82 • FORWARD 1 • Q1/5

3036

QUESTION ID

S82-F1-Q1

TYPE

Multiple choice

CORRECT ANSWER

A. Medians and IQRs

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q1

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 82 • FORWARD 1 • Q1/5

True or false?

When two samples are symmetric with no outliers, comparing their means is appropriate.

- A.** True
- B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 82 • FORWARD 1 • Q2/5

3037

QUESTION ID

S82-F1-Q2

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q2

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 82 • FORWARD 1 • Q2/5

Sample A: median 30, IQR 6. Sample B: median 45, IQR 6.

The difference in medians is how many times the IQR?

Times: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 82 • FORWARD 1 • Q3/5

3038

QUESTION ID

S82-F1-Q3

TYPE

Short answer

CORRECT ANSWER

2.5

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q3

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 82 • FORWARD 1 • Q3/5

Class A test scores have mean 78 and MAD 5. Class B test scores have mean 78 and MAD 12.

Which statement is true?

- A.** Class A scored higher.
- B.** Class B scored higher.
- C.** Class B's scores vary more.
- D.** Class A's scores vary more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 82 • FORWARD 1 • Q4/5

3039

QUESTION ID

S82-F1-Q4

TYPE

Multiple choice

CORRECT ANSWER

C. Class B's scores vary more.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q4

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 82 • FORWARD 1 • Q4/5

Random samples of puzzle times are both skewed right. Group X has a median of 14 minutes, and Group Y has a median of 20 minutes. Their IQRs are similar.

Which group typically takes longer?

Group: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 82 • FORWARD 1 • Q5/5

3040

QUESTION ID

S82-F1-Q5

TYPE

Short answer

CORRECT ANSWER

Group Y

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F1-Q5

Grade 7 • Statistics & Probability • 7.SP.B.4 • Set 82 • FORWARD 1 • Q5/5

FORWARD 2

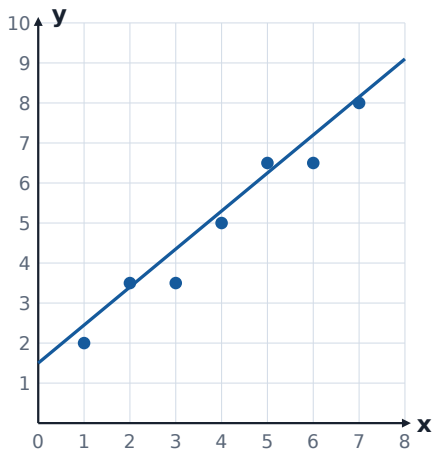
Judge how well a line fits the data • Grade 8 • Statistics & Probability • 8.SP.A.2

► No Grade 8 standard directly continues this skill. This branch uses the nearest related Grade 8 standard.

True or false?

The line is a good fit for the data.

- A.** True
- B.** False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 82 • FORWARD 2 • Q1/5 • NEAREST RELATED

3042

QUESTION ID

S82-F2-Q1

TYPE

True or false

CORRECT ANSWER

A. True

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

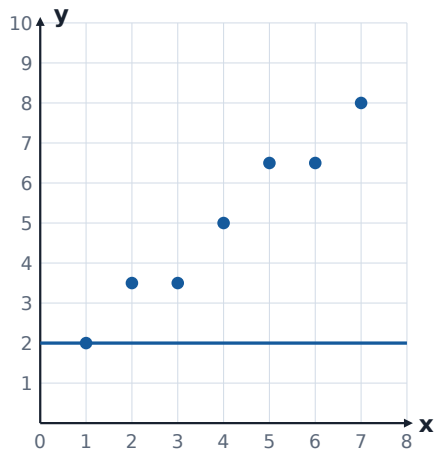
ID S82-F2-Q1

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 82 • FORWARD 2 • Q1/5 • NEAREST RELATED

True or false?

The line is a good fit for the data.

- A. True
- B. False



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 82 • FORWARD 2 • Q2/5 • NEAREST RELATED

3043

QUESTION ID

S82-F2-Q2

TYPE

True or false

CORRECT ANSWER

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q2

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 82 • FORWARD 2 • Q2/5 • NEAREST RELATED

A line of best fit should show ____.

- A. only the largest value
- B. only the smallest value
- C. only the first point
- D. the overall trend of the data

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 82 • FORWARD 2 • Q3/5 • NEAREST RELATED

3044

QUESTION ID

S82-F2-Q3

TYPE

Multiple choice

CORRECT ANSWER

**D. the overall trend
of the data**

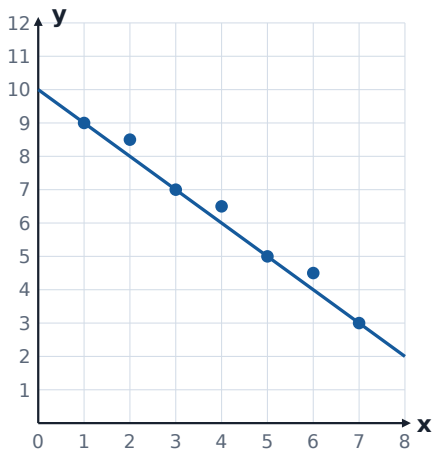
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q3

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 82 • FORWARD 2 • Q3/5 • NEAREST RELATED

Use the line of best fit to predict y when $x = 2$.

$y =$ _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 82 • FORWARD 2 • Q4/5 • NEAREST RELATED

3045

QUESTION ID

S82-F2-Q4

TYPE

Short answer

CORRECT ANSWER

8

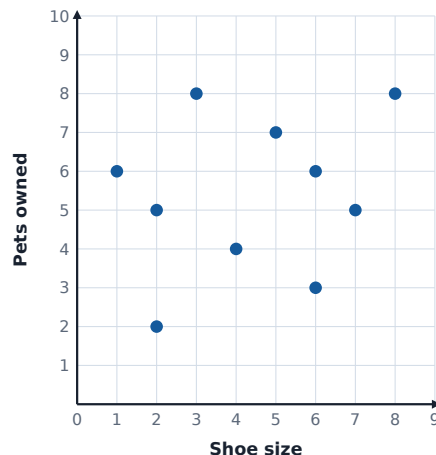
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q4

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 82 • FORWARD 2 • Q4/5 • NEAREST RELATED

Why should a scatter plot with no association NOT be modeled with a straight line?

- A. There are too many points.
- B. Lines must pass through the origin.
- C. The axes are labeled.
- D. There is no linear trend to model.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 82 • FORWARD 2 • Q5/5 • NEAREST RELATED

3046

QUESTION ID

S82-F2-Q5

TYPE

Multiple choice

CORRECT ANSWER

D. There is no linear trend to model.

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S82-F2-Q5

Grade 8 • Statistics & Probability • 8.SP.A.2 • Set 82 • FORWARD 2 • Q5/5 • NEAREST RELATED